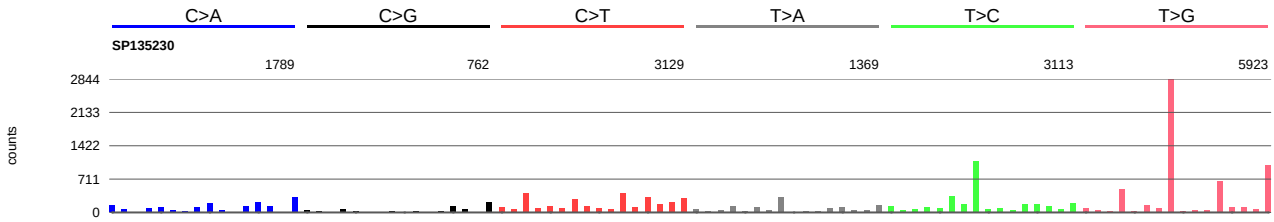
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SP135230
1789
762
3129
1369
3113
5923
counts
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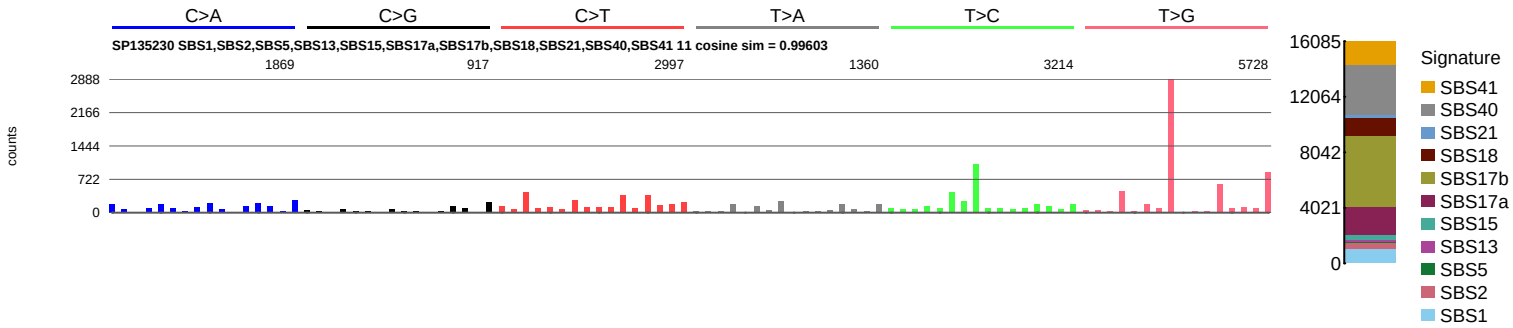
2133
1422
711
0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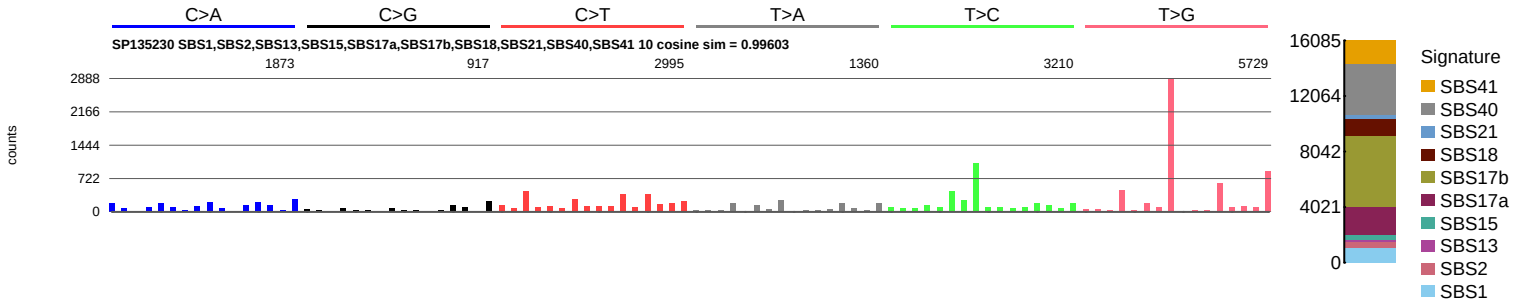
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99603
1869
917
2997
1360
3214
5728
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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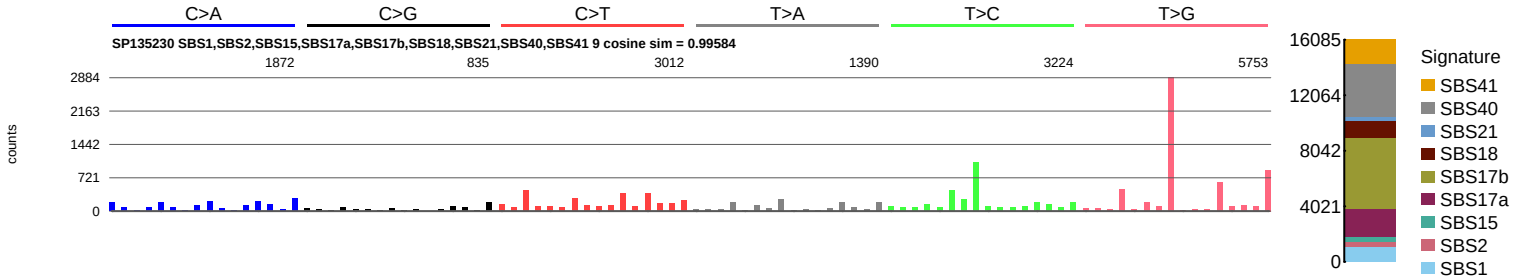
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99603
1873
917
2995
1360
3210
5729
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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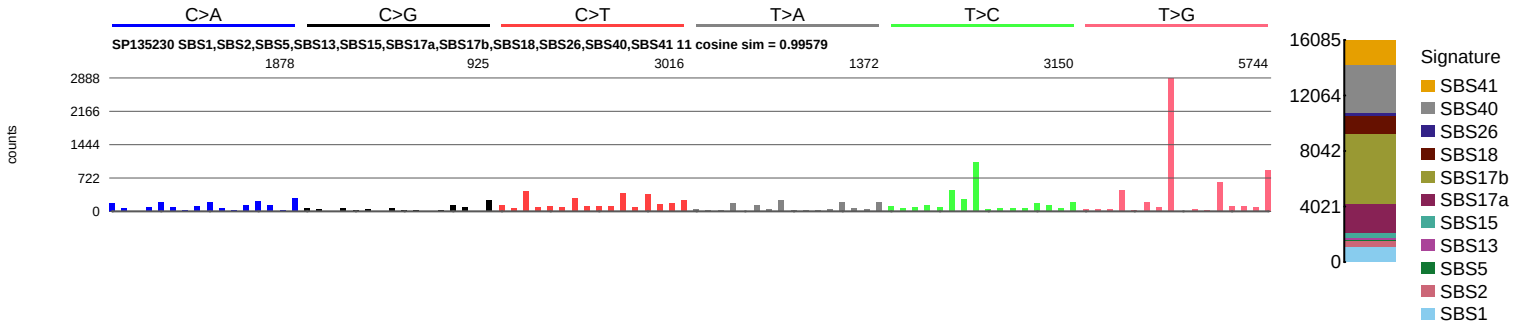
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99584
1872
835
3012
1390
3224
5753
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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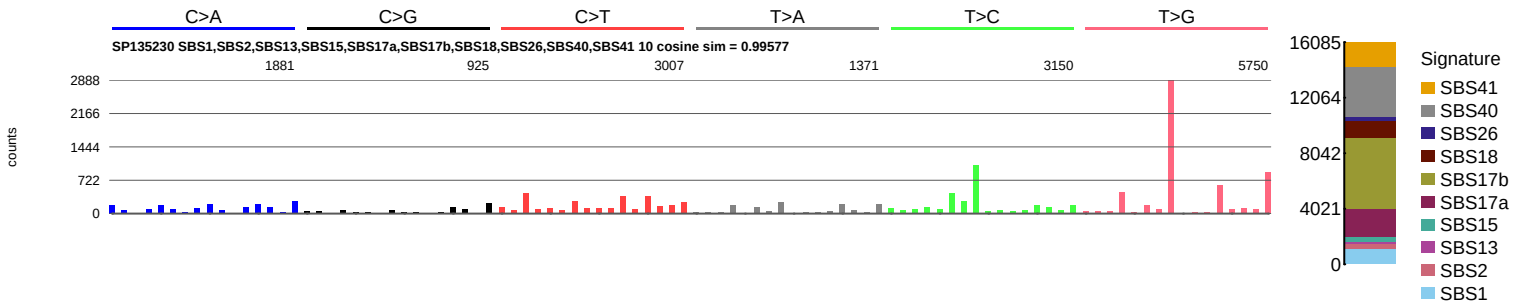
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.99579
1878
925
3016
1372
3150
5744
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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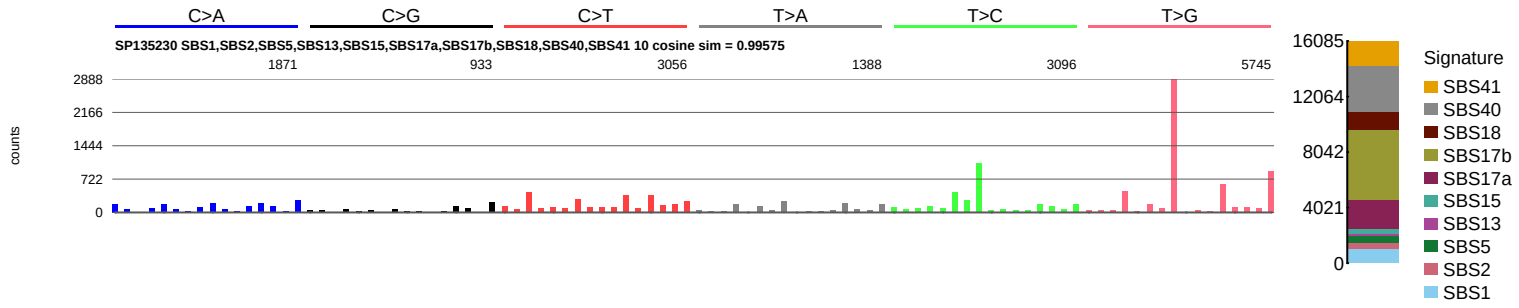
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99577
1881
925
3007
1371
3150
5750
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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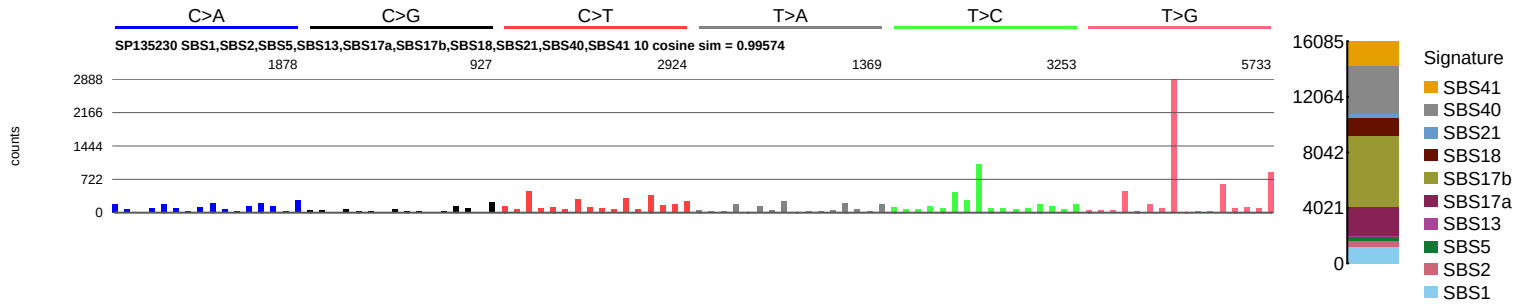
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.99575
1871
933
3056
1388
3096
5745
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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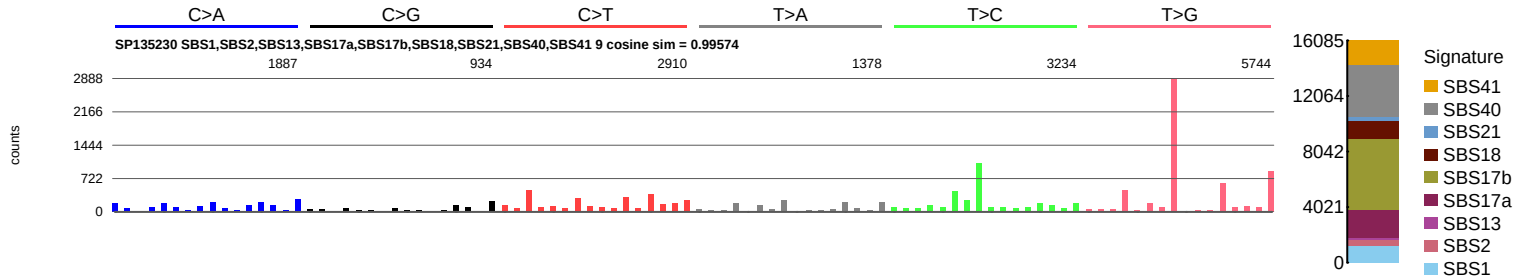
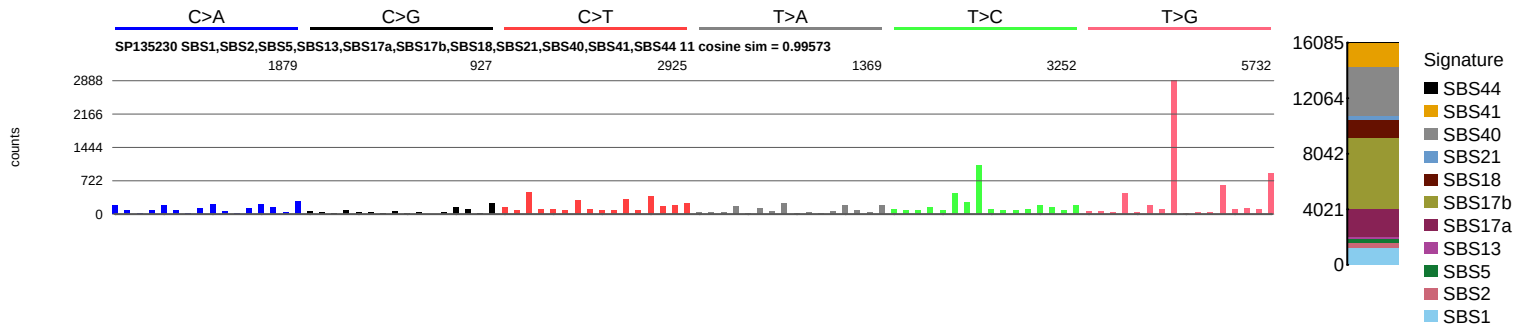
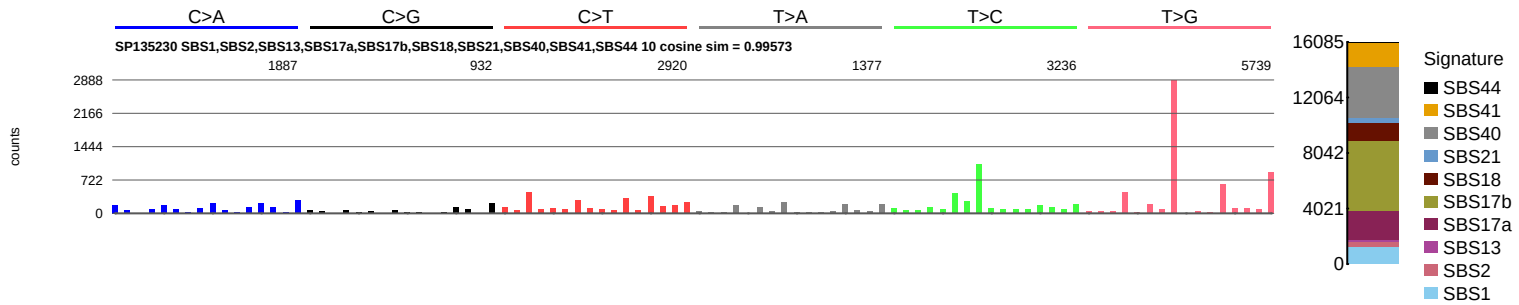
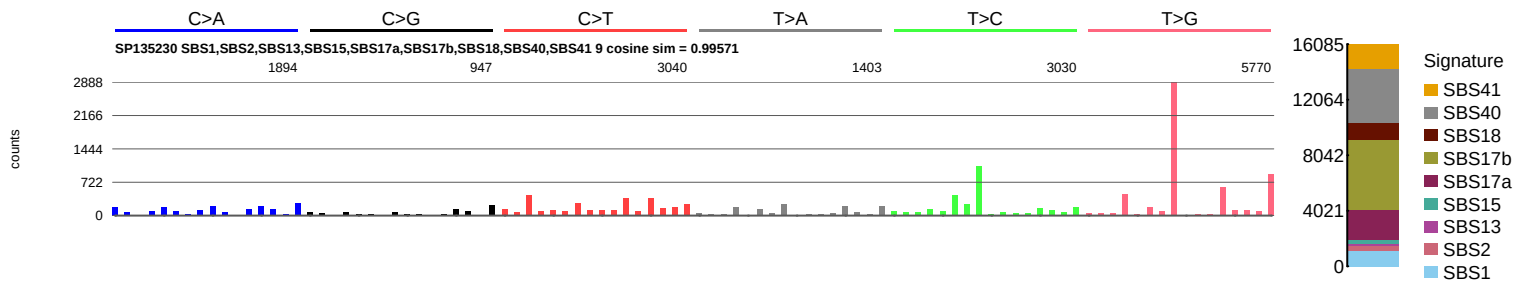
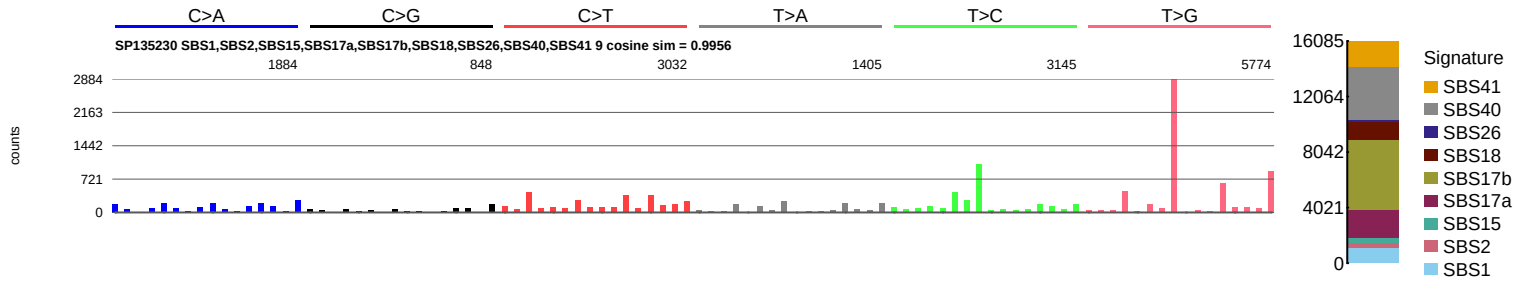
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99574
1878
927
2924
1369
3253
5733
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

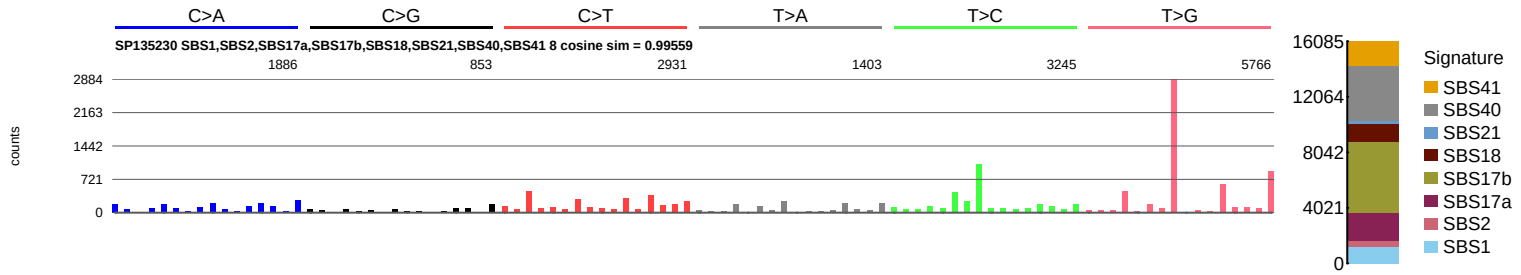

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99574
1887
934
2910
1378
3234
5744
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

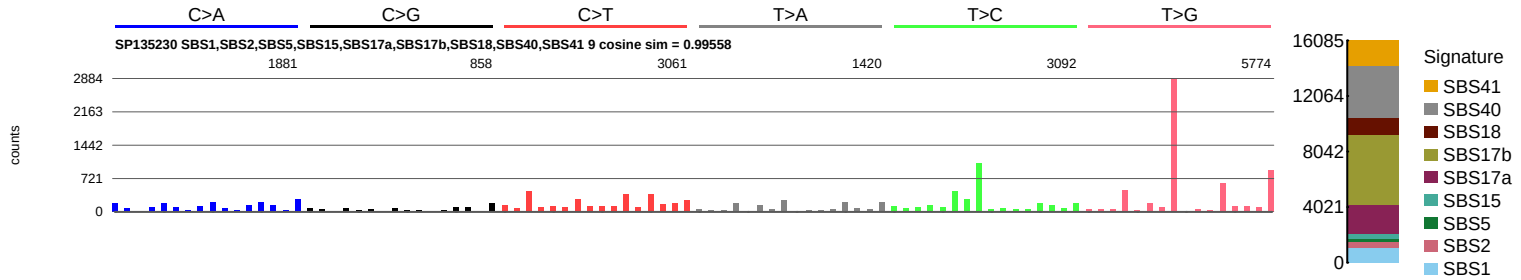

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99573
1879
927
2925
1369
3252
5732
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

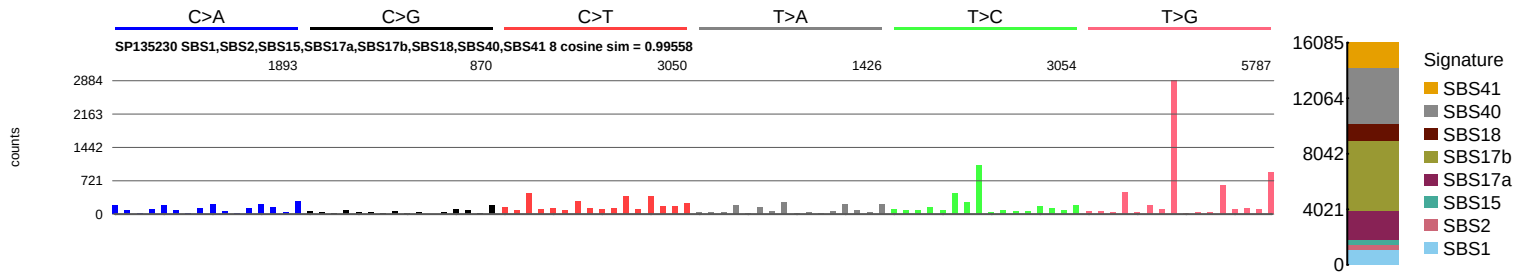

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99573
1887
932
2920
1377
3236
5739
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99571
1894
947
3040
1403
3030
5770
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.9956
1884
848
3032
1405
3145
5774
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99559
1886
853
2931
1403
3245
5766
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99558
1881
858
3061
1420
3092
5774
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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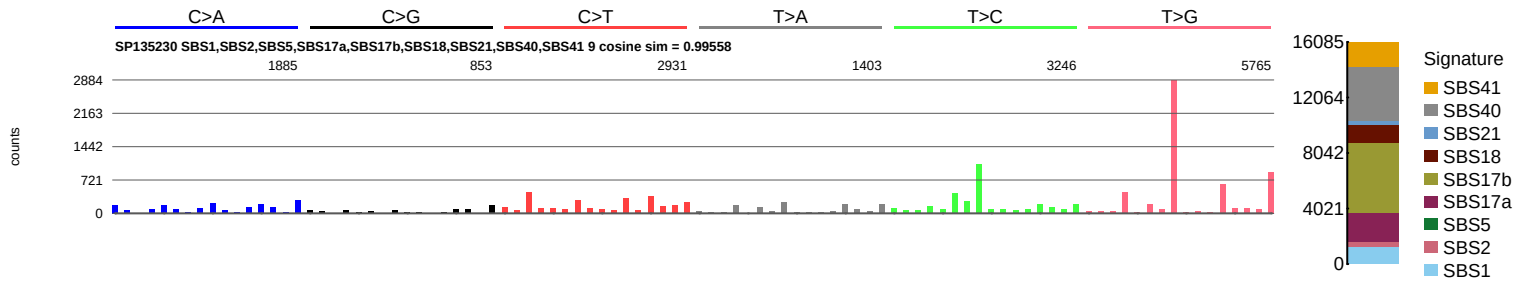
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99558
1893
870
3050
1426
3054
5787
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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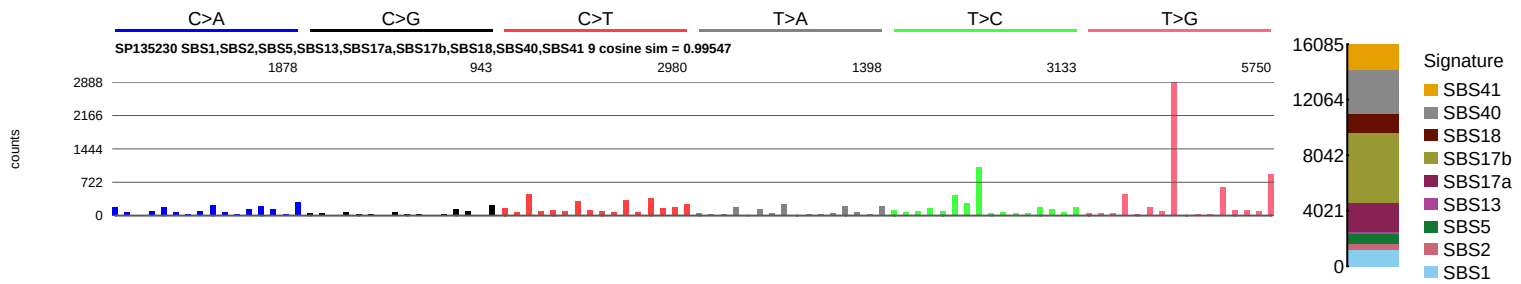
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99558
1885
853
2931
1403
3246
5765
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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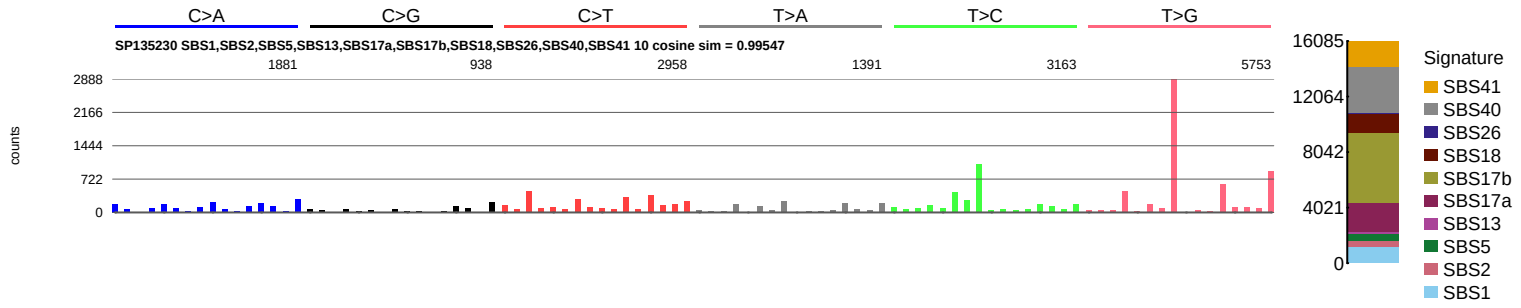
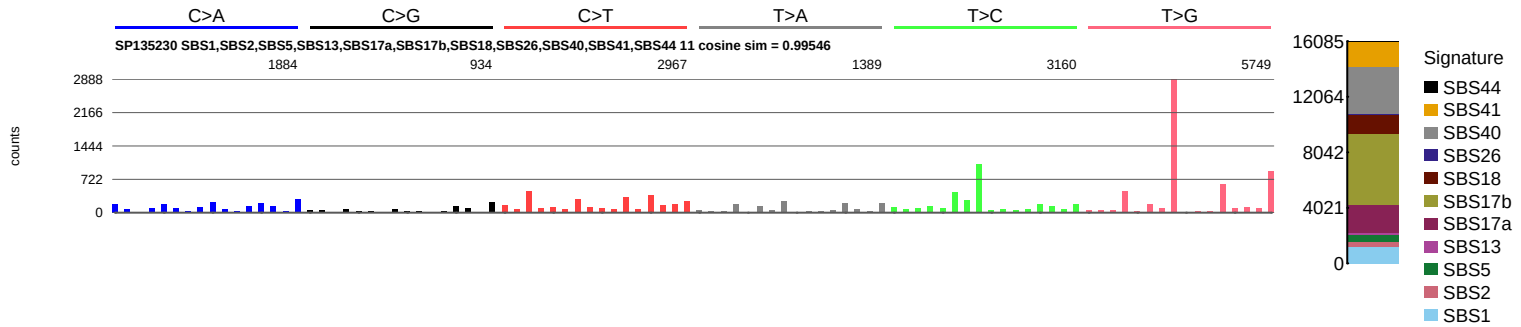
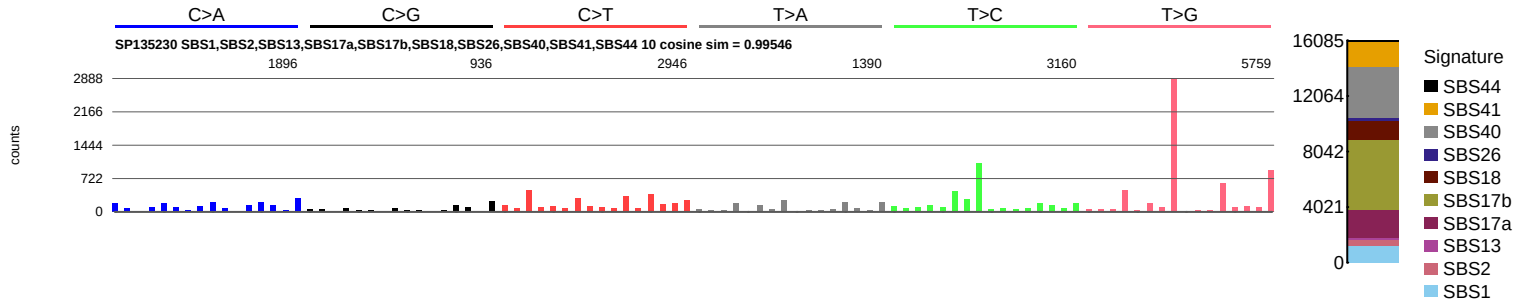
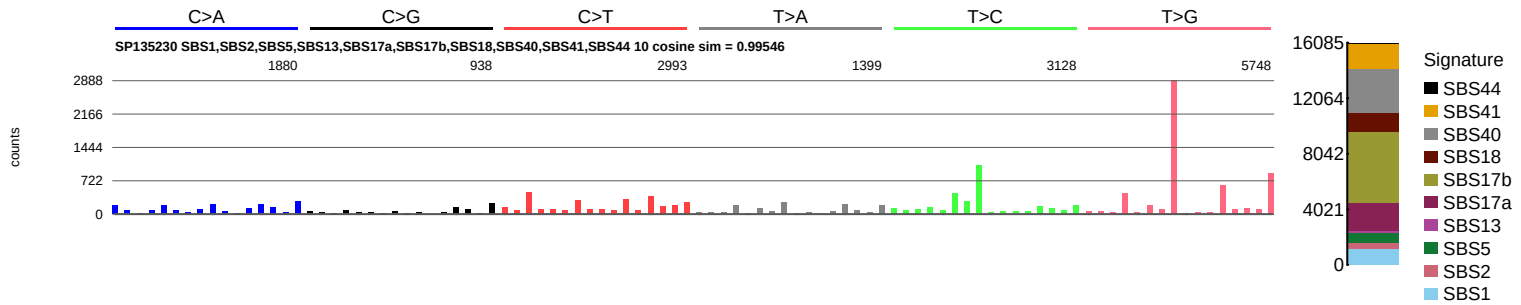
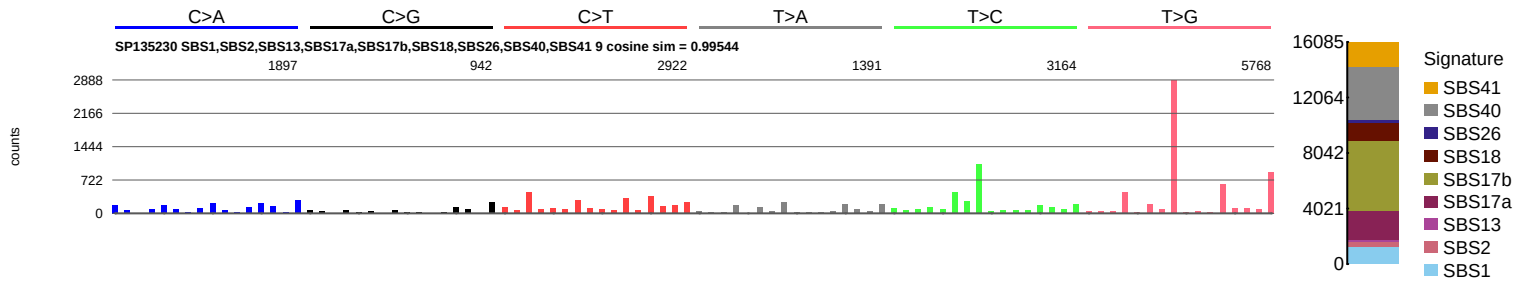
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99547
1878
943
2980
1398
3133
5750
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

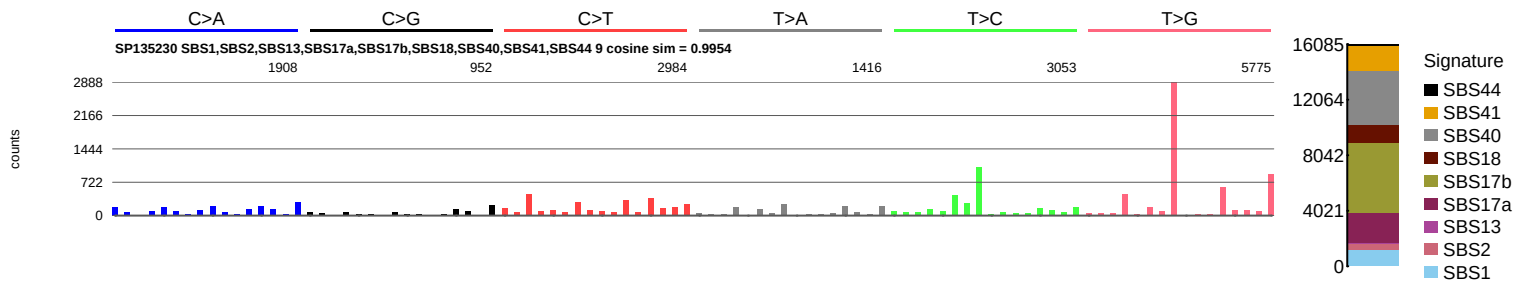

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99547
1881
938
2958
1391
3163
5753
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

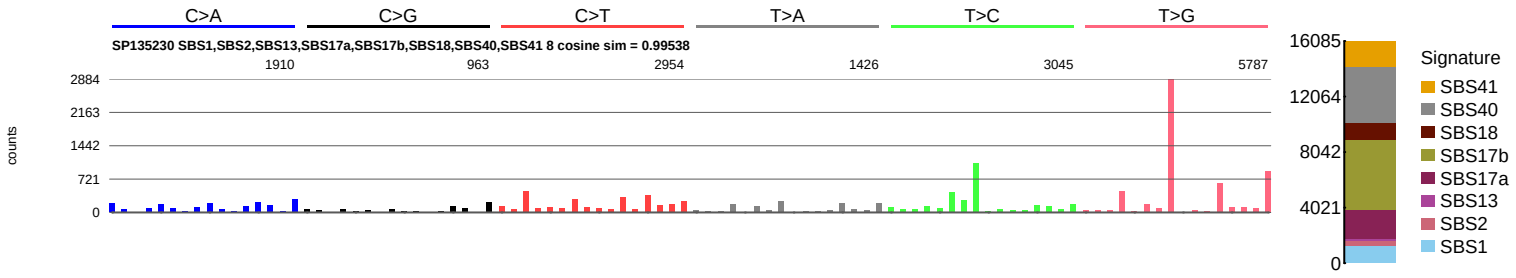

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99546
1884
934
2967
1389
3160
5749
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99546
1896
936
2946
1390
3160
5759
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99546
1880
938
2993
1399
3128
5748
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99544
1897
942
2922
1391
3164
5768
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.9954
1908
952
2984
1416
3053
5775
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
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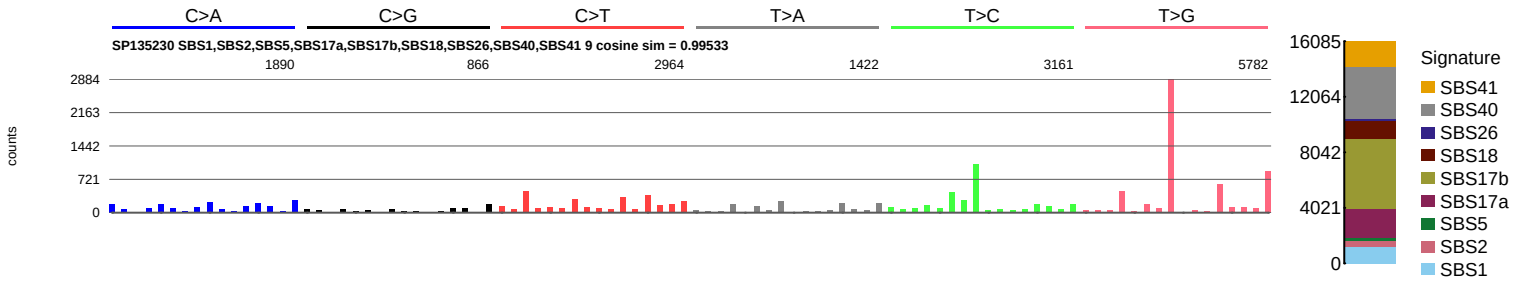
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99538
1910
963
2954
1426
3045
5787
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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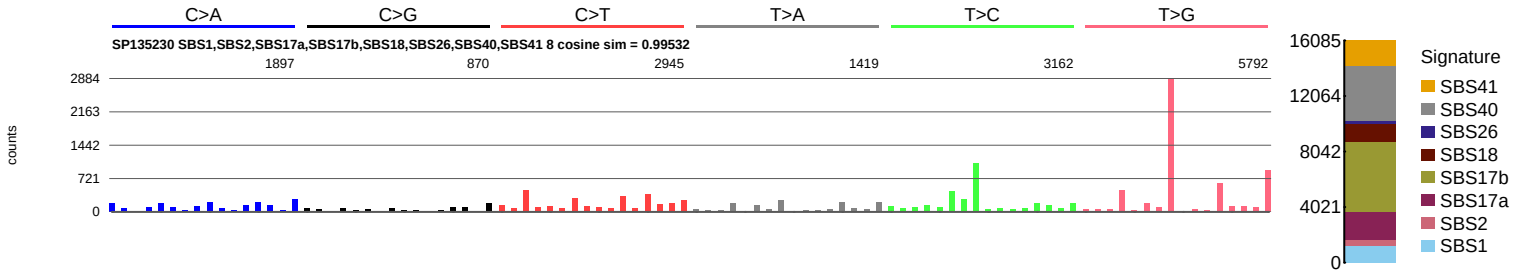
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99533
1890
866
2964
1422
3161
5782
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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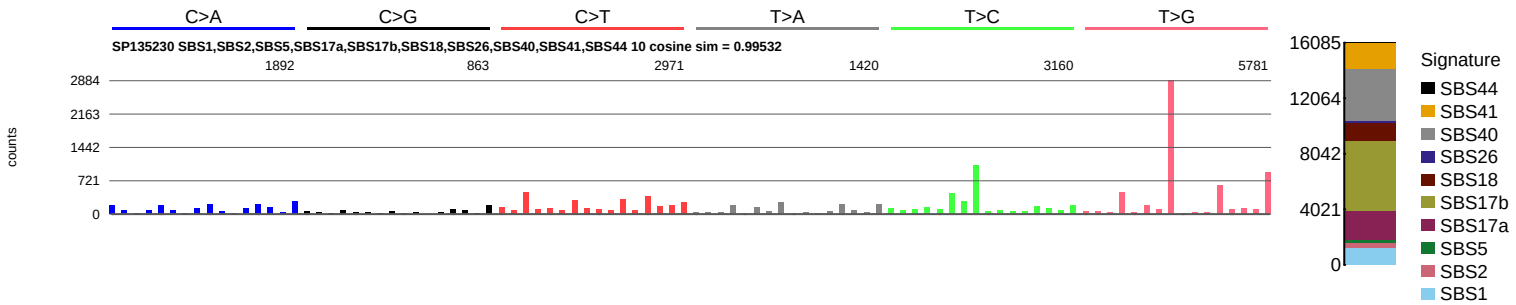
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.99532
1897
870
2945
1419
3162
5792
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
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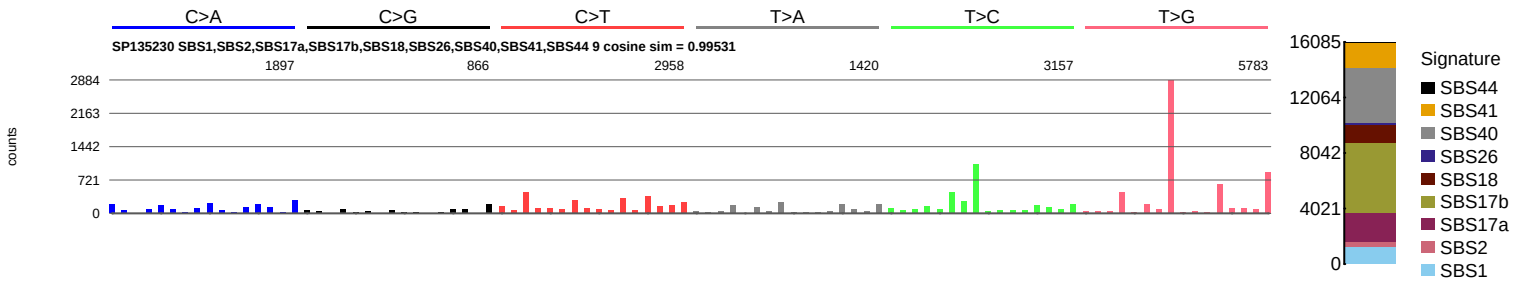
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99532
1892
863
2971
1420
3160
5781
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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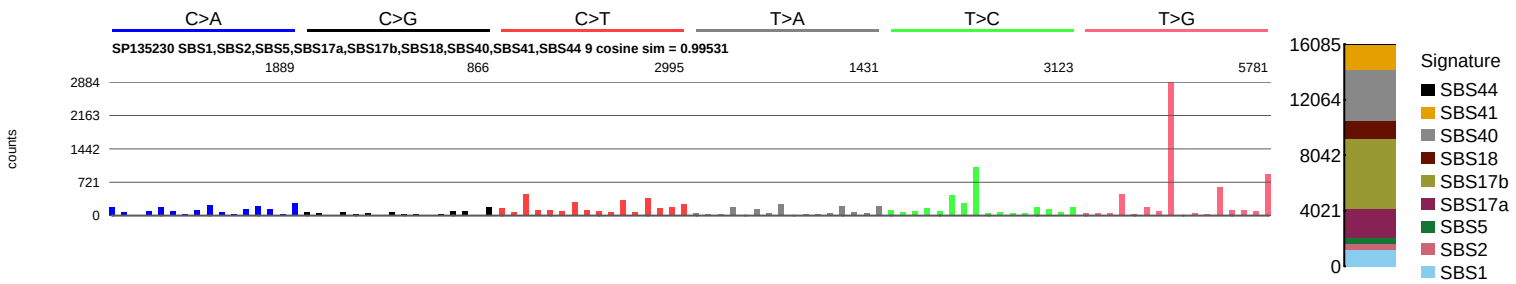
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99531
1897
866
2958
1420
3157
5783
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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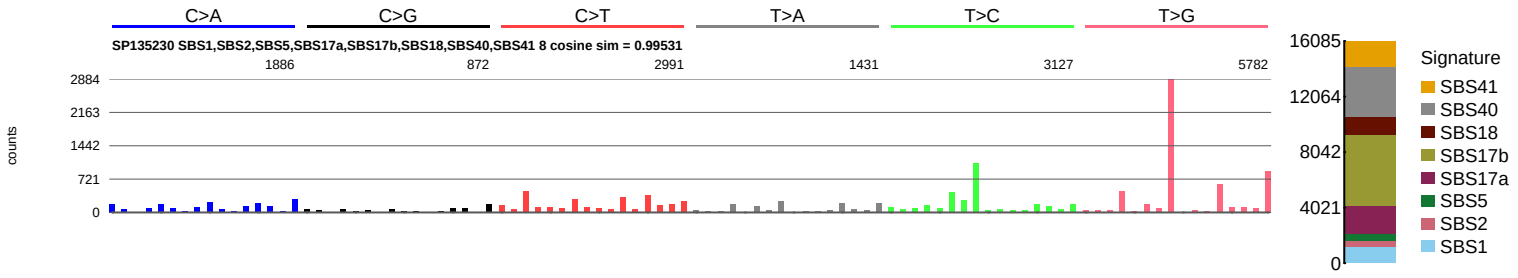
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99531
1889
866
2995
1431
3123
5781
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99531
1886
872
2991
1431
3127
5782
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
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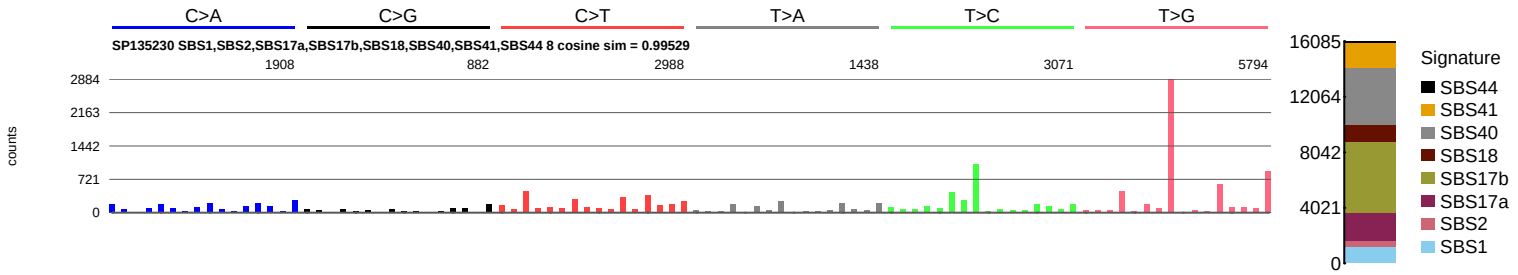
SBS17b
SBS17a
SBS5
SBS2
SBS1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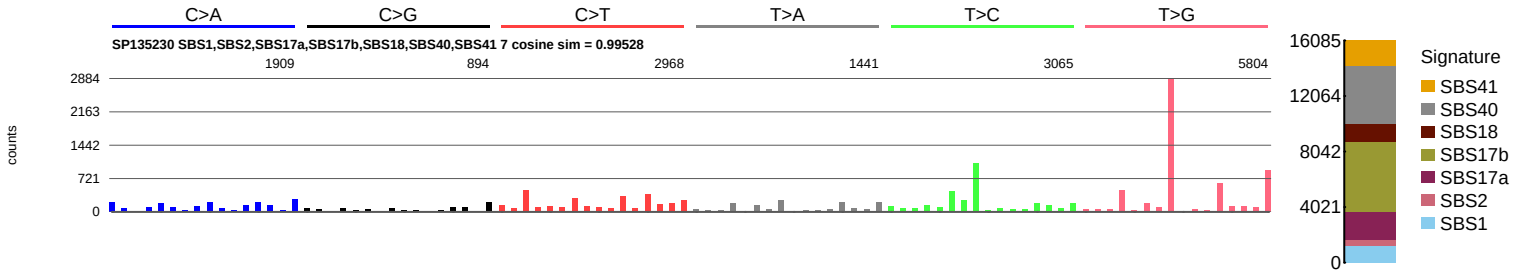
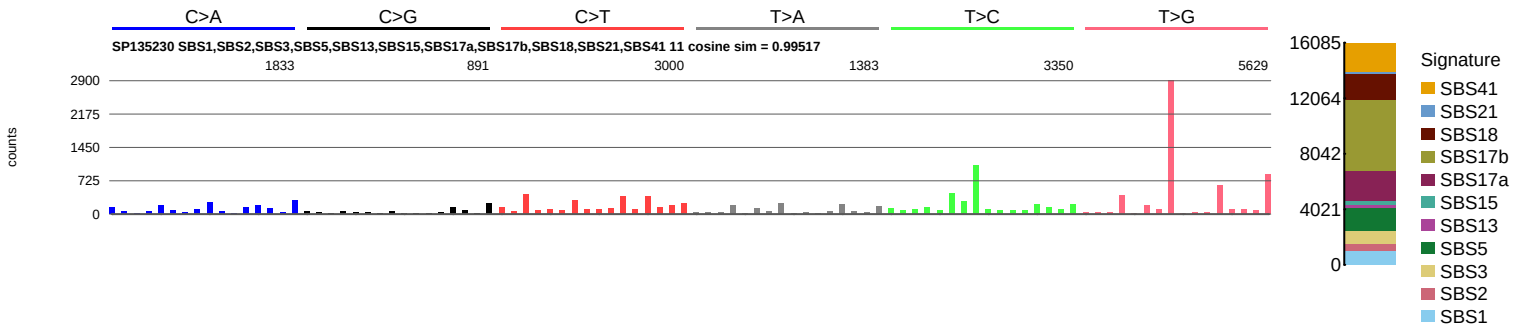
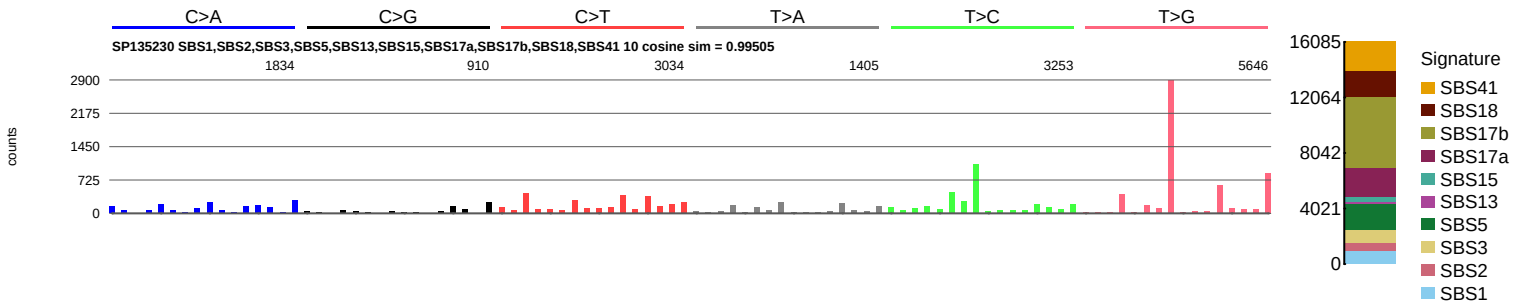
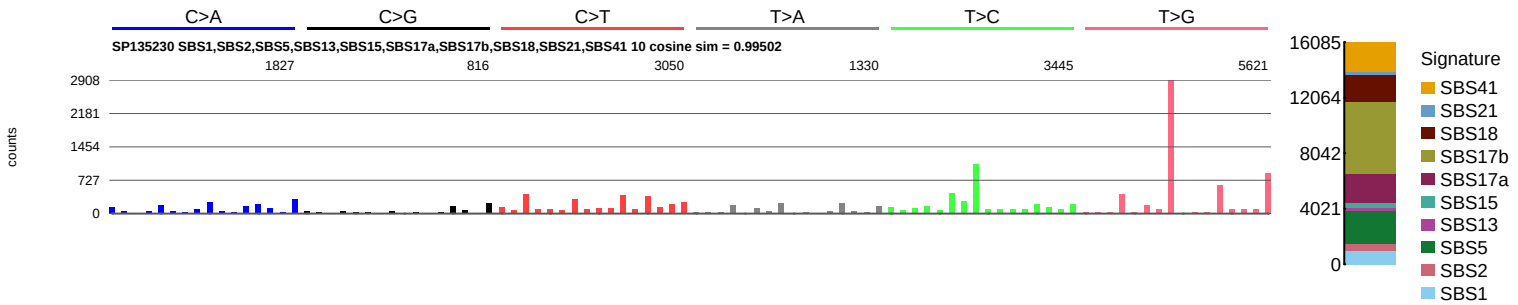
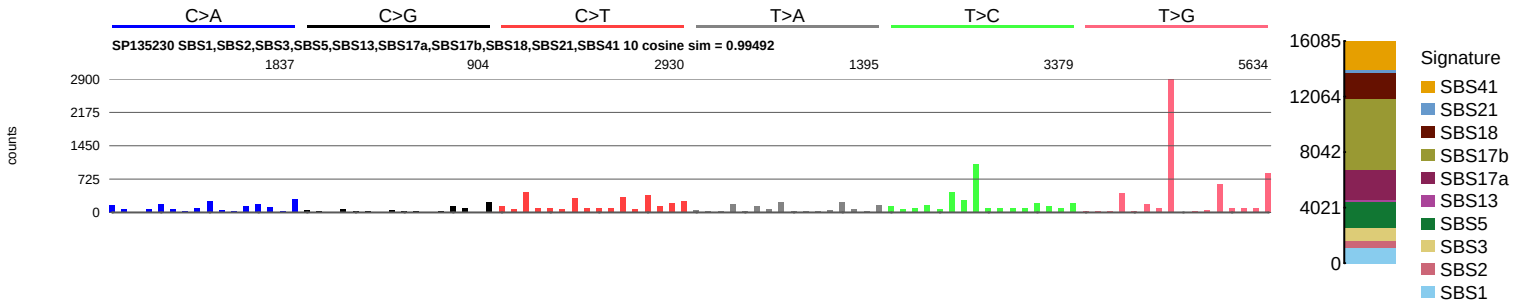
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.99529
1908
882
2988
1438
3071
5794
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

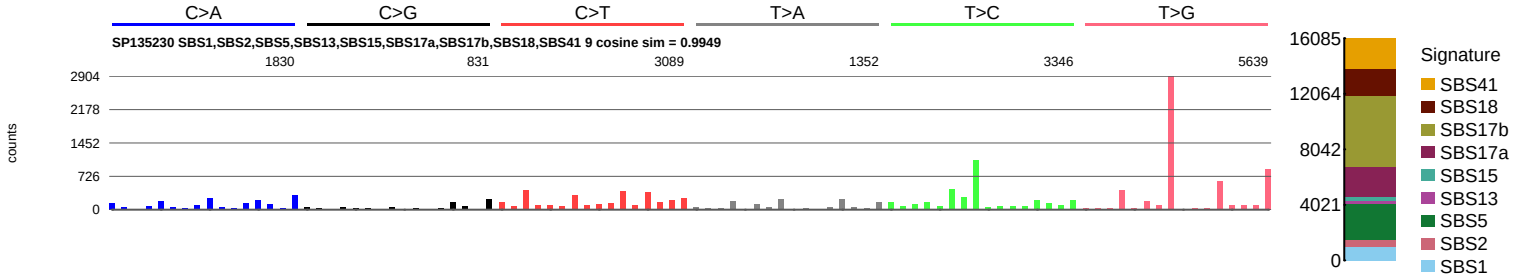

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99528
1909
894
2968
1441
3065
5804
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

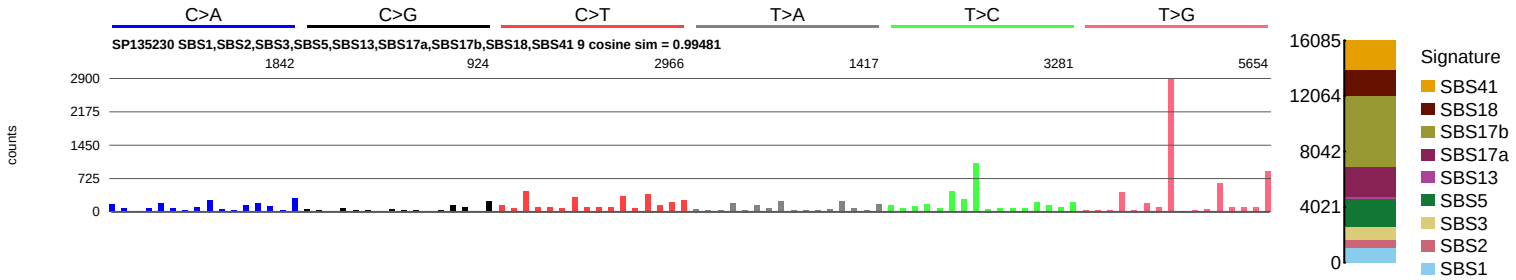

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.99517
1833
891
3000
1383
3350
5629
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

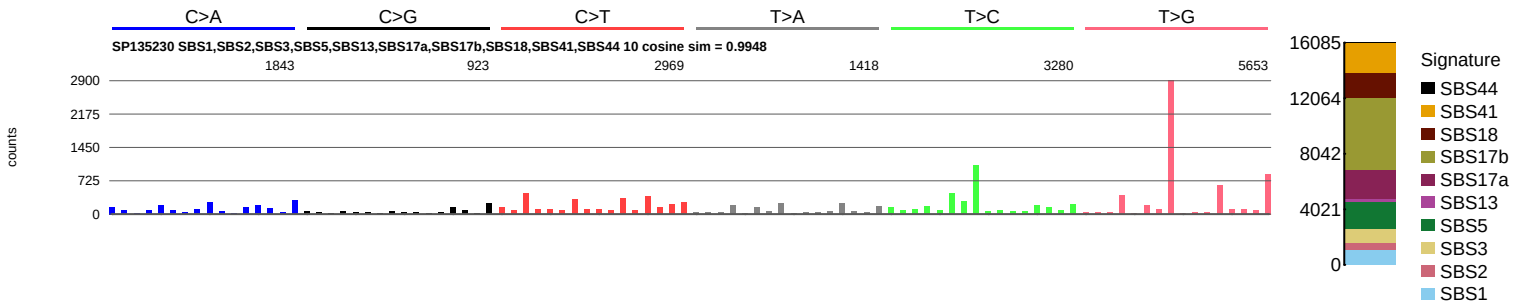

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.99505
1834
910
3034
1405
3253
5646
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

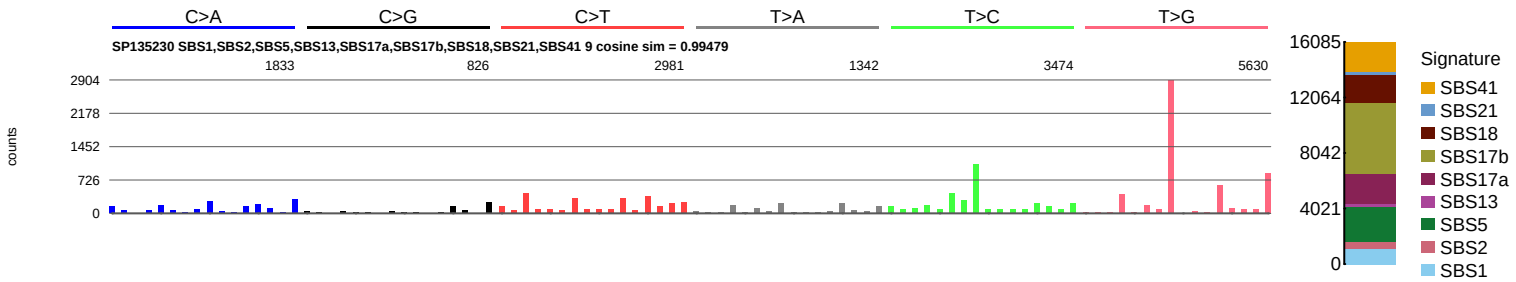

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99502
1827
816
3050
1330
3445
5621
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

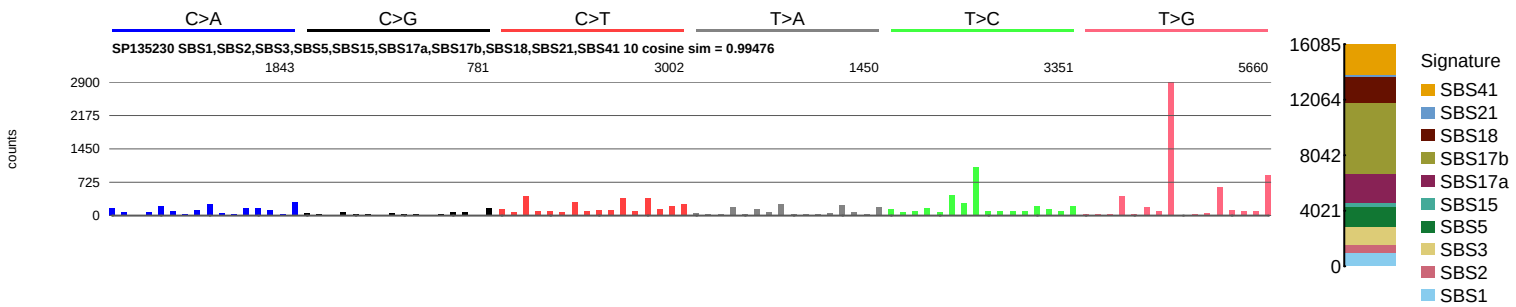

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99492
1837
904
2930
1395
3379
5634
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

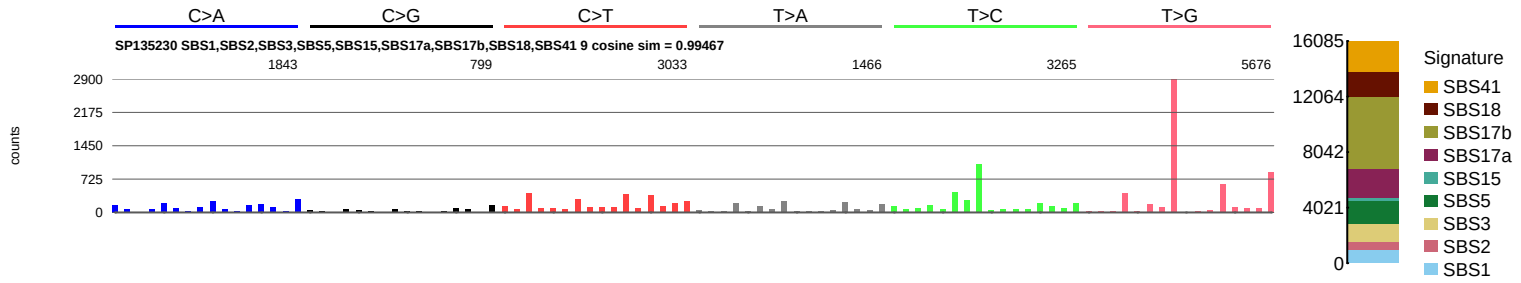

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.9949
1830
831
3089
1352
3346
5639
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99481
1842
924
2966
1417
3281
5654
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.9948
1843
923
2969
1418
3280
5653
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99479
1833
826
2981
1342
3474
5630
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99476
1843
781
3002
1450
3351
5660
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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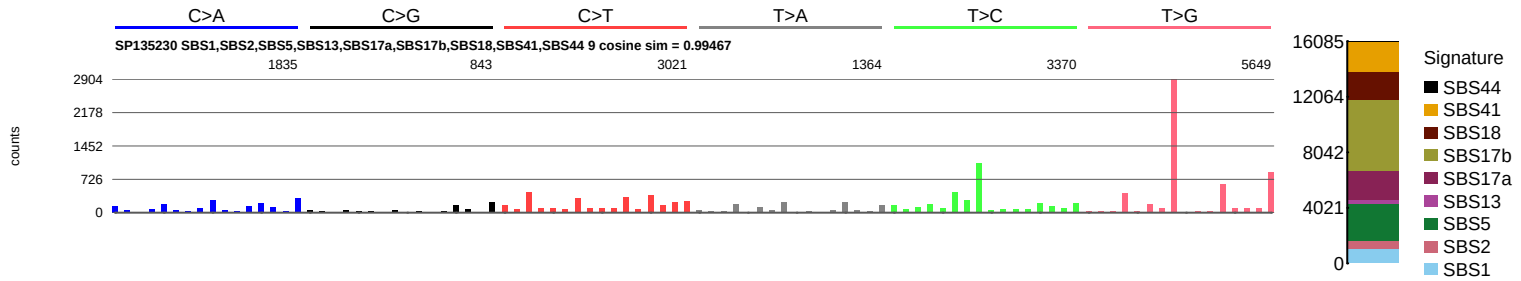
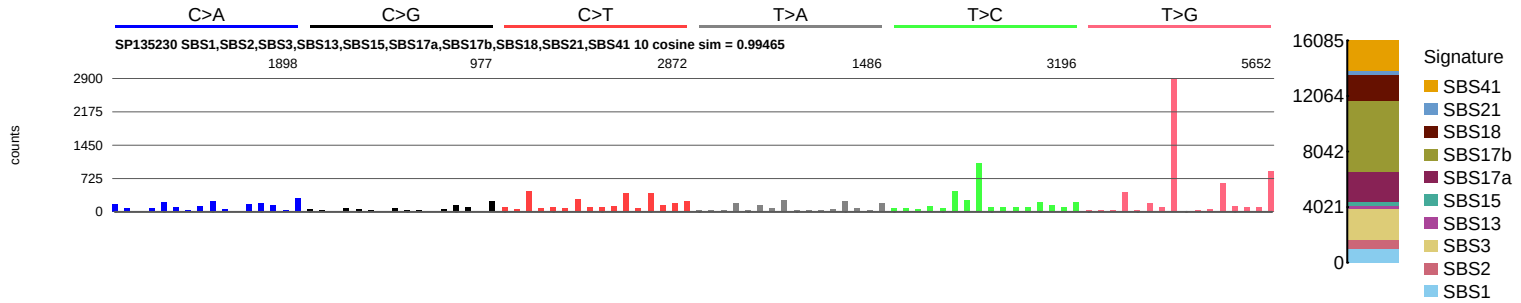
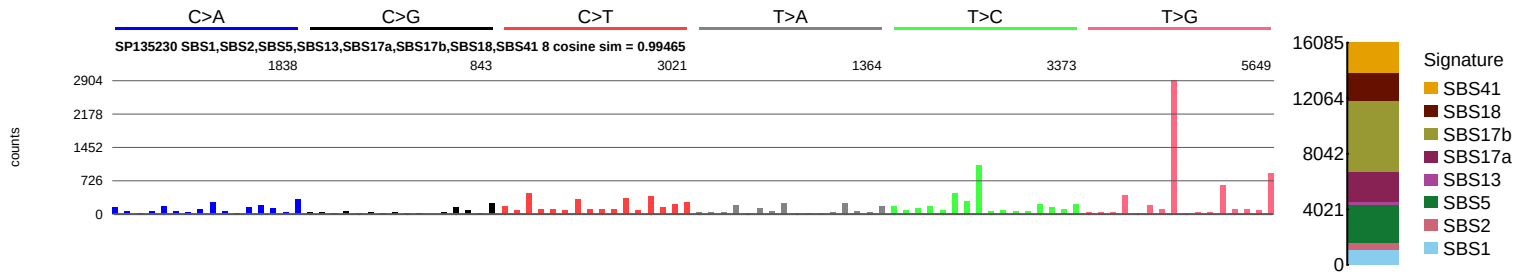
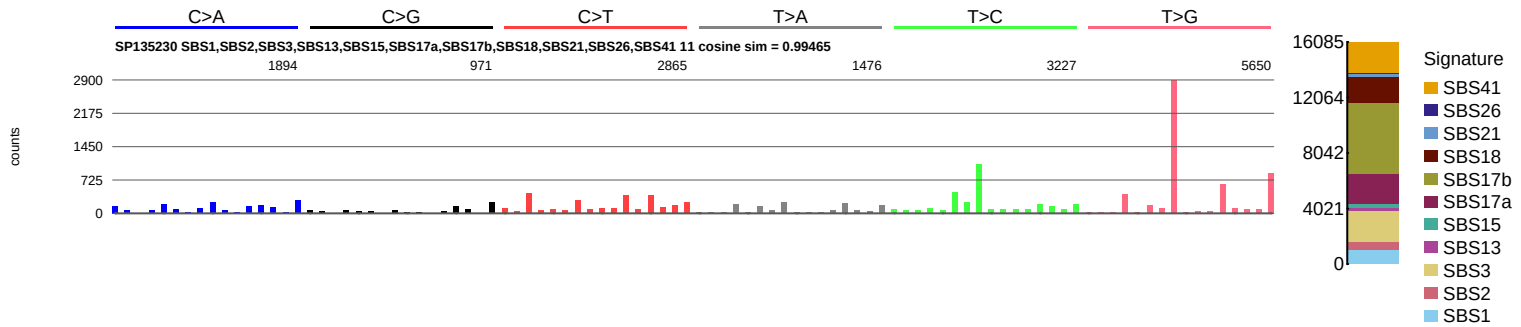
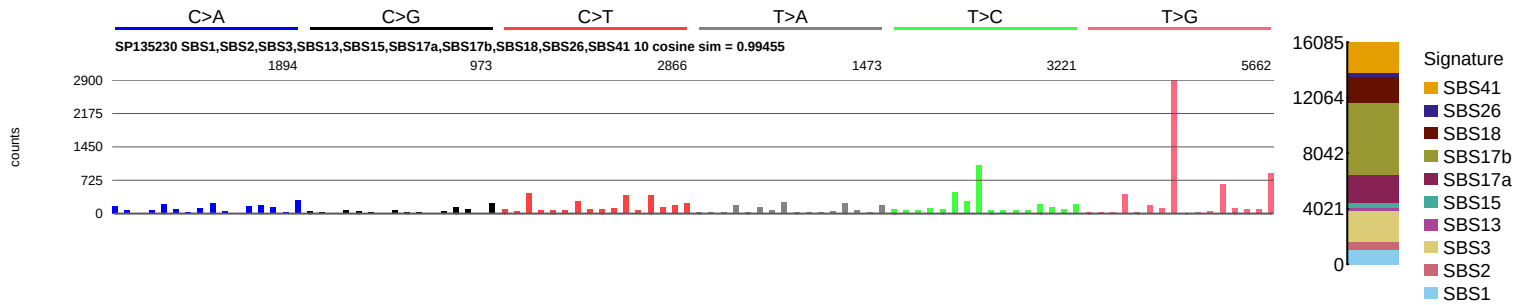
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99467
1843
799
3033
1466
3265
5676
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99467
1835
843
3021
1364
3370
5649
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99465
1898
977
2872
1486
3196
5652
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99465
1838
843
3021
1364
3373
5649
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 11 cosine sim = 0.99465
1894
971
2865
1476
3227
5650
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.99455
1894
973
2866
1473
3221
5662
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
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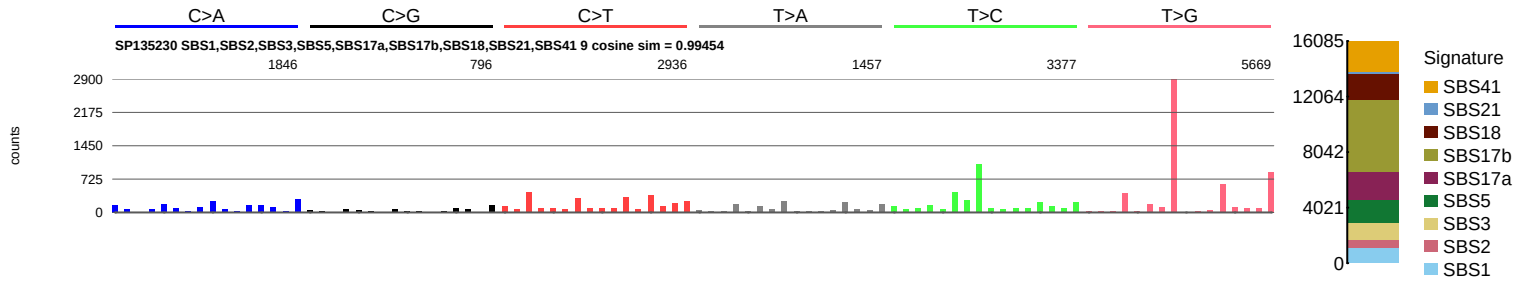
SBS1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SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99454
1846
796
2936
1457
3377
5669
counts
2900
2175
1450
725
0
16085
12064
8042
4021
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SBS41
SBS21
SBS18
SBS17b
SBS17a
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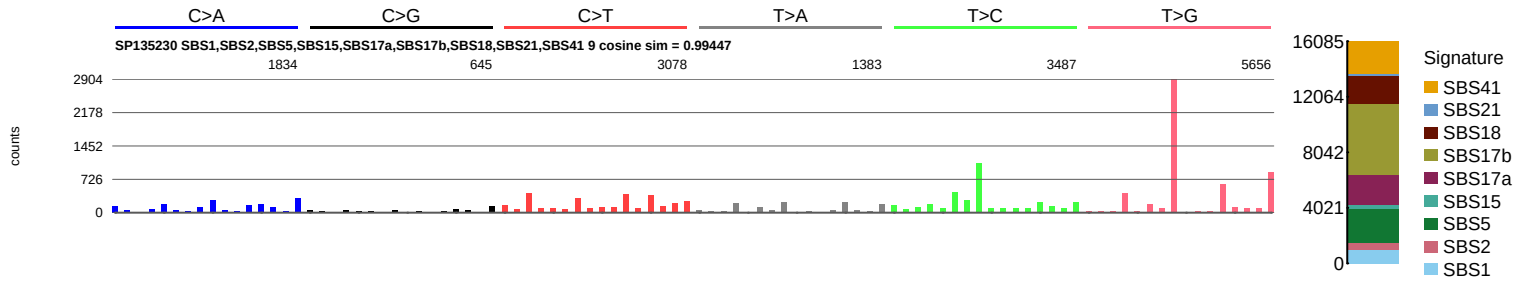
SBS1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C>G
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T>A
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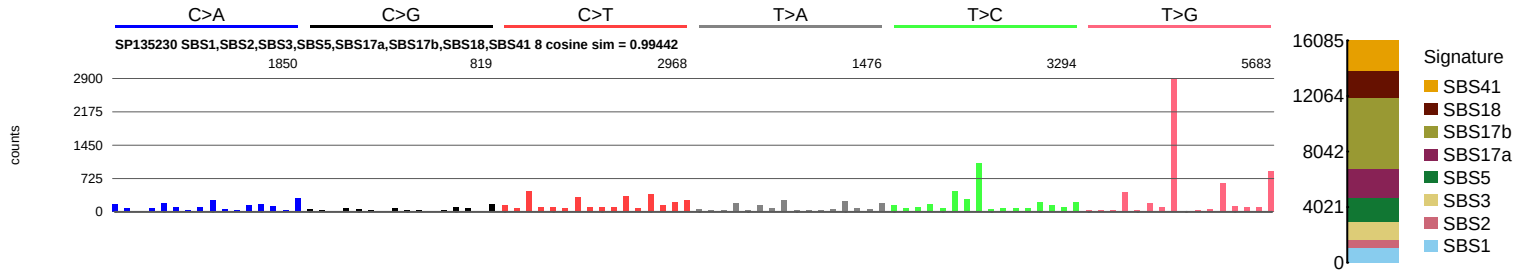
T>G
SP135230 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99447
1834
645
3078
1383
3487
5656
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>T
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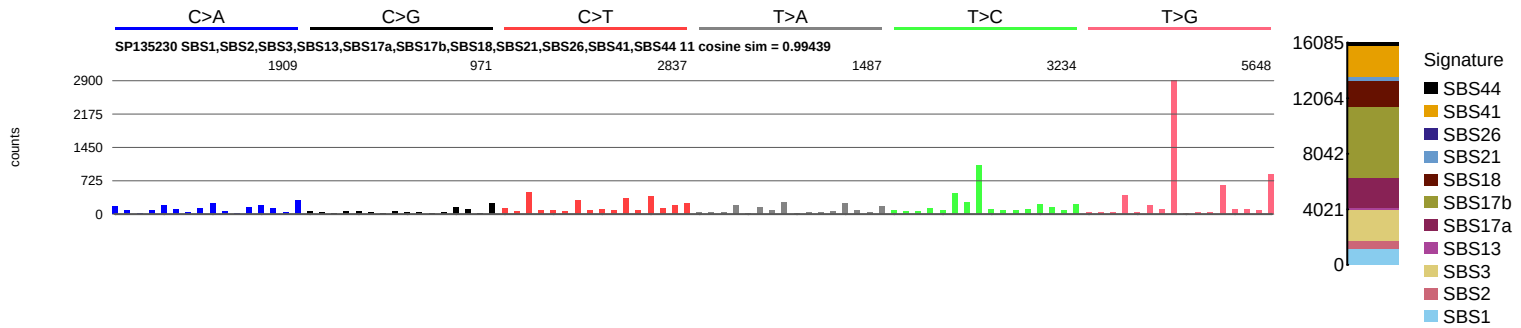
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99442
1850
819
2968
1476
3294
5683
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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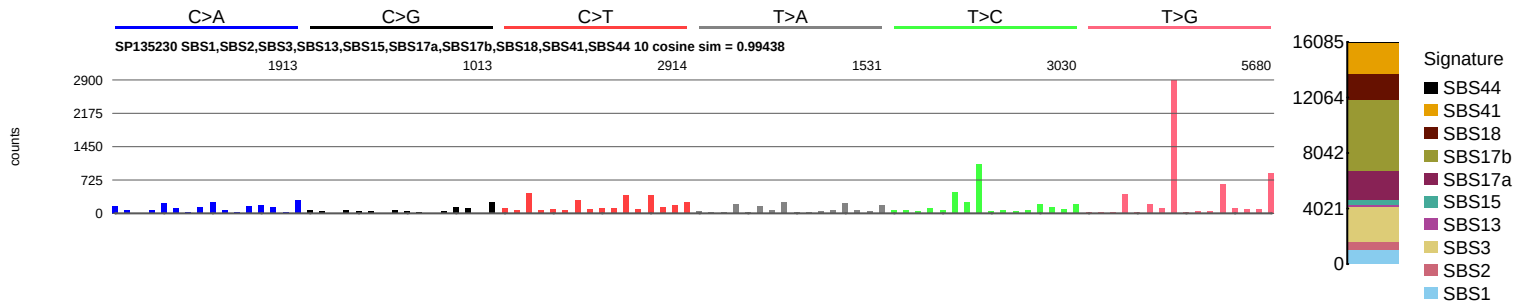
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.99439
1909
971
2837
1487
3234
5648
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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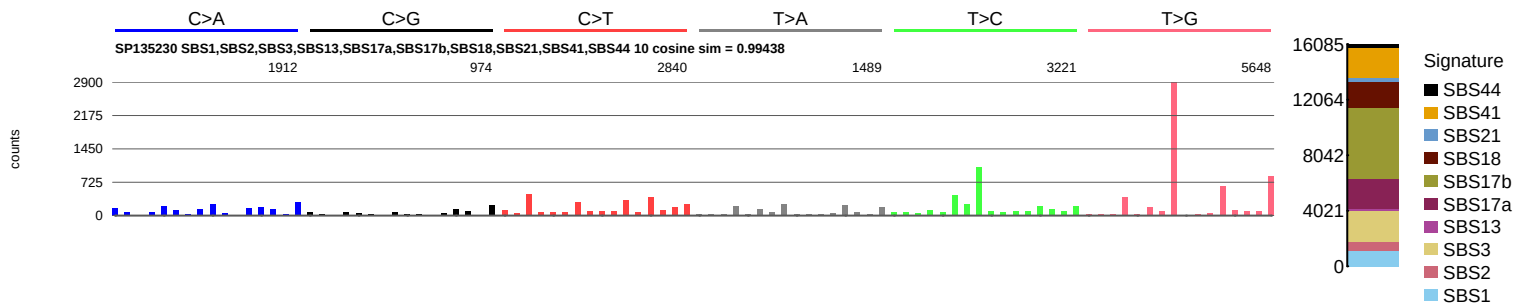
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.99438
1913
1013
2914
1531
3030
5680
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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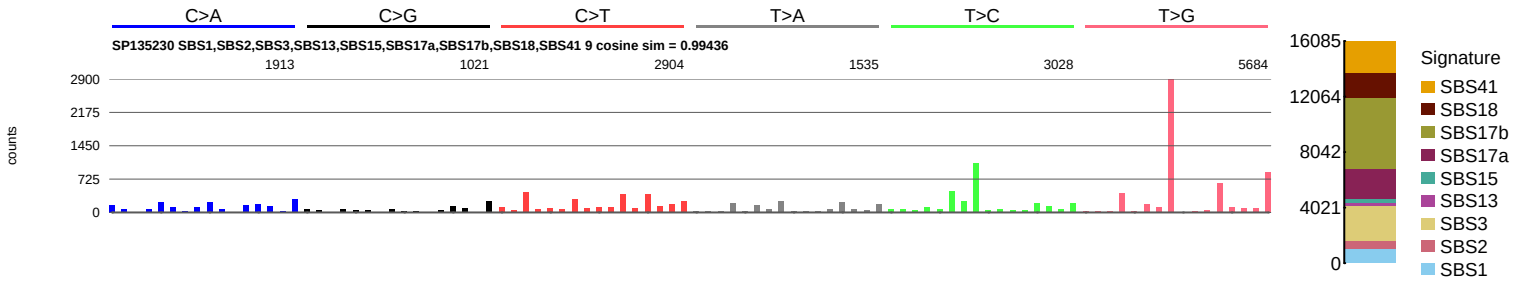
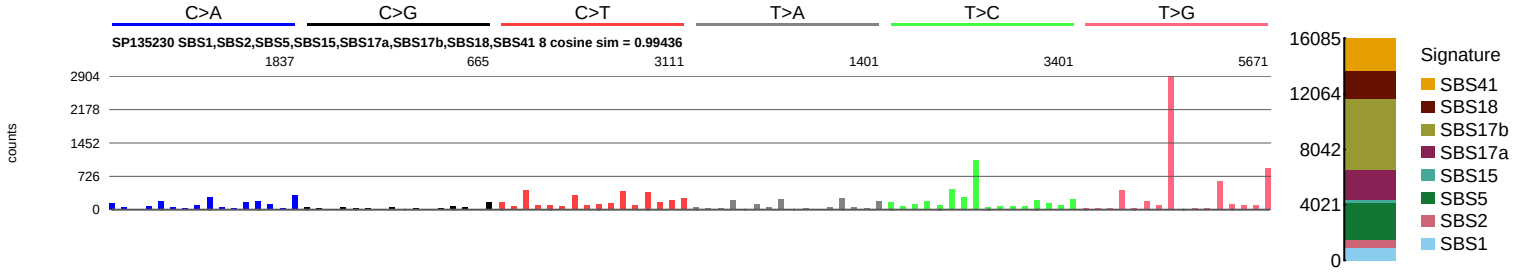
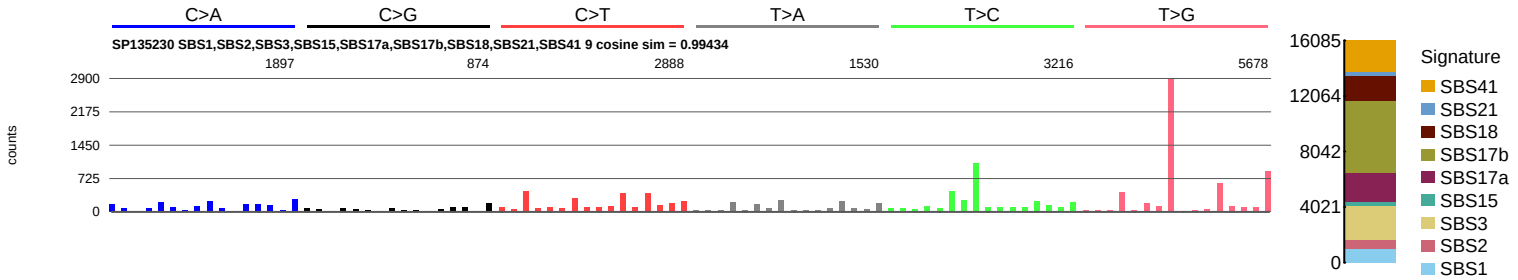
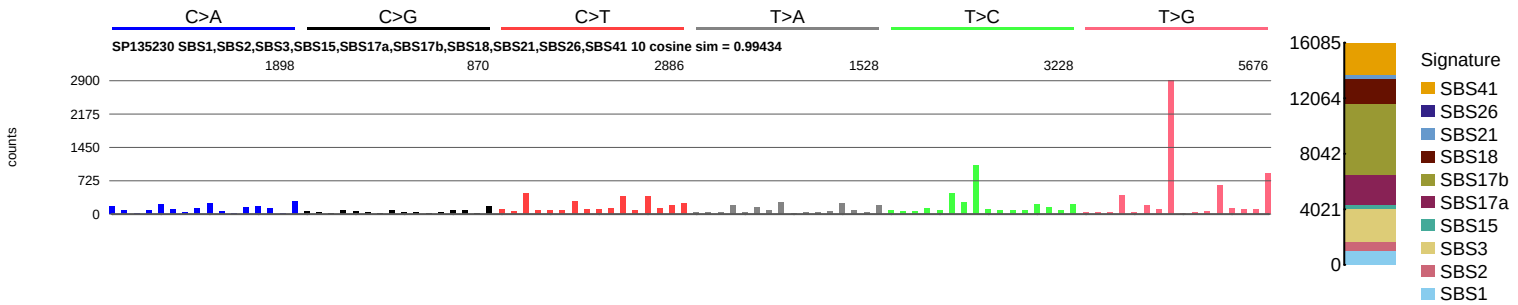
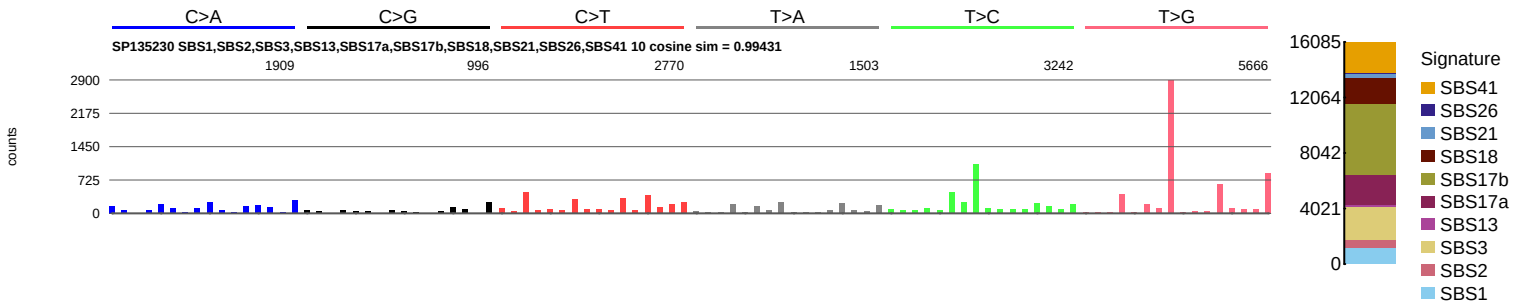
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.99438
1912
974
2840
1489
3221
5648
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

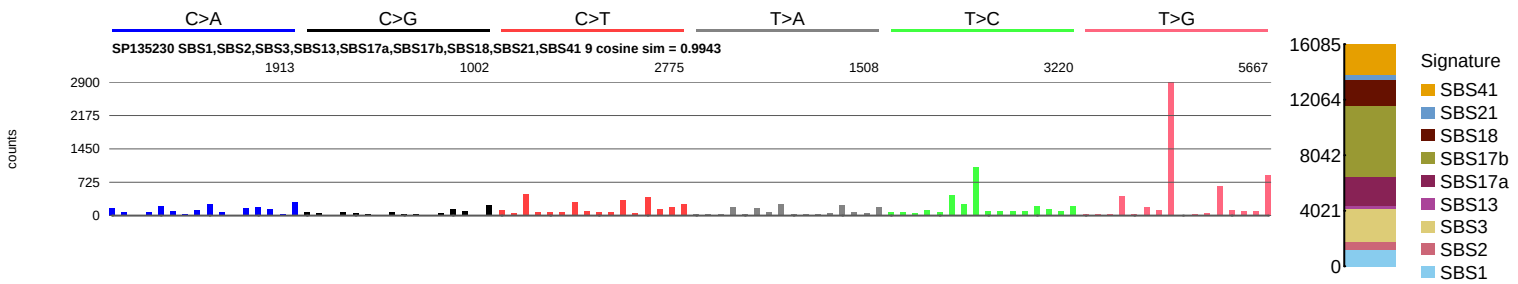

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99436
1913
1021
2904
1535
3028
5684
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

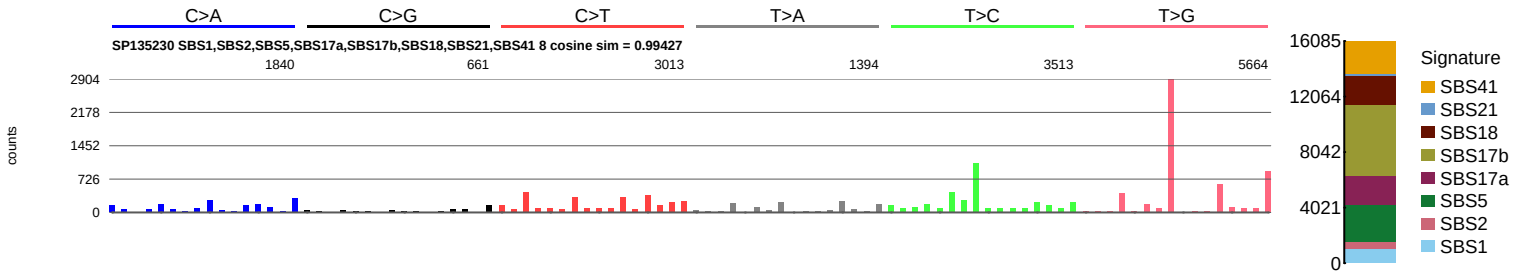

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99436
1837
665
3111
1401
3401
5671
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

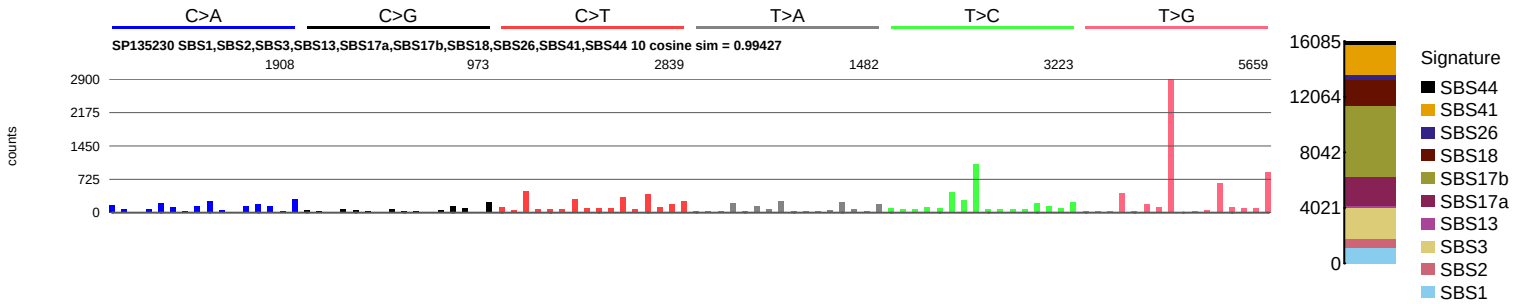

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99434
1897
874
2888
1530
3216
5678
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

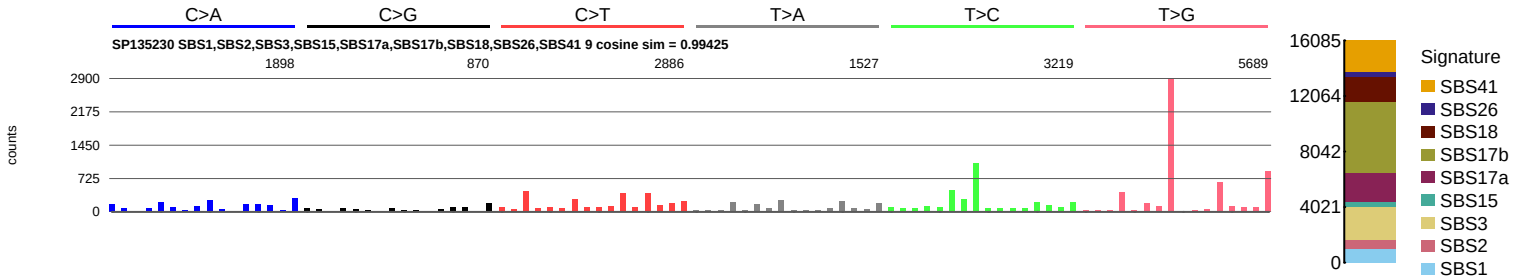

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99434
1898
870
2886
1528
3228
5676
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99431
1909
996
2770
1503
3242
5666
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9943
1913
1002
2775
1508
3220
5667
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99427
1840
661
3013
1394
3513
5664
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99427
1908
973
2839
1482
3223
5659
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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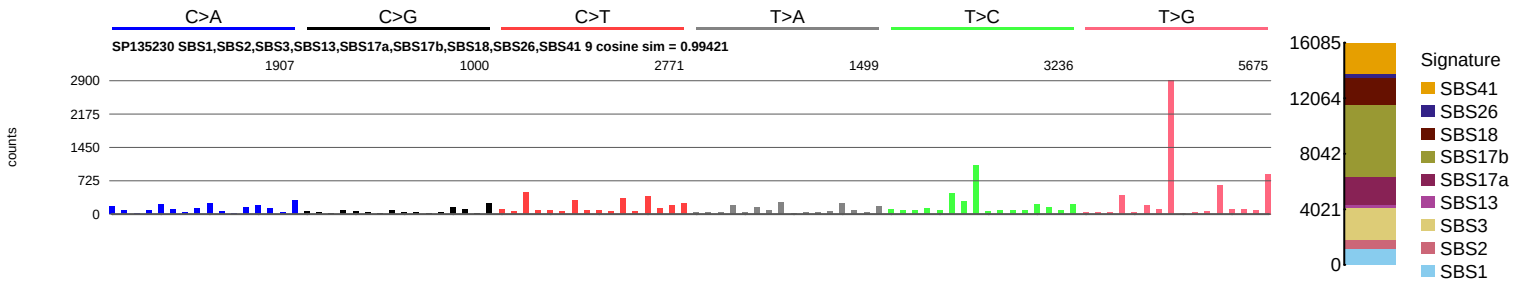
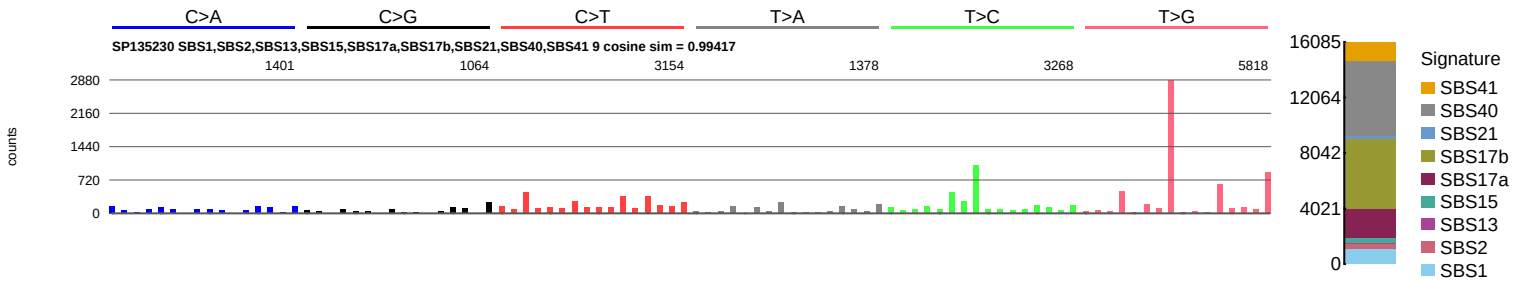
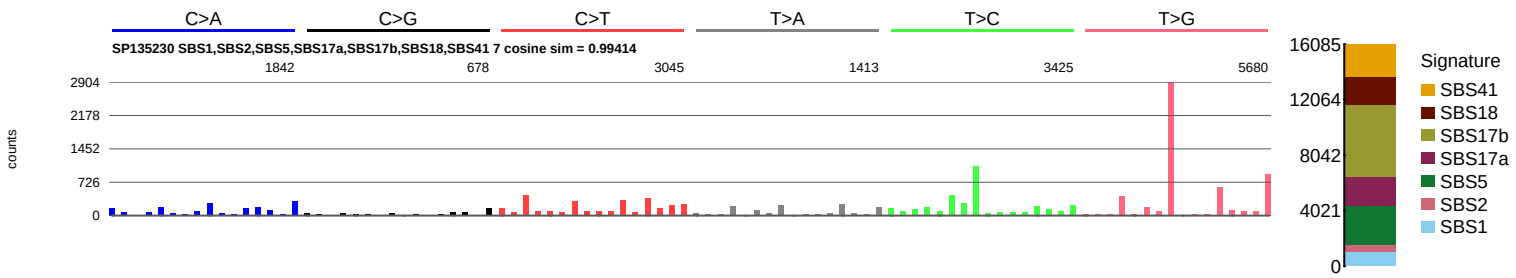
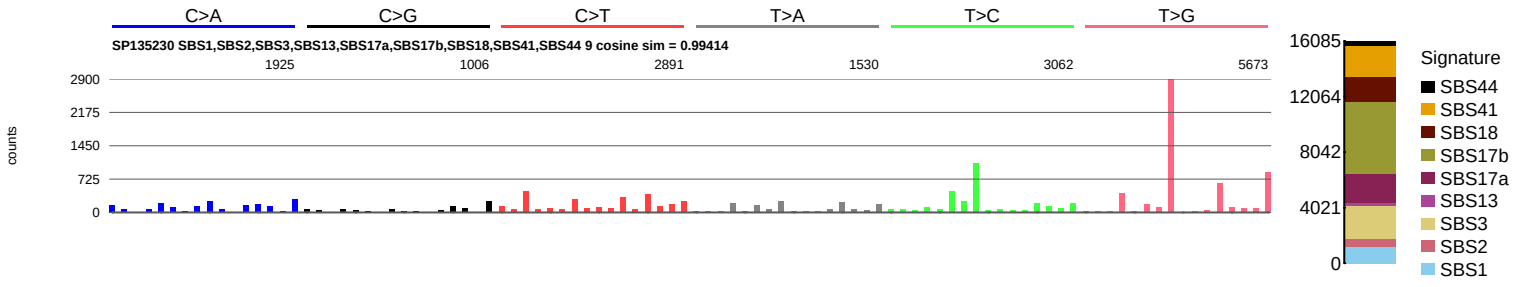
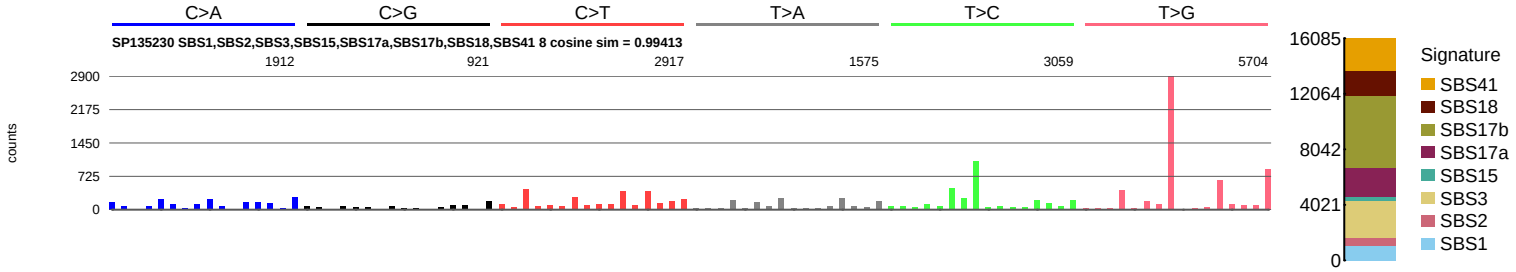
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99425
1898
870
2886
1527
3219
5689
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99421
1907
1000
2771
1499
3236
5675
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.99417
1401
1064
3154
1378
3268
5818
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99414
1842
678
3045
1413
3425
5680
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99414
1925
1006
2891
1530
3062
5673
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99413
1912
921
2917
1575
3059
5704
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
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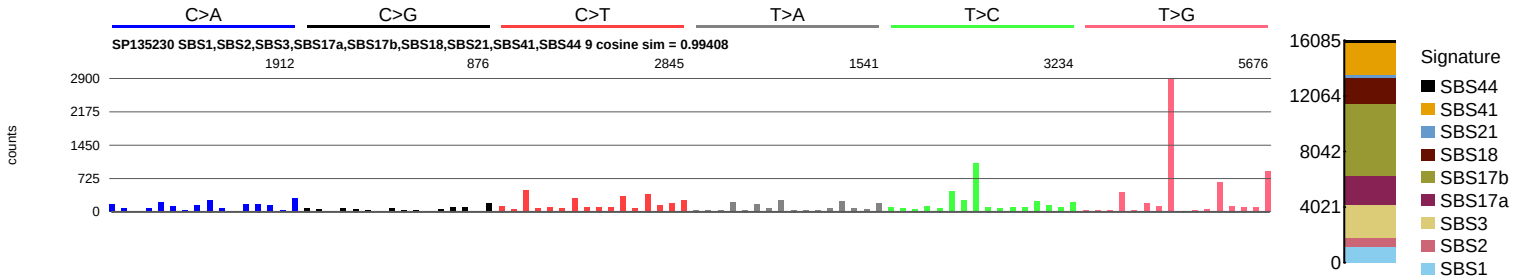
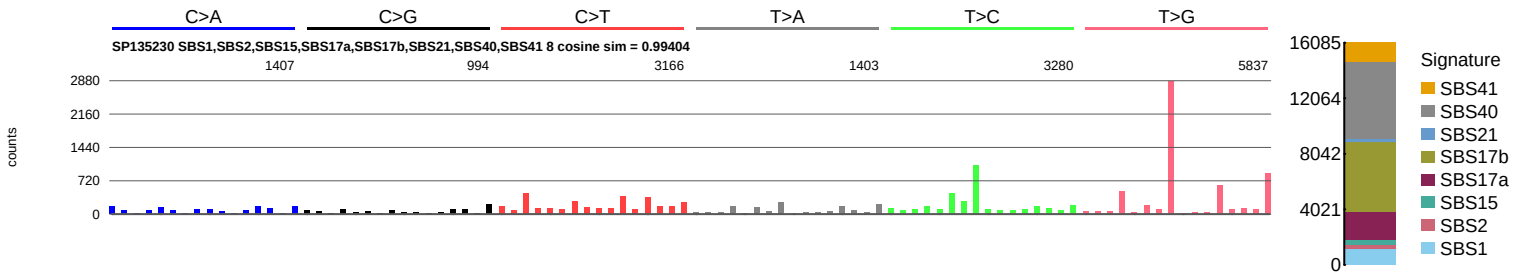
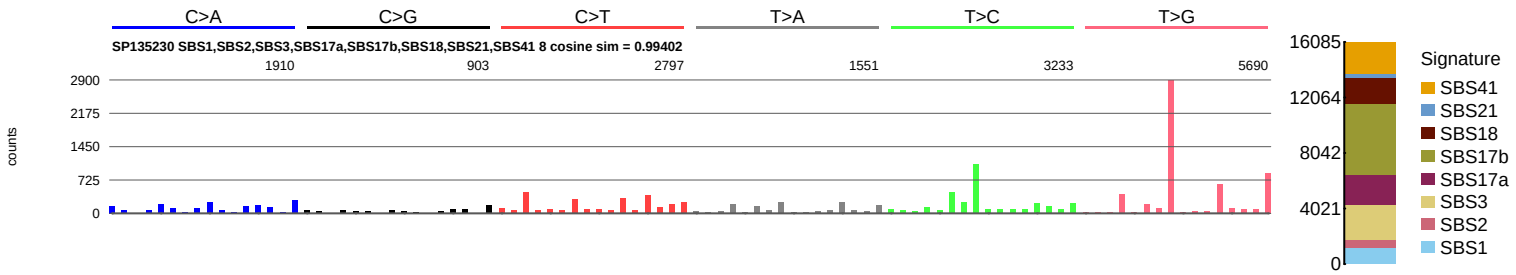
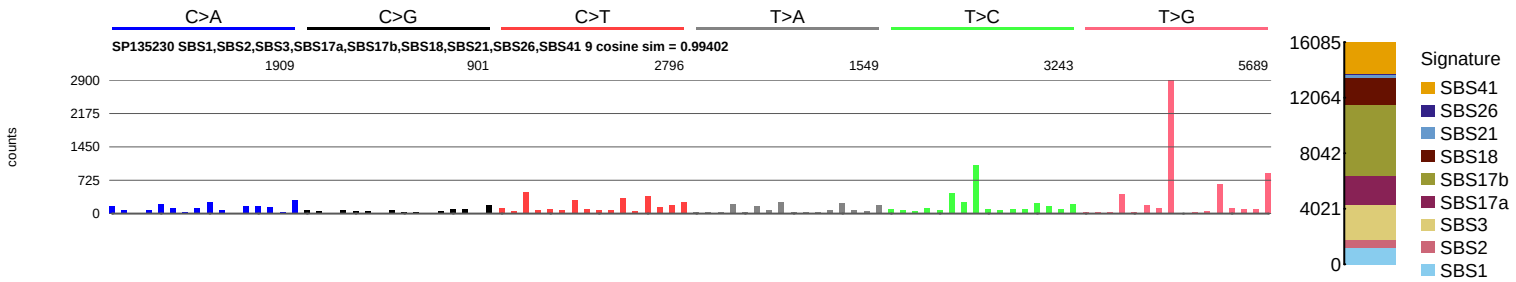
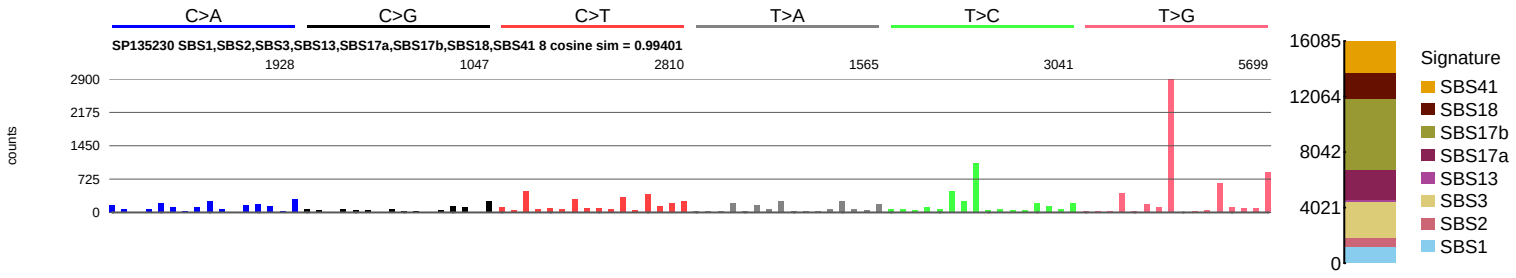
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99408
1912
876
2845
1541
3234
5676
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99404
1407
994
3166
1403
3280
5837
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99402
1910
903
2797
1551
3233
5690
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99402
1909
901
2796
1549
3243
5689
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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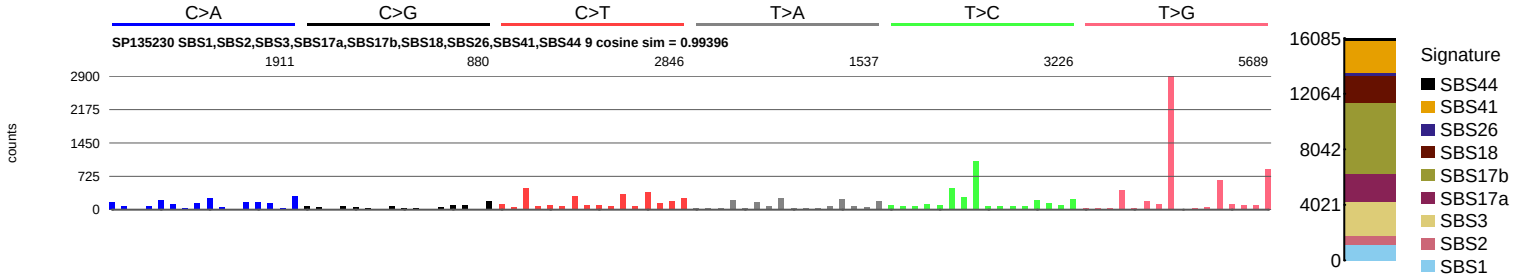
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99401
1928
1047
2810
1565
3041
5699
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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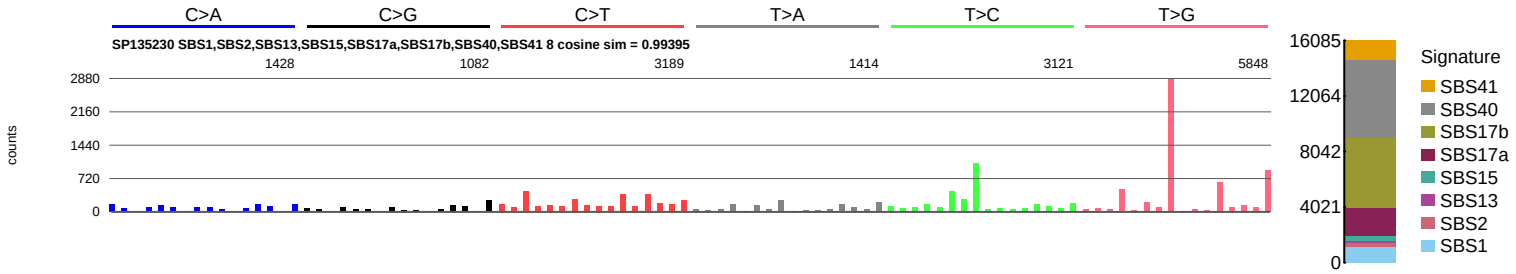
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99396
1911
880
2846
1537
3226
5689
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
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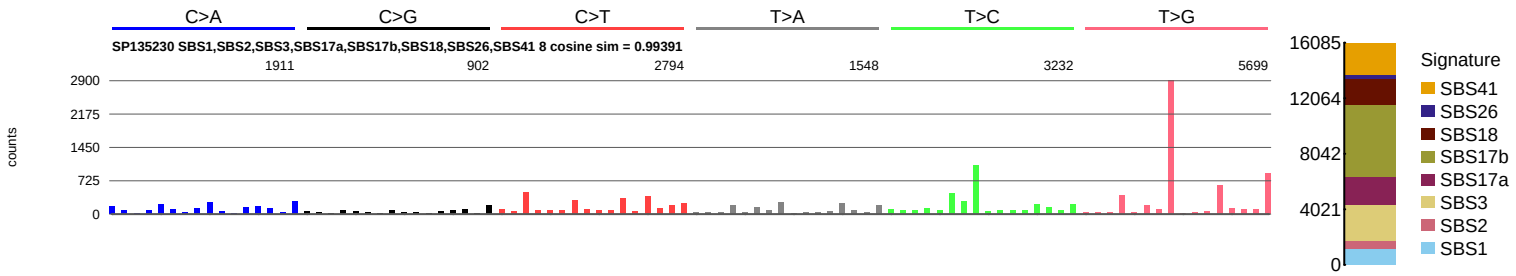
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.99395
1428
1082
3189
1414
3121
5848
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
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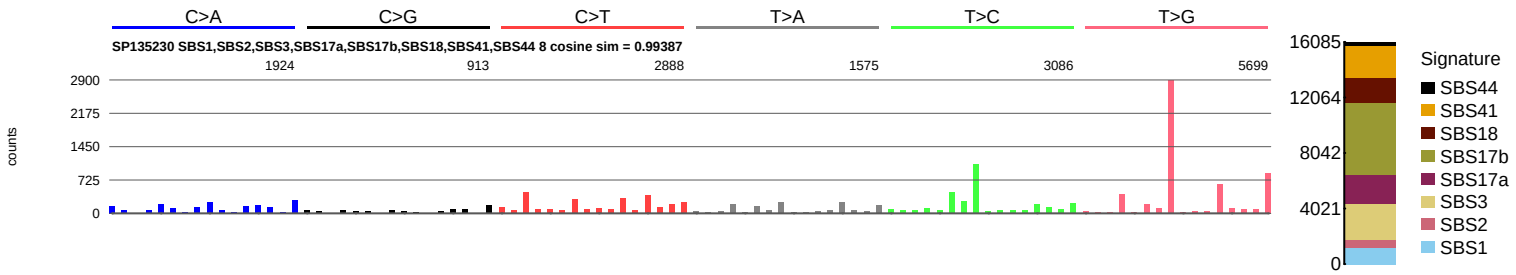
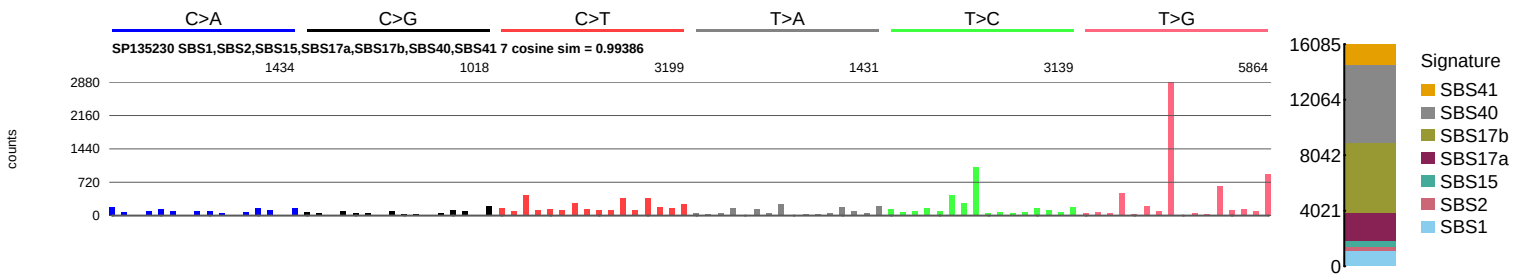
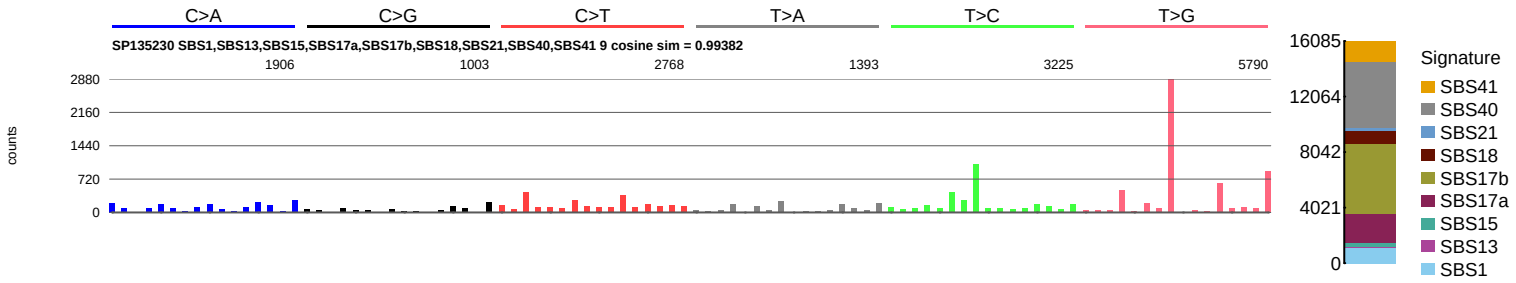
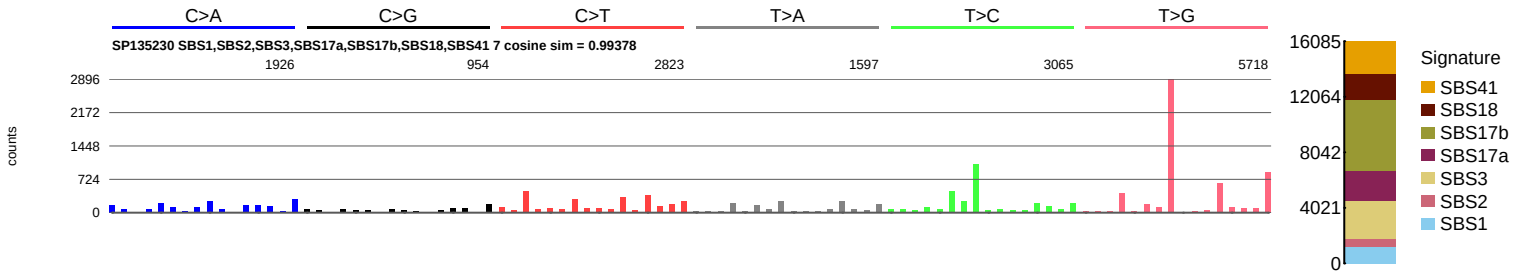
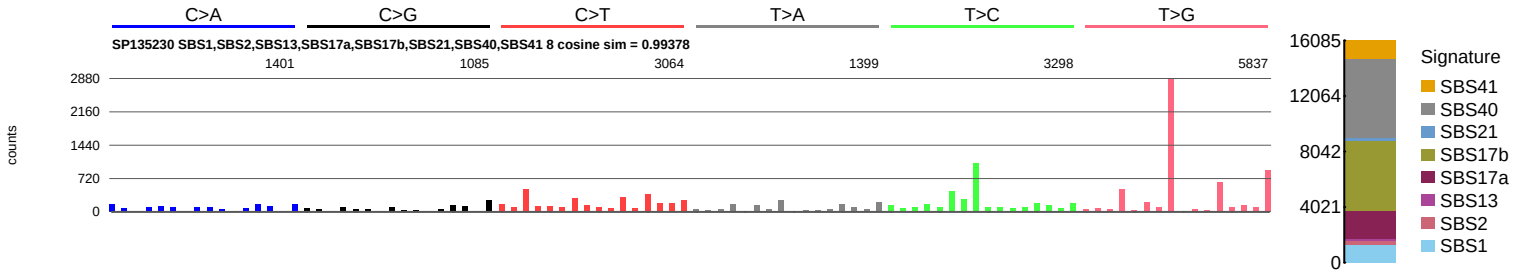
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99391
1911
902
2794
1548
3232
5699
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

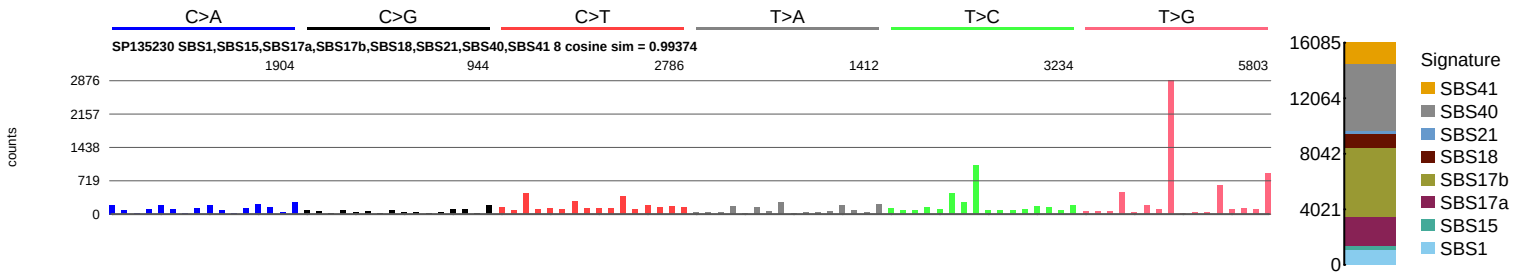

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99387
1924
913
2888
1575
3086
5699
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

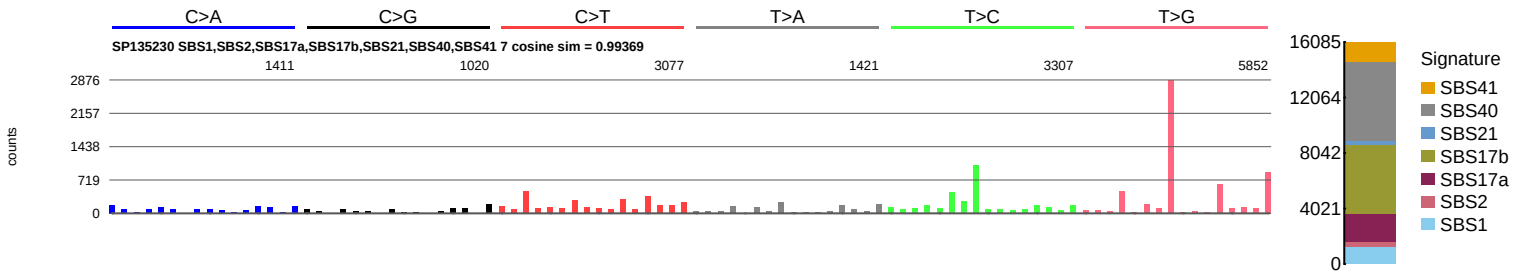

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99386
1434
1018
3199
1431
3139
5864
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1

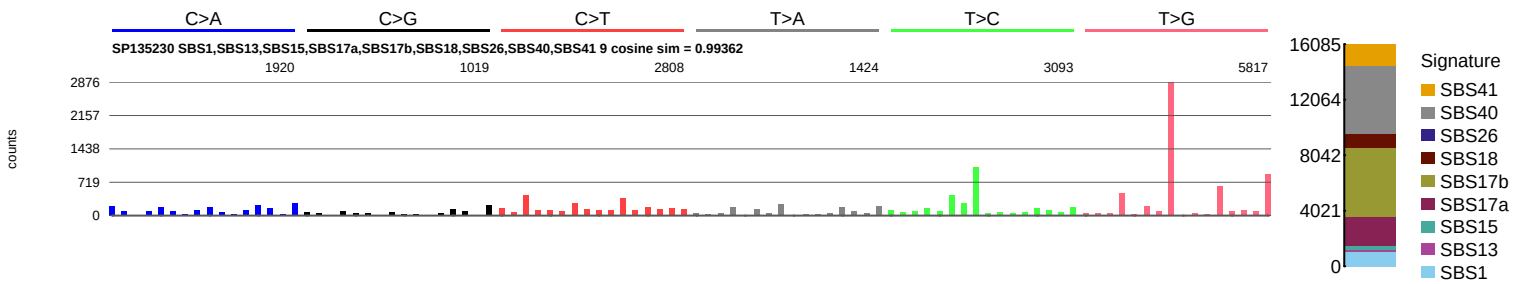

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99382
1906
1003
2768
1393
3225
5790
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

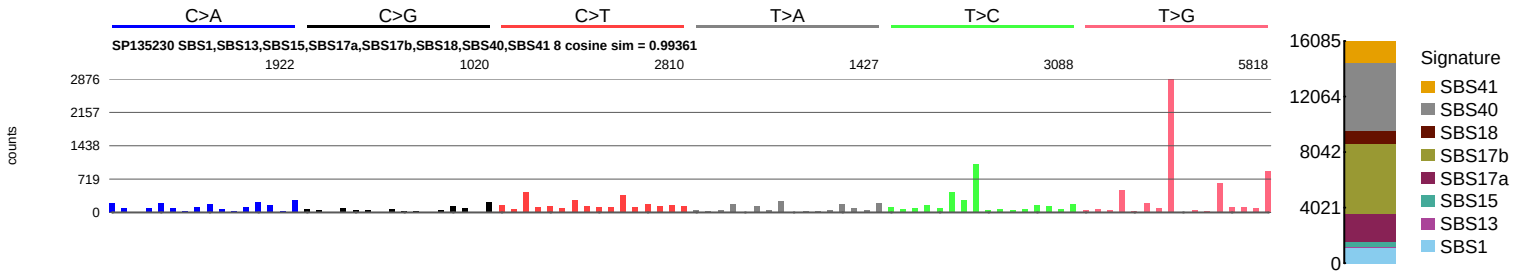

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99378
1926
954
2823
1597
3065
5718
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

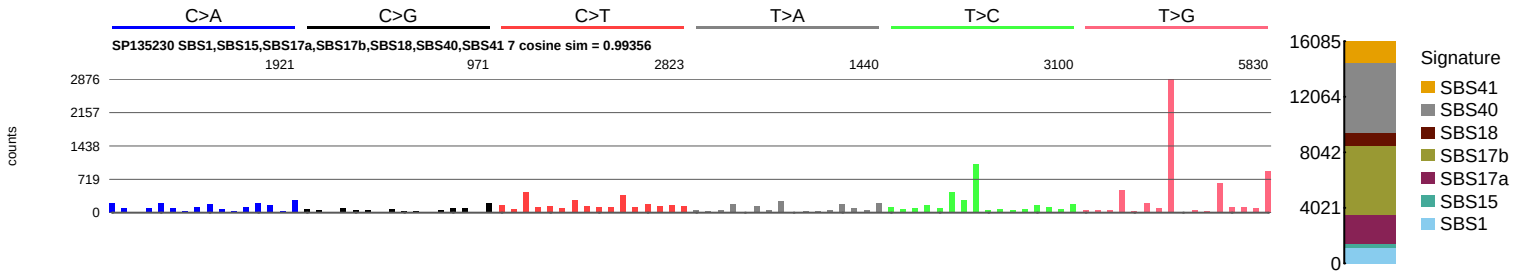

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99378
1401
1085
3064
1399
3298
5837
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

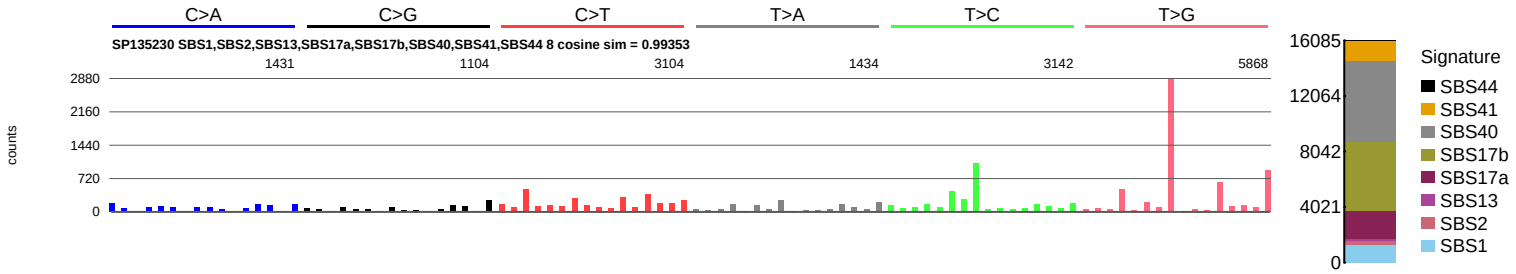

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99374
1904
944
2786
1412
3234
5803
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

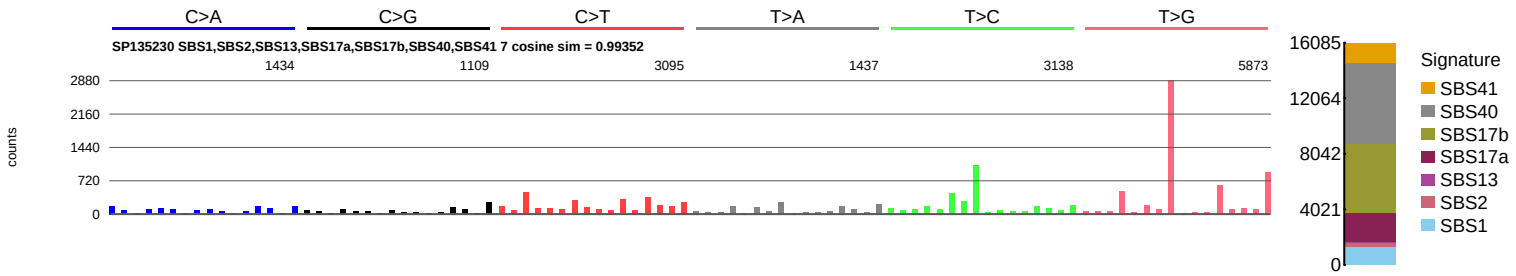

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99369
1411
1020
3077
1421
3307
5852
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1

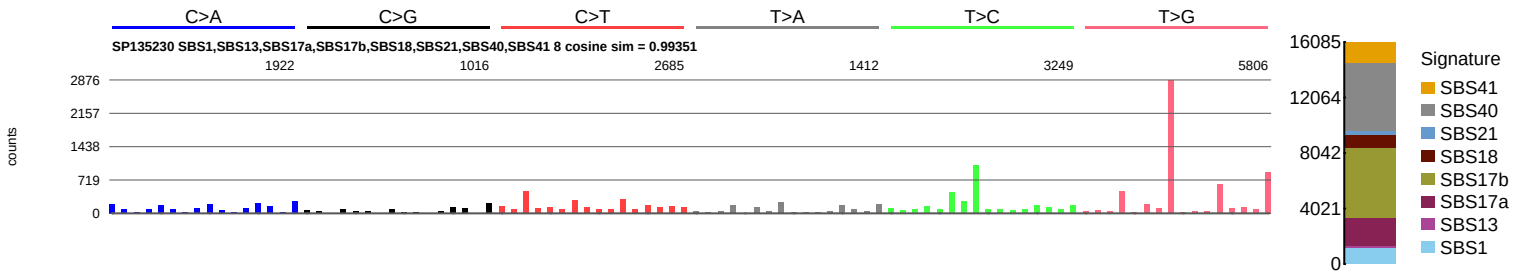

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99362
1920
1019
2808
1424
3093
5817
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

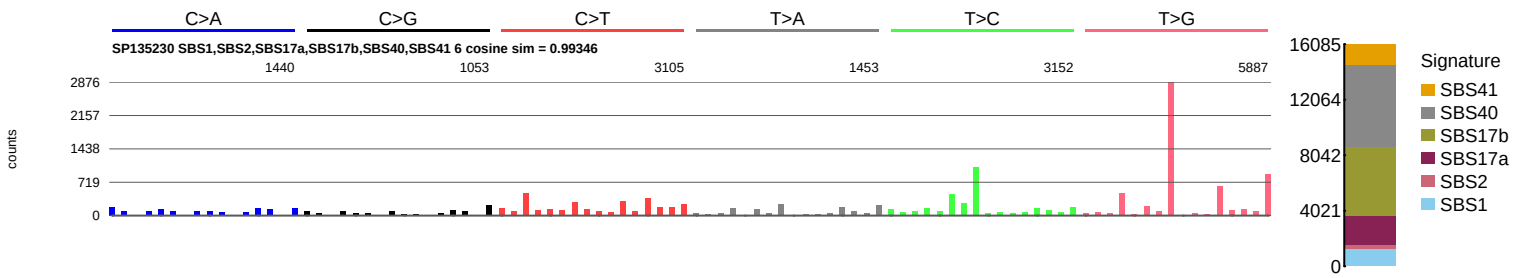

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99361
1922
1020
2810
1427
3088
5818
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

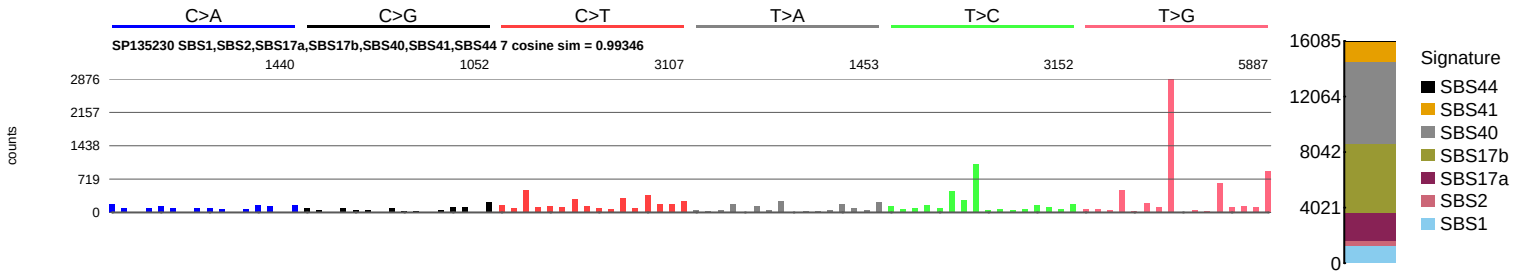

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99356
1921
971
2823
1440
3100
5830
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

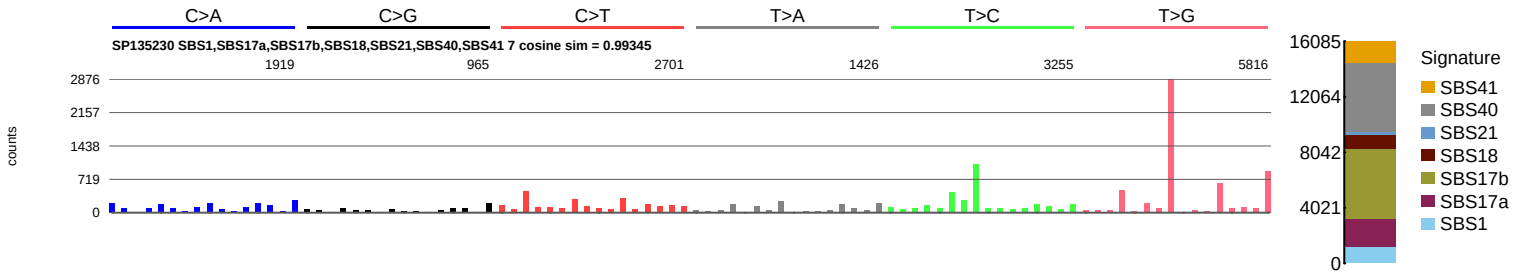

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.99353
1431
1104
3104
1434
3142
5868
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1

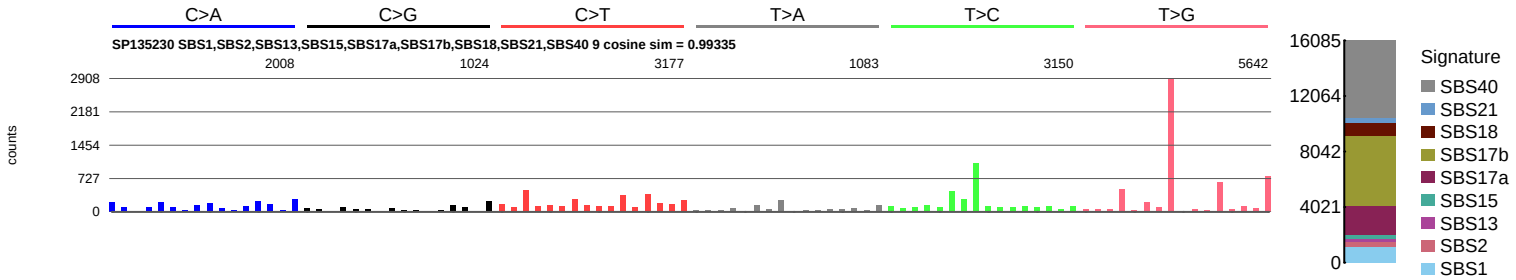

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99352
1434
1109
3095
1437
3138
5873
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1

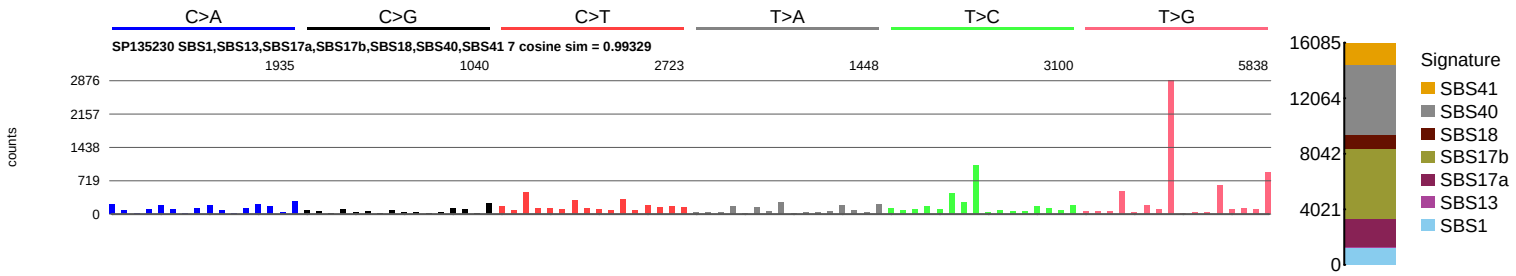

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99351
1922
1016
2685
1412
3249
5806
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

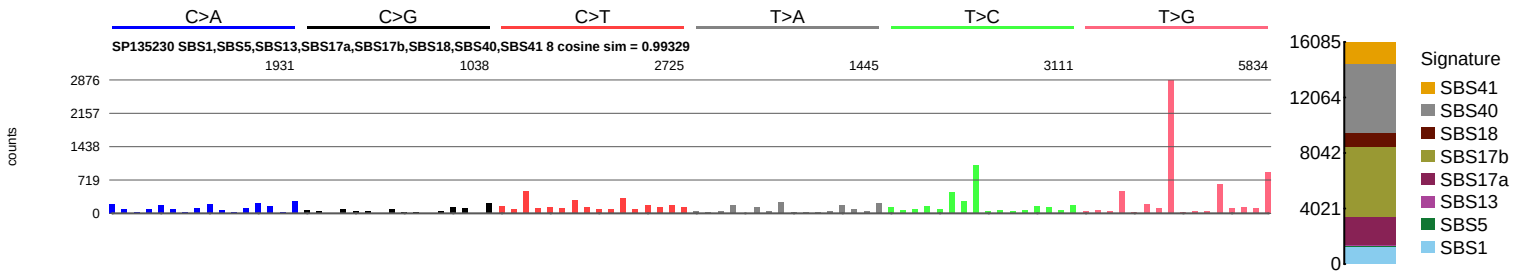

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99346
1440
1053
3105
1453
3152
5887
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99346
1440
1052
3107
1453
3152
5887
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.99345
1919
965
2701
1426
3255
5816
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.99335
2008
1024
3177
1083
3150
5642
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99329
1935
1040
2723
1448
3100
5838
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99329
1931
1038
2725
1445
3111
5834
counts
2876
2157
1438
719
0
16085
12064
8042
4021
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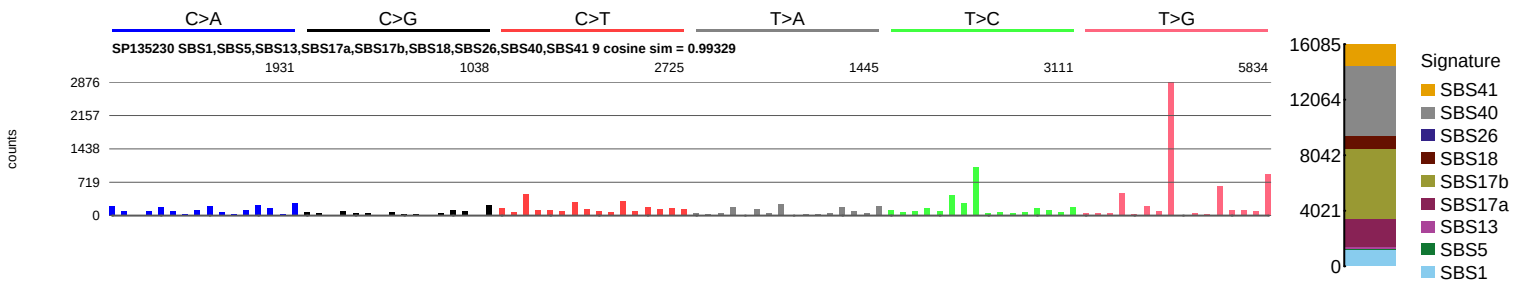
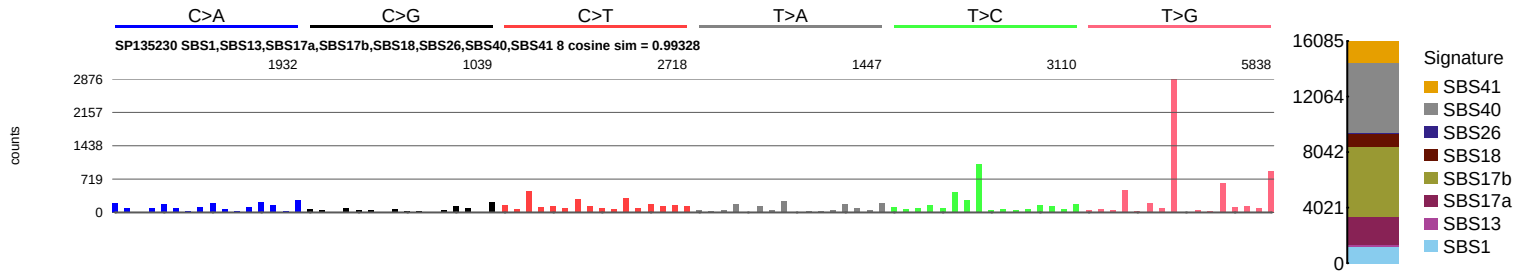
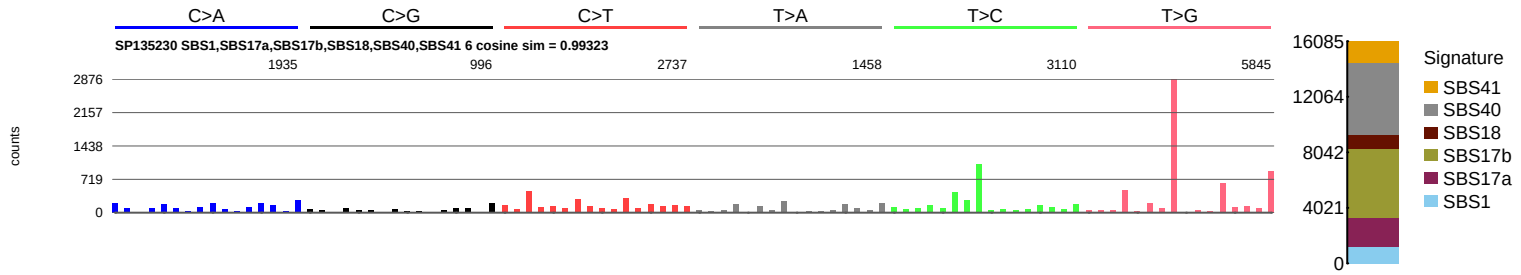
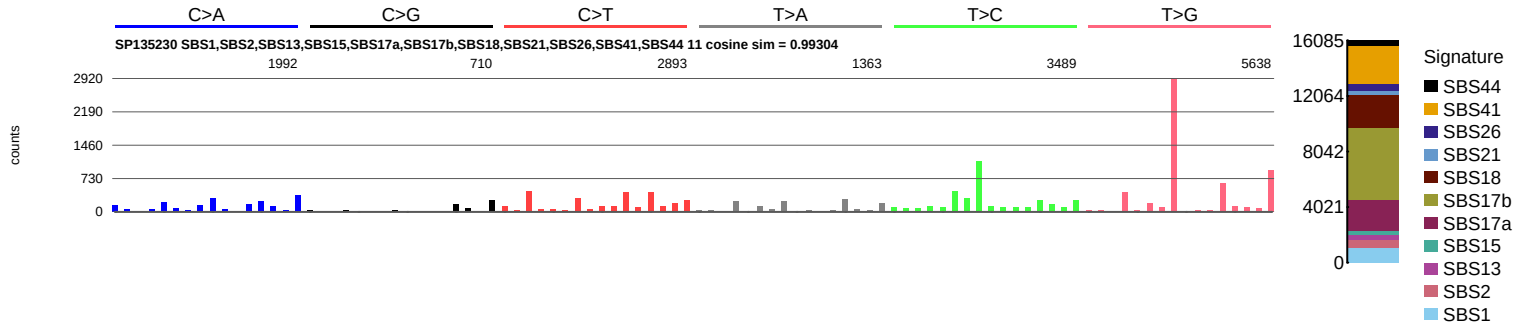
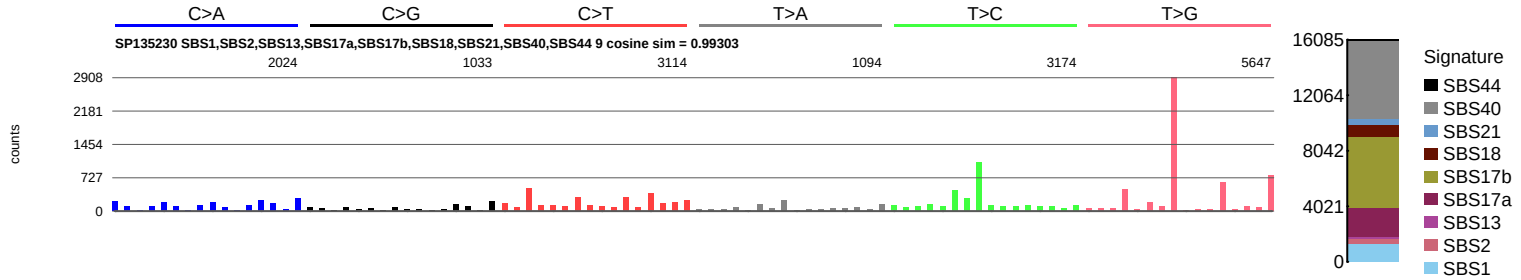
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

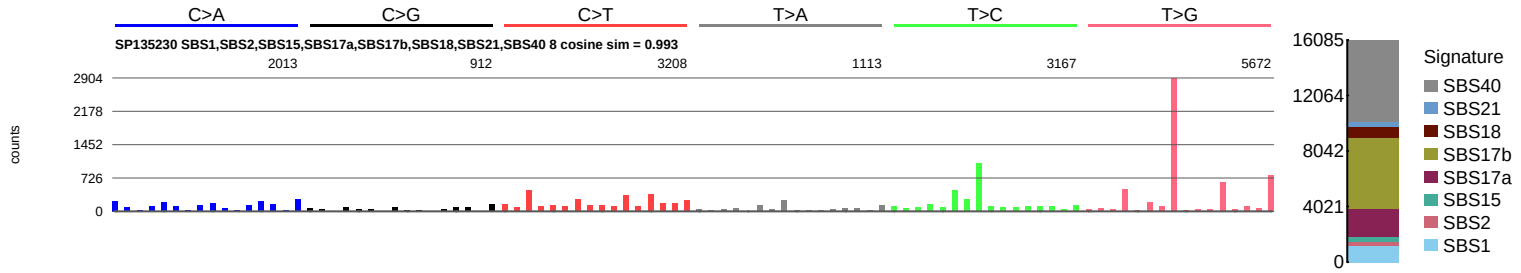

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99329
1931
1038
2725
1445
3111
5834
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

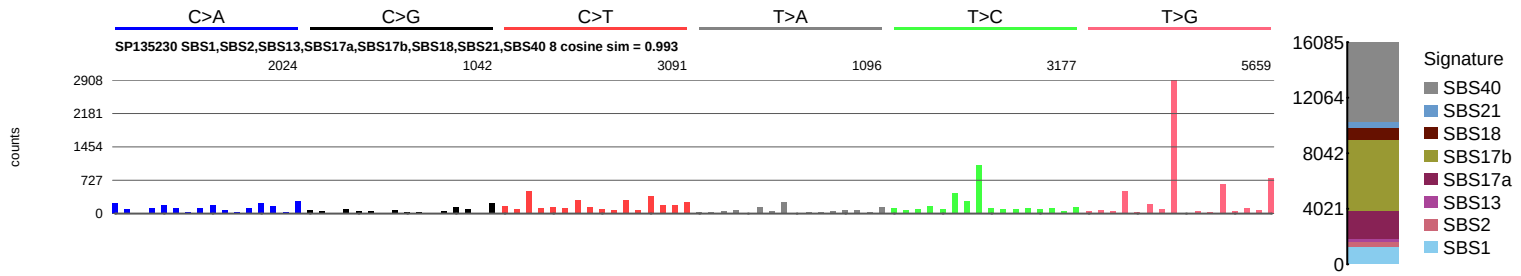

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.99328
1932
1039
2718
1447
3110
5838
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

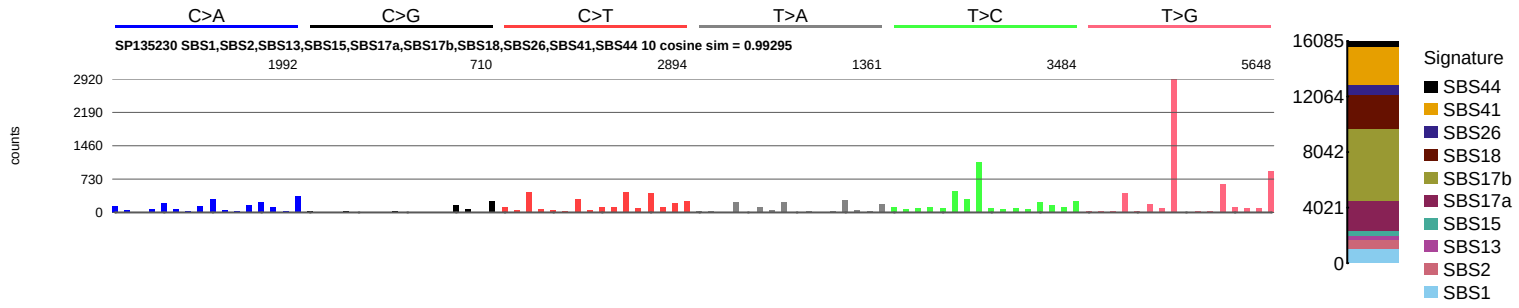

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.99323
1935
996
2737
1458
3110
5845
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1

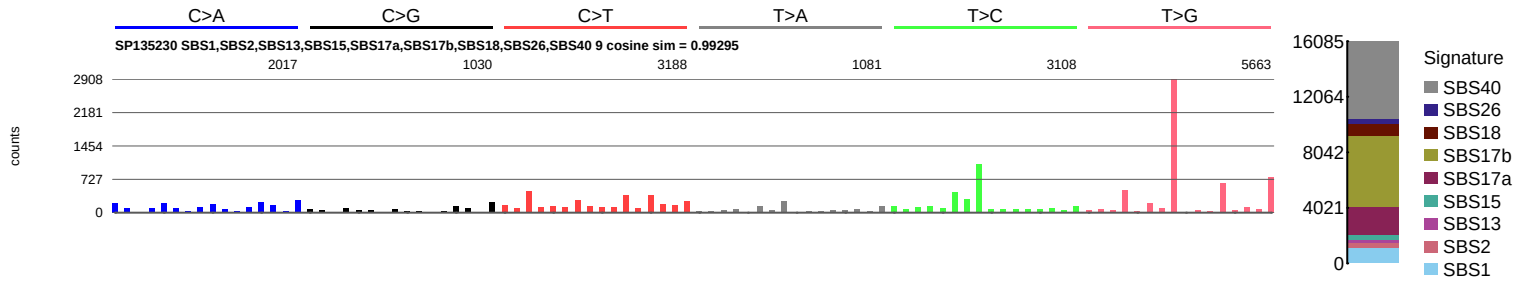

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.99304
1992
710
2893
1363
3489
5638
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

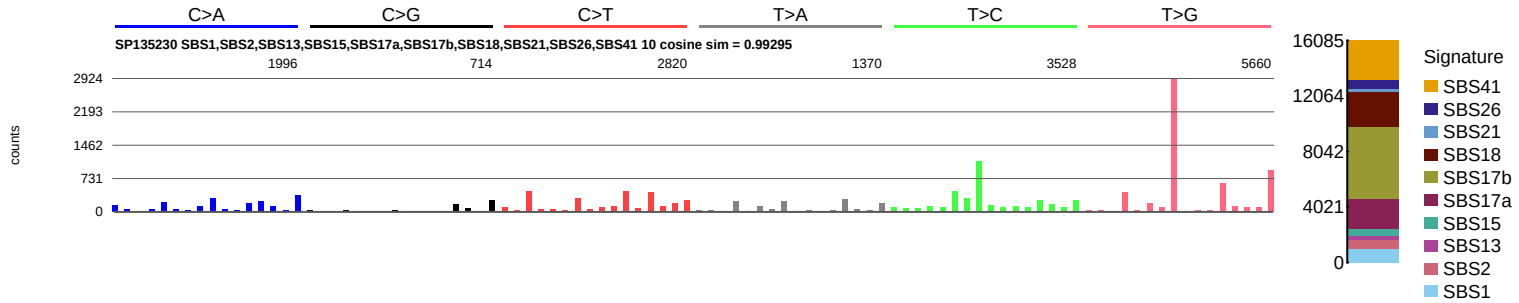

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.99303
2024
1033
3114
1094
3174
5647
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

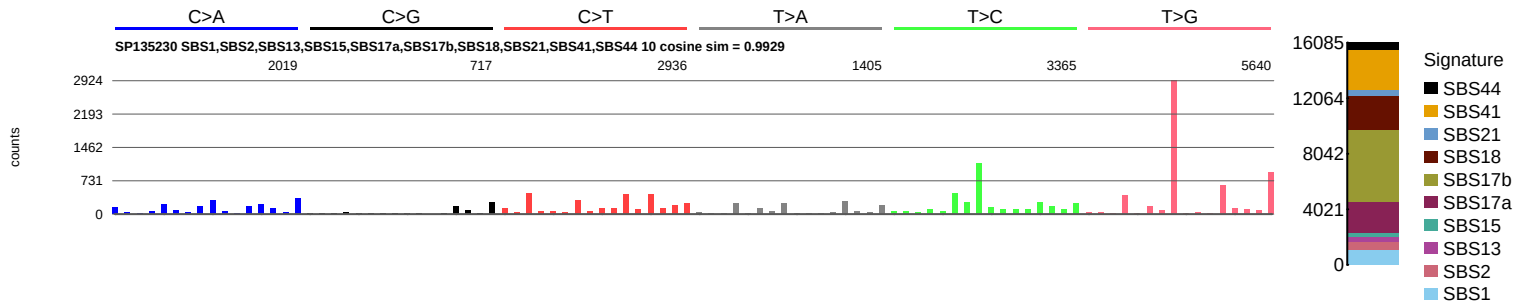

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.993
2013
912
3208
1113
3167
5672
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

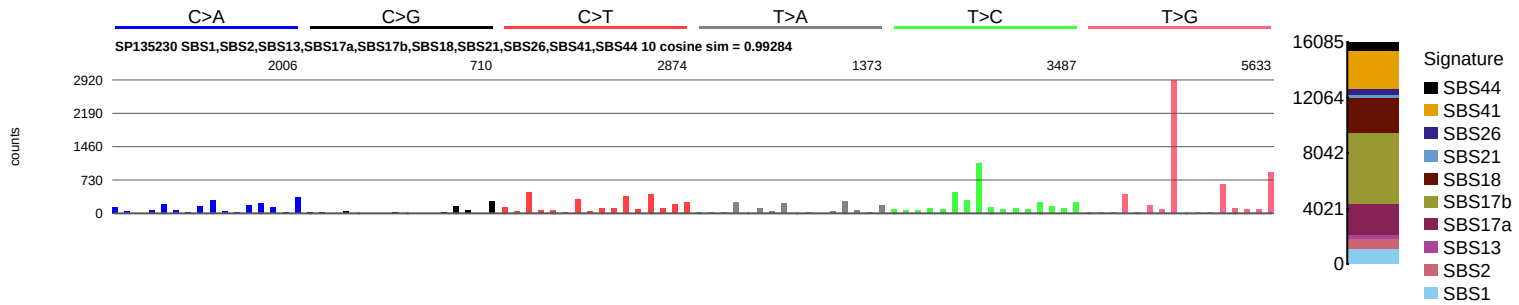

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.993
2024
1042
3091
1096
3177
5659
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

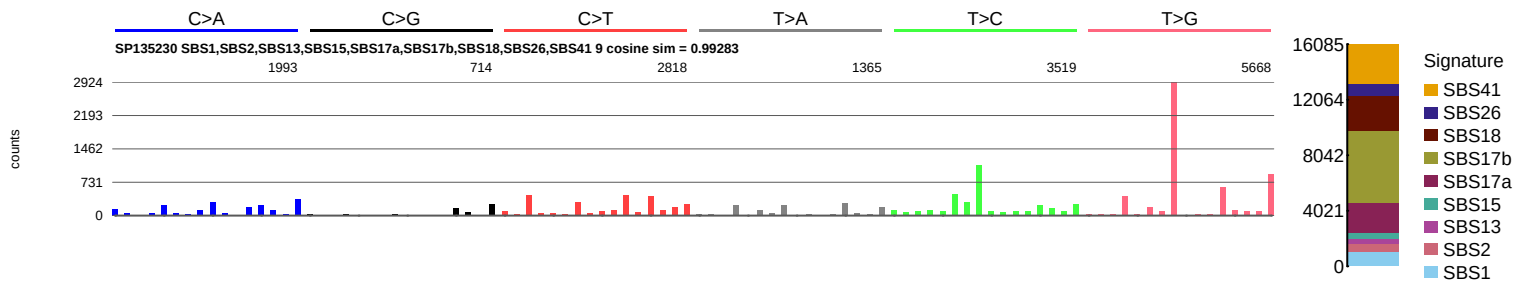

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99295
1992
710
2894
1361
3484
5648
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

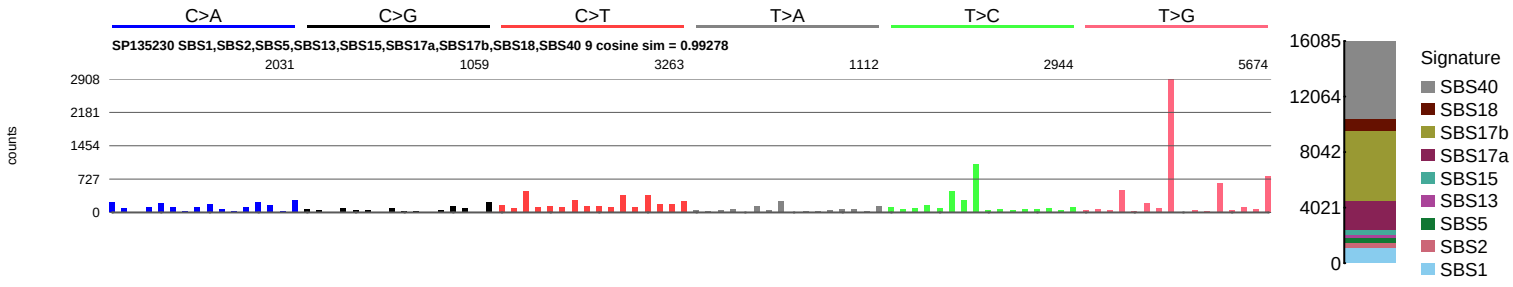

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.99295
2017
1030
3188
1081
3108
5663
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

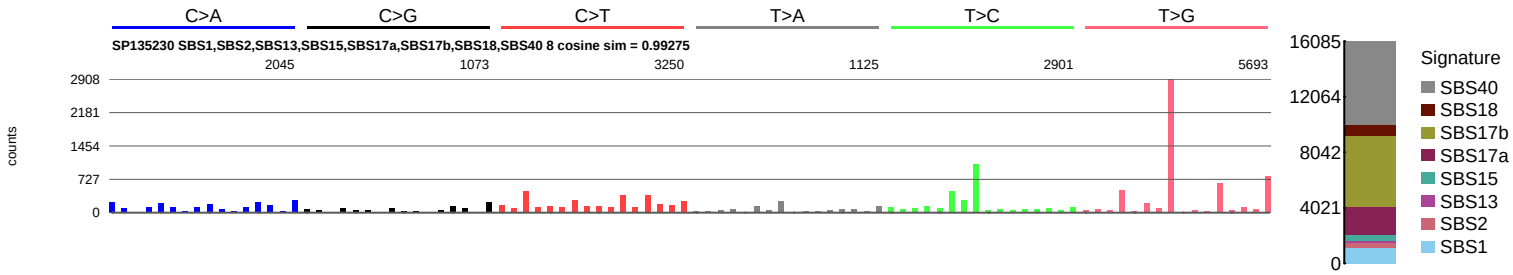

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99295
1996
714
2820
1370
3528
5660
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

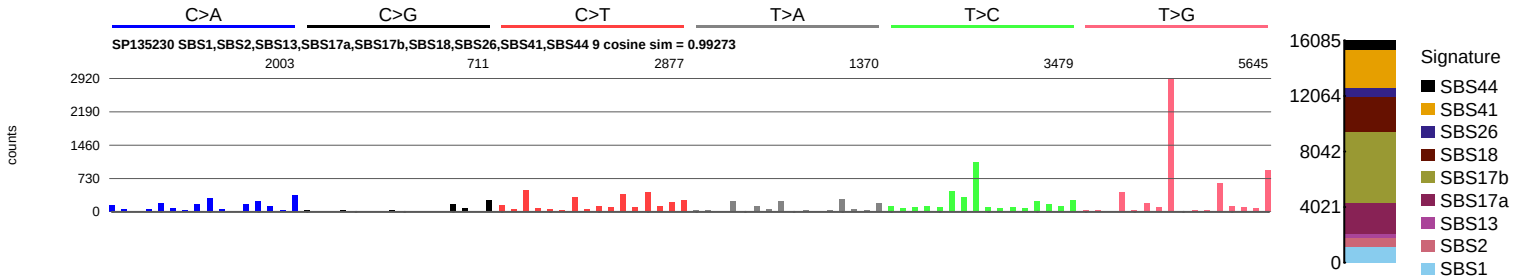

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9929
2019
717
2936
1405
3365
5640
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

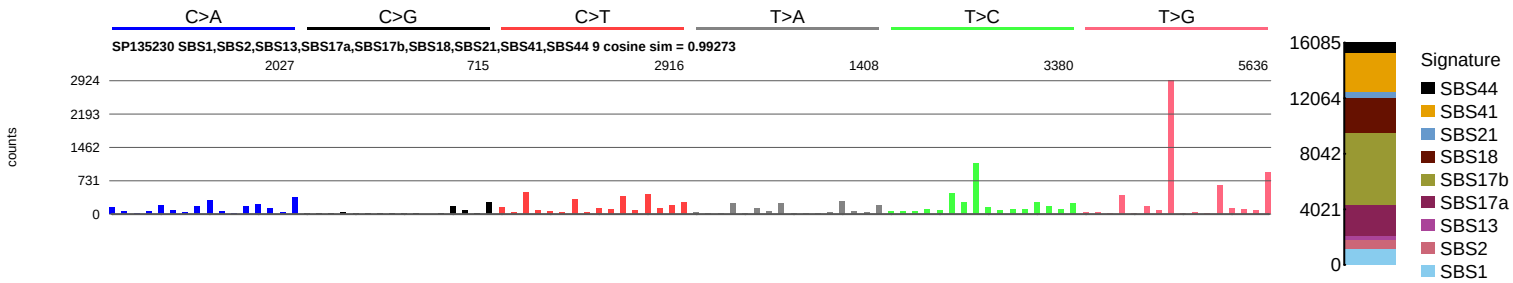

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99284
2006
710
2874
1373
3487
5633
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

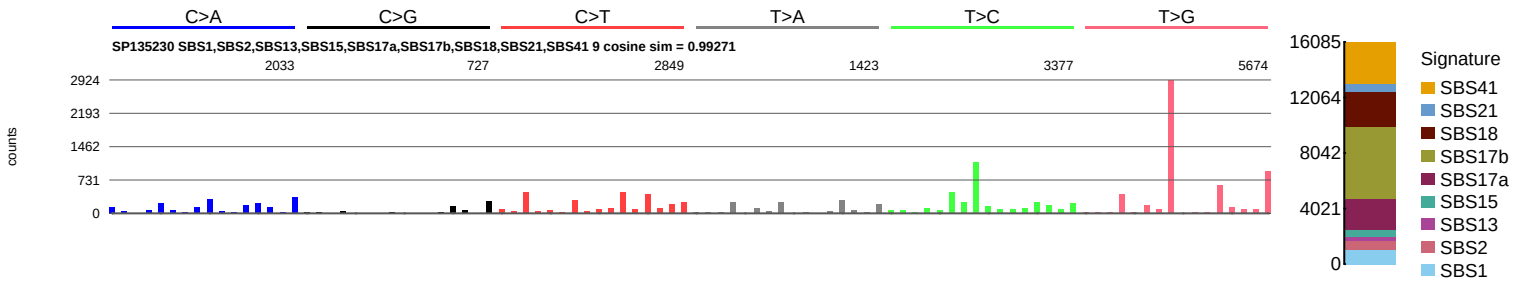

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99283
1993
714
2818
1365
3519
5668
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

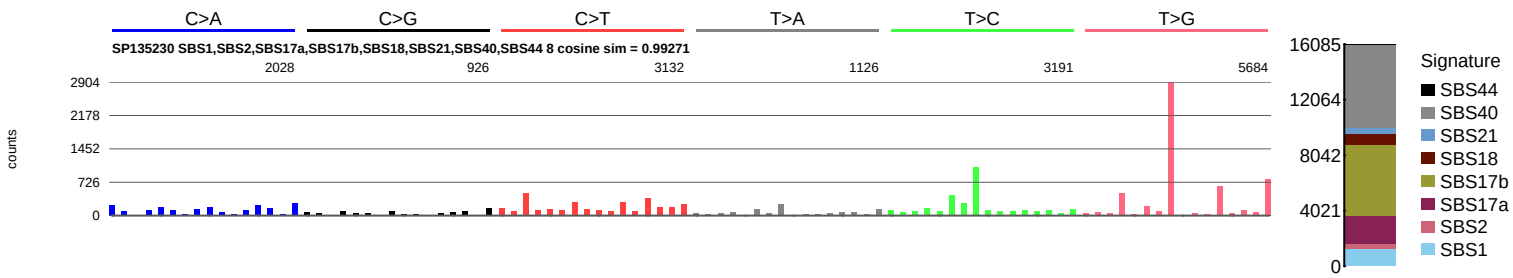

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.99278
2031
1059
3263
1112
2944
5674
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

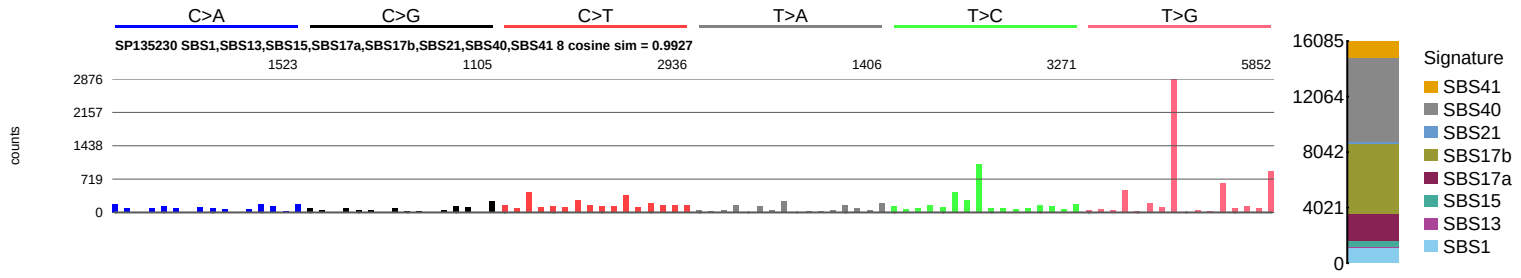

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.99275
2045
1073
3250
1125
2901
5693
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

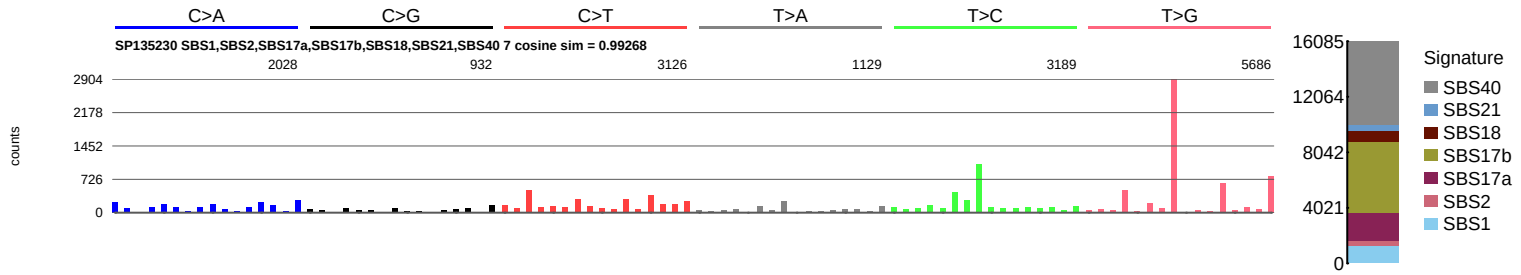

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99273
2003
711
2877
1370
3479
5645
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

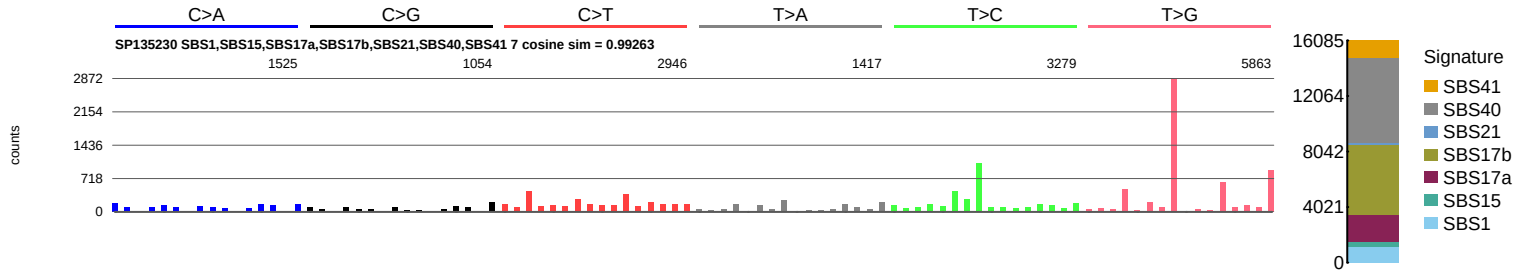

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99273
2027
715
2916
1408
3380
5636
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

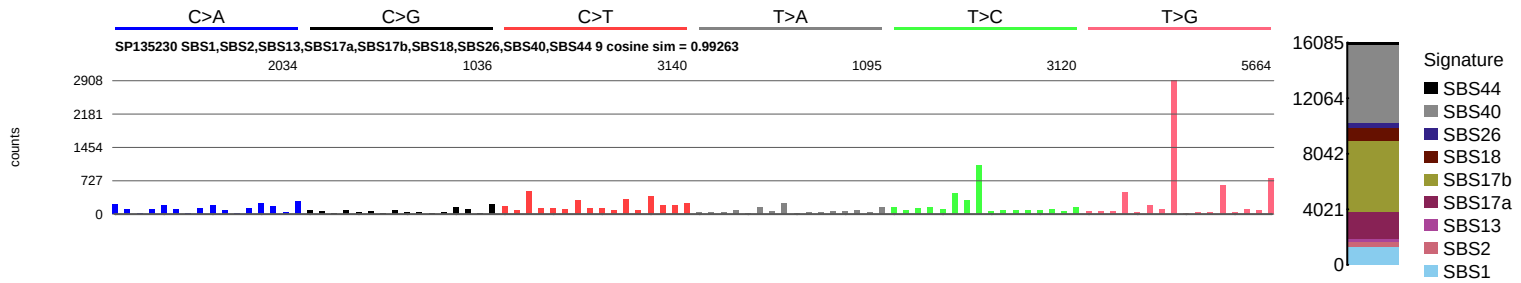

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99271
2033
727
2849
1423
3377
5674
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

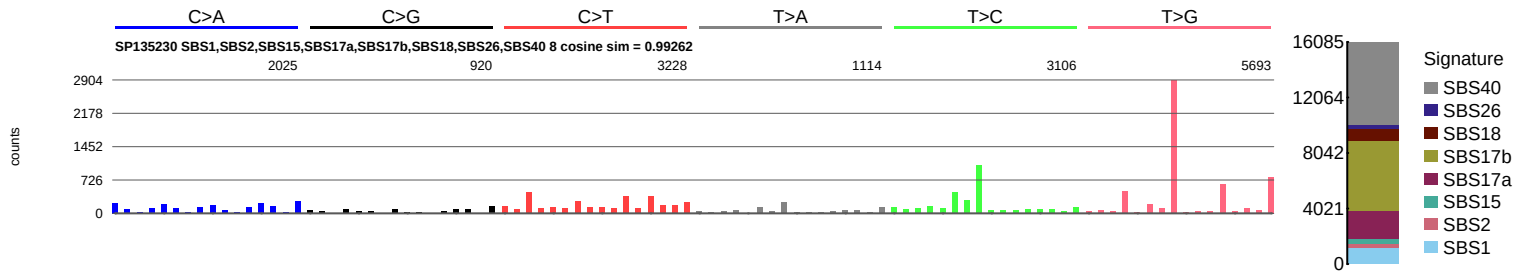

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.99271
2028
926
3132
1126
3191
5684
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.9927
1523
1105
2936
1406
3271
5852
counts
2876
2157
1438
719
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.99268
2028
932
3126
1129
3189
5686
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99263
1525
1054
2946
1417
3279
5863
counts
2872
2154
1436
718
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99263
2034
1036
3140
1095
3120
5664
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99262
2025
920
3228
1114
3106
5693
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
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SBS17b
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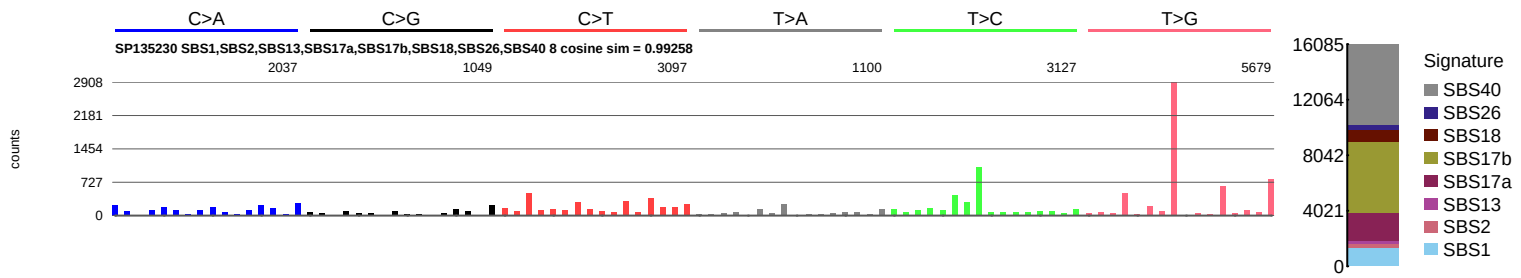
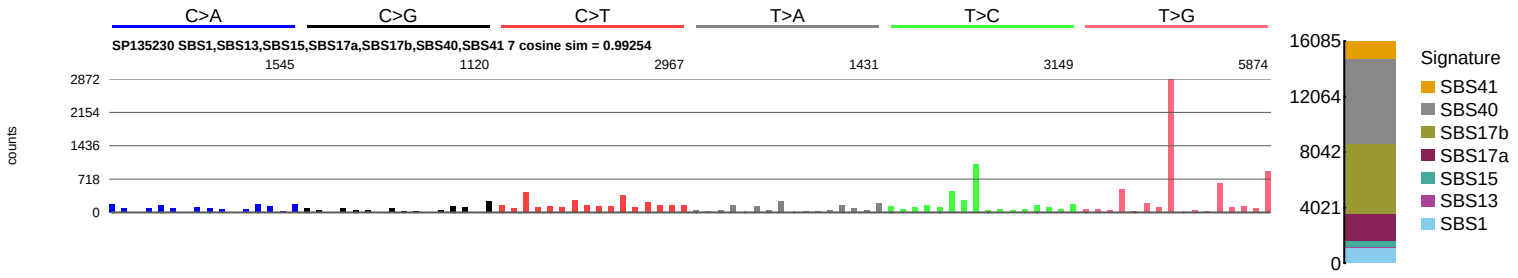
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99258
2037
1049
3097
1100
3127
5679
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99254
1545
1120
2967
1431
3149
5874
counts
2872
2154
1436
718
0
16085
12064
8042
4021
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SBS41
SBS40
SBS17b
SBS17a
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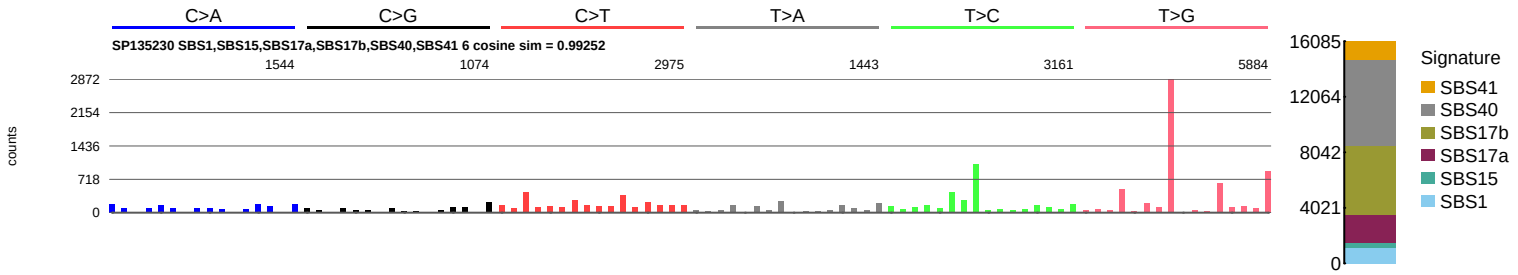
SBS13
SBS1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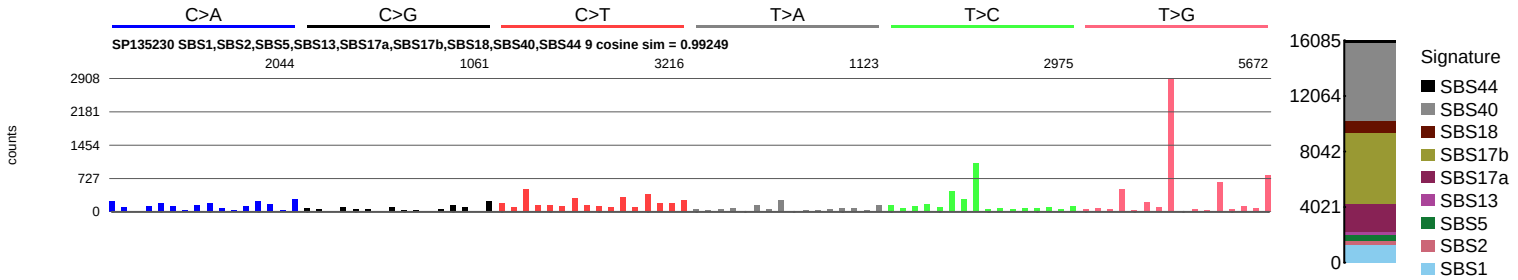
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99252
1544
1074
2975
1443
3161
5884
counts
2872
2154
1436
718
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1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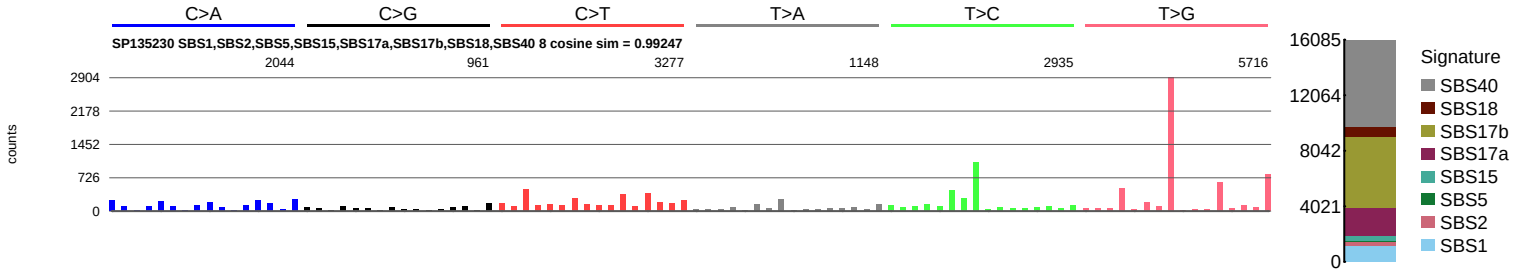
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.99249
2044
1061
3216
1123
2975
5672
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
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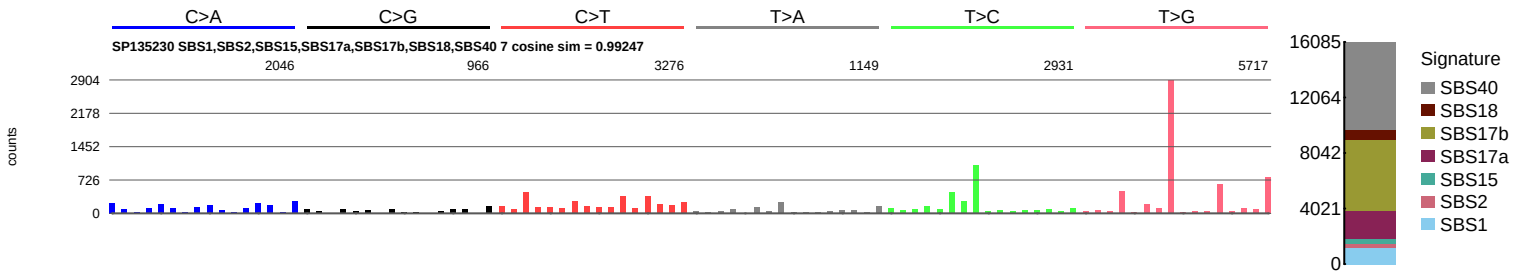
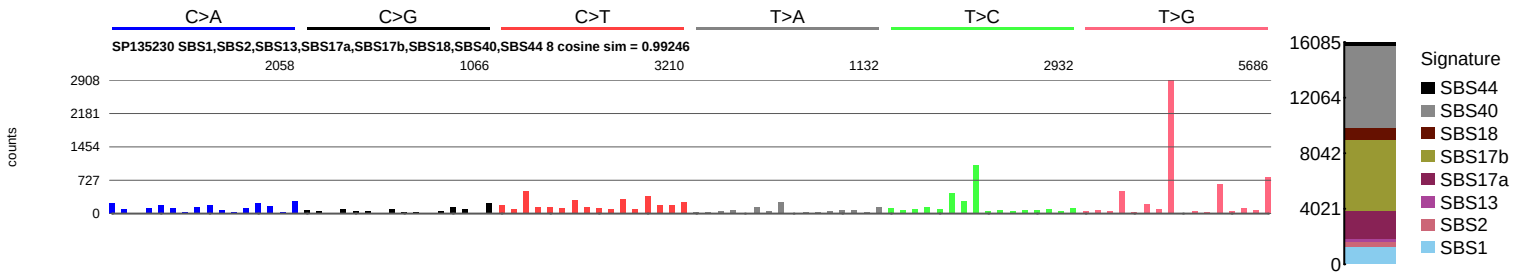
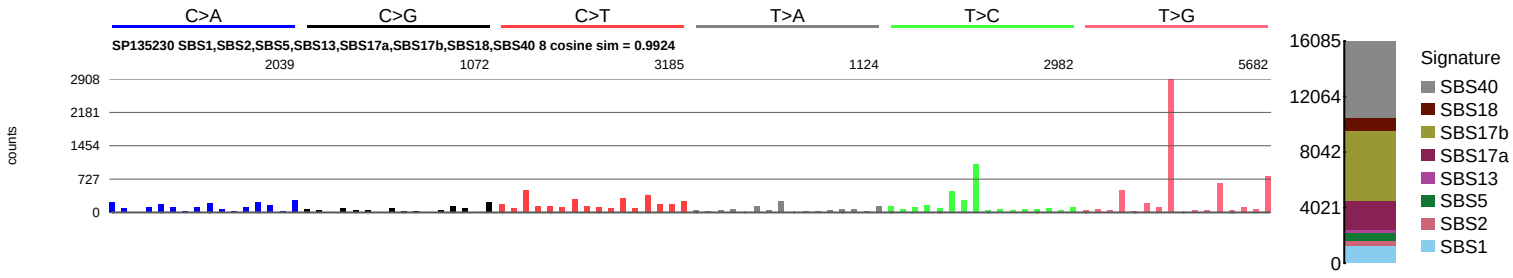
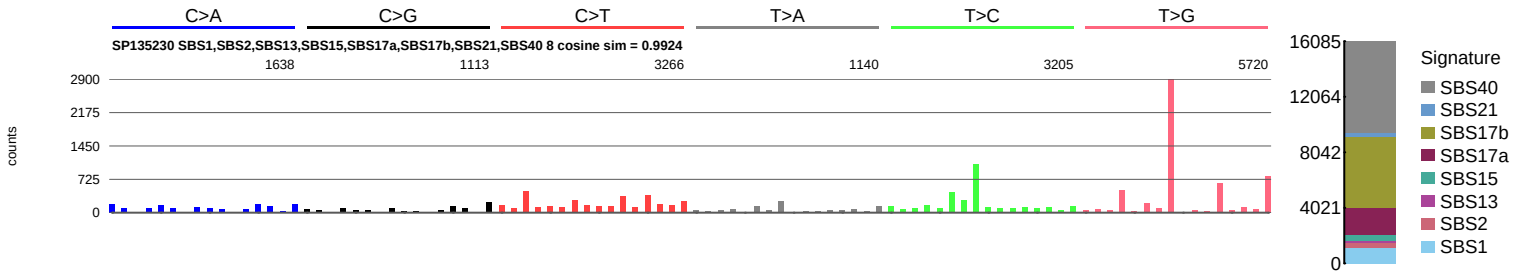
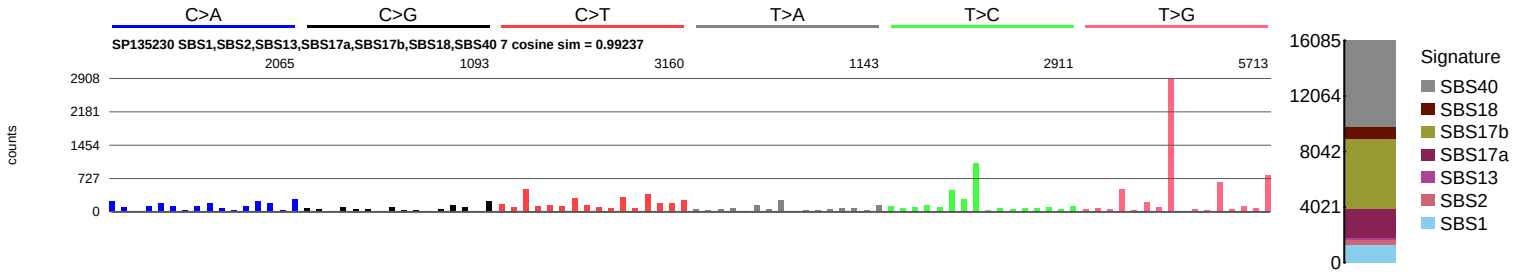
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.99247
2044
961
3277
1148
2935
5716
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

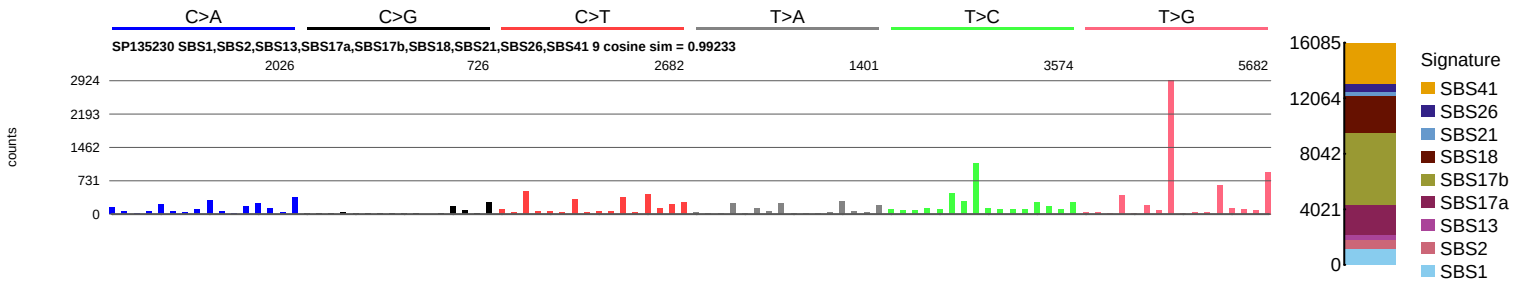

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99247
2046
966
3276
1149
2931
5717
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

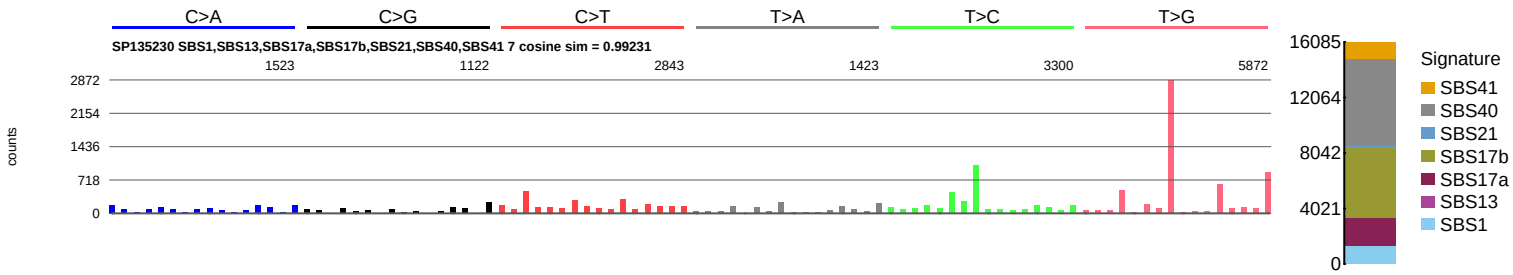

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.99246
2058
1066
3210
1132
2932
5686
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

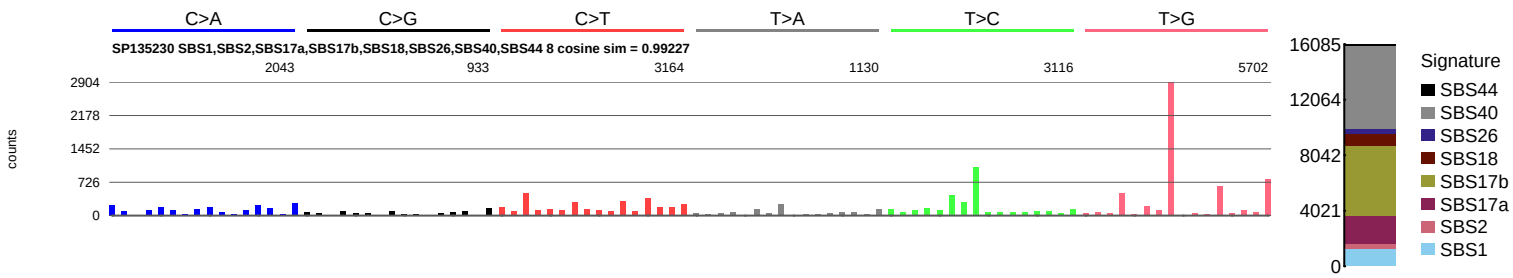

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.9924
2039
1072
3185
1124
2982
5682
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

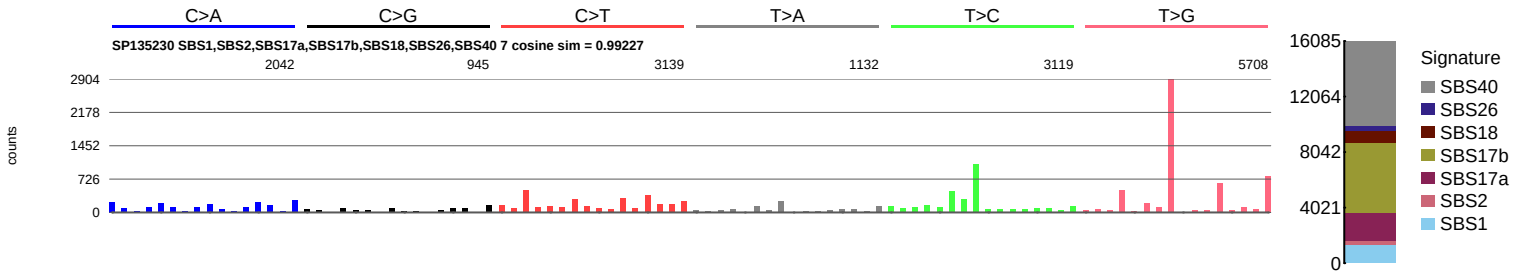

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.9924
1638
1113
3266
1140
3205
5720
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

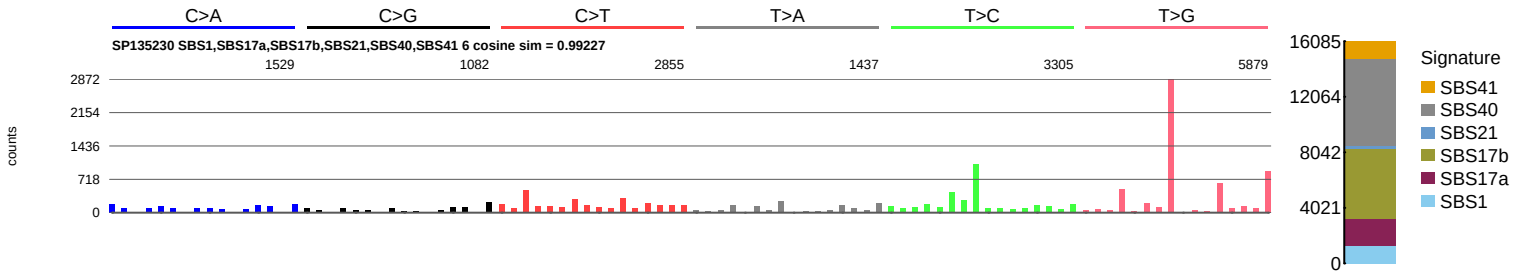

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99237
2065
1093
3160
1143
2911
5713
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

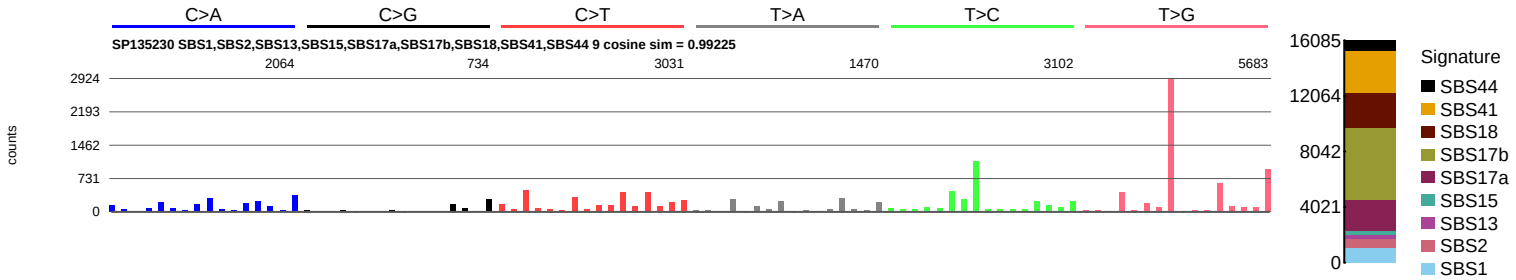

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99233
2026
726
2682
1401
3574
5682
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

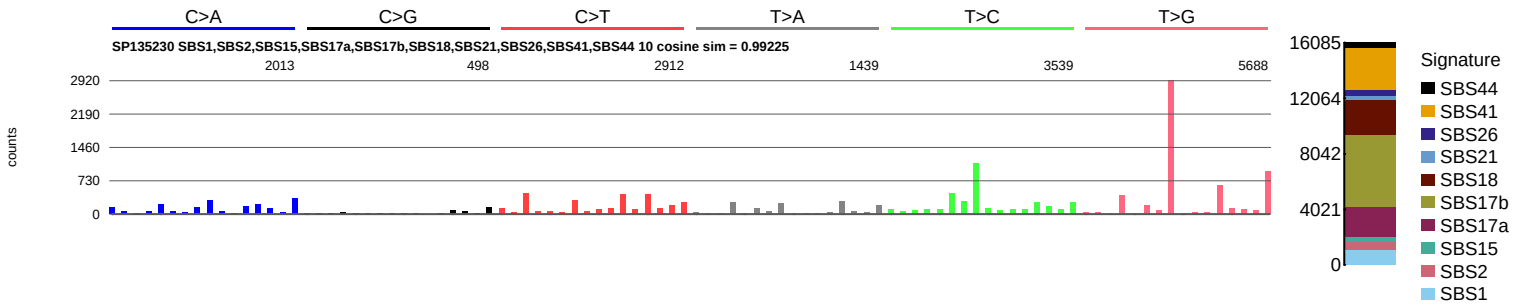

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99231
1523
1122
2843
1423
3300
5872
counts
2872
2154
1436
718
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1

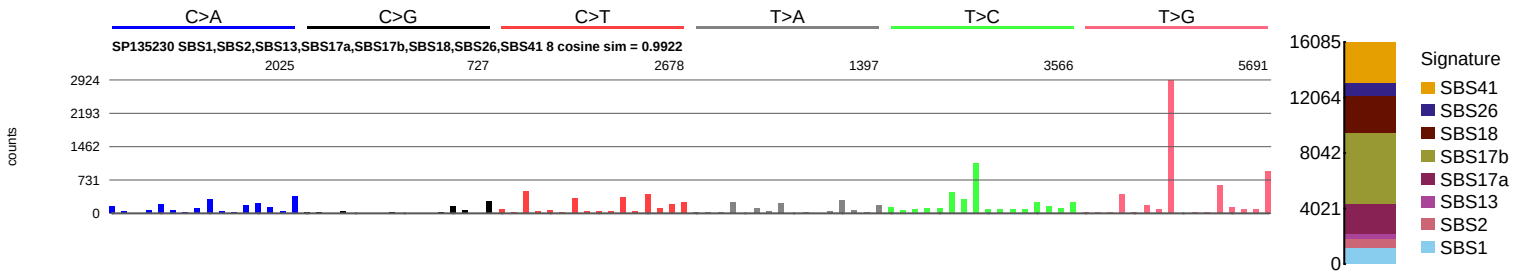

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.99227
2043
933
3164
1130
3116
5702
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

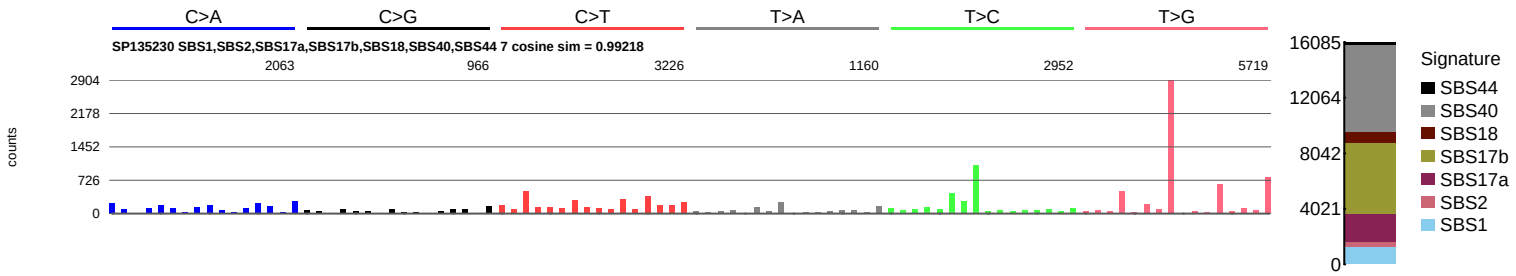

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.99227
2042
945
3139
1132
3119
5708
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

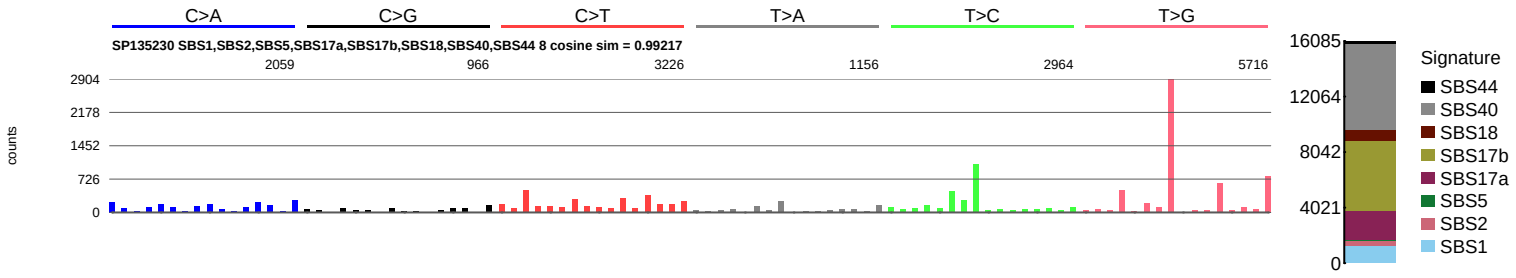

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.99227
1529
1082
2855
1437
3305
5879
counts
2872
2154
1436
718
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1

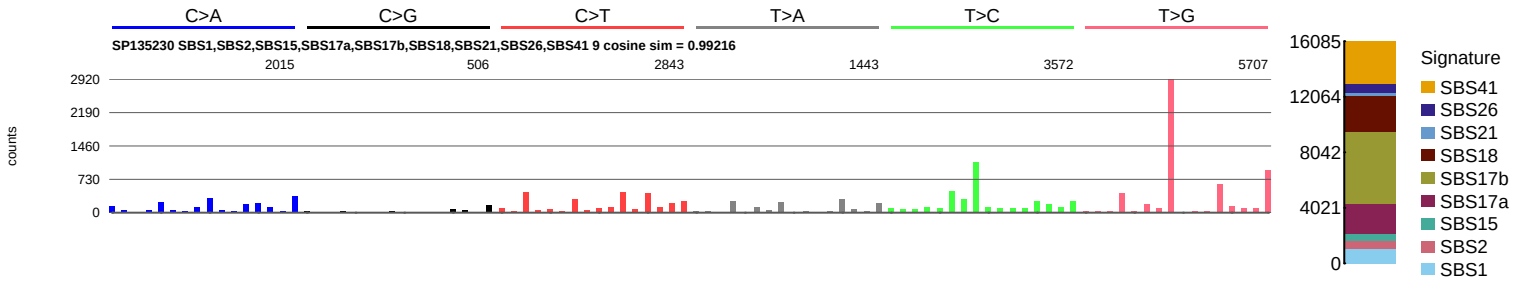

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99225
2064
734
3031
1470
3102
5683
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

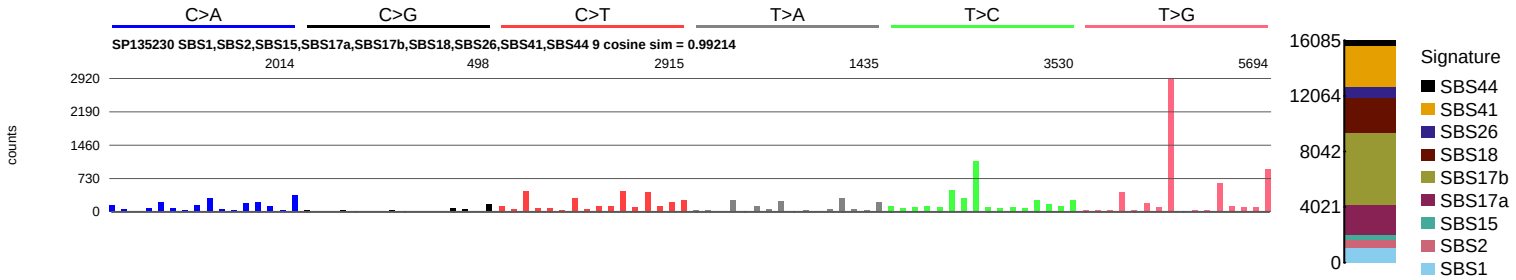

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99225
2013
498
2912
1439
3539
5688
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

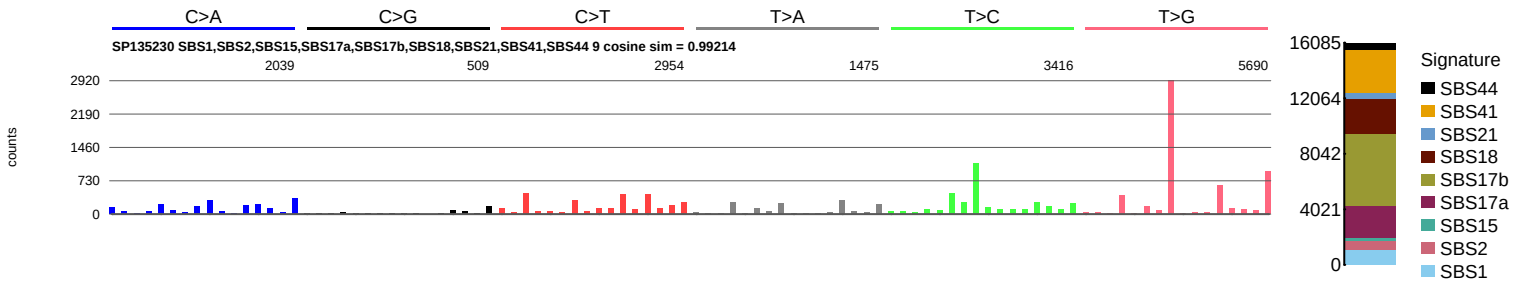

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.9922
2025
727
2678
1397
3566
5691
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

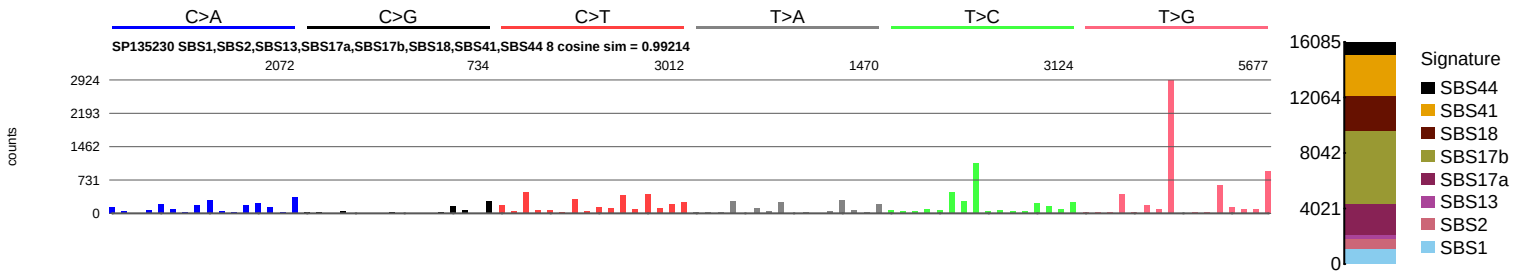

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.99218
2063
966
3226
1160
2952
5719
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

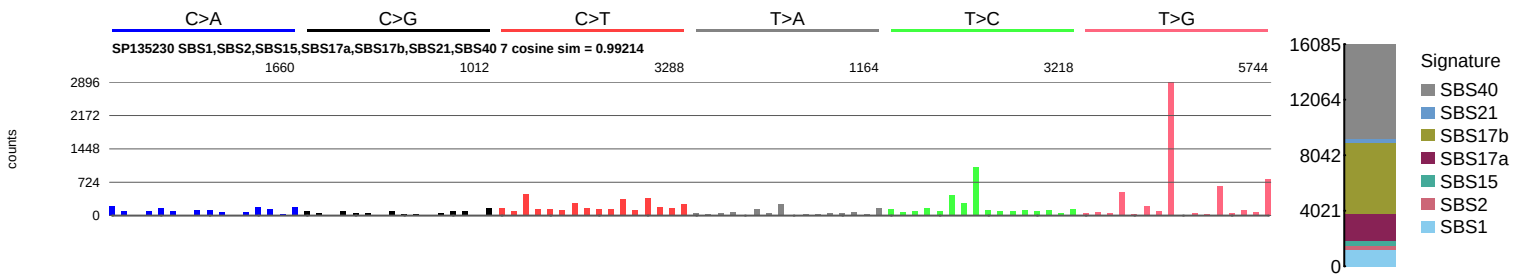

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.99217
2059
966
3226
1156
2964
5716
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

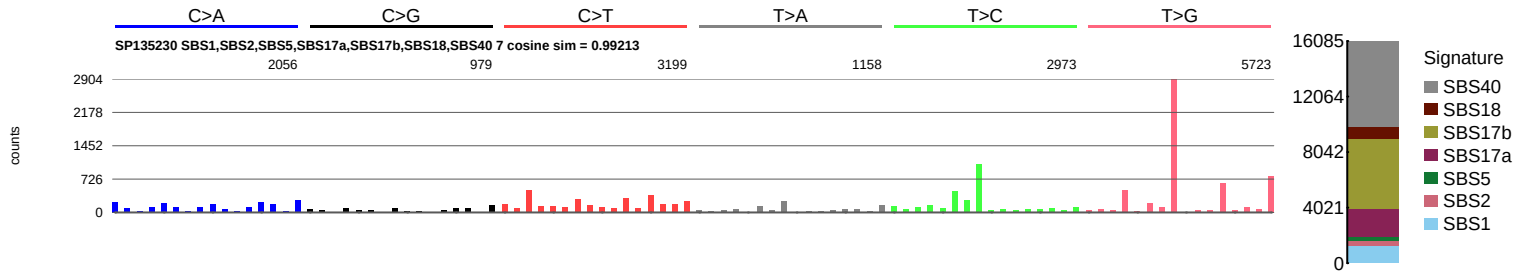

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99216
2015
506
2843
1443
3572
5707
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

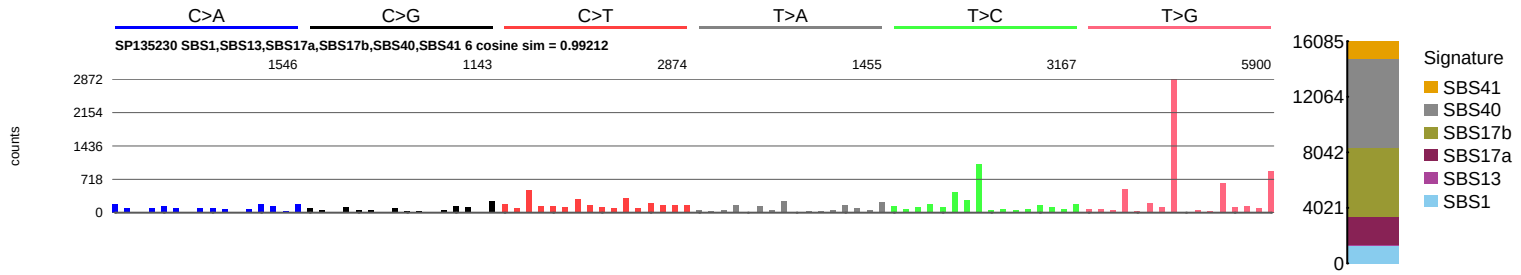

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99214
2014
498
2915
1435
3530
5694
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

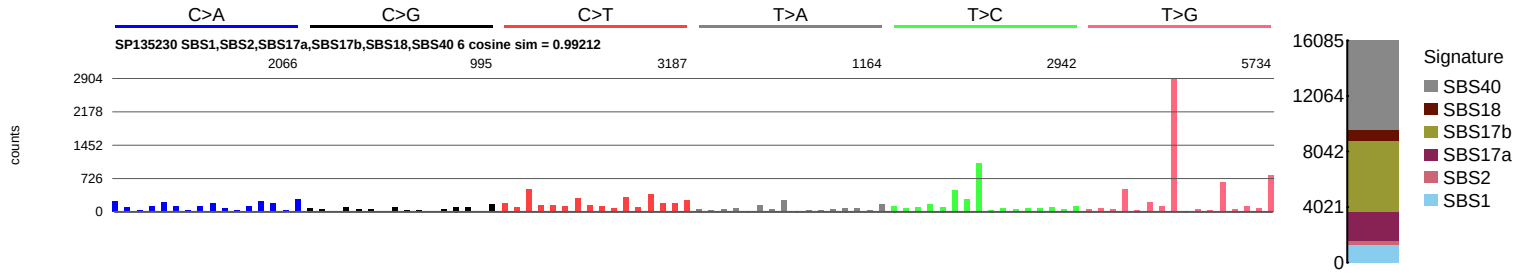

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99214
2039
509
2954
1475
3416
5690
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

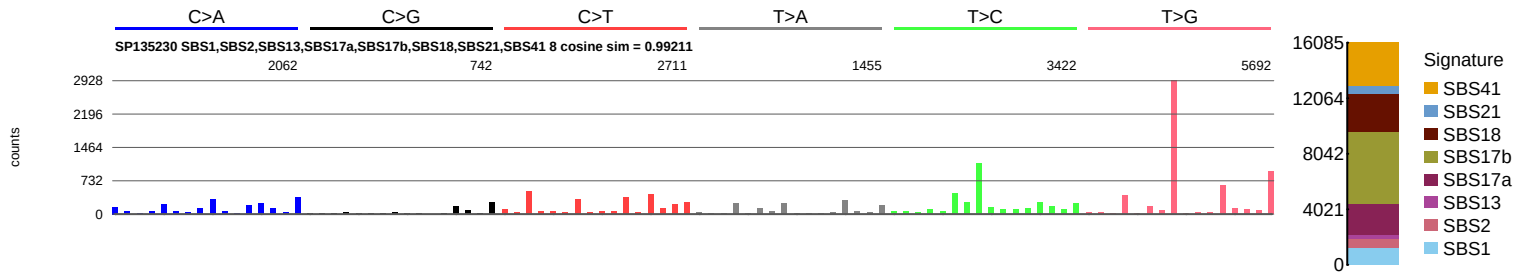

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99214
2072
734
3012
1470
3124
5677
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.99214
1660
1012
3288
1164
3218
5744
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99213
2056
979
3199
1158
2973
5723
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99212
1546
1143
2874
1455
3167
5900
counts
2872
2154
1436
718
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.99212
2066
995
3187
1164
2942
5734
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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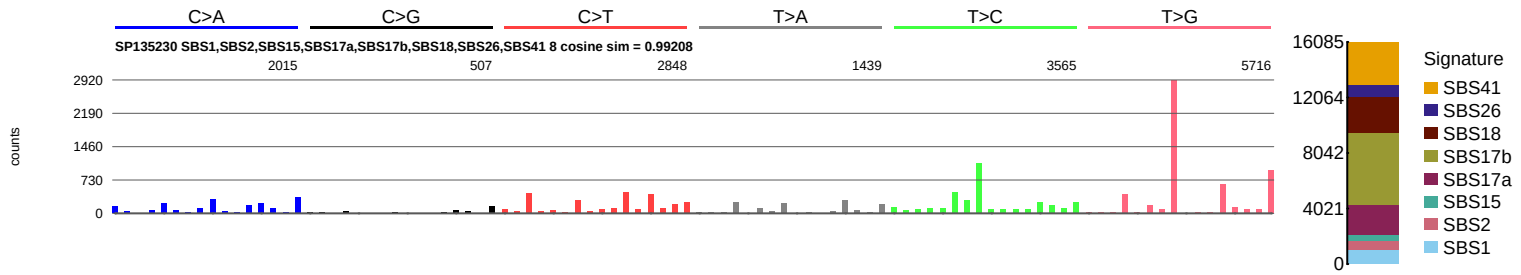
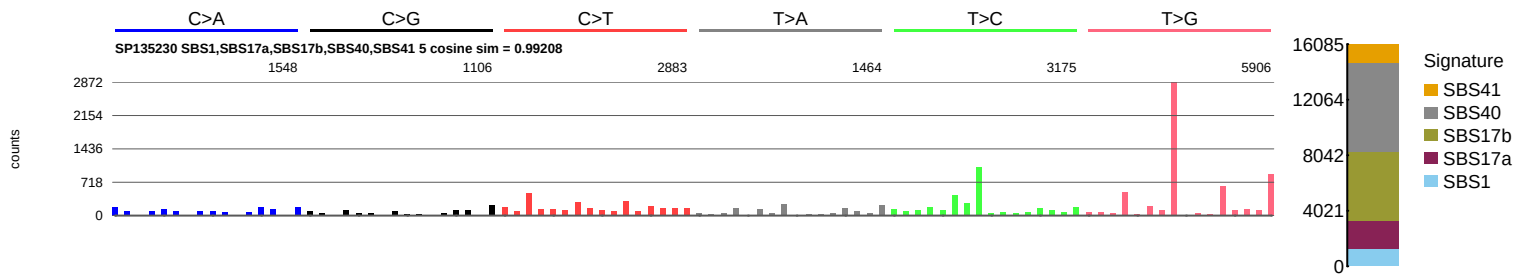
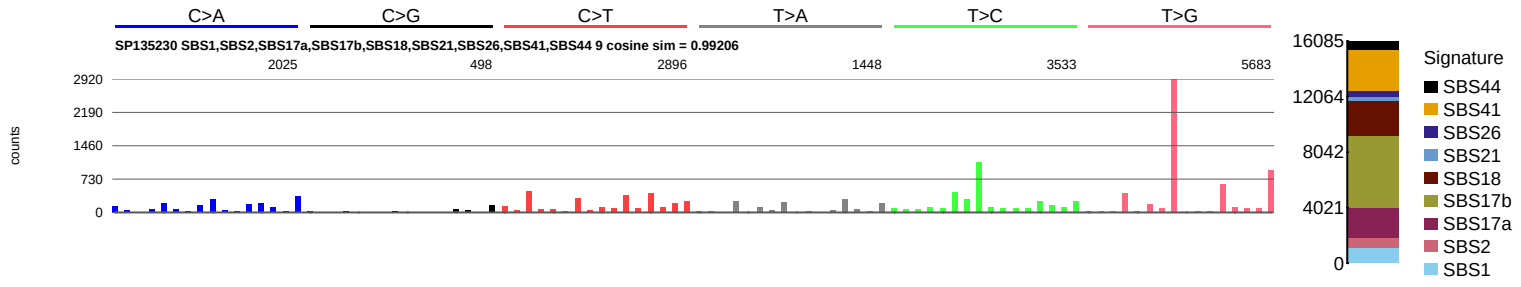
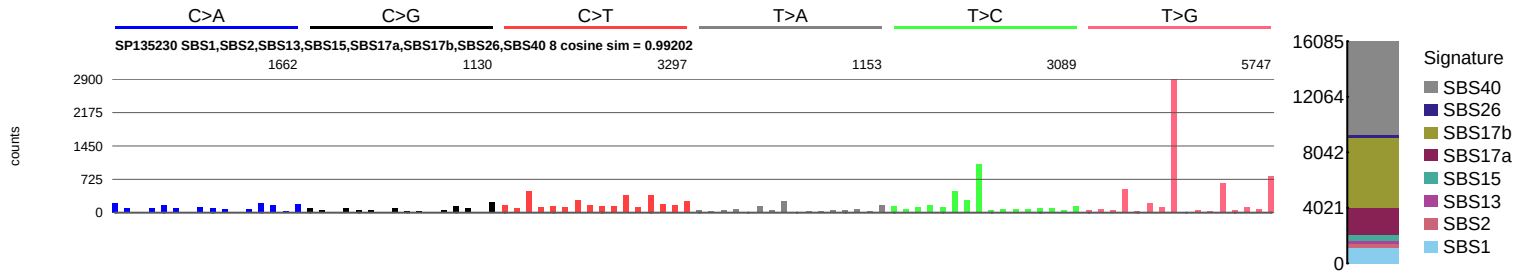
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99211
2062
742
2711
1455
3422
5692
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99208
2015
507
2848
1439
3565
5716
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.99208
1548
1106
2883
1464
3175
5906
counts
2872
2154
1436
718
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.99206
2025
498
2896
1448
3533
5683
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.99202
1662
1130
3297
1153
3089
5747
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS40
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SBS17b
SBS17a
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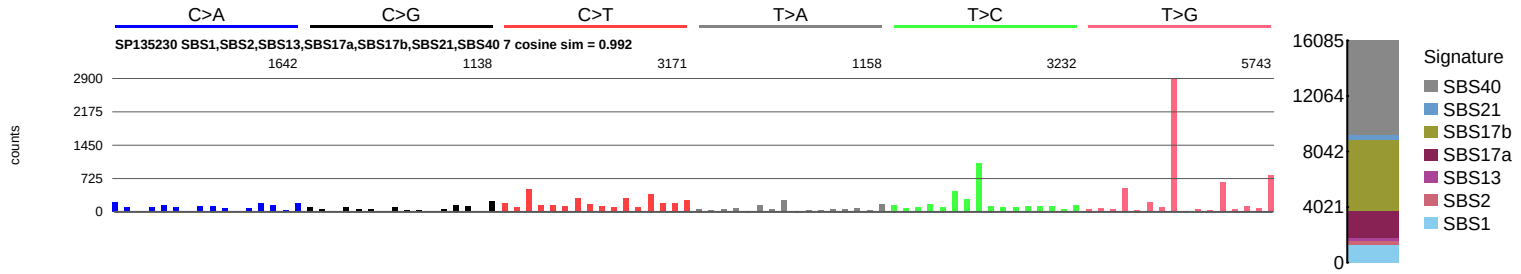
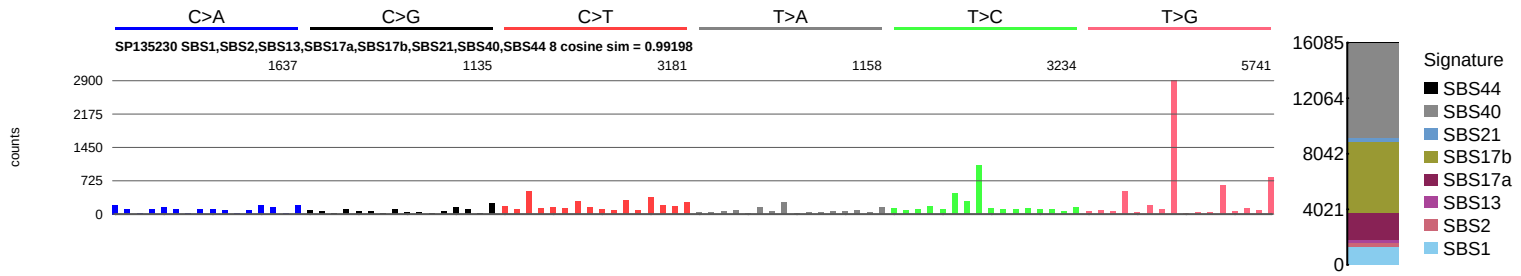
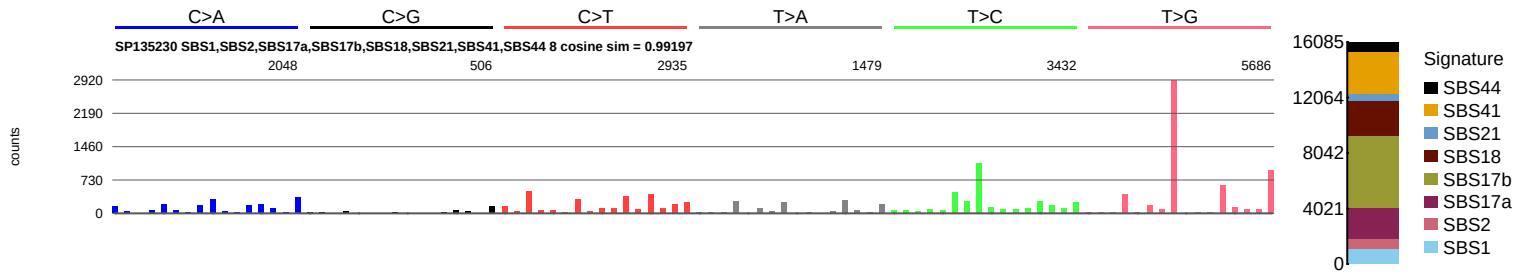
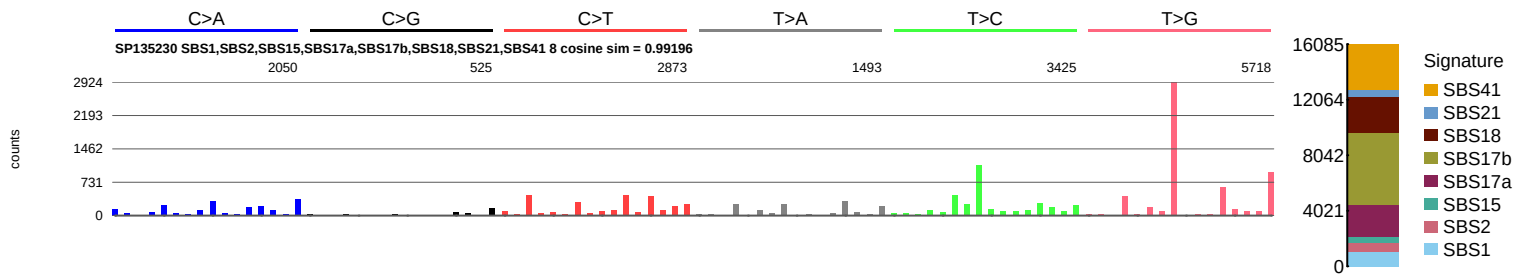
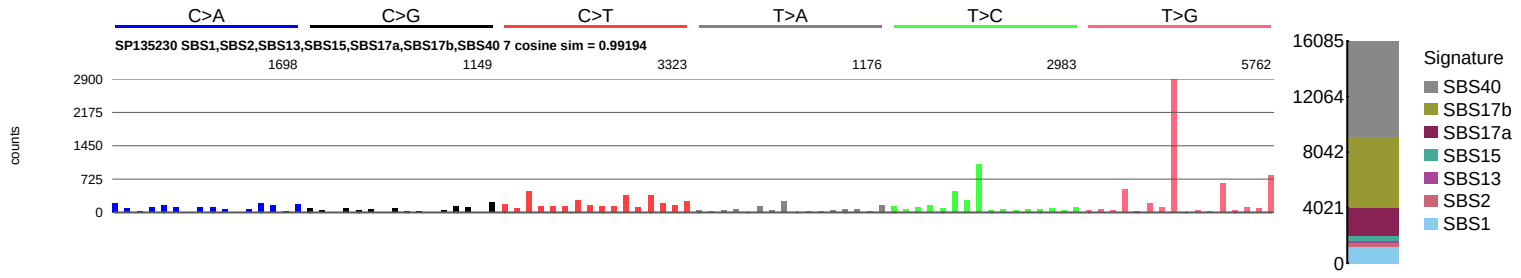
SBS13
SBS2
SBS1

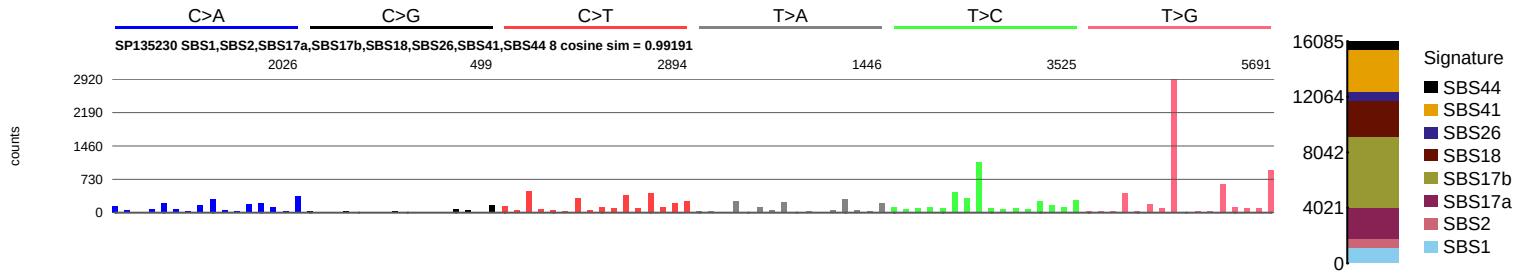

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.992
1642
1138
3171
1158
3232
5743
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

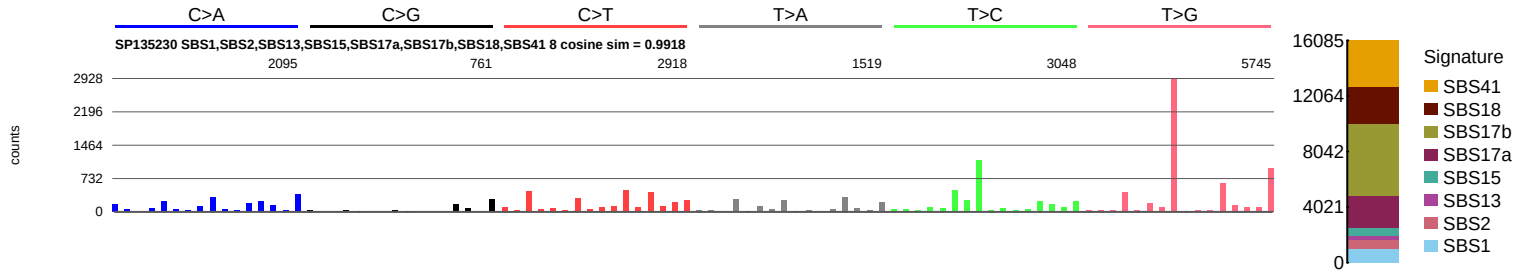

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.99198
1637
1135
3181
1158
3234
5741
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

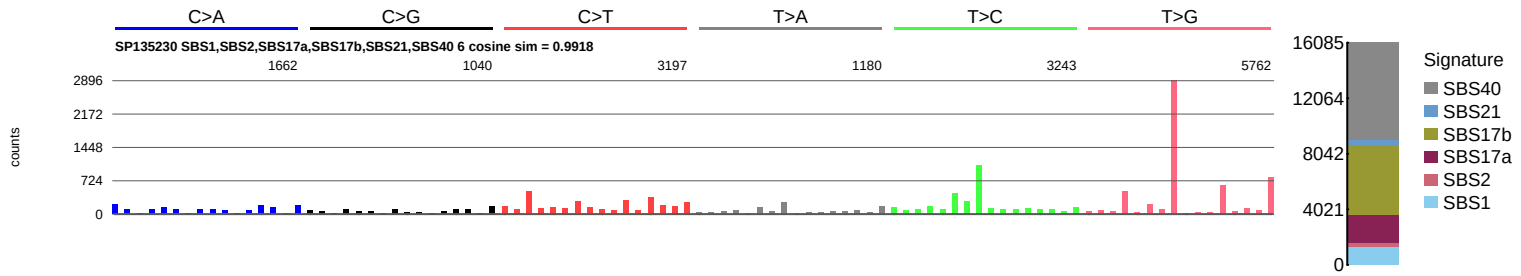

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.99197
2048
506
2935
1479
3432
5686
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

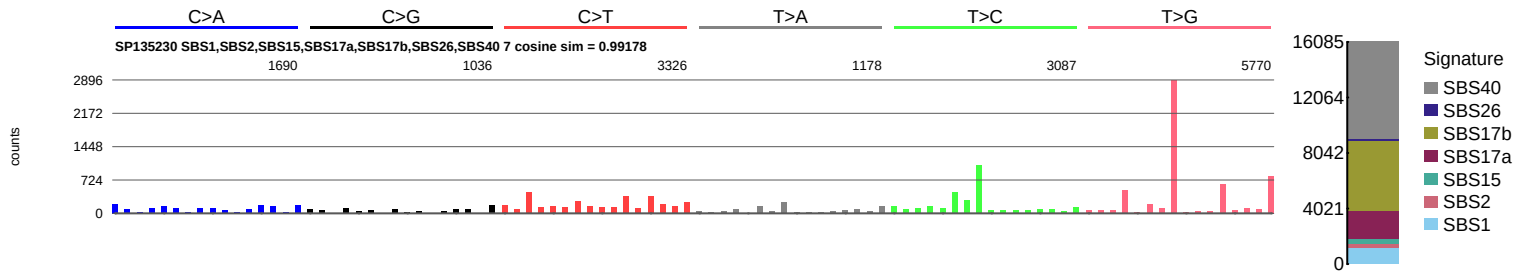

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99196
2050
525
2873
1493
3425
5718
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

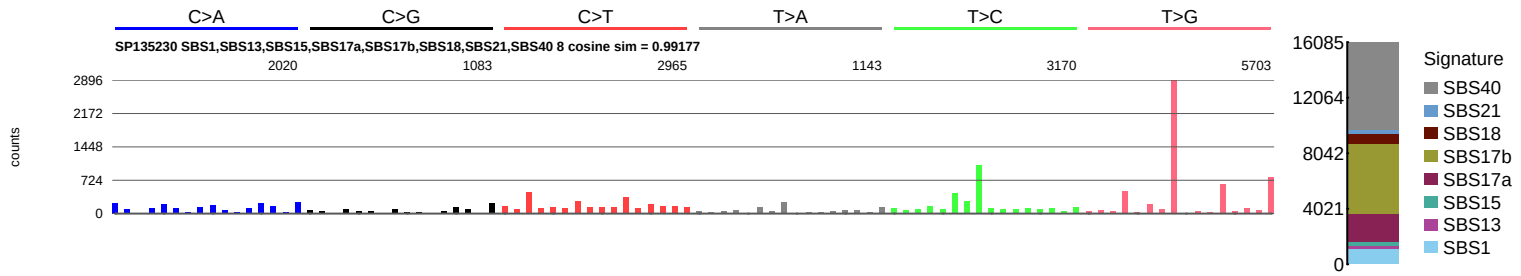

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.99194
1698
1149
3323
1176
2983
5762
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

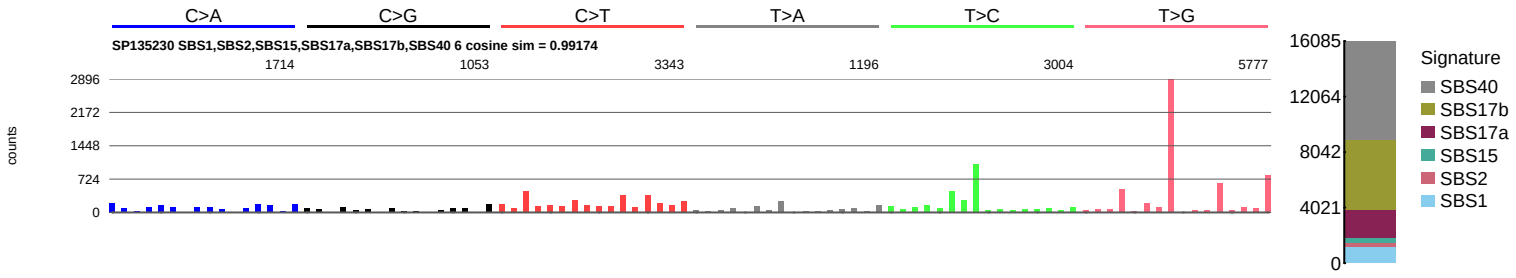

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.99191
2026
499
2894
1446
3525
5691
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

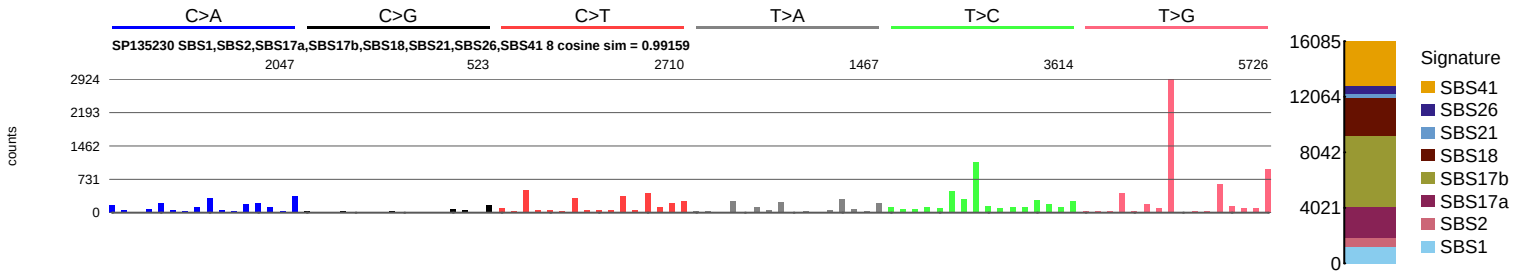

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9918
2095
761
2918
1519
3048
5745
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

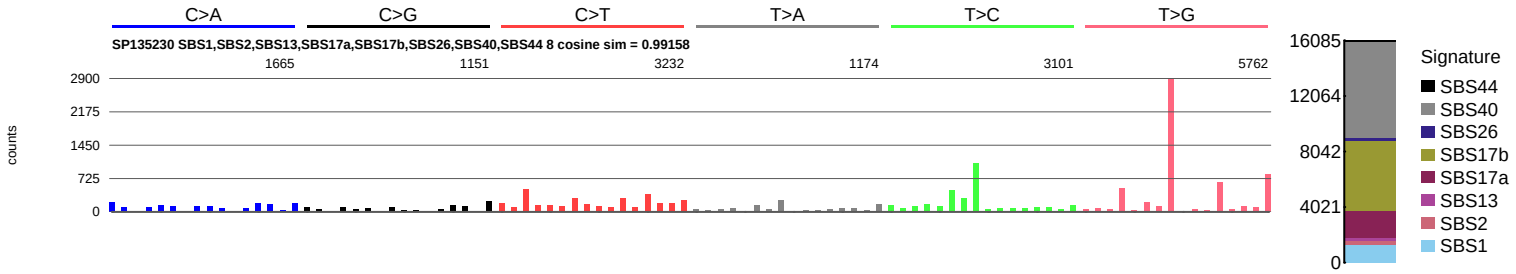

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.9918
1662
1040
3197
1180
3243
5762
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1

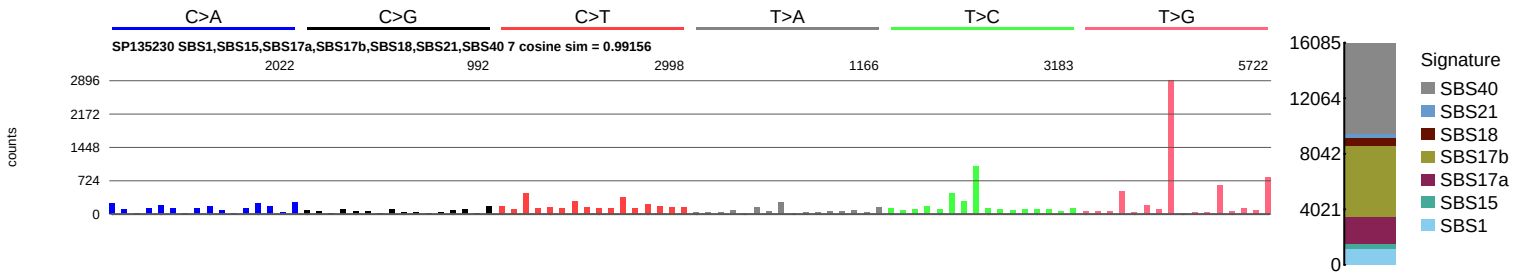

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.99178
1690
1036
3326
1178
3087
5770
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1

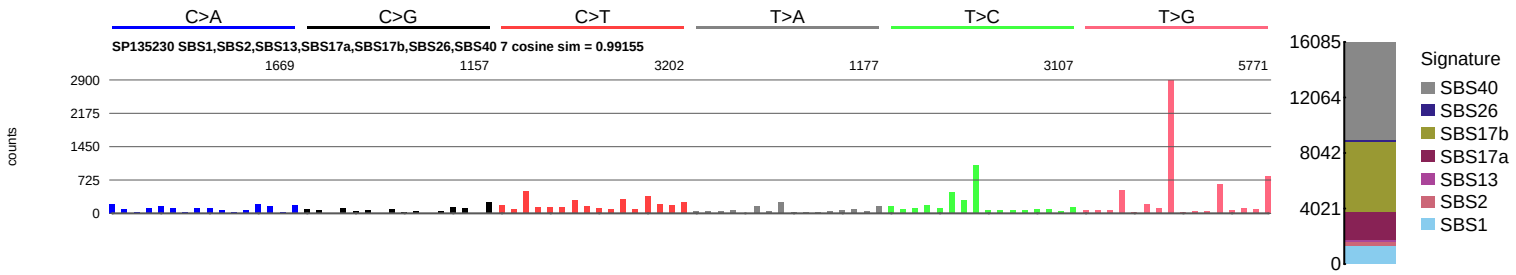

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99177
2020
1083
2965
1143
3170
5703
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

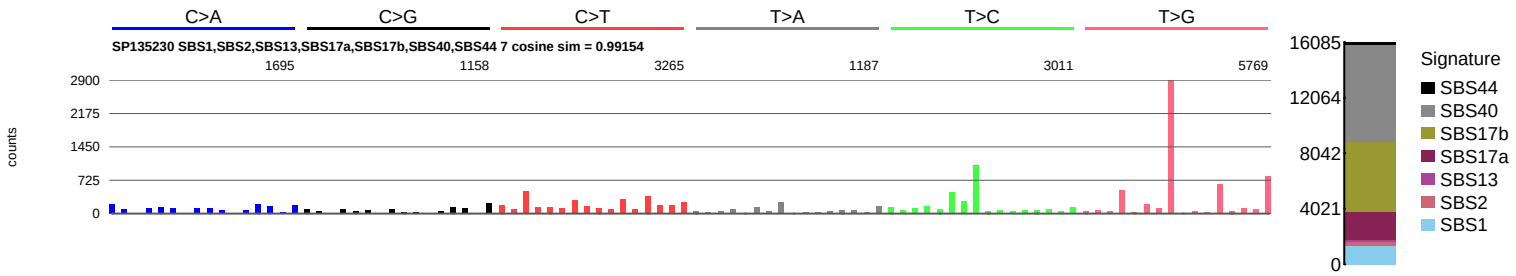

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.99174
1714
1053
3343
1196
3004
5777
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1

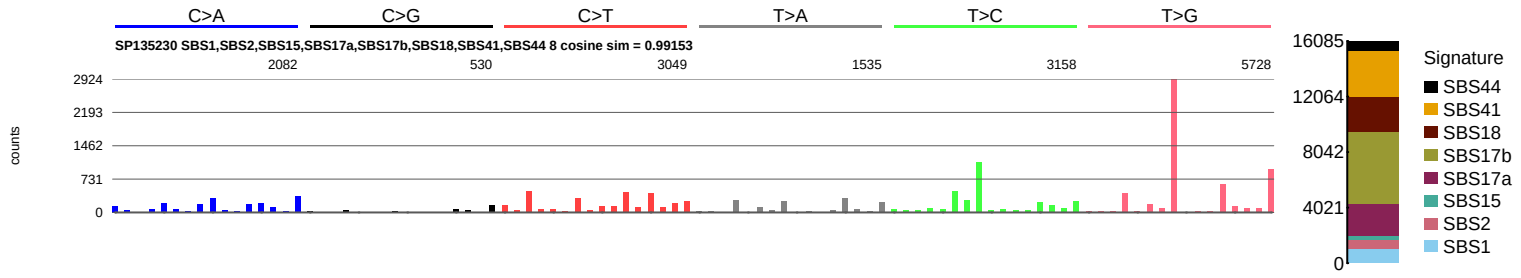

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.99159
2047
523
2710
1467
3614
5726
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

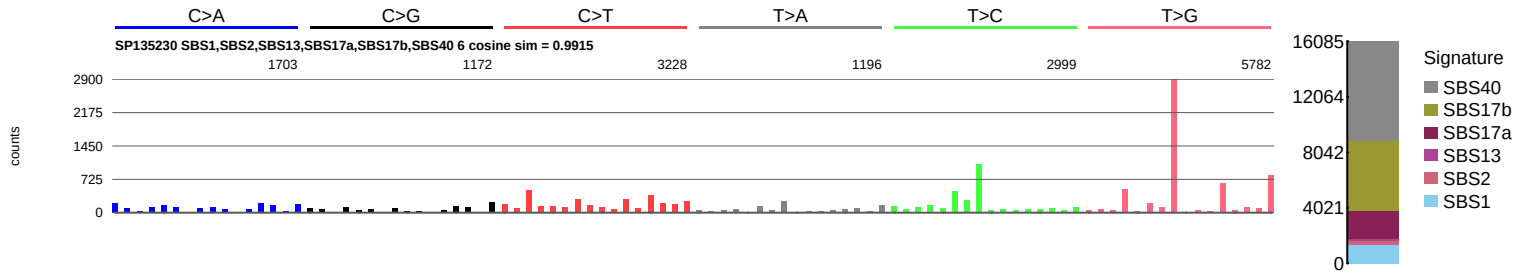

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.99158
1665
1151
3232
1174
3101
5762
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.99156
2022
992
2998
1166
3183
5722
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.99155
1669
1157
3202
1177
3107
5771
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.99154
1695
1158
3265
1187
3011
5769
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99153
2082
530
3049
1535
3158
5728
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40 6 cosine sim = 0.9915
1703
1172
3228
1196
2999
5782
counts
2900
2175
1450
725
0
16085
12064
8042
4021
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Signature
SBS40
SBS17b
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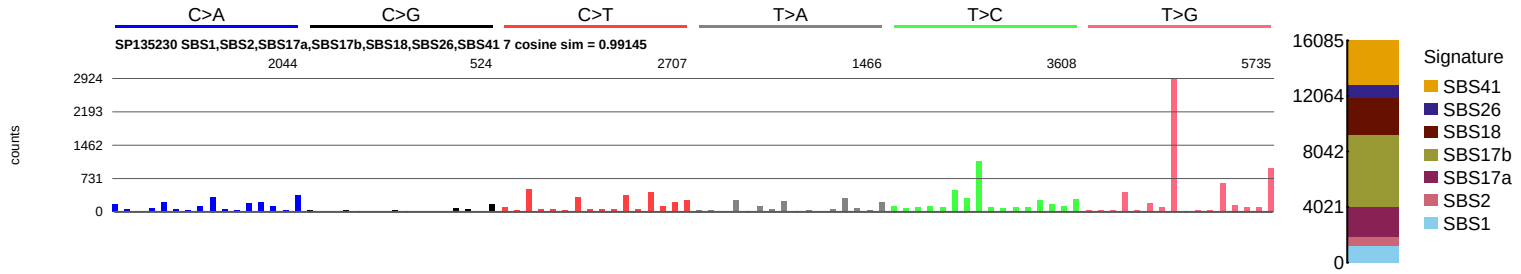
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.99145
2044
524
2707
1466
3608
5735
counts
2924
2193
1462
731
0
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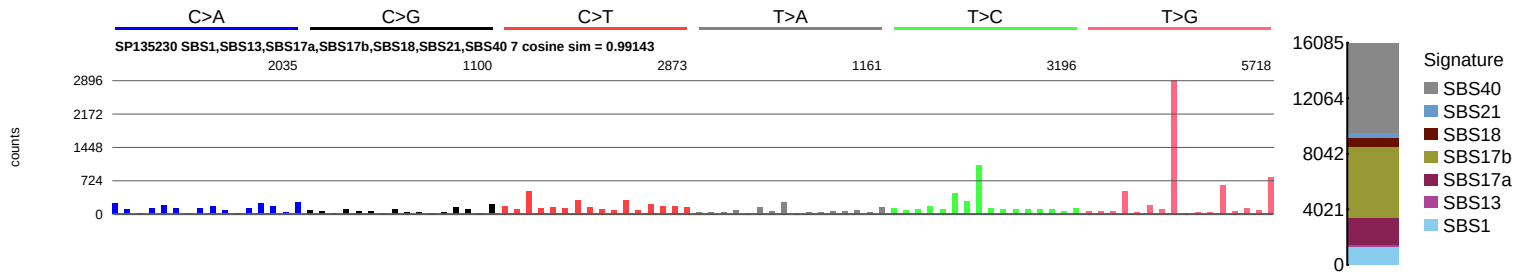
SBS2
SBS1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C>G
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SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.99143
2035
1100
2873
1161
3196
5718
counts
2896
2172
1448
724
0
16085
12064
8042
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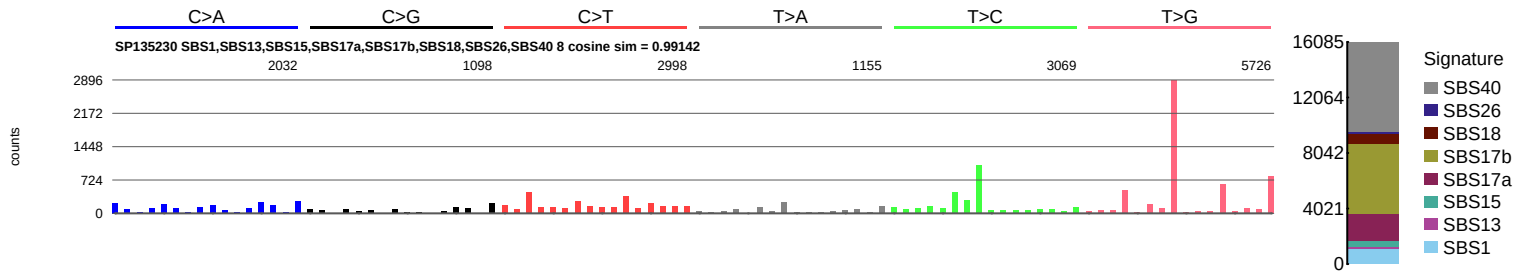
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C>T
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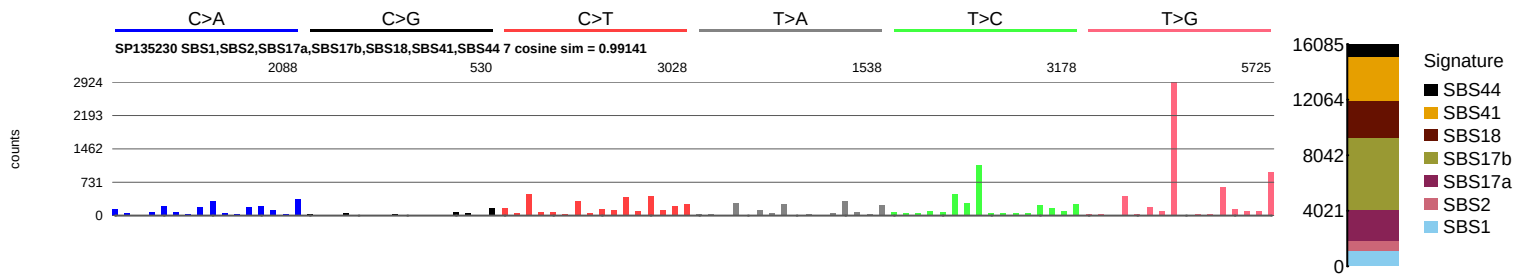
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99142
2032
1098
2998
1155
3069
5726
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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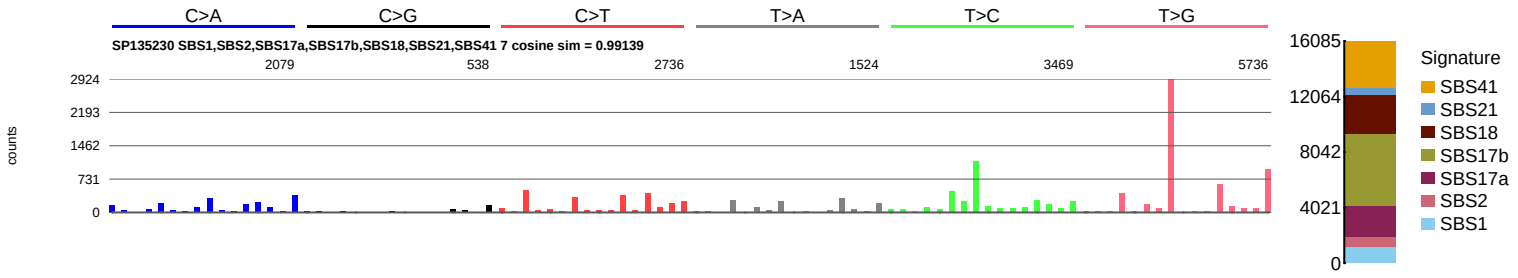
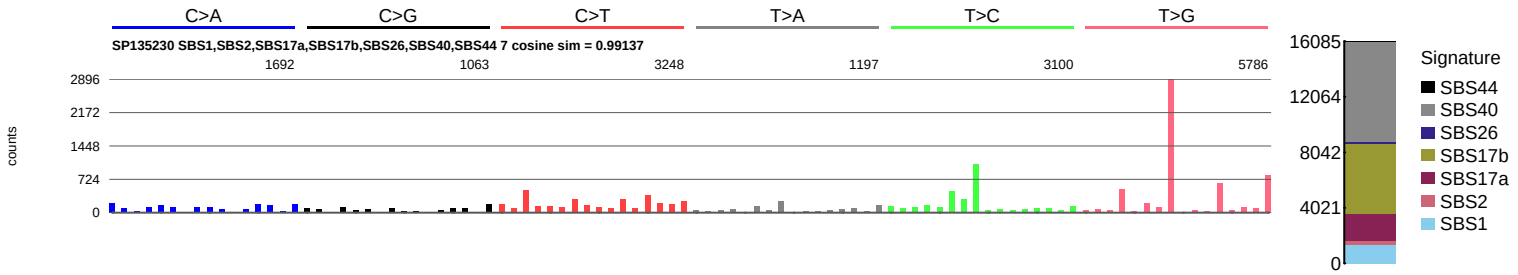
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.99141
2088
530
3028
1538
3178
5725
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99139
2079
538
2736
1524
3469
5736
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.99137
1692
1063
3248
1197
3100
5786
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
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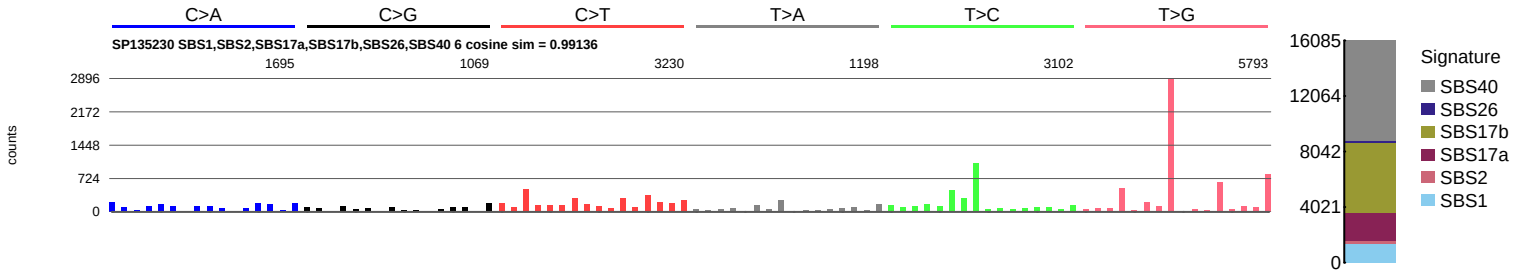
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.99136
1695
1069
3230
1198
3102
5793
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
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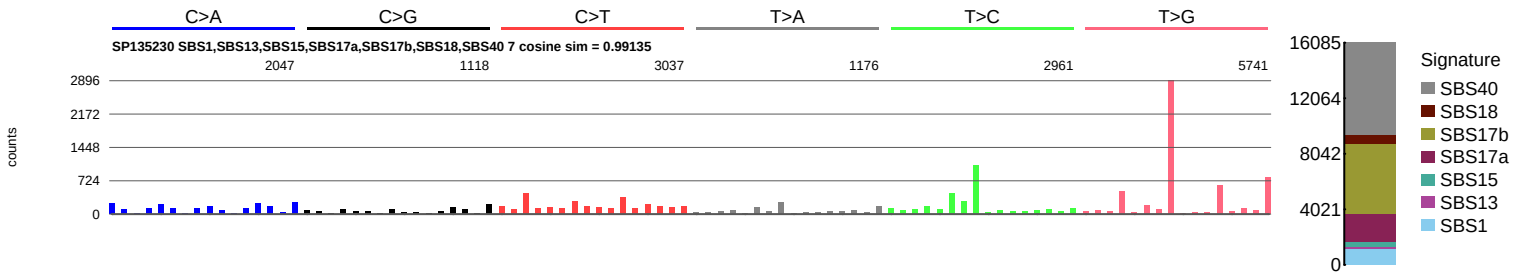
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99135
2047
1118
3037
1176
2961
5741
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
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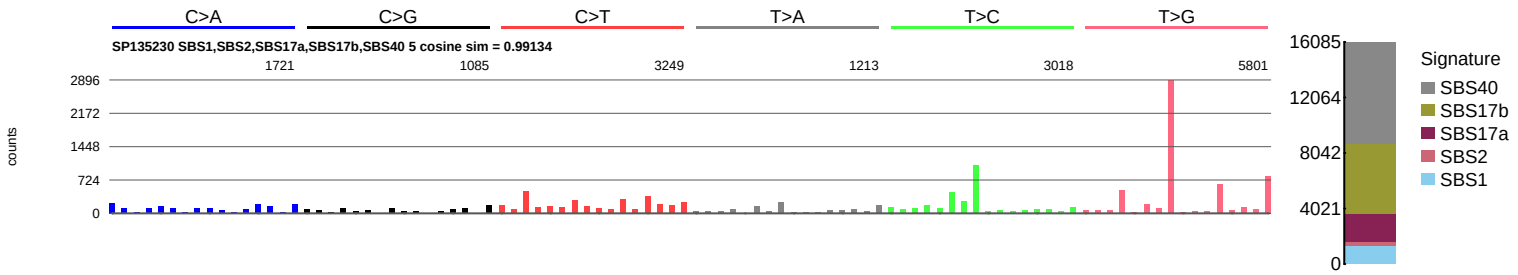
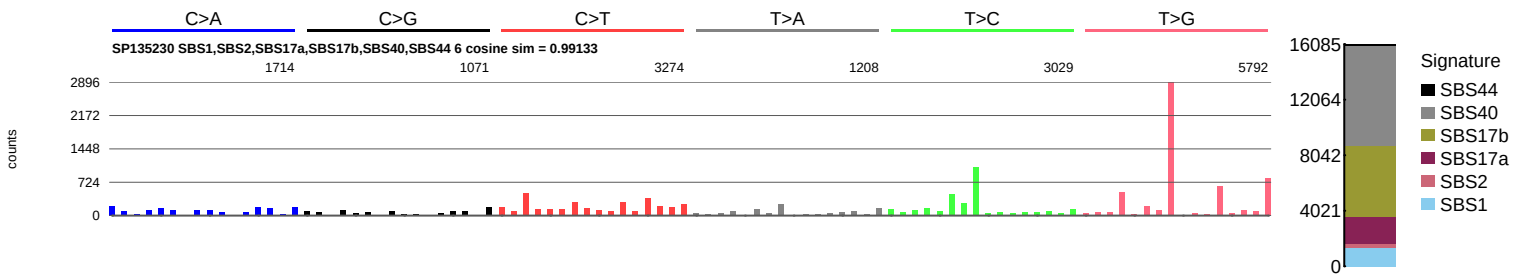
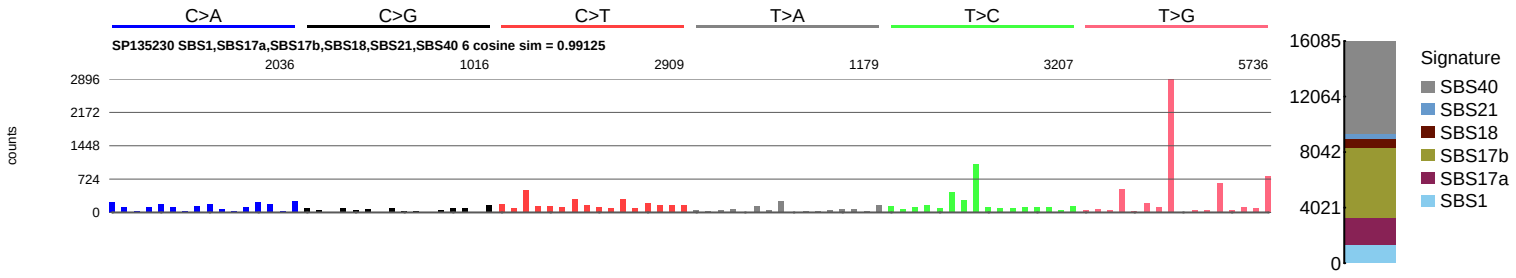
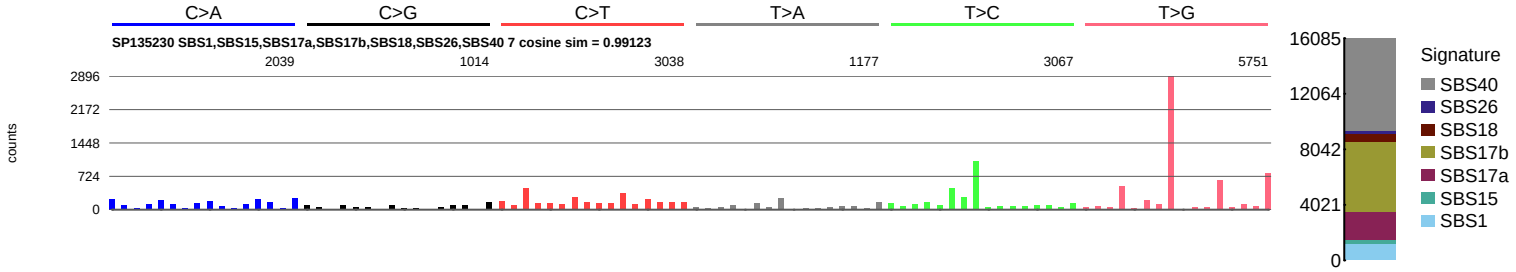
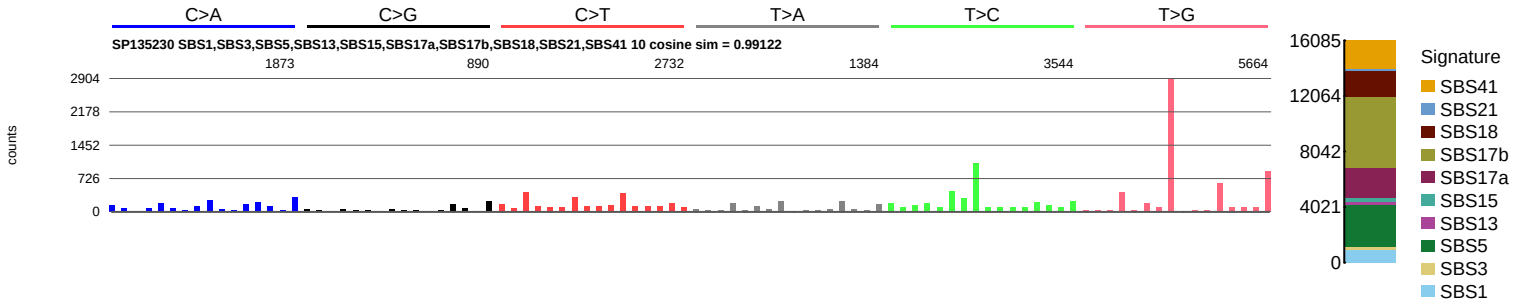
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

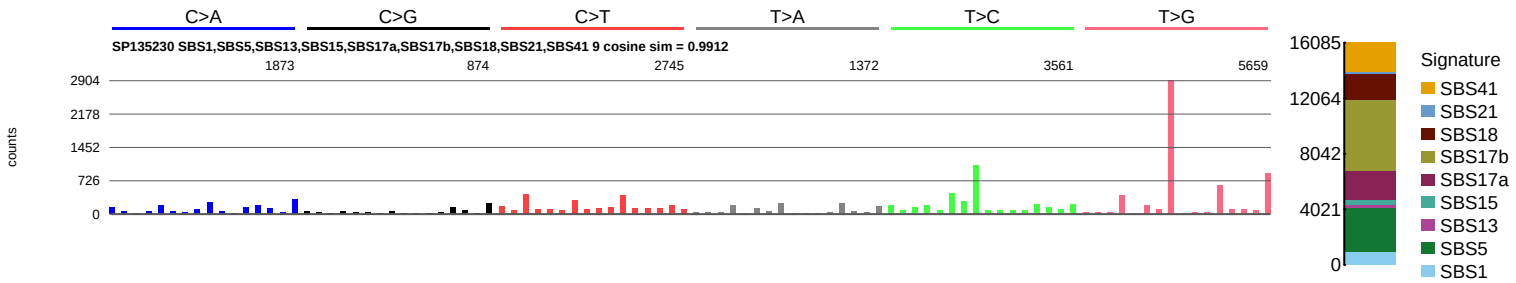

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS40 5 cosine sim = 0.99134
1721
1085
3249
1213
3018
5801
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS2
SBS1

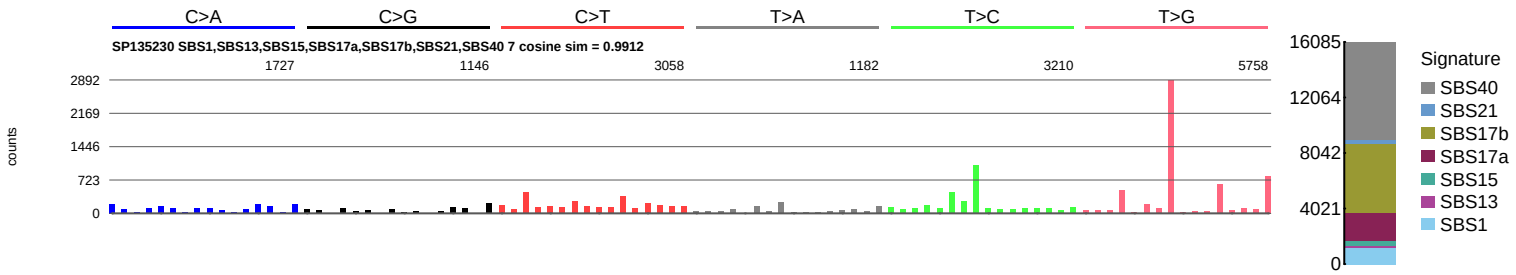

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.99133
1714
1071
3274
1208
3029
5792
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS2
SBS1

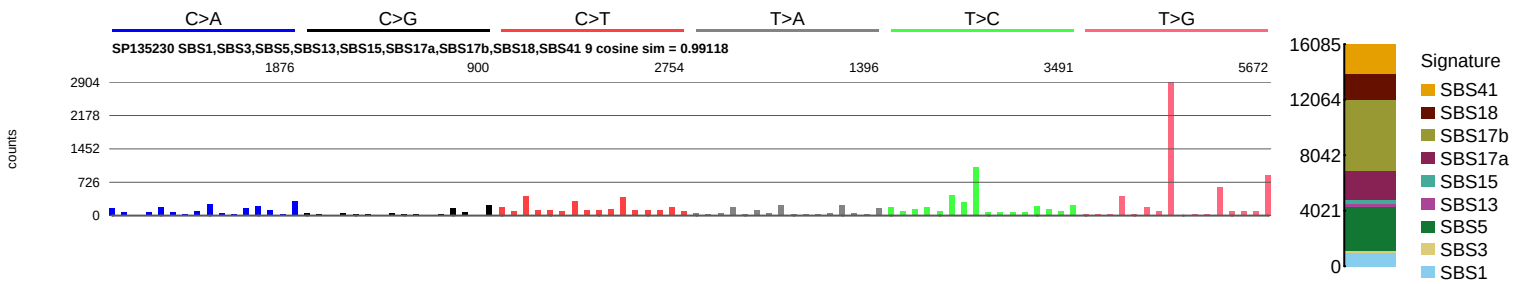

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.99125
2036
1016
2909
1179
3207
5736
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.99123
2039
1014
3038
1177
3067
5751
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99122
1873
890
2732
1384
3544
5664
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9912
1873
874
2745
1372
3561
5659
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.9912
1727
1146
3058
1182
3210
5758
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1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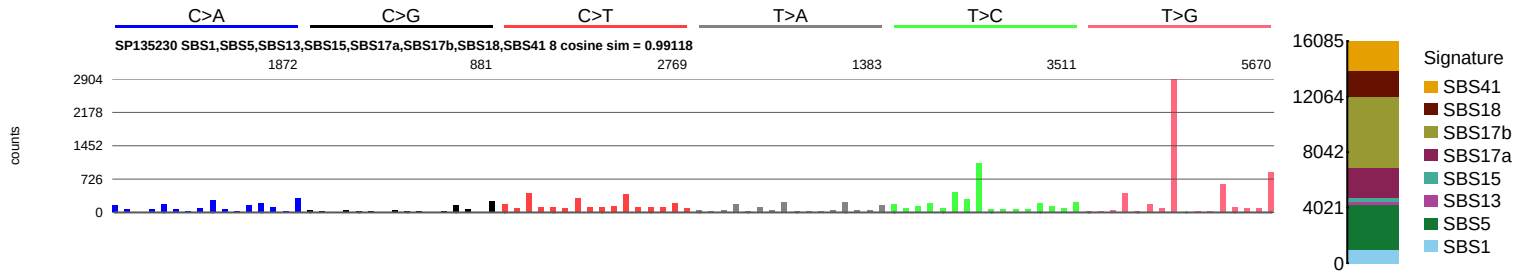
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99118
1876
900
2754
1396
3491
5672
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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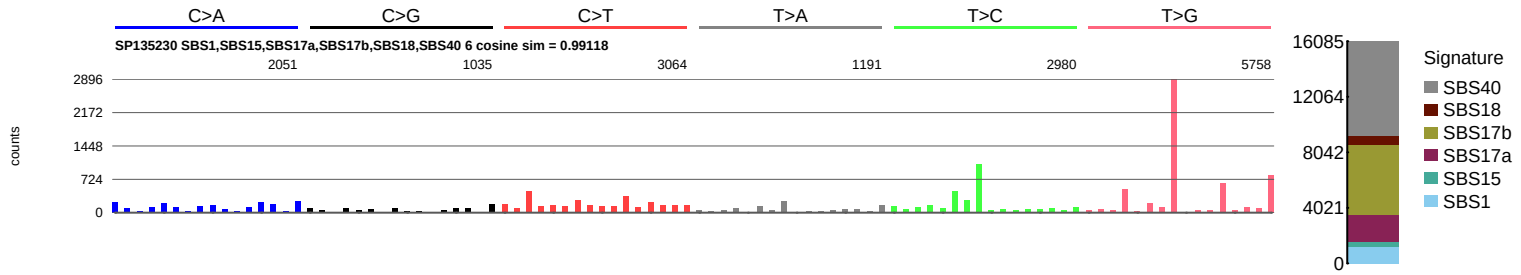
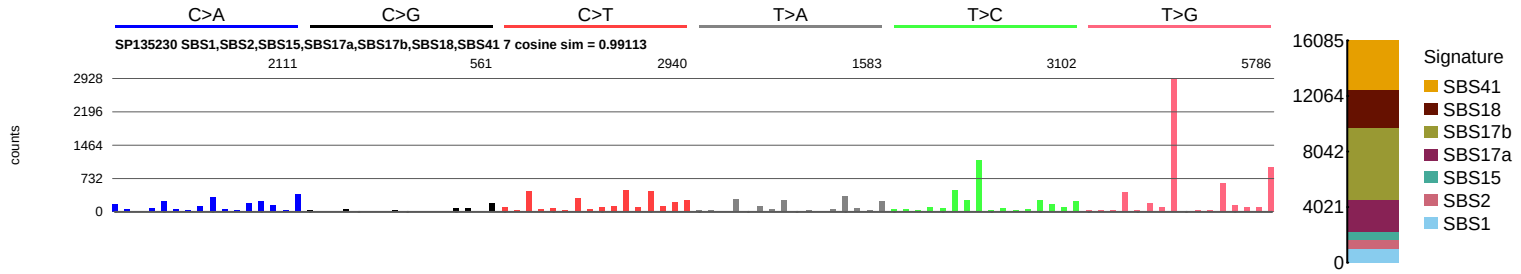
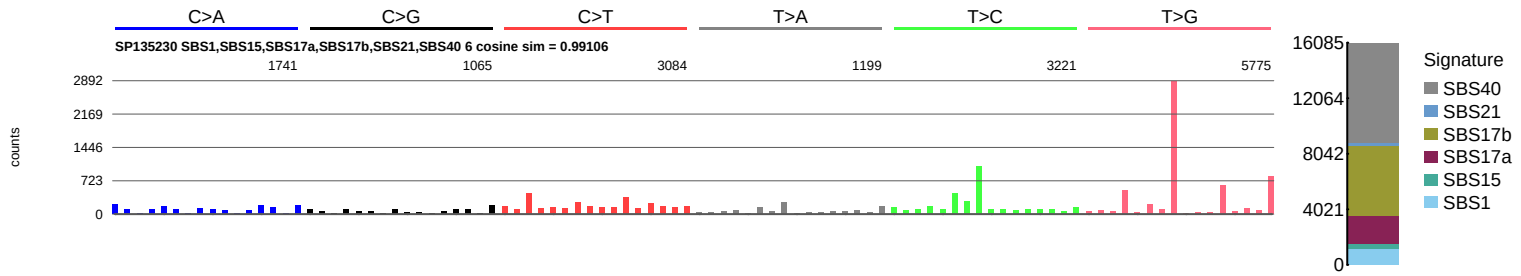
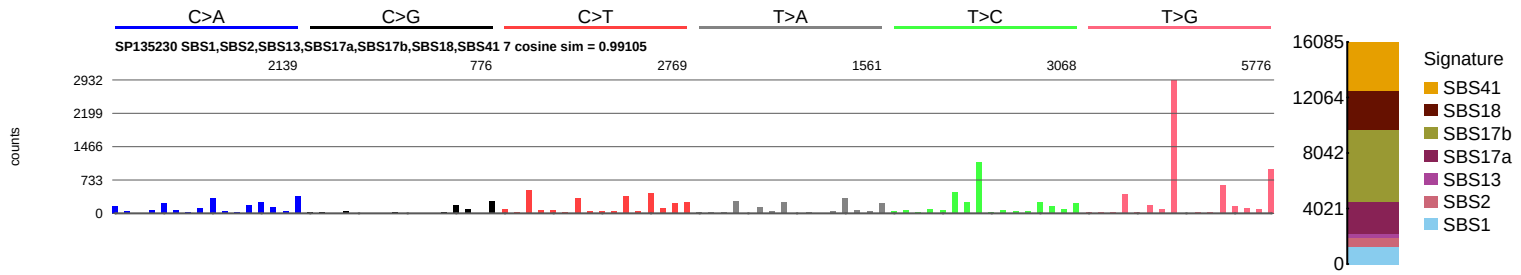
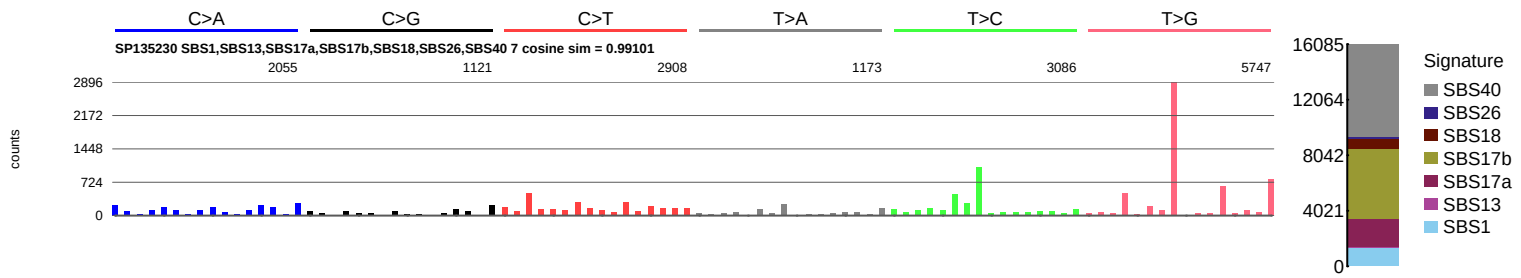
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99118
1872
881
2769
1383
3511
5670
2904
2178
1452
726
0
counts
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

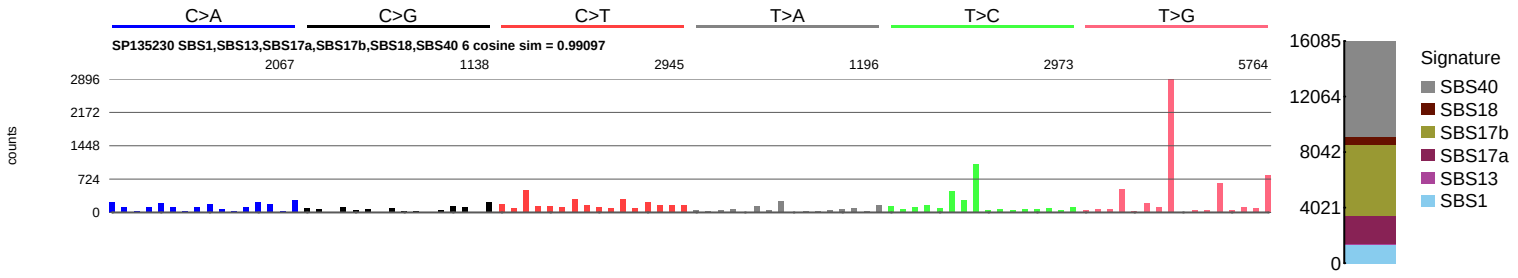

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.99118
2051
1035
3064
1191
2980
5758
2896
2172
1448
724
0
counts
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

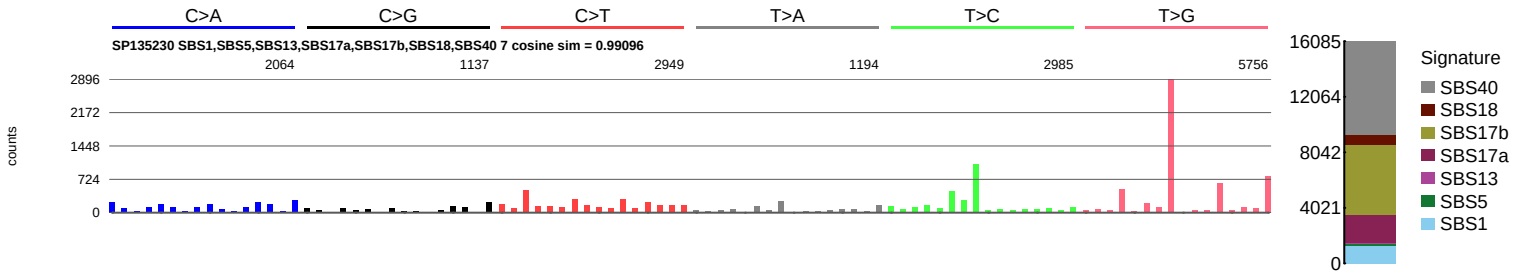

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99113
2111
561
2940
1583
3102
5786
2928
2196
1464
732
0
counts
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.99106
1741
1065
3084
1199
3221
5775
2892
2169
1446
723
0
counts
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99105
2139
776
2769
1561
3068
5776
2932
2199
1466
733
0
counts
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.99101
2055
1121
2908
1173
3086
5747
2896
2172
1448
724
0
counts
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.99097
2067
1138
2945
1196
2973
5764
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99096
2064
1137
2949
1194
2985
5756
counts
2896
2172
1448
724
0
16085
12064
8042
4021
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SBS40
SBS18
SBS17b
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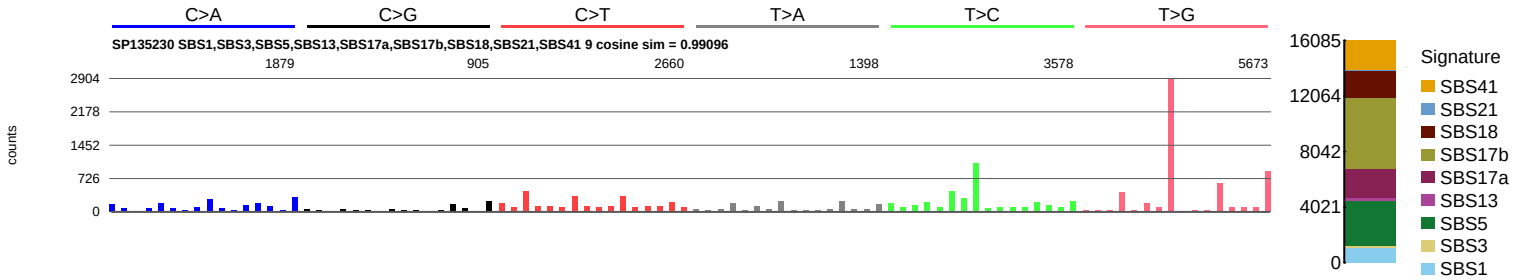
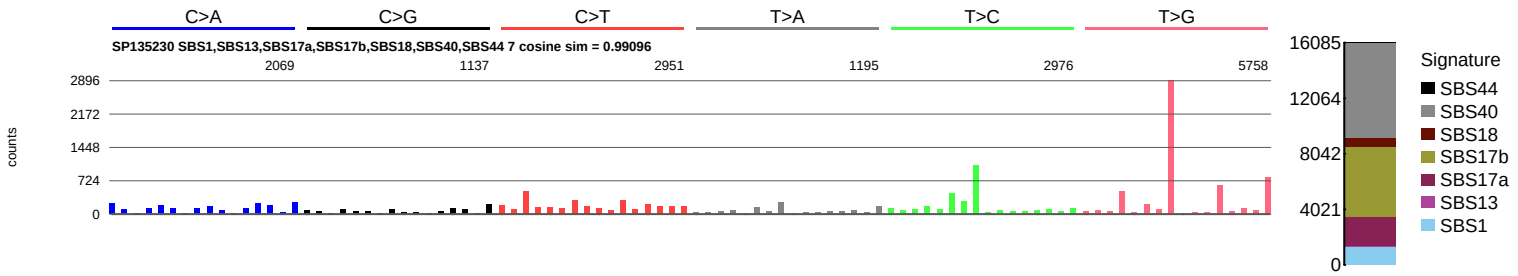
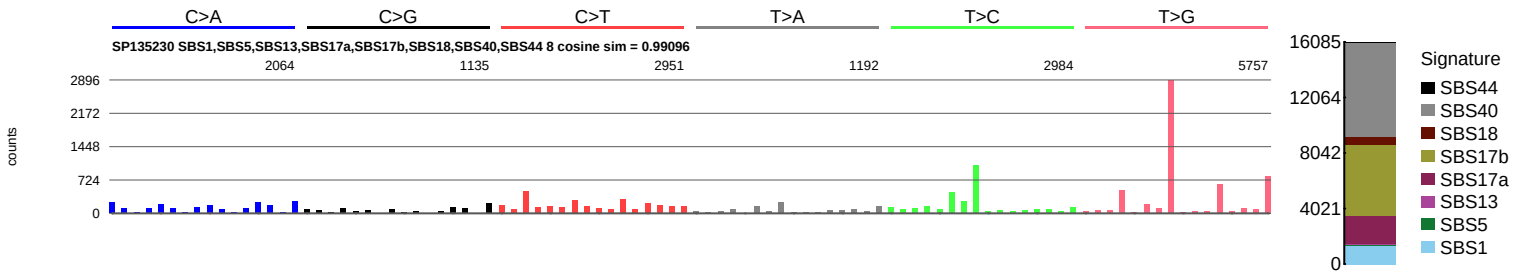
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99096
1879
905
2660
1398
3578
5673
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.99096
2069
1137
2951
1195
2976
5758
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1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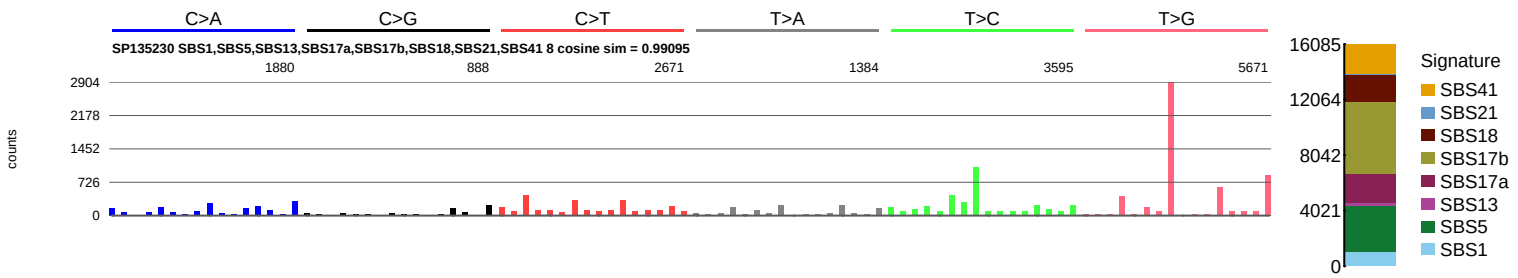
SP135230 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.99096
2064
1135
2951
1192
2984
5757
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
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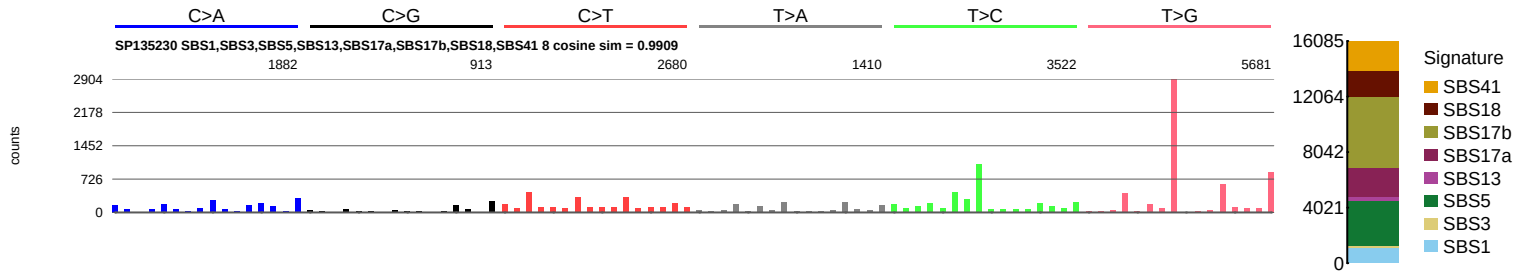
T>G
SP135230 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99095
1880
888
2671
1384
3595
5671
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
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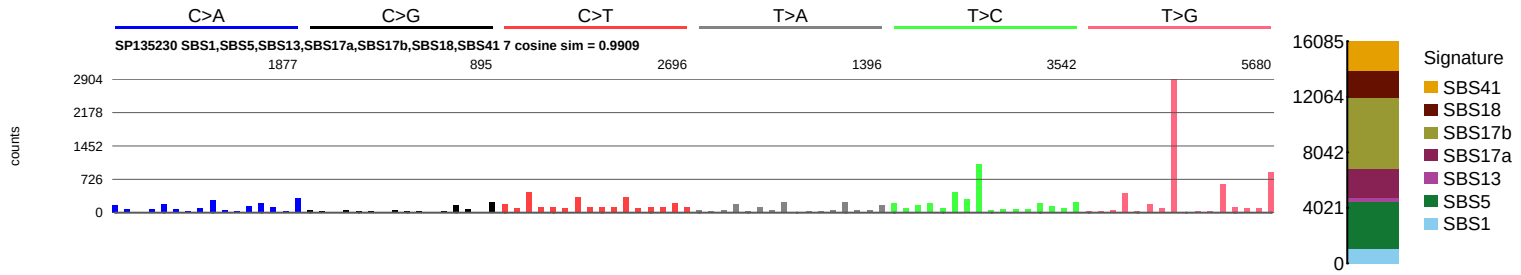
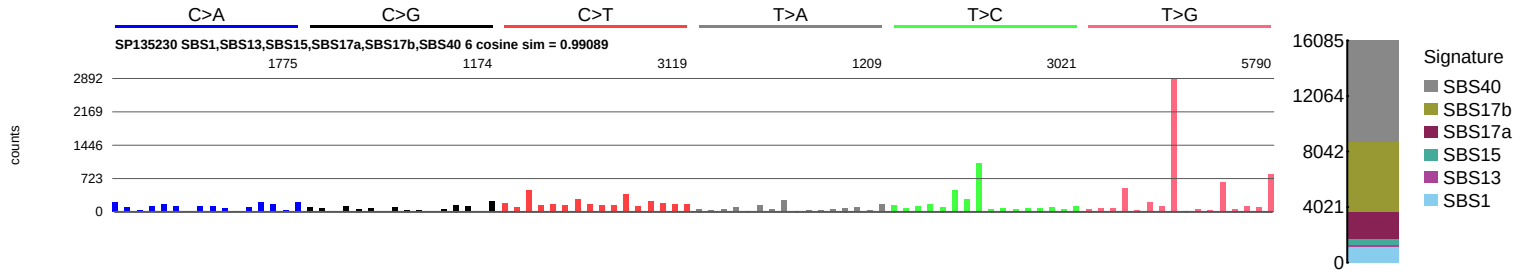
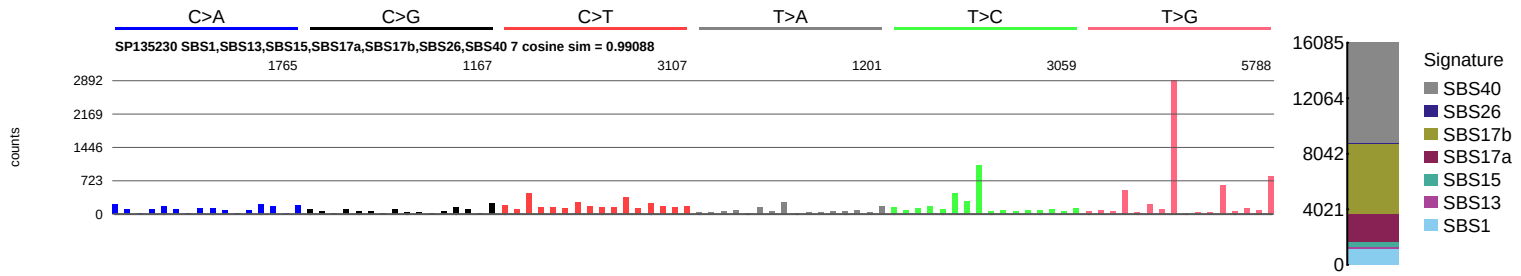
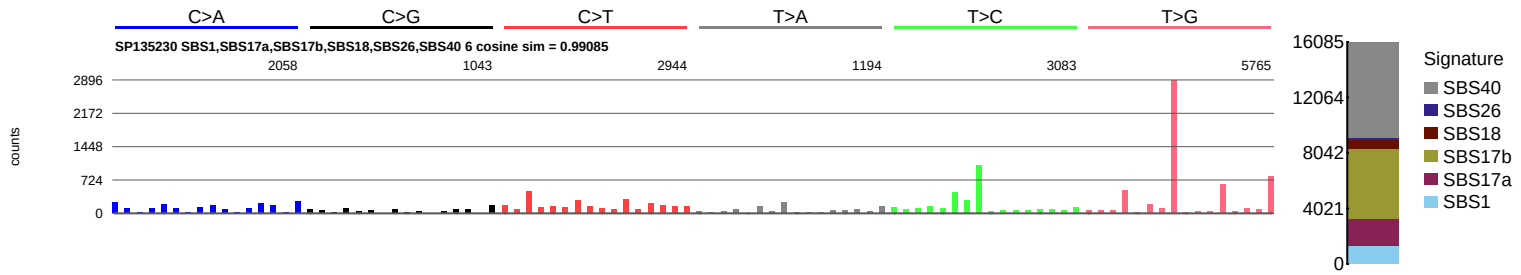
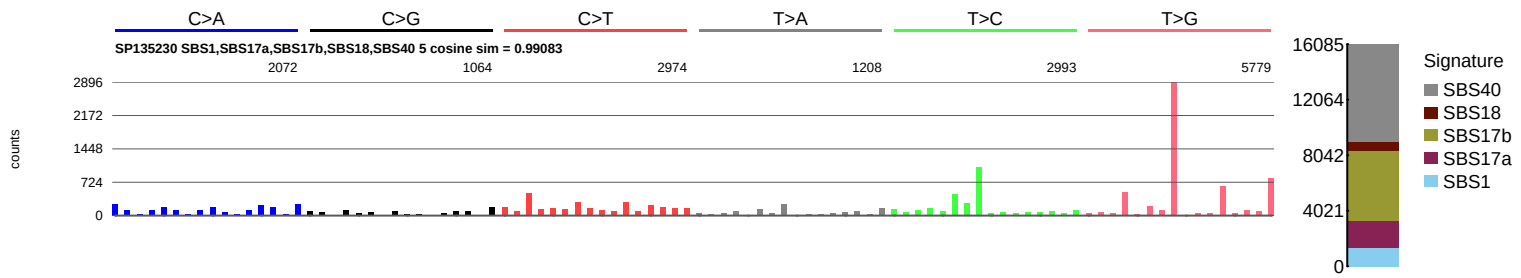
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9909
1882
913
2680
1410
3522
5681
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

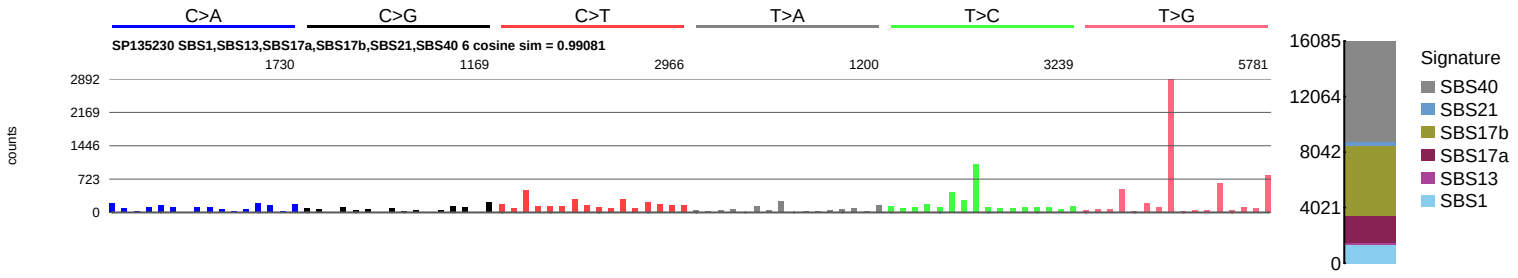

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9909
1877
895
2696
1396
3542
5680
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

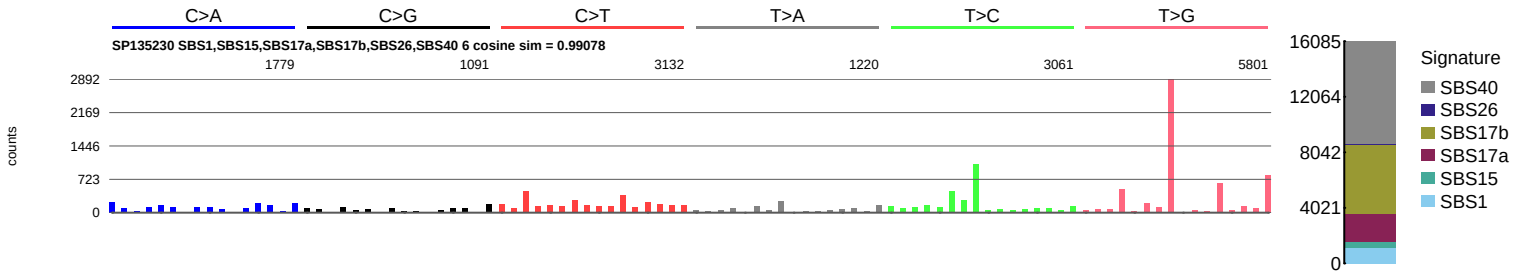

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.99089
1775
1174
3119
1209
3021
5790
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS1

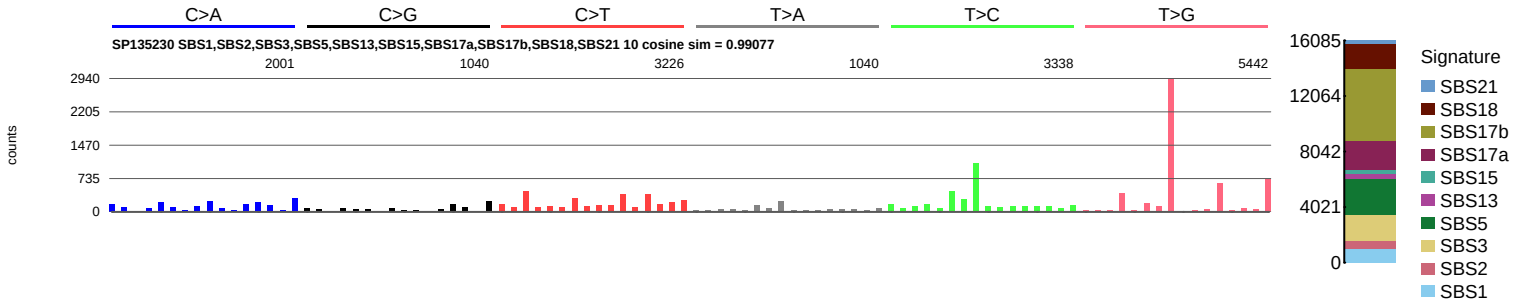

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.99088
1765
1167
3107
1201
3059
5788
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1

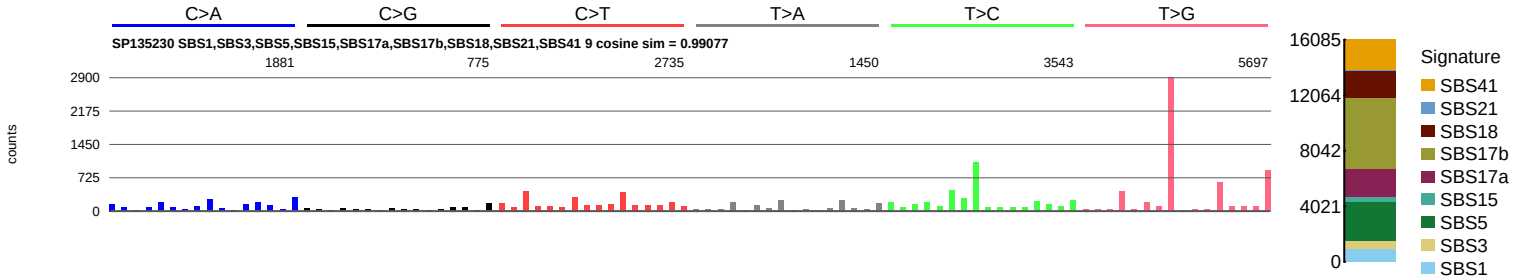

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.99085
2058
1043
2944
1194
3083
5765
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

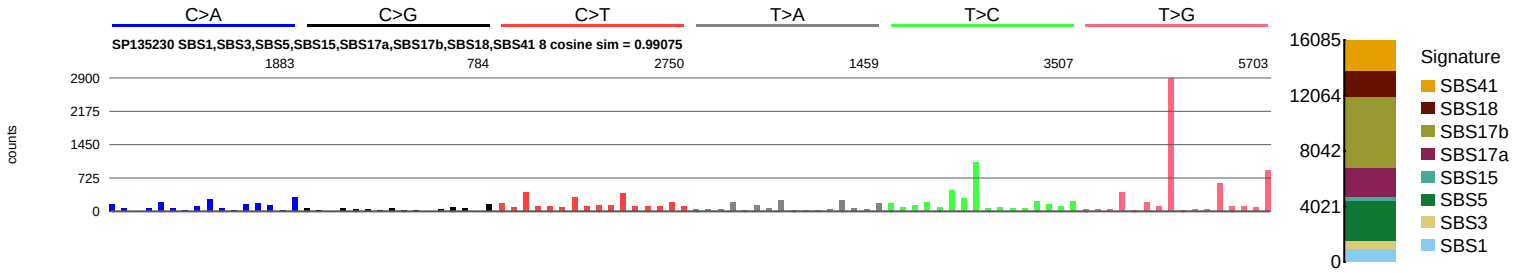

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.99083
2072
1064
2974
1208
2993
5779
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS1

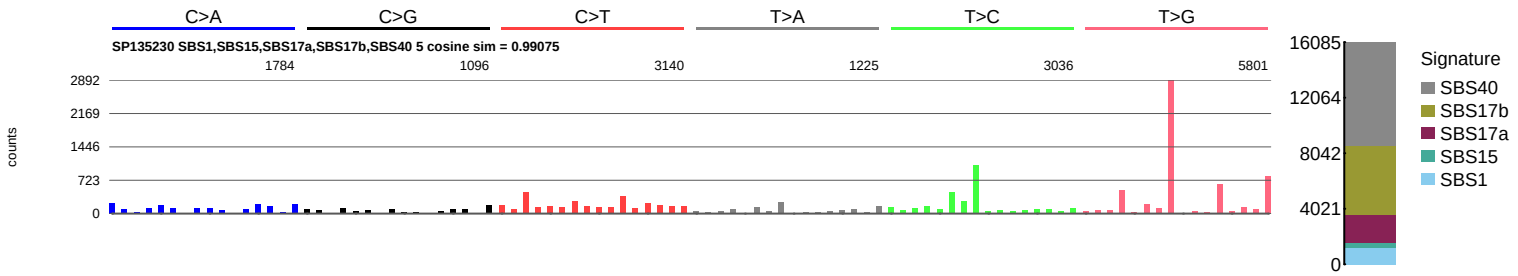

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.99081
1730
1169
2966
1200
3239
5781
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1

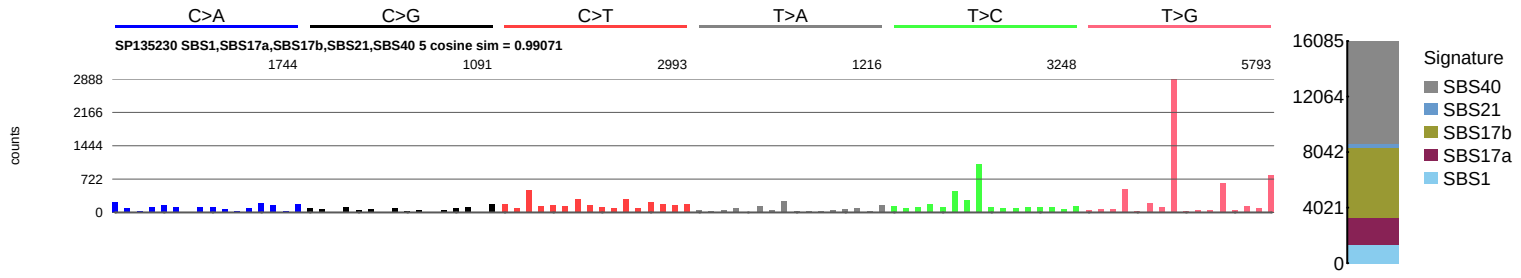

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.99078
1779
1091
3132
1220
3061
5801
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1

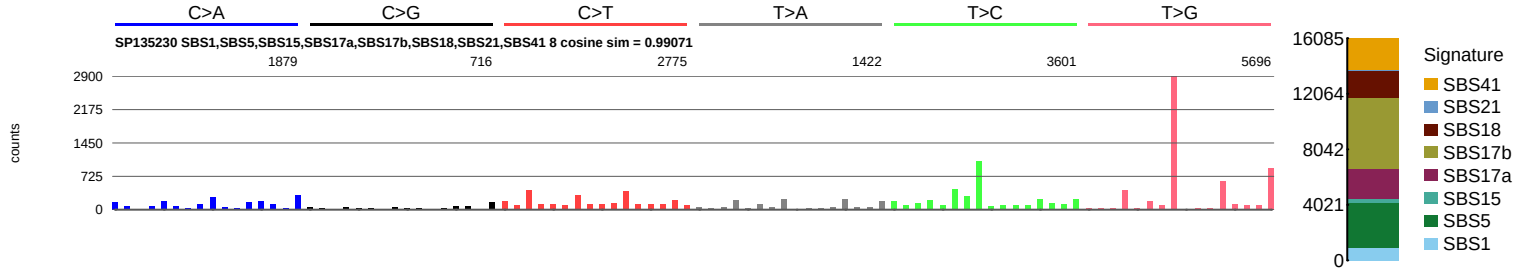

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.99077
2001
1040
3226
1040
3338
5442
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

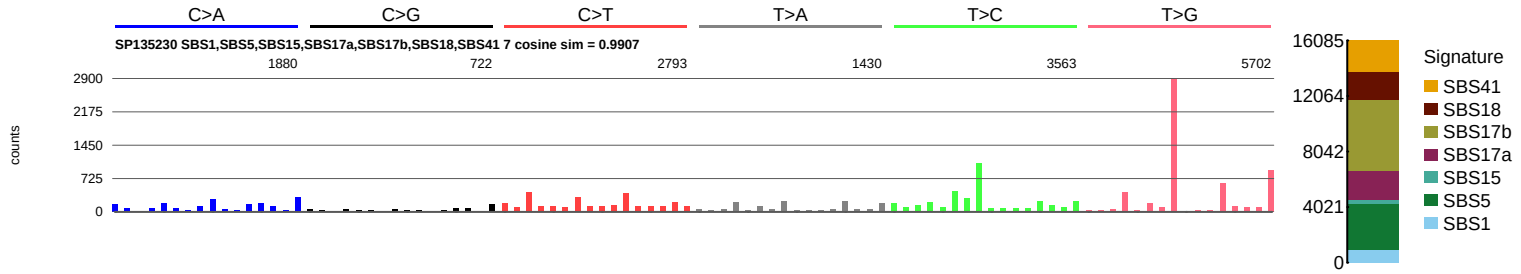

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99077
1881
775
2735
1450
3543
5697
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

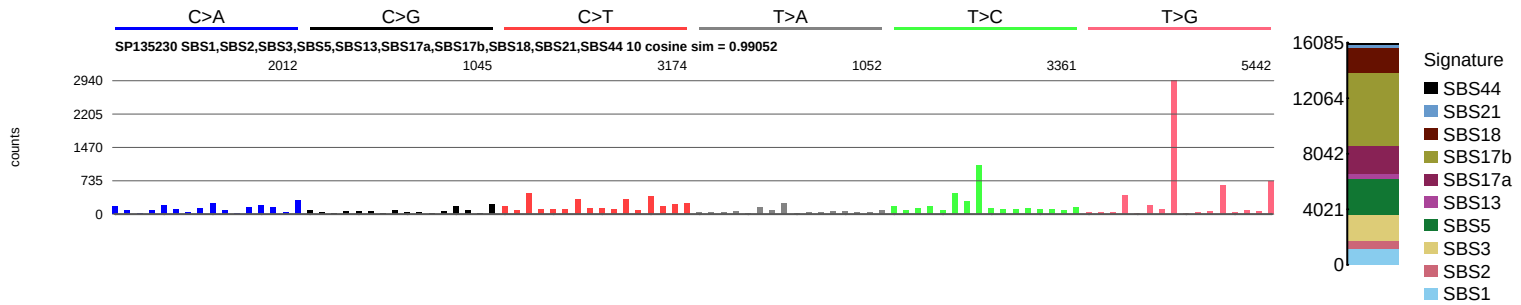

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99075
1883
784
2750
1459
3507
5703
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

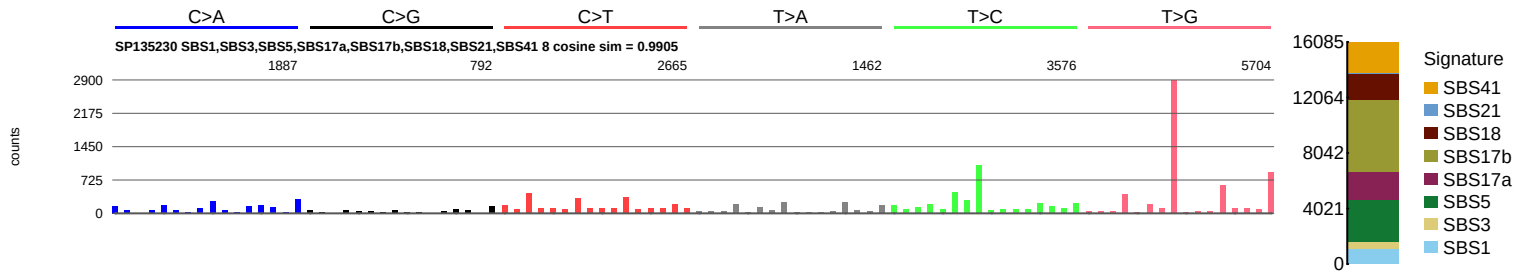

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.99075
1784
1096
3140
1225
3036
5801
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS1

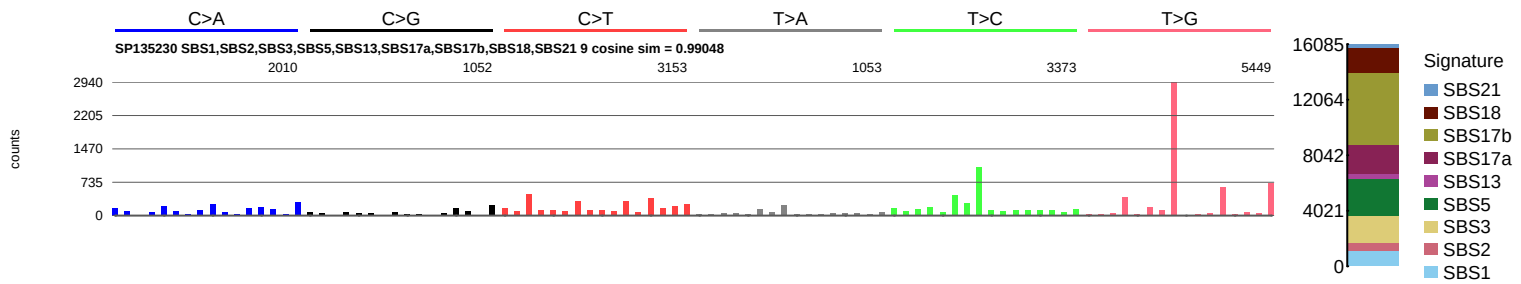

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.99071
1744
1091
2993
1216
3248
5793
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS1

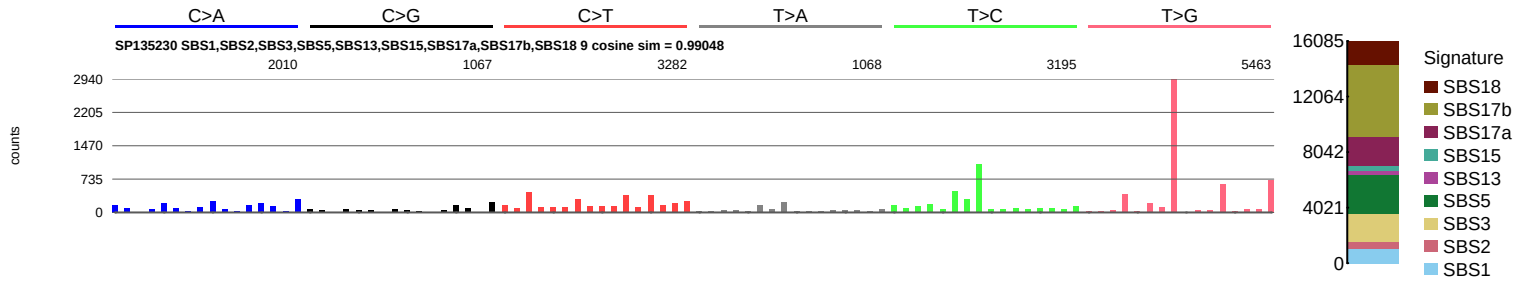

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99071
1879
716
2775
1422
3601
5696
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

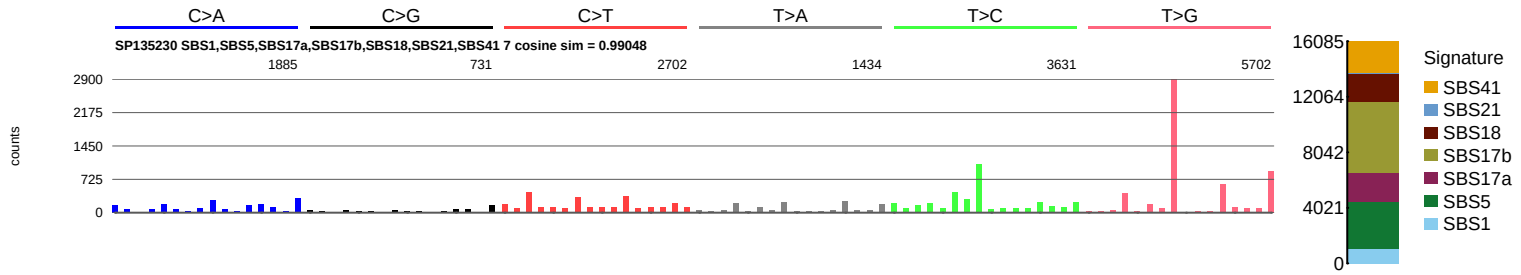

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9907
1880
722
2793
1430
3563
5702
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

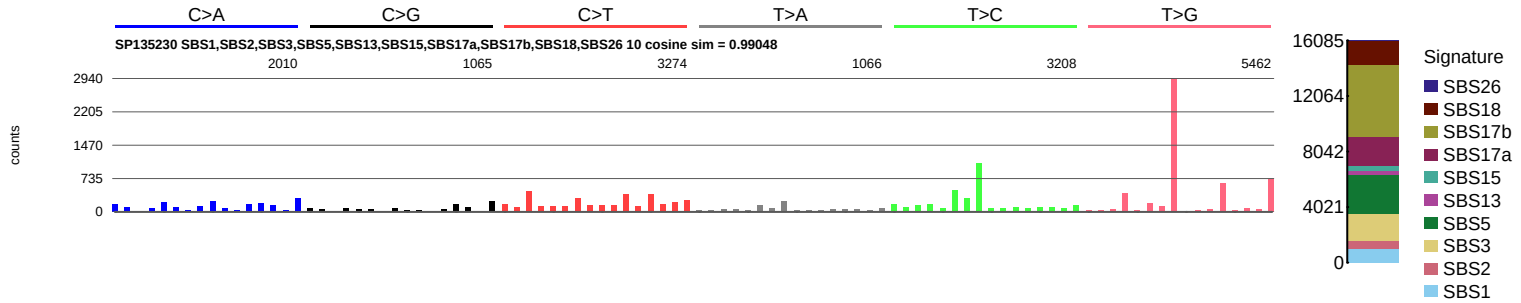

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.99052
2012
1045
3174
1052
3361
5442
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

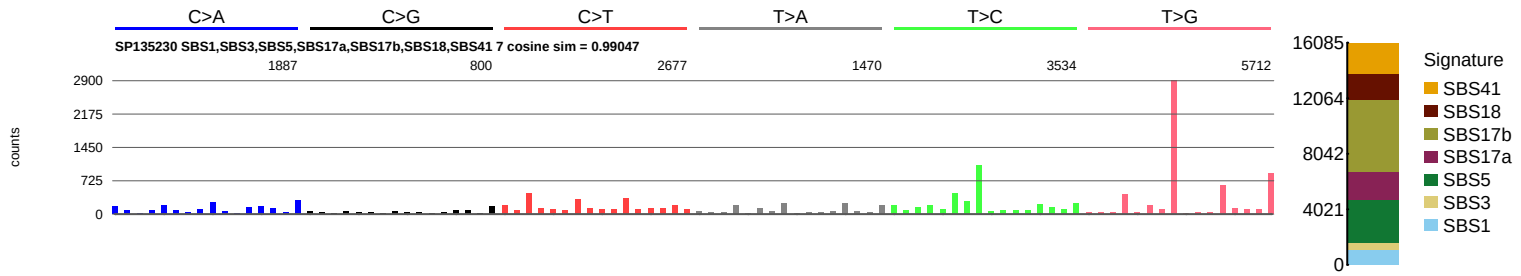

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9905
1887
792
2665
1462
3576
5704
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

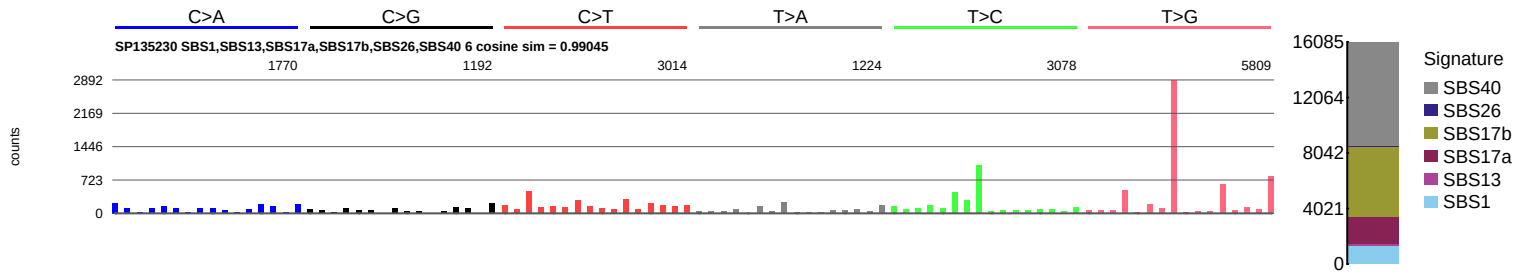

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99048
2010
1052
3153
1053
3373
5449
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

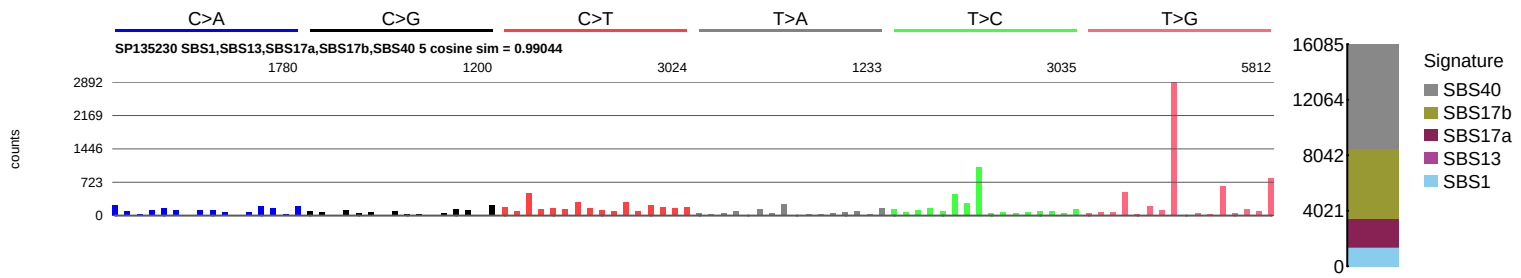

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.99048
2010
1067
3282
1068
3195
5463
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99048
1885
731
2702
1434
3631
5702
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 10 cosine sim = 0.99048
2010
1065
3274
1066
3208
5462
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99047
1887
800
2677
1470
3534
5712
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.99045
1770
1192
3014
1224
3078
5809
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS1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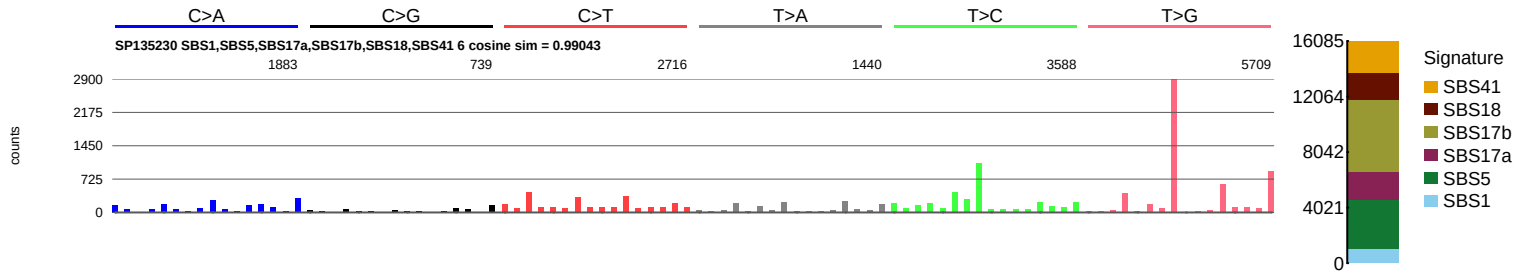
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS40 5 cosine sim = 0.99044
1780
1200
3024
1233
3035
5812
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS1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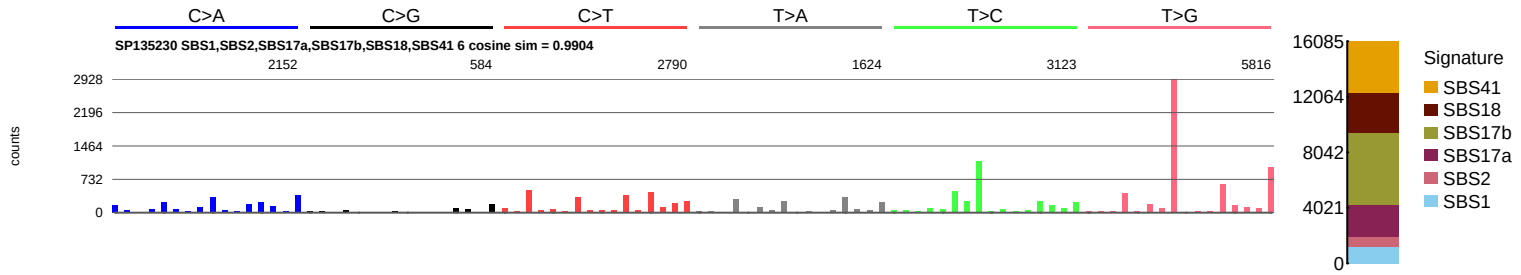
SP135230 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.99043
1883
739
2716
1440
3588
5709
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
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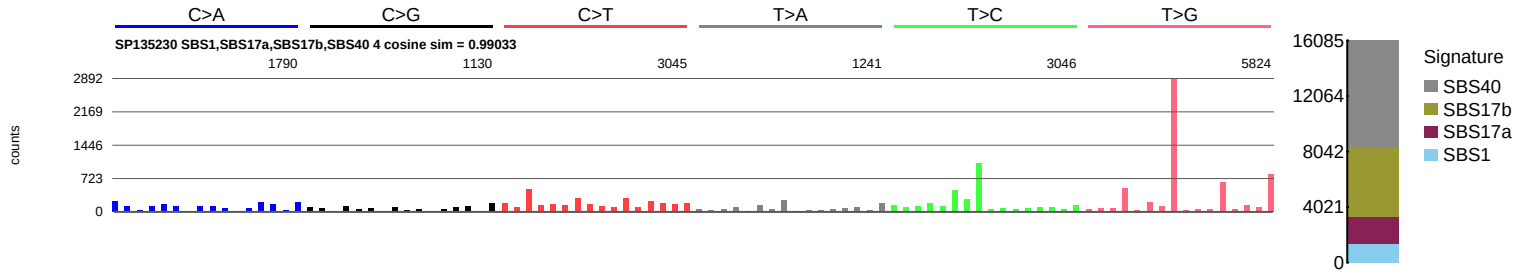
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.9904
2152
584
2790
1624
3123
5816
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1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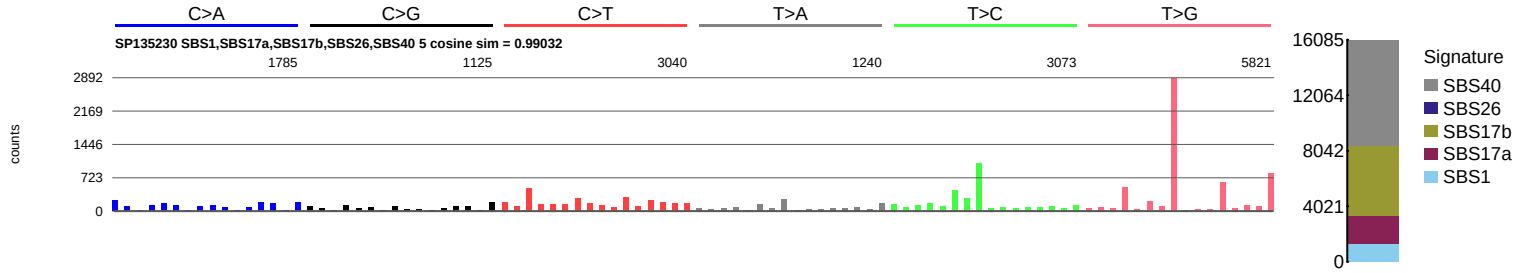
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS40 4 cosine sim = 0.99033
1790
1130
3045
1241
3046
5824
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS1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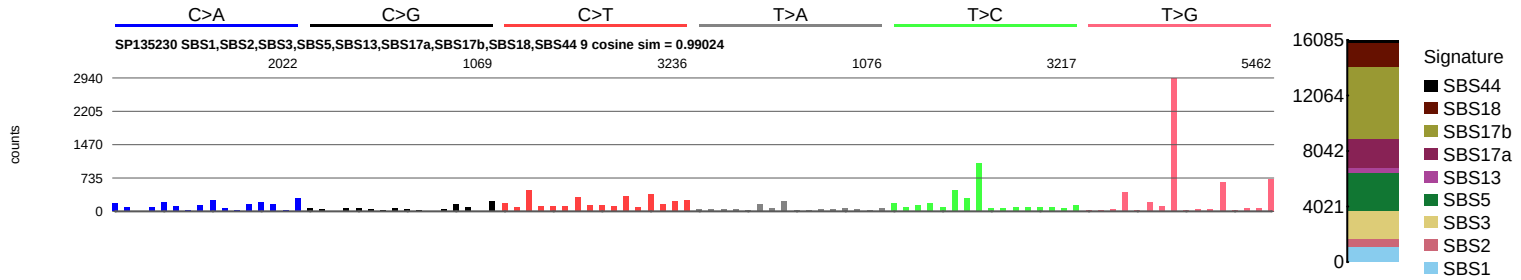
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.99032
1785
1125
3040
1240
3073
5821
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS1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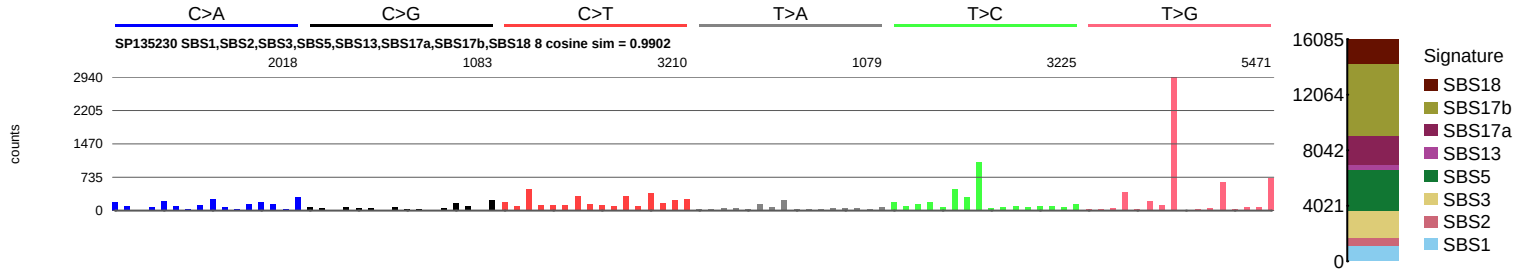
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.99024
2022
1069
3236
1076
3217
5462
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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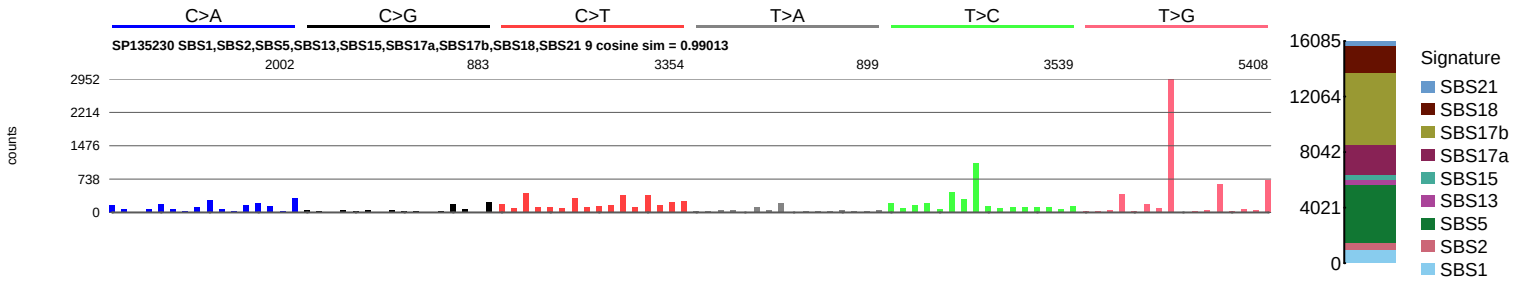
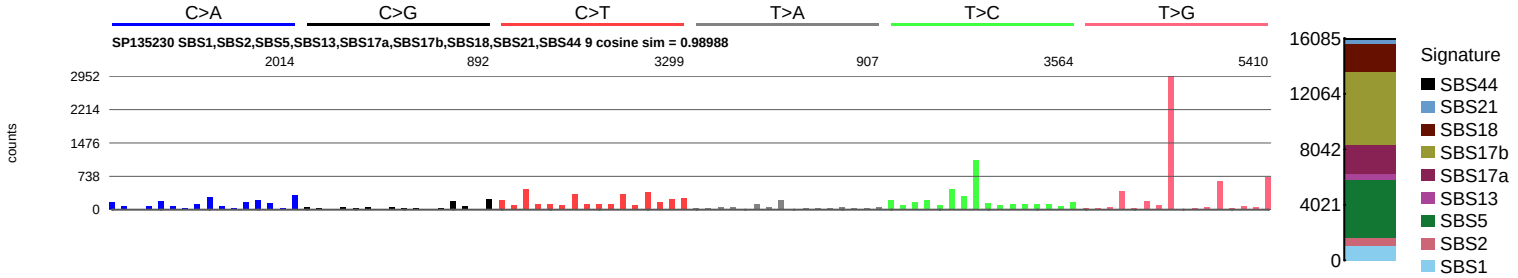
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.9902
2018
1083
3210
1079
3225
5471
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99013
2002
883
3354
899
3539
5408
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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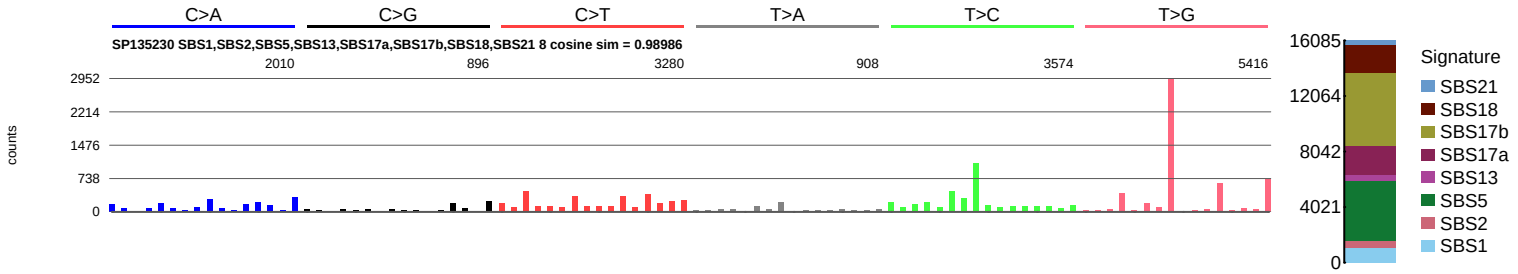
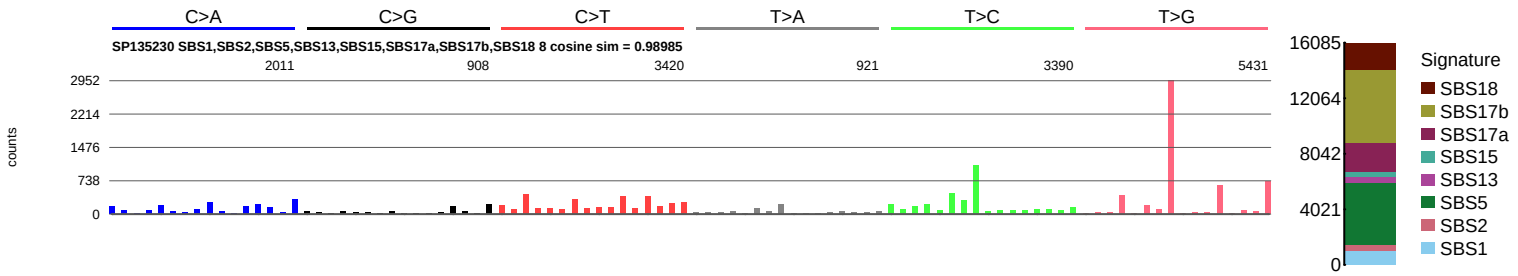
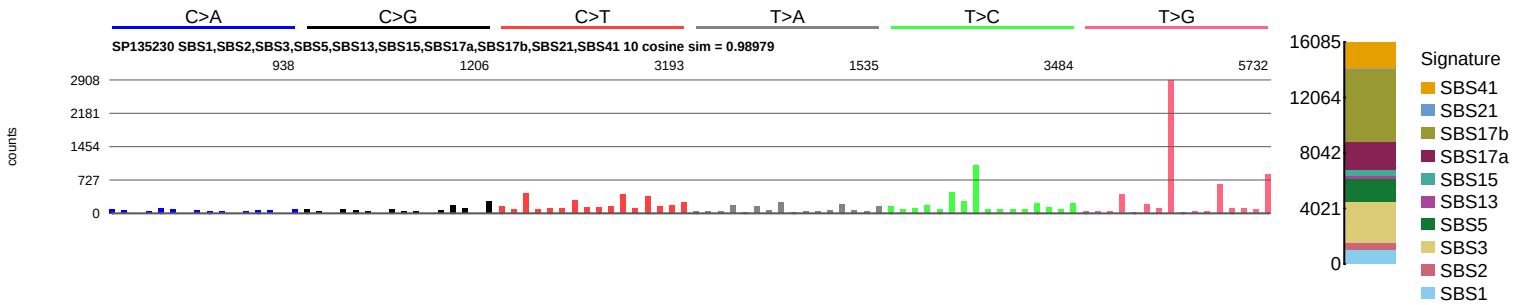
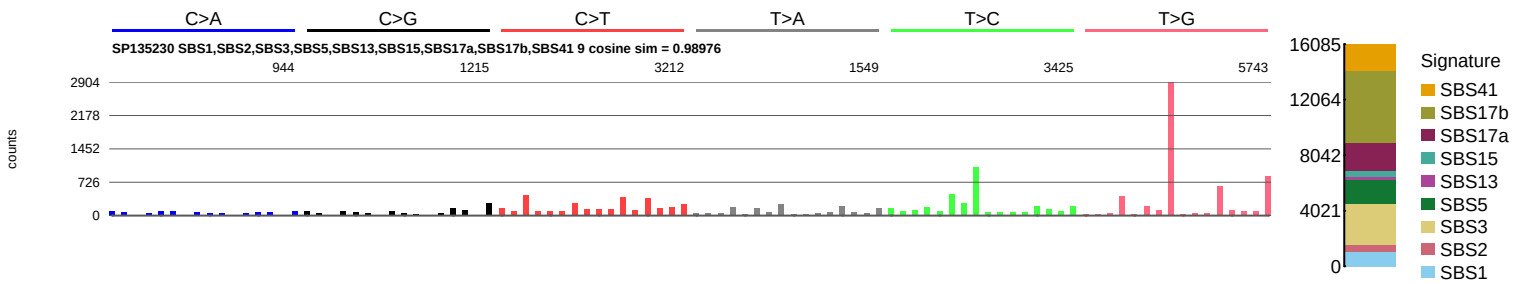
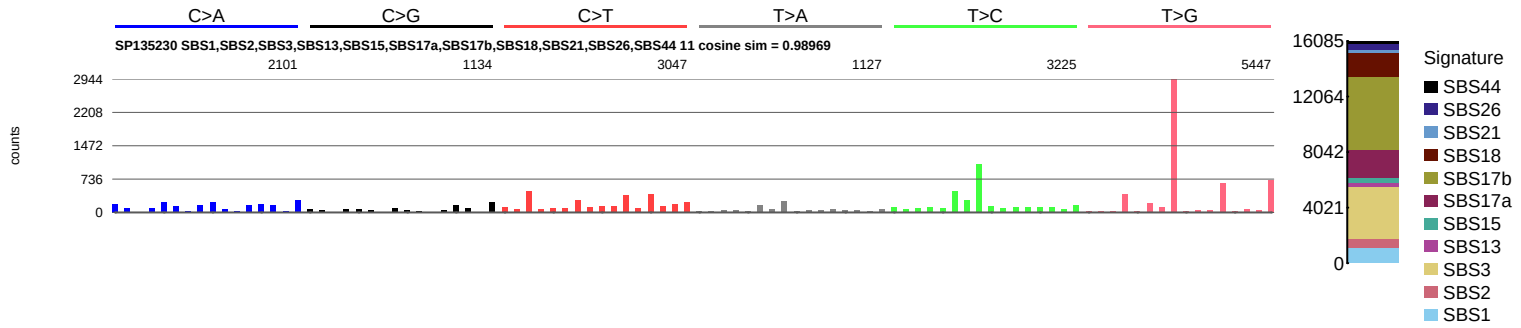
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98988
2014
892
3299
907
3564
5410
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

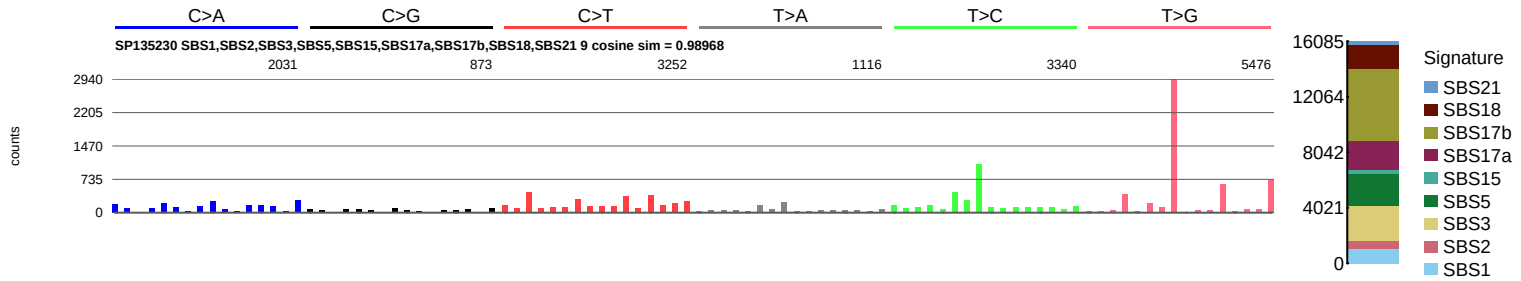

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98986
2010
896
3280
908
3574
5416
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98985
2011
908
3420
921
3390
5431
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 10 cosine sim = 0.98979
938
1206
3193
1535
3484
5732
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 9 cosine sim = 0.98976
944
1215
3212
1549
3425
5743
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.98969
2101
1134
3047
1127
3225
5447
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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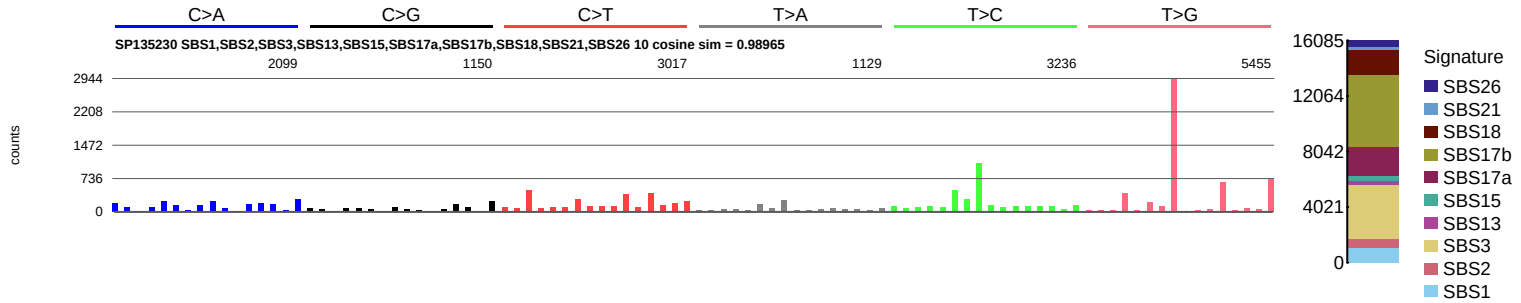
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98968
2031
873
3252
1116
3340
5476
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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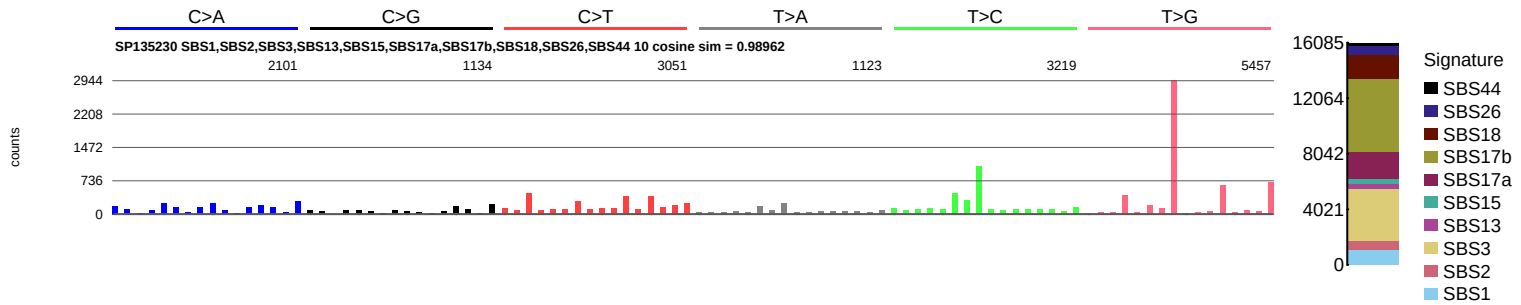
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98965
2099
1150
3017
1129
3236
5455
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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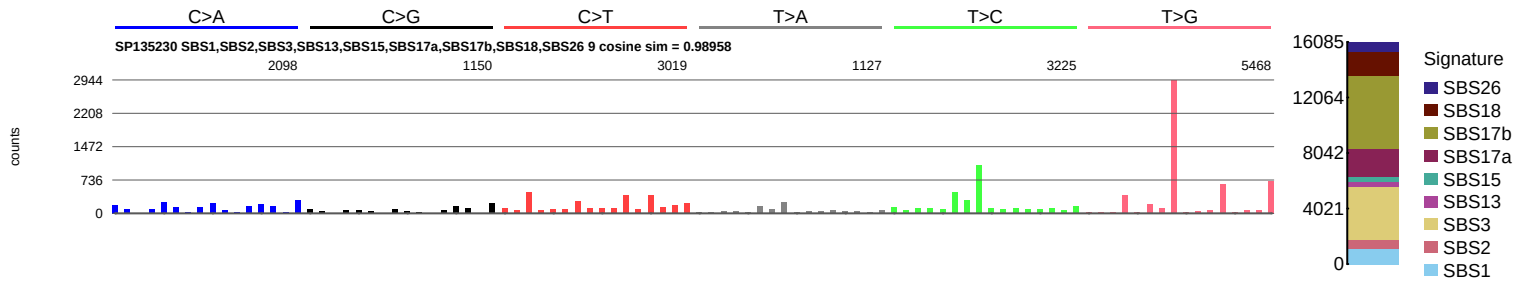
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98962
2101
1134
3051
1123
3219
5457
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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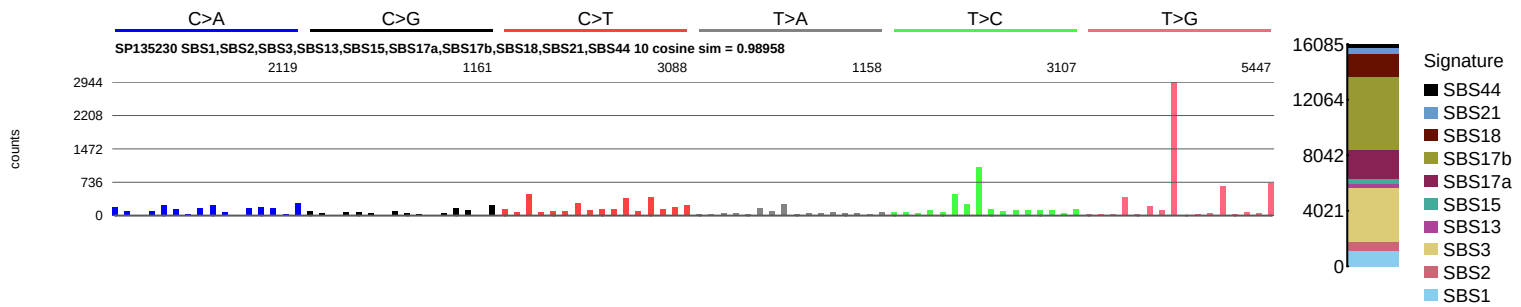
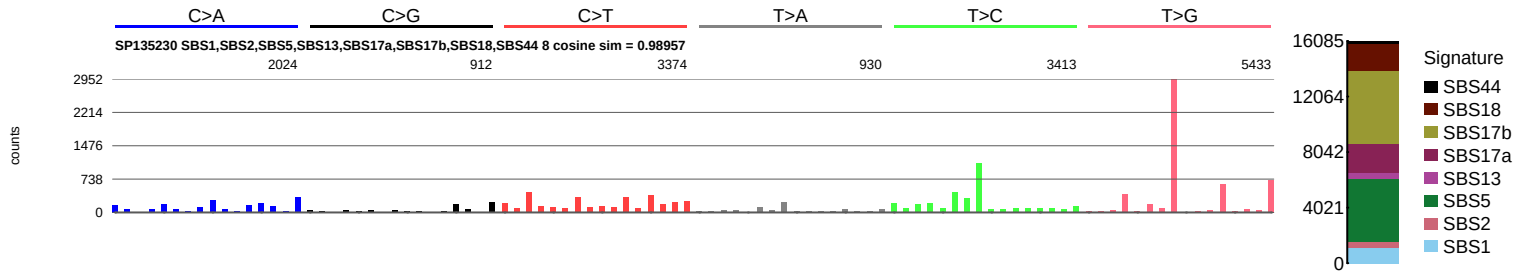
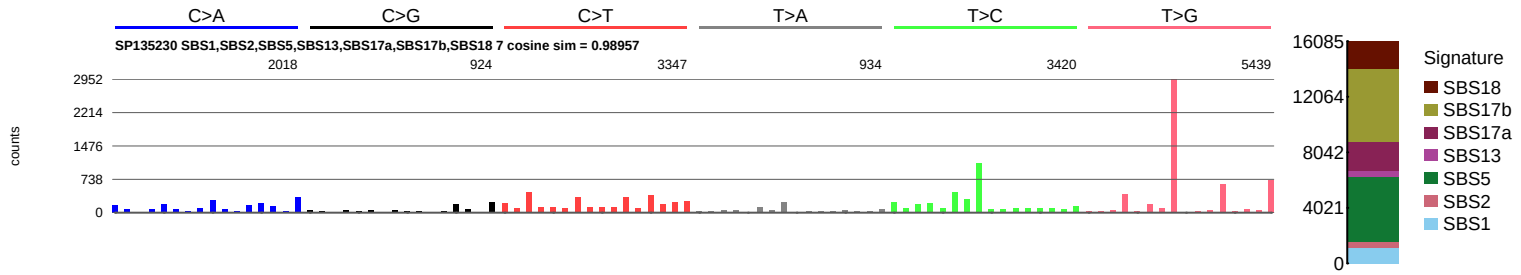
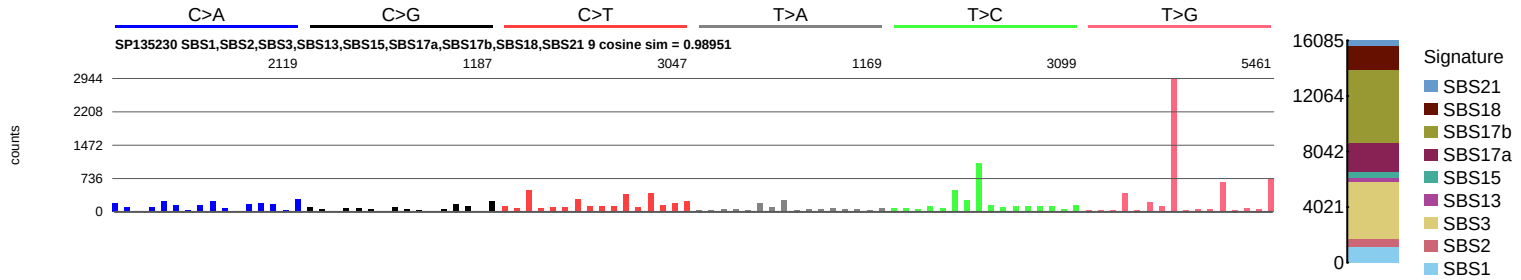
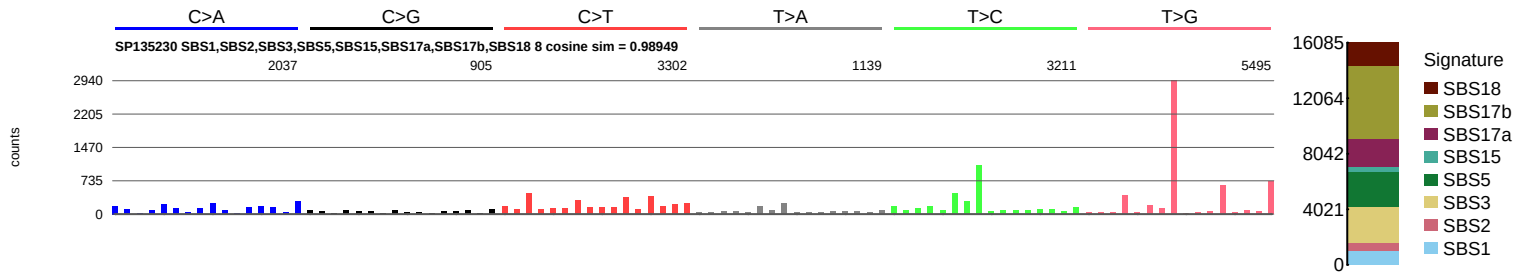
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98958
2098
1150
3019
1127
3225
5468
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

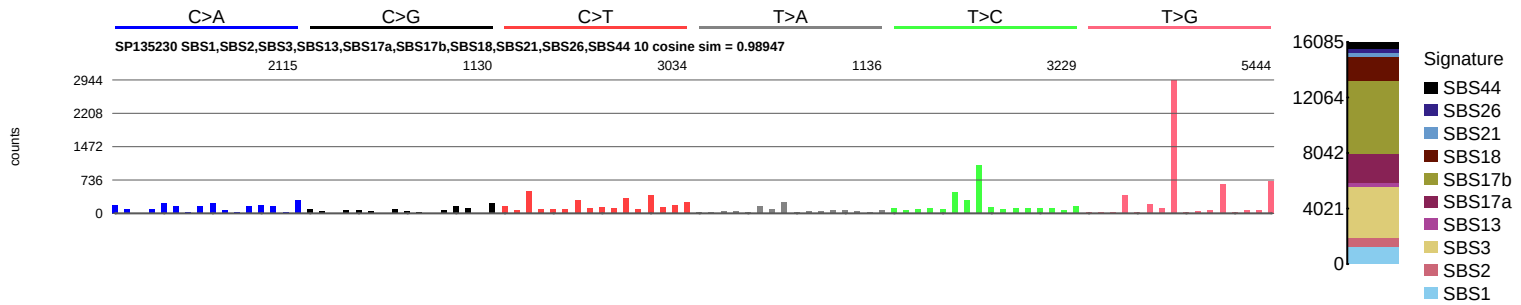

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98958
2119
1161
3088
1158
3107
5447
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

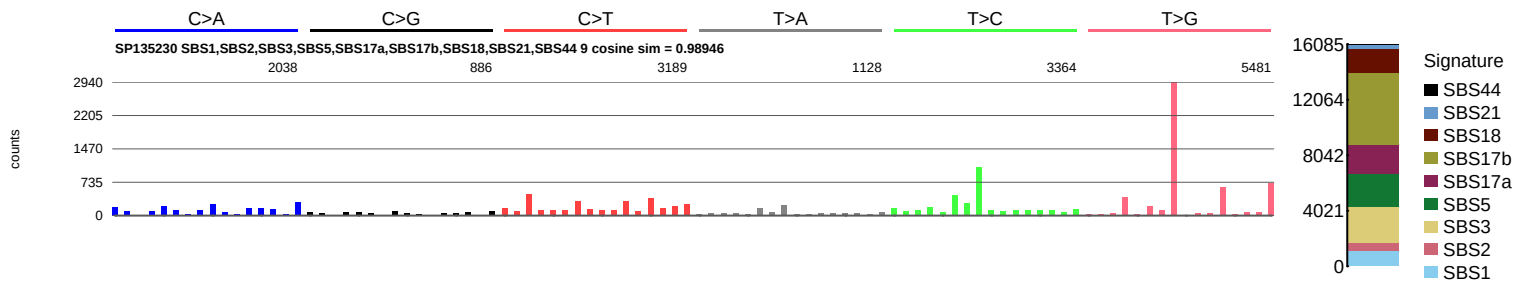

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98957
2024
912
3374
930
3413
5433
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

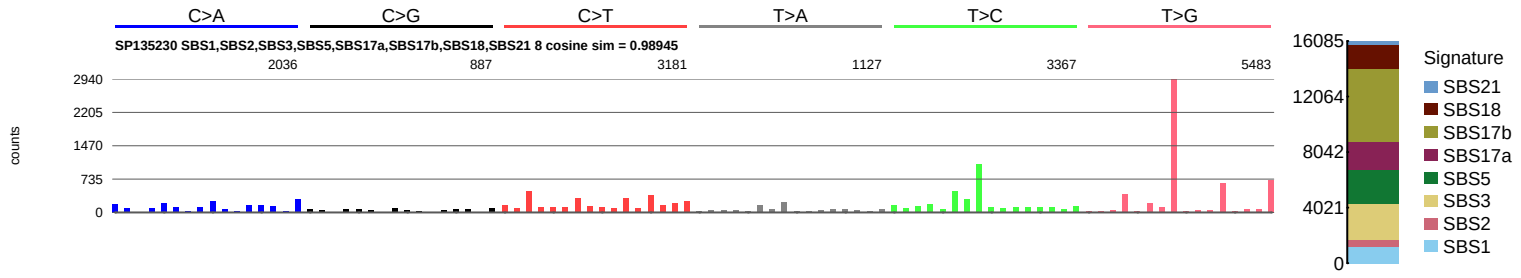

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98957
2018
924
3347
934
3420
5439
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

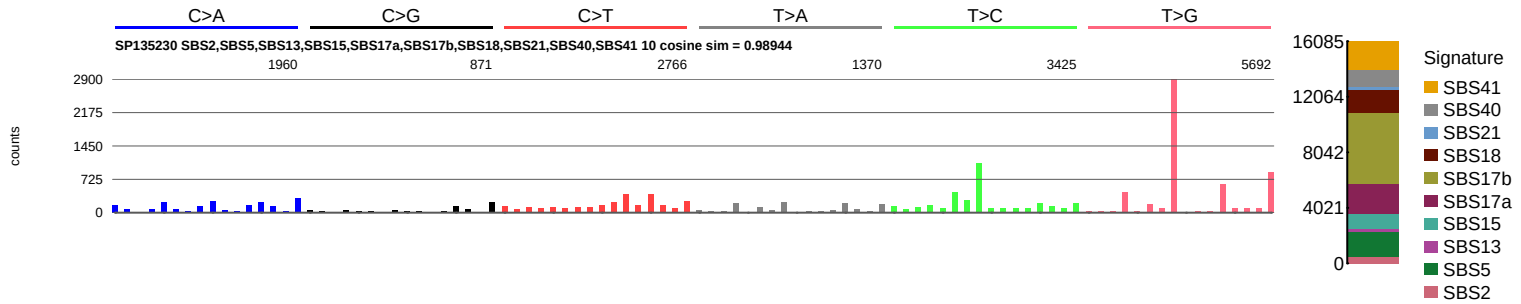

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98951
2119
1187
3047
1169
3099
5461
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

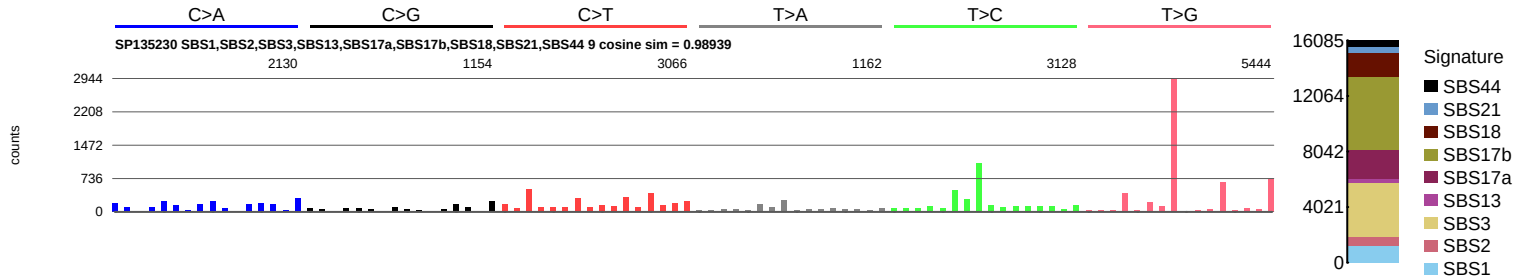

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98949
2037
905
3302
1139
3211
5495
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

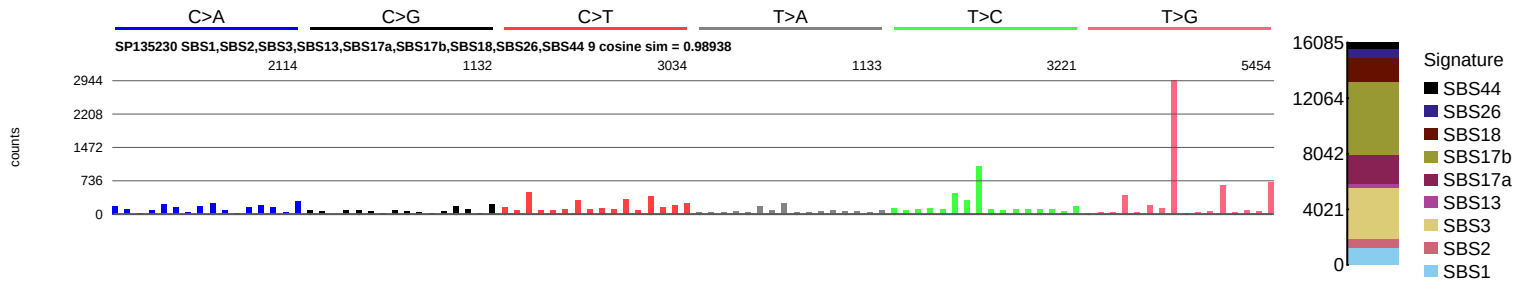

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98947
2115
1130
3034
1136
3229
5444
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

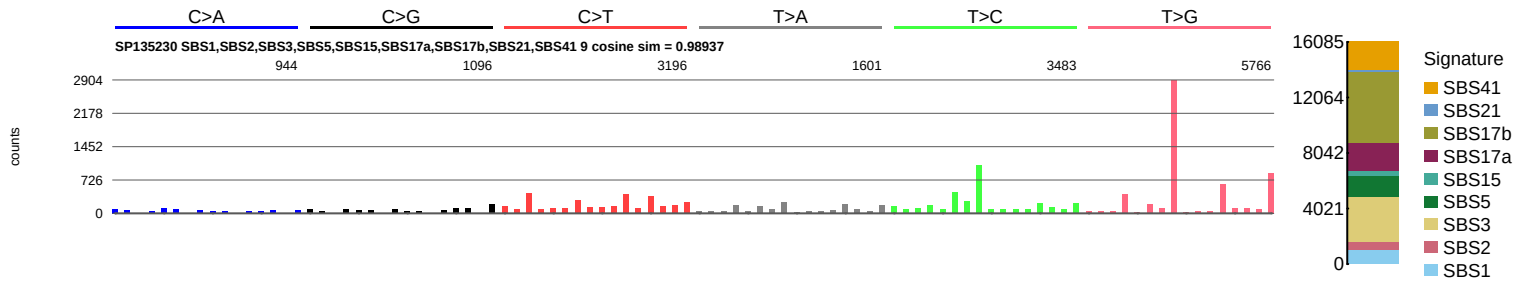

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98946
2038
886
3189
1128
3364
5481
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

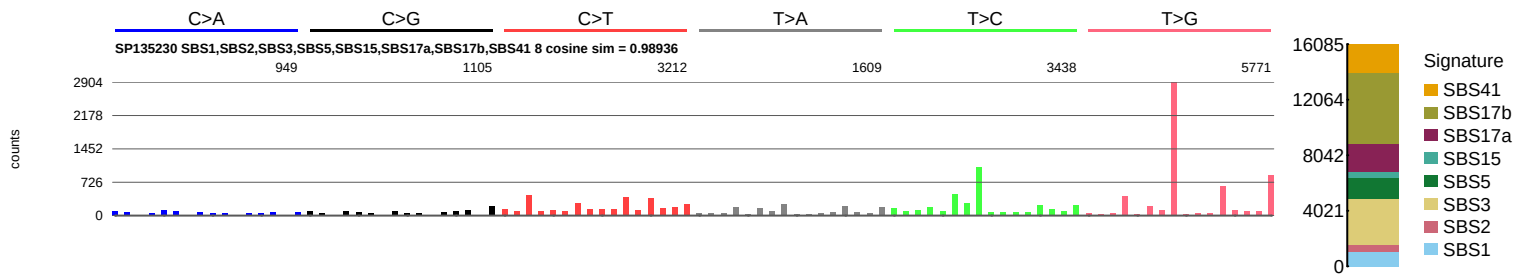

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98945
2036
887
3181
1127
3367
5483
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98944
1960
871
2766
1370
3425
5692
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98939
2130
1154
3066
1162
3128
5444
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98938
2114
1132
3034
1133
3221
5454
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98937
944
1096
3196
1601
3483
5766
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98936
949
1105
3212
1609
3438
5771
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
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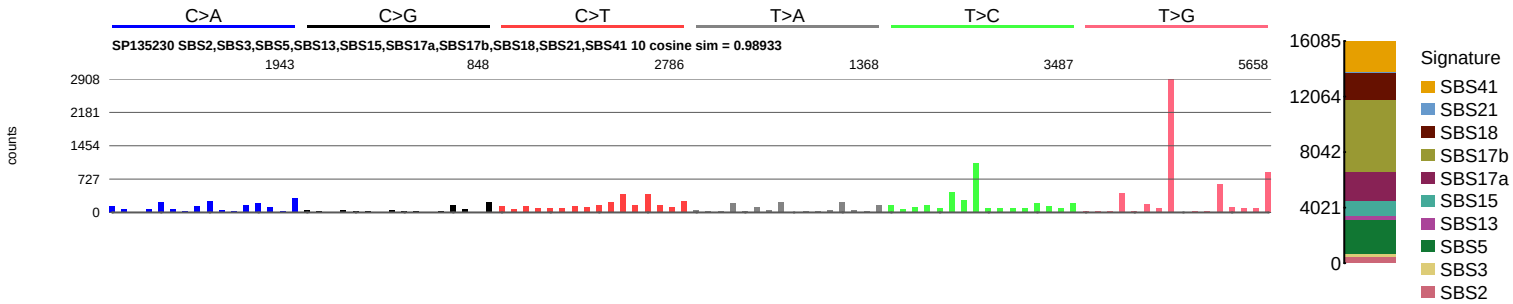
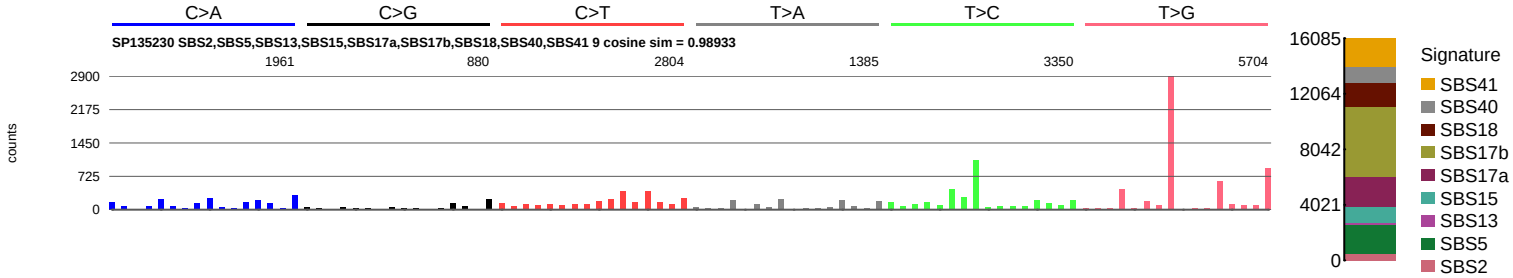
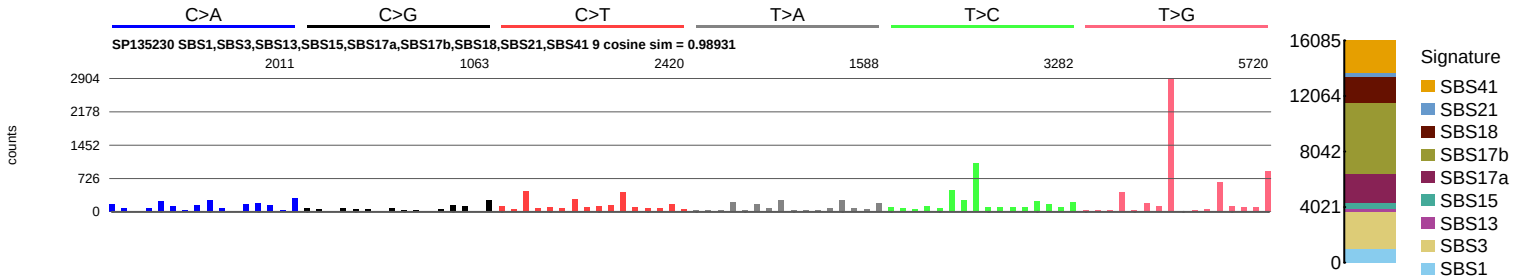
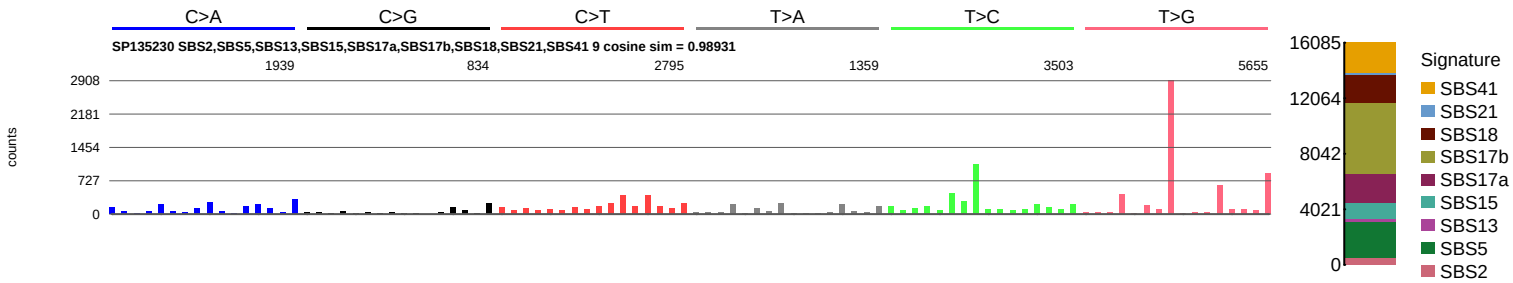
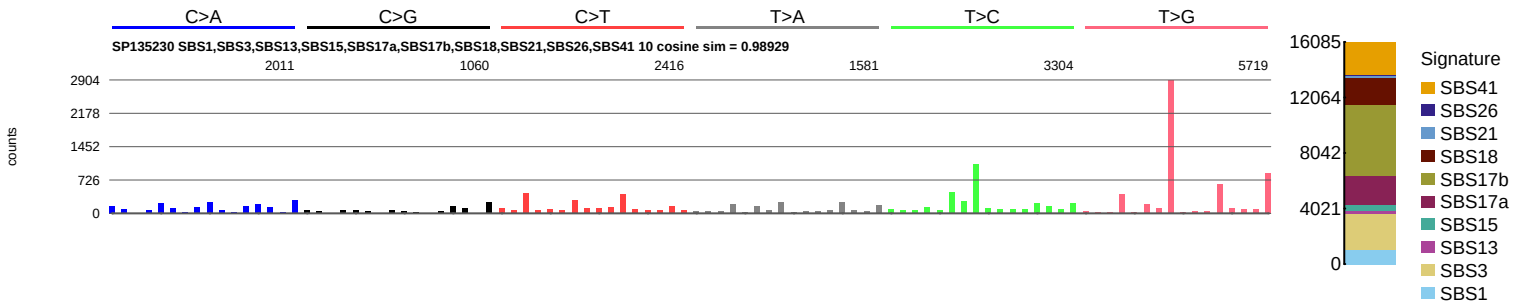
SBS3
SBS2
SBS1

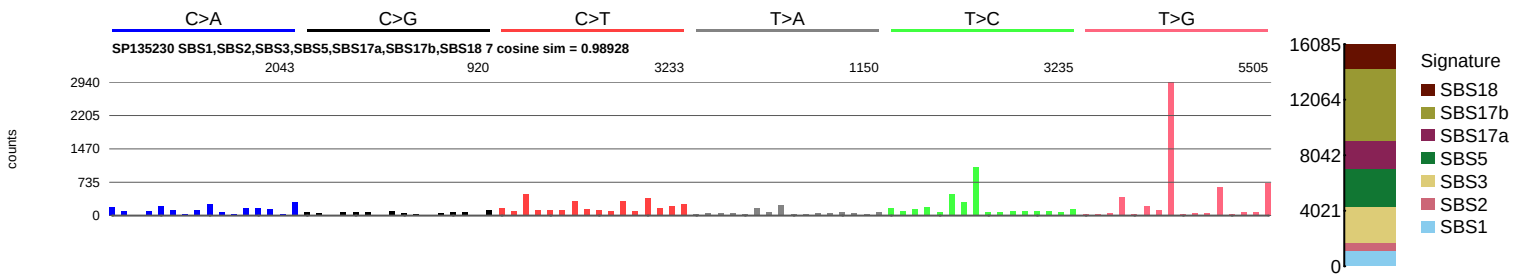

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98933
1943
848
2786
1368
3487
5658
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98933
1961
880
2804
1385
3350
5704
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98931
2011
1063
2420
1588
3282
5720
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98931
1939
834
2795
1359
3503
5655
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98929
2011
1060
2416
1581
3304
5719
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98928
2043
920
3233
1150
3235
5505
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
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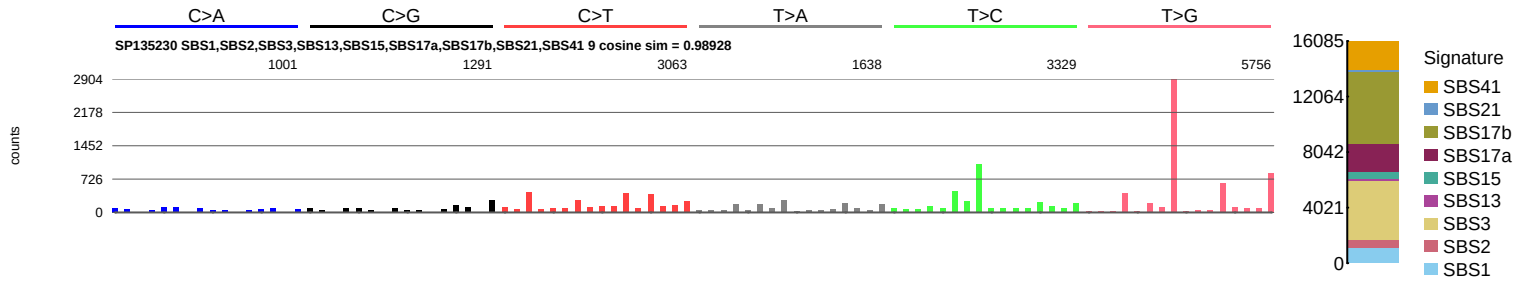
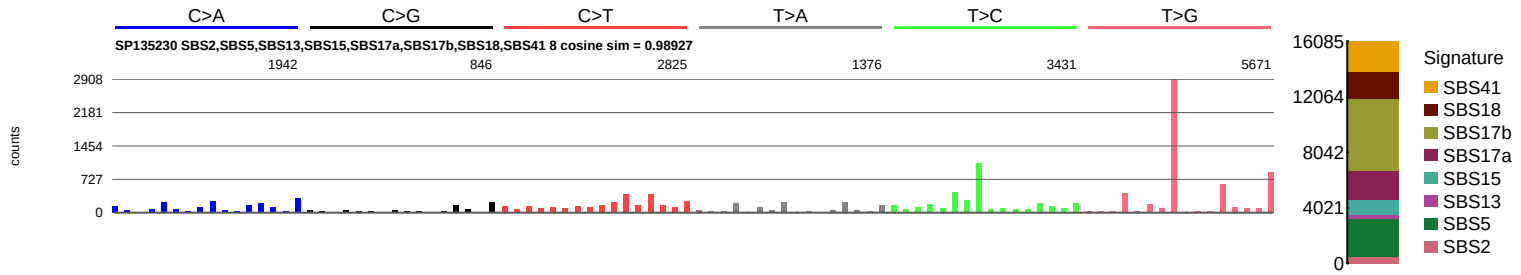
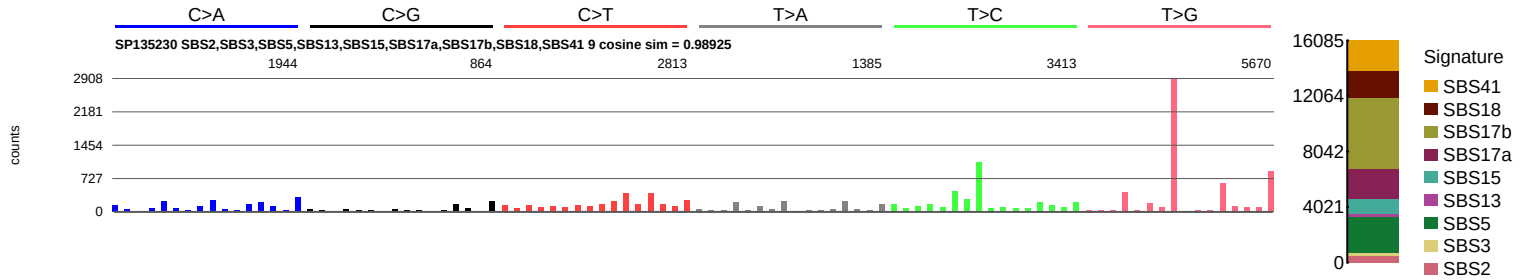
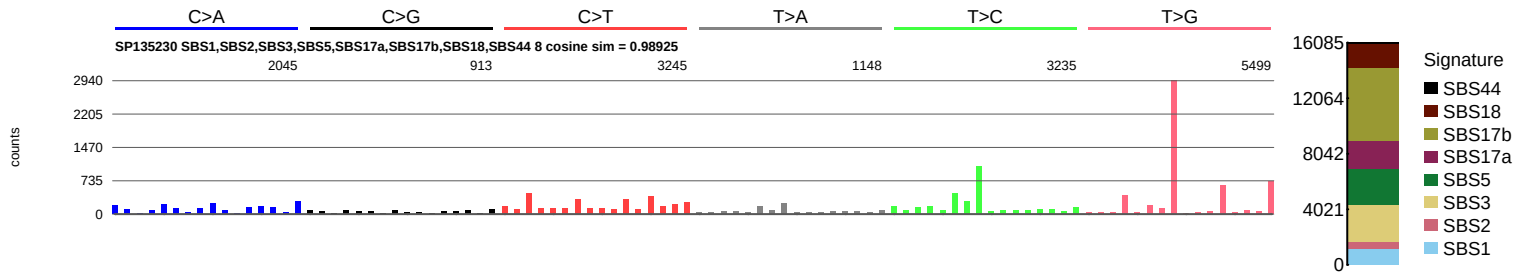
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98928
1001
1291
3063
1638
3329
5756
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98927
1942
846
2825
1376
3431
5671
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98925
1944
864
2813
1385
3413
5670
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98925
2045
913
3245
1148
3235
5499
counts
2940
2205
1470
735
0
16085
12064
8042
4021
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Signature
SBS44
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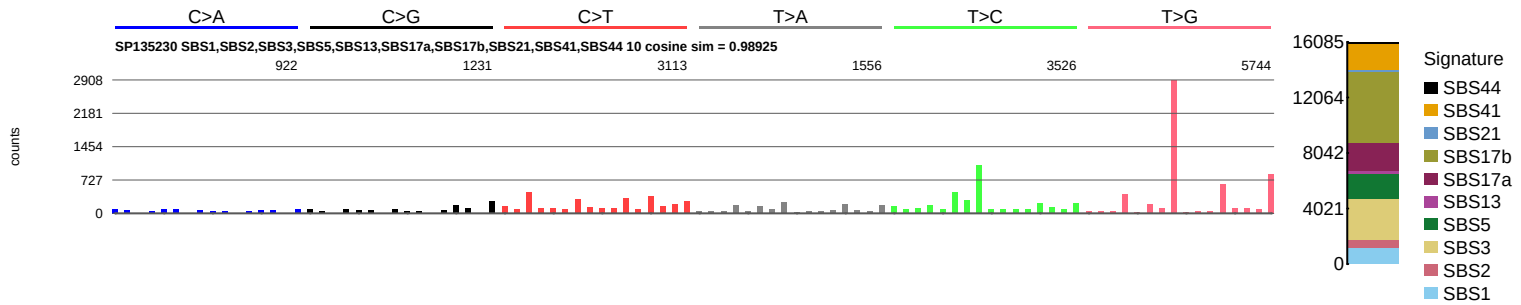
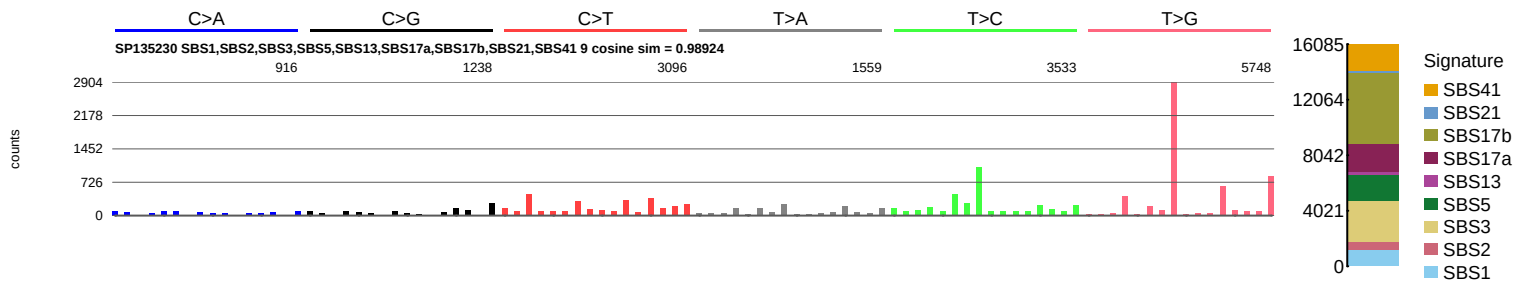
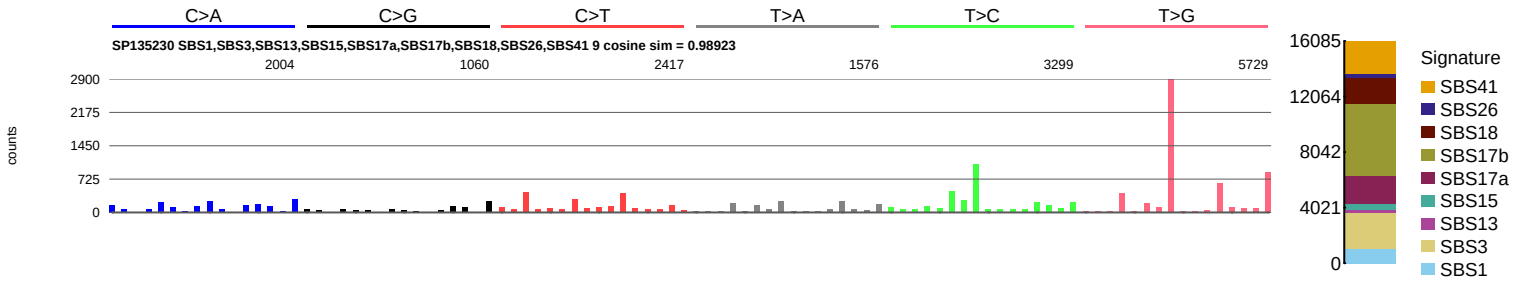
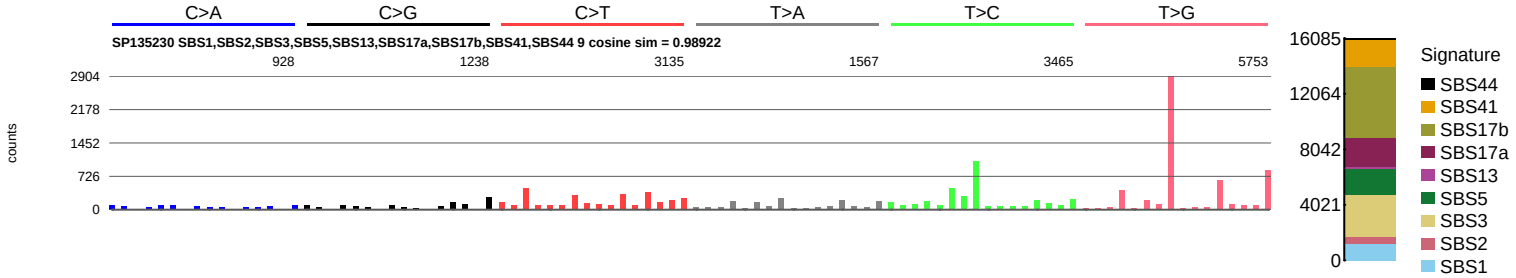
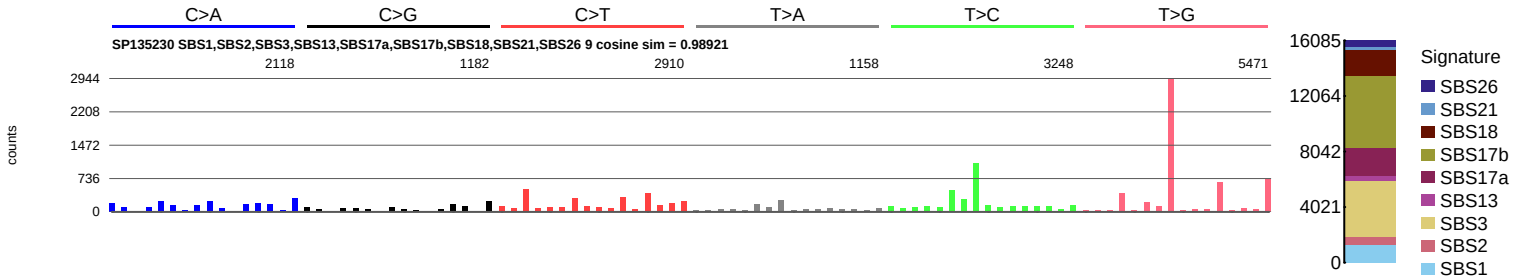
SBS3
SBS2
SBS1

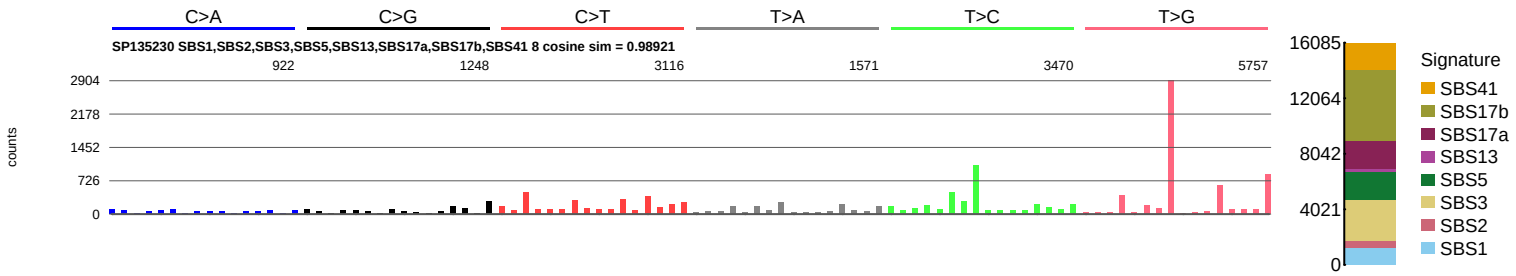

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 10 cosine sim = 0.98925
922
1231
3113
1556
3526
5744
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98924
916
1238
3096
1559
3533
5748
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98923
2004
1060
2417
1576
3299
5729
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.98922
928
1238
3135
1567
3465
5753
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98921
2118
1182
2910
1158
3248
5471
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98921
922
1248
3116
1571
3470
5757
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
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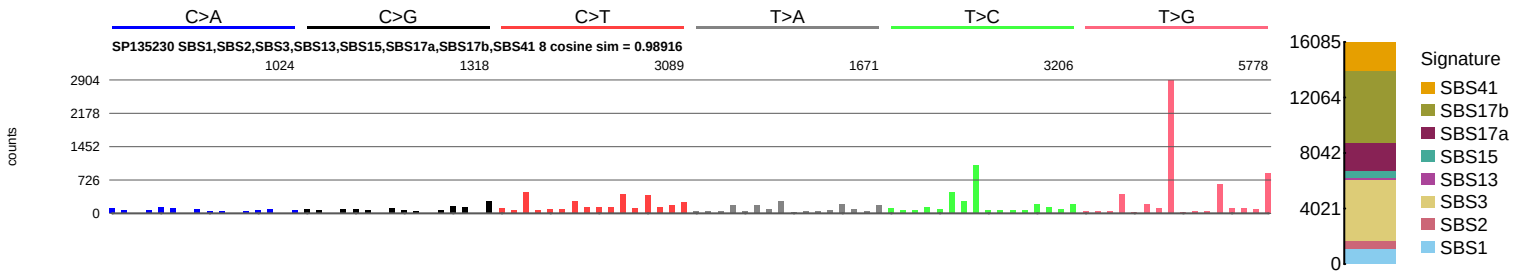
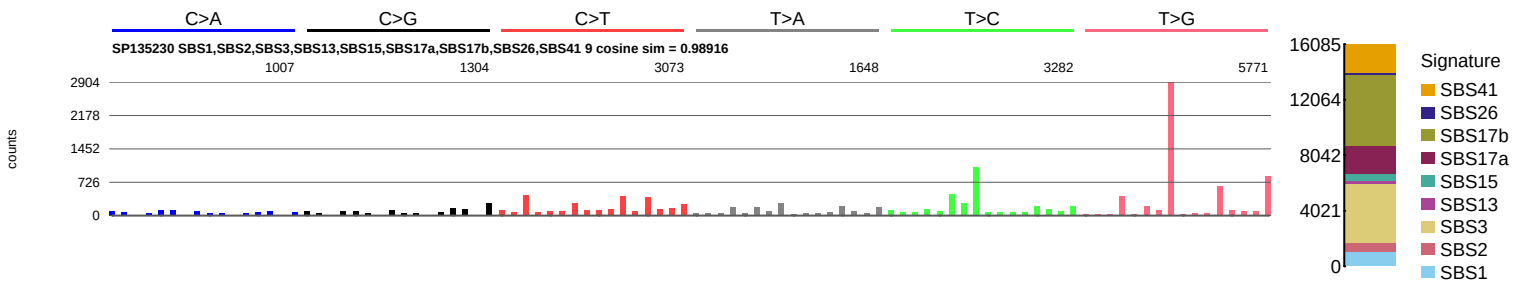
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98916
1024
1318
3089
1671
3206
5778
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 9 cosine sim = 0.98916
1007
1304
3073
1648
3282
5771
counts
2904
2178
1452
726
0
16085
12064
8042
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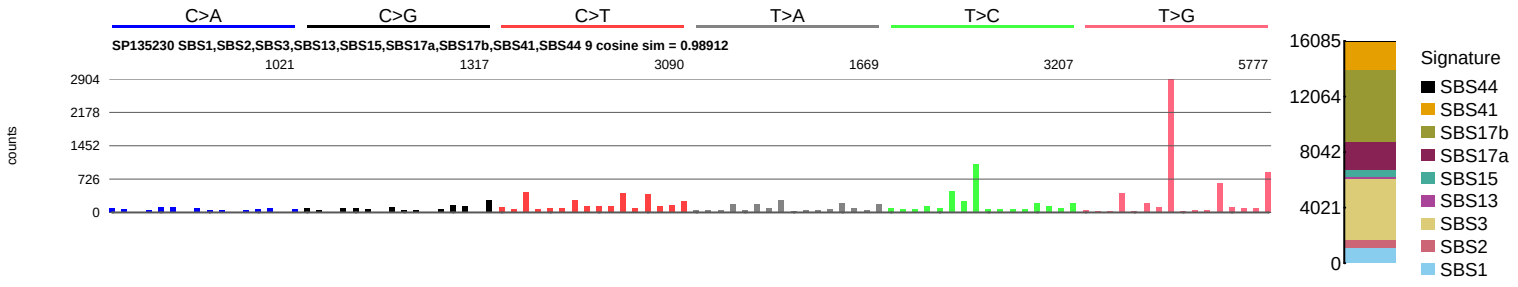
SBS2
SBS1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SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.98912
1021
1317
3090
1669
3207
5777
counts
2904
2178
1452
726
0
16085
12064
8042
4021
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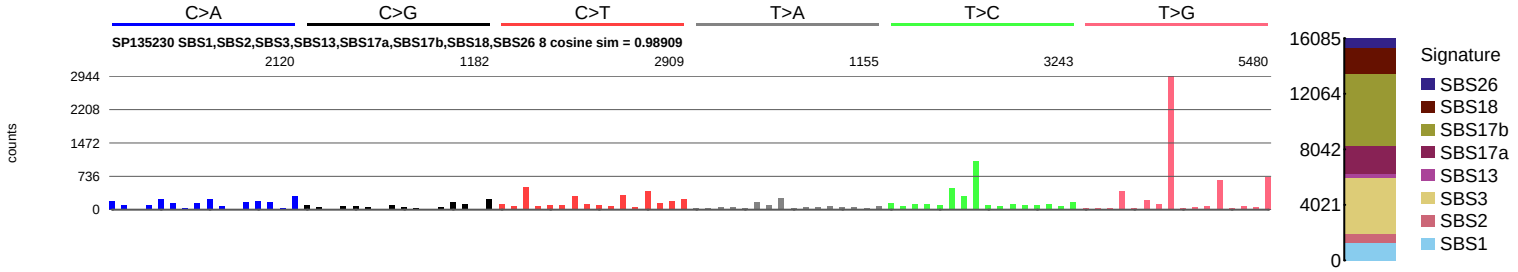
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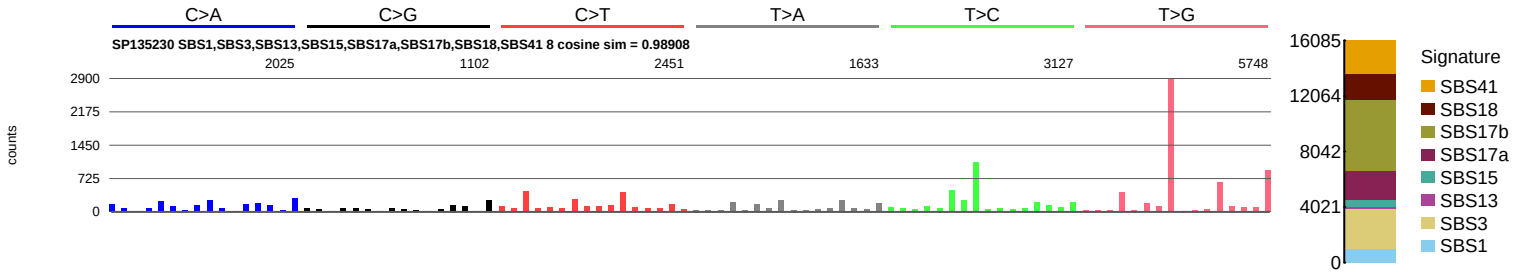
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98909
2120
1182
2909
1155
3243
5480
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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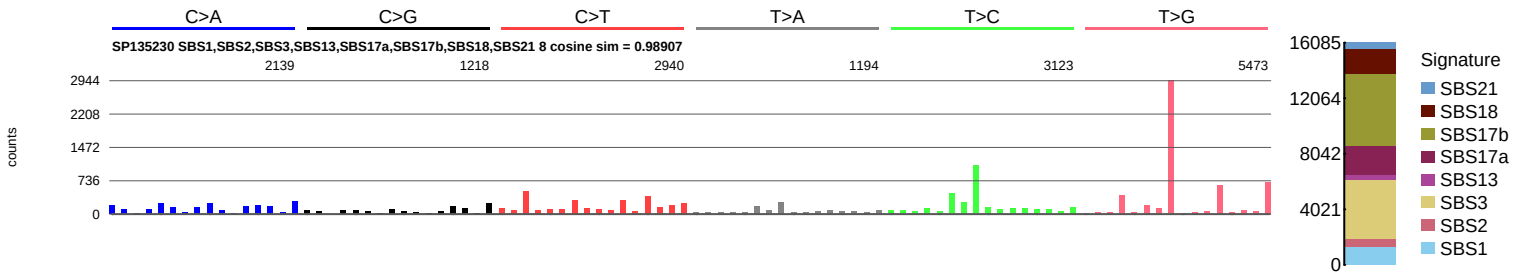
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98908
2025
1102
2451
1633
3127
5748
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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C>G
C>T
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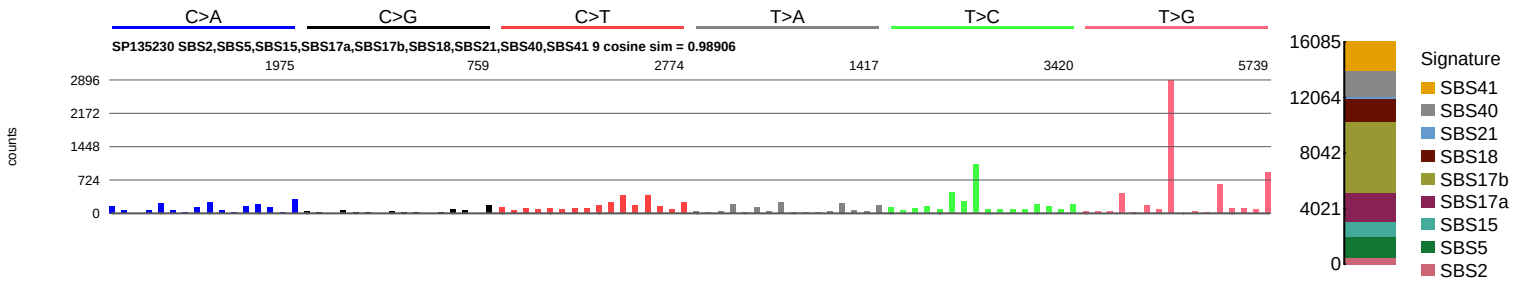
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98907
2139
1218
2940
1194
3123
5473
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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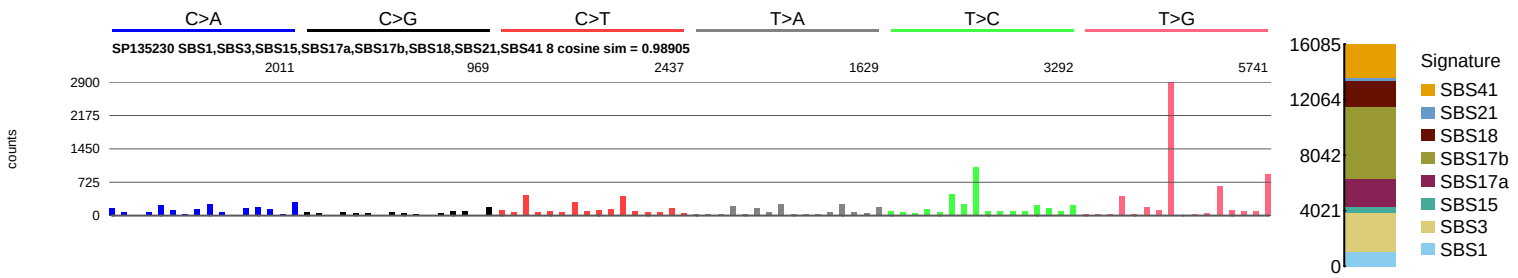
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98906
1975
759
2774
1417
3420
5739
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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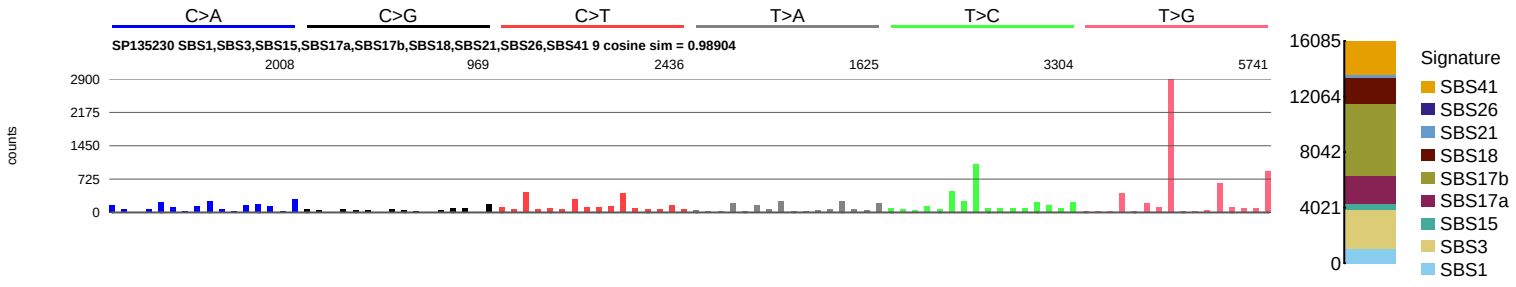
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98905
2011
969
2437
1629
3292
5741
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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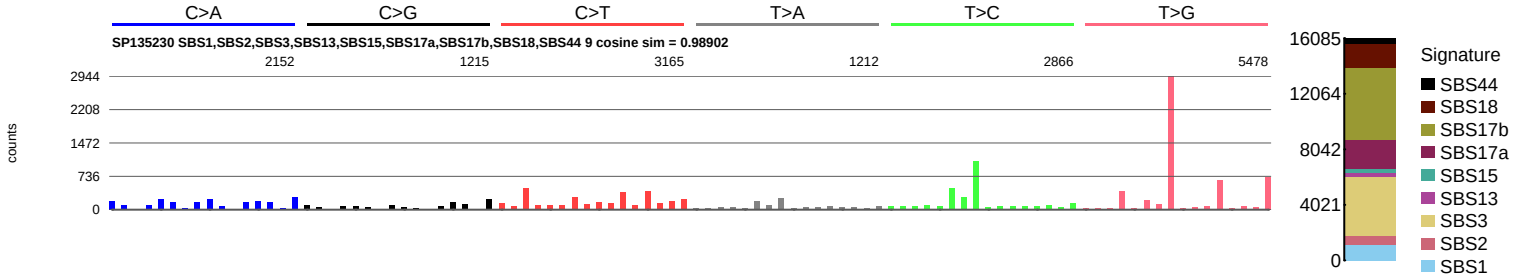
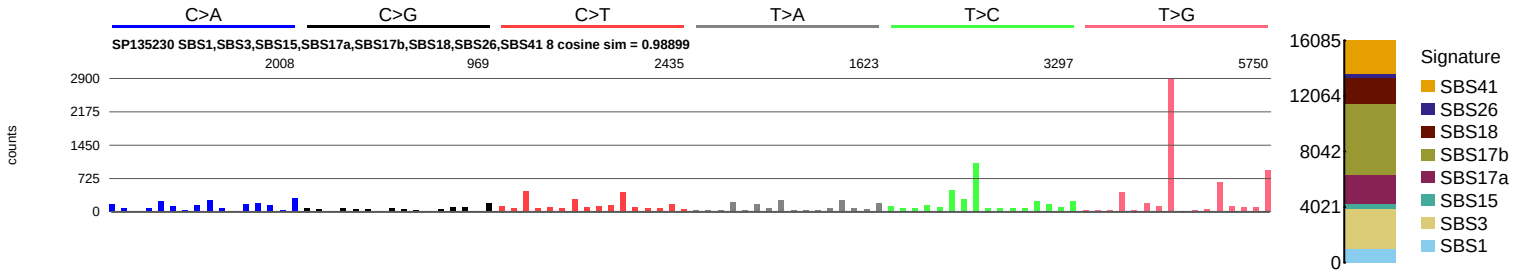
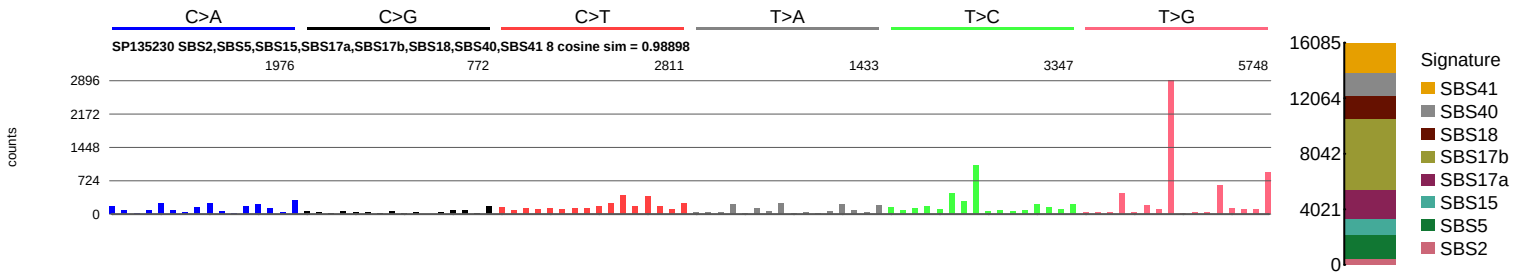
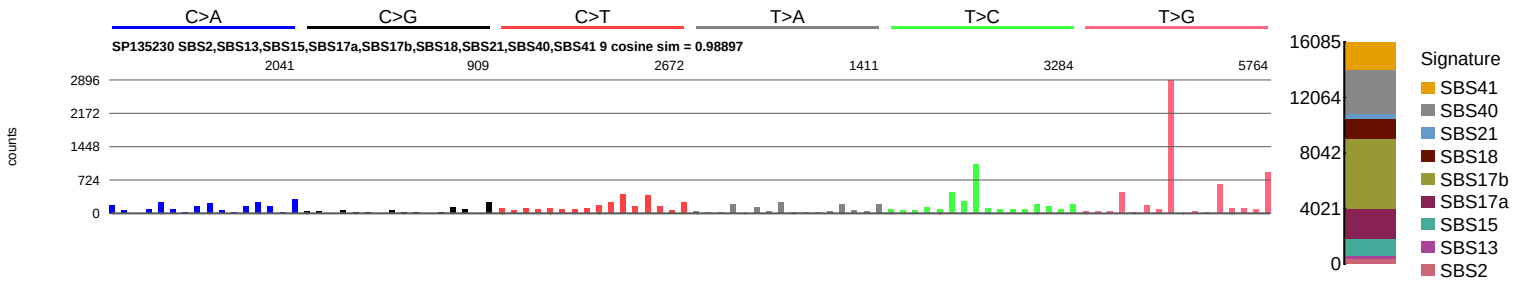
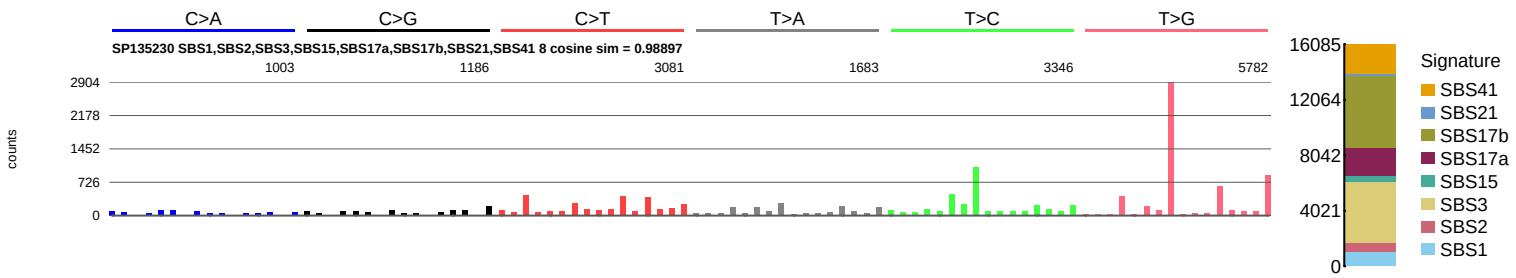
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98904
2008
969
2436
1625
3304
5741
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

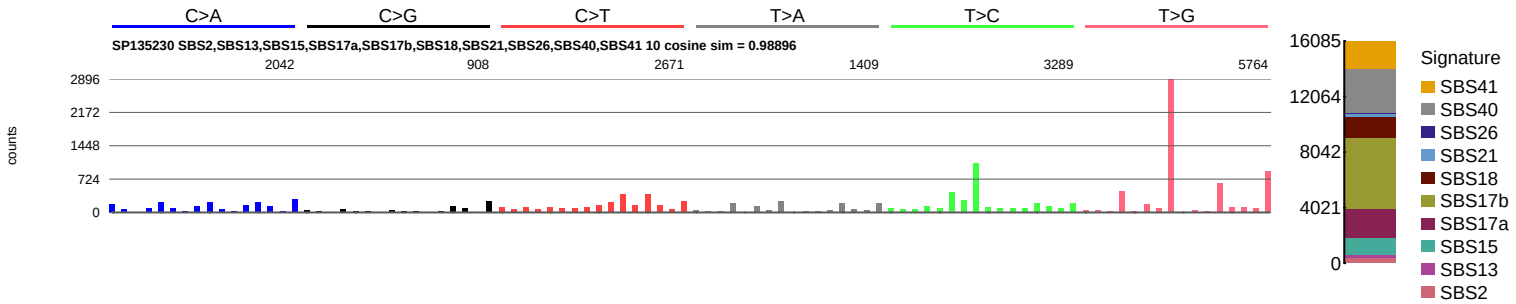

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98902
2152
1215
3165
1212
2866
5478
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

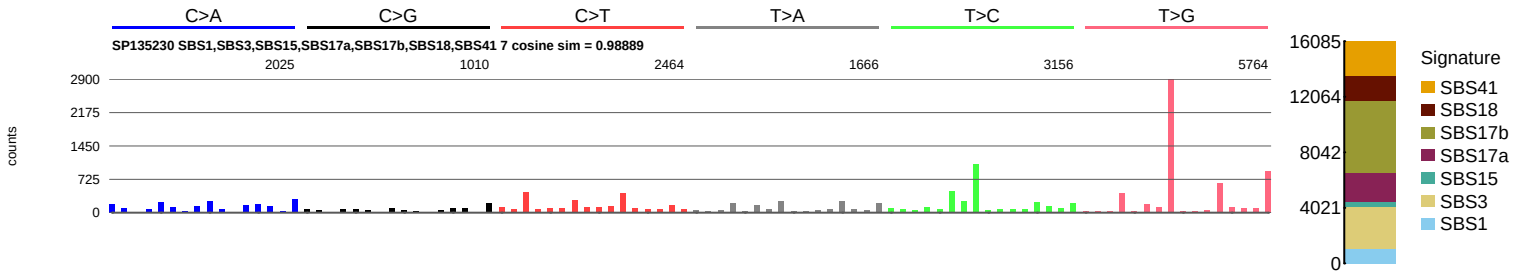

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98899
2008
969
2435
1623
3297
5750
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

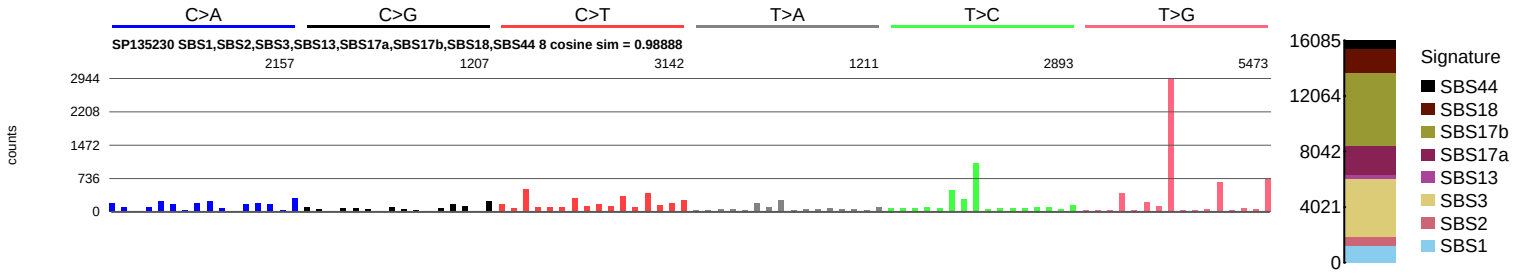

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98898
1976
772
2811
1433
3347
5748
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98897
2041
909
2672
1411
3284
5764
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98897
1003
1186
3081
1683
3346
5782
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98896
2042
908
2671
1409
3289
5764
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98889
2025
1010
2464
1666
3156
5764
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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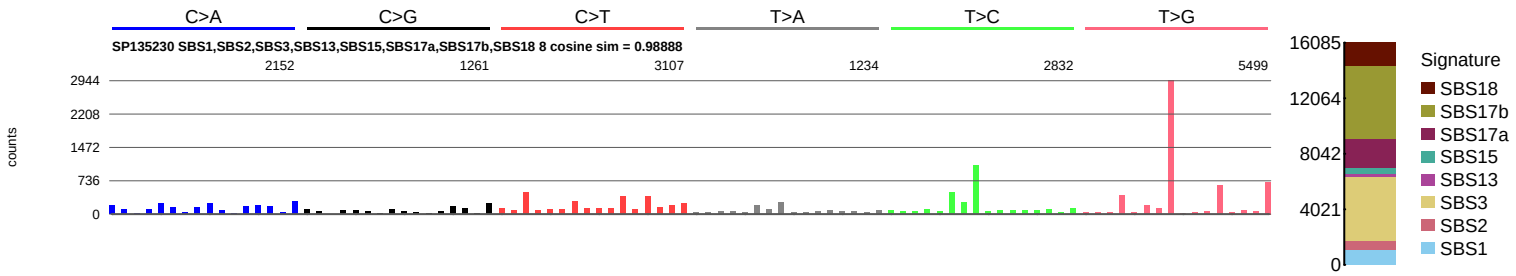
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98888
2157
1207
3142
1211
2893
5473
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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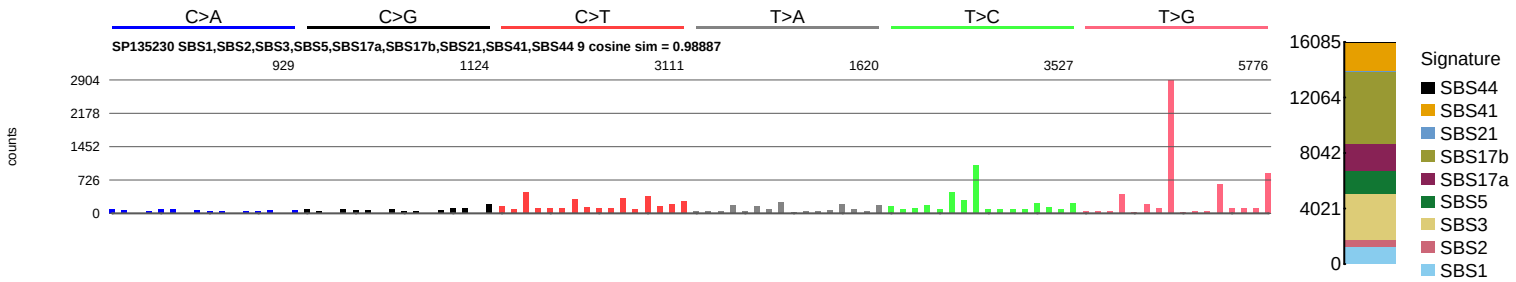
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98888
2152
1261
3107
1234
2832
5499
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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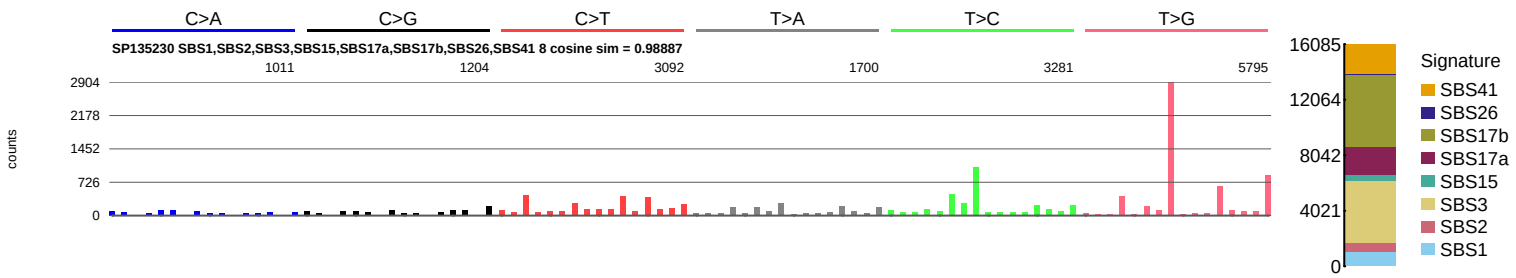
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98887
929
1124
3111
1620
3527
5776
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98887
1011
1204
3092
1700
3281
5795
counts
2904
2178
1452
726
0
16085
12064
8042
4021
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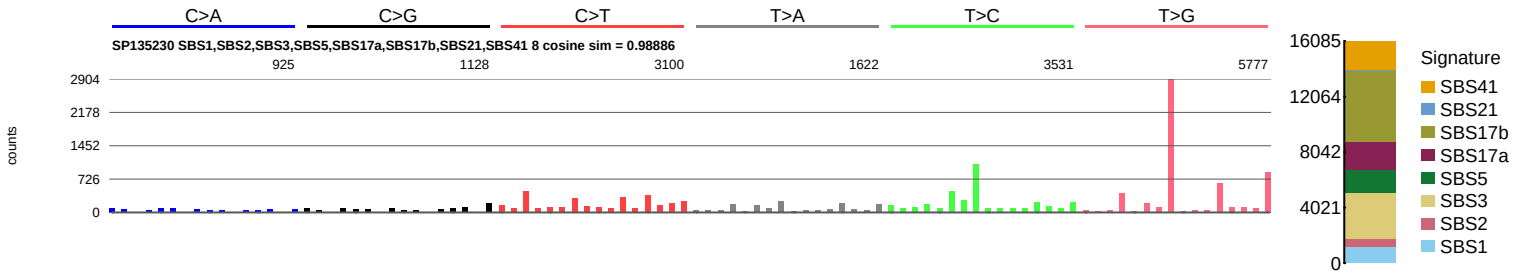
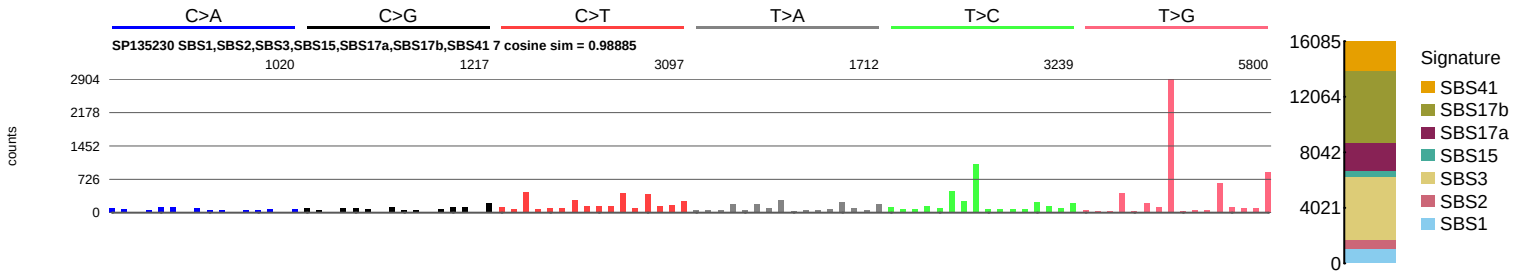
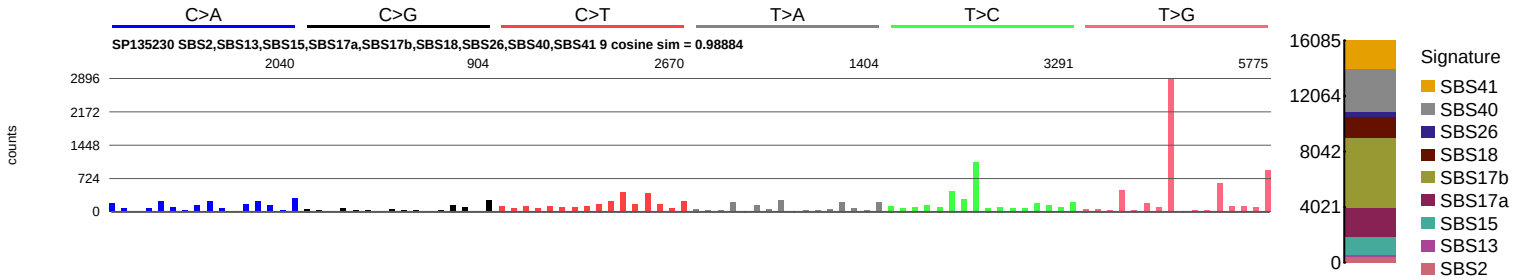
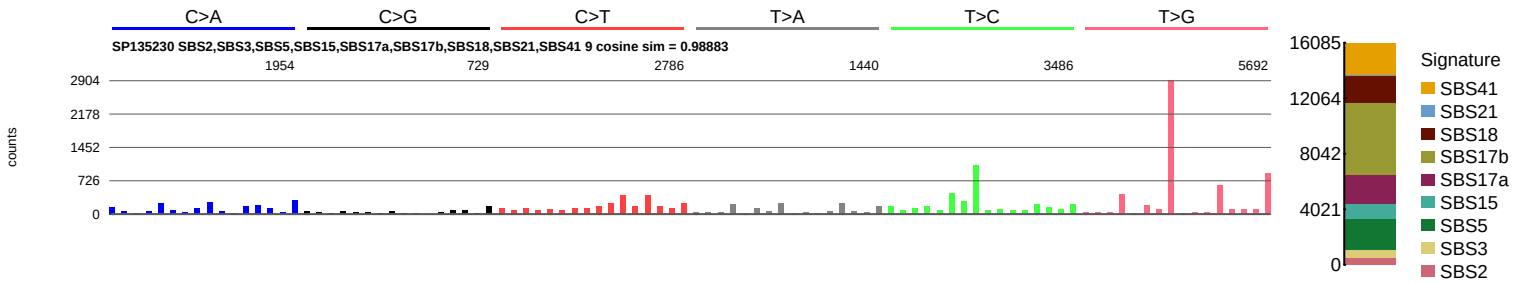
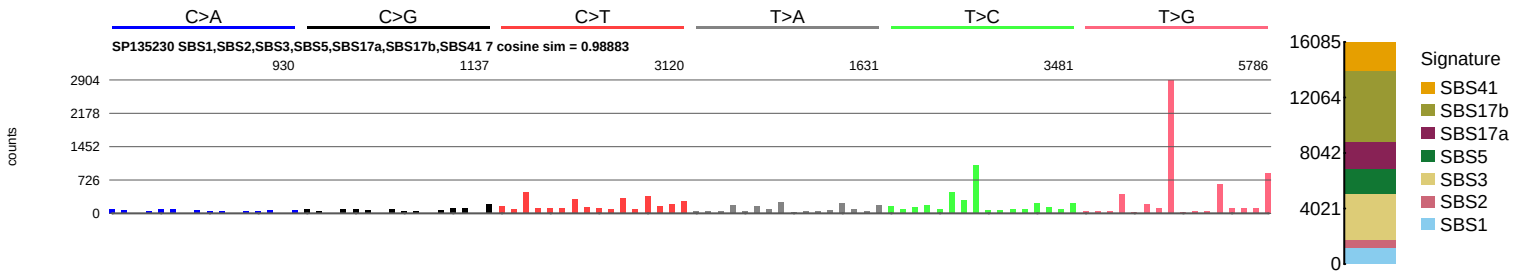
SBS3
SBS2
SBS1

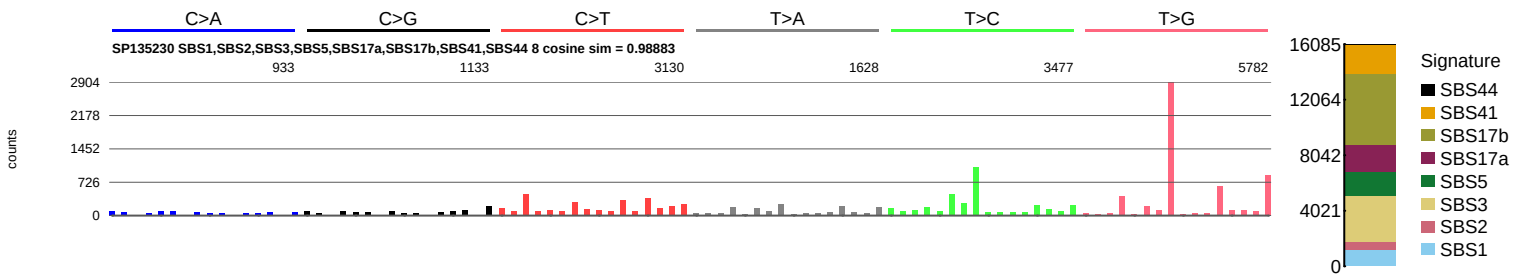

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98886
925
1128
3100
1622
3531
5777
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

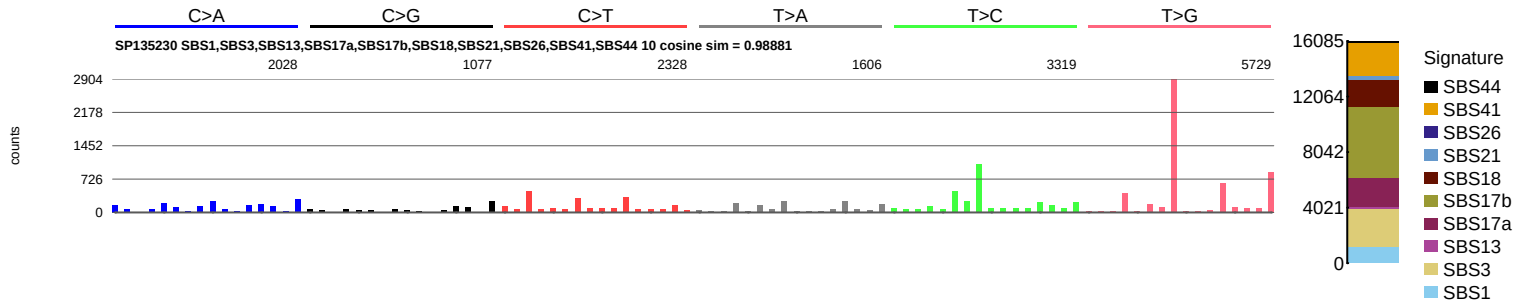

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98885
1020
1217
3097
1712
3239
5800
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98884
2040
904
2670
1404
3291
5775
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98883
1954
729
2786
1440
3486
5692
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98883
930
1137
3120
1631
3481
5786
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98883
933
1133
3130
1628
3477
5782
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98881
2028
1077
2328
1606
3319
5729
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
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SBS44
SBS41
SBS26
SBS21
SBS18
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SBS17a
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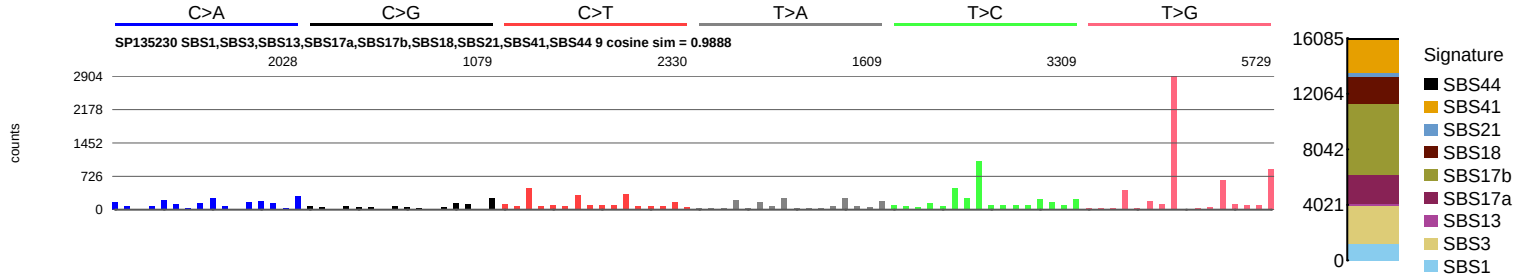
SBS3
SBS1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2028
1079
2330
1609
3309
5729
counts
2904
2178
1452
726
0
16085
12064
8042
4021
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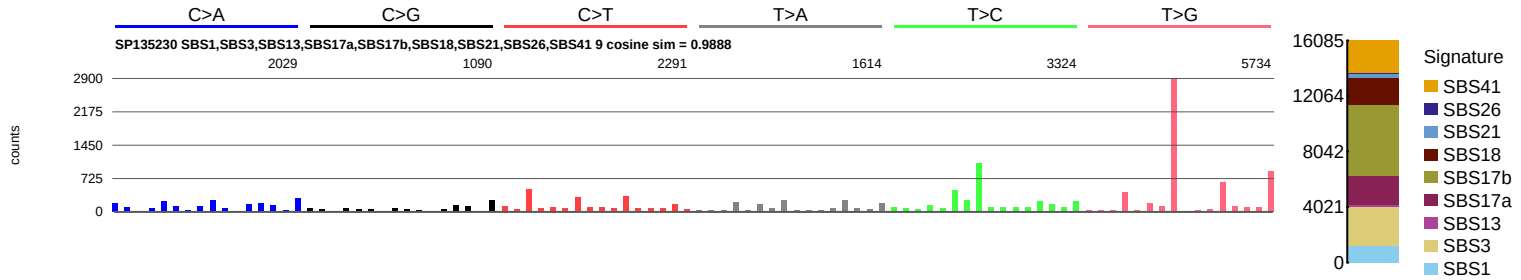
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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T>C
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SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.9888
2029
1090
2291
1614
3324
5734
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
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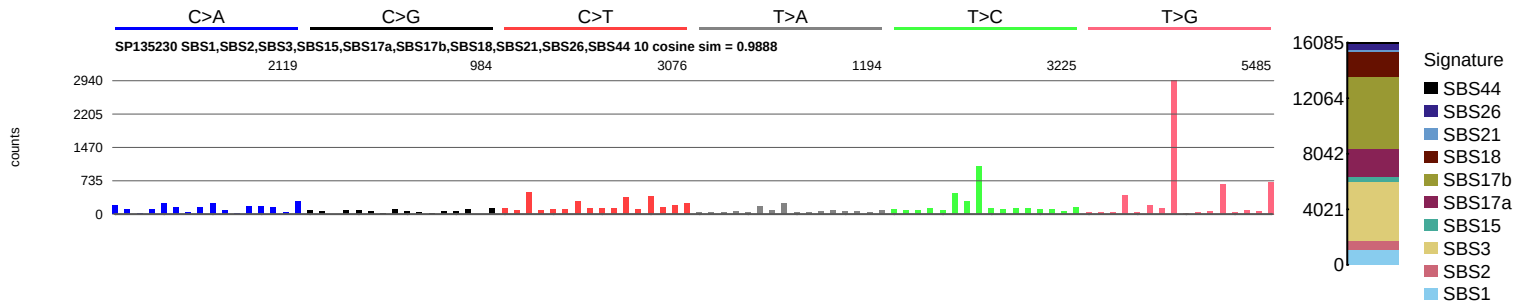
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.9888
2119
984
3076
1194
3225
5485
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
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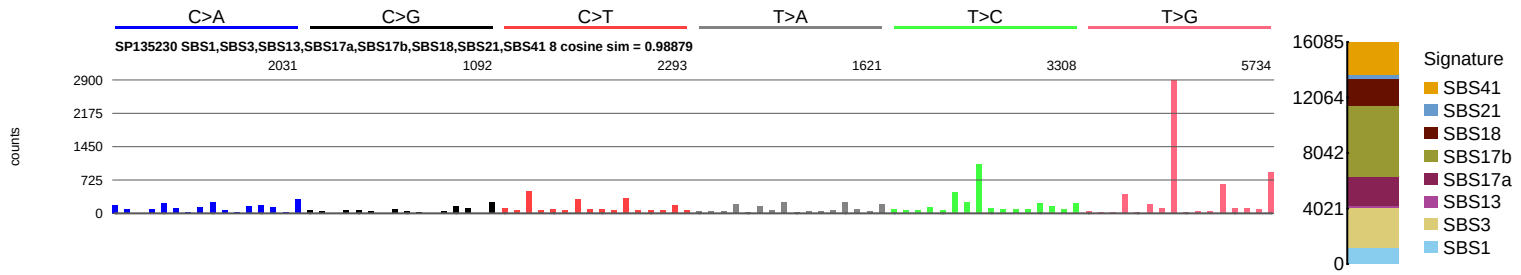
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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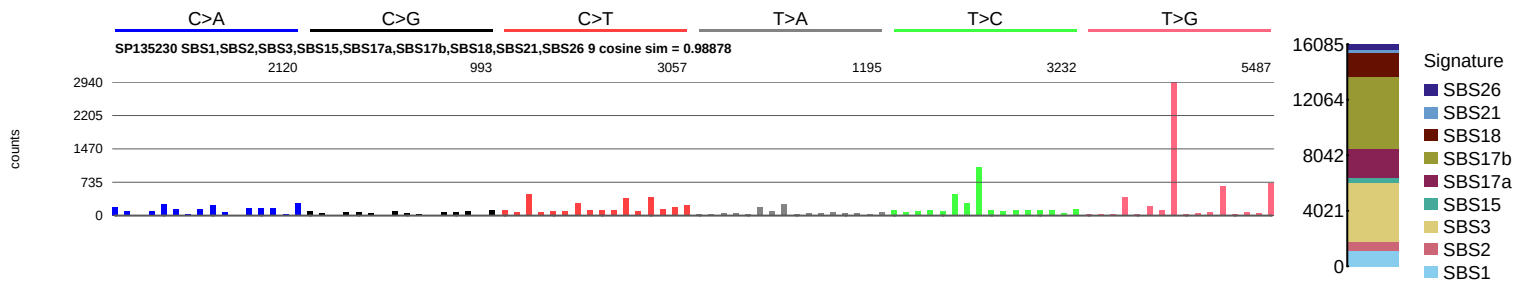
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98879
2031
1092
2293
1621
3308
5734
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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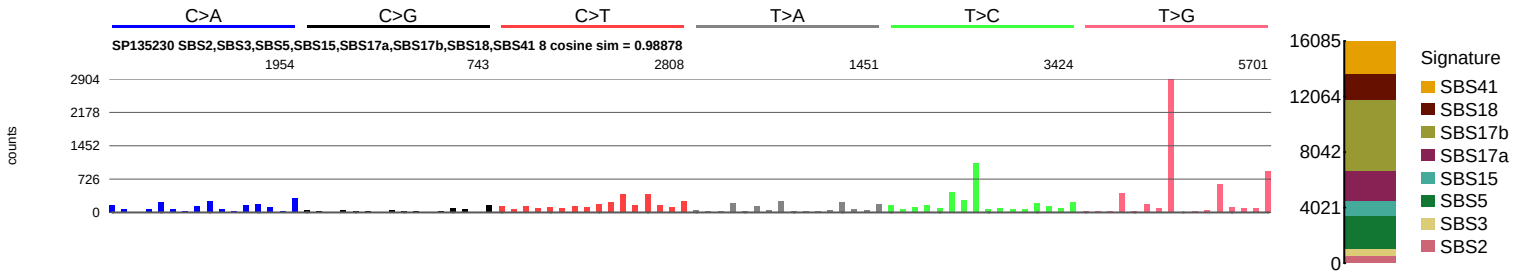
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98878
2120
993
3057
1195
3232
5487
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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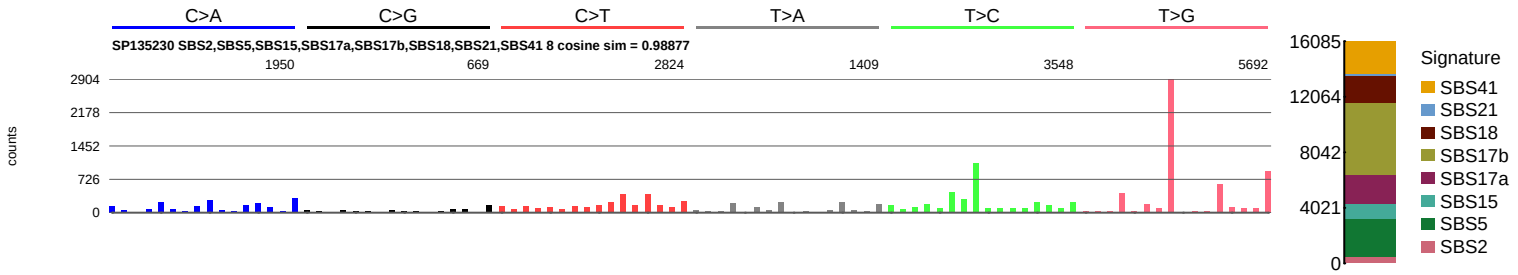
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98878
1954
743
2808
1451
3424
5701
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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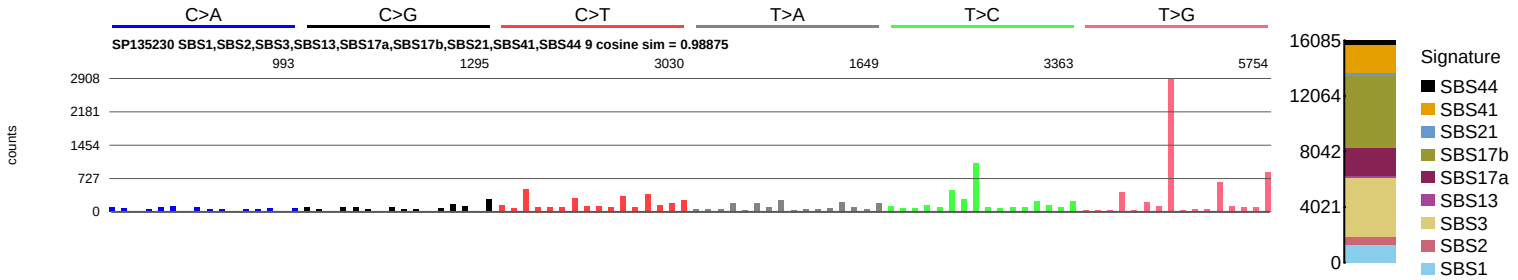
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98877
1950
669
2824
1409
3548
5692
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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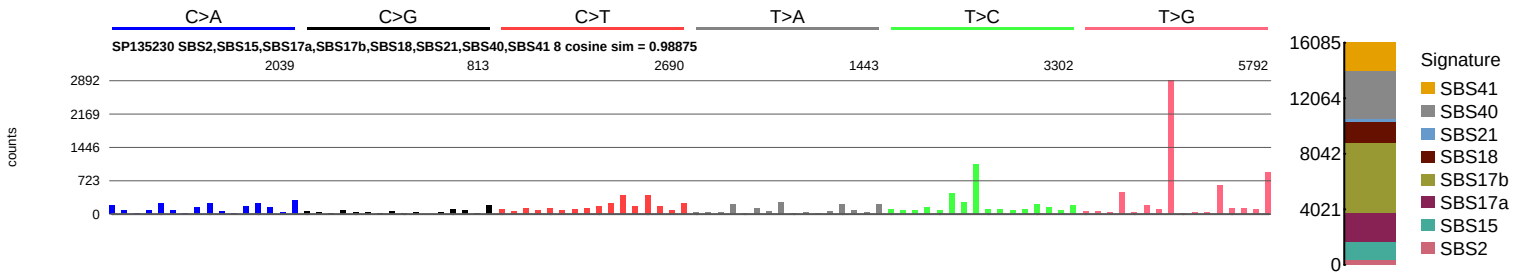
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98875
993
1295
3030
1649
3363
5754
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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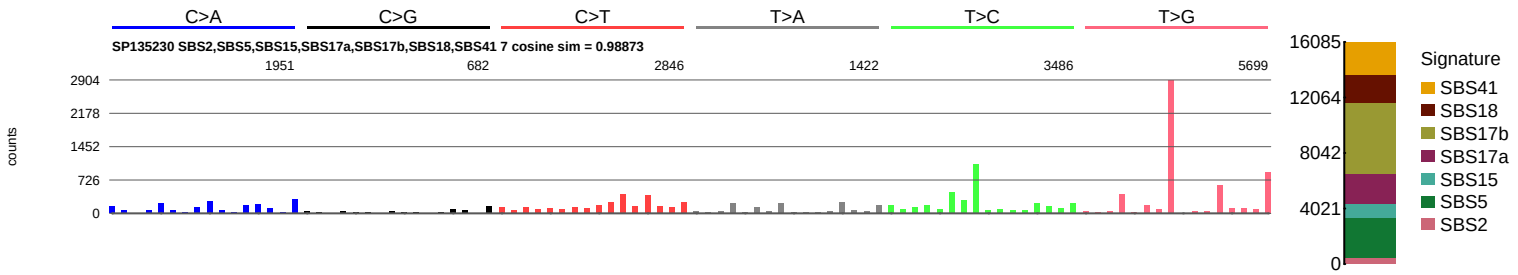
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98875
2039
813
2690
1443
3302
5792
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2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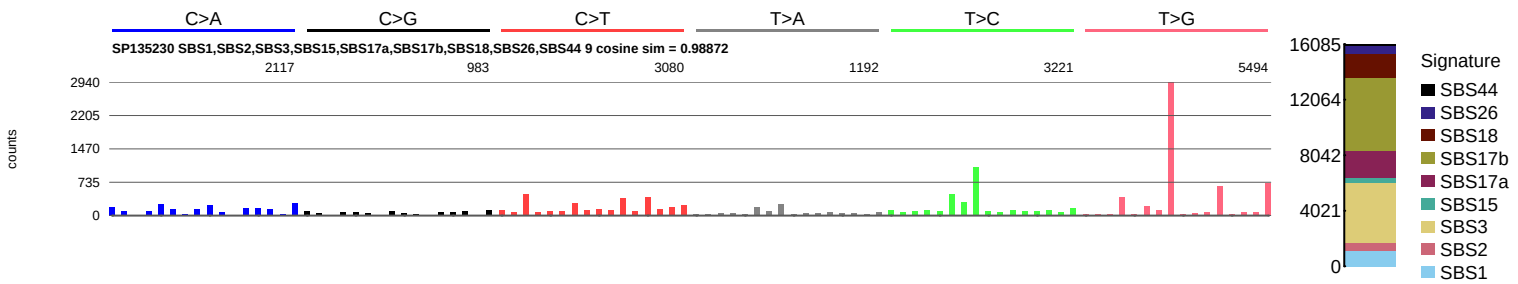
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98873
1951
682
2846
1422
3486
5699
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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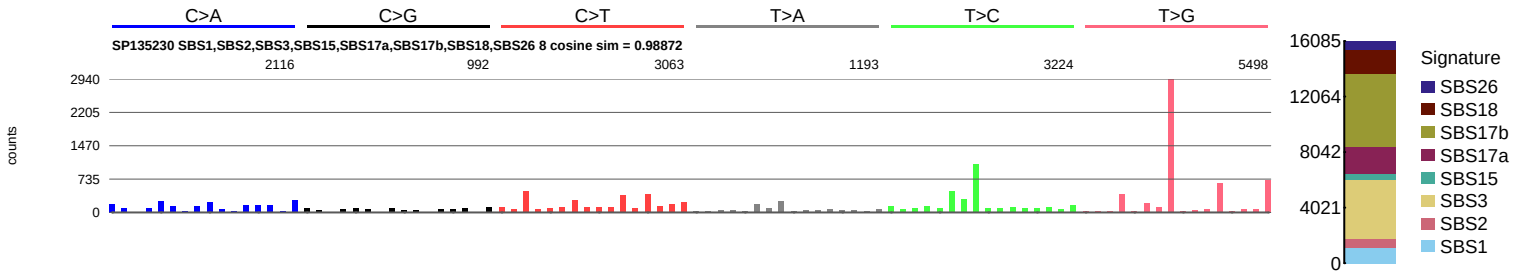
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98872
2117
983
3080
1192
3221
5494
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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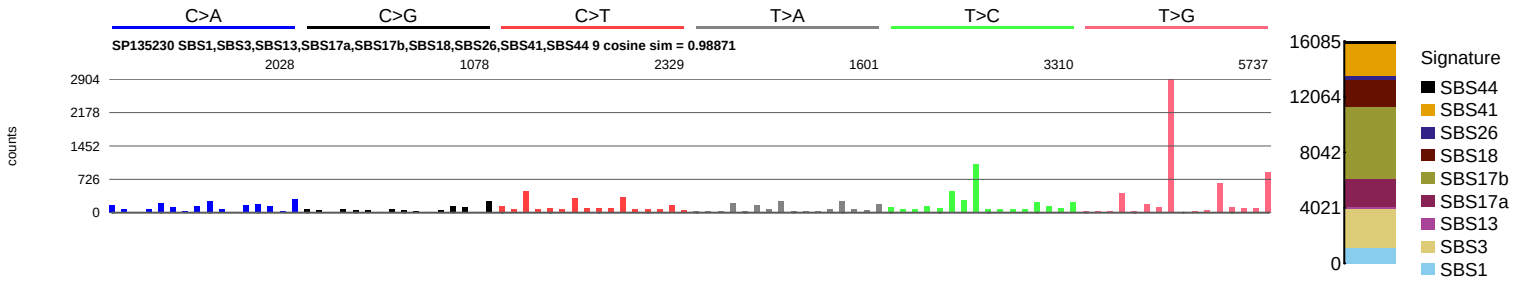
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98872
2116
992
3063
1193
3224
5498
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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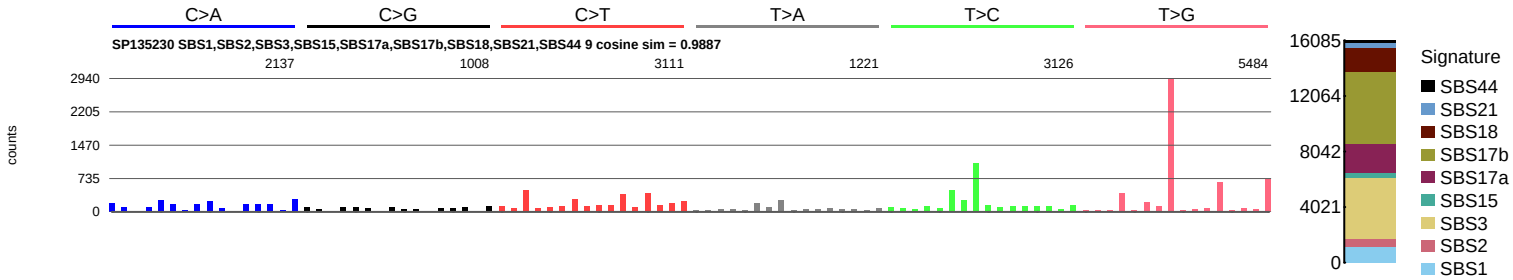
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98871
2028
1078
2329
1601
3310
5737
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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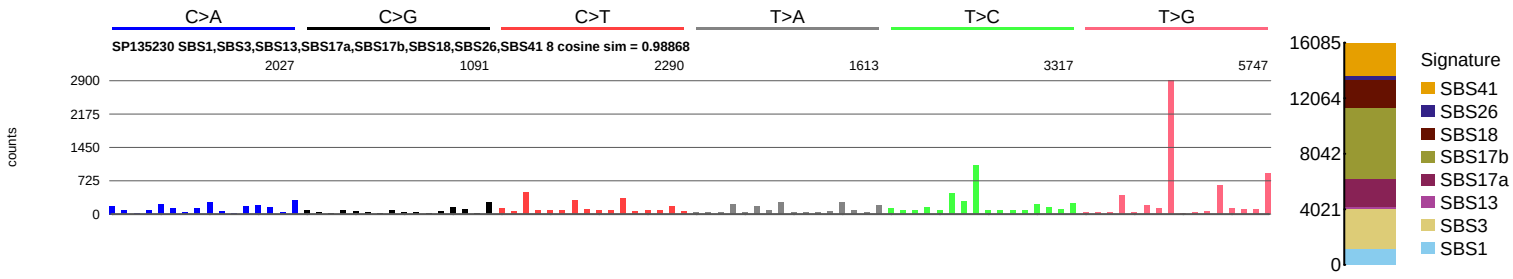
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.9887
2137
1008
3111
1221
3126
5484
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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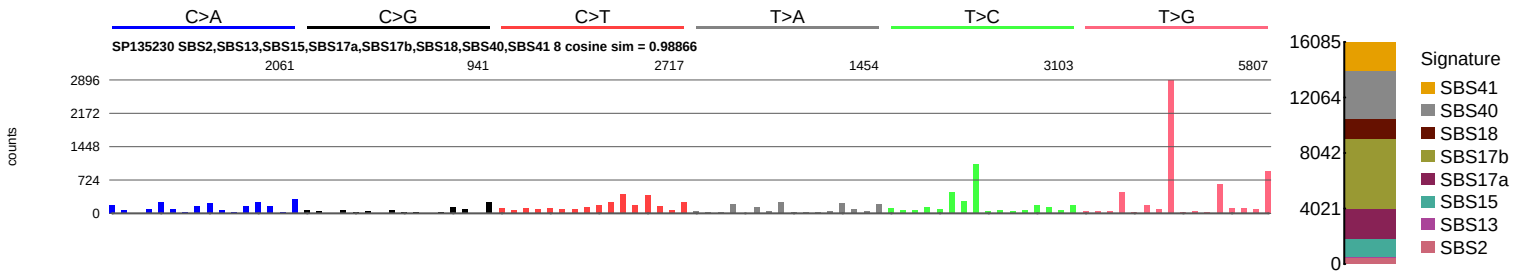
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98868
2027
1091
2290
1613
3317
5747
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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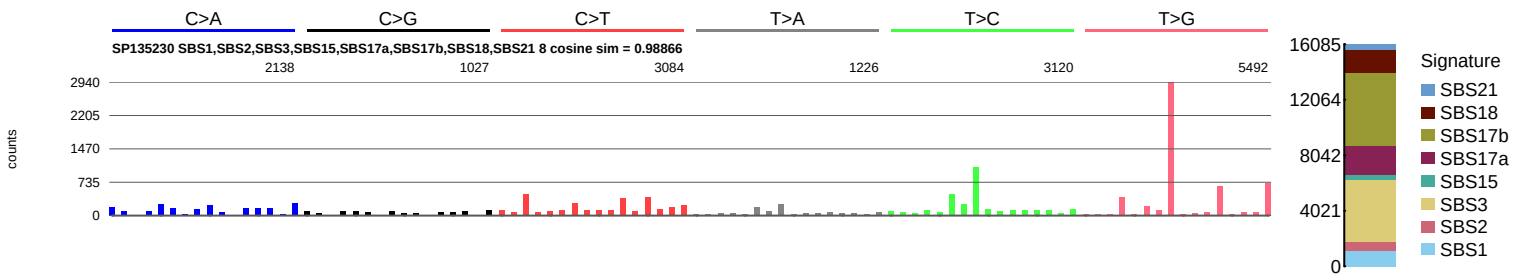
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98866
2061
941
2717
1454
3103
5807
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98866
2138
1027
3084
1226
3120
5492
counts
2940
2205
1470
735
0
16085
12064
8042
4021
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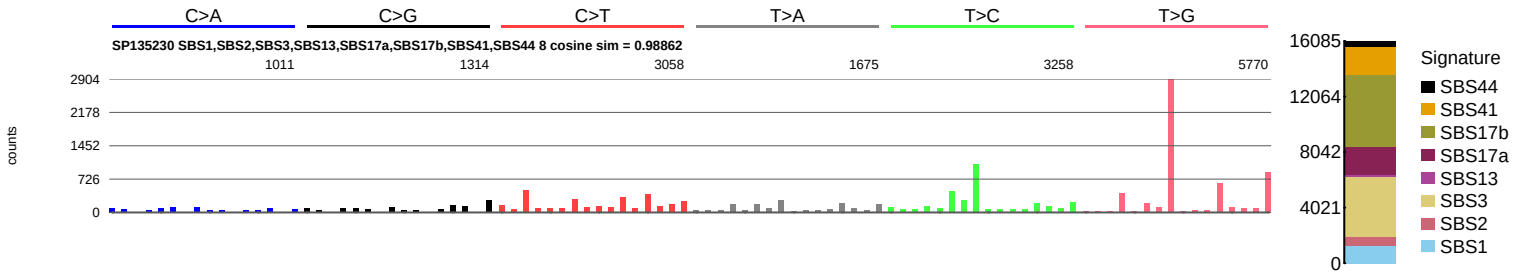
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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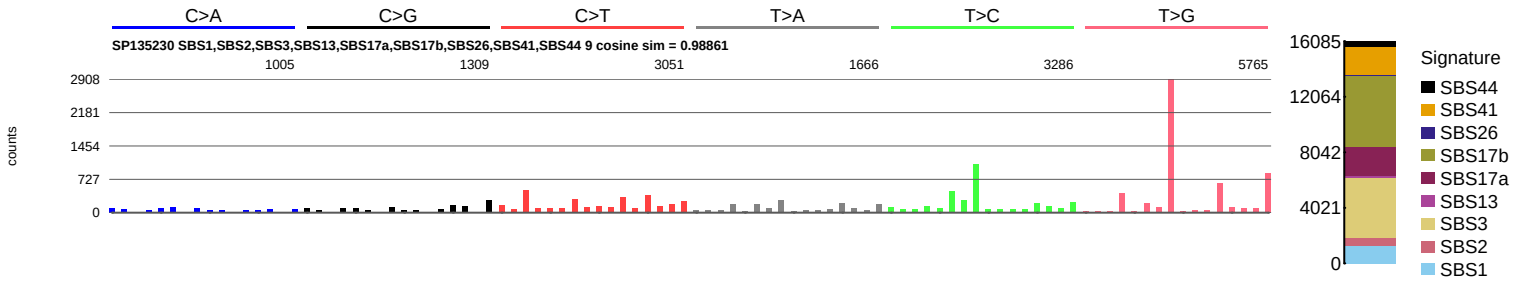
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98862
1011
1314
3058
1675
3258
5770
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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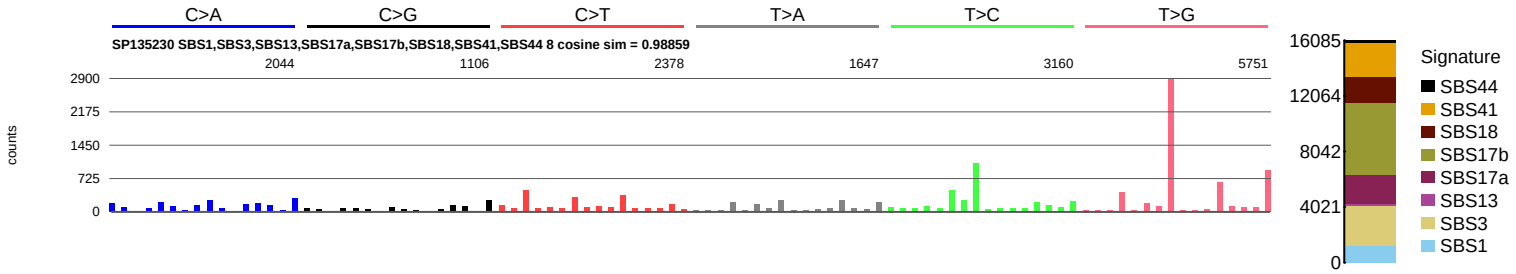
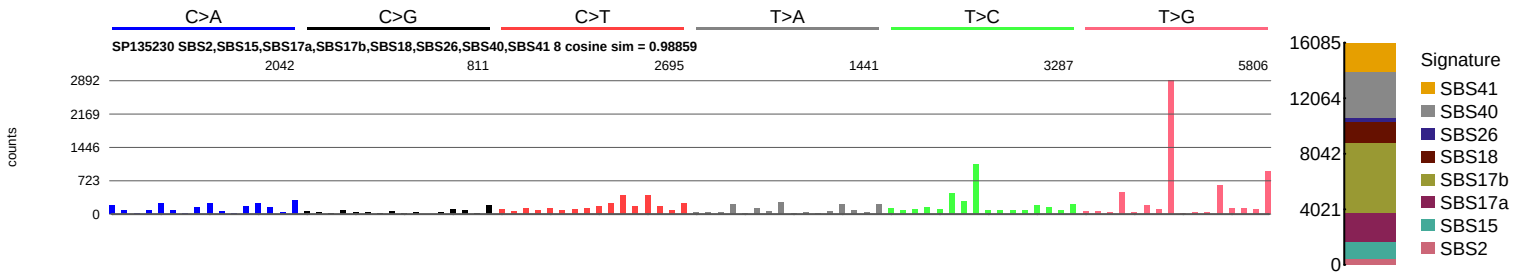
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98861
1005
1309
3051
1666
3286
5765
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98859
2044
1106
2378
1647
3160
5751
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98859
2042
811
2695
1441
3287
5806
counts
2892
2169
1446
723
0
16085
12064
8042
4021
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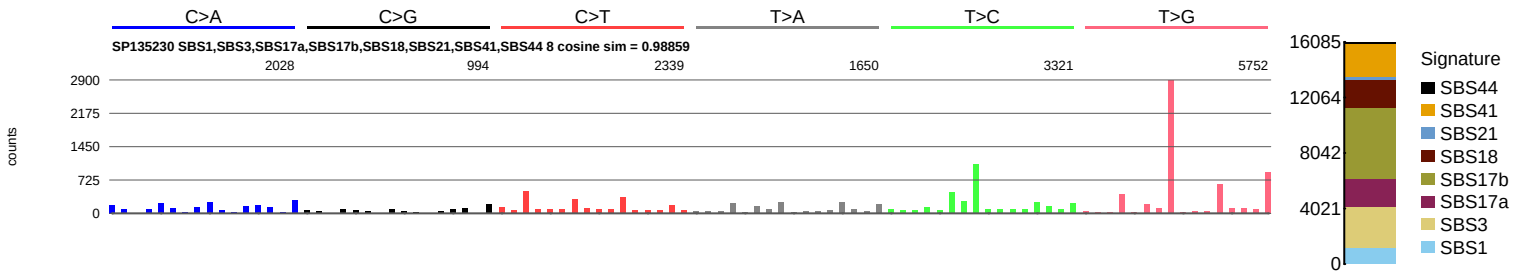
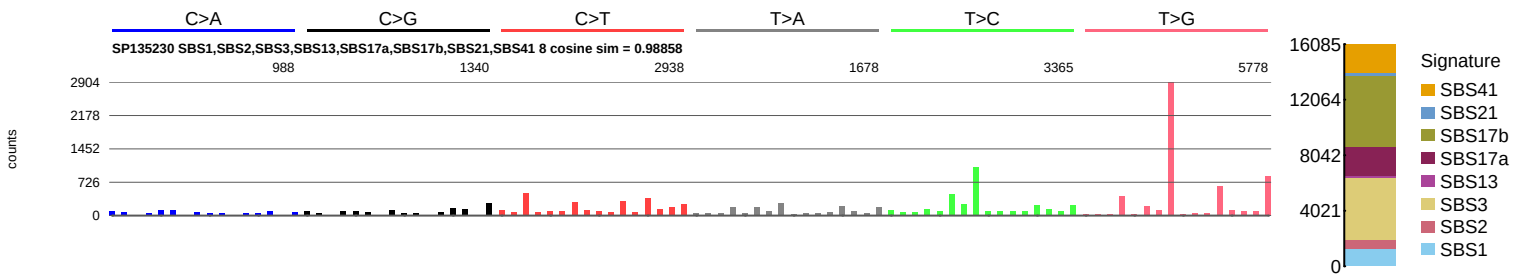
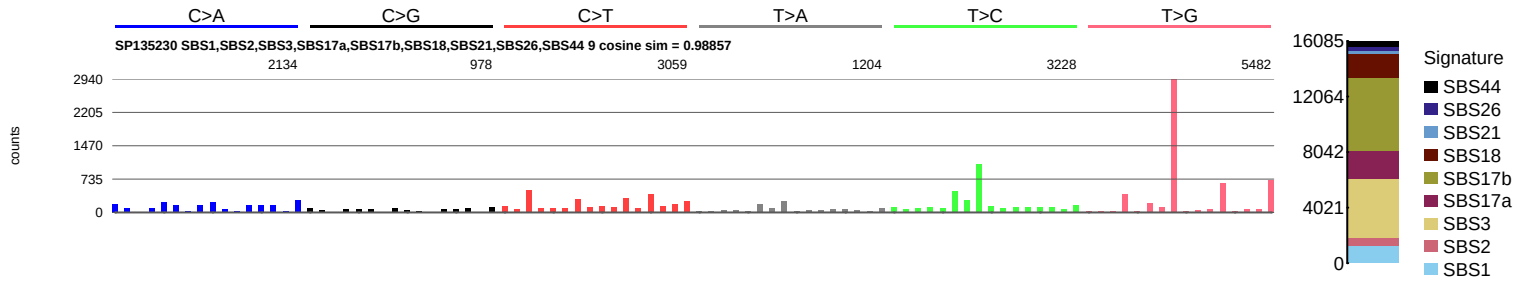
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98859
2028
994
2339
1650
3321
5752
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98858
988
1340
2938
1678
3365
5778
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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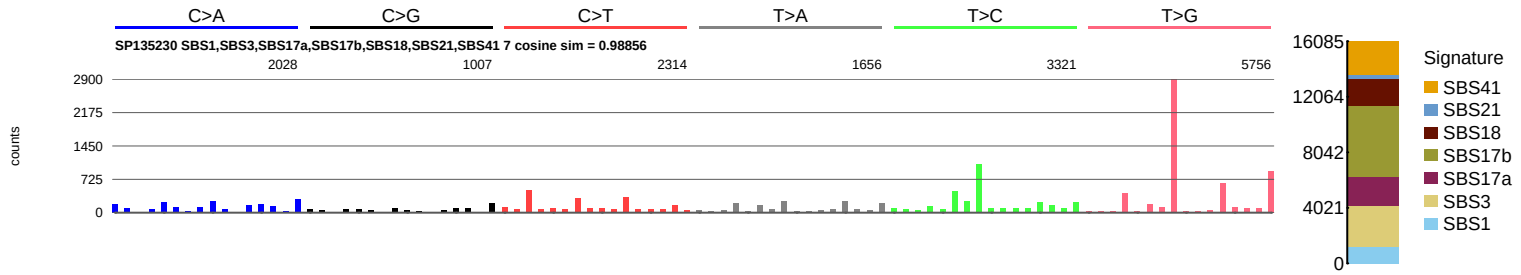
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98857
2134
978
3059
1204
3228
5482
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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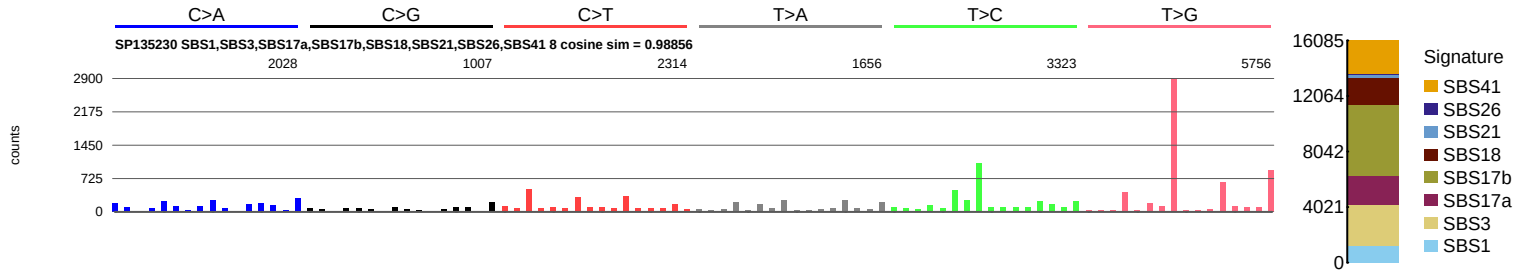
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98856
2028
1007
2314
1656
3321
5756
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>G
C>T
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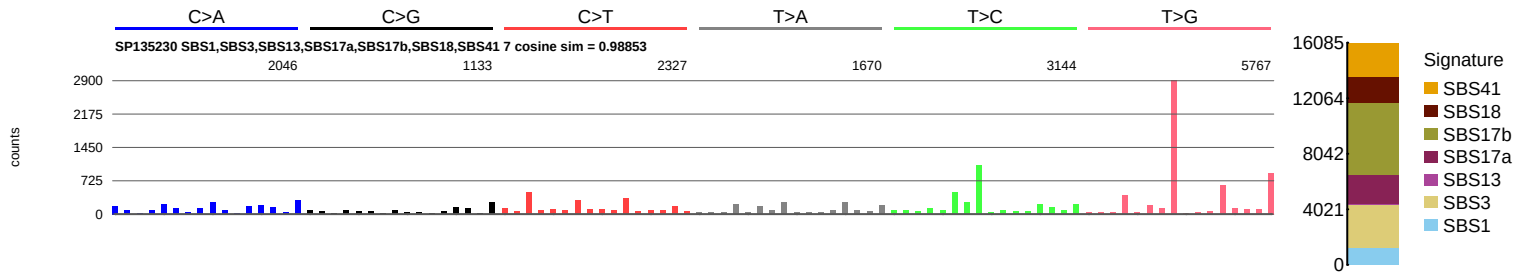
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98856
2028
1007
2314
1656
3323
5756
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>G
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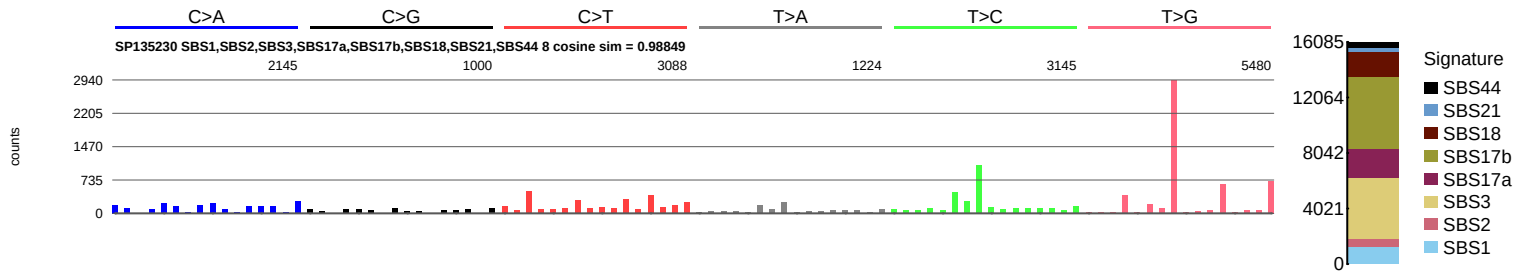
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98853
2046
1133
2327
1670
3144
5767
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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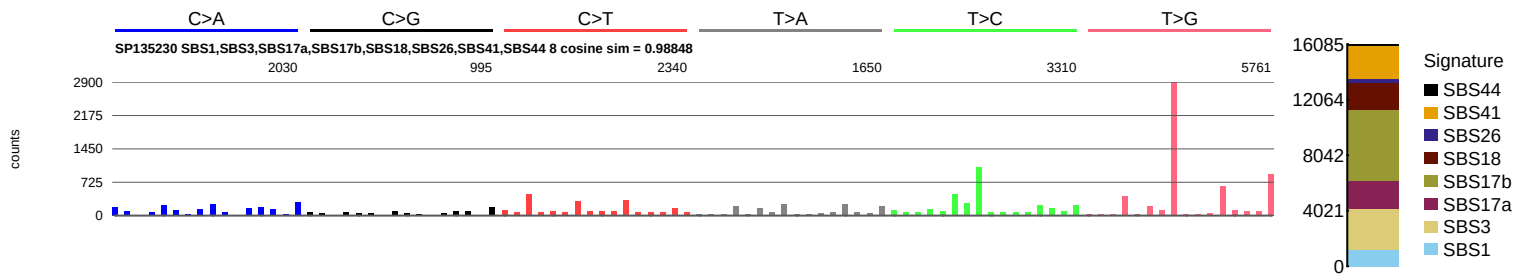
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98849
2145
1000
3088
1224
3145
5480
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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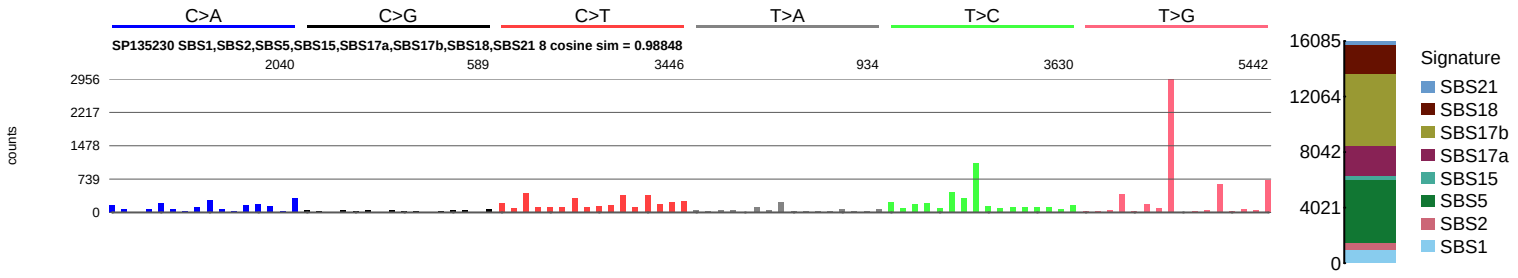
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98848
2030
995
2340
1650
3310
5761
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1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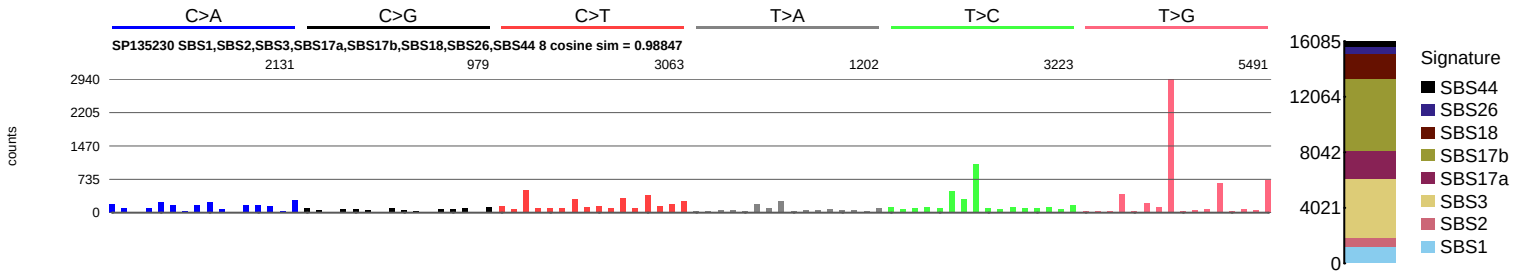
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98848
2040
589
3446
934
3630
5442
counts
2956
2217
1478
739
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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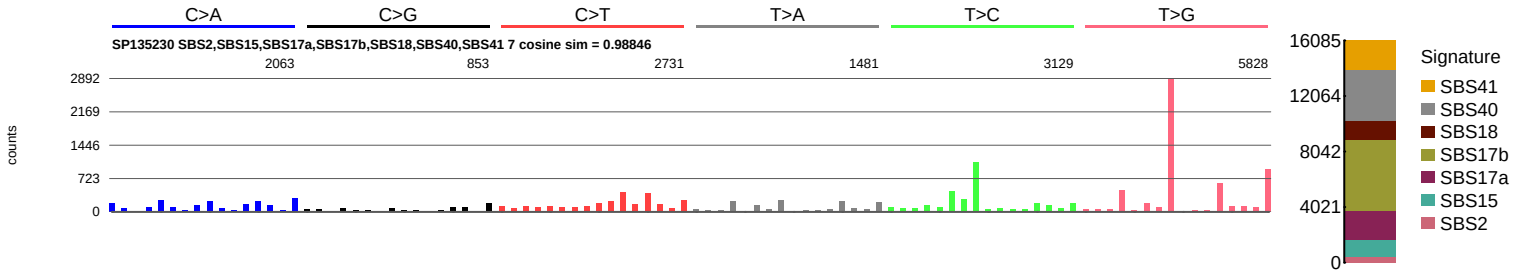
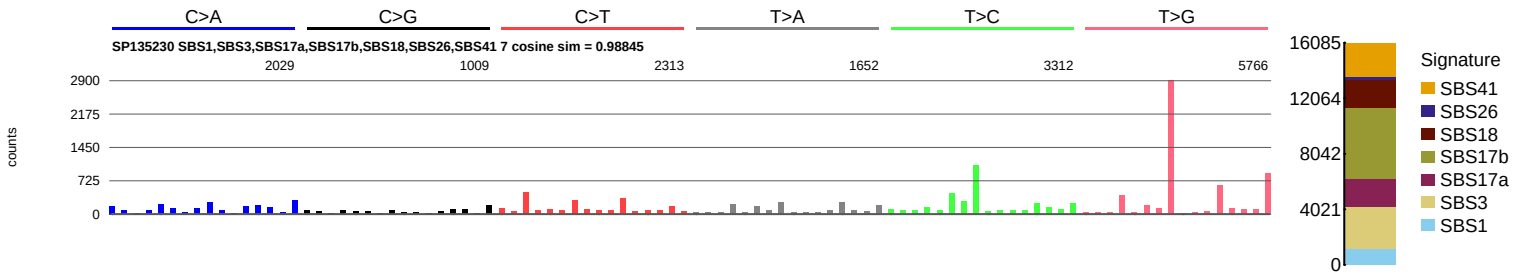
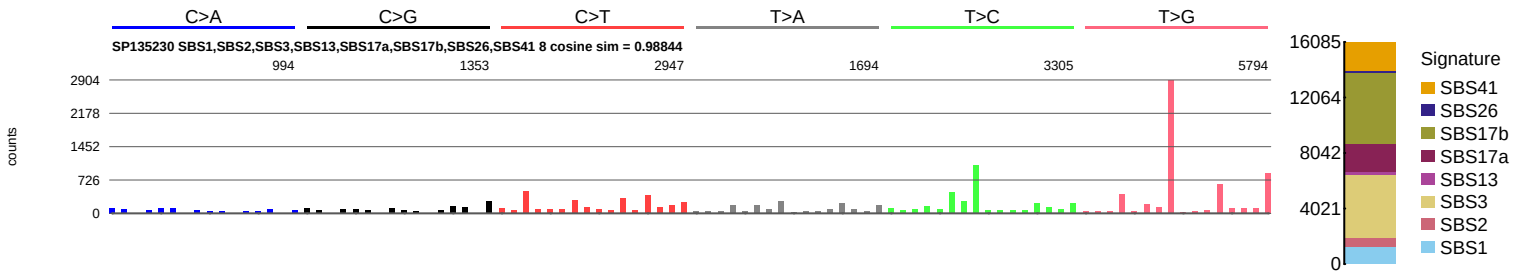
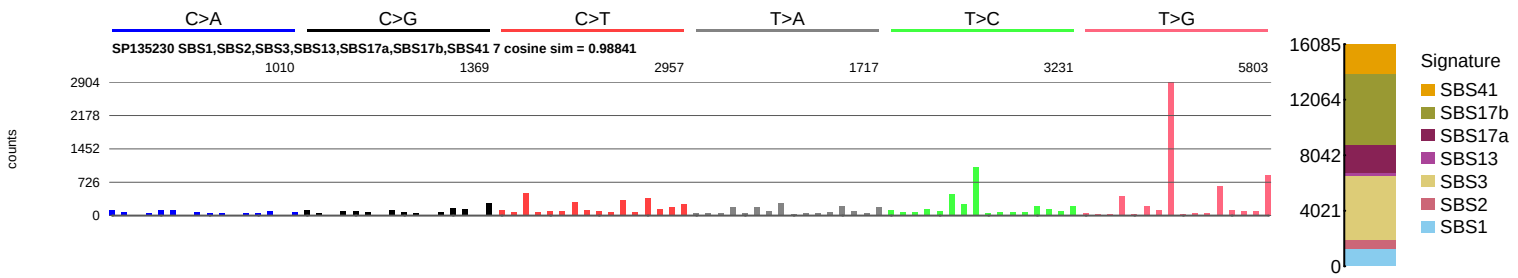
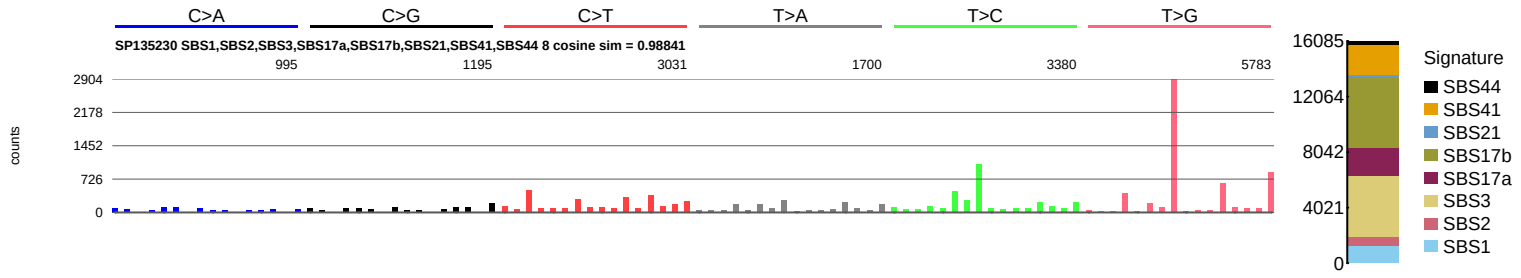
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98847
2131
979
3063
1202
3223
5491
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

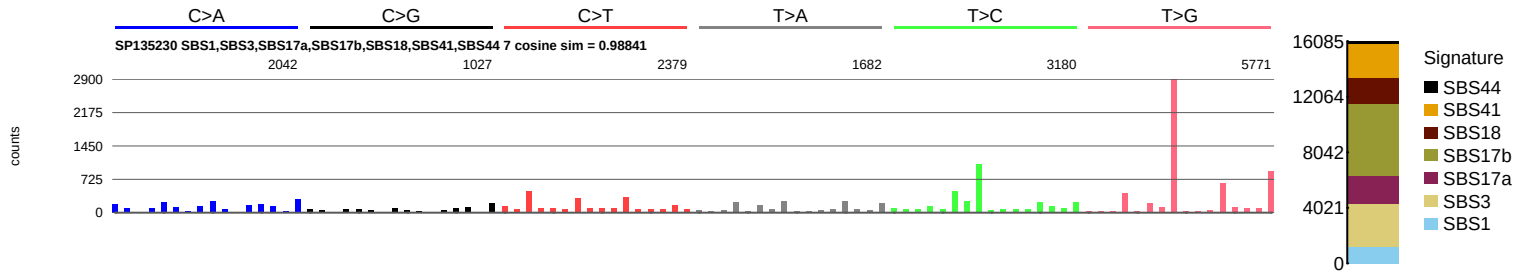

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98846
2063
853
2731
1481
3129
5828
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2

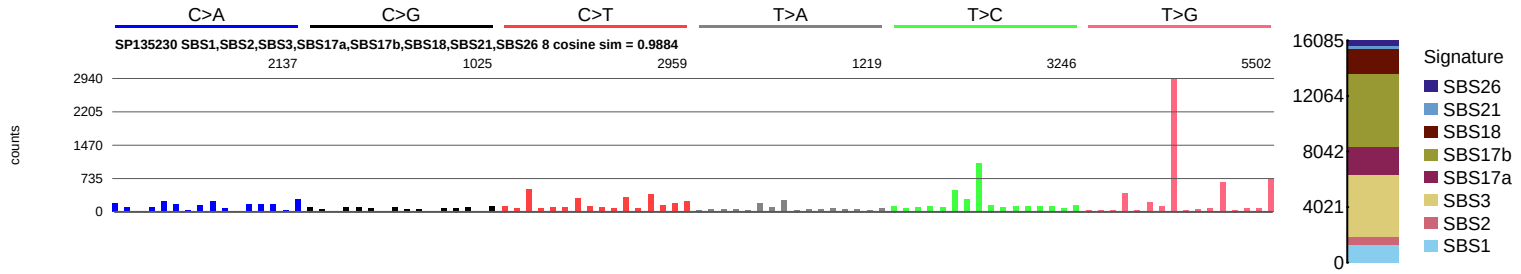

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98845
2029
1009
2313
1652
3312
5766
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98844
994
1353
2947
1694
3305
5794
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98841
1010
1369
2957
1717
3231
5803
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98841
995
1195
3031
1700
3380
5783
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98841
2042
1027
2379
1682
3180
5771
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1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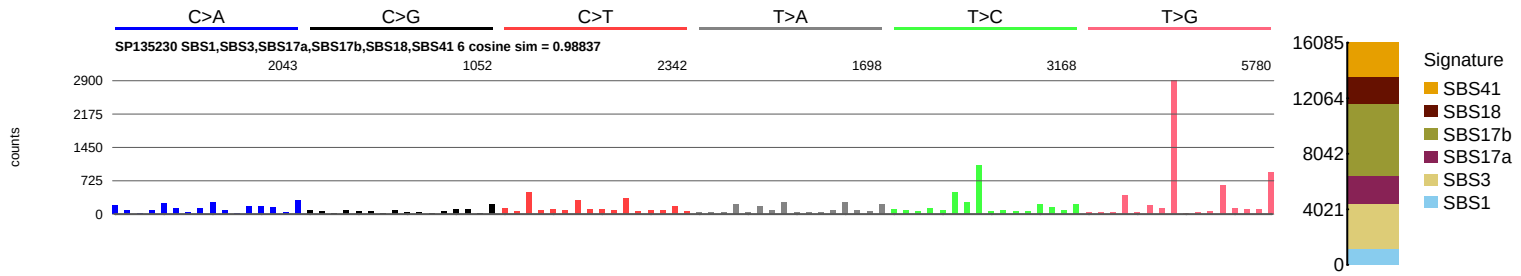
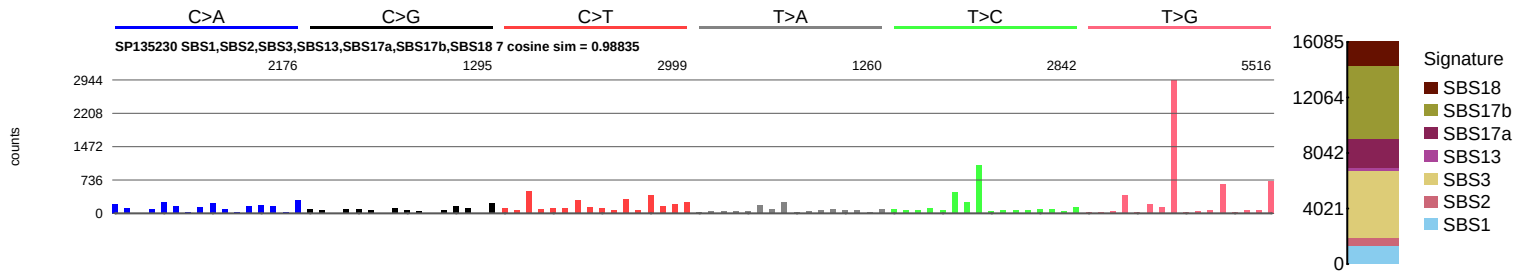
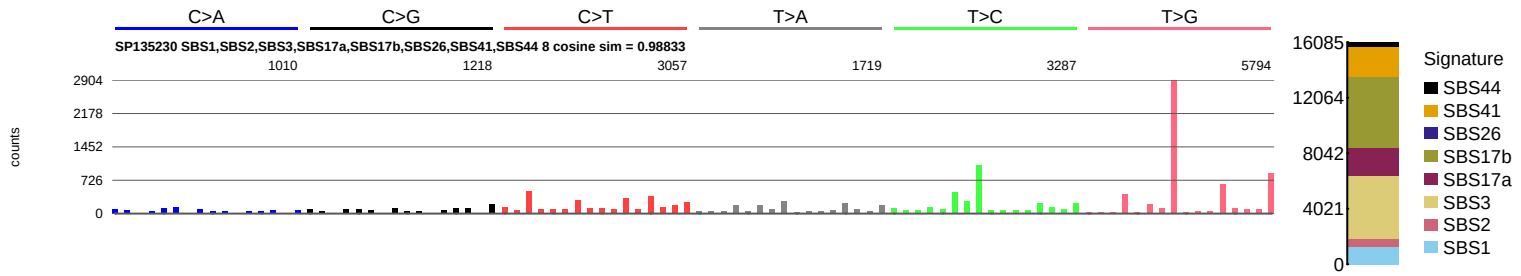
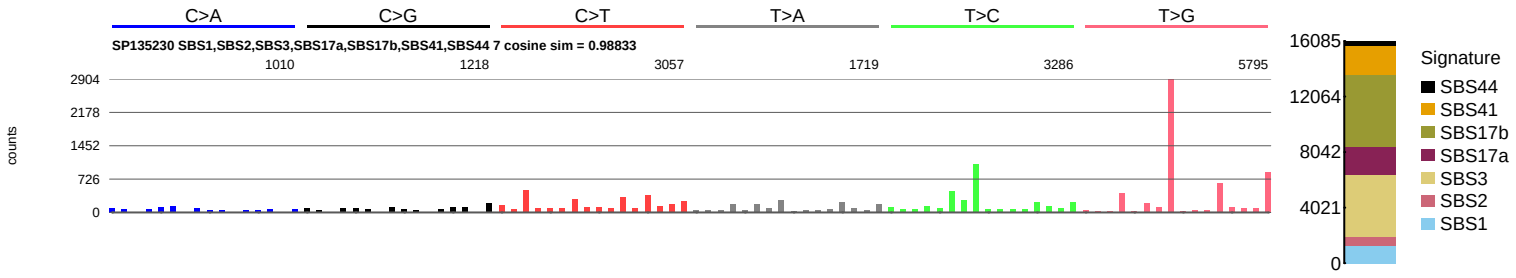
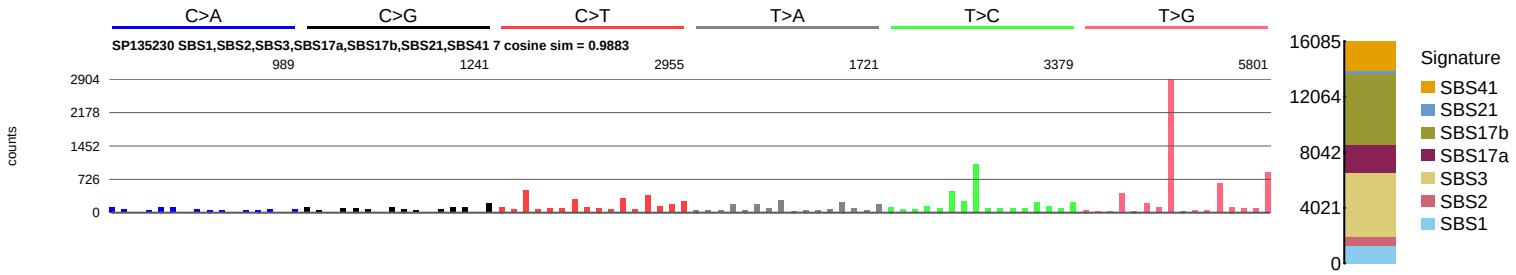
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.9884
2137
1025
2959
1219
3246
5502
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

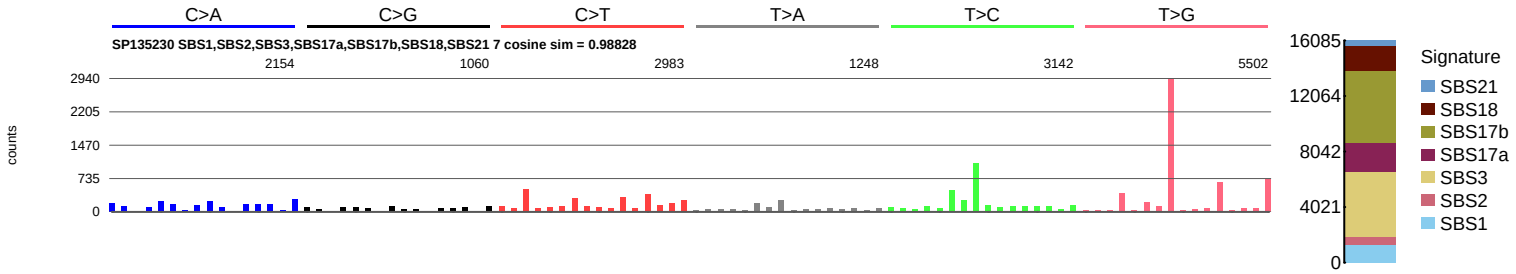

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98837
2043
1052
2342
1698
3168
5780
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1

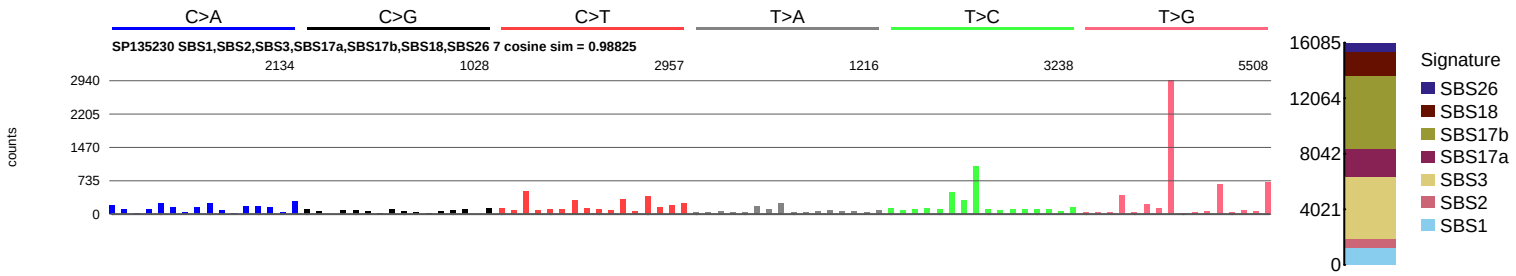

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98835
2176
1295
2999
1260
2842
5516
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

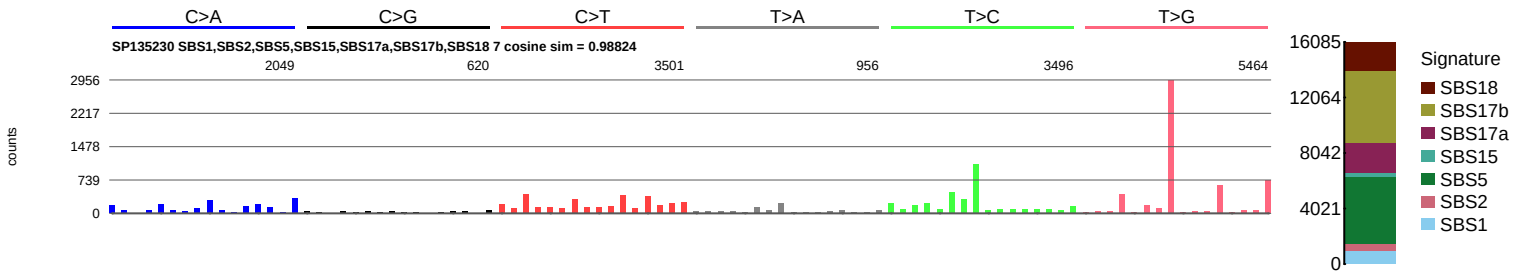

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98833
1010
1218
3057
1719
3287
5794
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

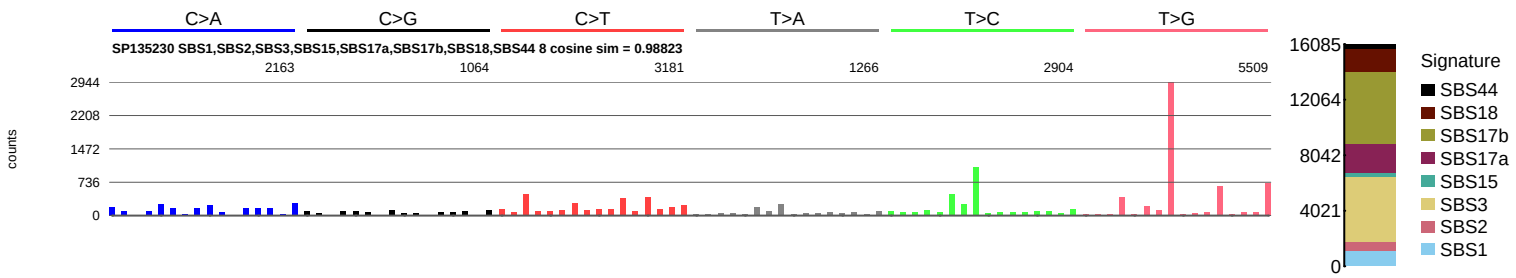

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98833
1010
1218
3057
1719
3286
5795
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1

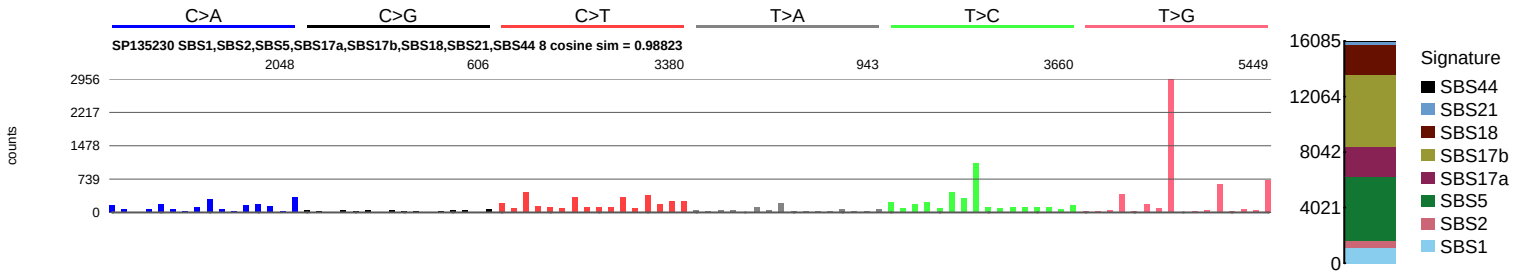

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.9883
989
1241
2955
1721
3379
5801
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

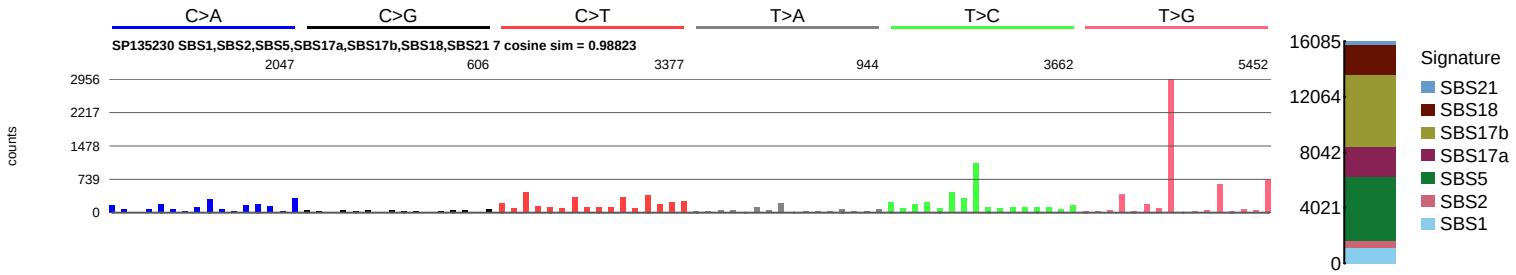

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98828
2154
1060
2983
1248
3142
5502
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98825
2134
1028
2957
1216
3238
5508
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98824
2049
620
3501
956
3496
5464
counts
2956
2217
1478
739
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98823
2163
1064
3181
1266
2904
5509
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98823
2048
606
3380
943
3660
5449
counts
2956
2217
1478
739
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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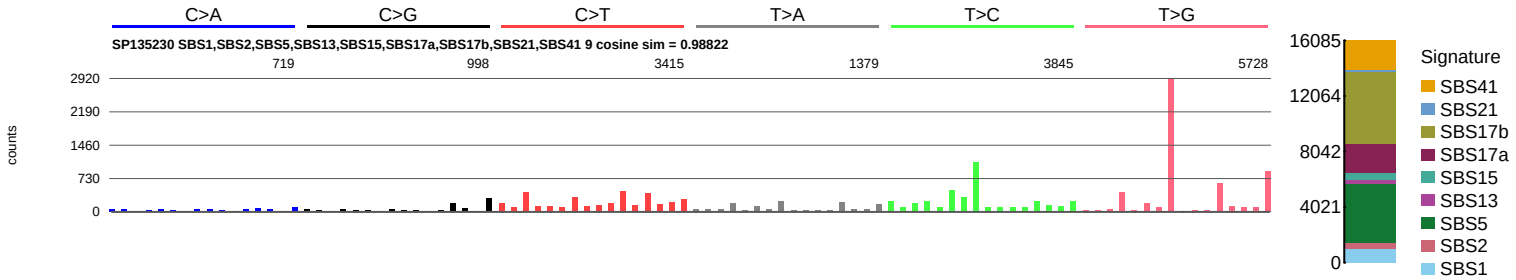
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98823
2047
606
3377
944
3662
5452
counts
2956
2217
1478
739
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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T>A
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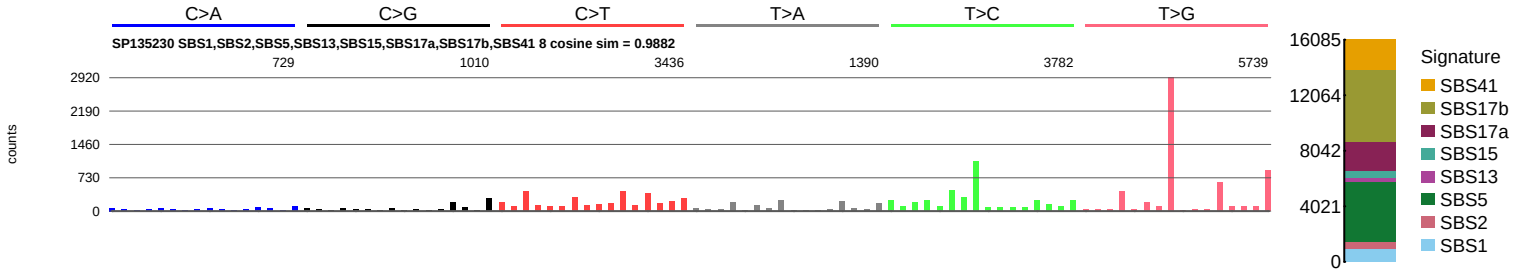
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98822
719
998
3415
1379
3845
5728
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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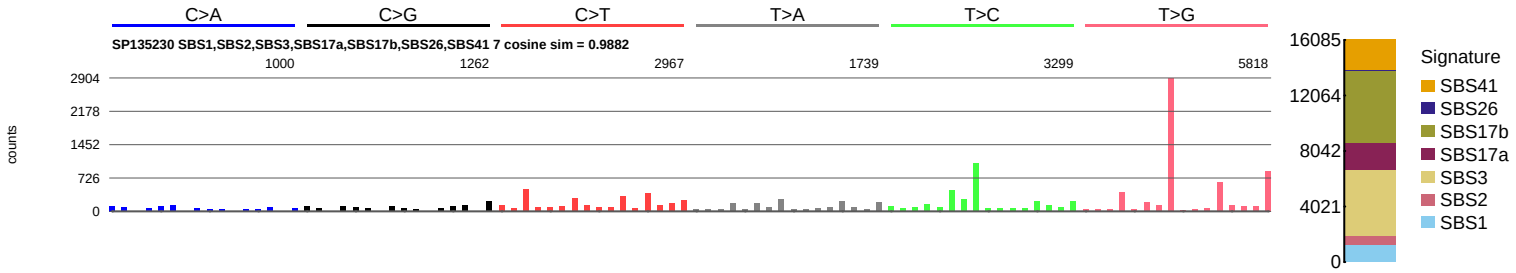
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.9882
729
1010
3436
1390
3782
5739
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
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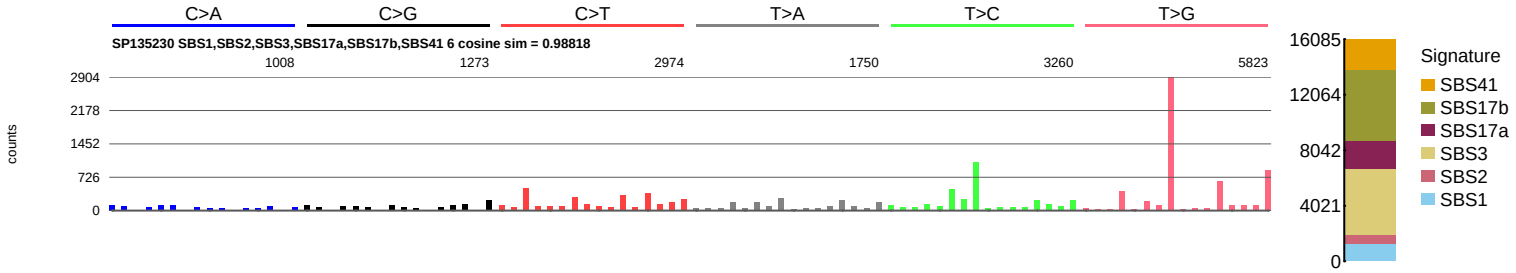
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.9882
1000
1262
2967
1739
3299
5818
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
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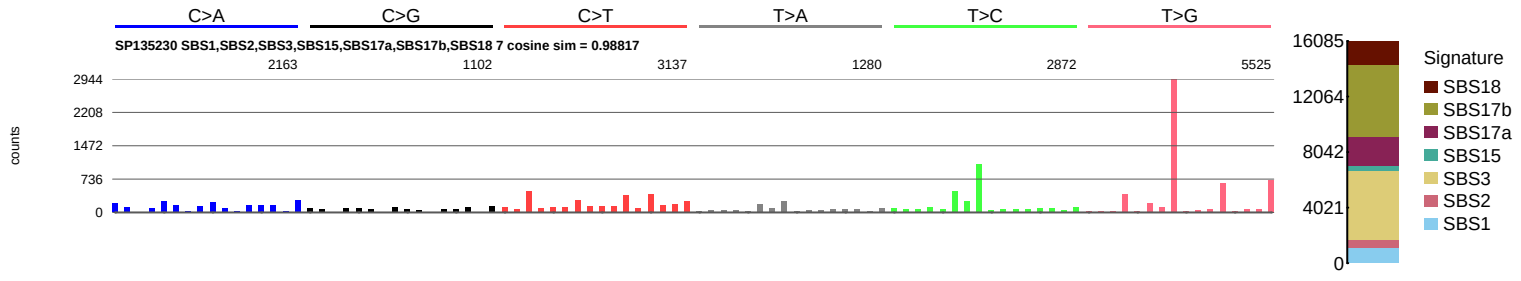
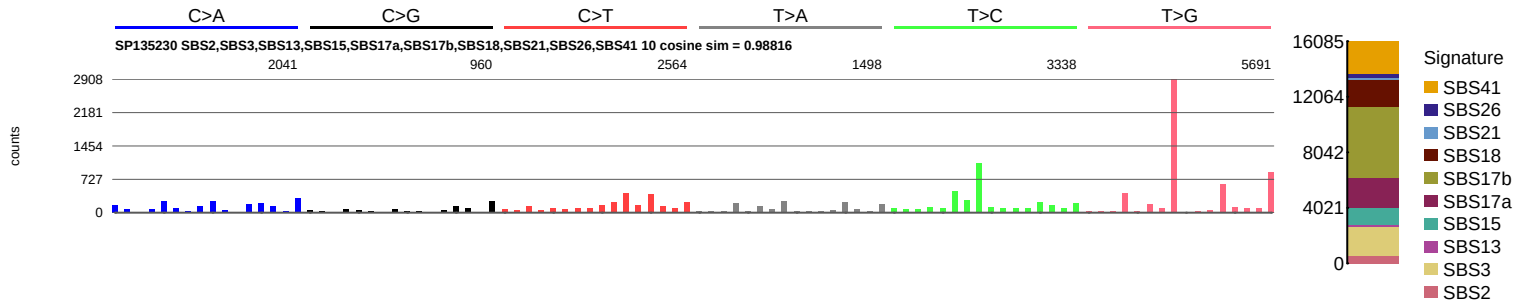
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98818
1008
1273
2974
1750
3260
5823
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98817
2163
1102
3137
1280
2872
5525
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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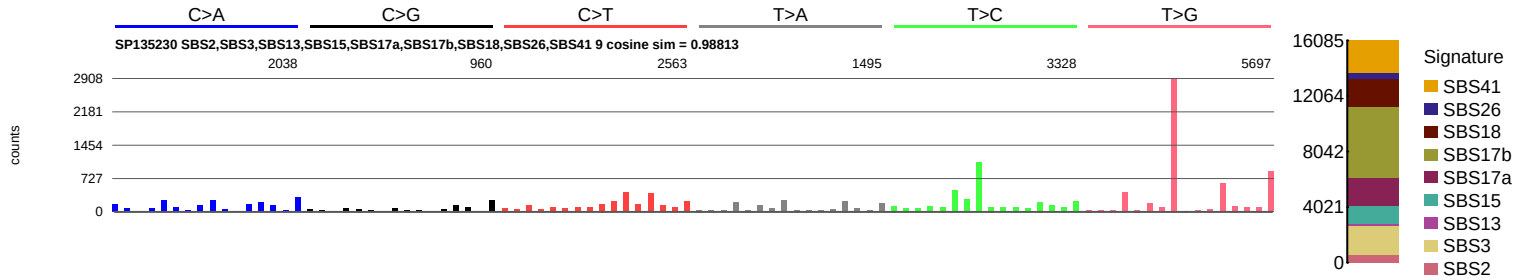
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98816
2041
960
2564
1498
3338
5691
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2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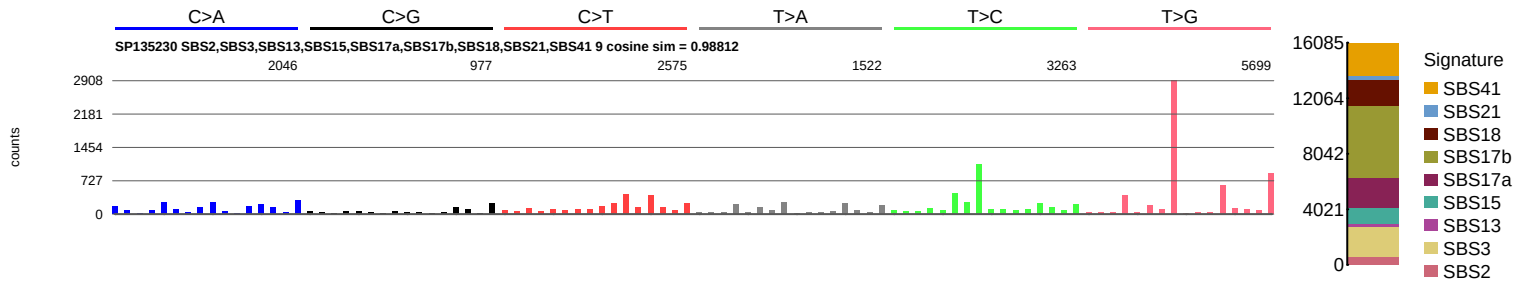
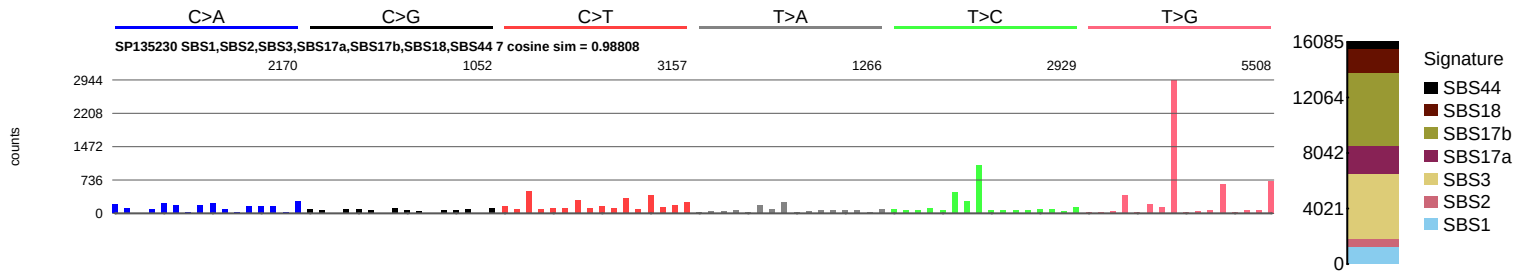
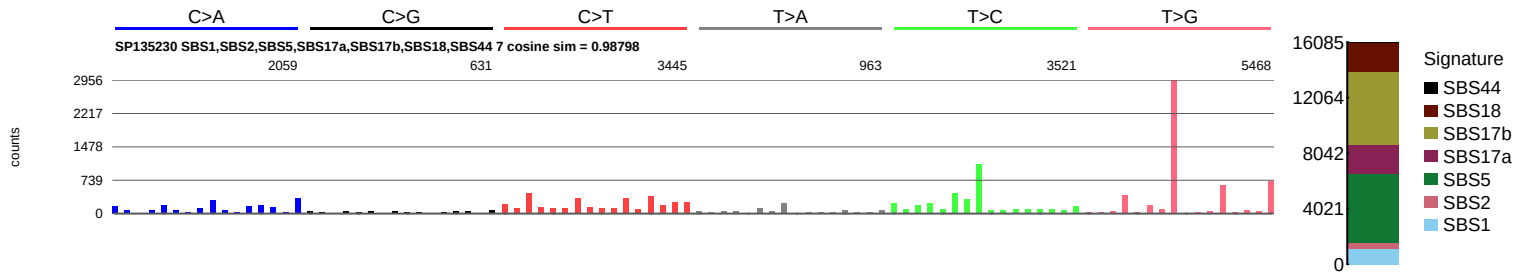
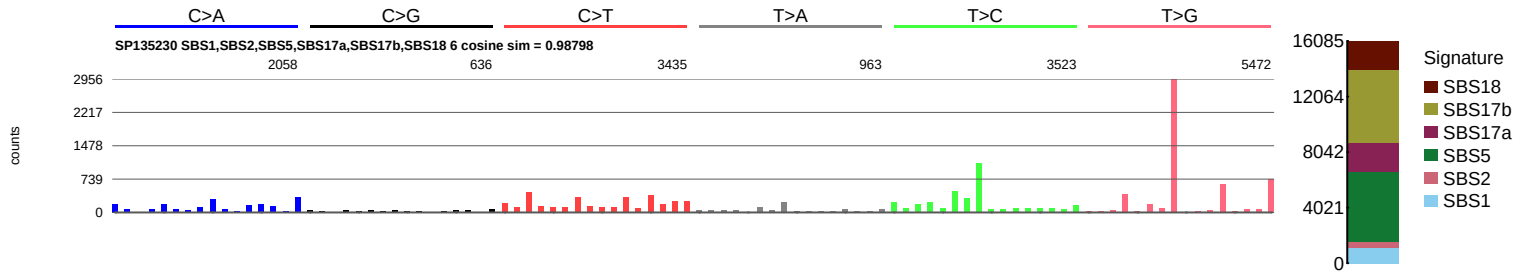
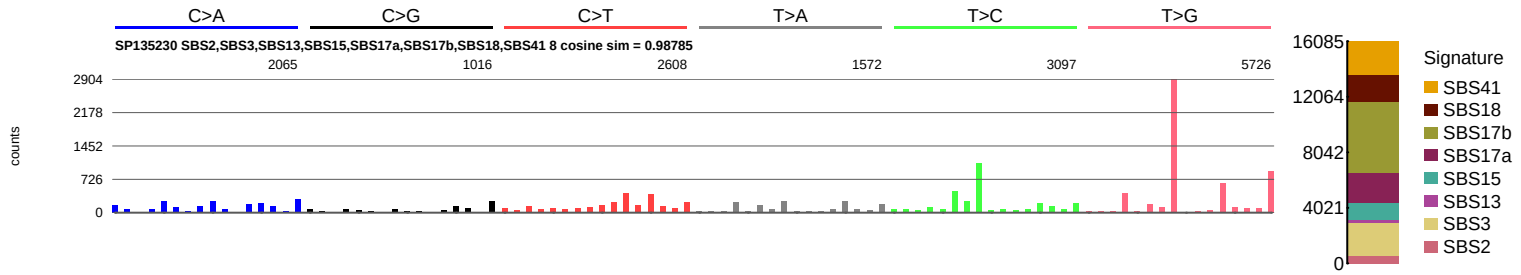
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98813
2038
960
2563
1495
3328
5697
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

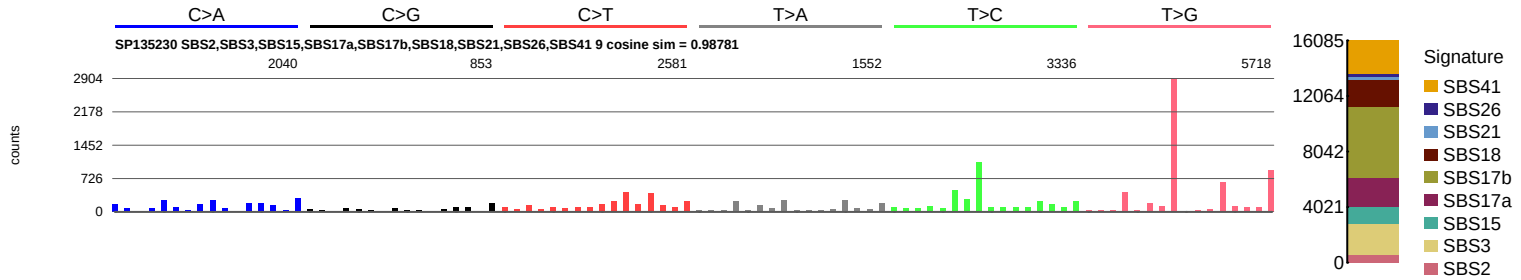

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98812
2046
977
2575
1522
3263
5699
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

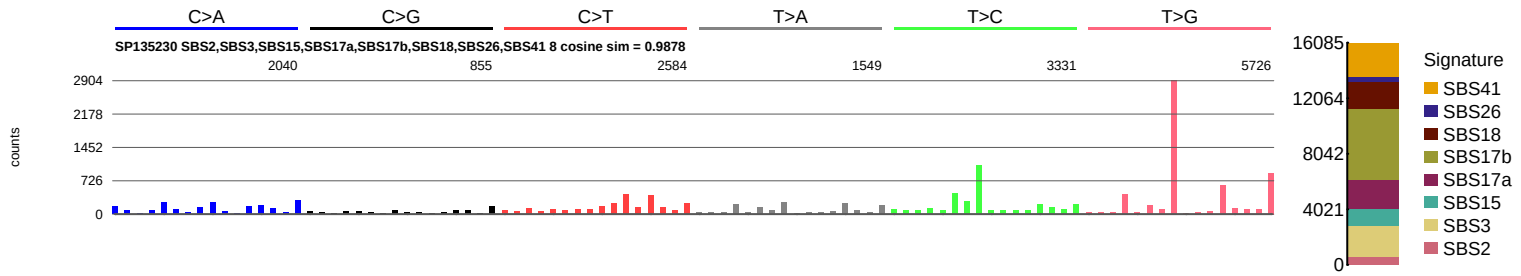

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98808
2170
1052
3157
1266
2929
5508
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

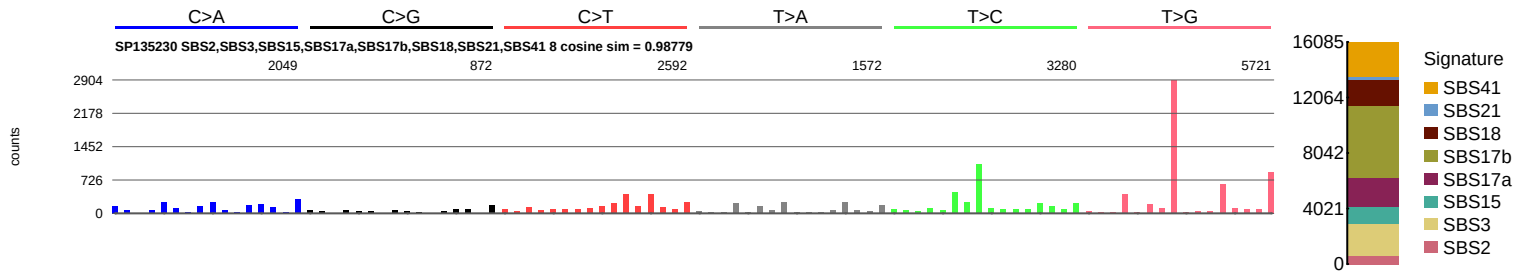

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98798
2059
631
3445
963
3521
5468
counts
2956
2217
1478
739
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98798
2058
636
3435
963
3523
5472
counts
2956
2217
1478
739
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98785
2065
1016
2608
1572
3097
5726
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98781
2040
853
2581
1552
3336
5718
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.9878
2040
855
2584
1549
3331
5726
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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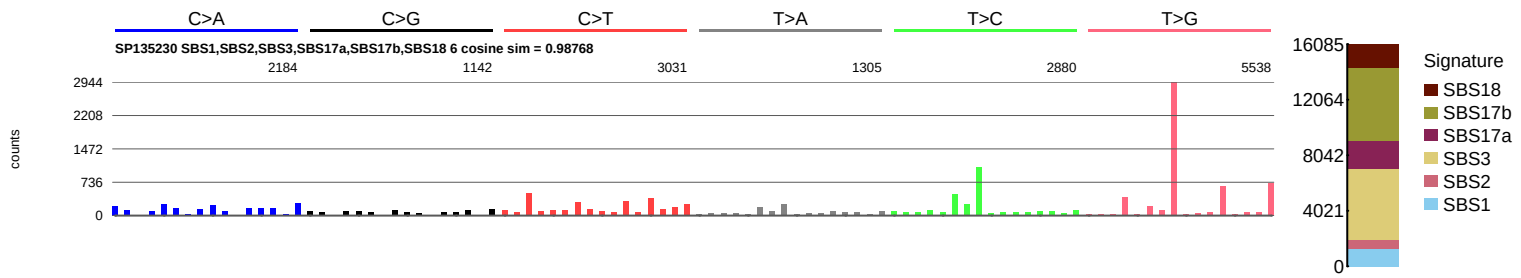
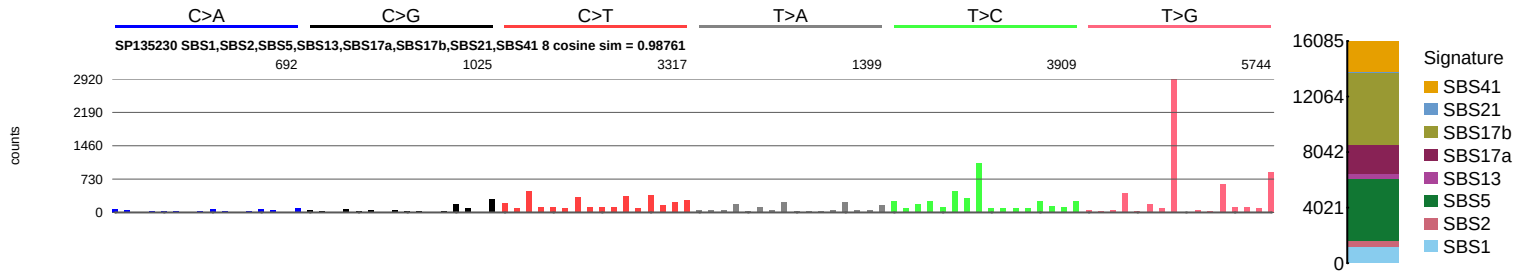
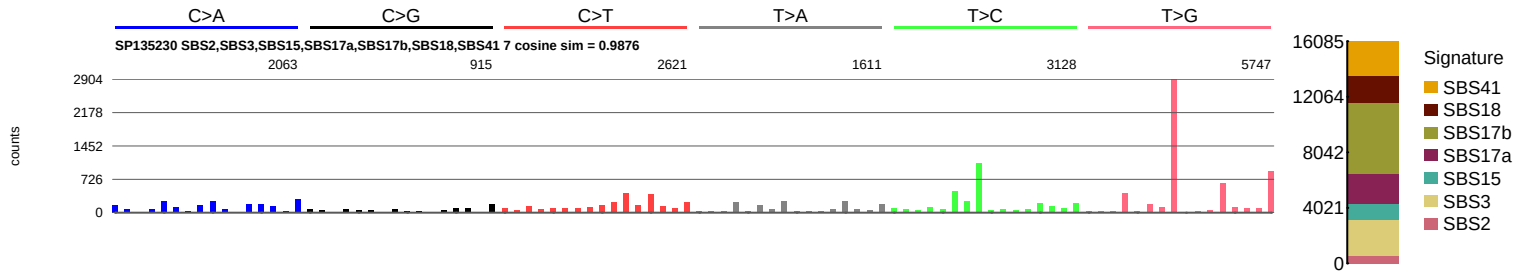
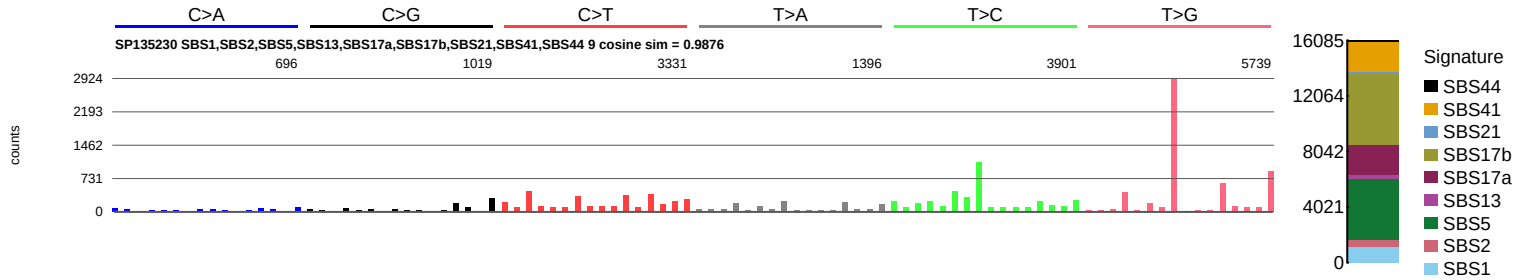
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98779
2049
872
2592
1572
3280
5721
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98768
2184
1142
3031
1305
2880
5538
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98761
692
1025
3317
1399
3909
5744
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9876
2063
915
2621
1611
3128
5747
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.9876
696
1019
3331
1396
3901
5739
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS44
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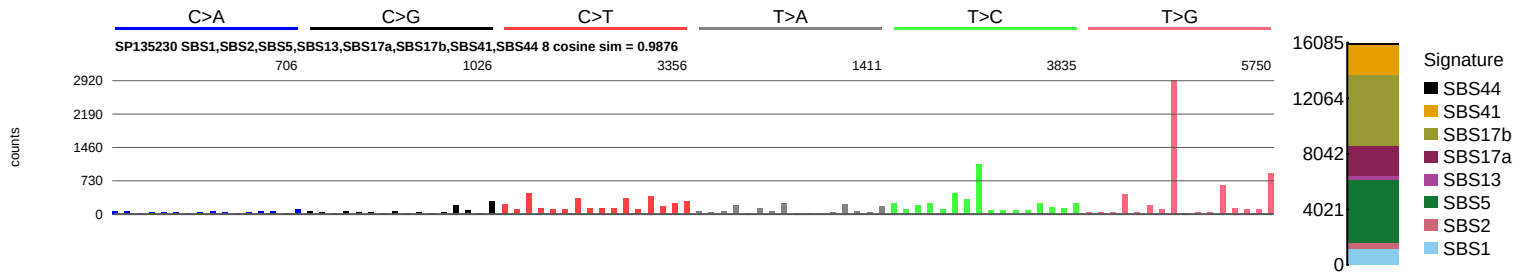
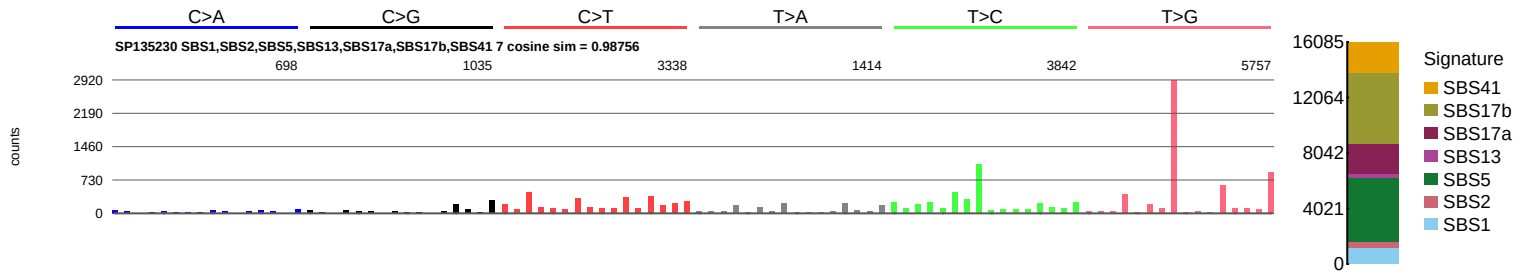
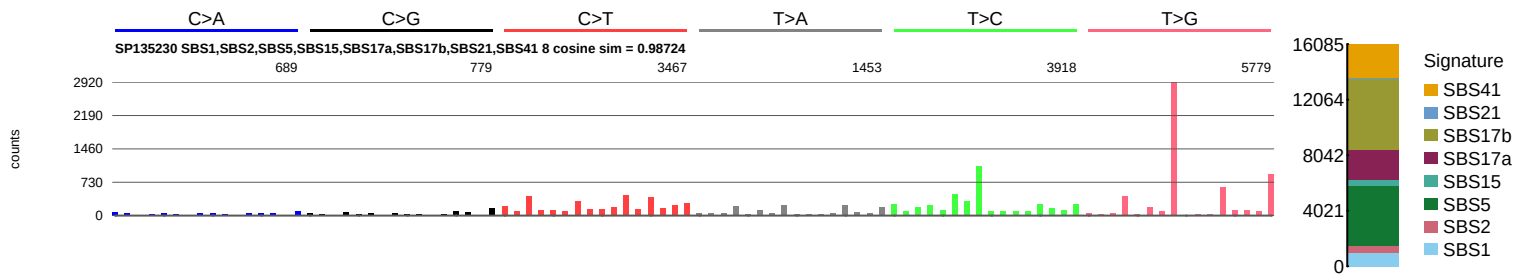
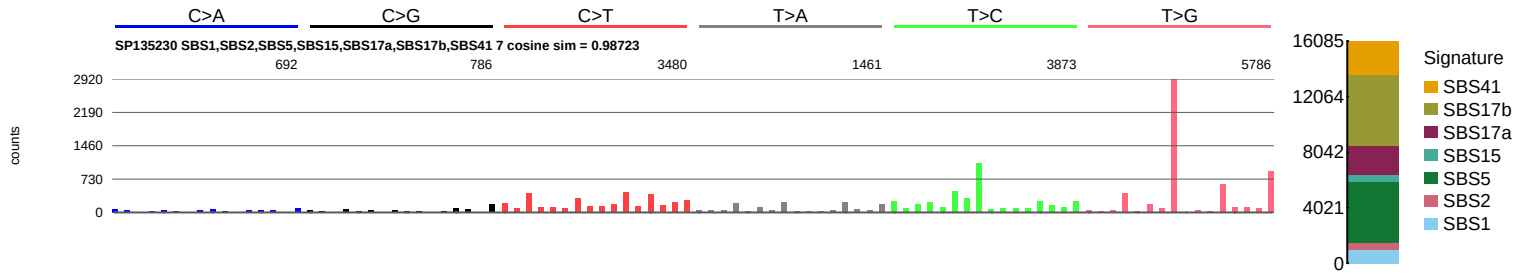
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.9876
706
1026
3356
1411
3835
5750
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98756
698
1035
3338
1414
3842
5757
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98724
689
779
3467
1453
3918
5779
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98723
692
786
3480
1461
3873
5786
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
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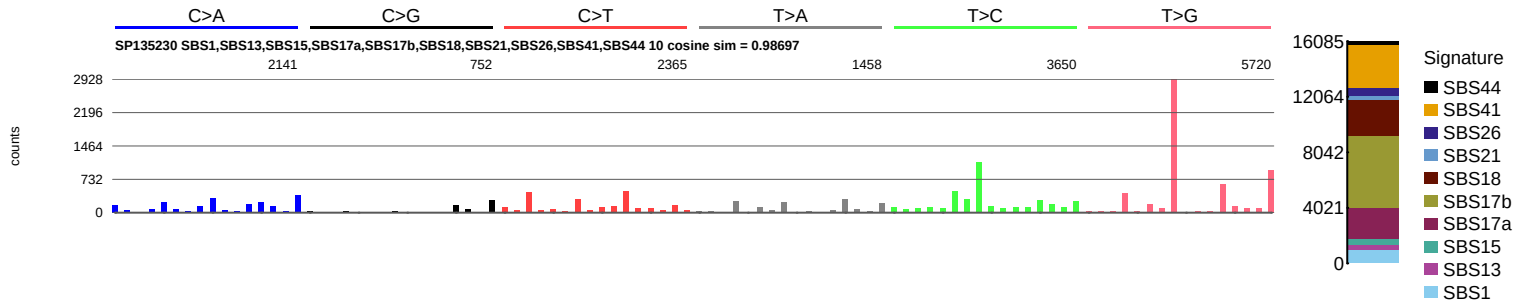
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98697
2141
752
2365
1458
3650
5720
counts
2928
2196
1464
732
0
16085
12064
8042
4021
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Signature
SBS44
SBS41
SBS26
SBS21
SBS18
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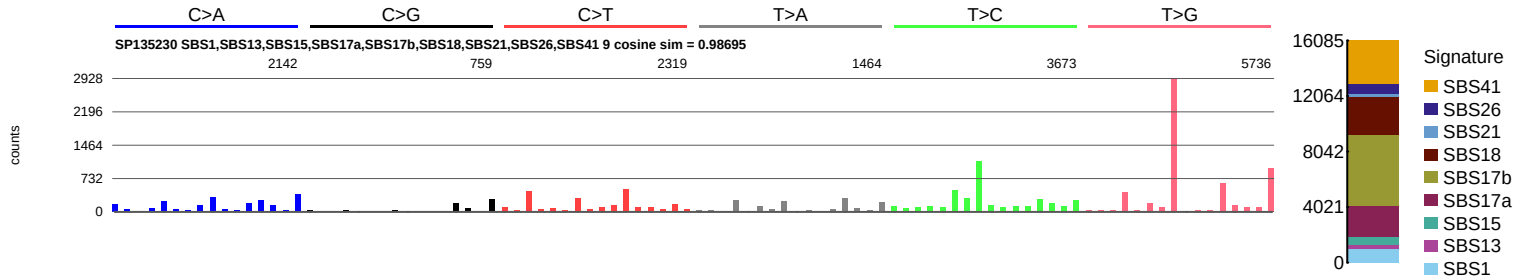
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
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T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98695
2142
759
2319
1464
3673
5736
counts
2928
2196
1464
732
0
16085
12064
8042
4021
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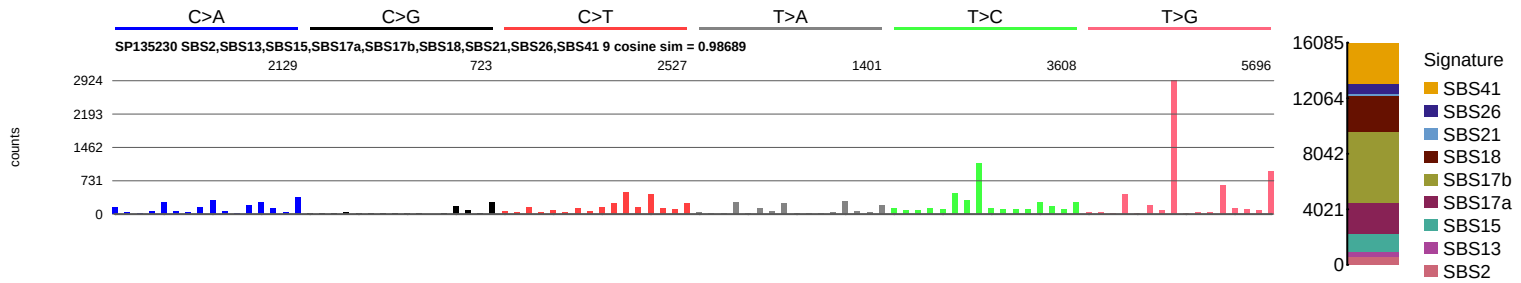
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
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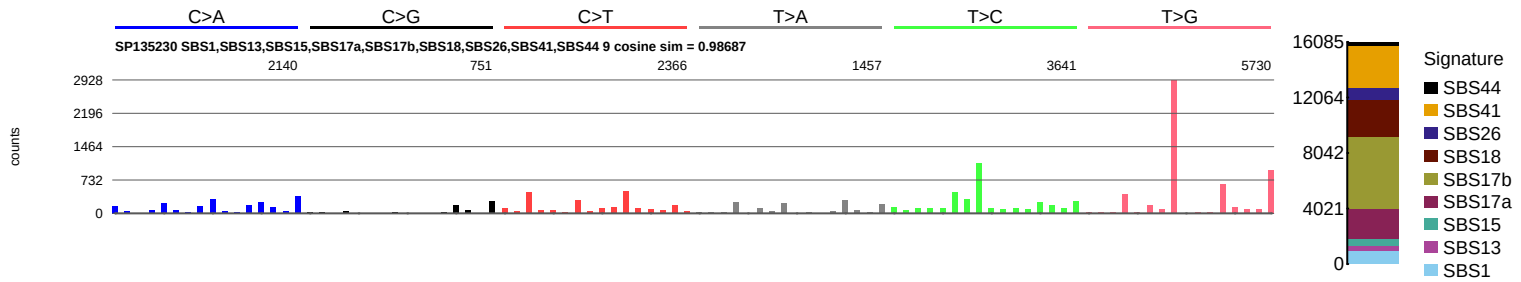
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98689
2129
723
2527
1401
3608
5696
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
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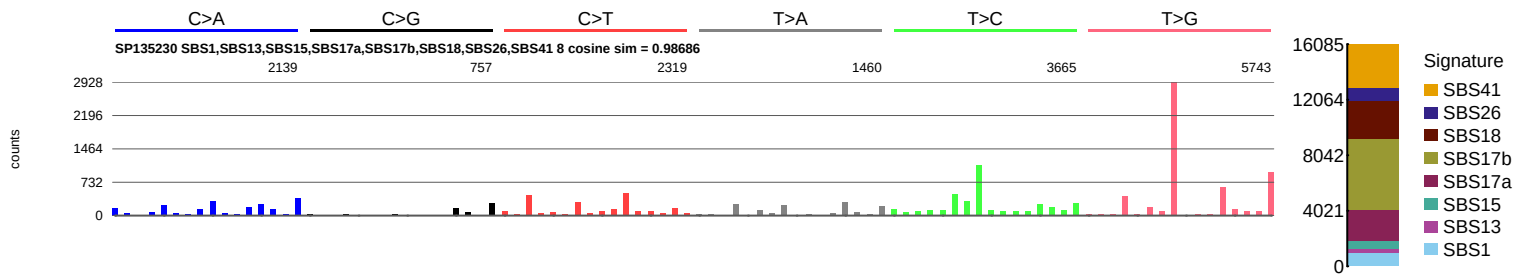
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98687
2140
751
2366
1457
3641
5730
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
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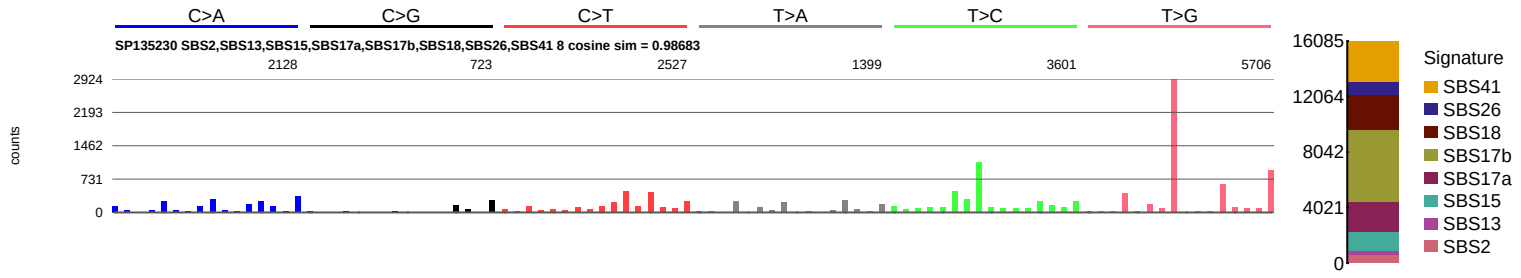
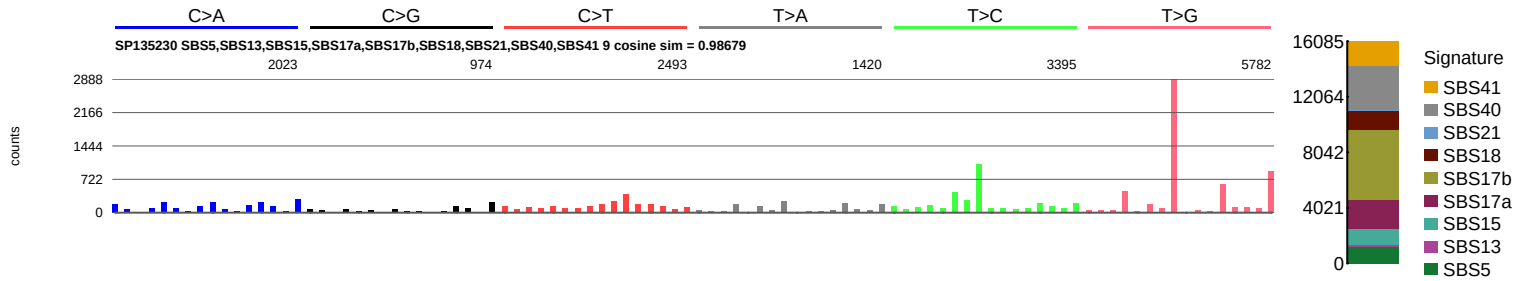
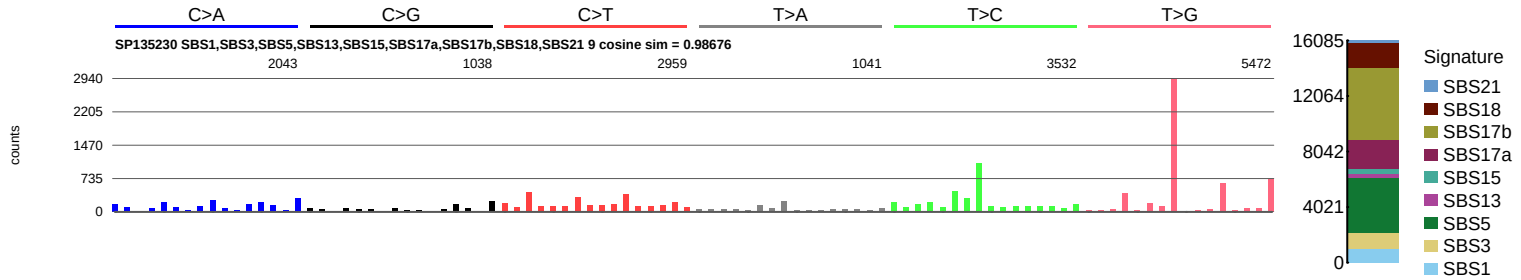
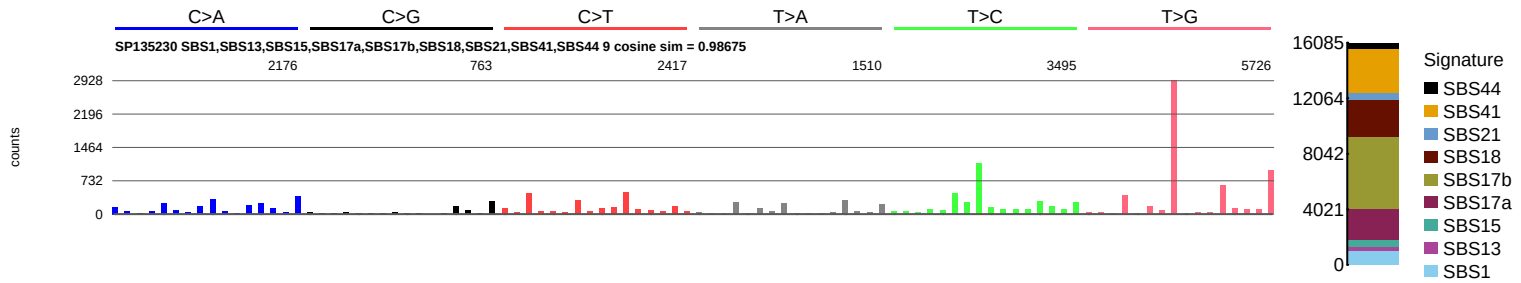
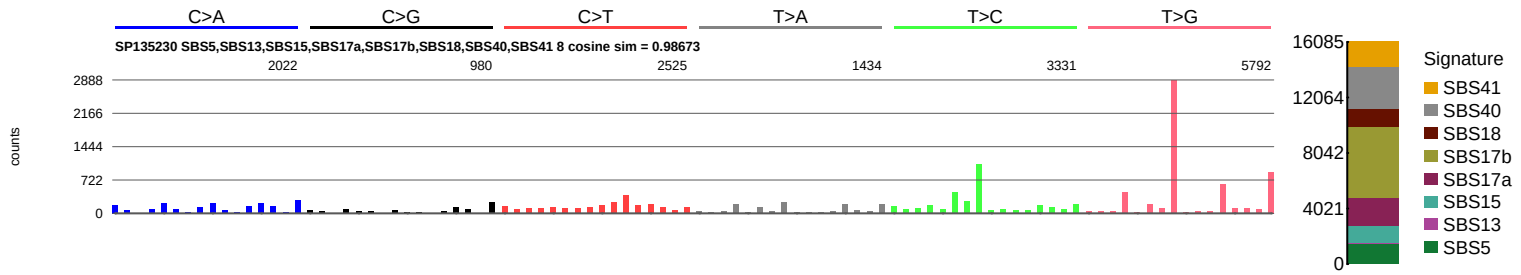
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98686
2139
757
2319
1460
3665
5743
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

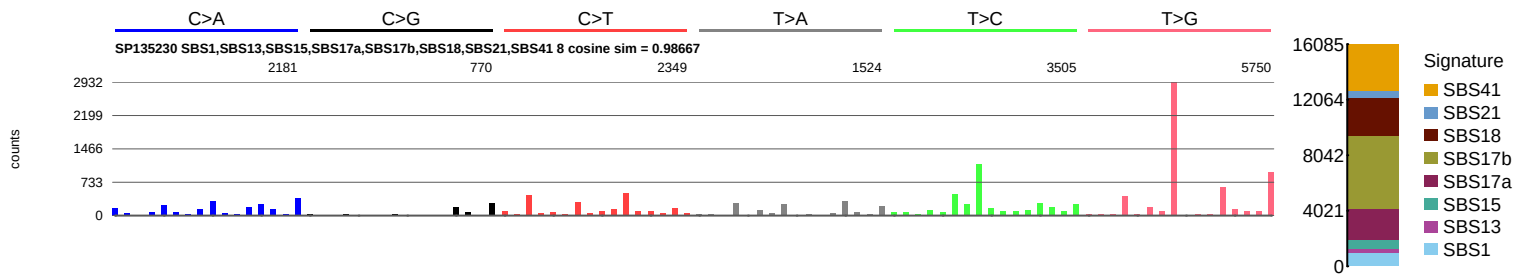

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98683
2128
723
2527
1399
3601
5706
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

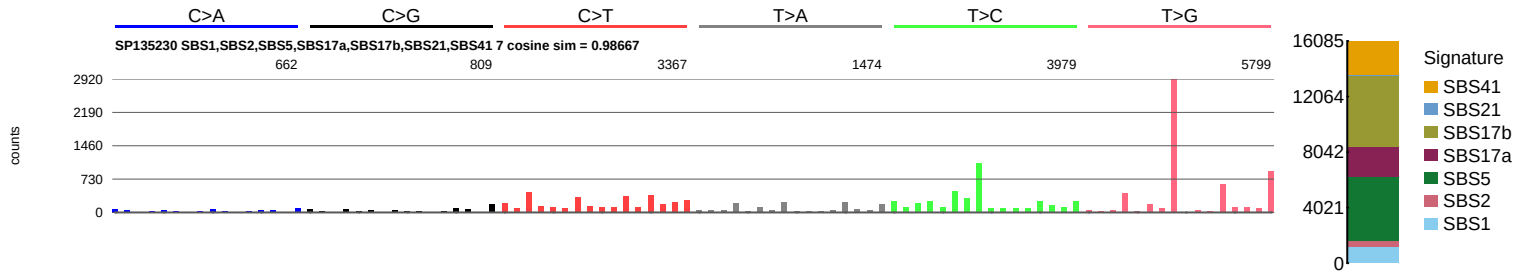

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98679
2023
974
2493
1420
3395
5782
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

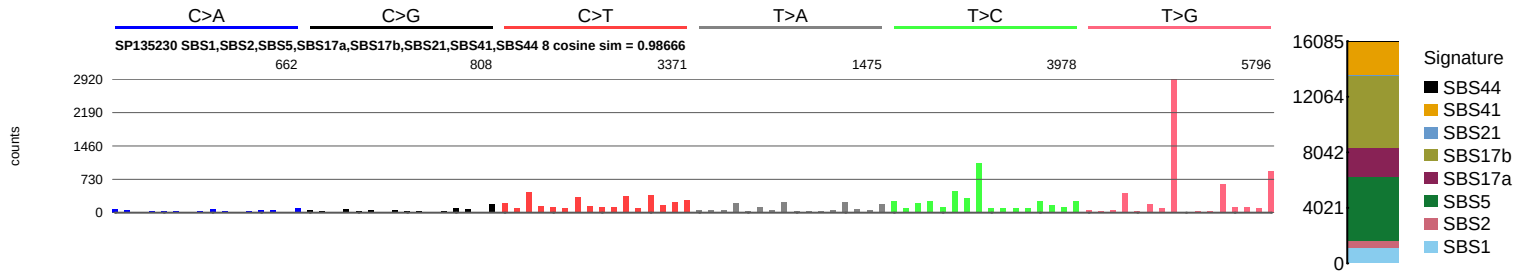

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98676
2043
1038
2959
1041
3532
5472
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

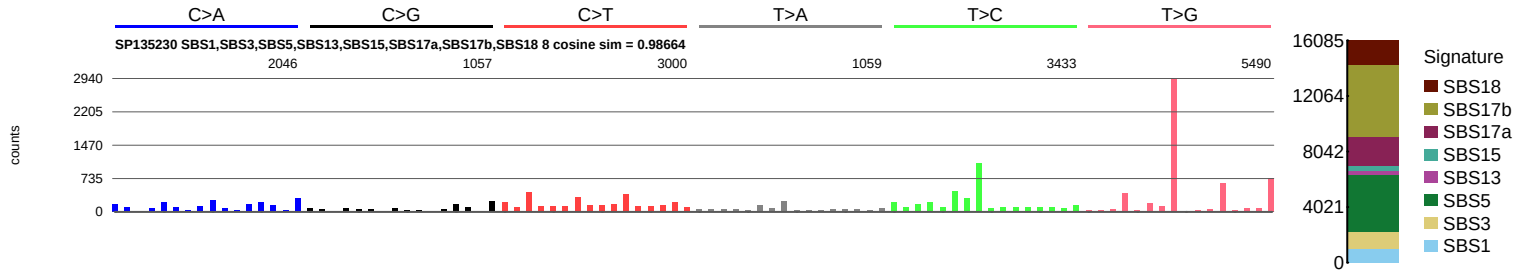

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98675
2176
763
2417
1510
3495
5726
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

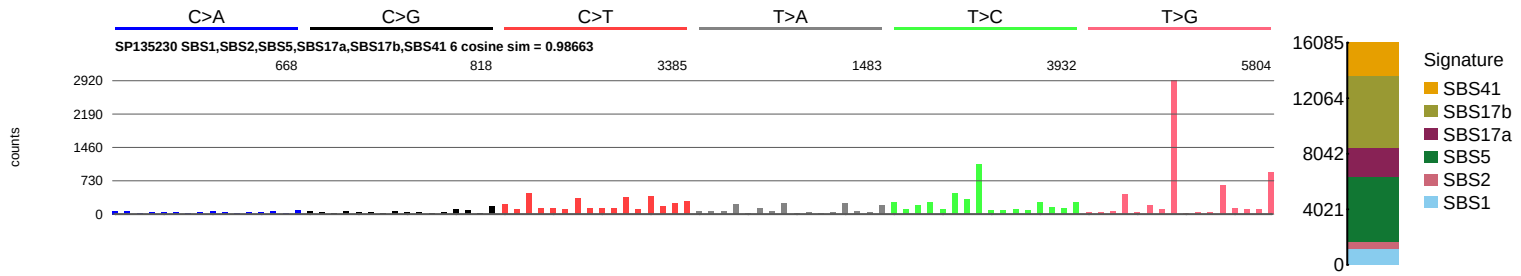

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98673
2022
980
2525
1434
3331
5792
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98667
2181
770
2349
1524
3505
5750
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98667
662
809
3367
1474
3979
5799
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98666
662
808
3371
1475
3978
5796
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98664
2046
1057
3000
1059
3433
5490
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98663
668
818
3385
1483
3932
5804
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
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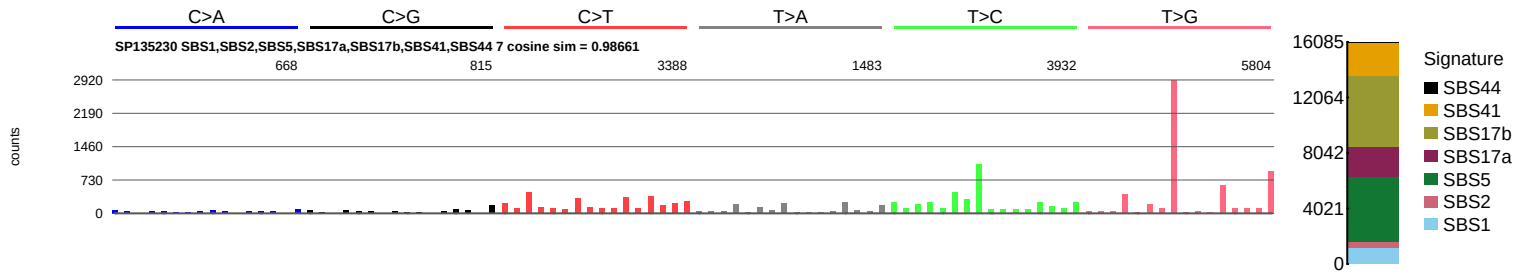
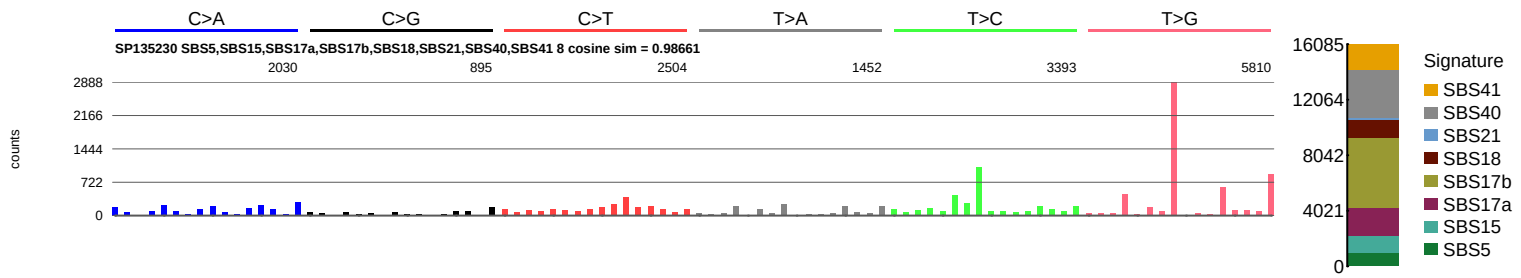
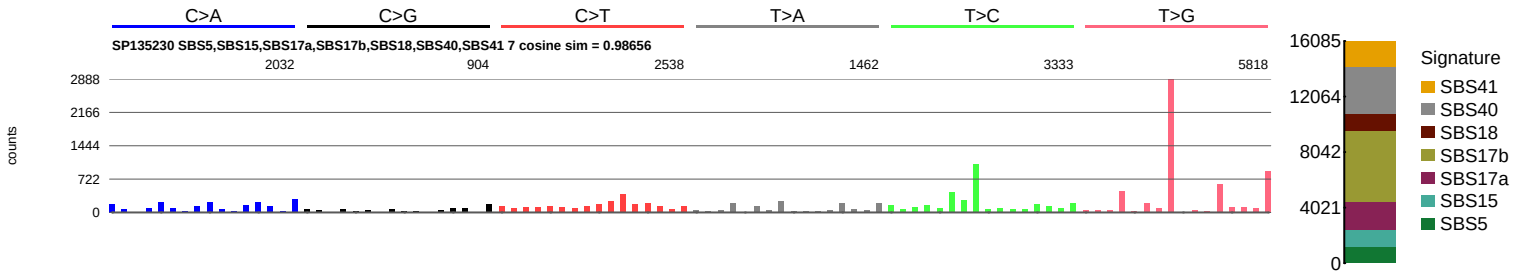
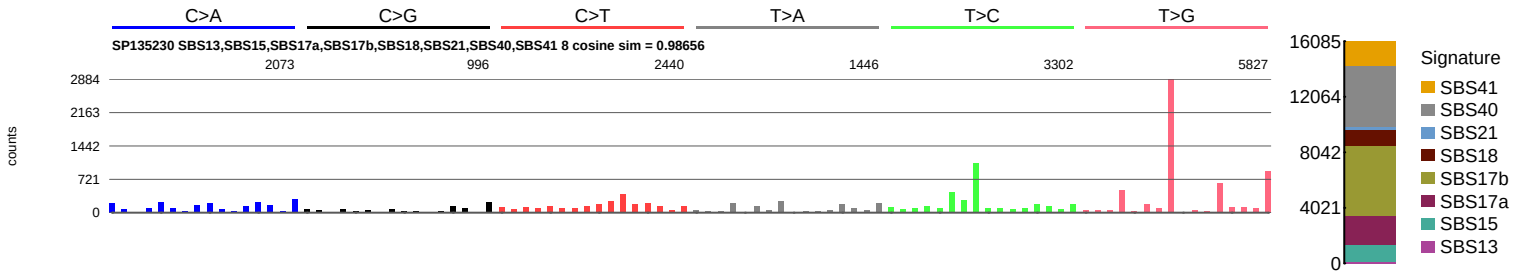
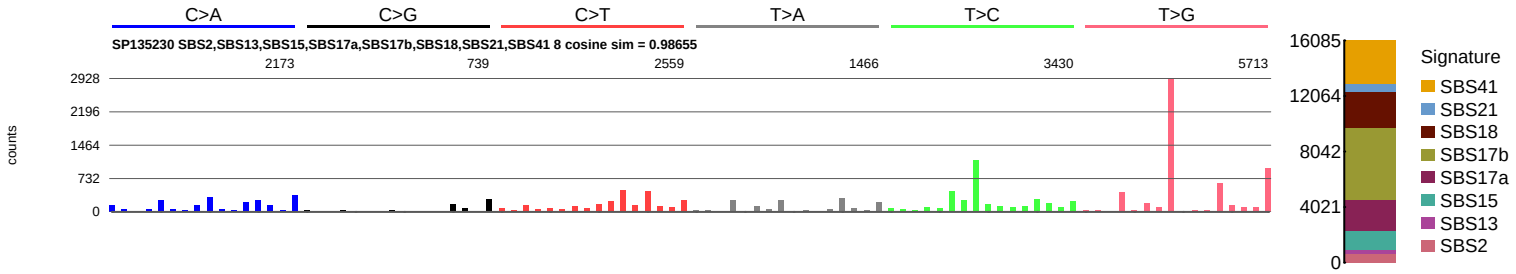
SBS5
SBS2
SBS1

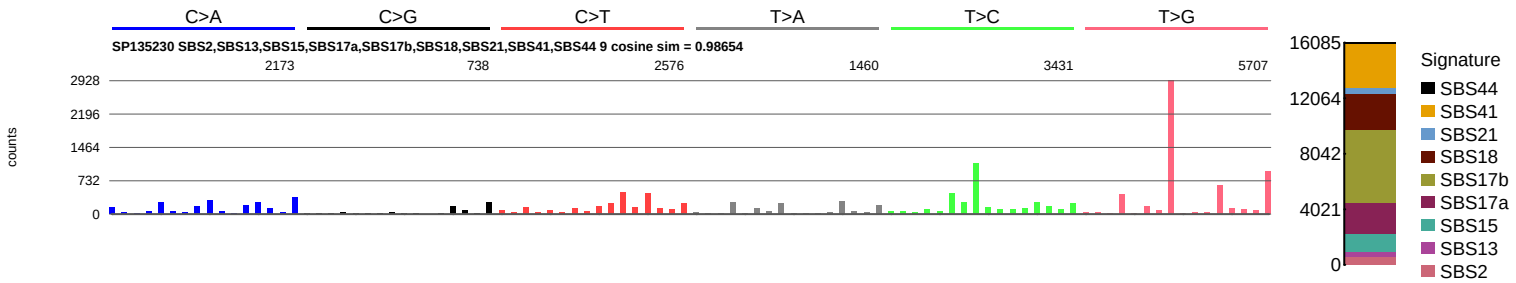

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98661
668
815
3388
1483
3932
5804
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS2
SBS1

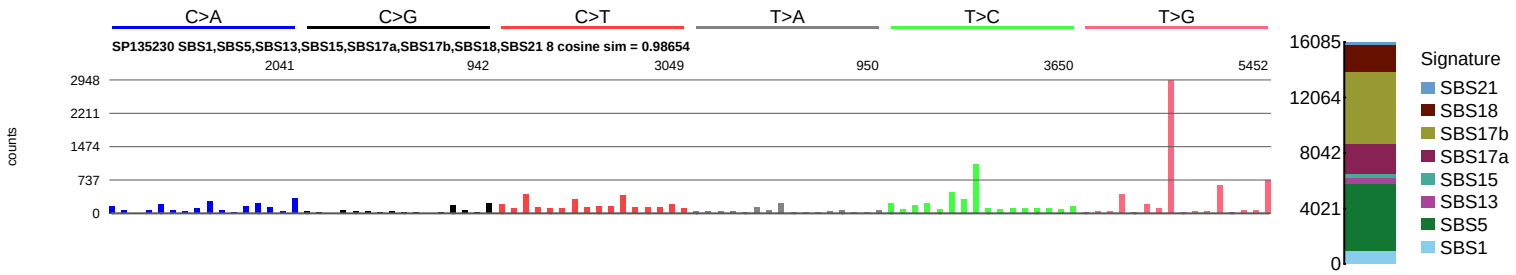

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98661
2030
895
2504
1452
3393
5810
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98656
2032
904
2538
1462
3333
5818
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98656
2073
996
2440
1446
3302
5827
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98655
2173
739
2559
1466
3430
5713
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98654
2173
738
2576
1460
3431
5707
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98654
2041
942
3049
950
3650
5452
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
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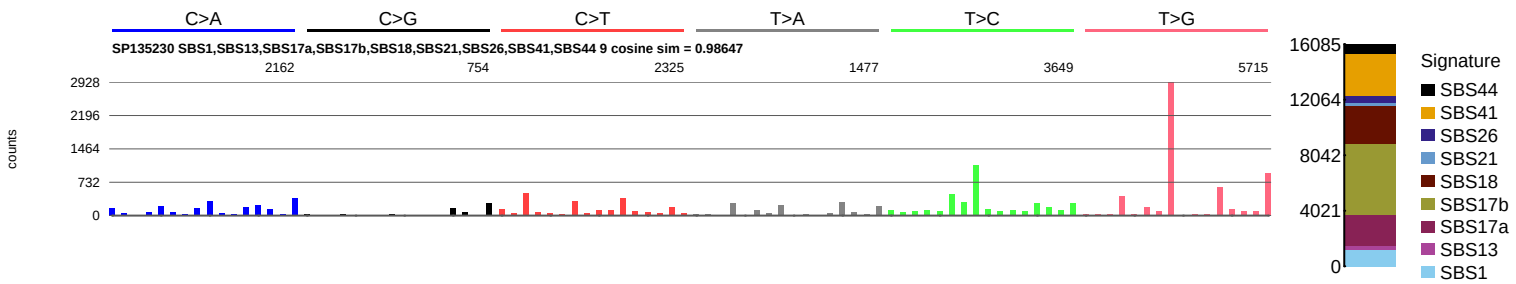
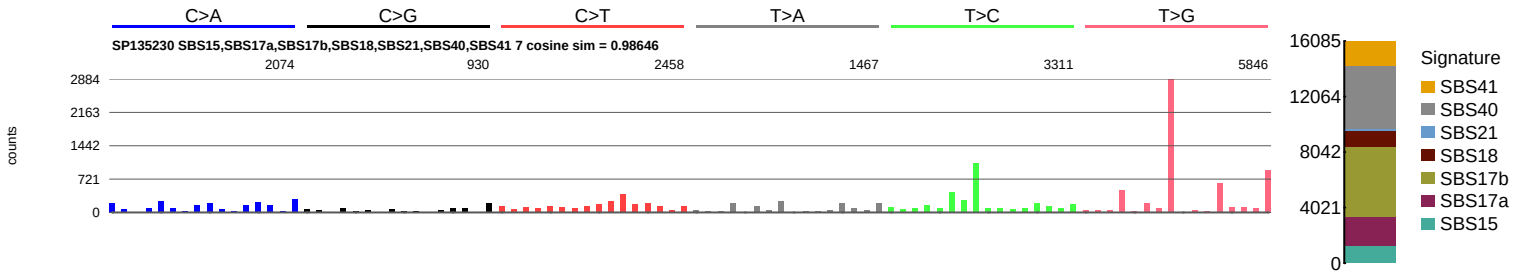
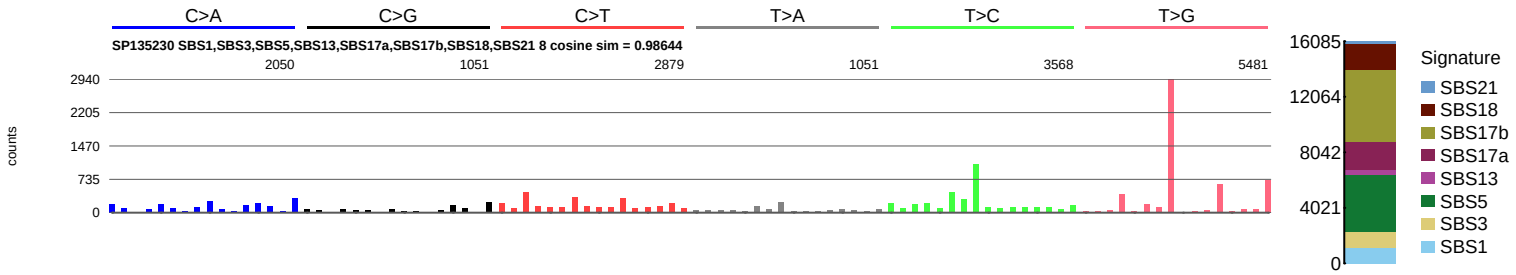
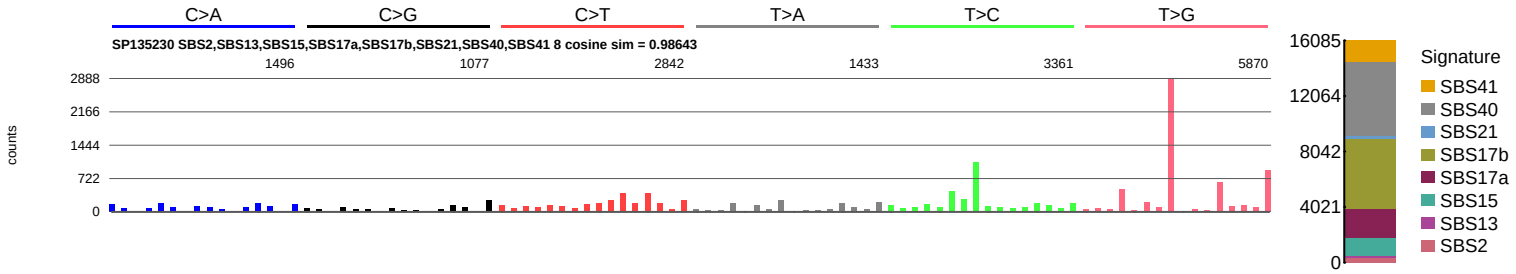
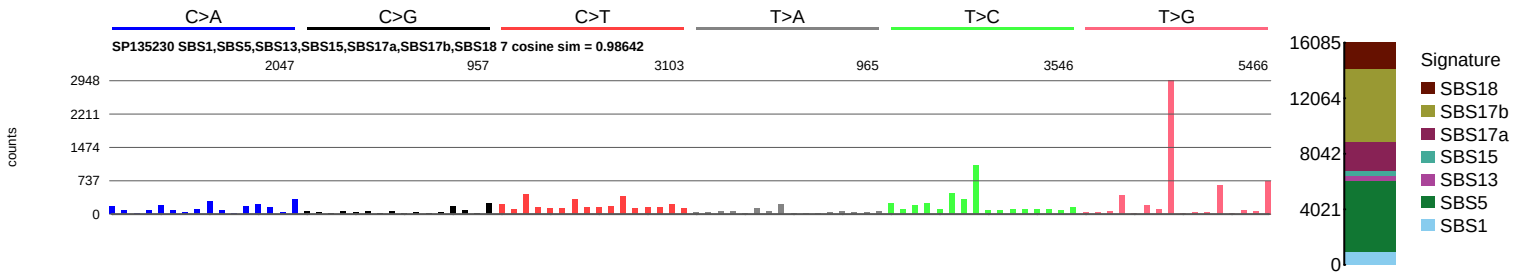
SBS13
SBS5
SBS1

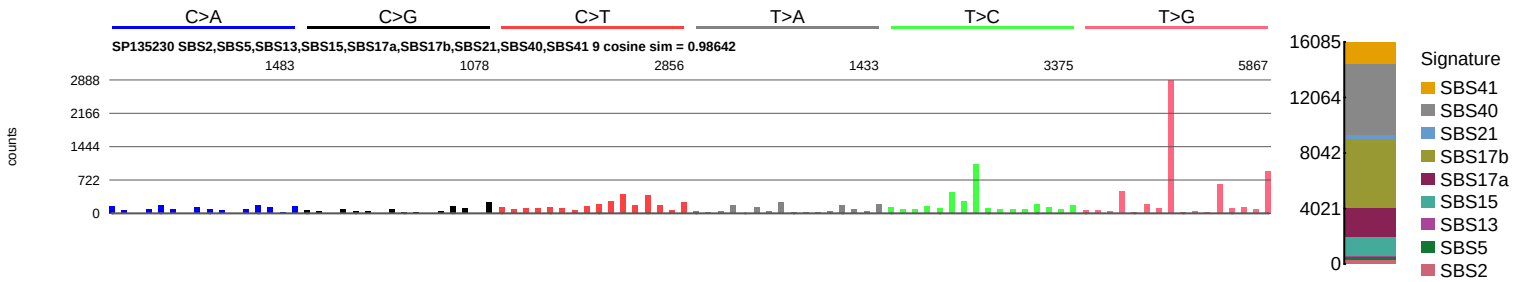

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98647
2162
754
2325
1477
3649
5715
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

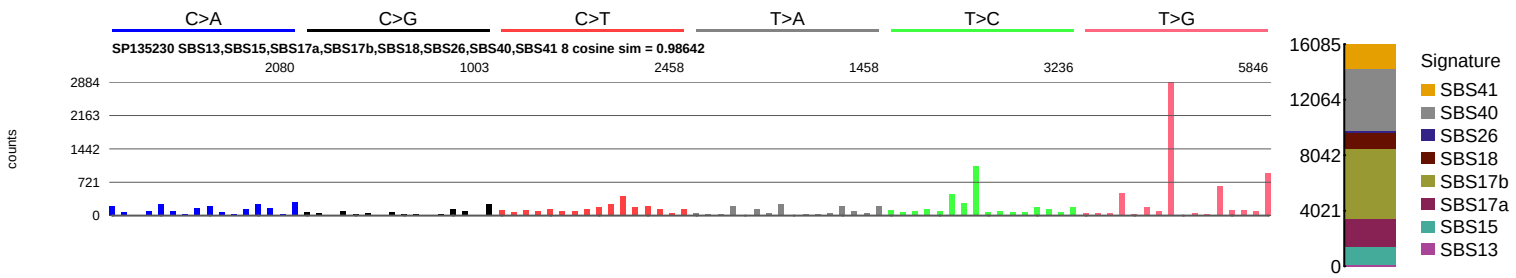

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98646
2074
930
2458
1467
3311
5846
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15

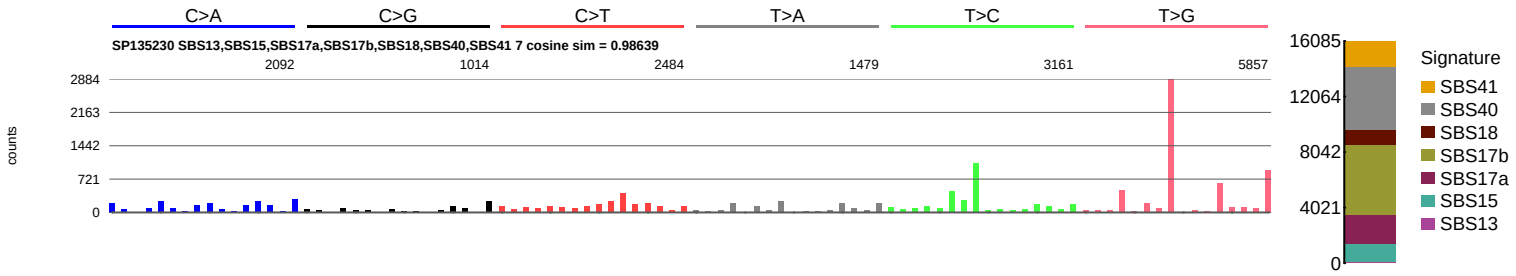

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98644
2050
1051
2879
1051
3568
5481
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

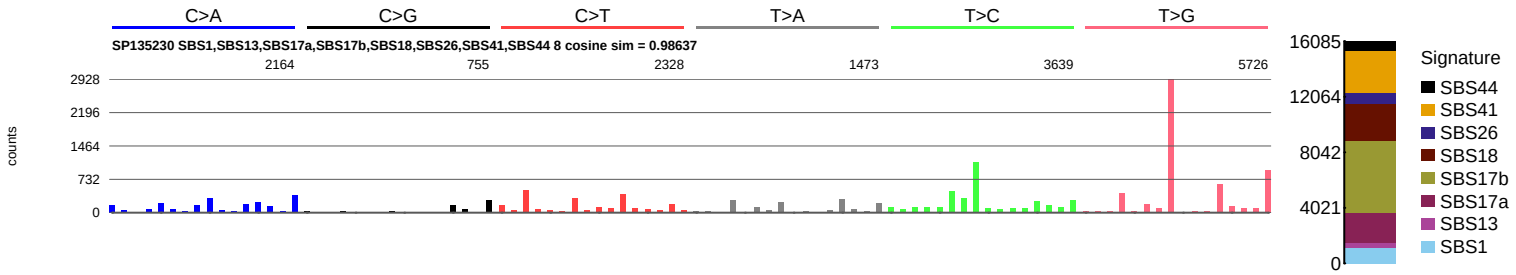

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98643
1496
1077
2842
1433
3361
5870
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2

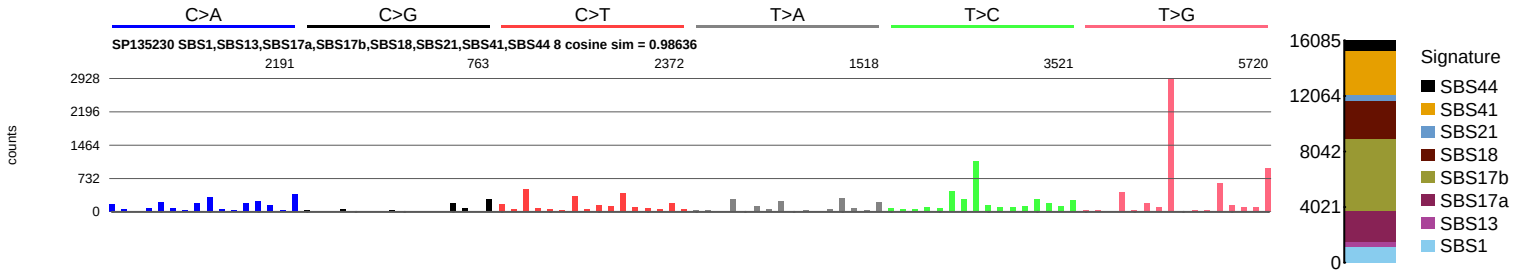

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98642
2047
957
3103
965
3546
5466
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

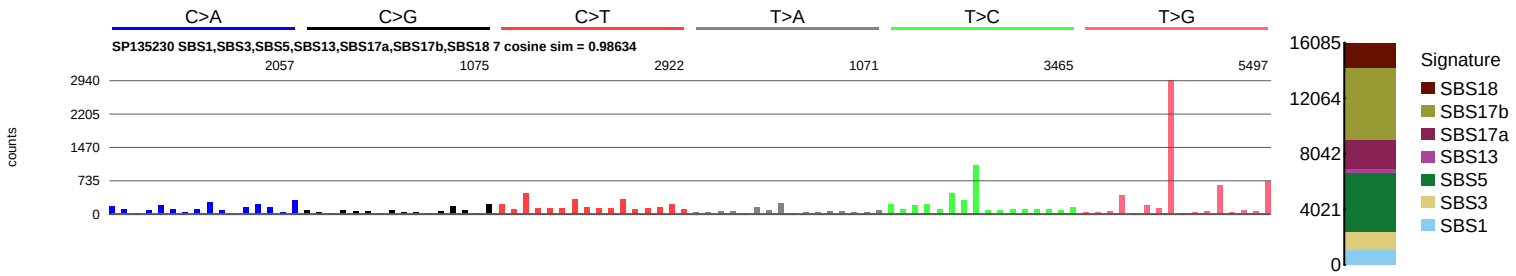

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98642
1483
1078
2856
1433
3375
5867
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

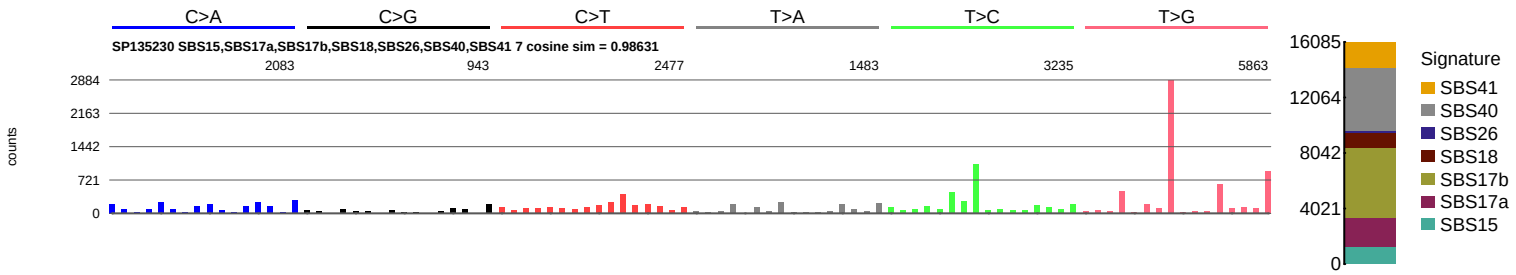

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98642
2080
1003
2458
1458
3236
5846
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13

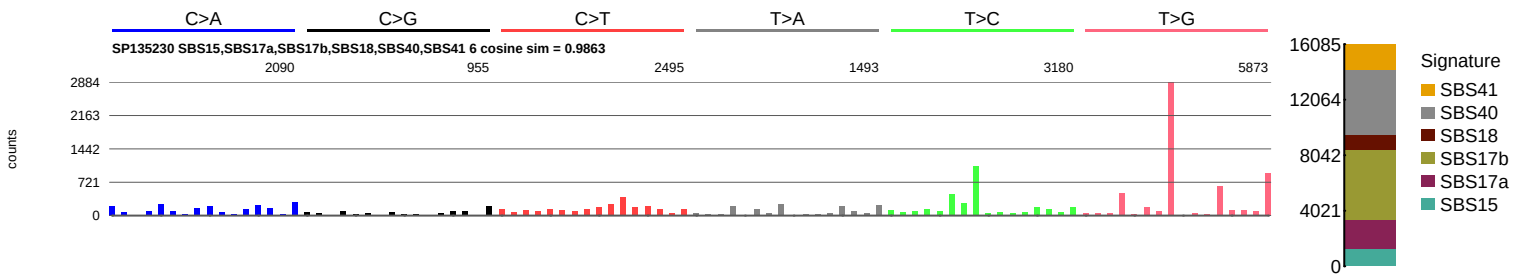

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98639
2092
1014
2484
1479
3161
5857
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13

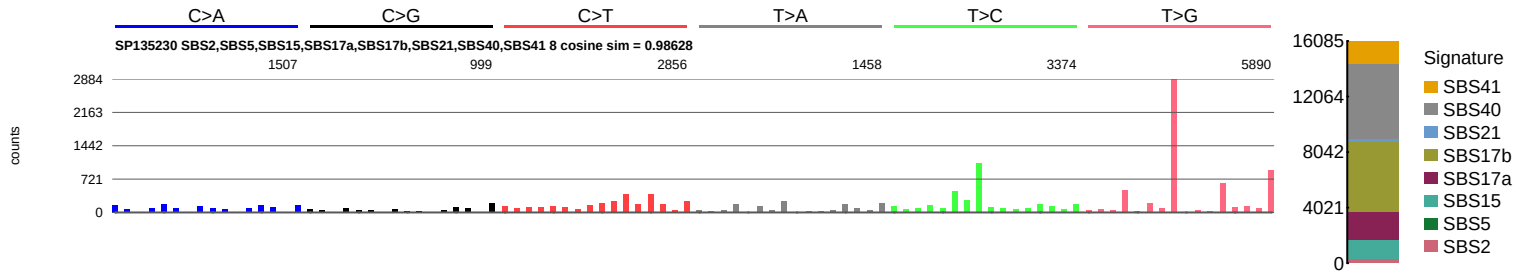

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98637
2164
755
2328
1473
3639
5726
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

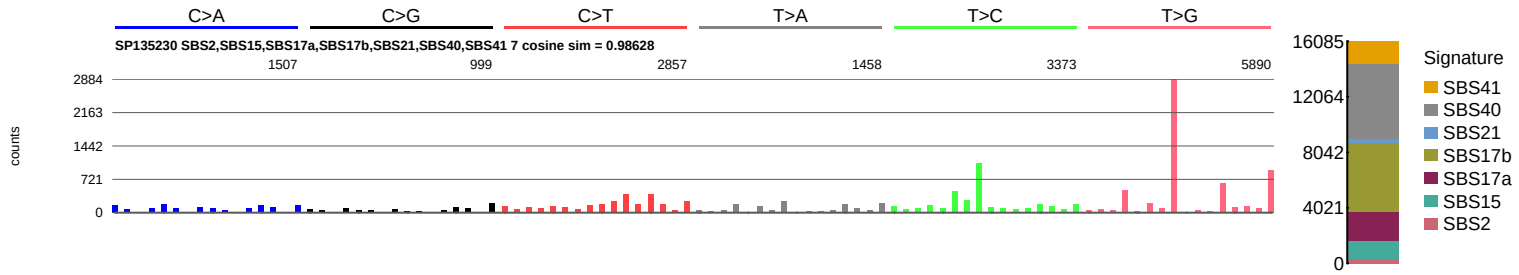

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98636
2191
763
2372
1518
3521
5720
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

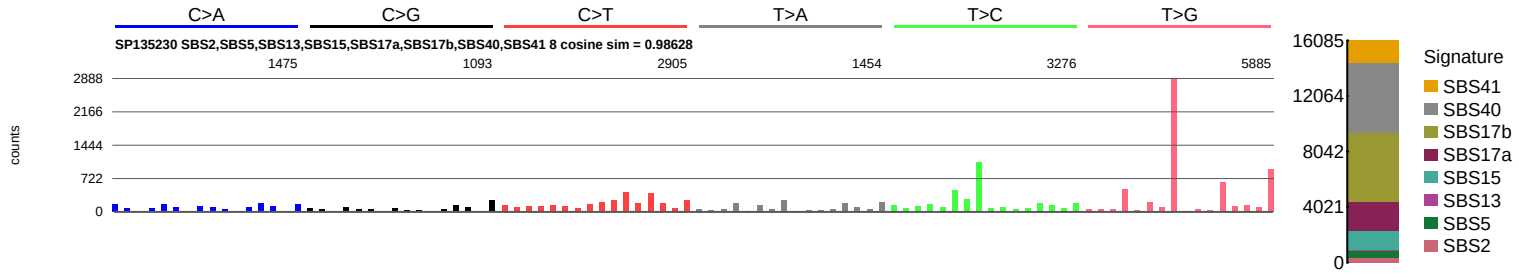

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98634
2057
1075
2922
1071
3465
5497
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

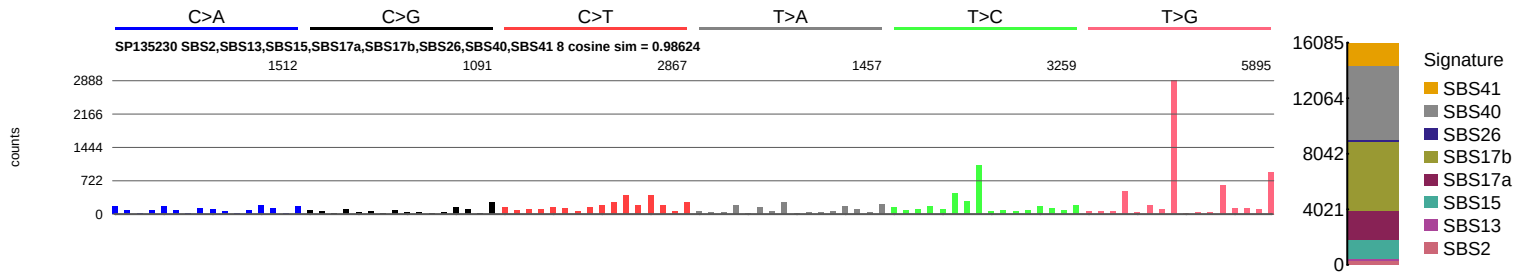

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98631
2083
943
2477
1483
3235
5863
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15

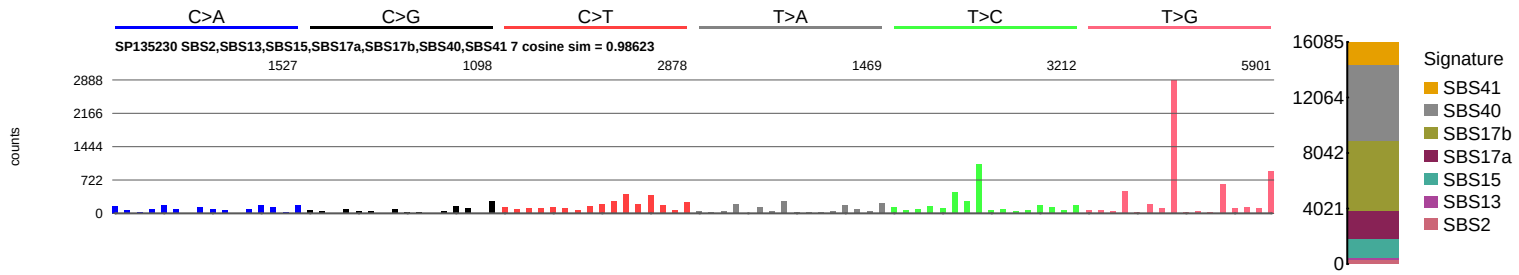

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.9863
2090
955
2495
1493
3180
5873
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15

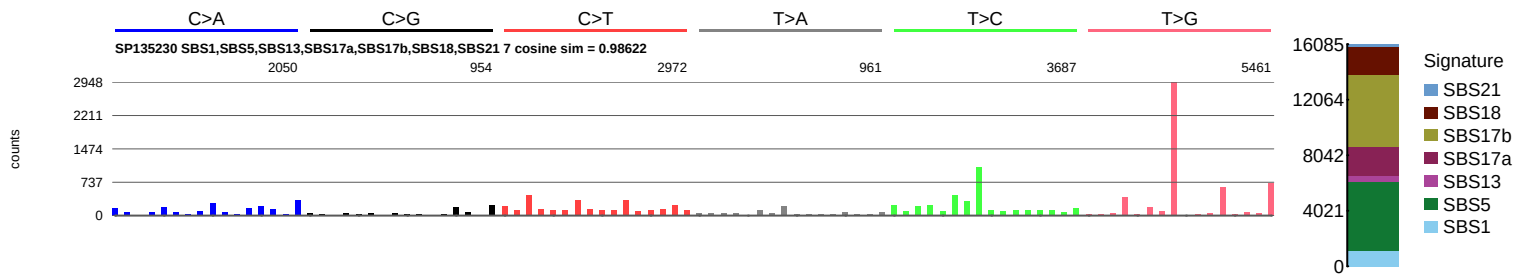

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98628
1507
999
2856
1458
3374
5890
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2

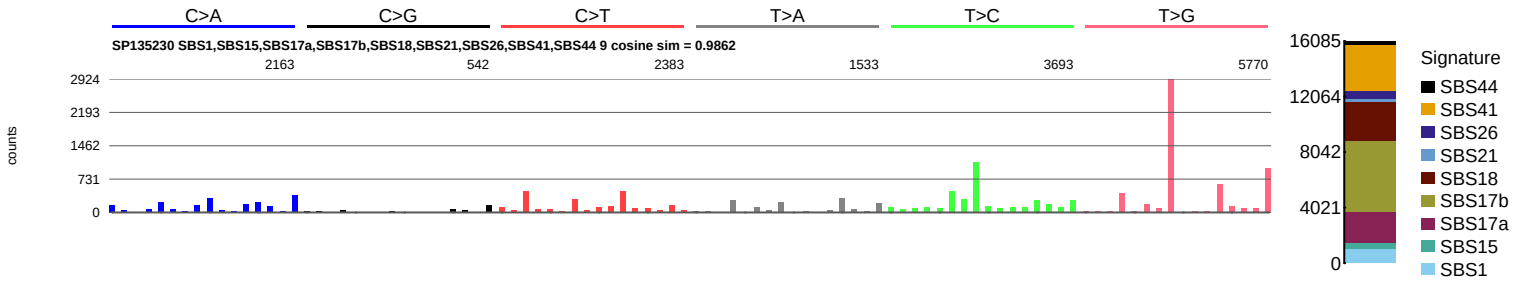

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98628
1507
999
2857
1458
3373
5890
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2

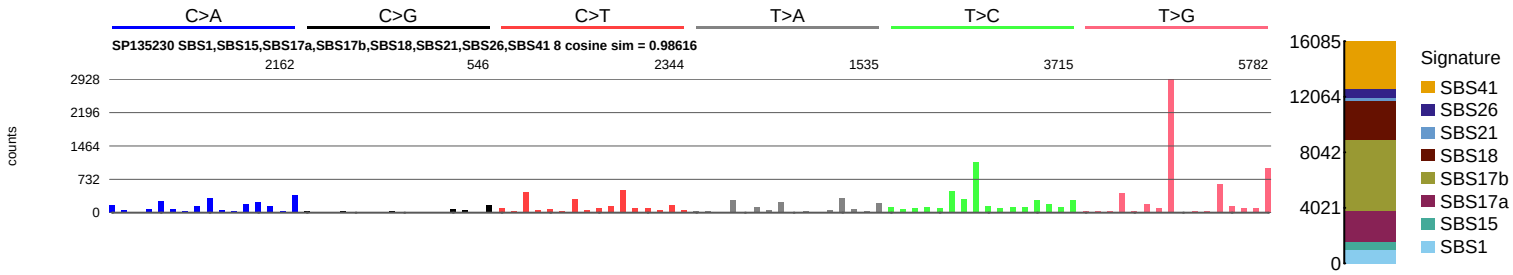

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98628
1475
1093
2905
1454
3276
5885
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

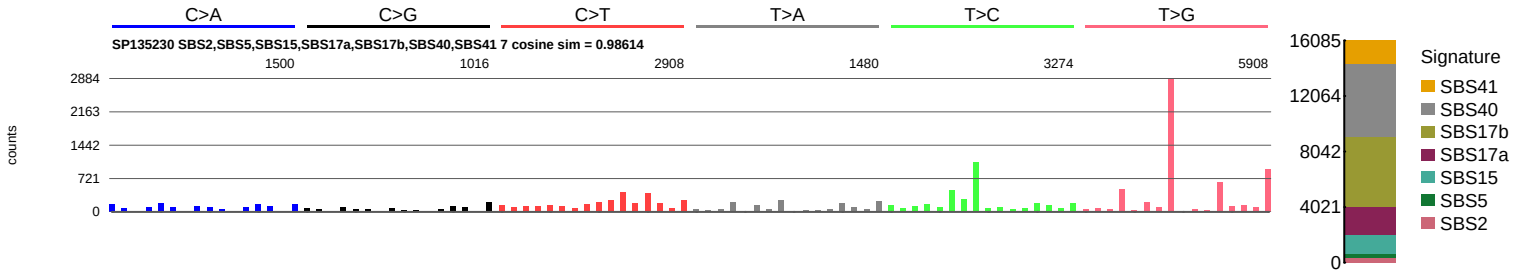

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98624
1512
1091
2867
1457
3259
5895
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2

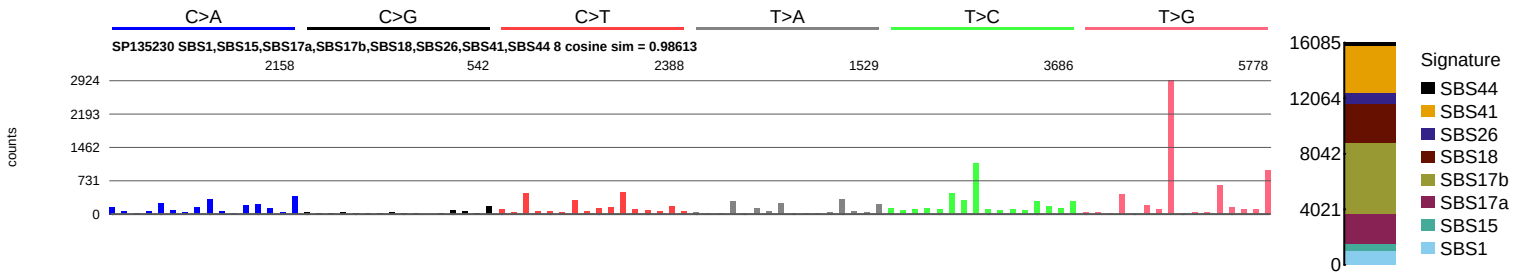

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98623
1527
1098
2878
1469
3212
5901
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2

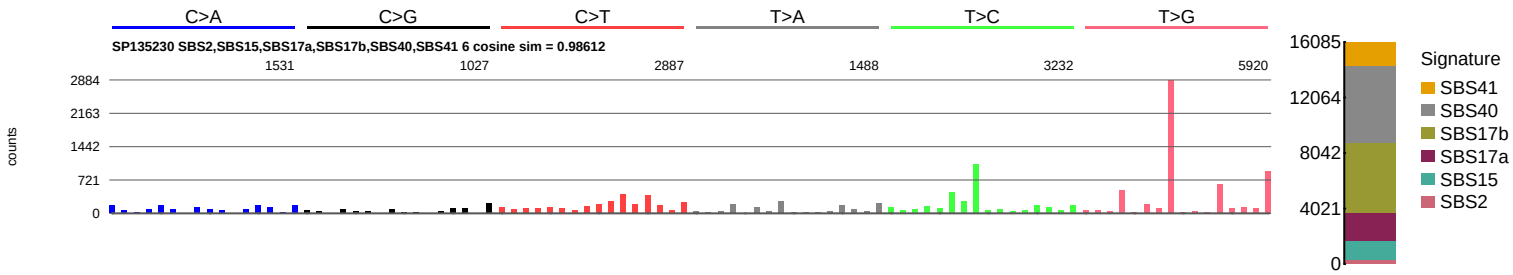

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98622
2050
954
2972
961
3687
5461
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

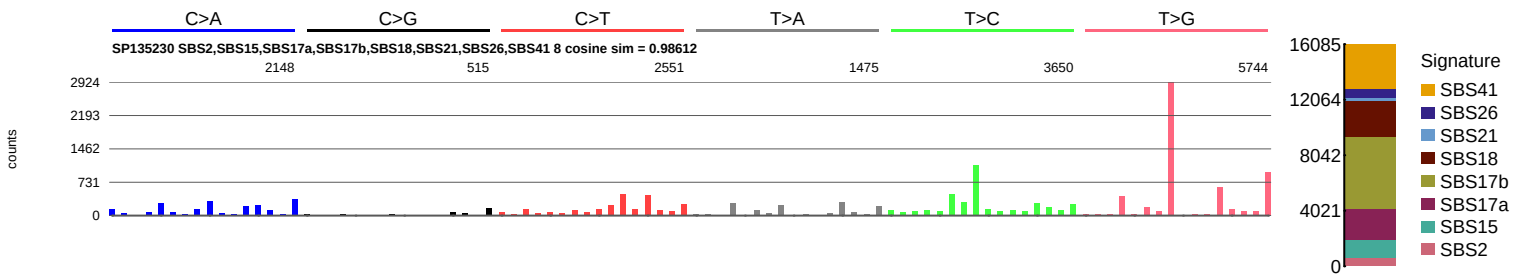

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.9862
2163
542
2383
1533
3693
5770
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

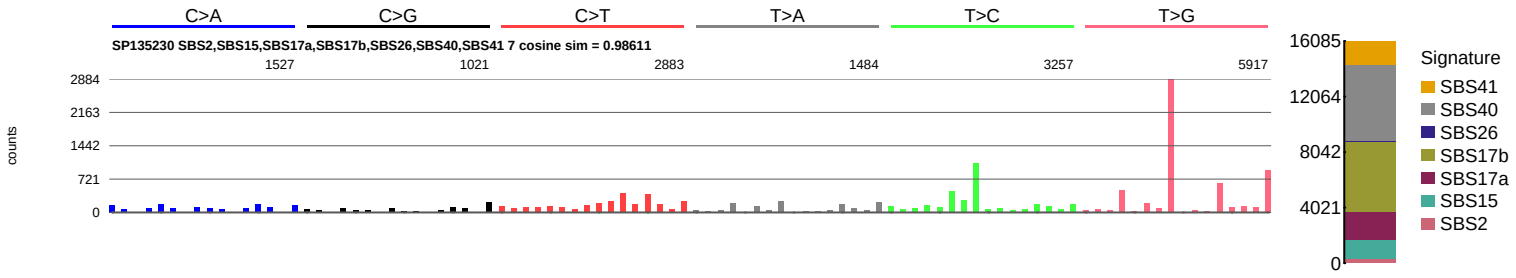

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98616
2162
546
2344
1535
3715
5782
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

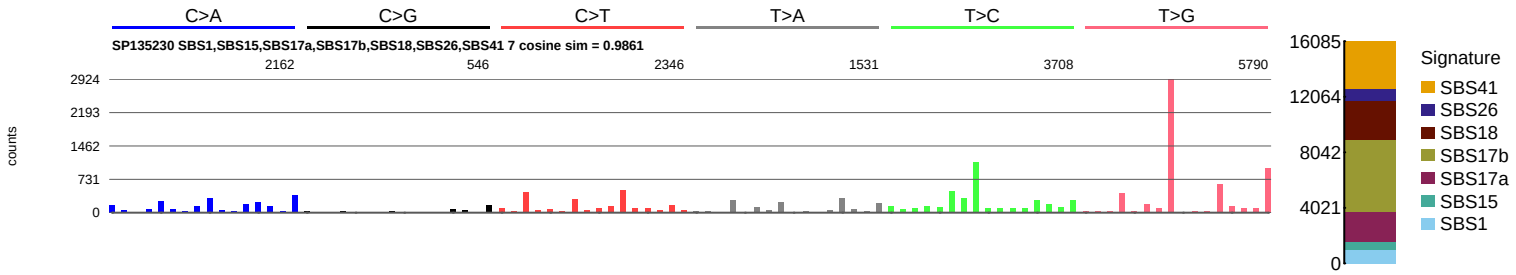

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98614
1500
1016
2908
1480
3274
5908
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2

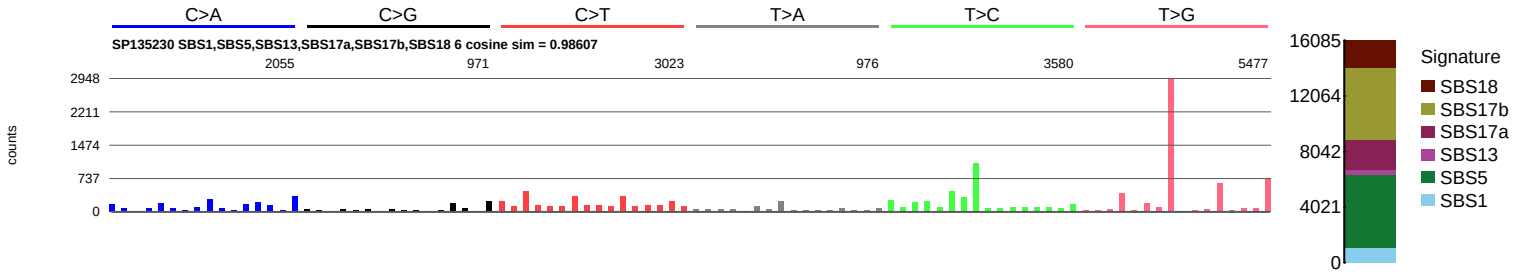

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98613
2158
542
2388
1529
3686
5778
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

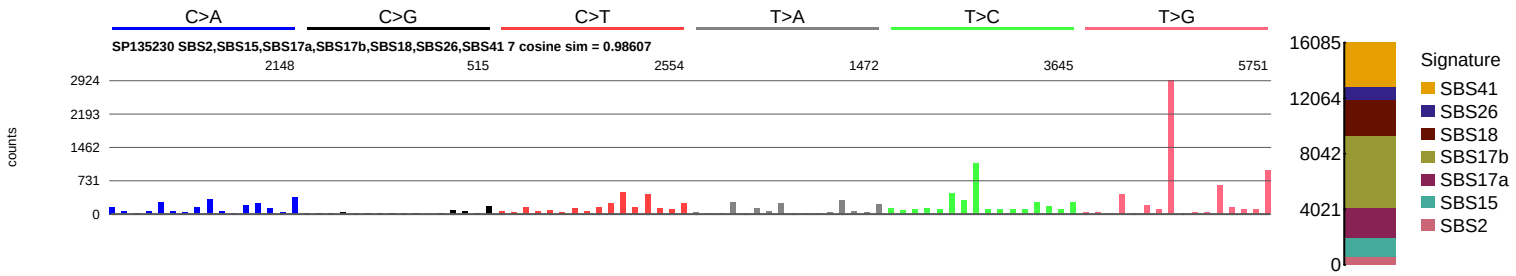

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98612
1531
1027
2887
1488
3232
5920
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98612
2148
515
2551
1475
3650
5744
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98611
1527
1021
2883
1484
3257
5917
counts
2884
2163
1442
721
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.9861
2162
546
2346
1531
3708
5790
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98607
2055
971
3023
976
3580
5477
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
C>T
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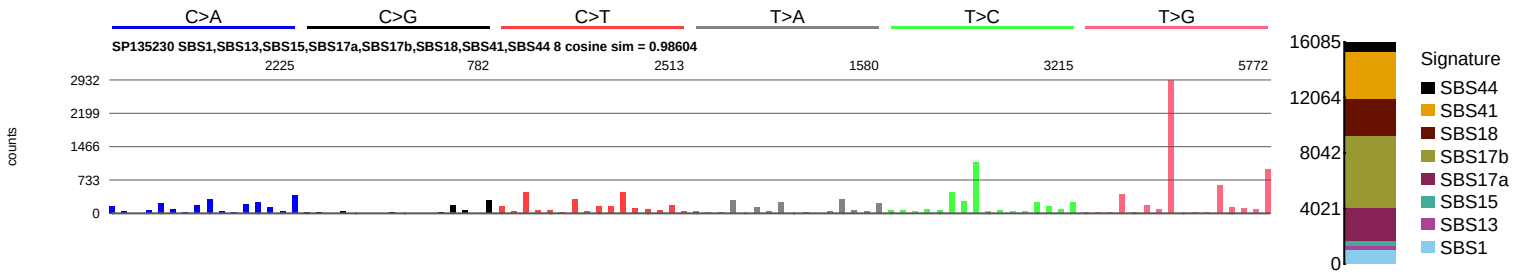
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98607
2148
515
2554
1472
3645
5751
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2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C>G
C>T
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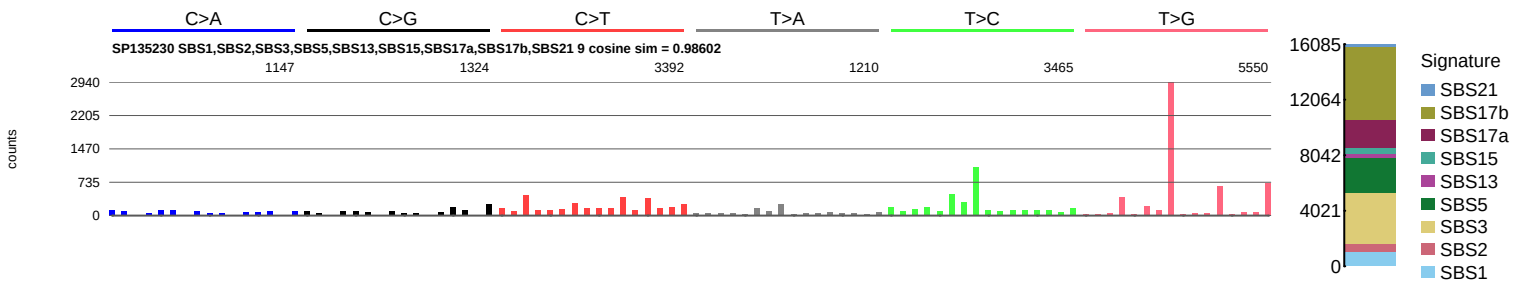
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98604
2225
782
2513
1580
3215
5772
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
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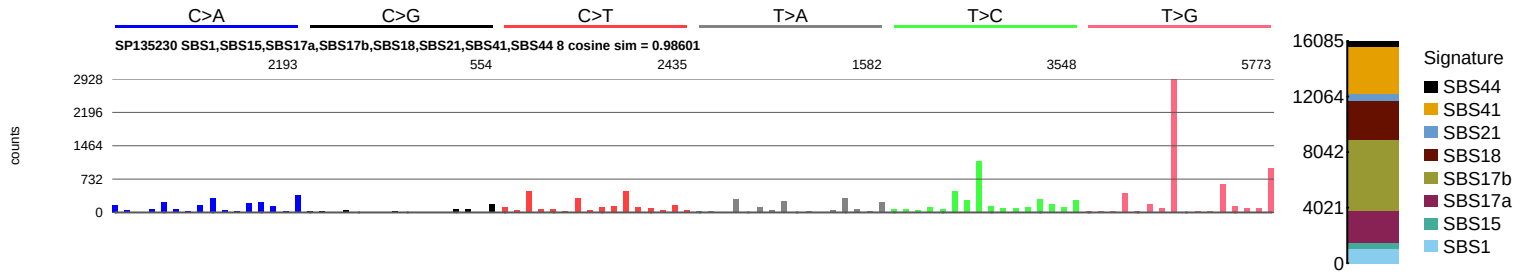
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 9 cosine sim = 0.98602
1147
1324
3392
1210
3465
5550
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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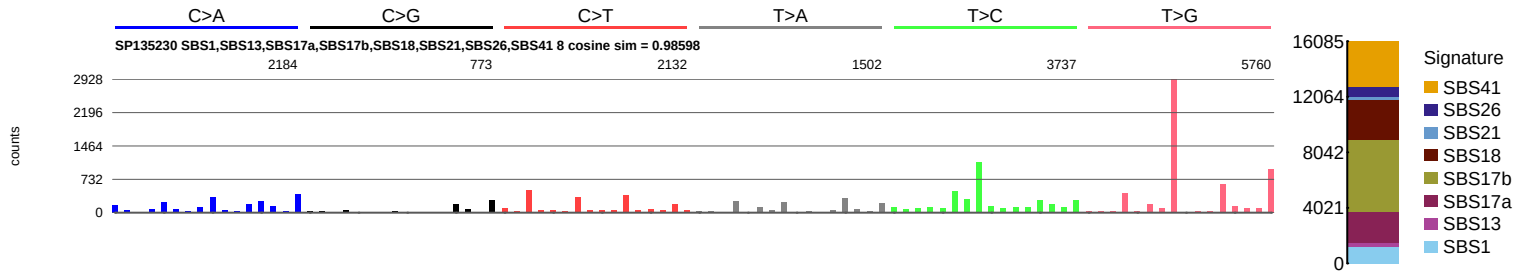
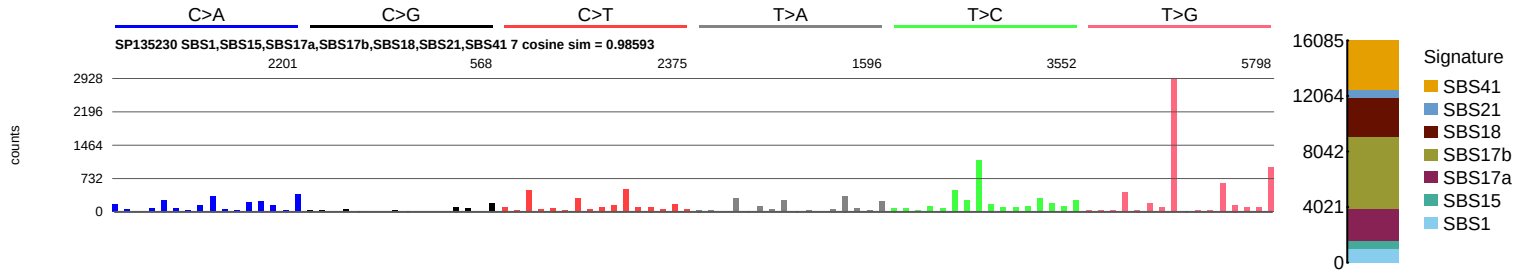
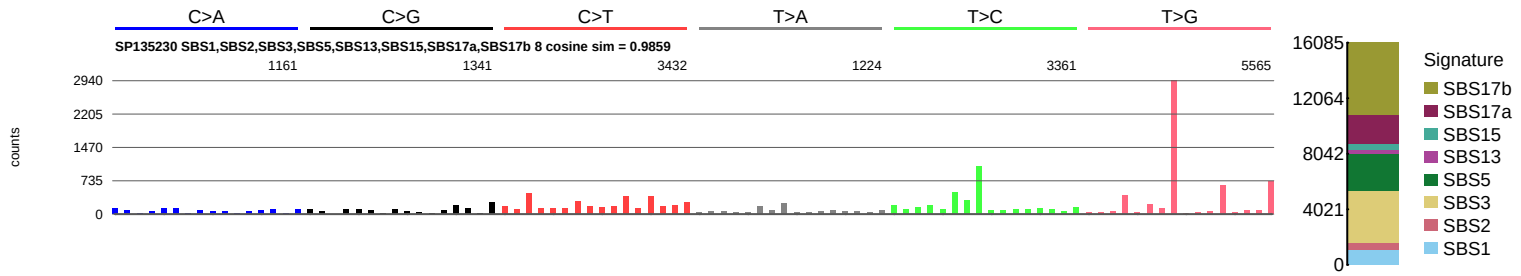
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98601
2193
554
2435
1582
3548
5773
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98598
2184
773
2132
1502
3737
5760
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98593
2201
568
2375
1596
3552
5798
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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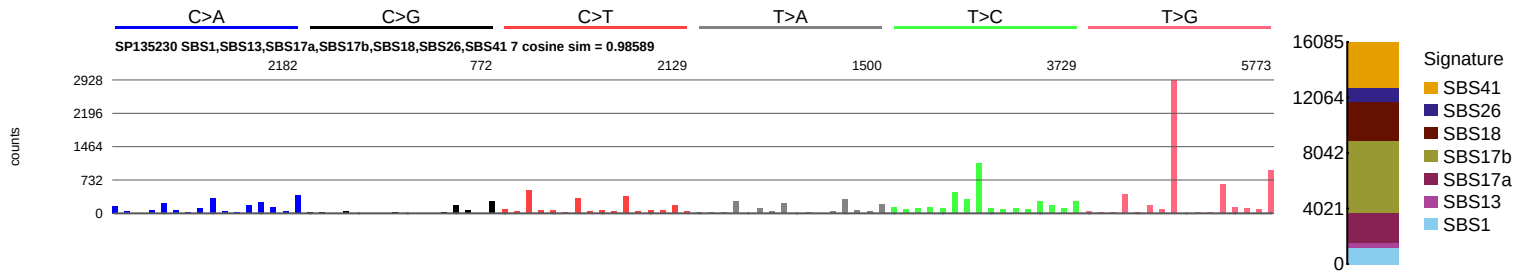
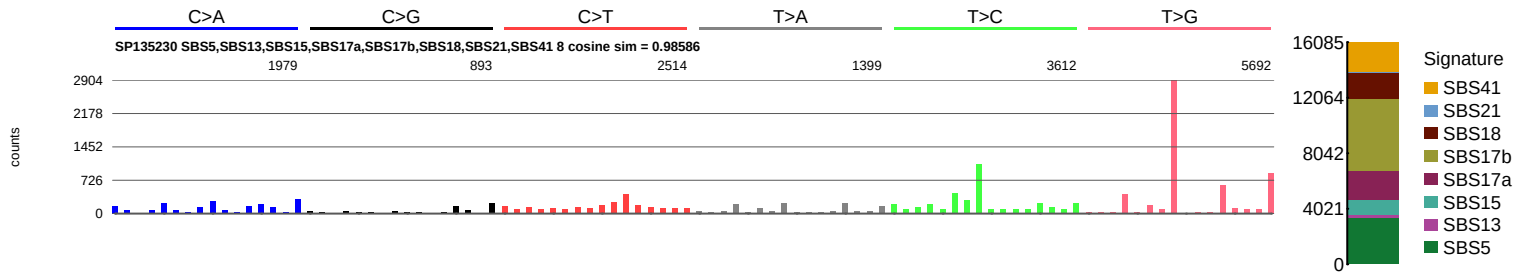
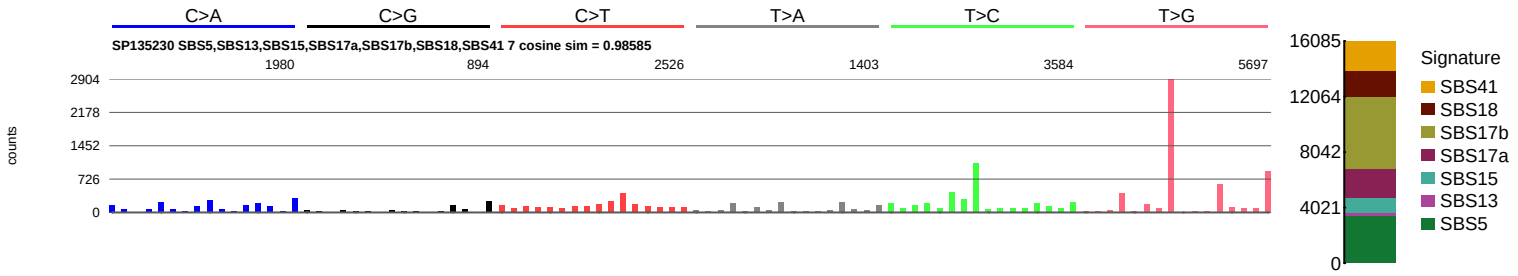
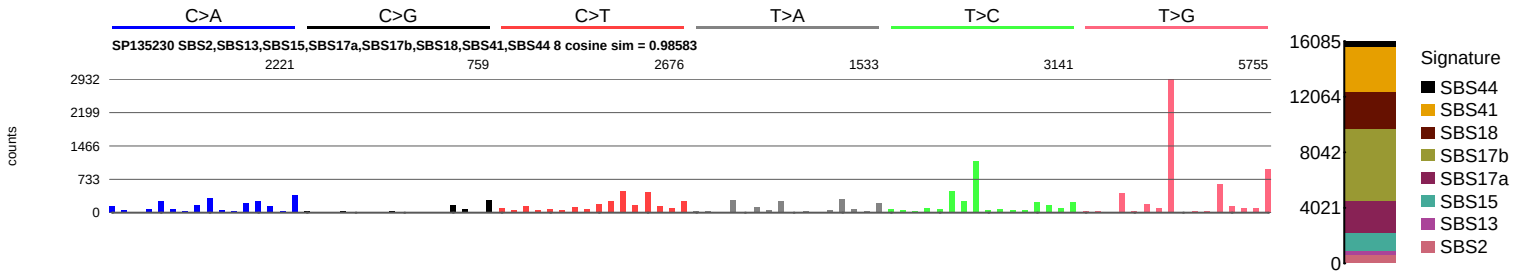
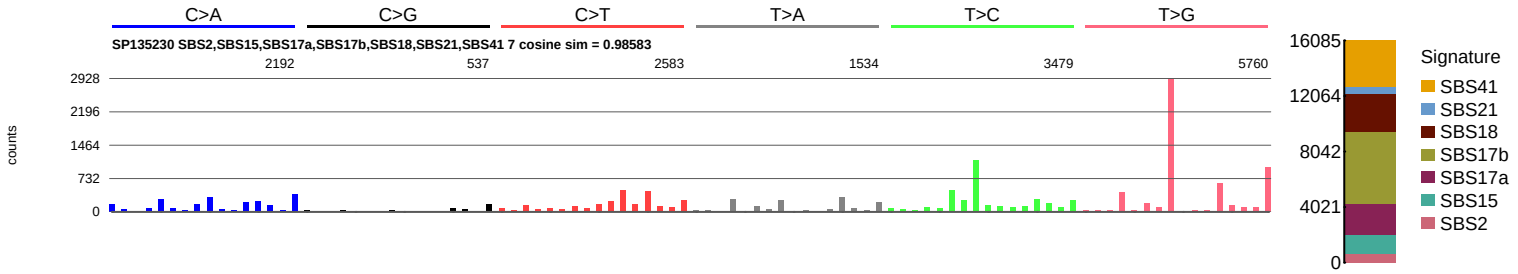
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 8 cosine sim = 0.9859
1161
1341
3432
1224
3361
5565
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

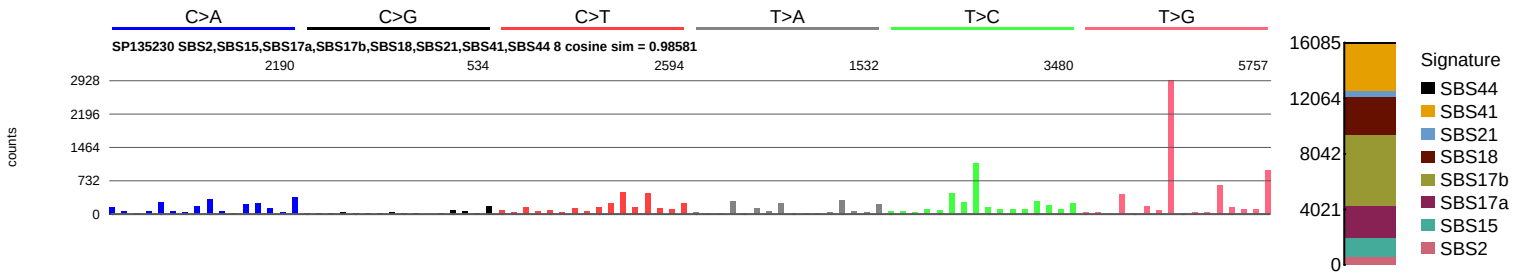

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98589
2182
772
2129
1500
3729
5773
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

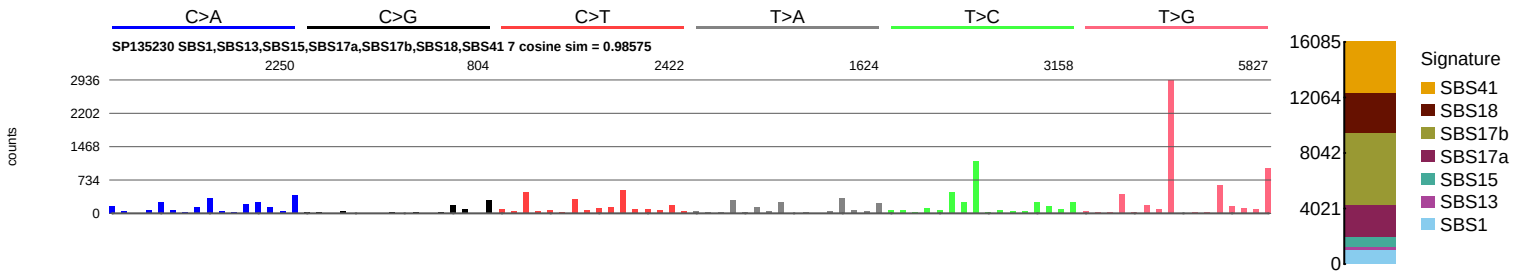

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98586
1979
893
2514
1399
3612
5692
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

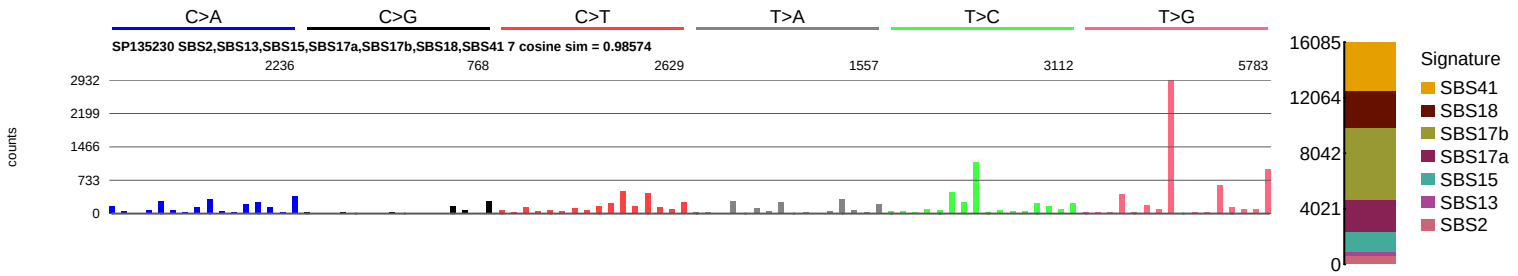

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98585
1980
894
2526
1403
3584
5697
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

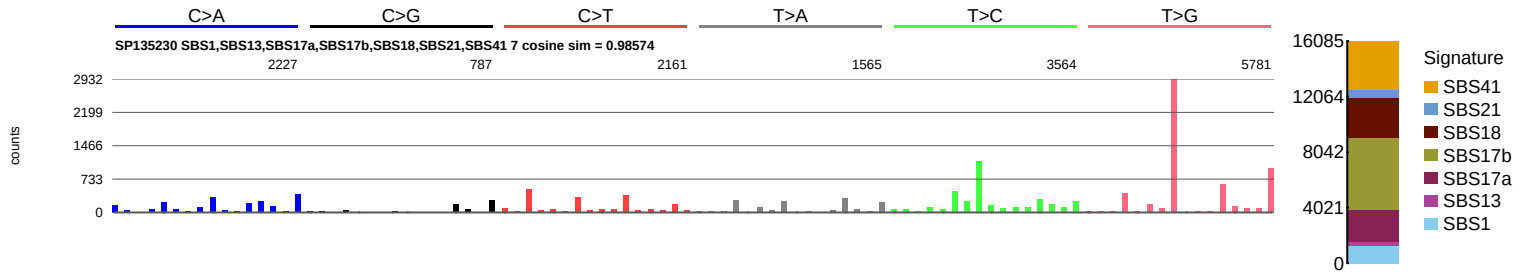

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98583
2221
759
2676
1533
3141
5755
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

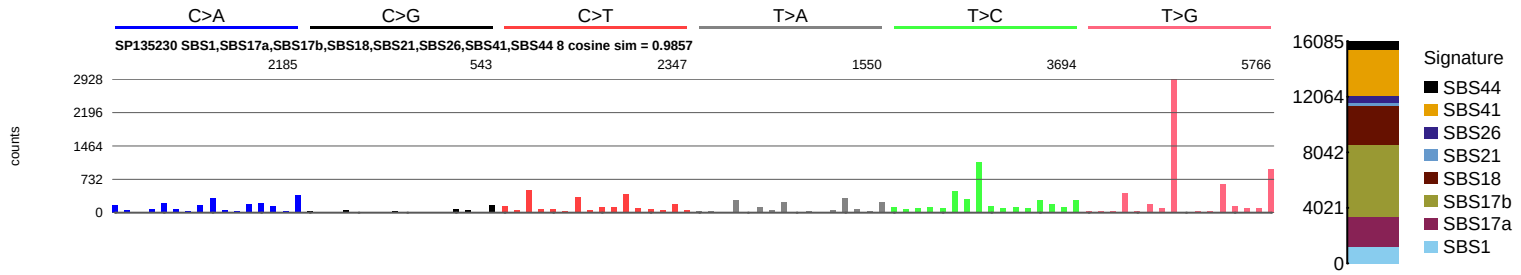

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98583
2192
537
2583
1534
3479
5760
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2

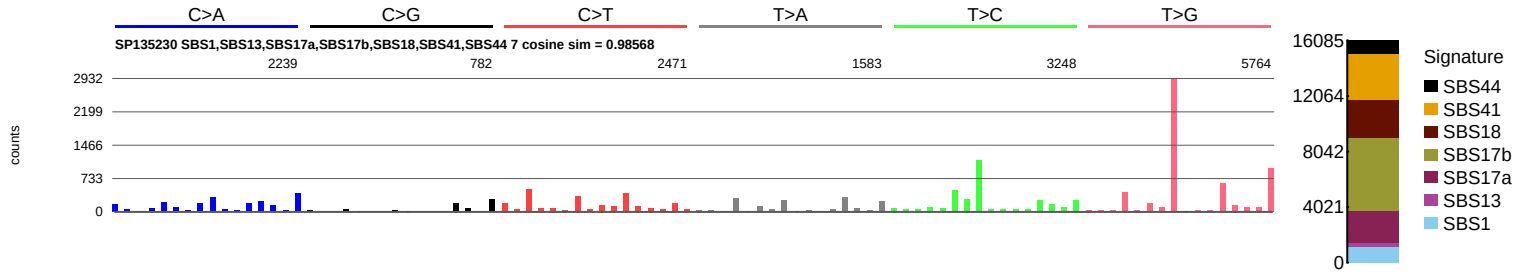

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98581
2190
534
2594
1532
3480
5757
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2

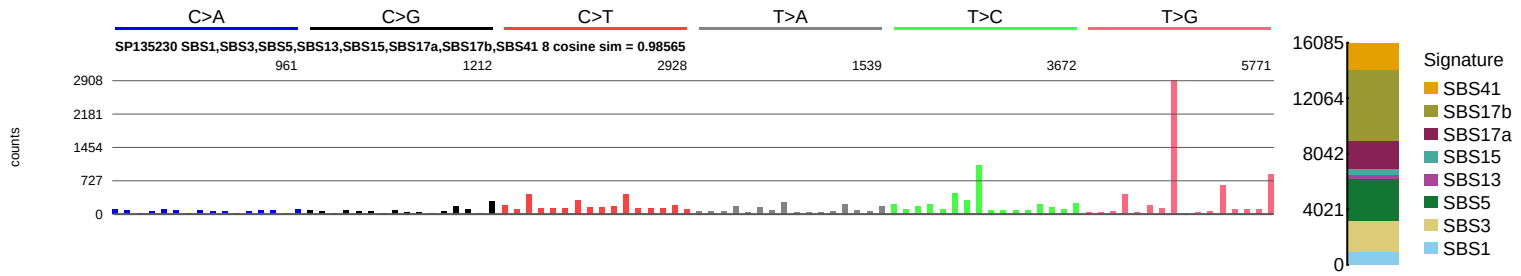

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98575
2250
804
2422
1624
3158
5827
counts
2936
2202
1468
734
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

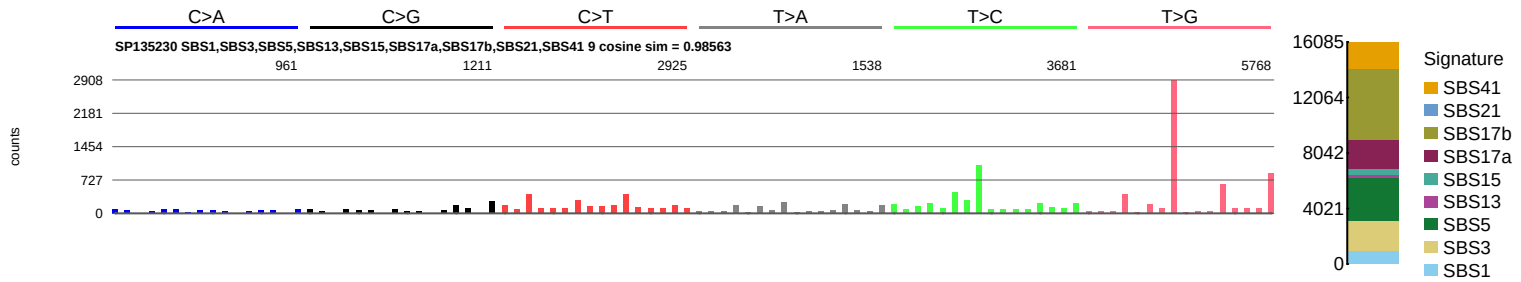

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98574
2236
768
2629
1557
3112
5783
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

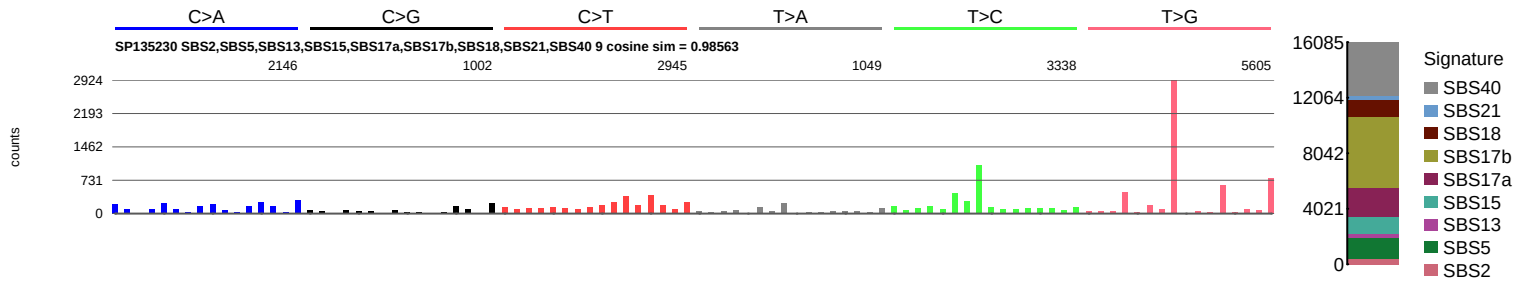

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98574
2227
787
2161
1565
3564
5781
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

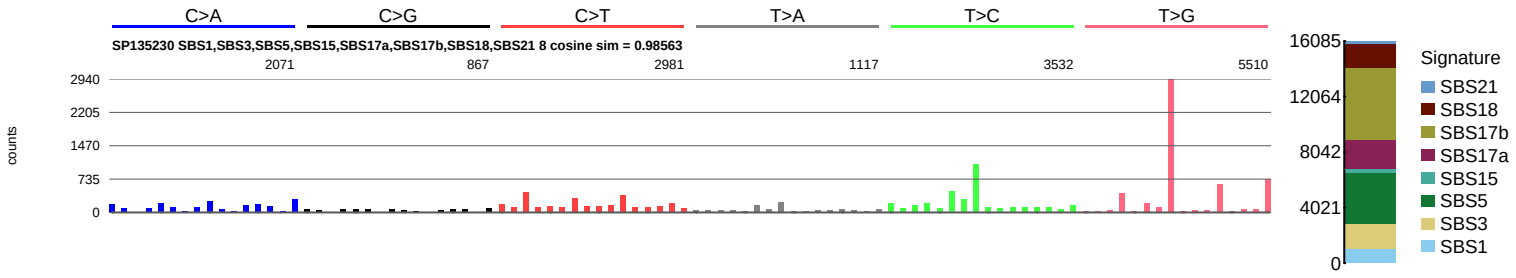

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.9857
2185
543
2347
1550
3694
5766
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1

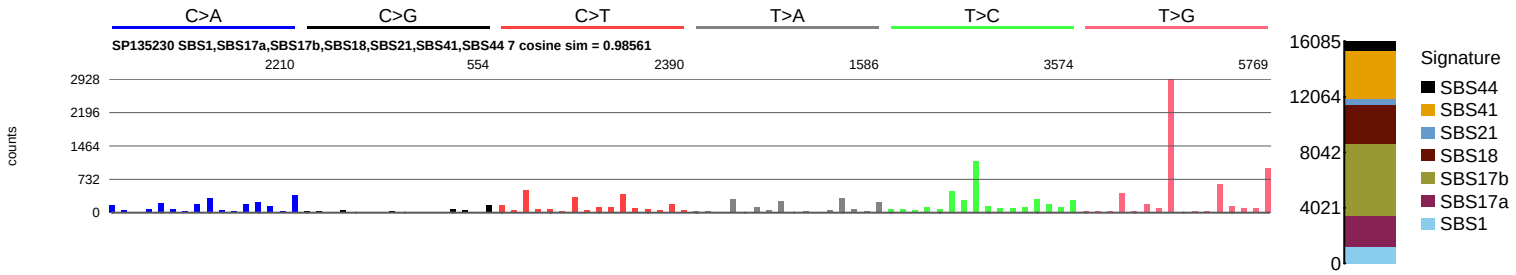

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98568
2239
782
2471
1583
3248
5764
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98565
961
1212
2928
1539
3672
5771
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98563
961
1211
2925
1538
3681
5768
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98563
2146
1002
2945
1049
3338
5605
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98563
2071
867
2981
1117
3532
5510
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98561
2210
554
2390
1586
3574
5769
counts
2928
2196
1464
732
0
16085
12064
8042
4021
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Signature
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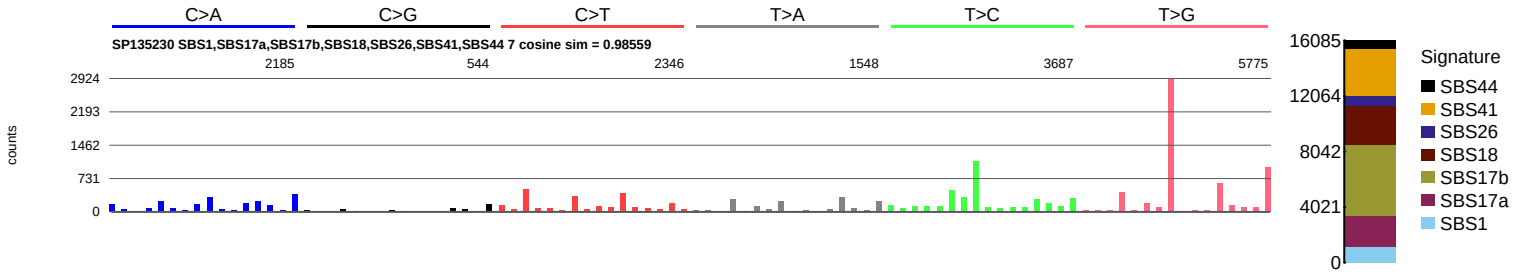
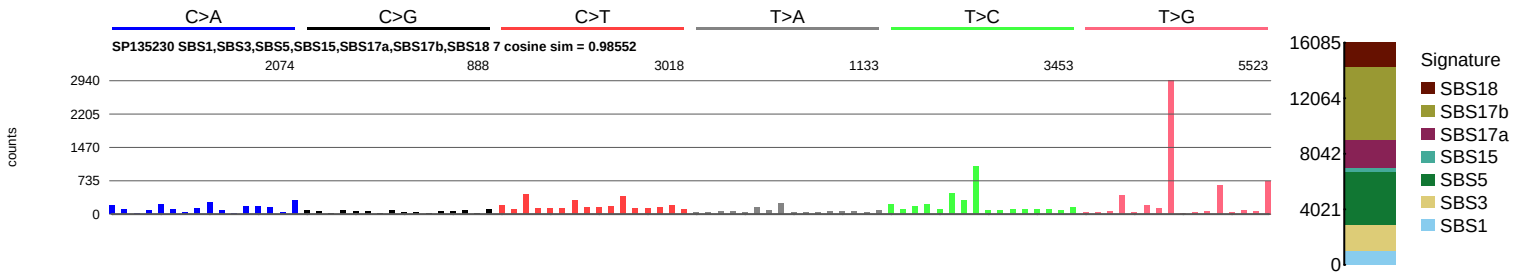
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98559
2185
544
2346
1548
3687
5775
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98552
2074
888
3018
1133
3453
5523
counts
2940
2205
1470
735
0
16085
12064
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4021
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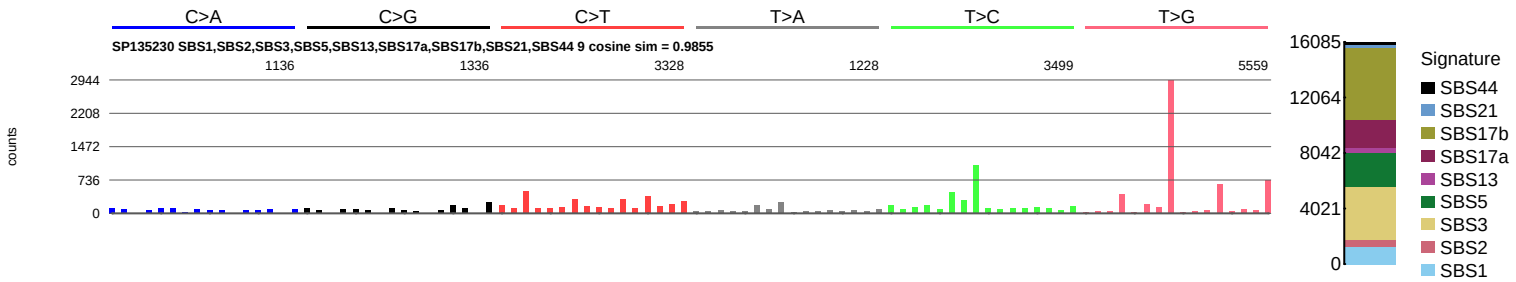
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
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T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.9855
1136
1336
3328
1228
3499
5559
counts
2944
2208
1472
736
0
16085
12064
8042
4021
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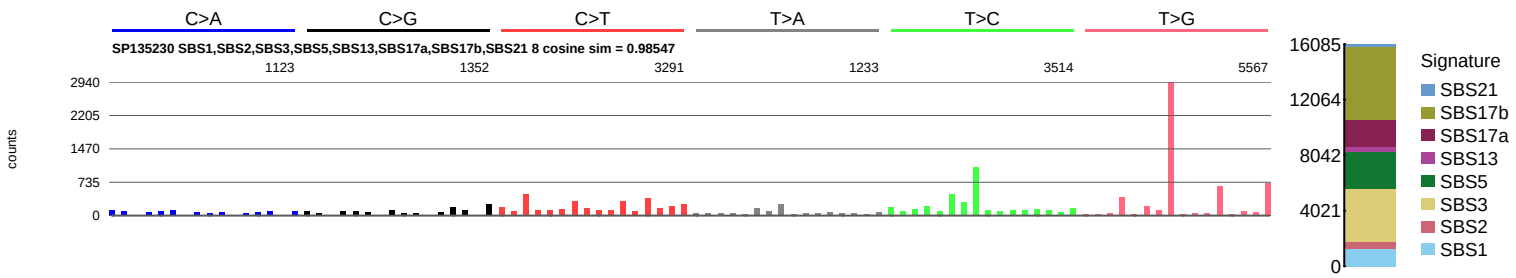
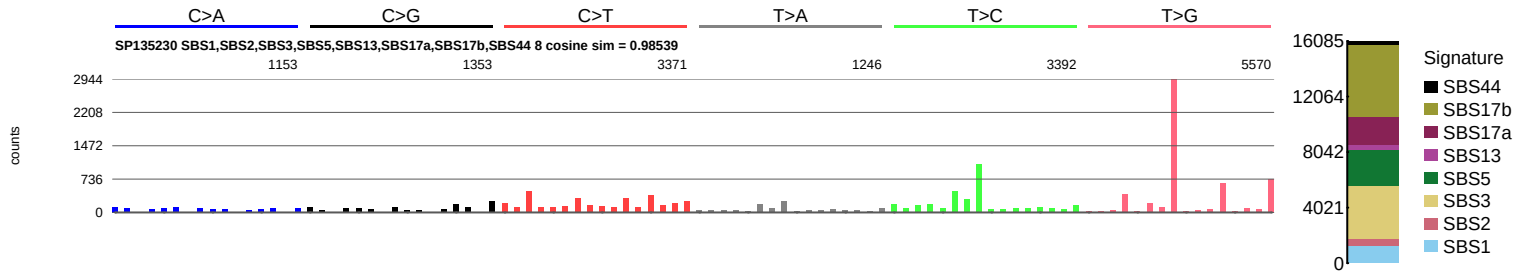
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98547
1123
1352
3291
1233
3514
5567
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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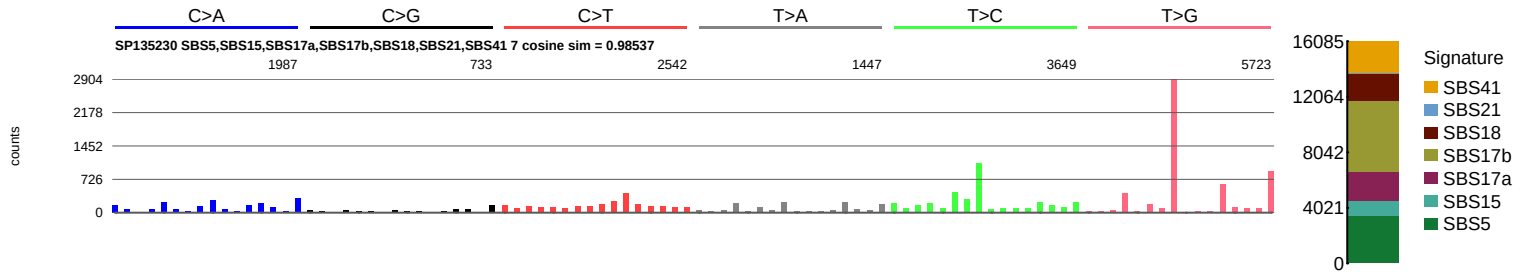
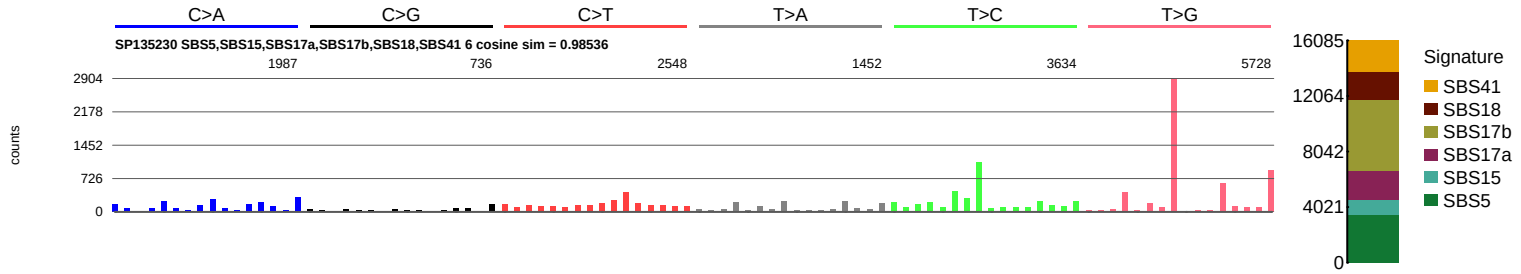
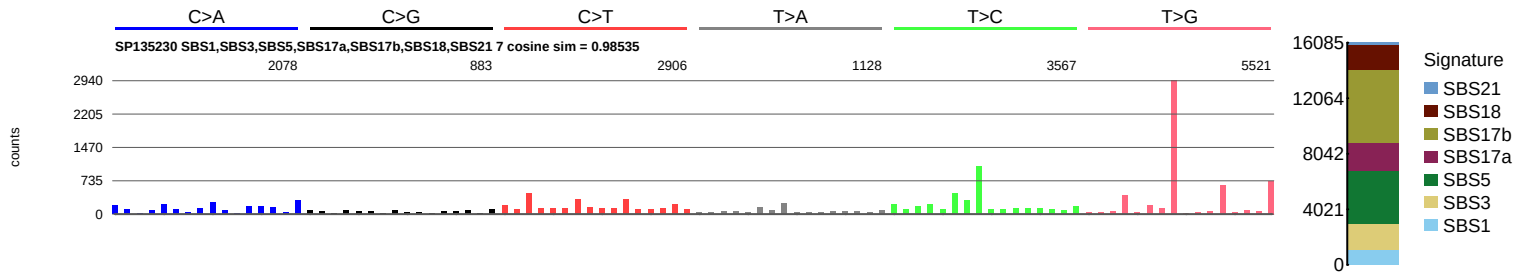
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98539
1153
1353
3371
1246
3392
5570
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98537
1987
733
2542
1447
3649
5723
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98536
1987
736
2548
1452
3634
5728
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98535
2078
883
2906
1128
3567
5521
counts
2940
2205
1470
735
0
16085
12064
8042
4021
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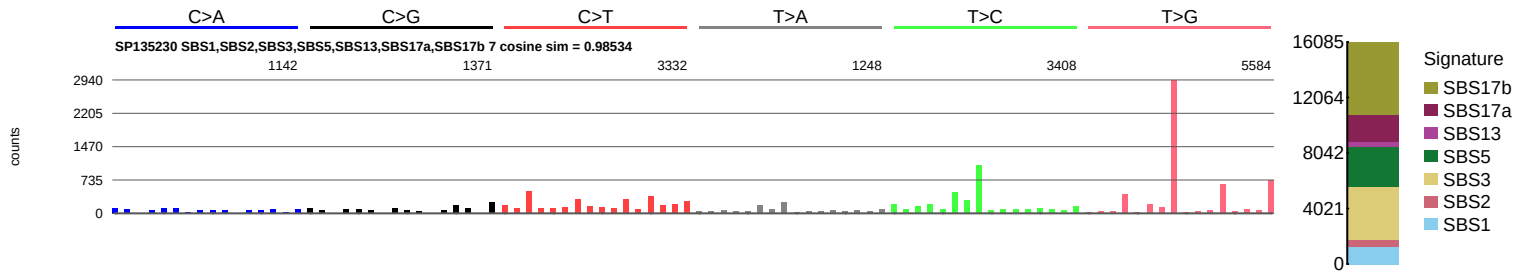
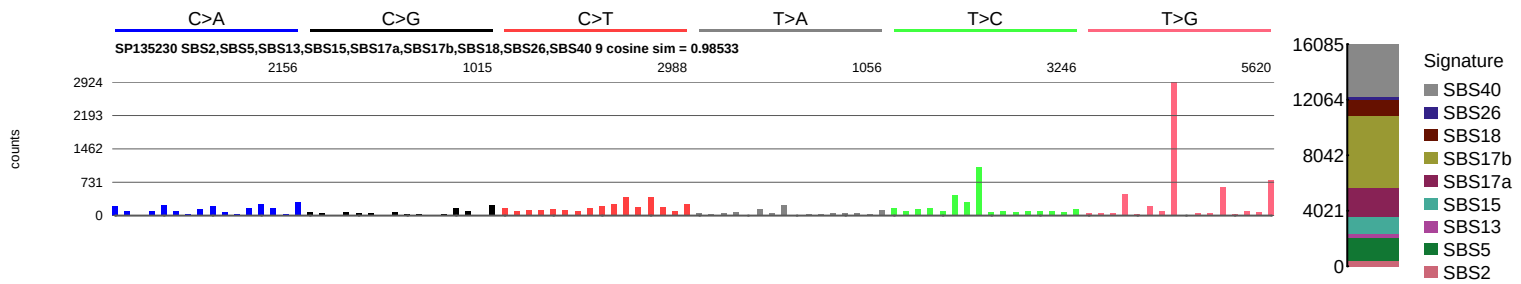
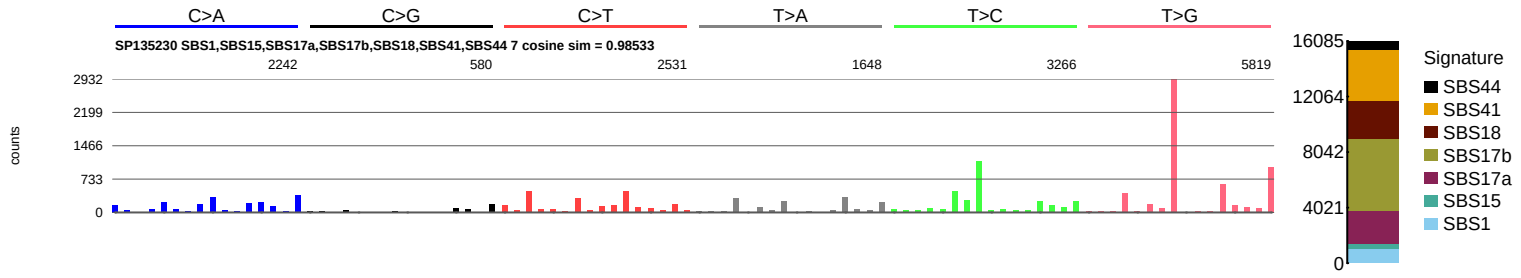
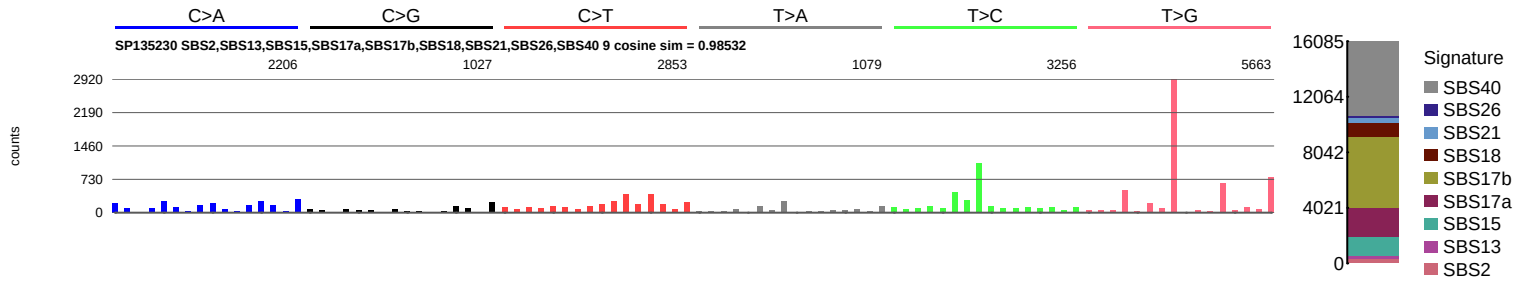
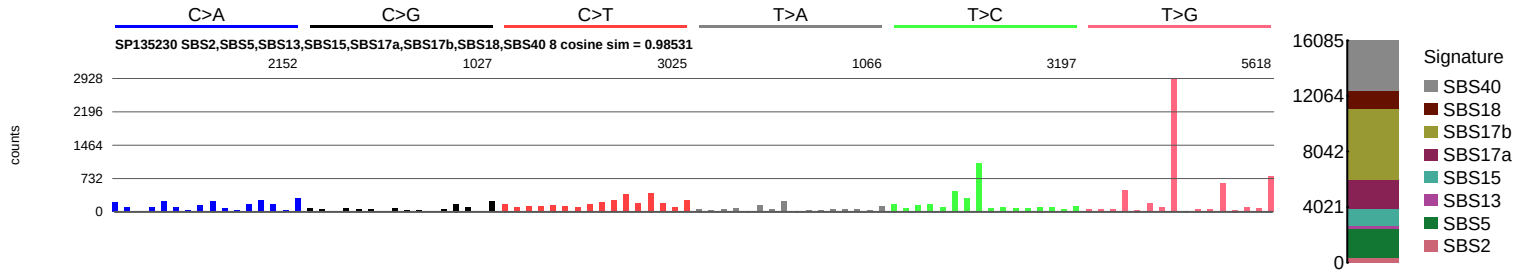
SBS3
SBS1

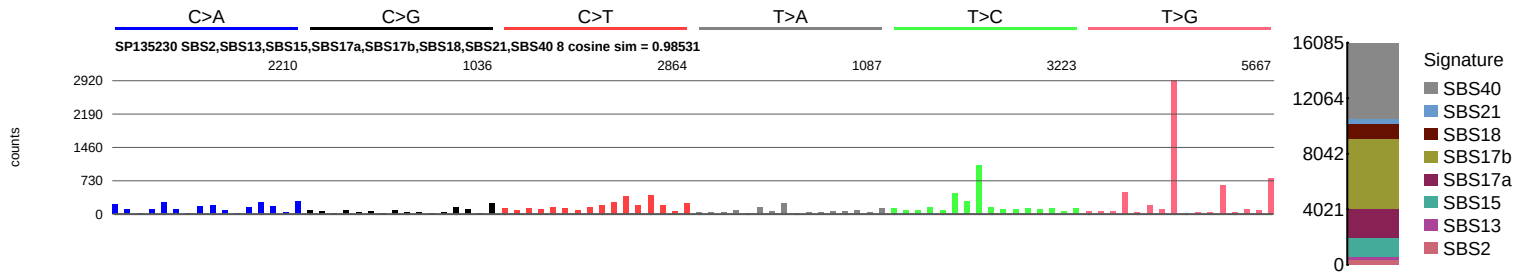

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b 7 cosine sim = 0.98534
1142
1371
3332
1248
3408
5584
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98533
2156
1015
2988
1056
3246
5620
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98533
2242
580
2531
1648
3266
5819
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98532
2206
1027
2853
1079
3256
5663
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98531
2152
1027
3025
1066
3197
5618
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98531
2210
1036
2864
1087
3223
5667
counts
2920
2190
1460
730
0
16085
12064
8042
4021
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Signature
SBS40
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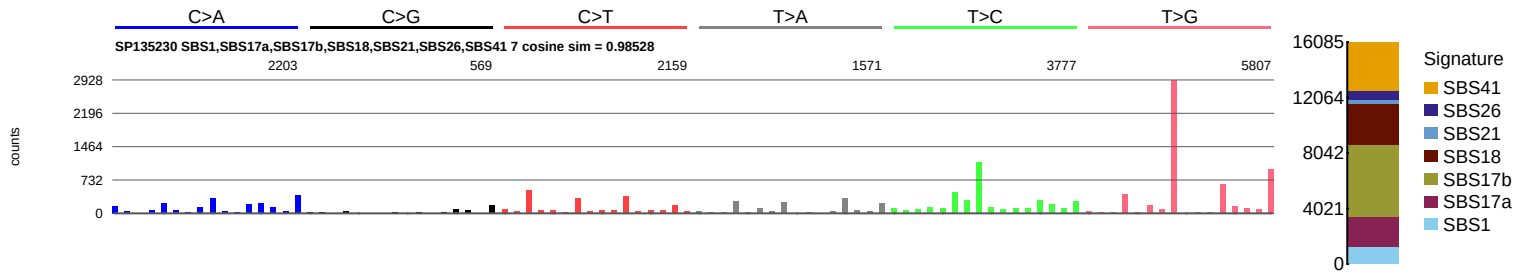
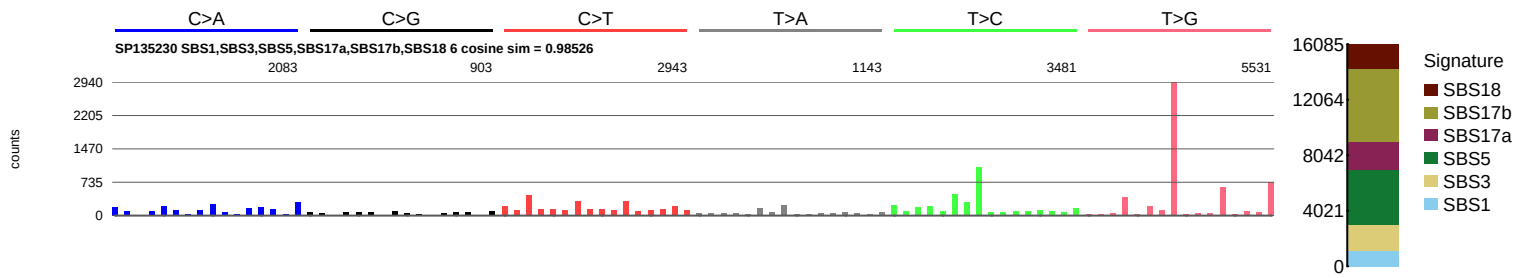
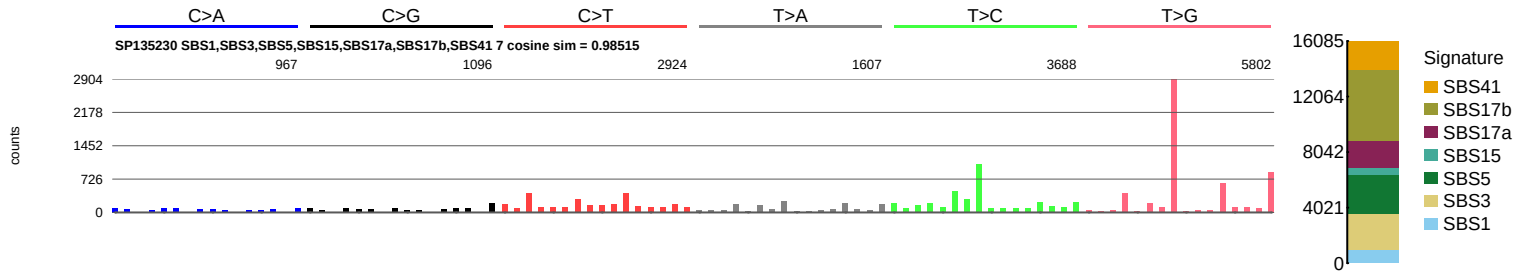
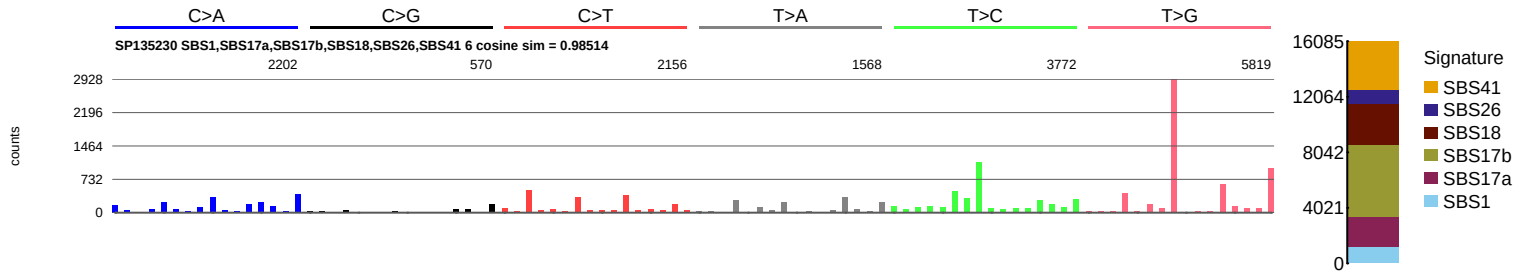
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.98528
2203
569
2159
1571
3777
5807
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98526
2083
903
2943
1143
3481
5531
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98515
967
1096
2924
1607
3688
5802
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98514
2202
570
2156
1568
3772
5819
counts
2928
2196
1464
732
0
16085
12064
8042
4021
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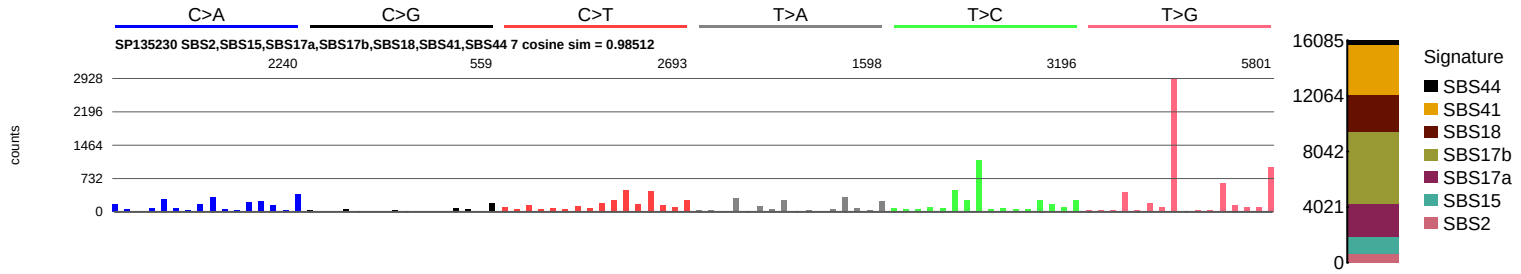
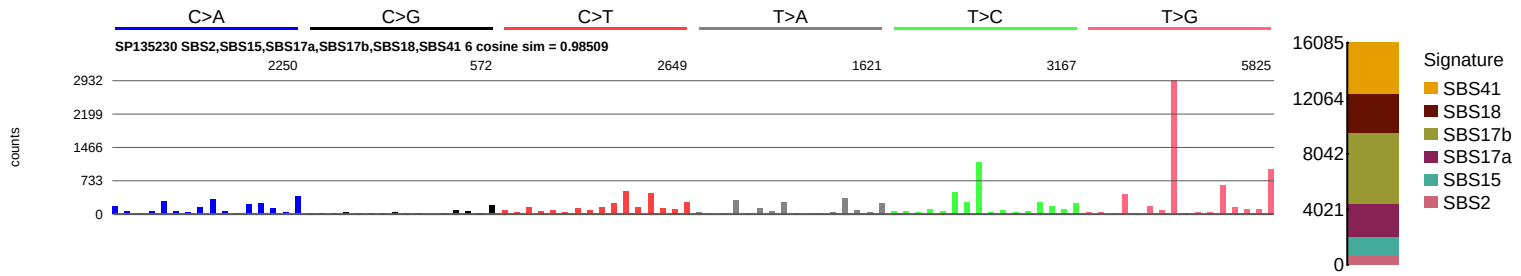
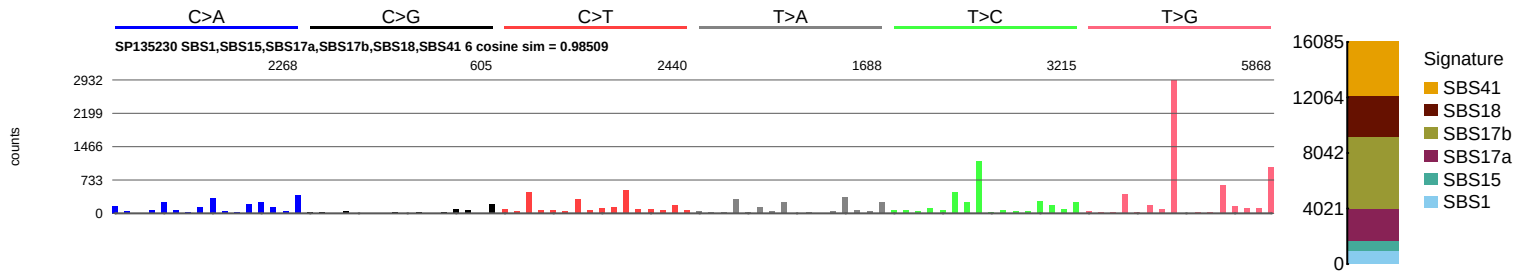
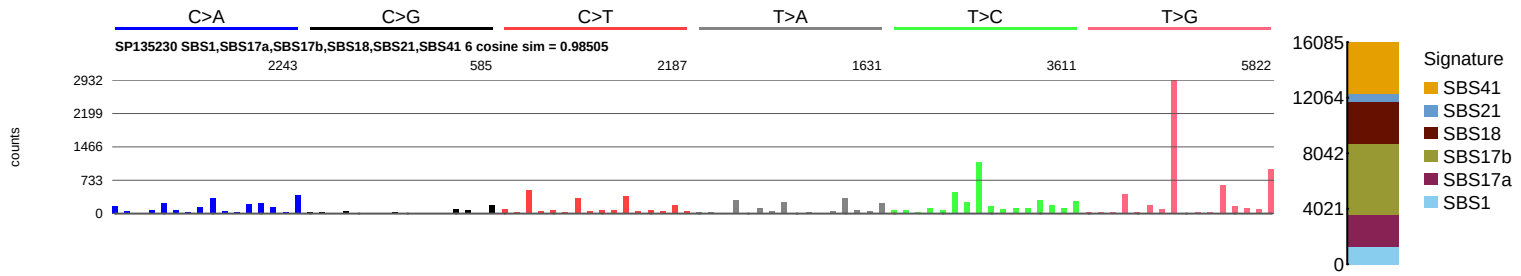
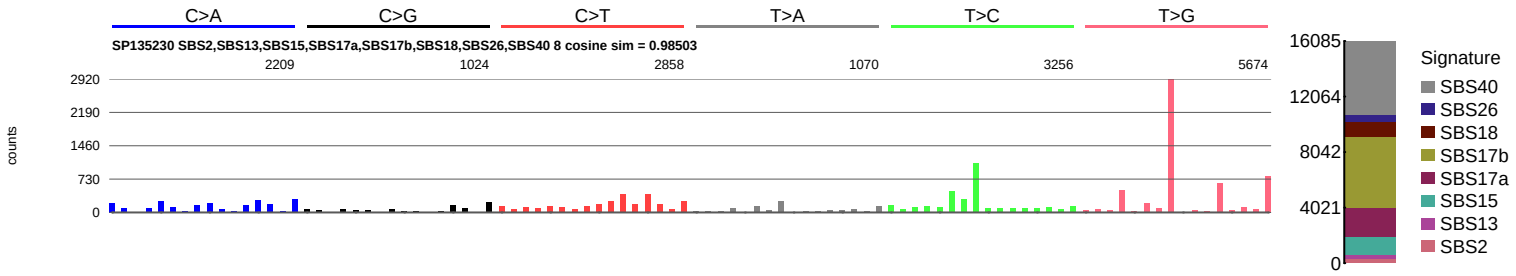
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1

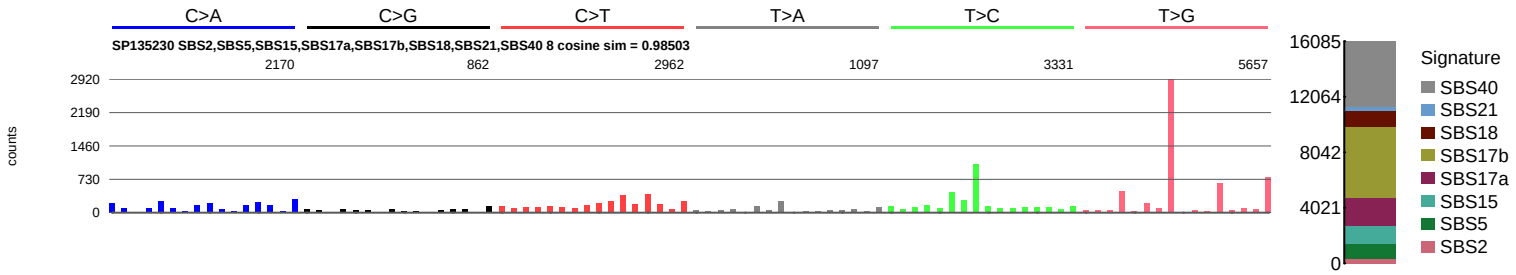

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98512
2240
559
2693
1598
3196
5801
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2

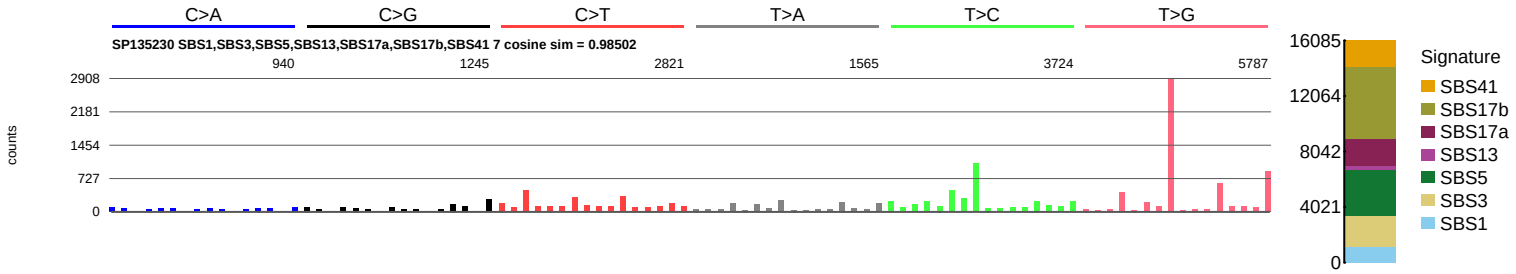

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98509
2250
572
2649
1621
3167
5825
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2

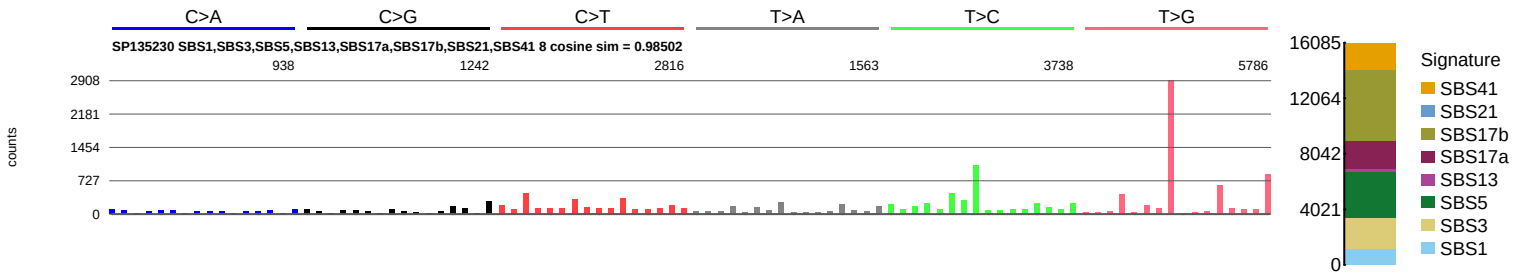

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98509
2268
605
2440
1688
3215
5868
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS1

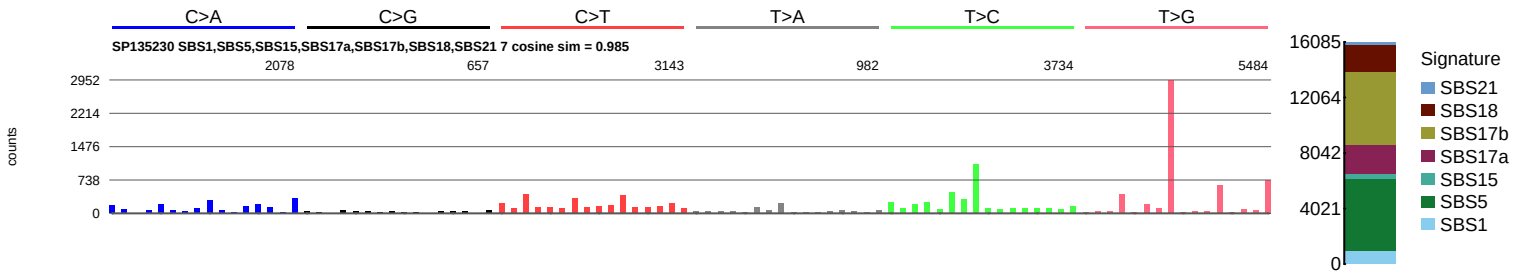

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98505
2243
585
2187
1631
3611
5822
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1

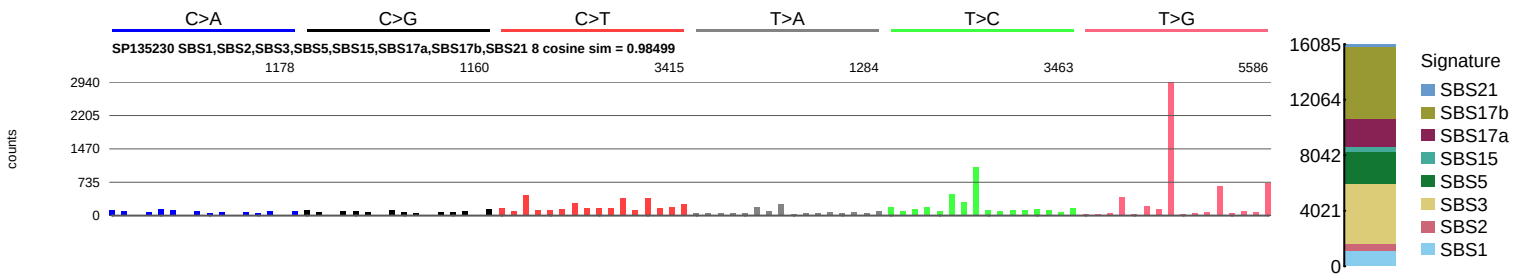

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98503
2209
1024
2858
1070
3256
5674
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98503
2170
862
2962
1097
3331
5657
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98502
940
1245
2821
1565
3724
5787
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98502
938
1242
2816
1563
3738
5786
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.985
2078
657
3143
982
3734
5484
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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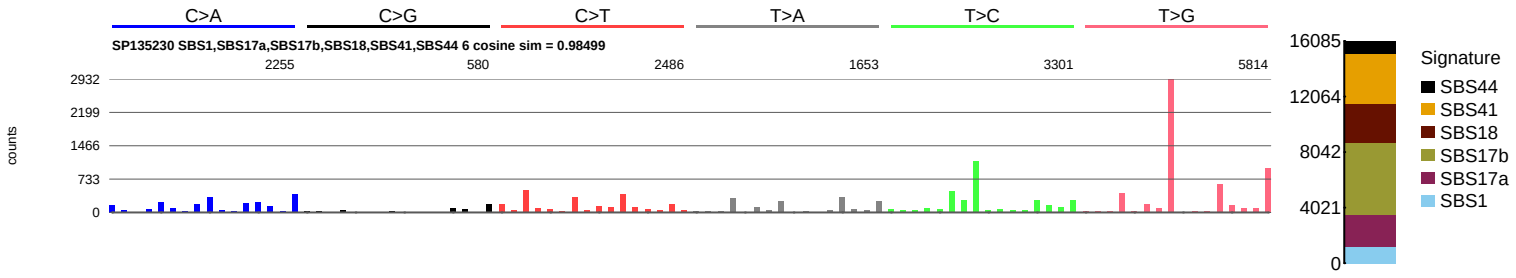
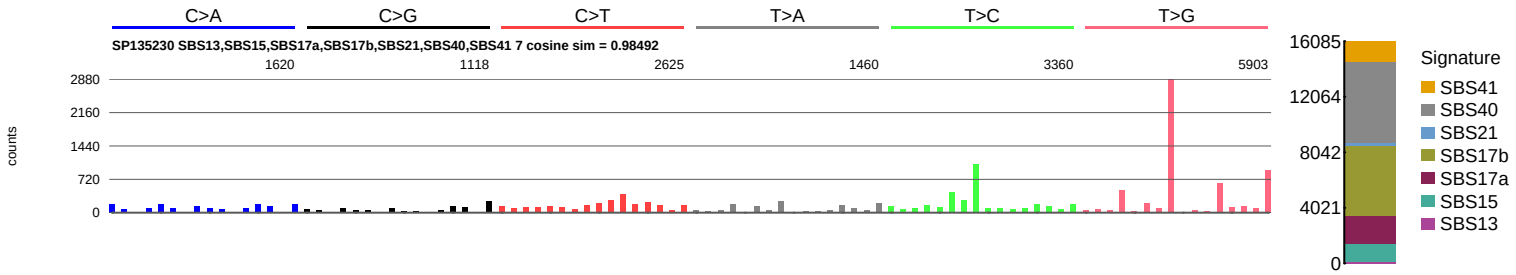
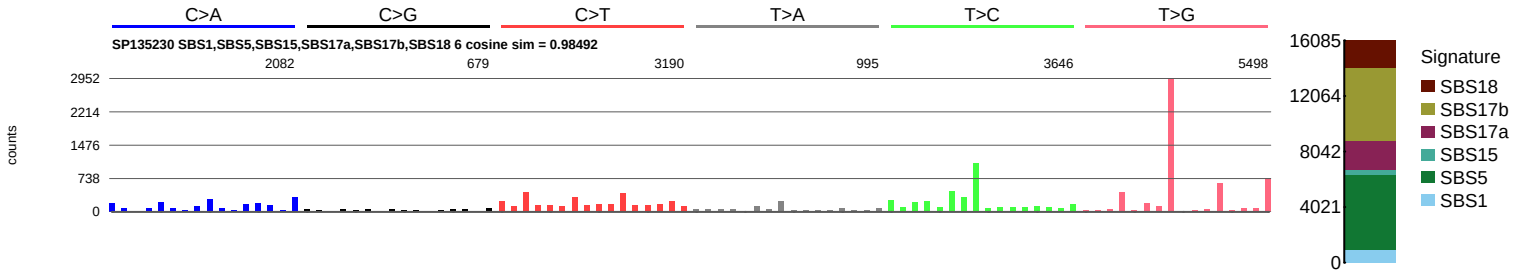
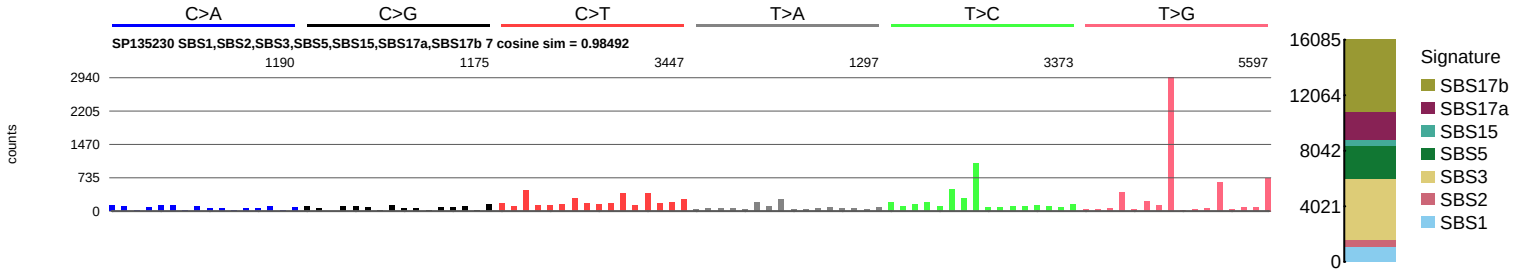
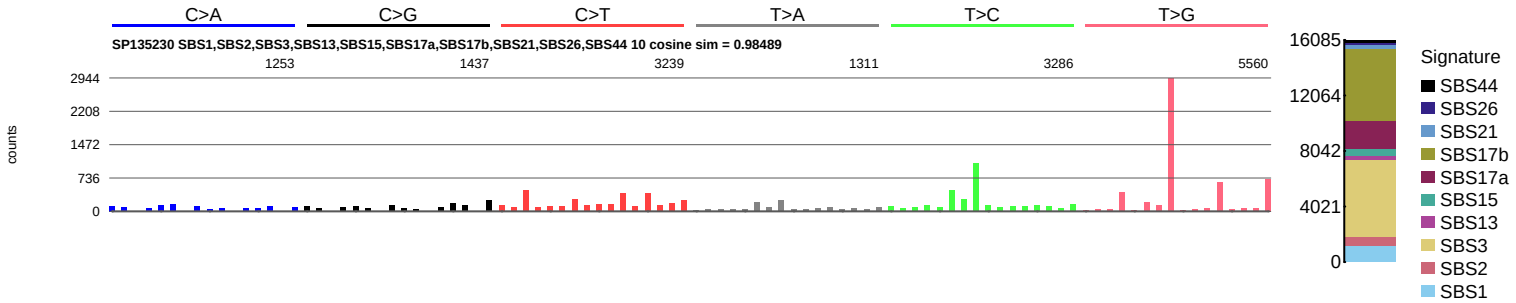
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98499
1178
1160
3415
1284
3463
5586
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

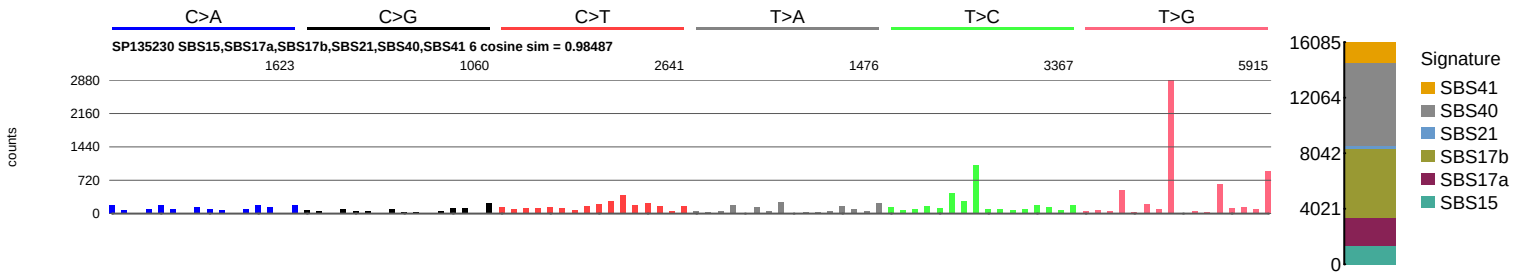

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98499
2255
580
2486
1653
3301
5814
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS1

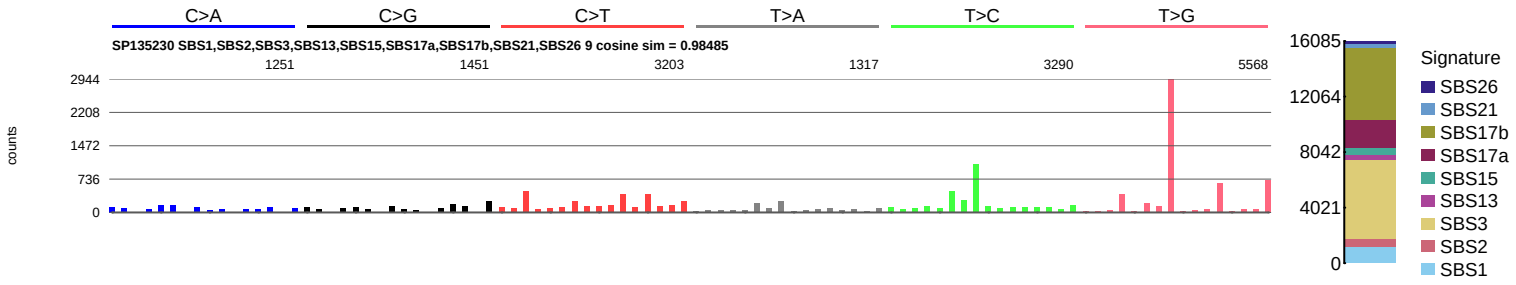

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98492
1620
1118
2625
1460
3360
5903
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13

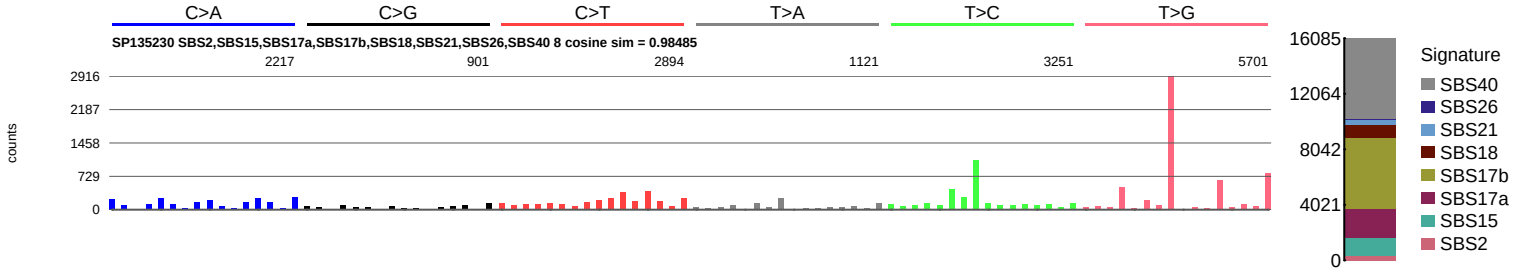

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98492
2082
679
3190
995
3646
5498
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

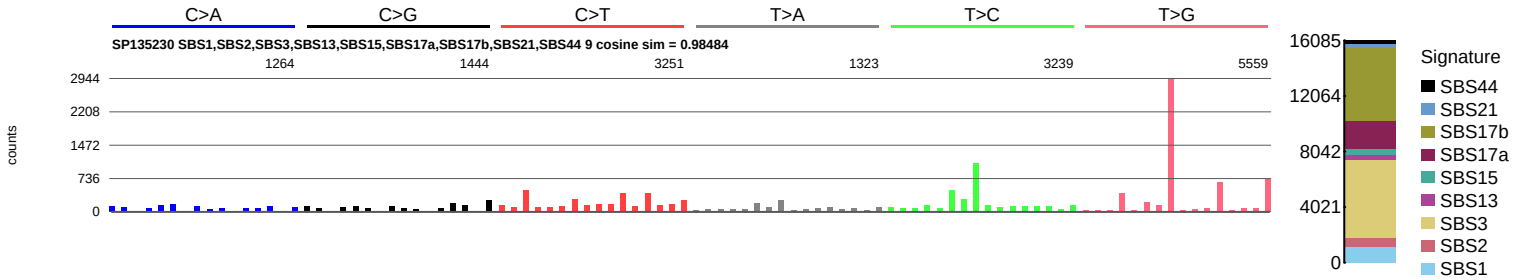

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98492
1190
1175
3447
1297
3373
5597
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

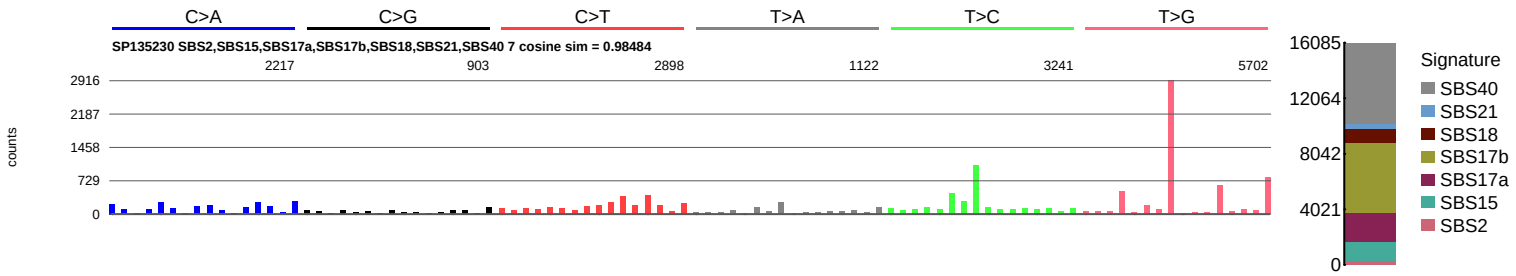

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 10 cosine sim = 0.98489
1253
1437
3239
1311
3286
5560
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

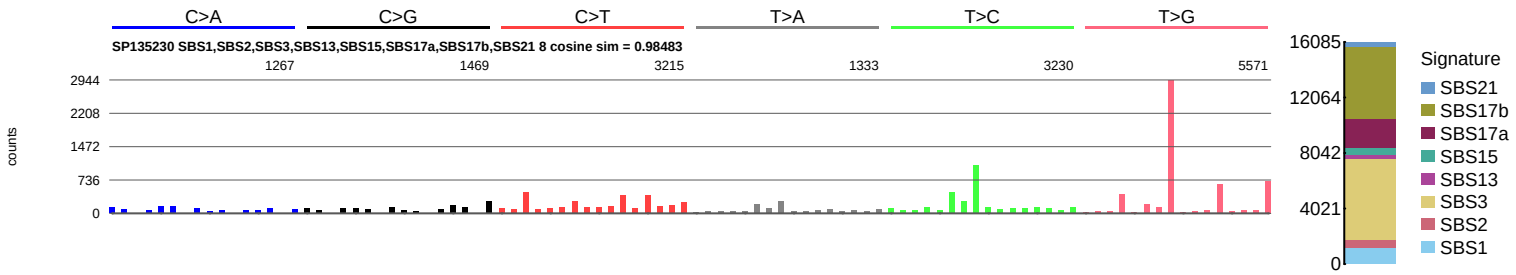

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98487
1623
1060
2641
1476
3367
5915
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15

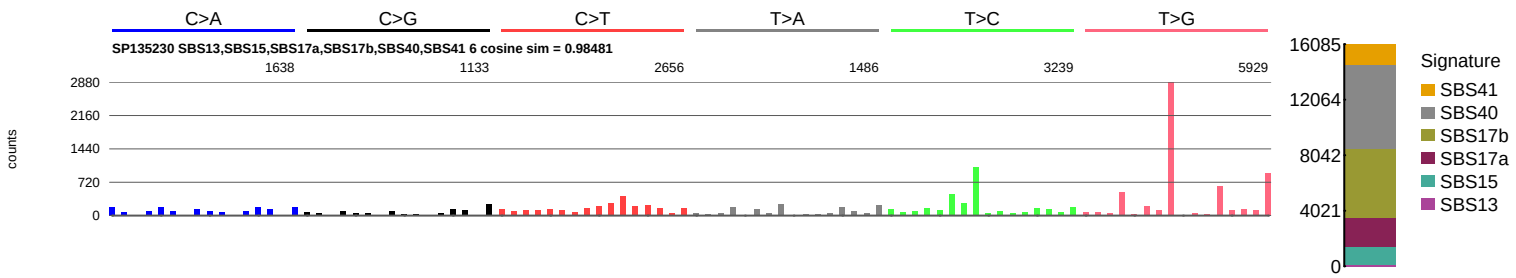

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26 9 cosine sim = 0.98485
1251
1451
3203
1317
3290
5568
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

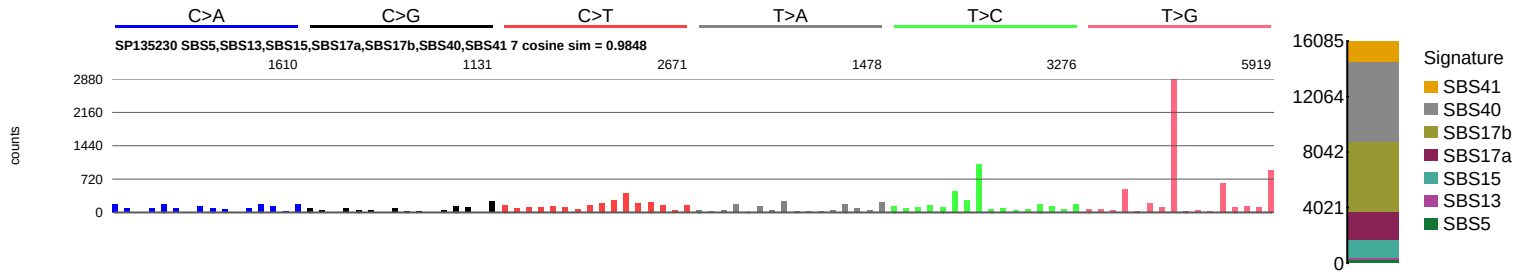

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98485
2217
901
2894
1121
3251
5701
counts
2916
2187
1458
729
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2

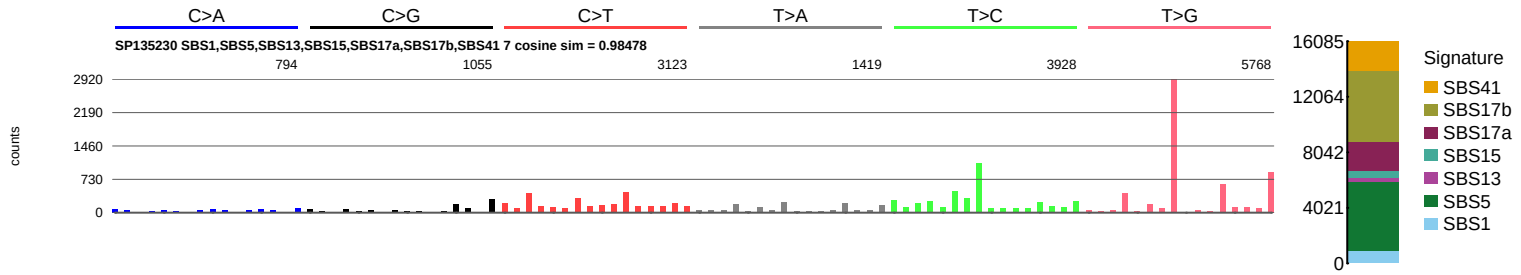

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98484
1264
1444
3251
1323
3239
5559
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

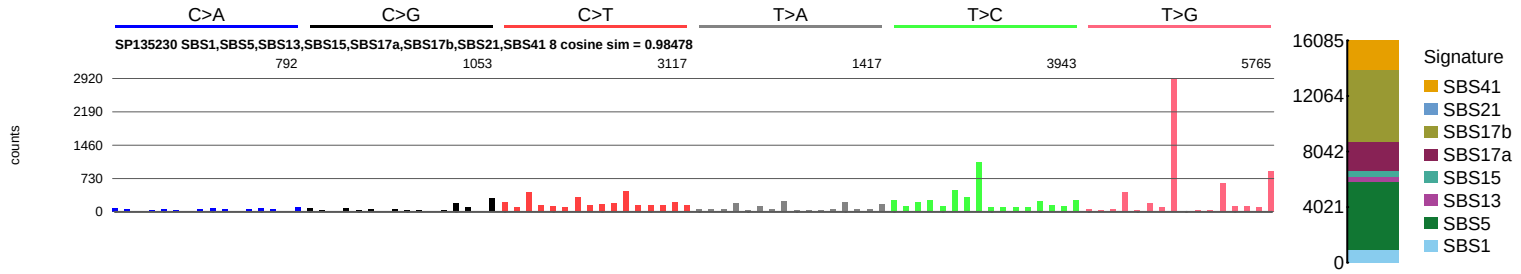

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98484
2217
903
2898
1122
3241
5702
counts
2916
2187
1458
729
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2

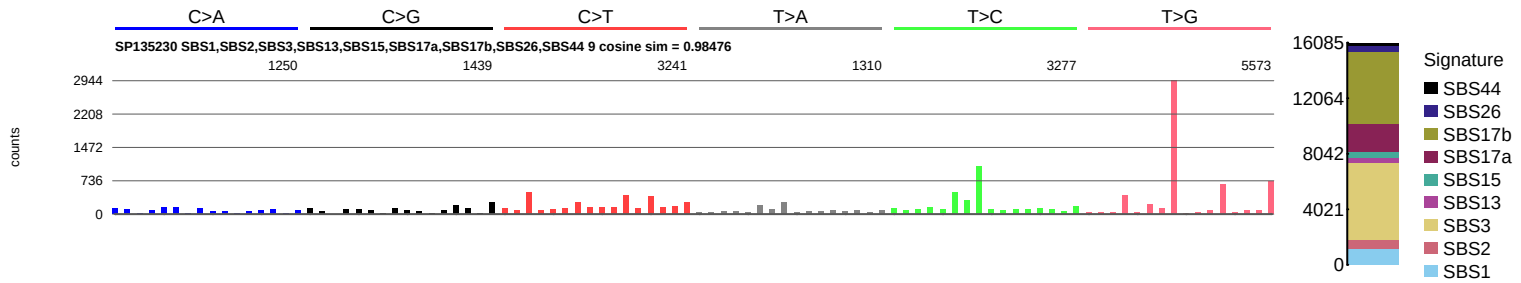

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98483
1267
1469
3215
1333
3230
5571
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

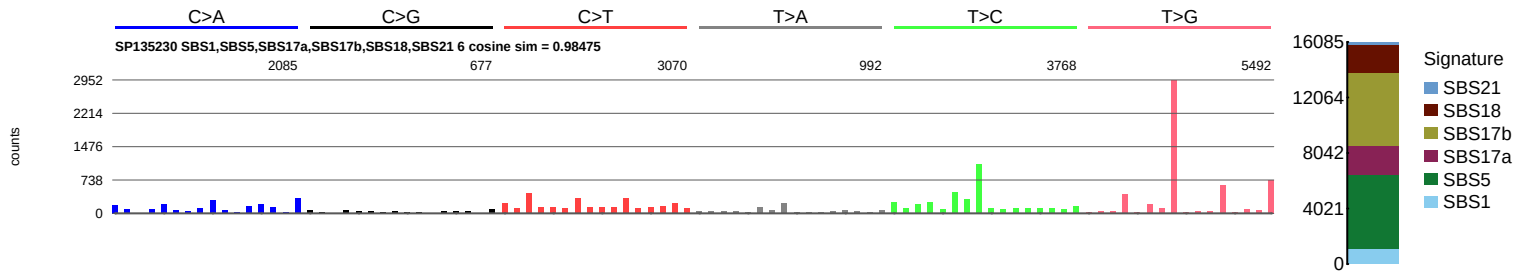

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98481
1638
1133
2656
1486
3239
5929
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13

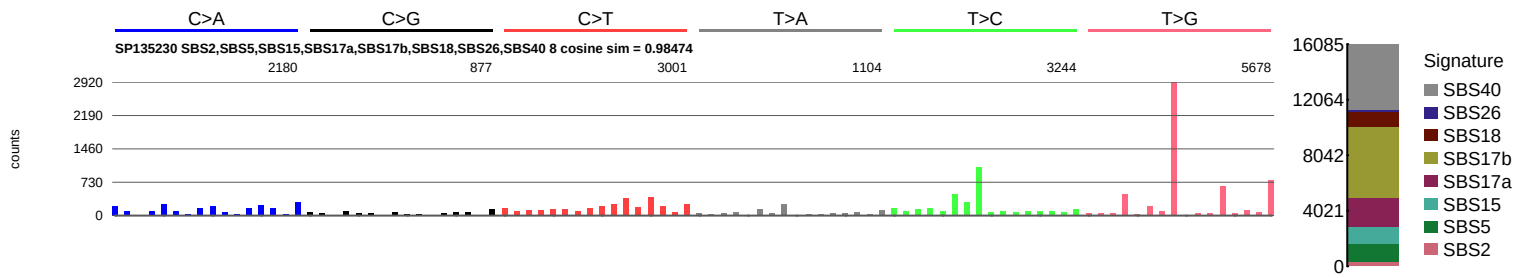

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.9848
1610
1131
2671
1478
3276
5919
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5

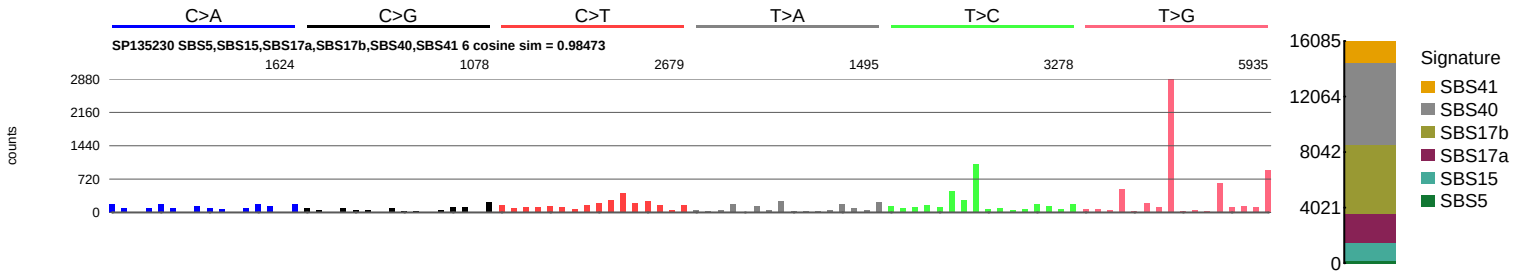

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98478
794
1055
3123
1419
3928
5768
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

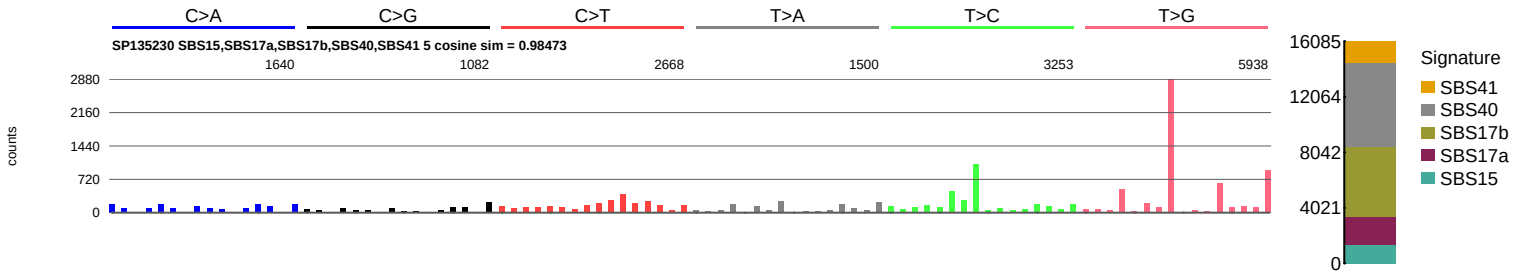

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98478
792
1053
3117
1417
3943
5765
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

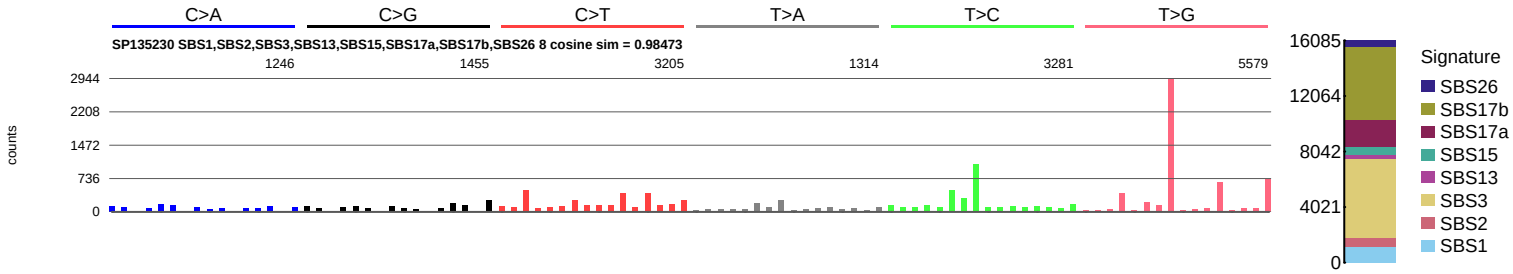

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98476
1250
1439
3241
1310
3277
5573
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

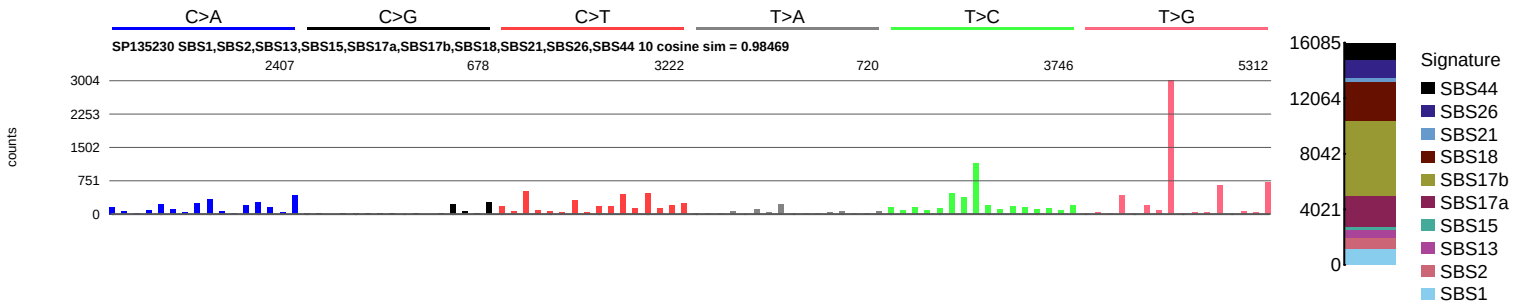

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98475
2085
677
3070
992
3768
5492
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

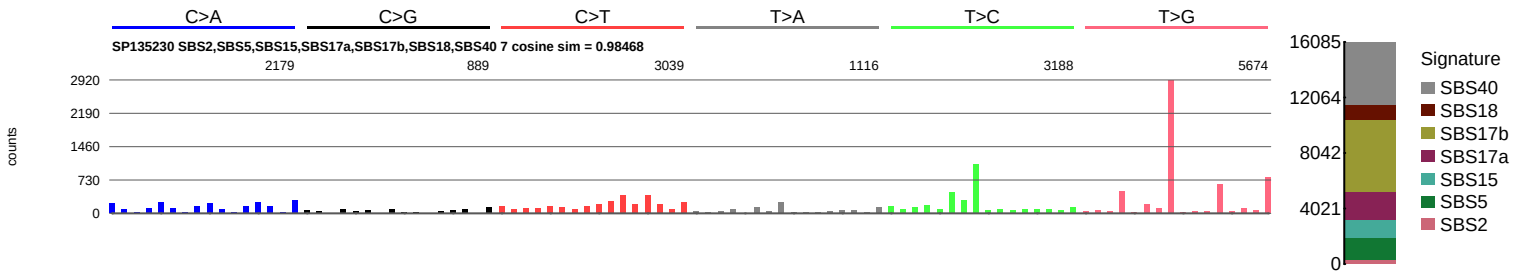

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98474
2180
877
3001
1104
3244
5678
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

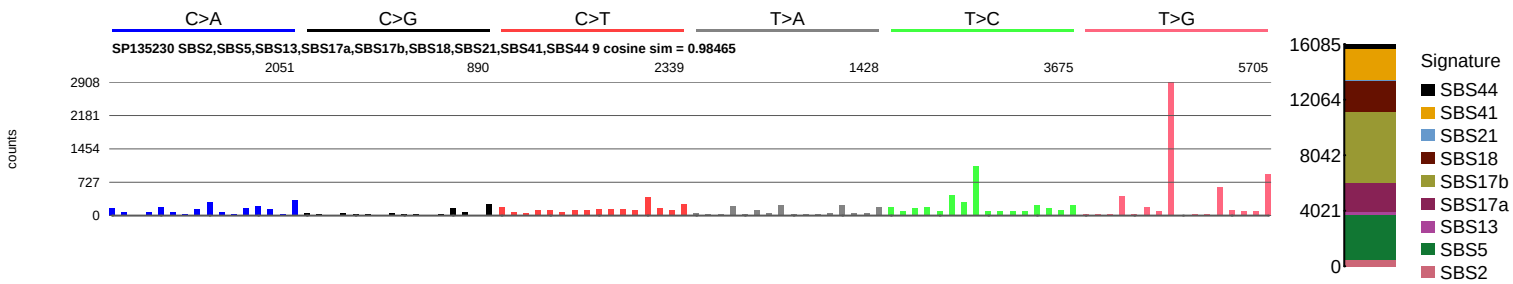

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98473
1624
1078
2679
1495
3278
5935
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5

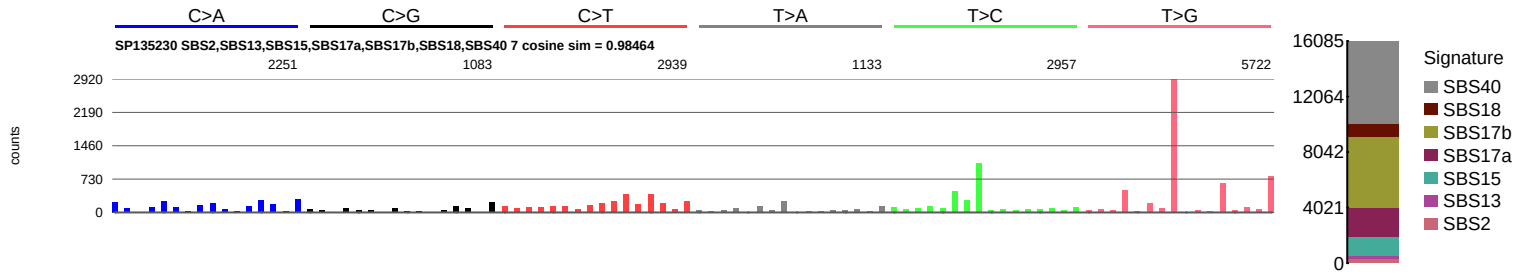

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98473
1640
1082
2668
1500
3253
5938
counts
2880
2160
1440
720
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98473
1246
1455
3205
1314
3281
5579
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98469
2407
678
3222
720
3746
5312
counts
3004
2253
1502
751
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98468
2179
889
3039
1116
3188
5674
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98465
2051
890
2339
1428
3675
5705
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
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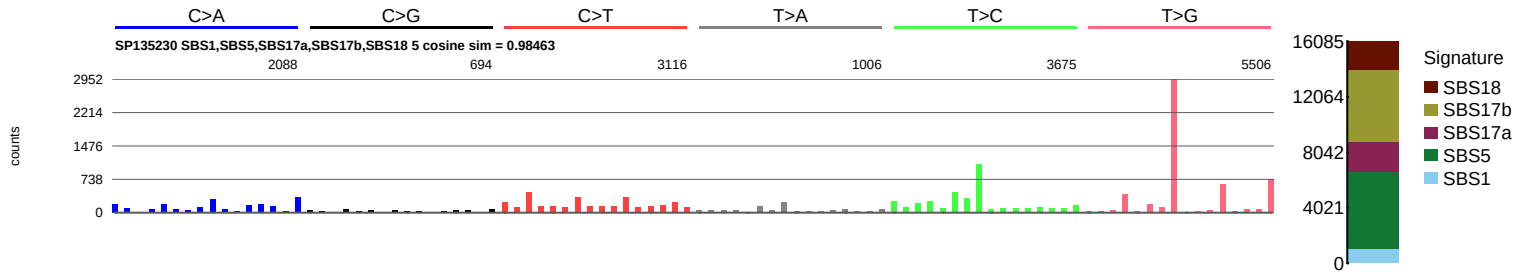
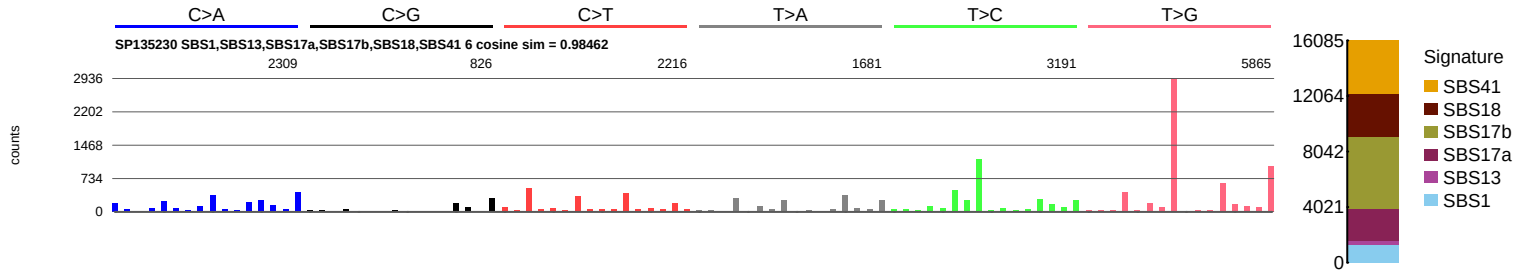
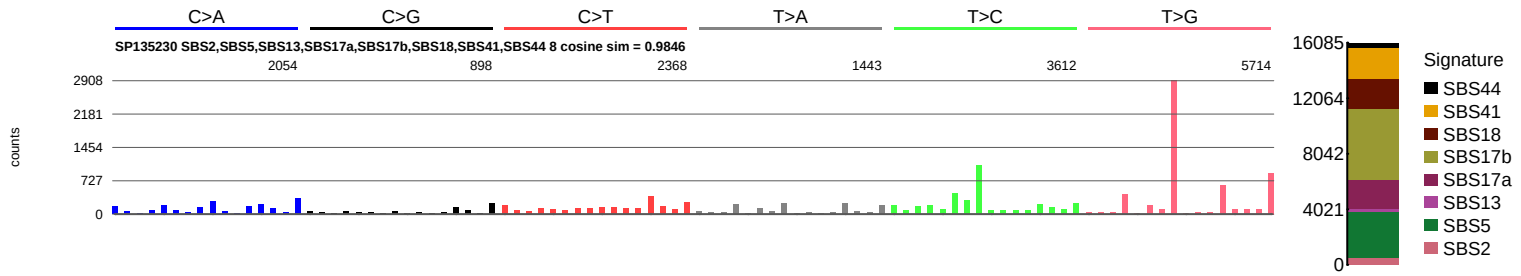
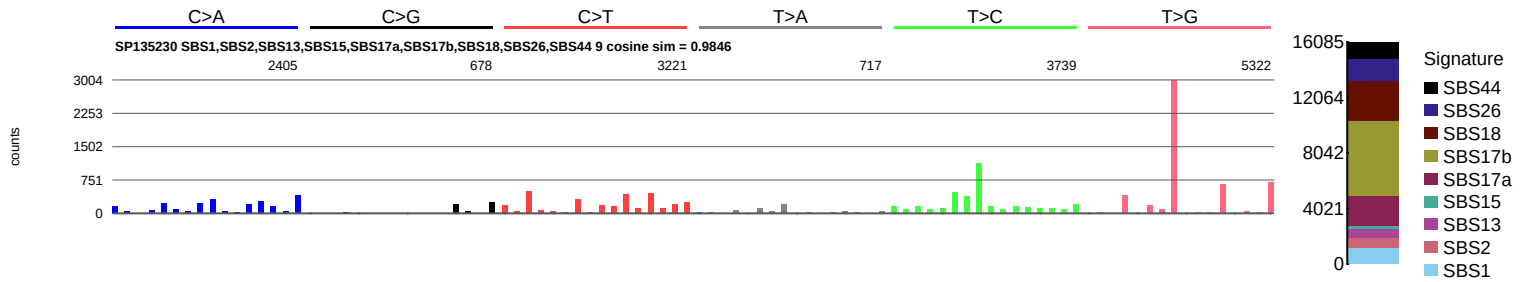
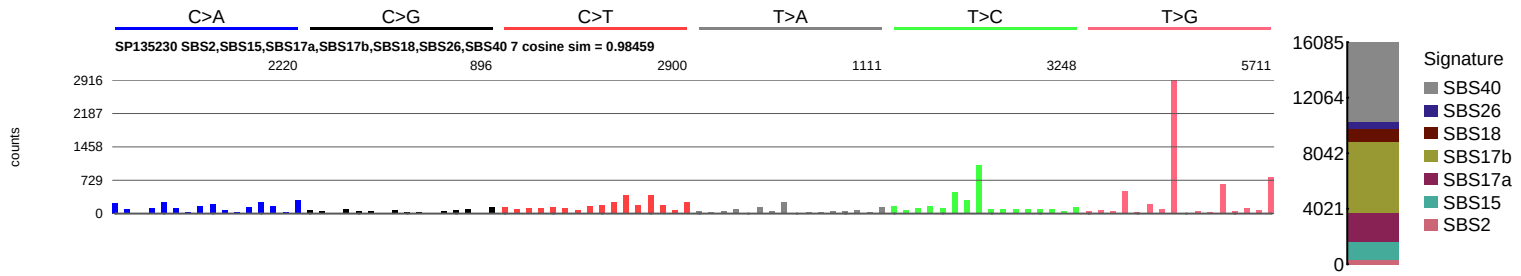
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98464
2251
1083
2939
1133
2957
5722
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

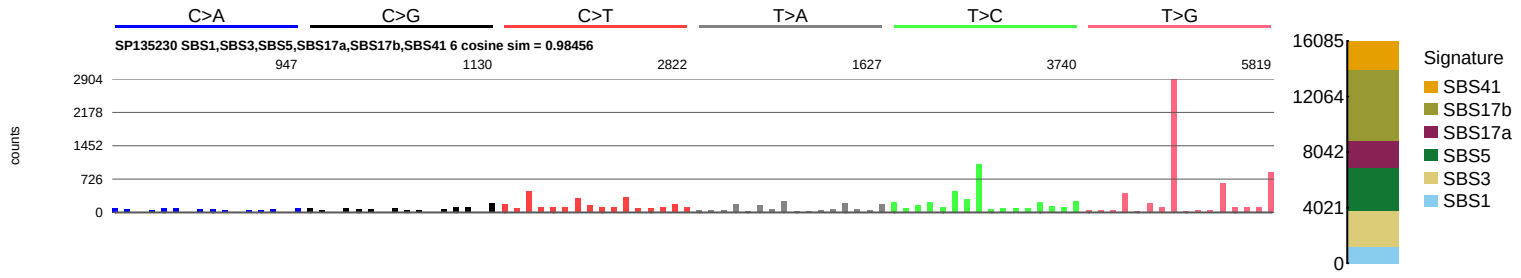

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98463
2088
694
3116
1006
3675
5506
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS1

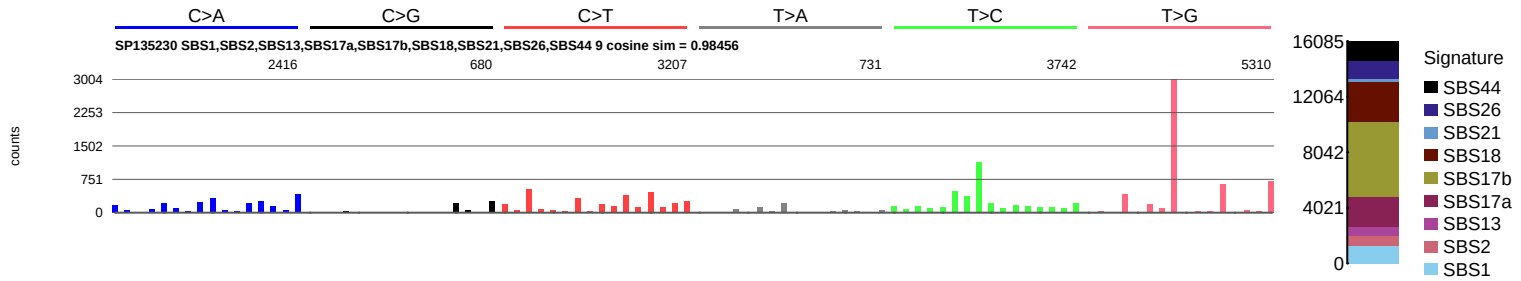

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98462
2309
826
2216
1681
3191
5865
counts
2936
2202
1468
734
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS1

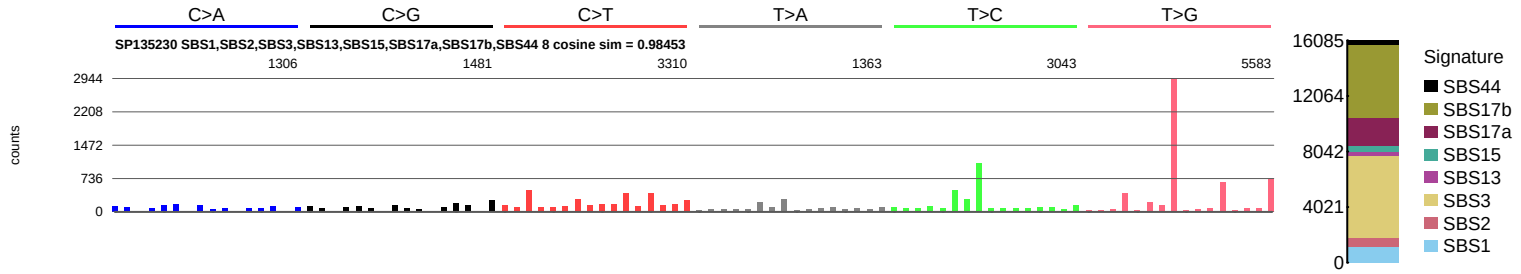

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.9846
2054
898
2368
1443
3612
5714
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

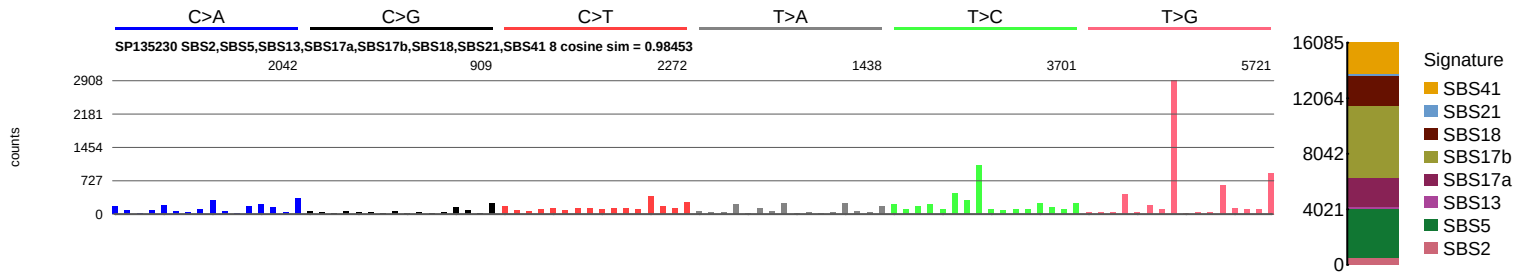

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.9846
2405
678
3221
717
3739
5322
counts
3004
2253
1502
751
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

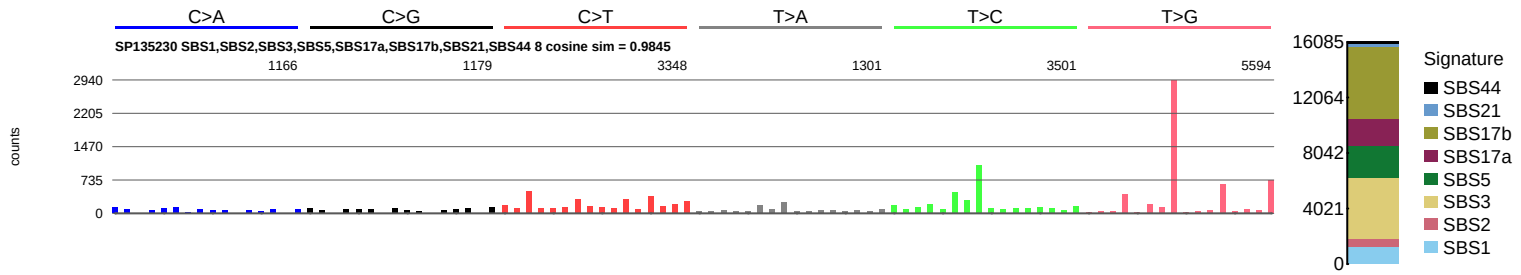

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98459
2220
896
2900
1111
3248
5711
counts
2916
2187
1458
729
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2

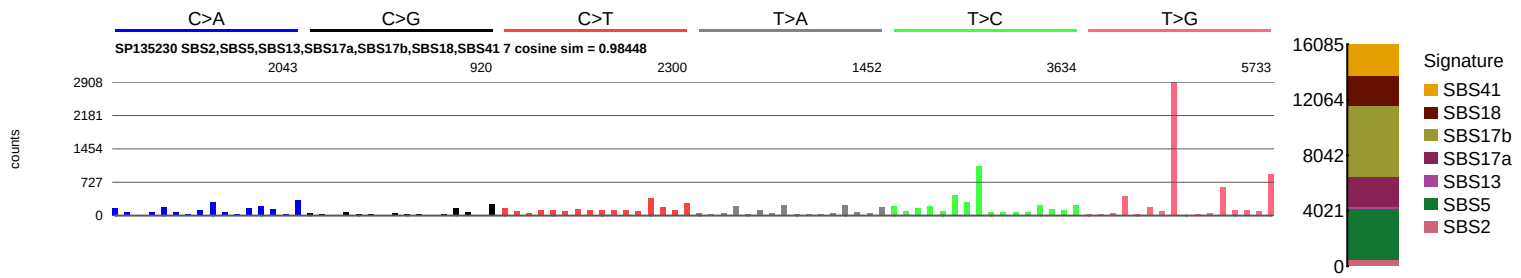

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98456
947
1130
2822
1627
3740
5819
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS1

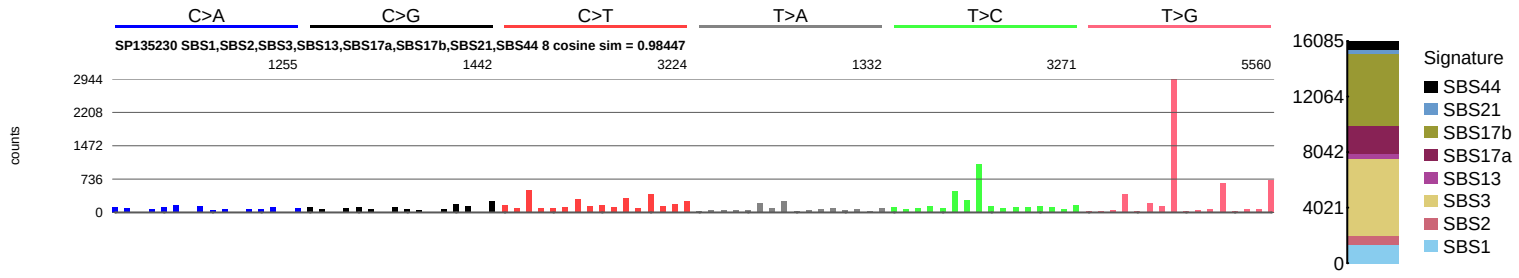

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98456
2416
680
3207
731
3742
5310
counts
3004
2253
1502
751
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

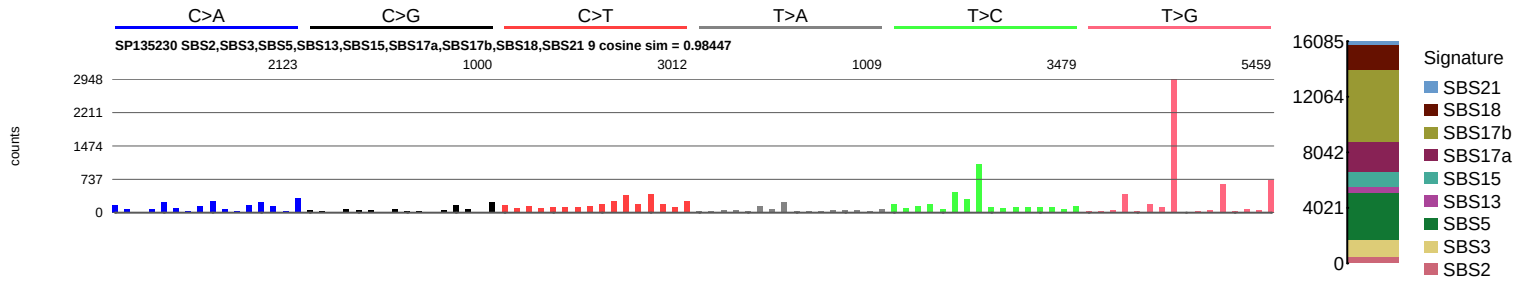

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98453
1306
1481
3310
1363
3043
5583
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

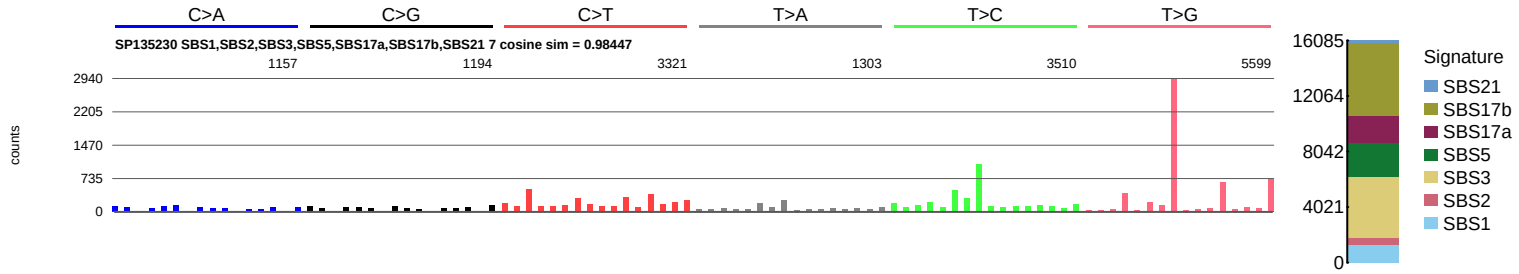

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98453
2042
909
2272
1438
3701
5721
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

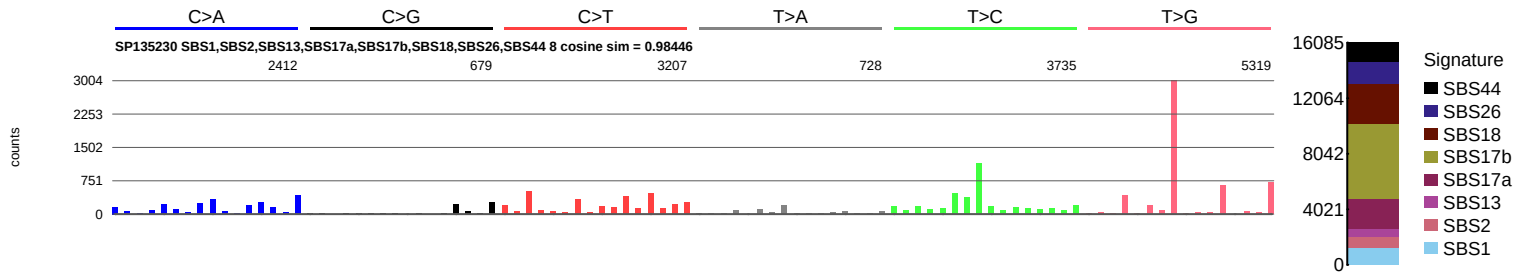

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.9845
1166
1179
3348
1301
3501
5594
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

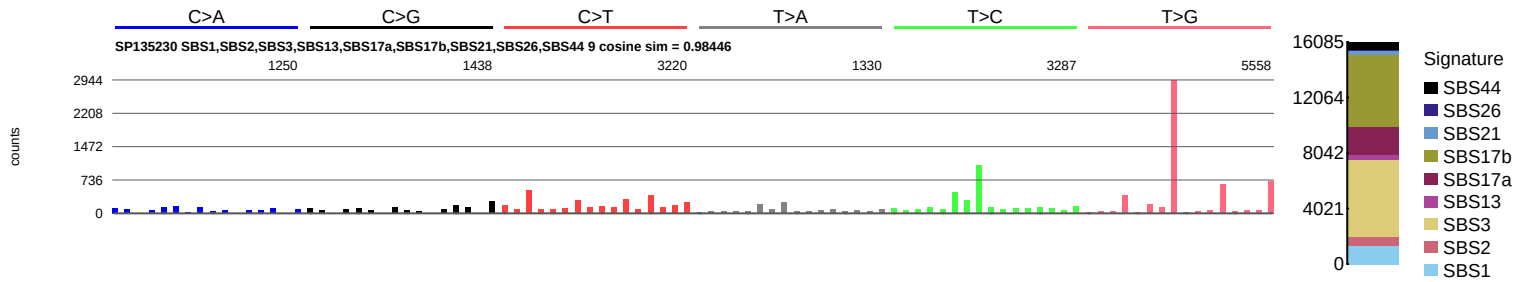

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98448
2043
920
2300
1452
3634
5733
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

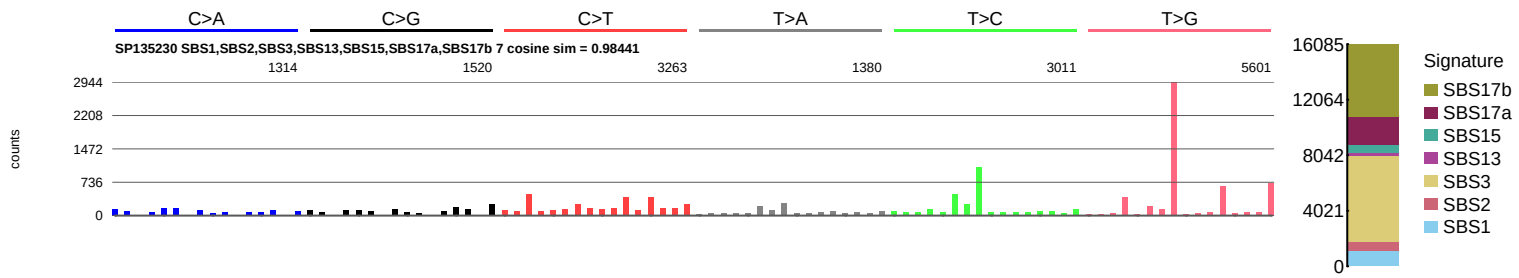

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98447
1255
1442
3224
1332
3271
5560
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

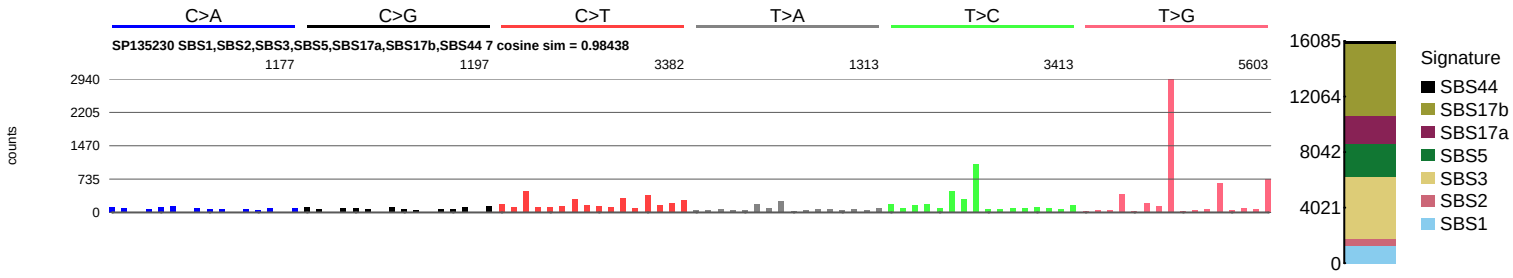

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98447
2123
1000
3012
1009
3479
5459
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

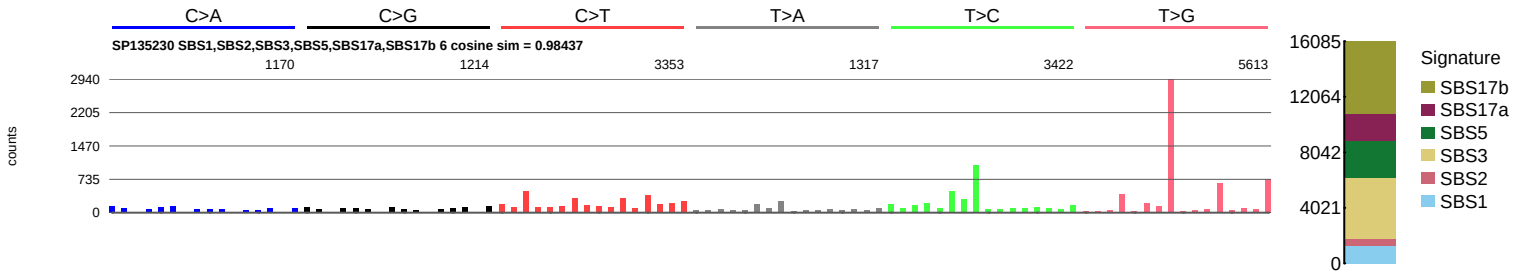

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98447
1157
1194
3321
1303
3510
5599
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98446
2412
679
3207
728
3735
5319
counts
3004
2253
1502
751
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.98446
1250
1438
3220
1330
3287
5558
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98441
1314
1520
3263
1380
3011
5601
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98438
1177
1197
3382
1313
3413
5603
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b 6 cosine sim = 0.98437
1170
1214
3353
1317
3422
5613
counts
2940
2205
1470
735
0
16085
12064
8042
4021
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Signature
SBS17b
SBS17a
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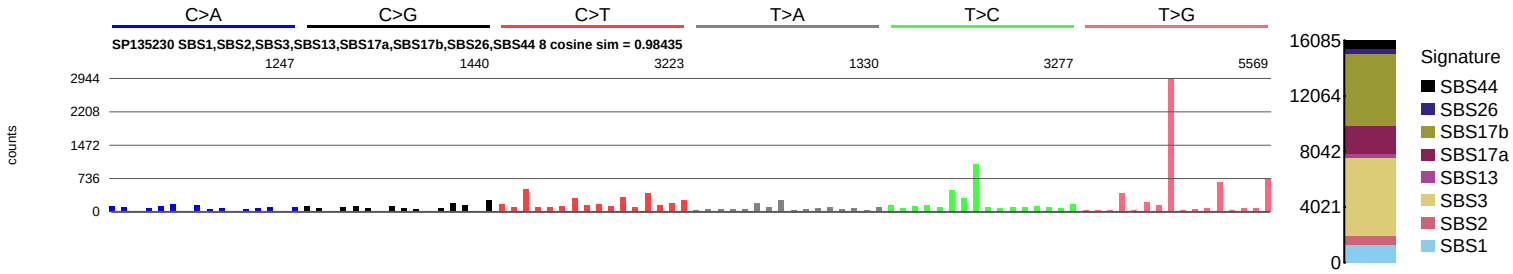
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98435
1247
1440
3223
1330
3277
5569
counts
2944
2208
1472
736
0
16085
12064
8042
4021
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Signature
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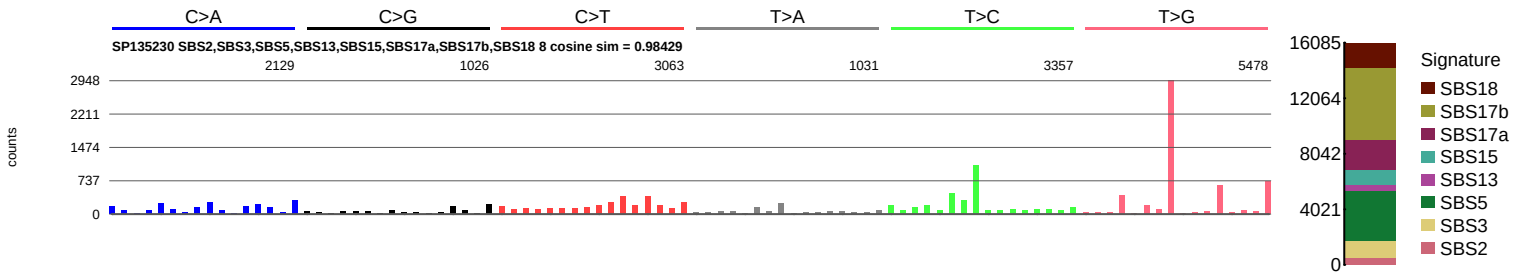
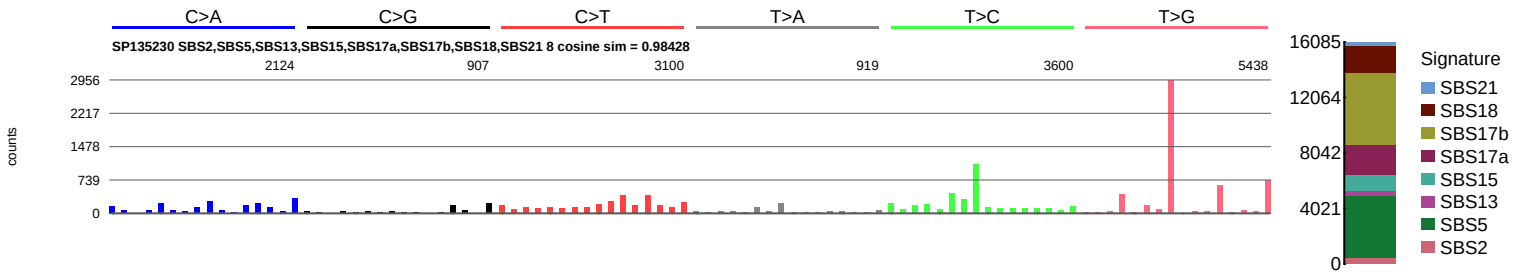
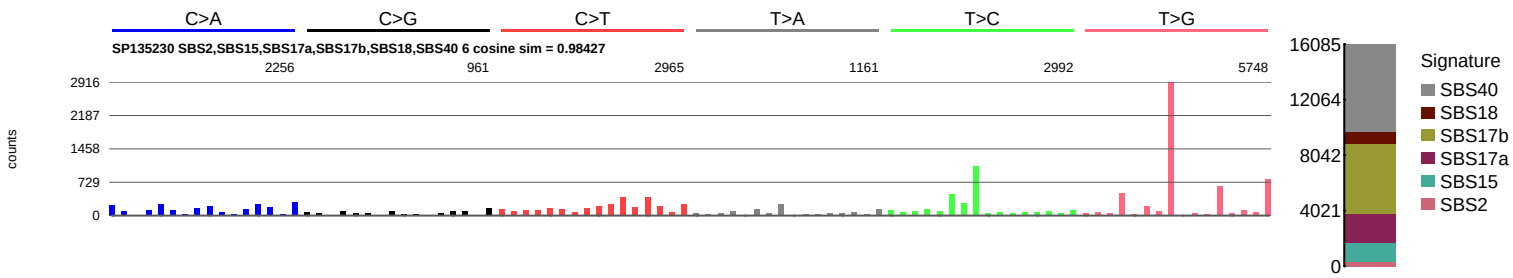
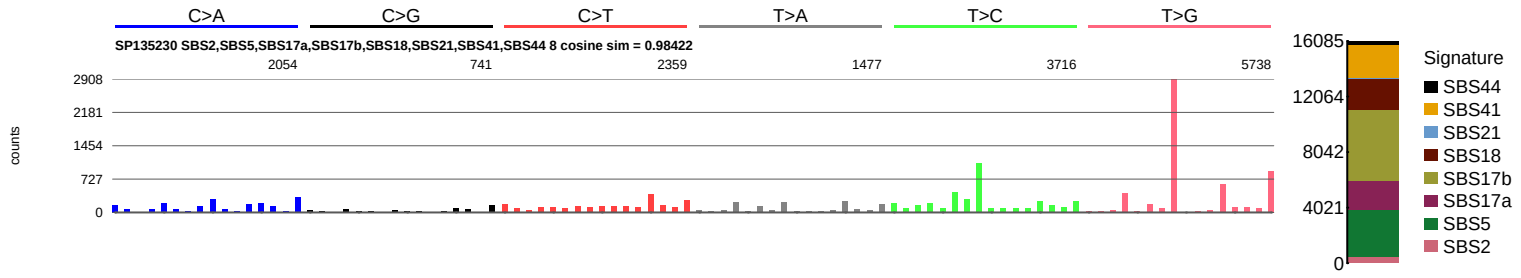
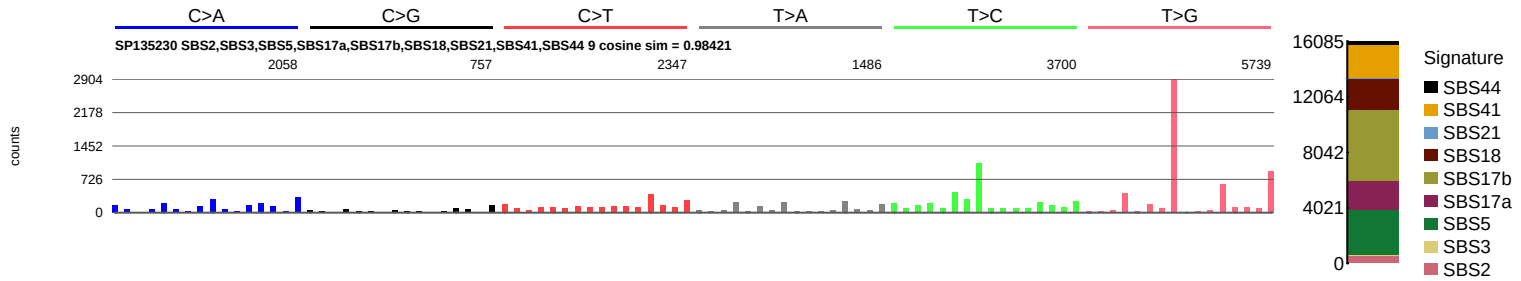
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98429
2129
1026
3063
1031
3357
5478
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98428
2124
907
3100
919
3600
5438
counts
2956
2217
1478
739
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98427
2256
961
2965
1161
2992
5748
counts
2916
2187
1458
729
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98422
2054
741
2359
1477
3716
5738
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2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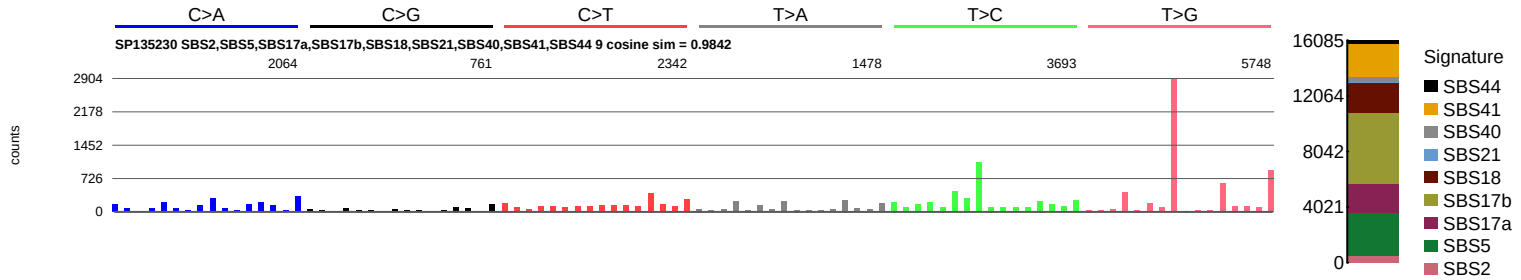
SP135230 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98421
2058
757
2347
1486
3700
5739
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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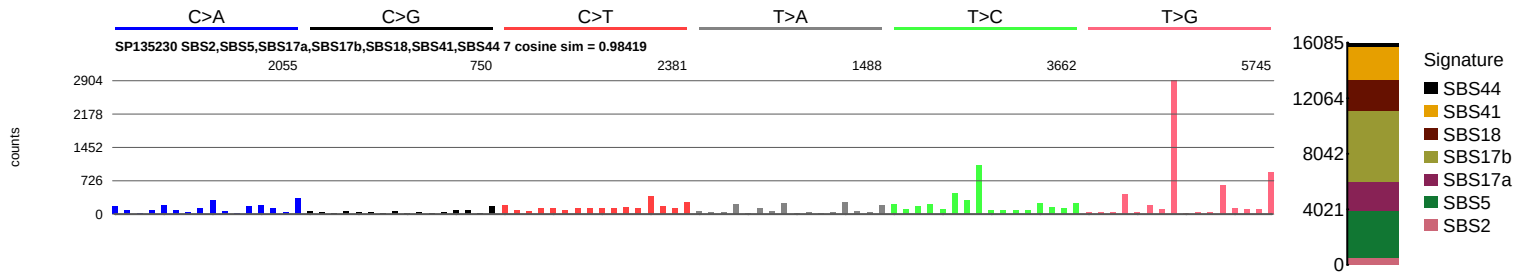
SP135230 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9842
2064
761
2342
1478
3693
5748
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2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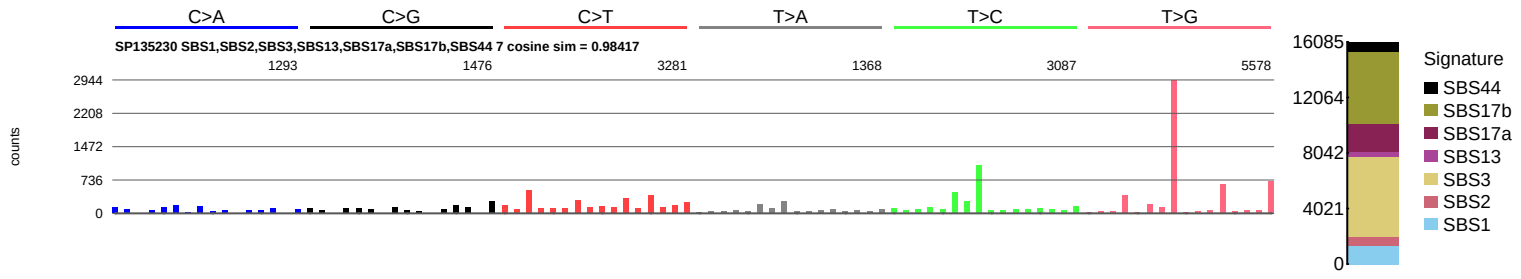
SP135230 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98419
2055
750
2381
1488
3662
5745
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2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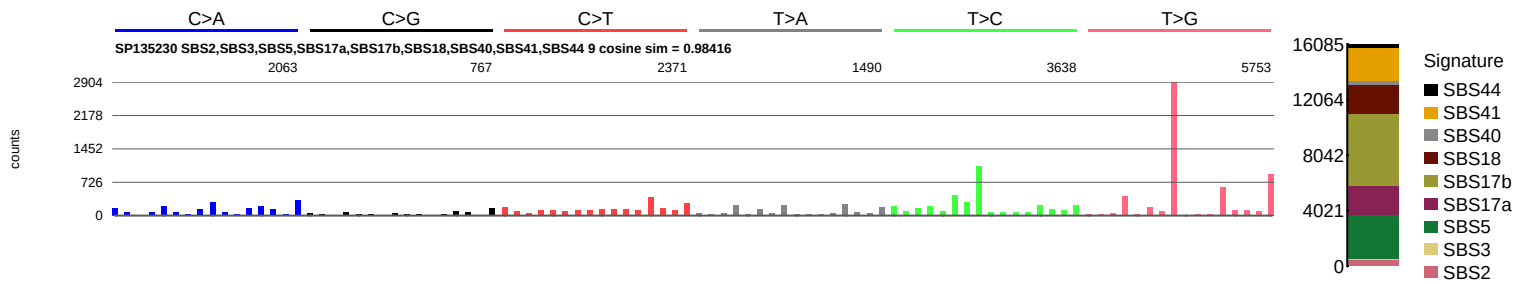
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98417
1293
1476
3281
1368
3087
5578
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>T
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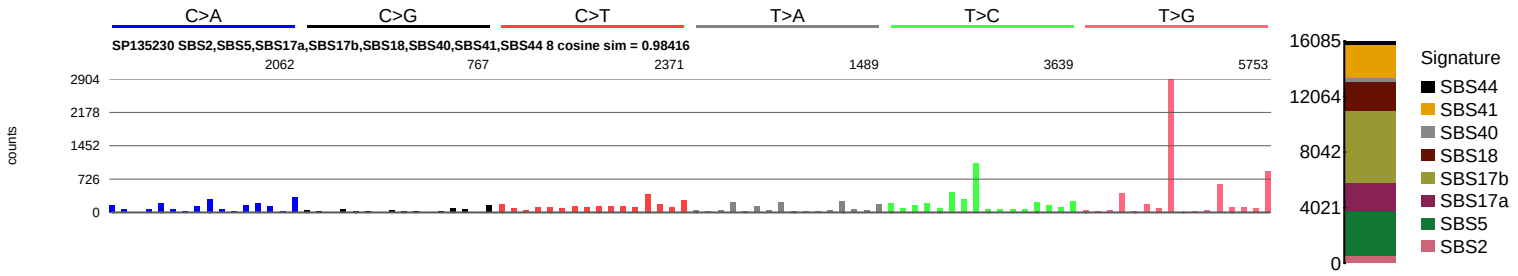
SP135230 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98416
2063
767
2371
1490
3638
5753
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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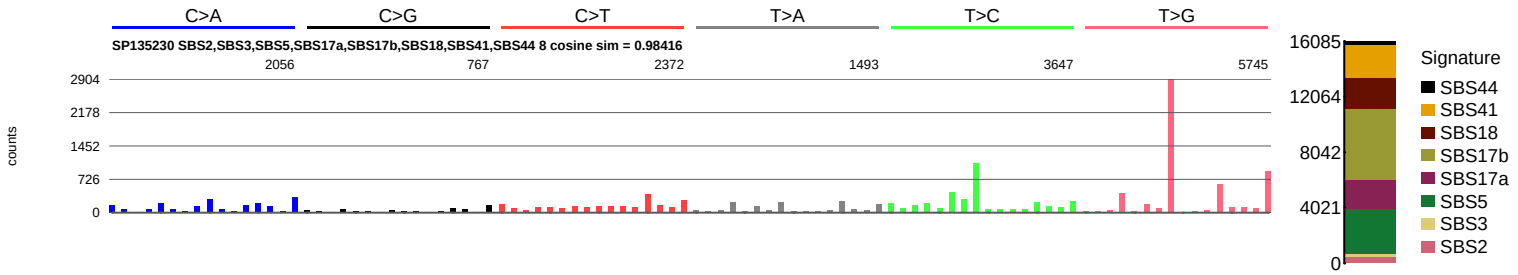
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98416
2062
767
2371
1489
3639
5753
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2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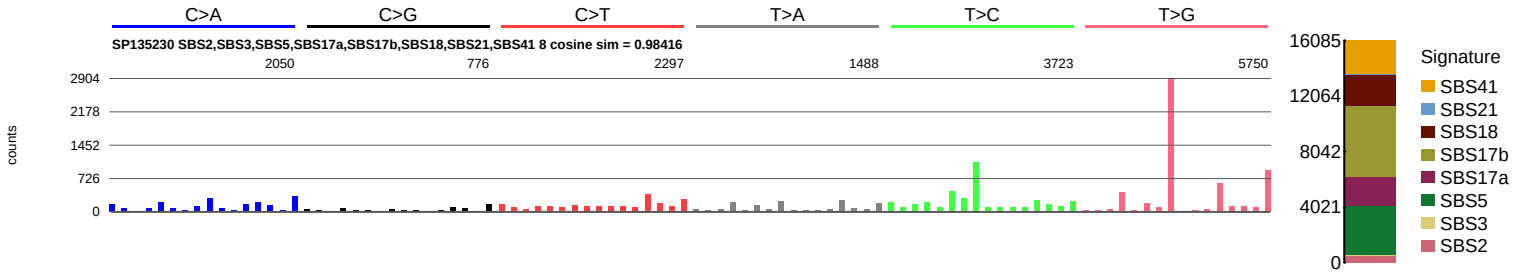
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98416
2056
767
2372
1493
3647
5745
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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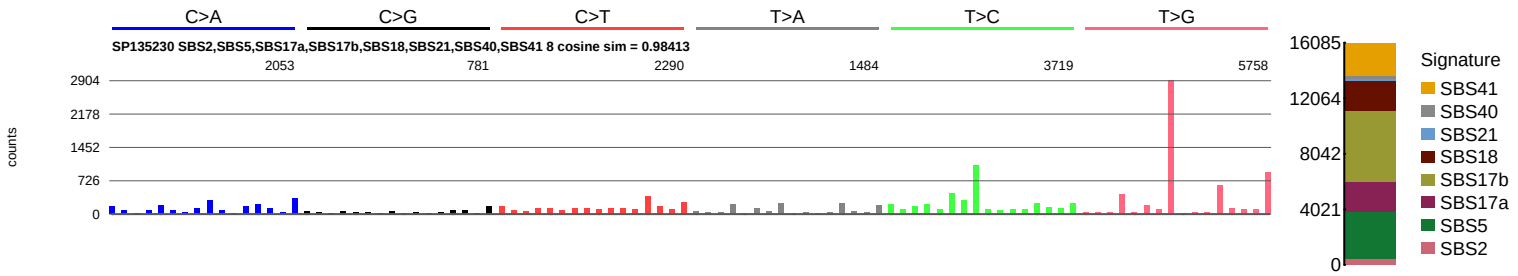
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98416
2050
776
2297
1488
3723
5750
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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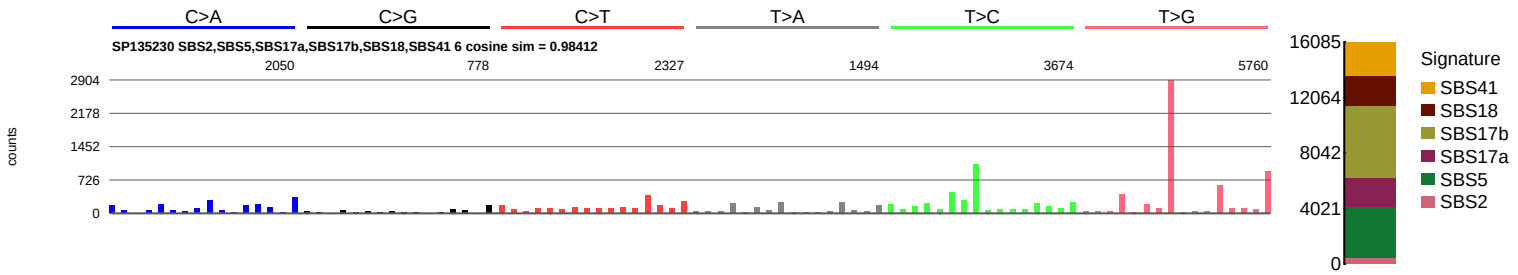
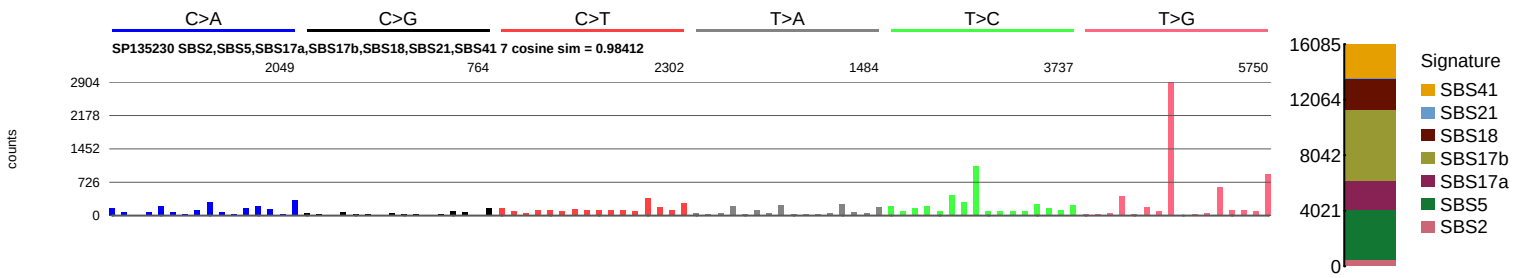
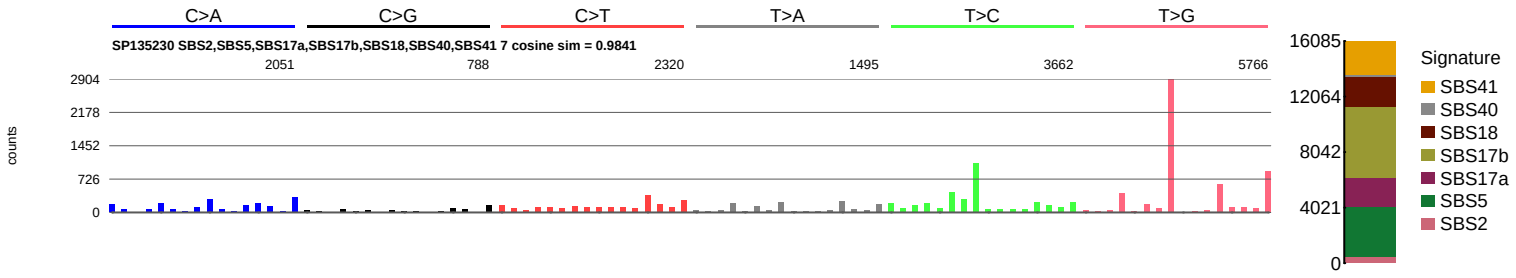
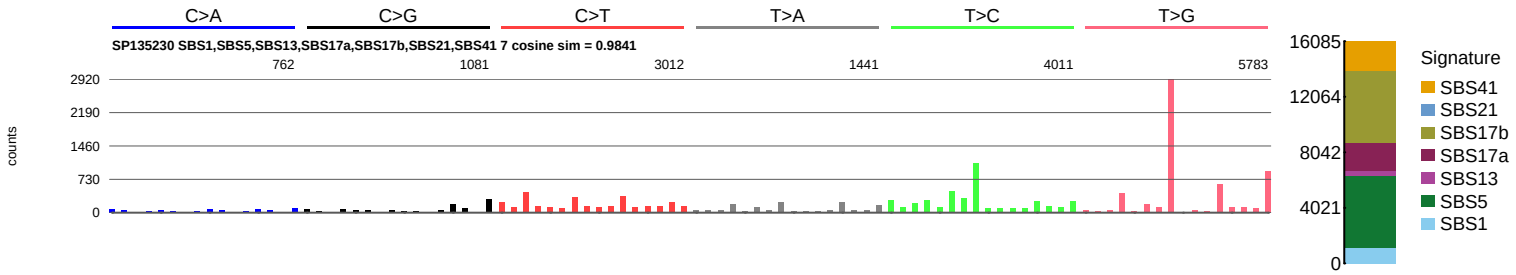
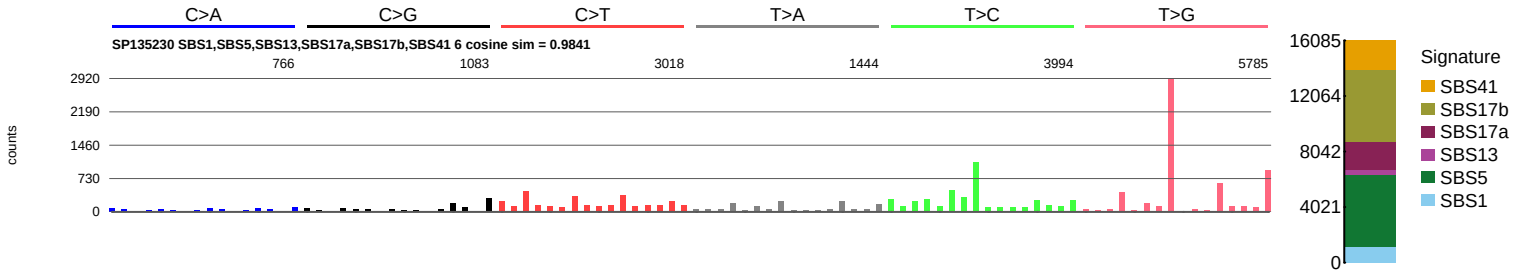
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98413
2053
781
2290
1484
3719
5758
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

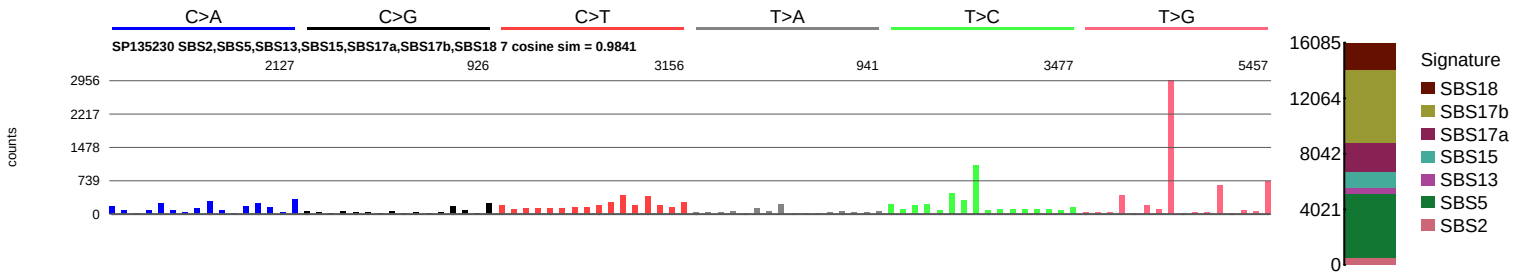

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98412
2050
778
2327
1494
3674
5760
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2

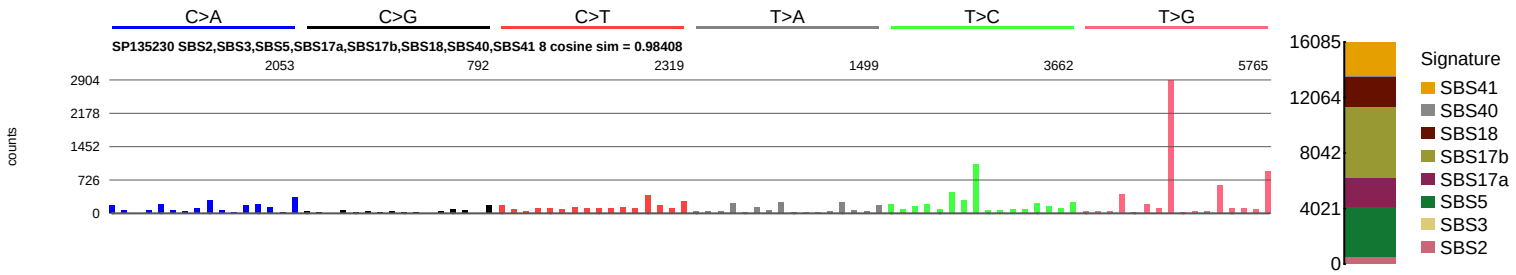

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98412
2049
764
2302
1484
3737
5750
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.9841
2051
788
2320
1495
3662
5766
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.9841
762
1081
3012
1441
4011
5783
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.9841
766
1083
3018
1444
3994
5785
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.9841
2127
926
3156
941
3477
5457
counts
2956
2217
1478
739
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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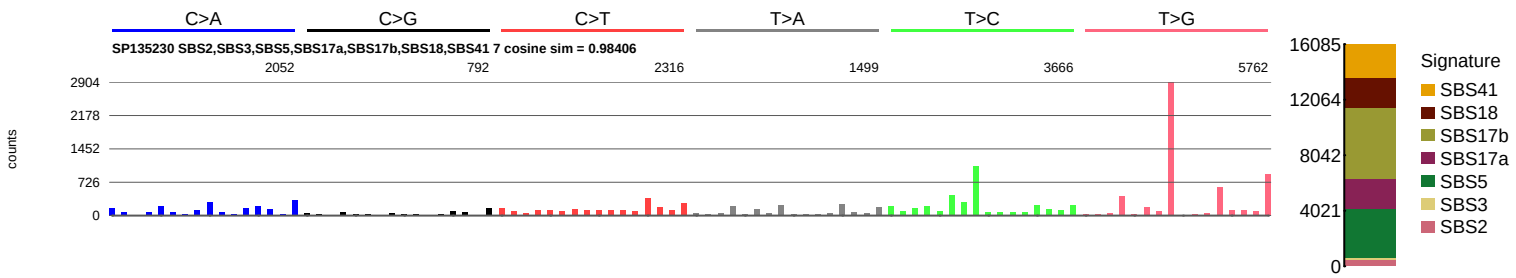
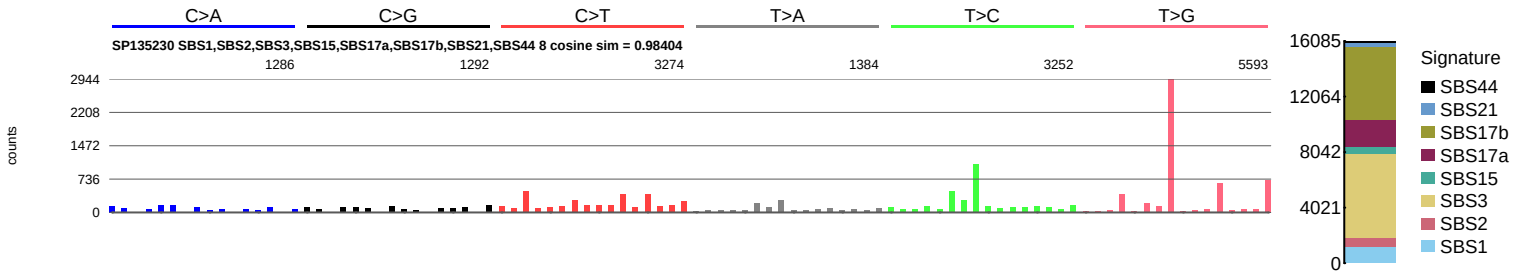
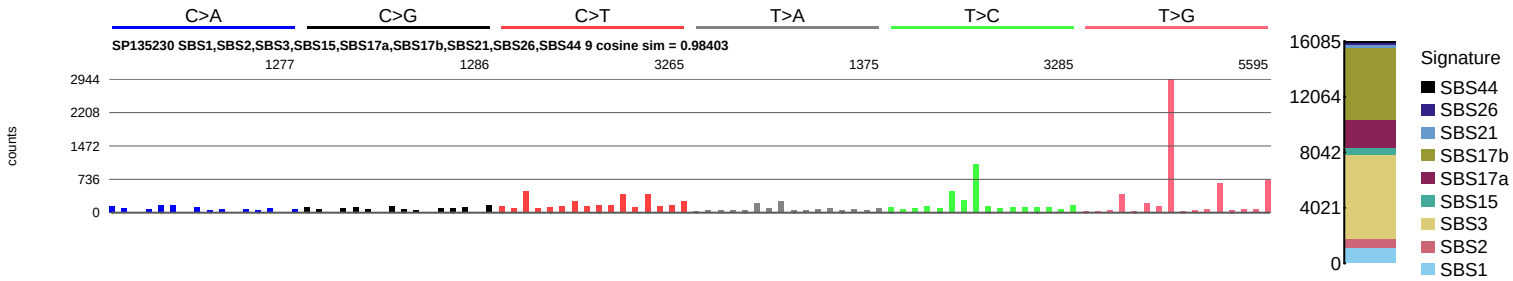
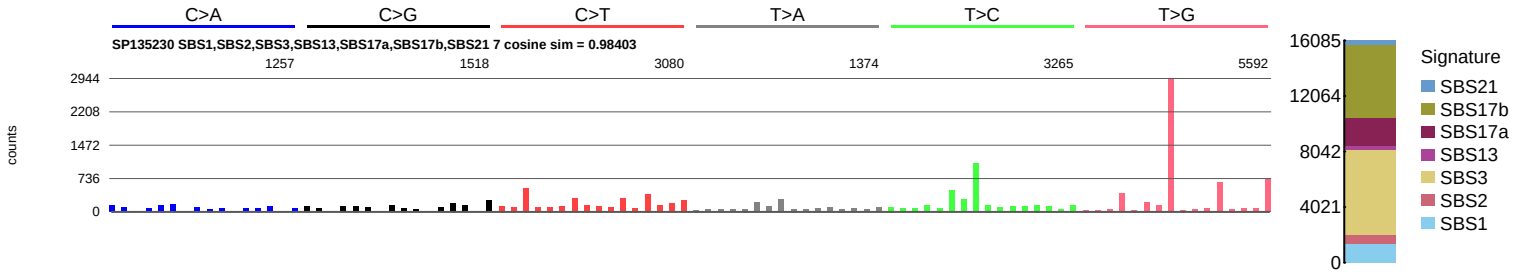
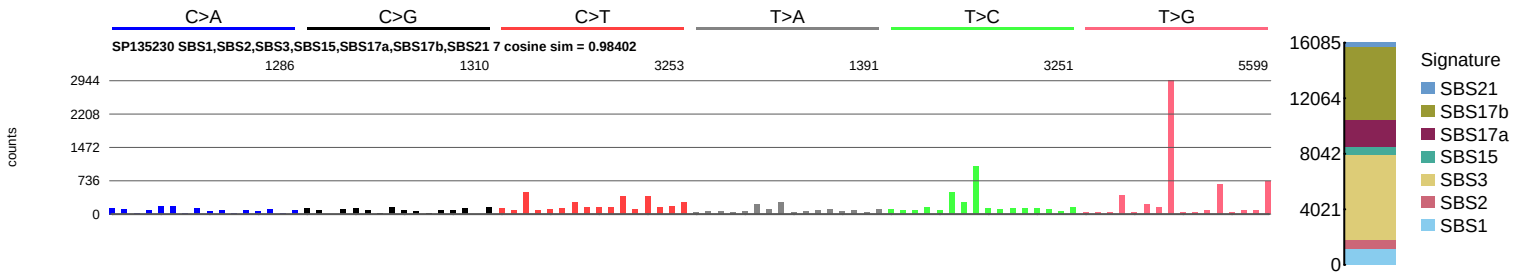
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98408
2053
792
2319
1499
3662
5765
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

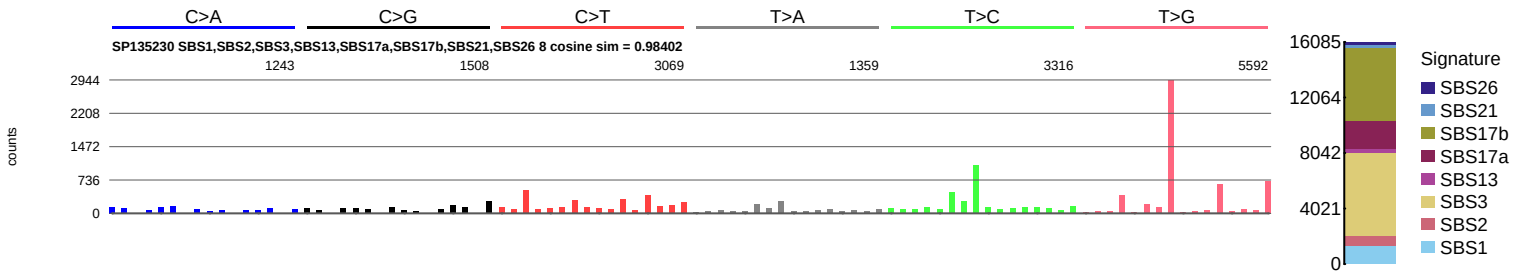

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98406
2052
792
2316
1499
3666
5762
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

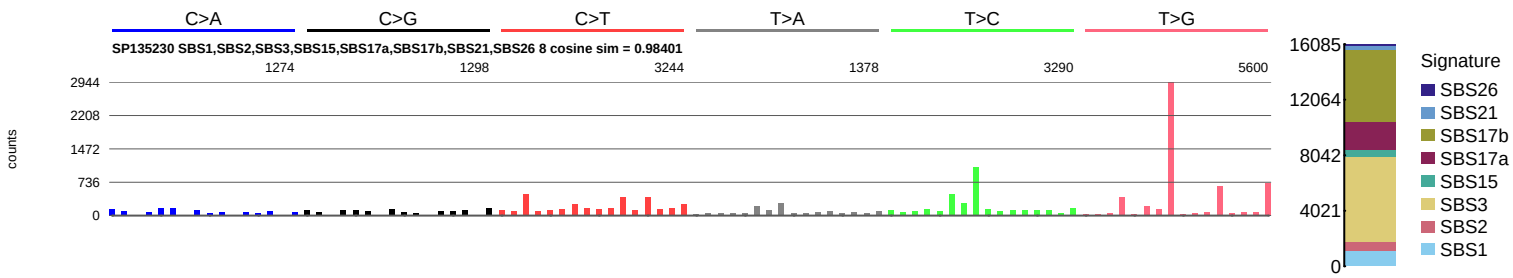

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98404
1286
1292
3274
1384
3252
5593
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.98403
1277
1286
3265
1375
3285
5595
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98403
1257
1518
3080
1374
3265
5592
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98402
1286
1310
3253
1391
3251
5599
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98402
1243
1508
3069
1359
3316
5592
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98401
1274
1298
3244
1378
3290
5600
counts
2944
2208
1472
736
0
16085
12064
8042
4021
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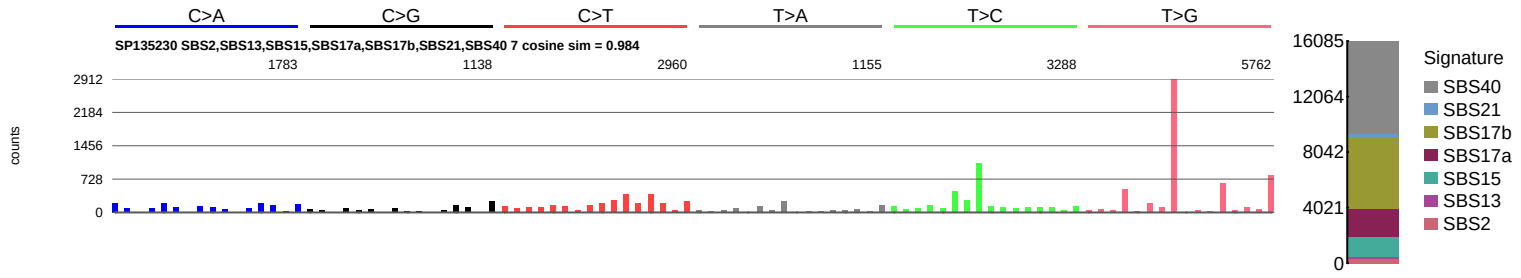
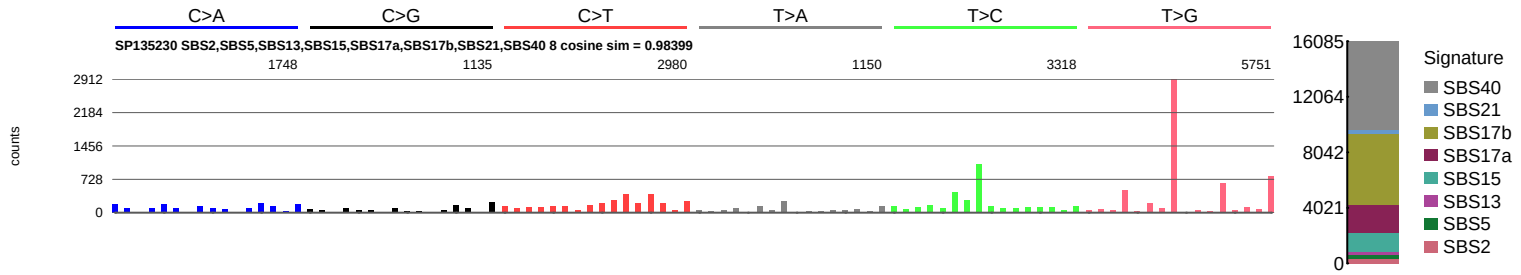
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.984
1783
1138
2960
1155
3288
5762
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98399
1748
1135
2980
1150
3318
5751
counts
2912
2184
1456
728
0
16085
12064
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SBS40
SBS21
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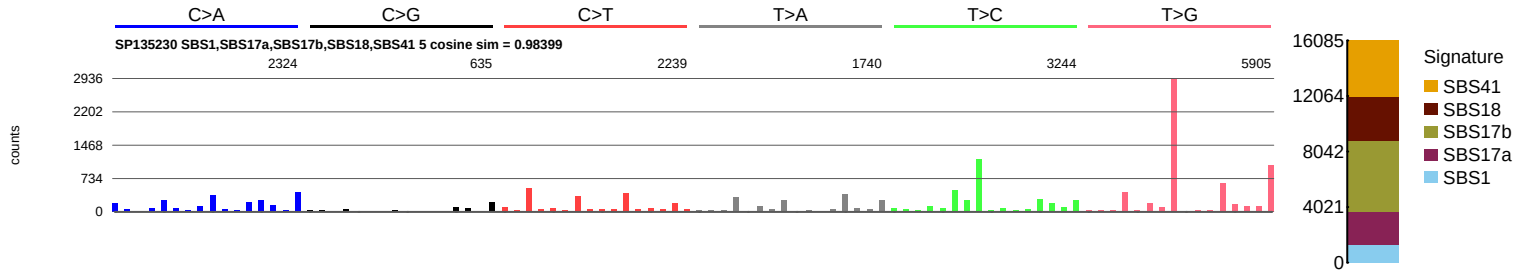
SBS2


C>A
C>G
C>T
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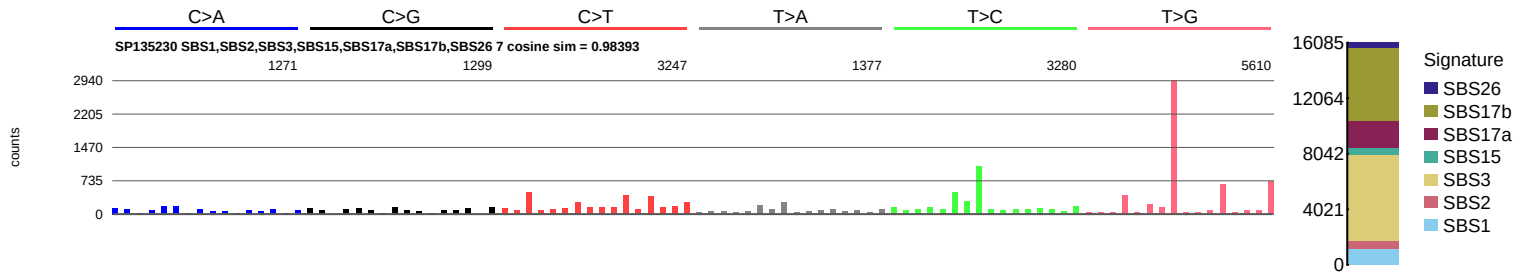
T>C
T>G
SP135230 SBS1,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98399
2324
635
2239
1740
3244
5905
counts
2936
2202
1468
734
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
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T>C
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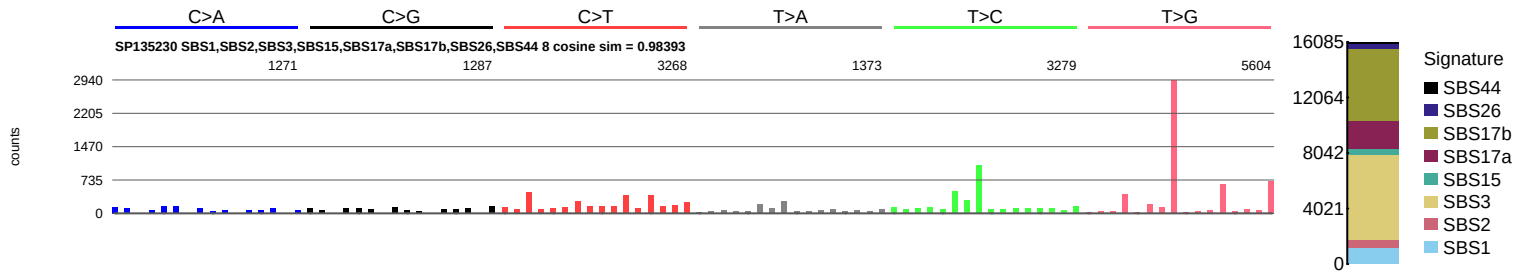
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98393
1271
1299
3247
1377
3280
5610
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
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C>T
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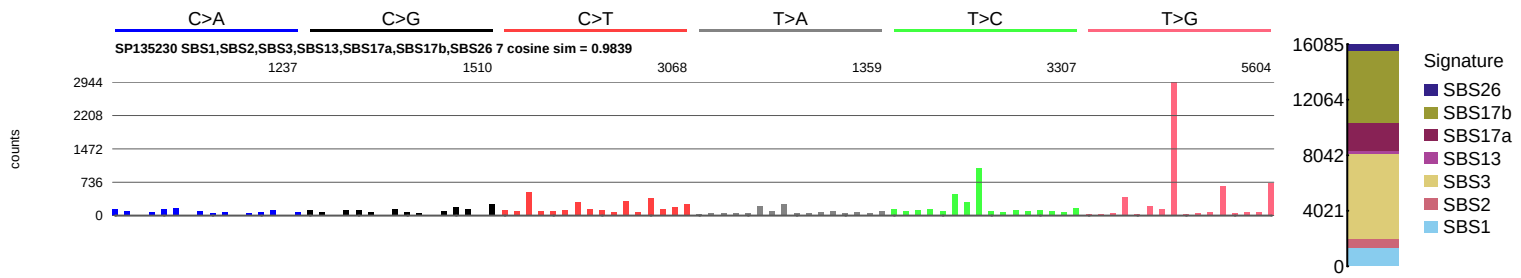
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98393
1271
1287
3268
1373
3279
5604
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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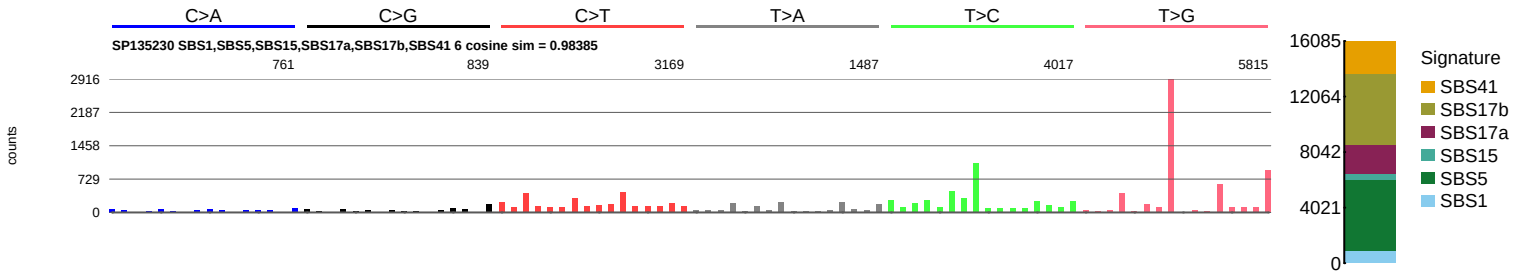
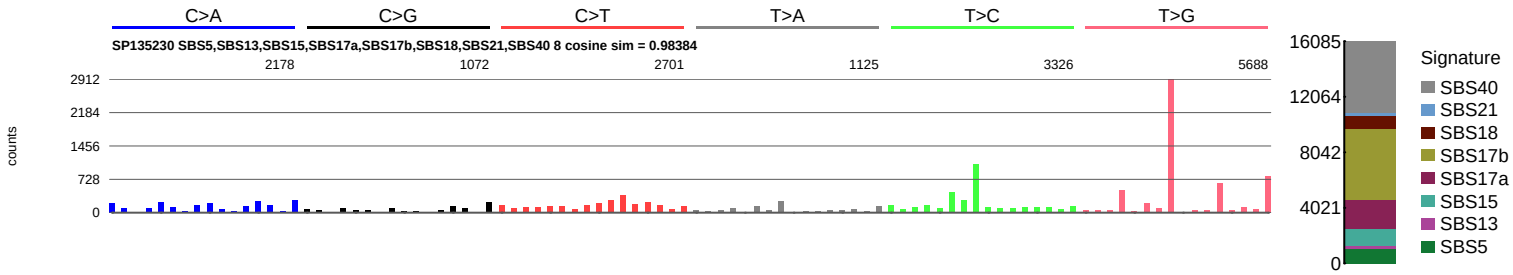
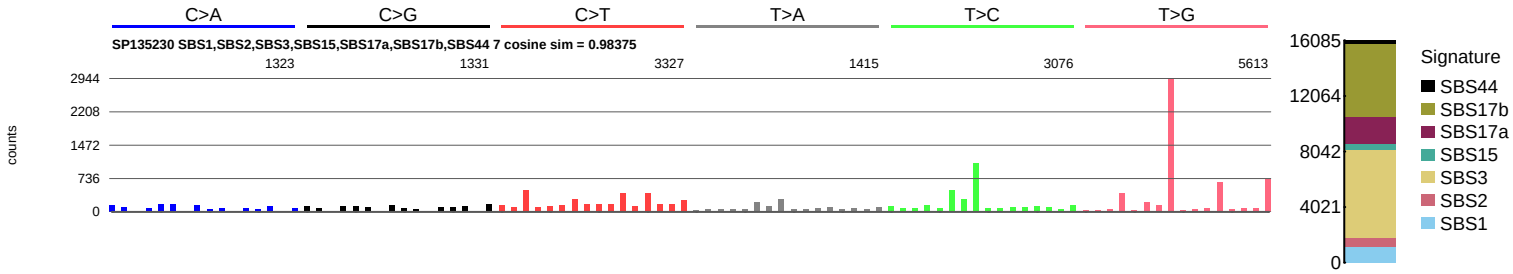
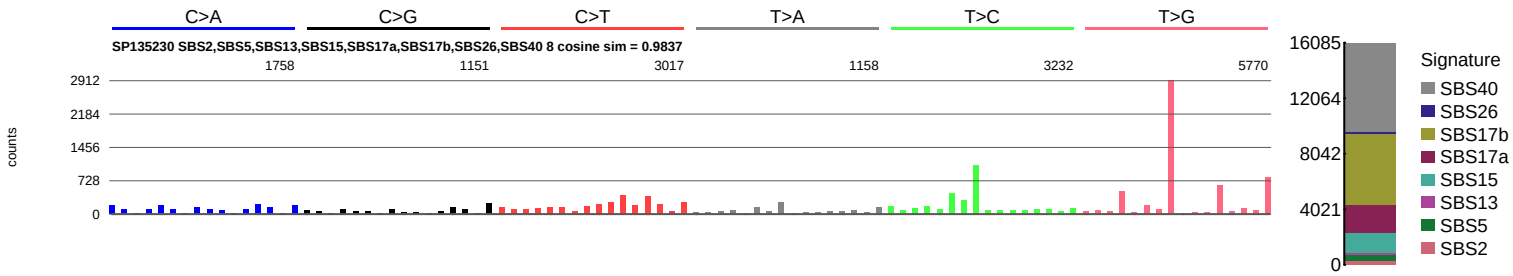
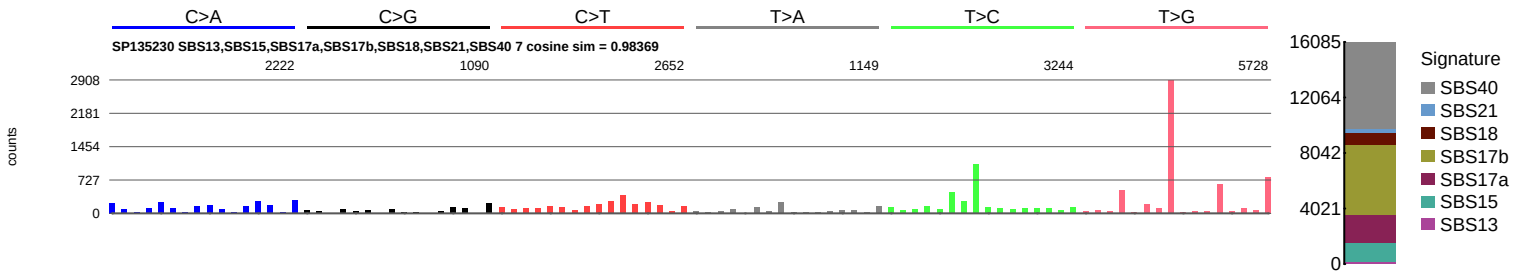
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26 7 cosine sim = 0.9839
1237
1510
3068
1359
3307
5604
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

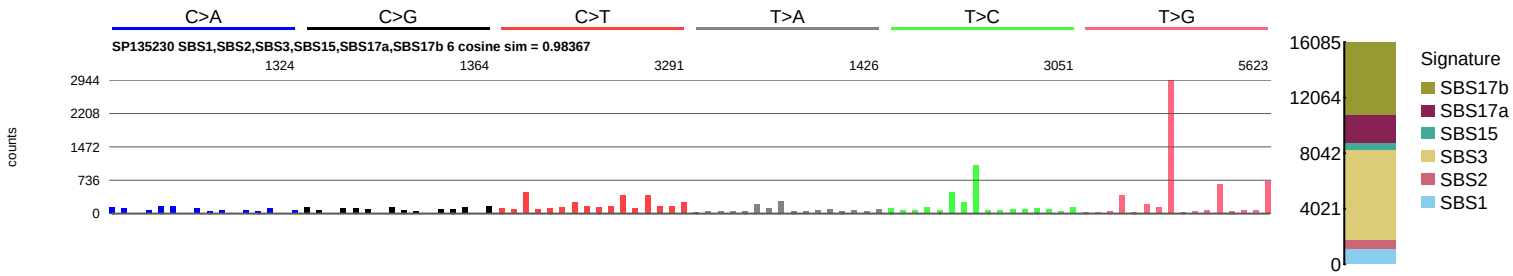

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98385
761
839
3169
1487
4017
5815
counts
2916
2187
1458
729
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS1

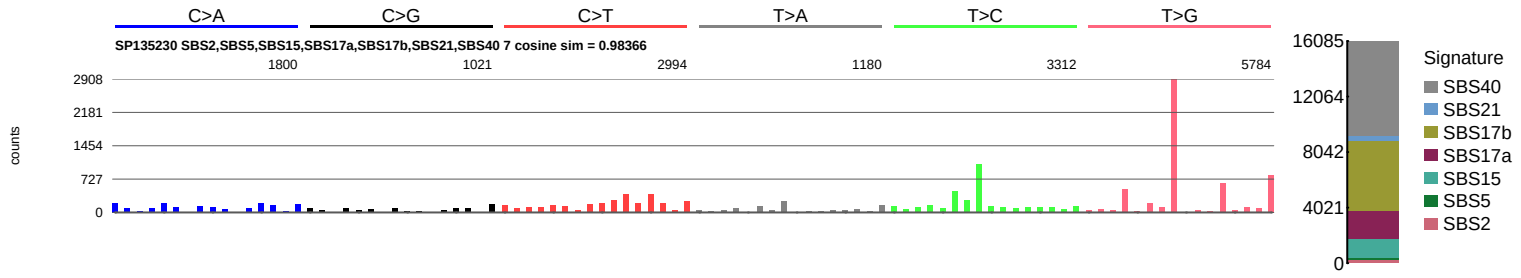

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98384
2178
1072
2701
1125
3326
5688
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

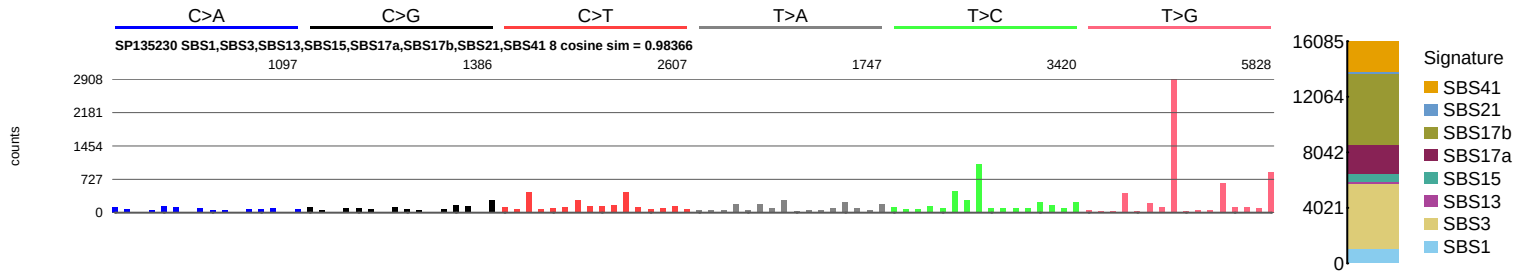

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98375
1323
1331
3327
1415
3076
5613
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

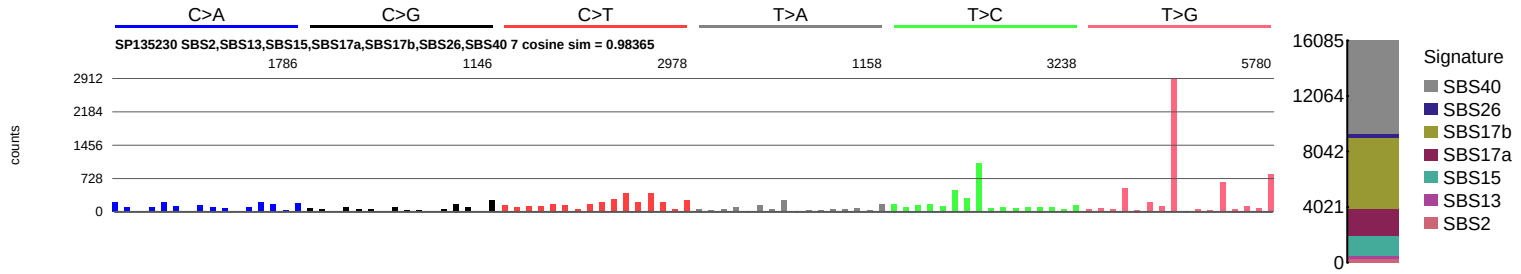

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.9837
1758
1151
3017
1158
3232
5770
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

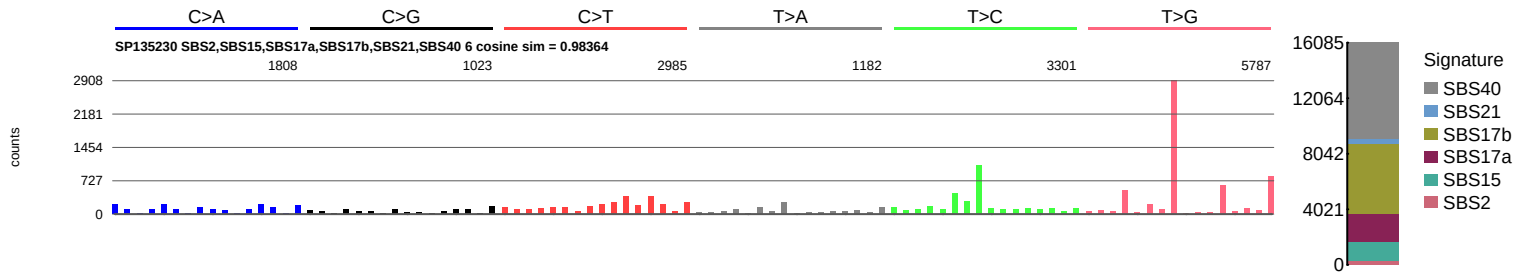

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98369
2222
1090
2652
1149
3244
5728
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13

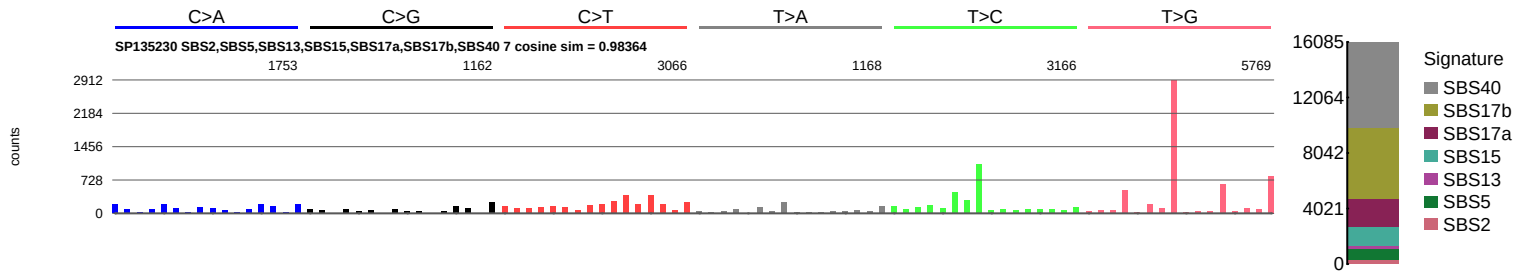

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98367
1324
1364
3291
1426
3051
5623
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

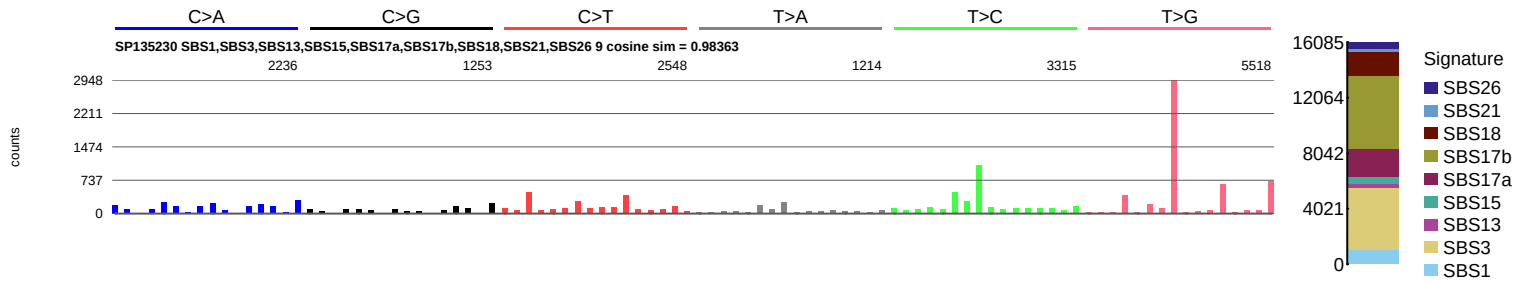

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98366
1800
1021
2994
1180
3312
5784
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2

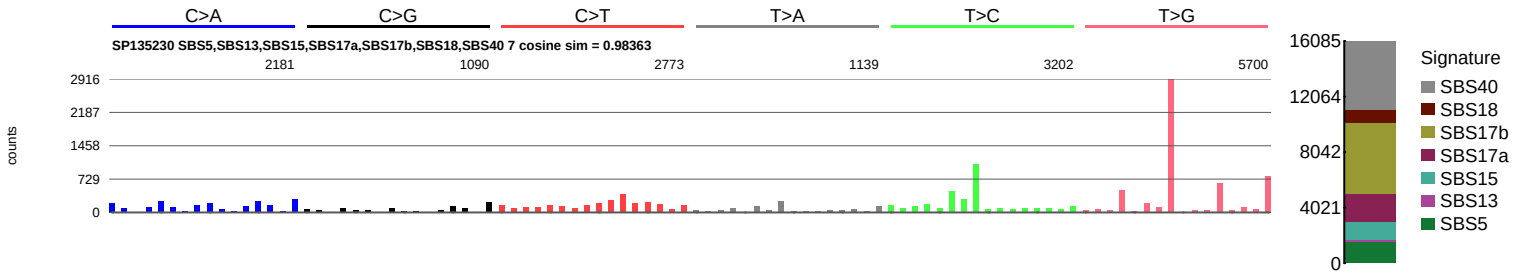

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98366
1097
1386
2607
1747
3420
5828
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

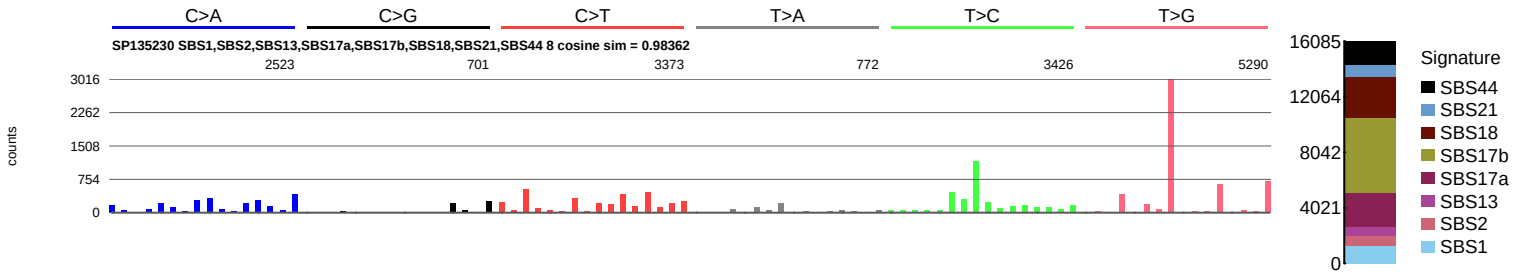

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98365
1786
1146
2978
1158
3238
5780
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2

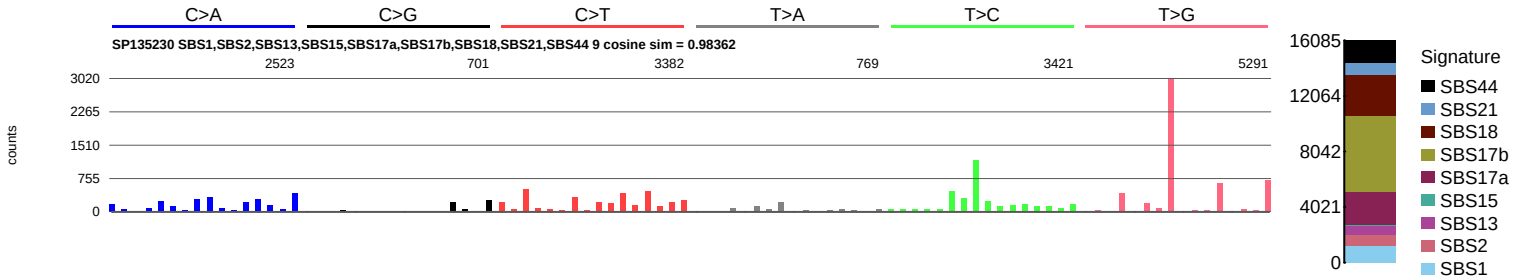

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98364
1808
1023
2985
1182
3301
5787
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2

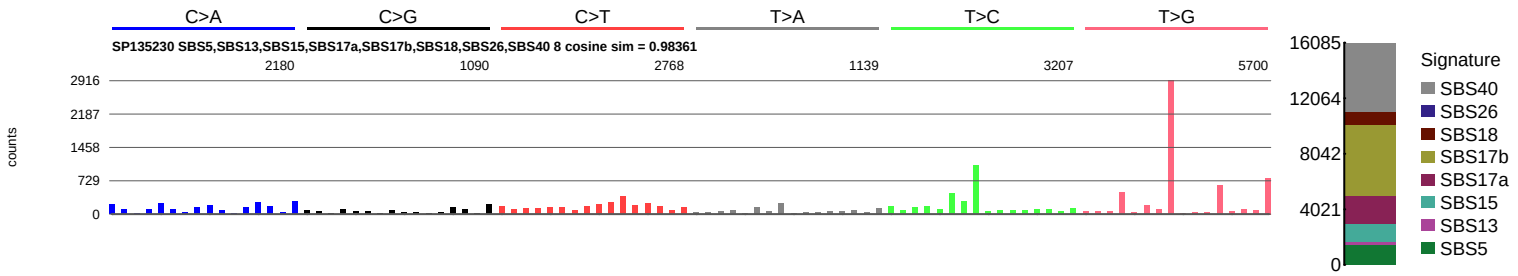

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98364
1753
1162
3066
1168
3166
5769
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

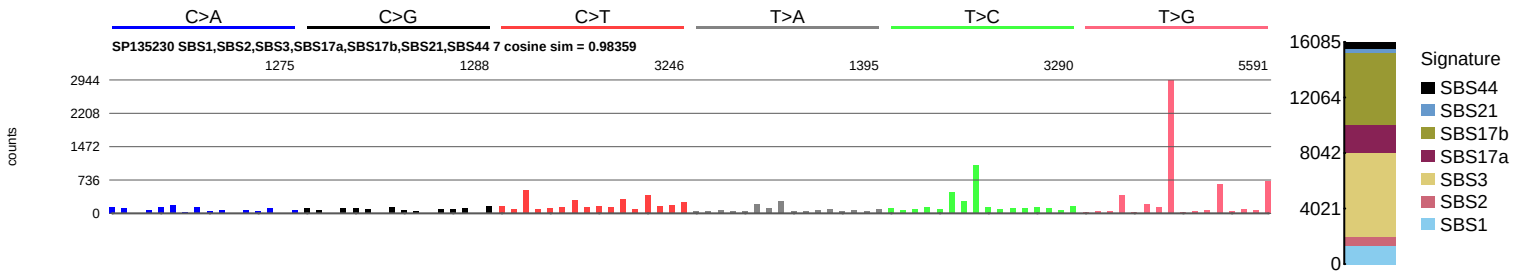

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98363
2236
1253
2548
1214
3315
5518
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98363
2181
1090
2773
1139
3202
5700
counts
2916
2187
1458
729
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98362
2523
701
3373
772
3426
5290
counts
3016
2262
1508
754
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98362
2523
701
3382
769
3421
5291
counts
3020
2265
1510
755
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98361
2180
1090
2768
1139
3207
5700
counts
2916
2187
1458
729
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98359
1275
1288
3246
1395
3290
5591
counts
2944
2208
1472
736
0
16085
12064
8042
4021
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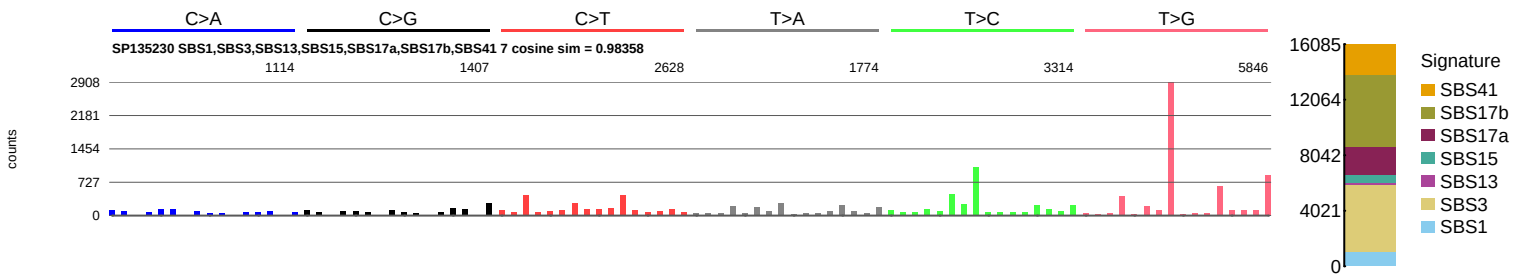
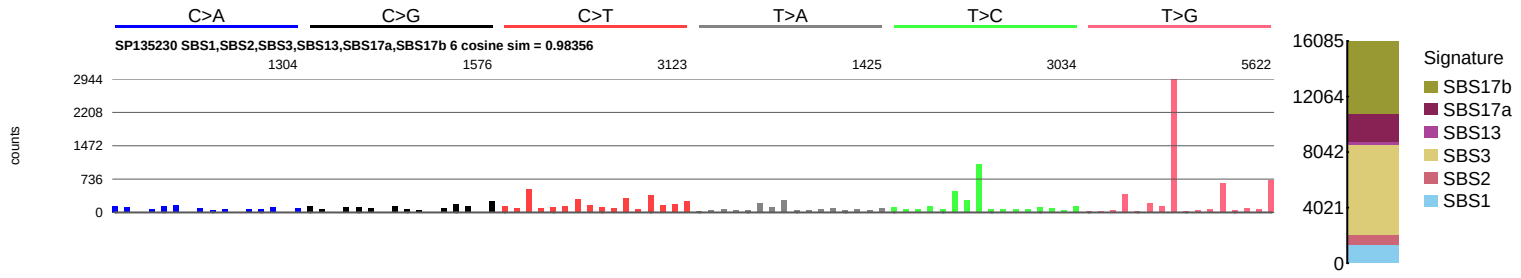
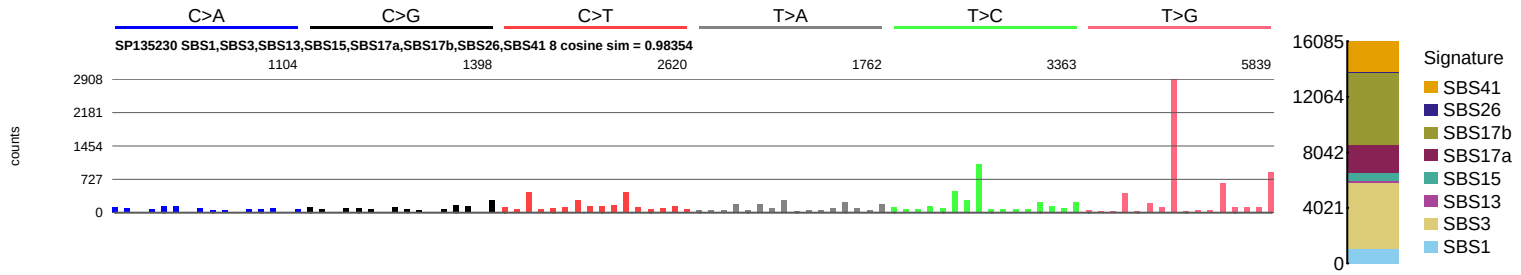
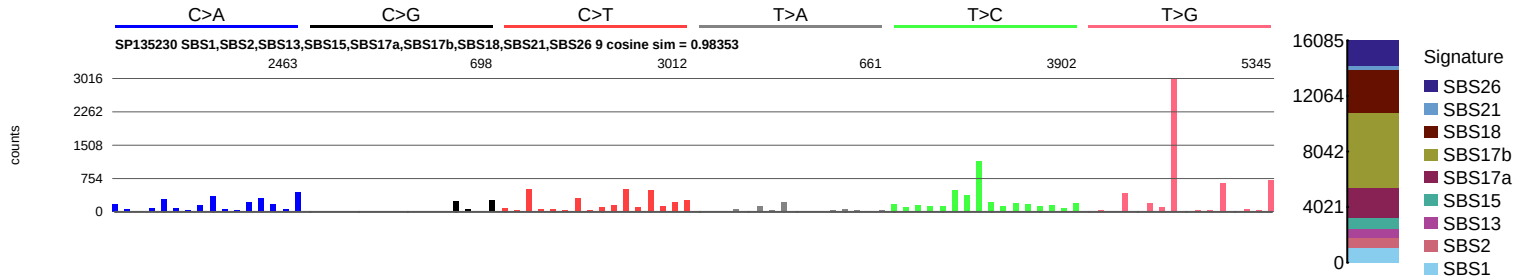
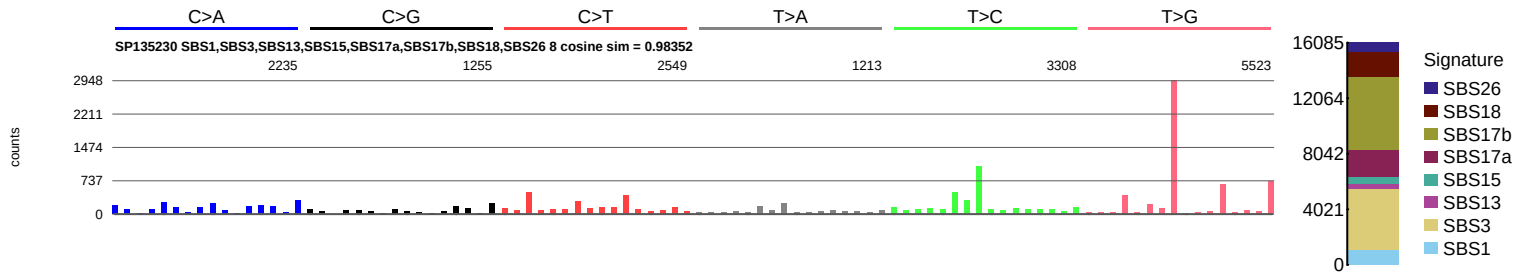
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

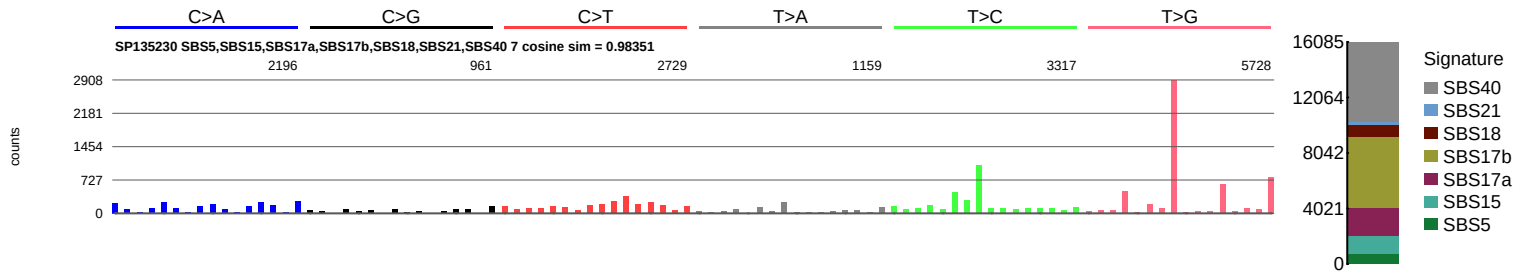

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98358
1114
1407
2628
1774
3314
5846
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

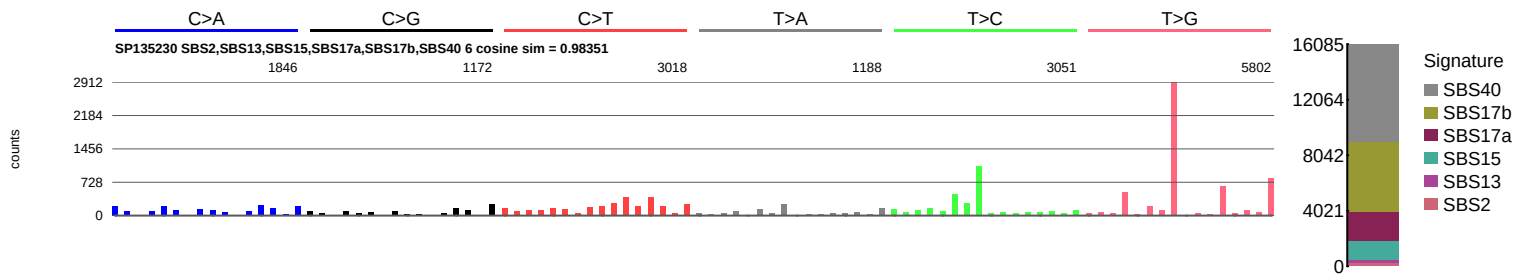

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98356
1304
1576
3123
1425
3034
5622
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

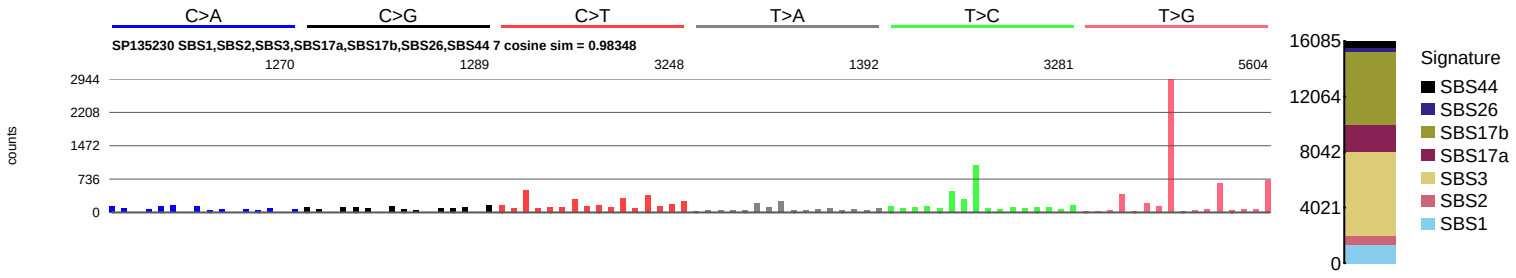

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98354
1104
1398
2620
1762
3363
5839
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

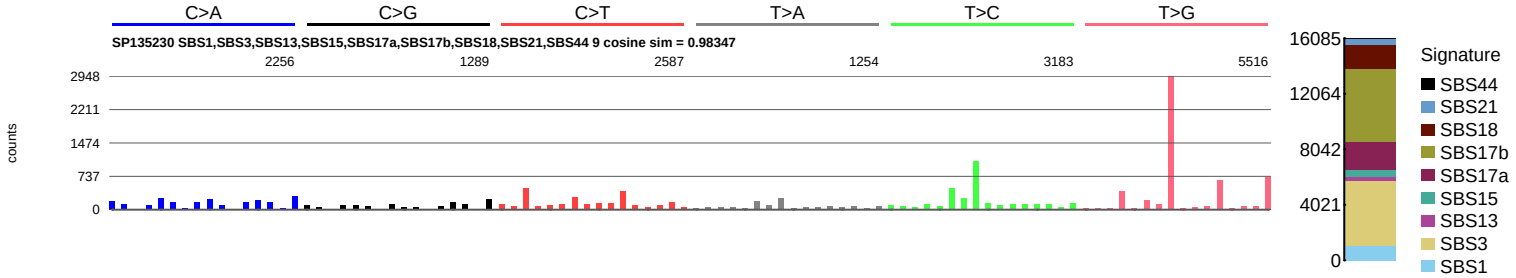

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98353
2463
698
3012
661
3902
5345
counts
3016
2262
1508
754
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

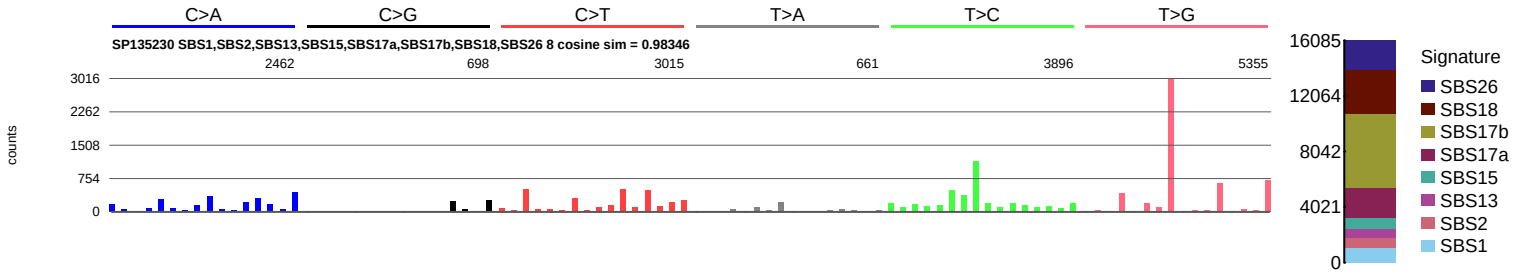

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98352
2235
1255
2549
1213
3308
5523
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

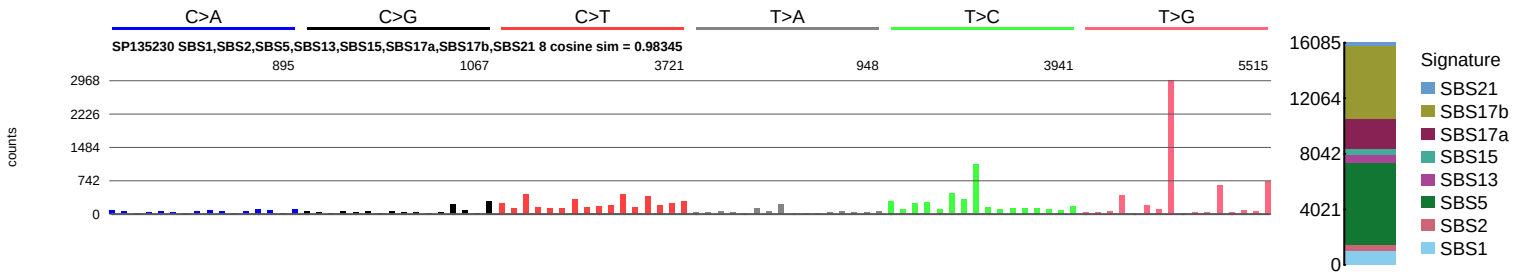

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98351
2196
961
2729
1159
3317
5728
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5

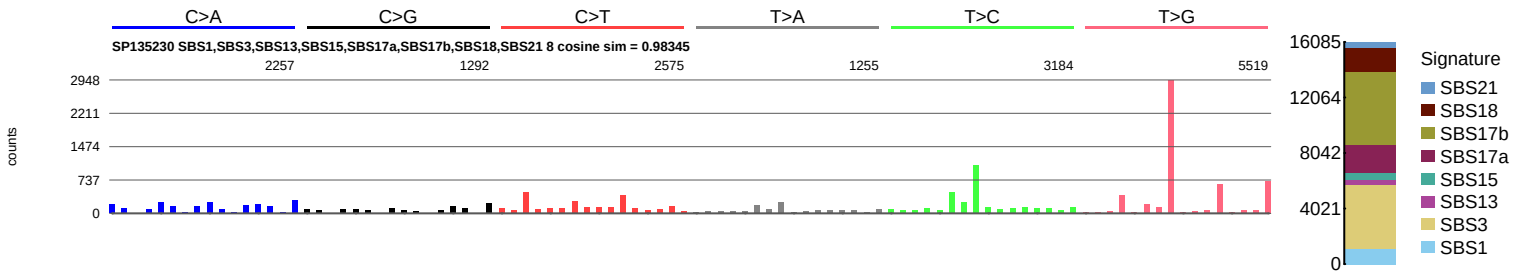

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98351
1846
1172
3018
1188
3051
5802
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2

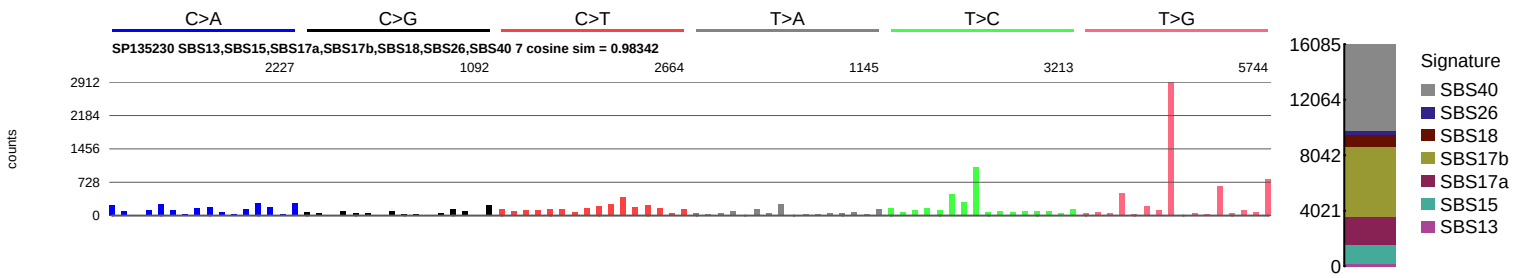

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98348
1270
1289
3248
1392
3281
5604
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

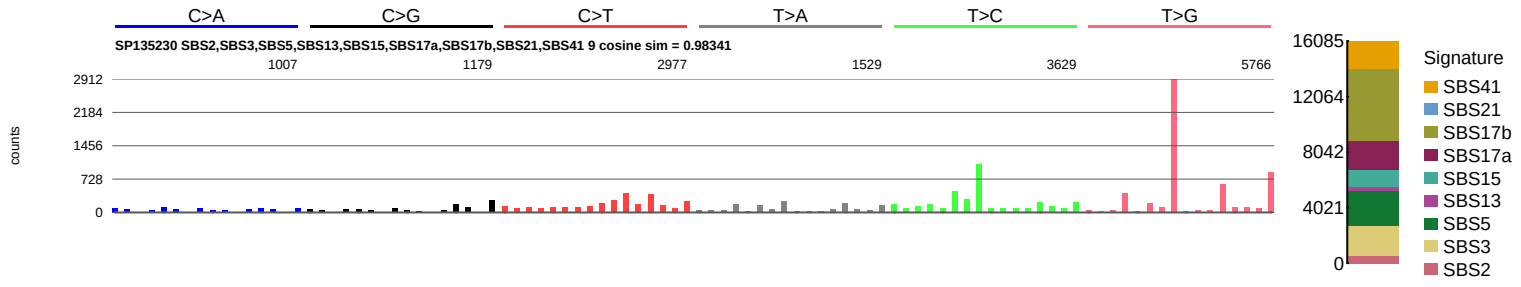

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98347
2256
1289
2587
1254
3183
5516
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

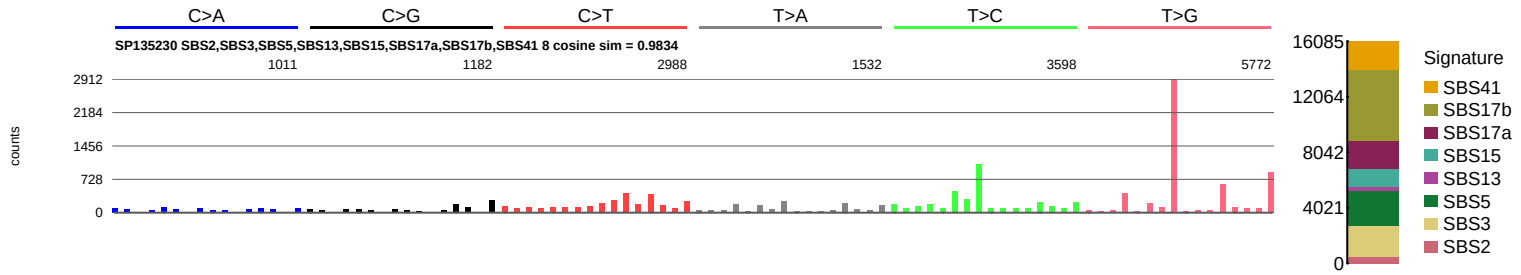

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98346
2462
698
3015
661
3896
5355
counts
3016
2262
1508
754
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

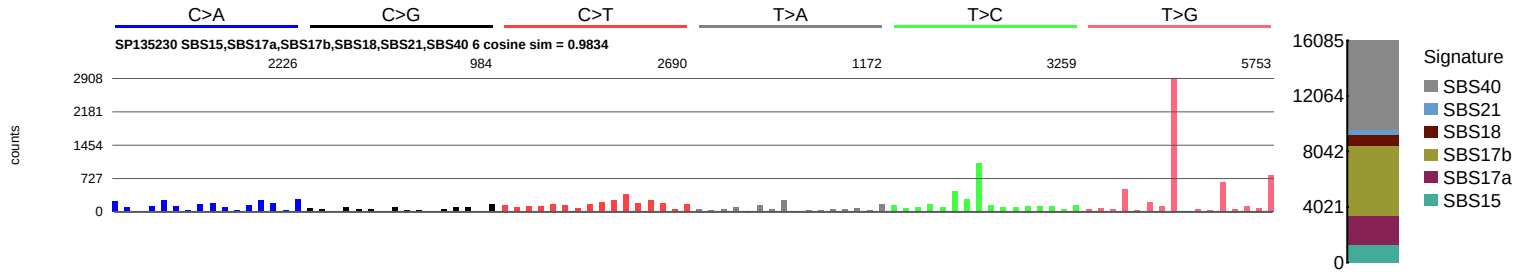

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98345
895
1067
3721
948
3941
5515
counts
2968
2226
1484
742
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

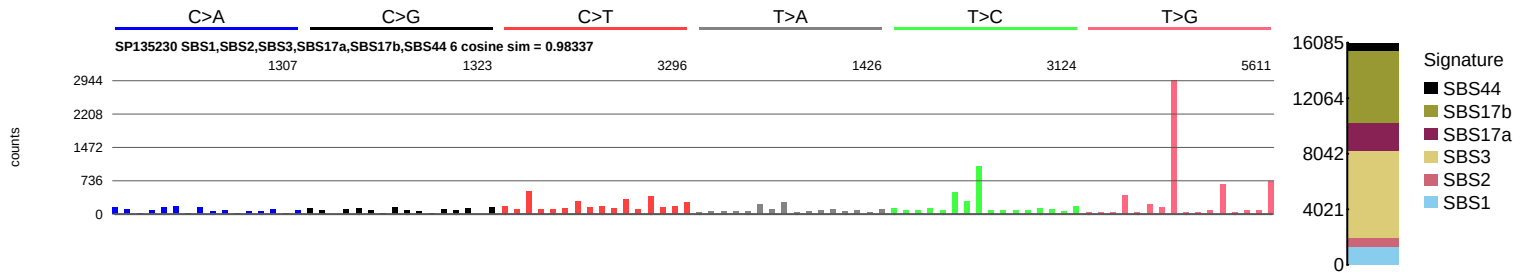

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98345
2257
1292
2575
1255
3184
5519
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

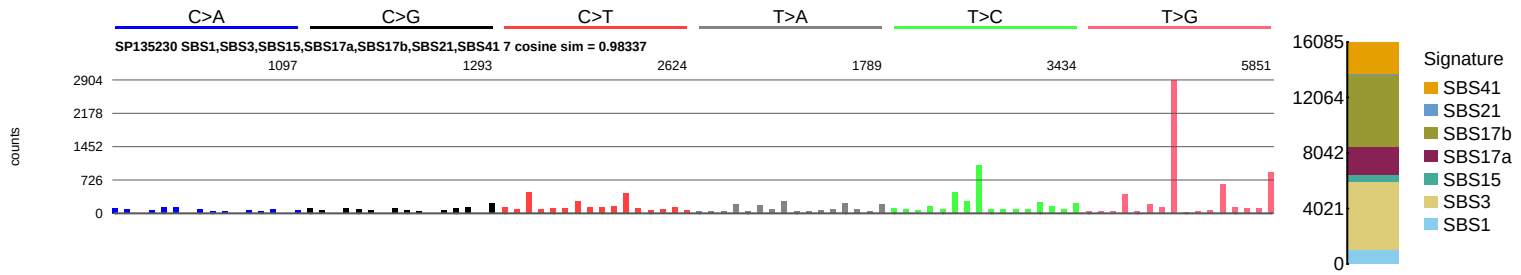

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98342
2227
1092
2664
1145
3213
5744
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13

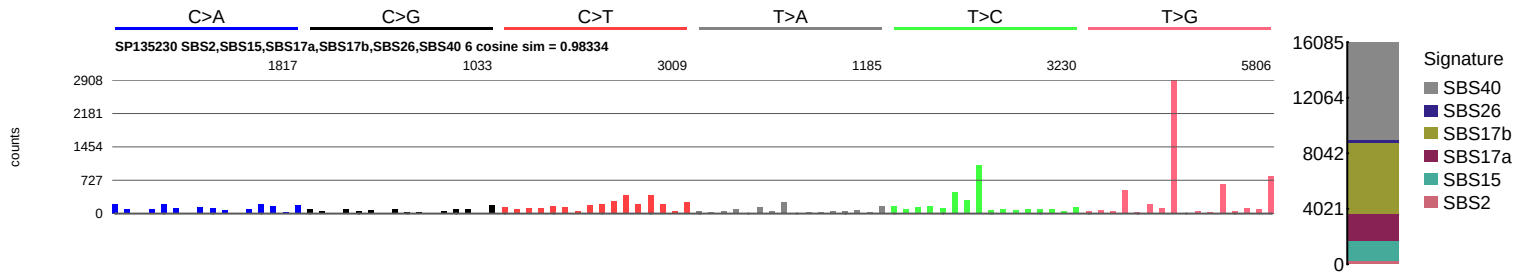

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98341
1007
1179
2977
1529
3629
5766
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

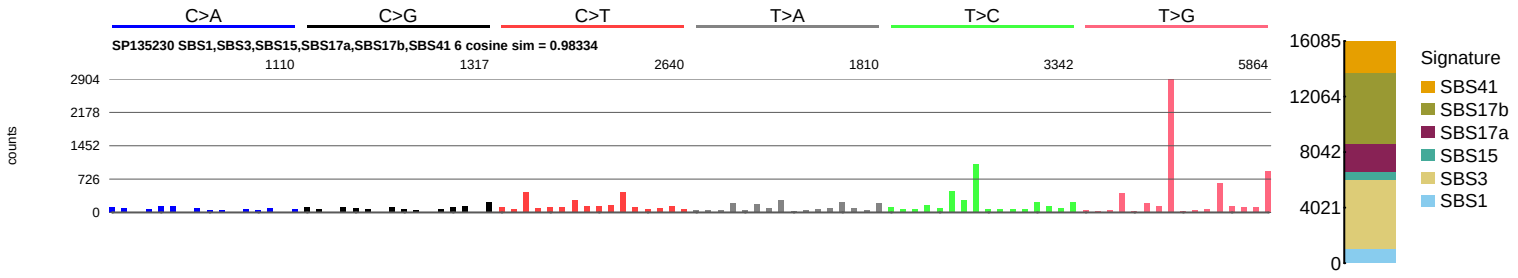

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.9834
1011
1182
2988
1532
3598
5772
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

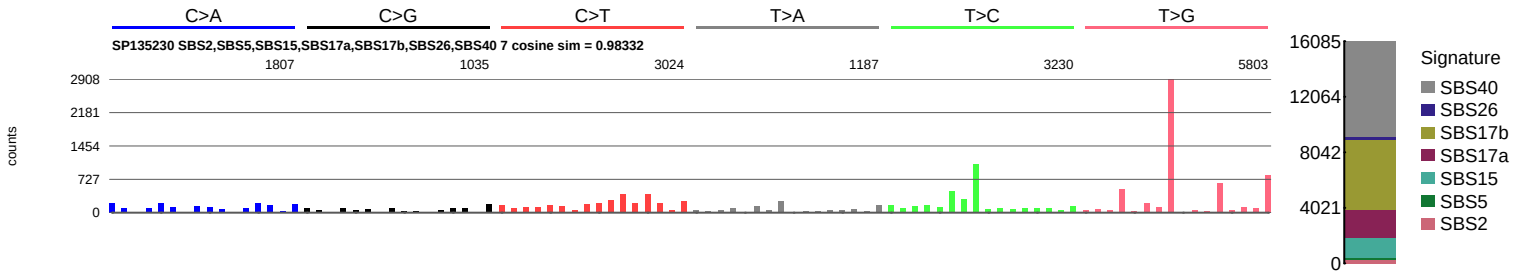

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.9834
2226
984
2690
1172
3259
5753
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15

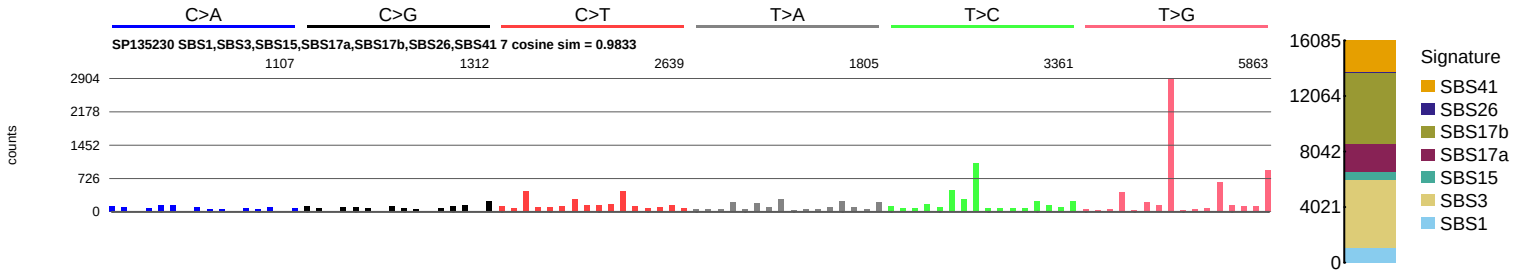

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98337
1307
1323
3296
1426
3124
5611
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS17b
SBS17a
SBS3
SBS2
SBS1

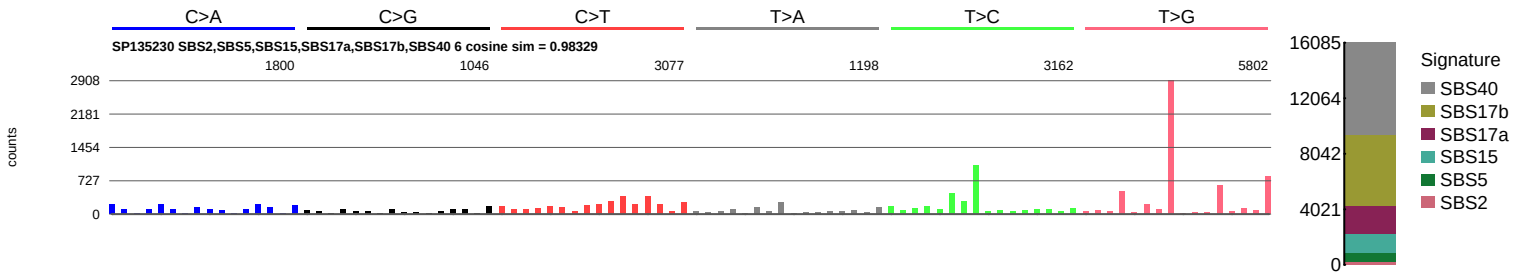

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98337
1097
1293
2624
1789
3434
5851
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

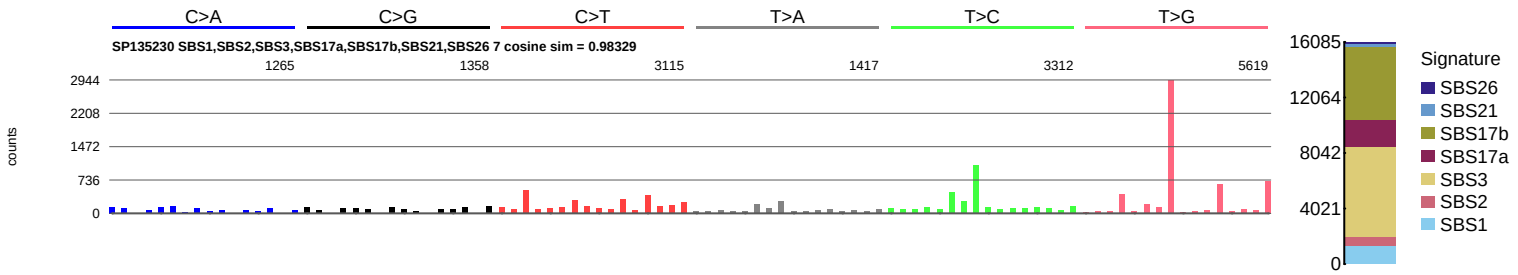

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98334
1817
1033
3009
1185
3230
5806
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2

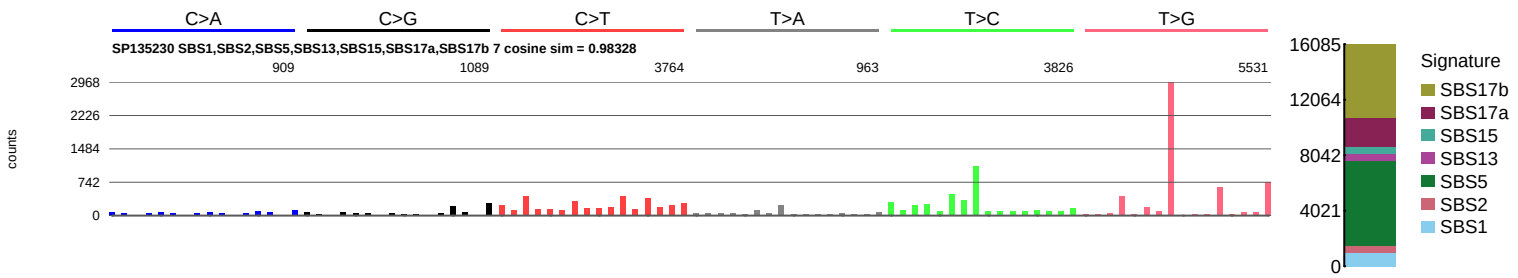

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98334
1110
1317
2640
1810
3342
5864
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS1

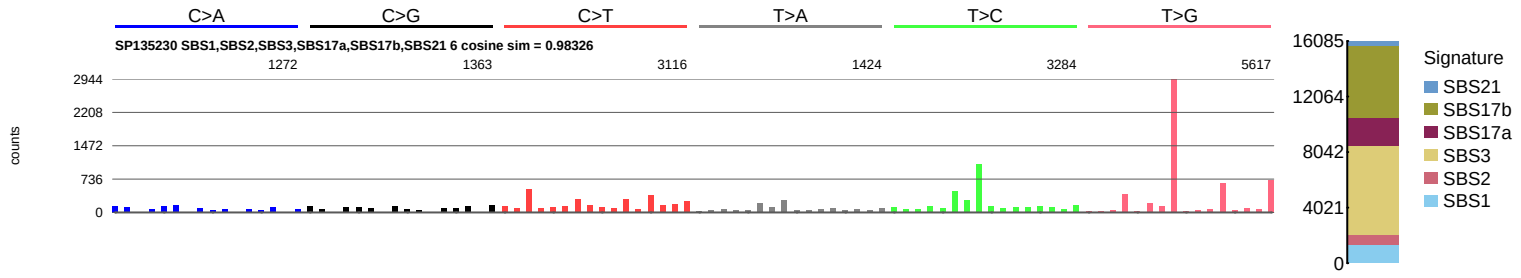

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98332
1807
1035
3024
1187
3230
5803
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2

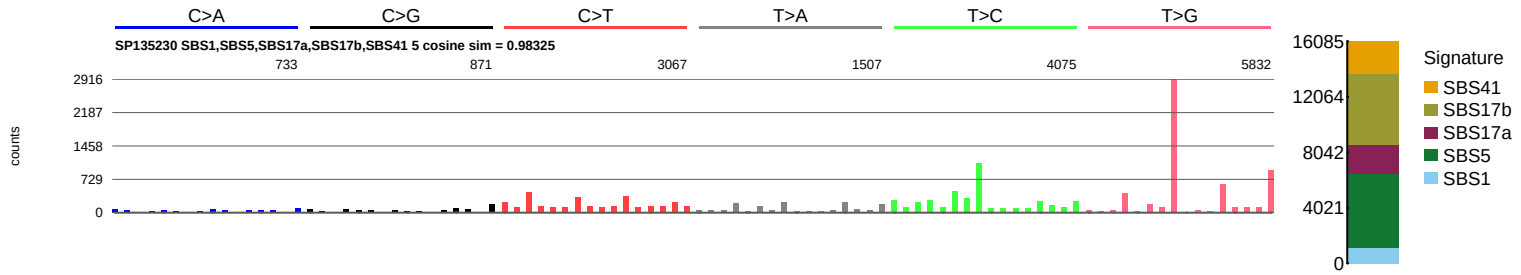

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.9833
1107
1312
2639
1805
3361
5863
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1

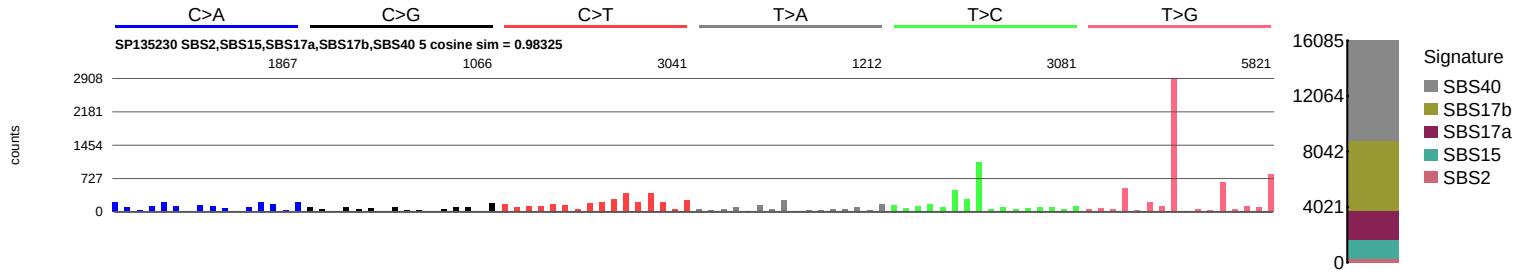

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98329
1800
1046
3077
1198
3162
5802
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2

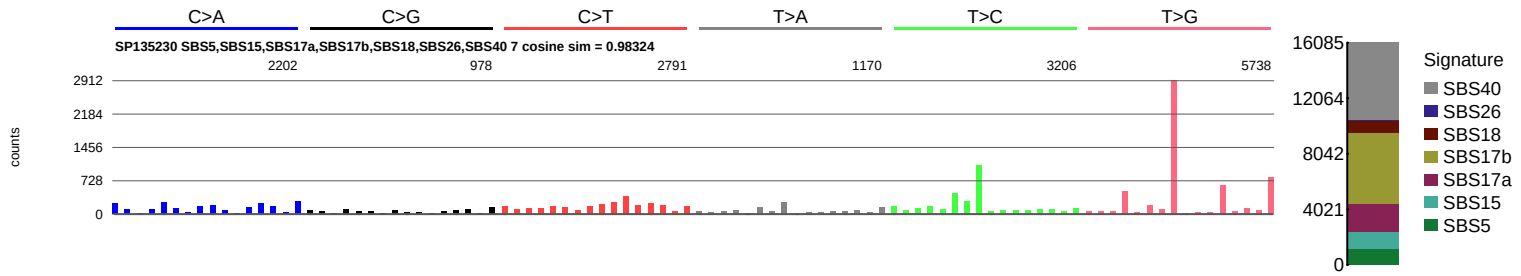

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26 7 cosine sim = 0.98329
1265
1358
3115
1417
3312
5619
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

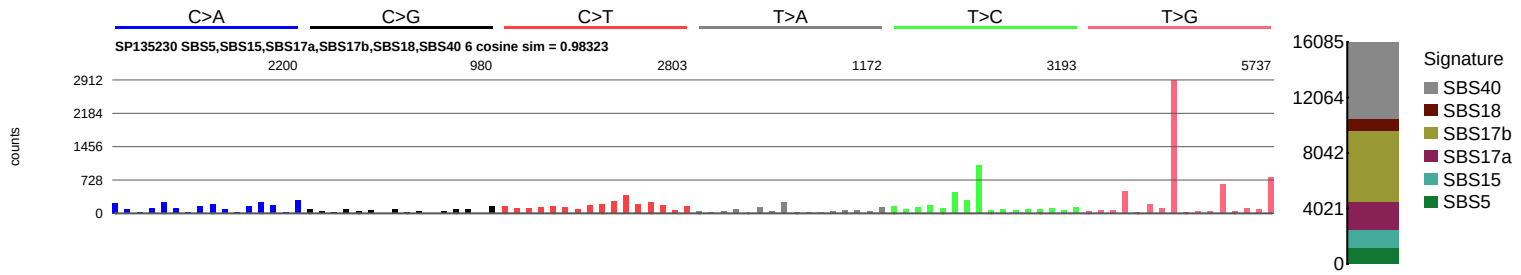

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98328
909
1089
3764
963
3826
5531
counts
2968
2226
1484
742
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98326
1272
1363
3116
1424
3284
5617
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98325
733
871
3067
1507
4075
5832
counts
2916
2187
1458
729
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98325
1867
1066
3041
1212
3081
5821
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98324
2202
978
2791
1170
3206
5738
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
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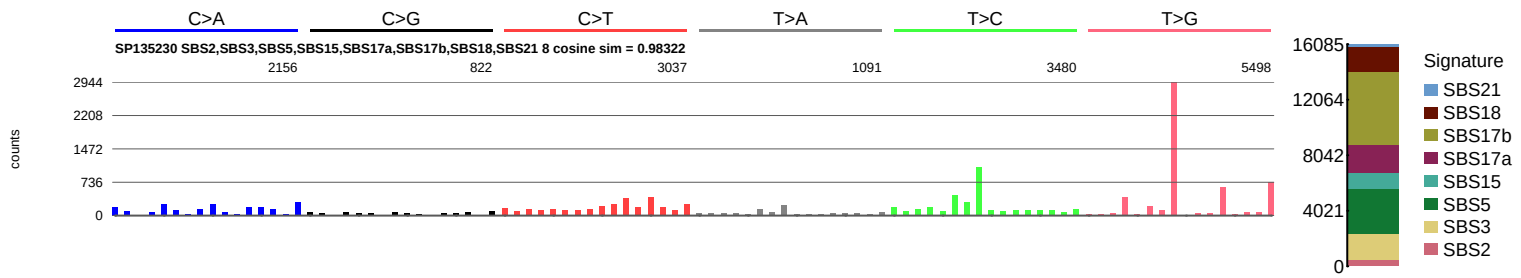
T>G
SP135230 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98323
2200
980
2803
1172
3193
5737
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5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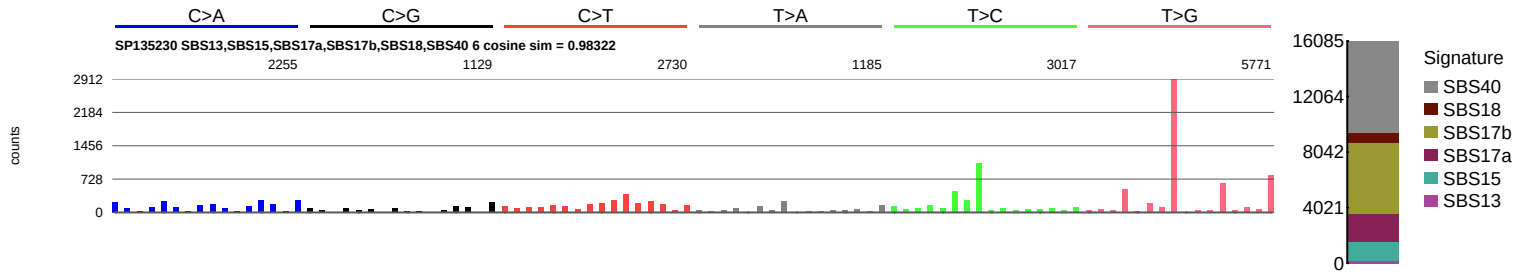
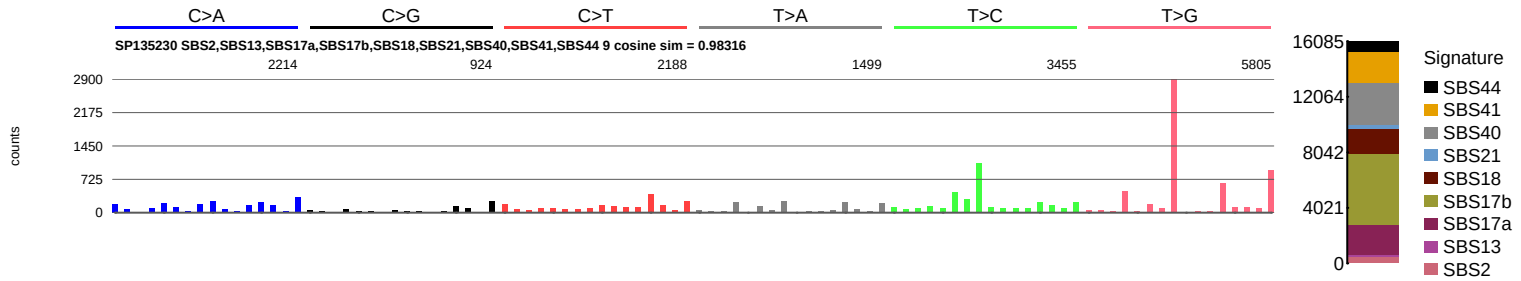
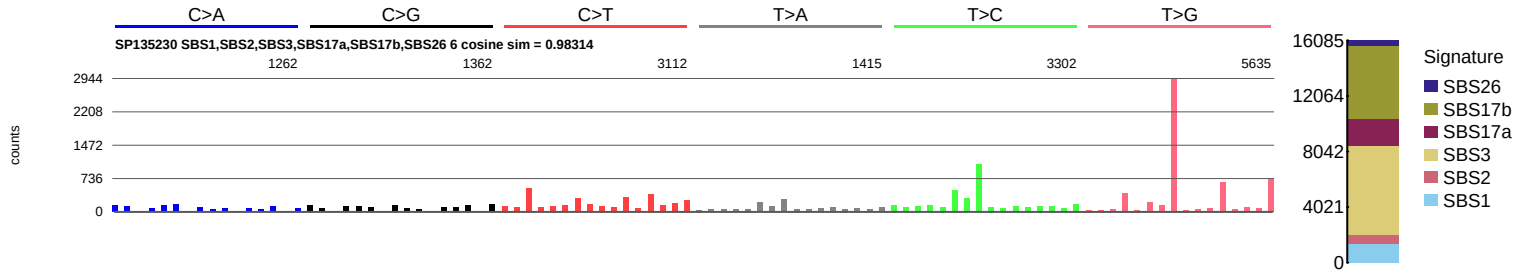
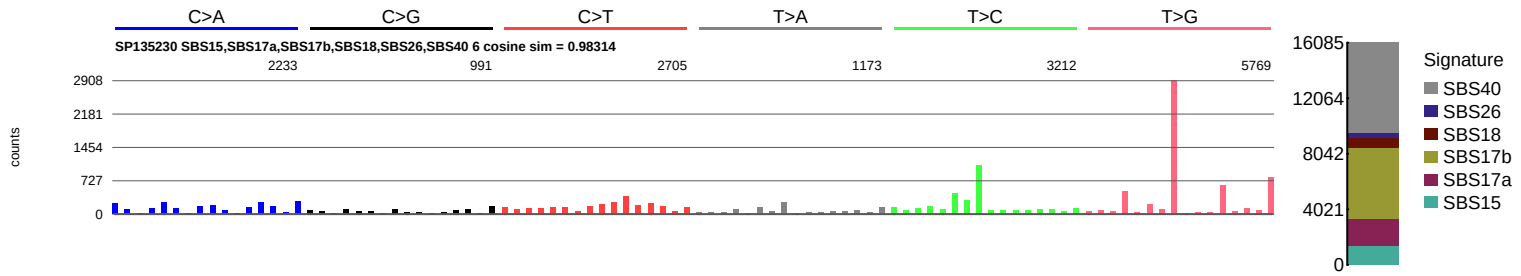
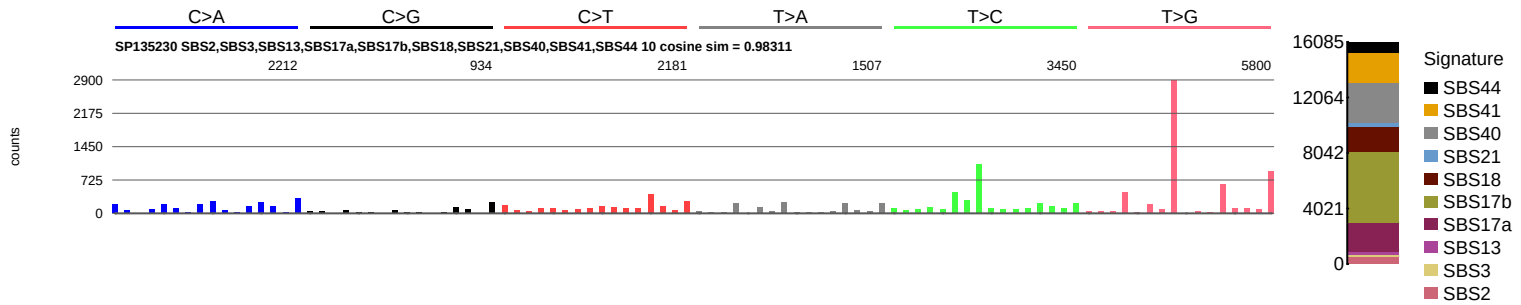
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98322
2156
822
3037
1091
3480
5498
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98322
2255
1129
2730
1185
3017
5771
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98316
2214
924
2188
1499
3455
5805
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98314
1262
1362
3112
1415
3302
5635
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98314
2233
991
2705
1173
3212
5769
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15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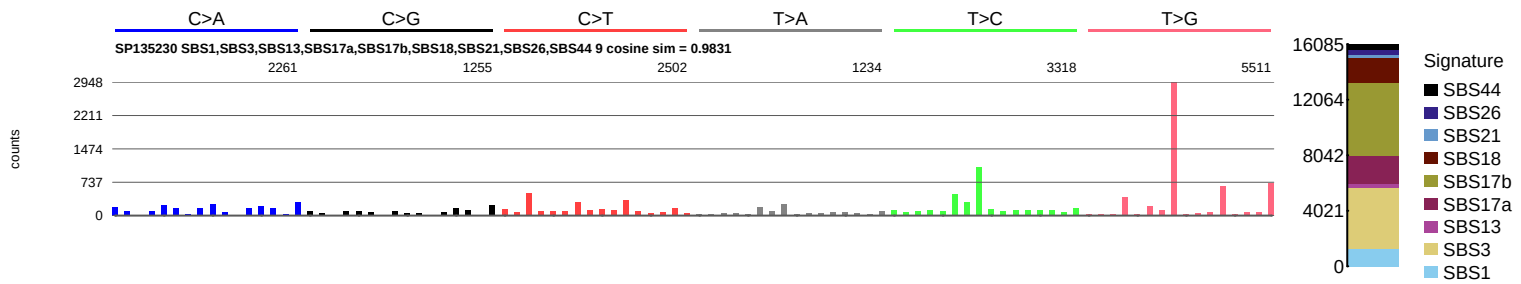
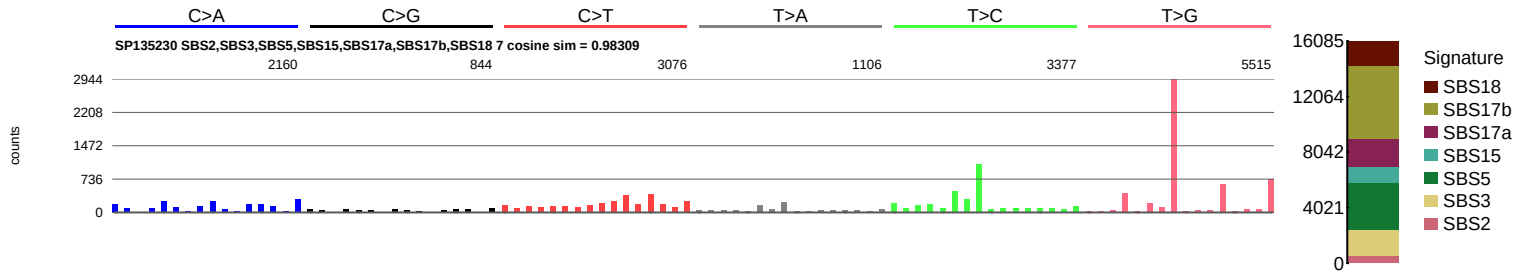
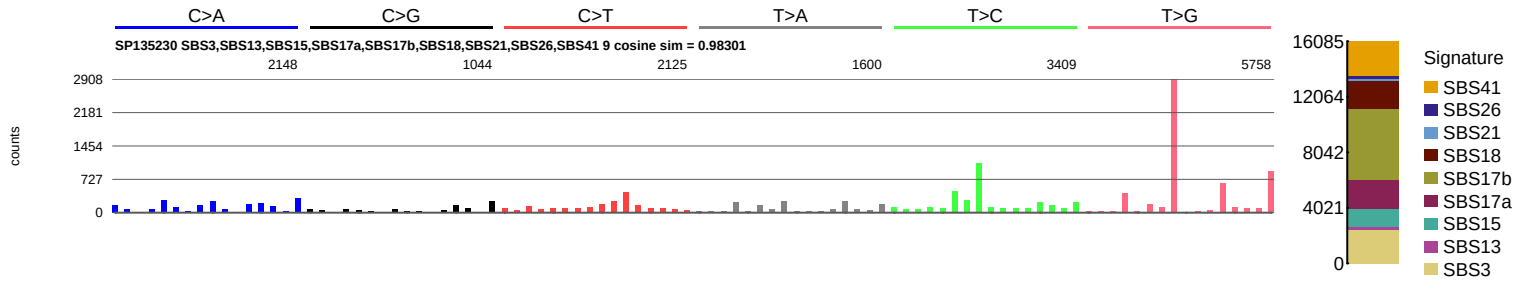
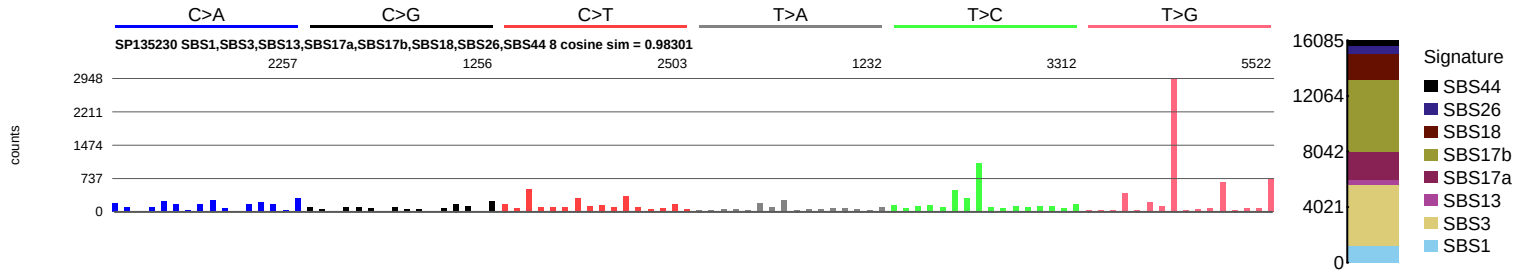
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98311
2212
934
2181
1507
3450
5800
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.9831
2261
1255
2502
1234
3318
5511
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98309
2160
844
3076
1106
3377
5515
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98301
2148
1044
2125
1600
3409
5758
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98301
2257
1256
2503
1232
3312
5522
counts
2948
2211
1474
737
0
16085
12064
8042
4021
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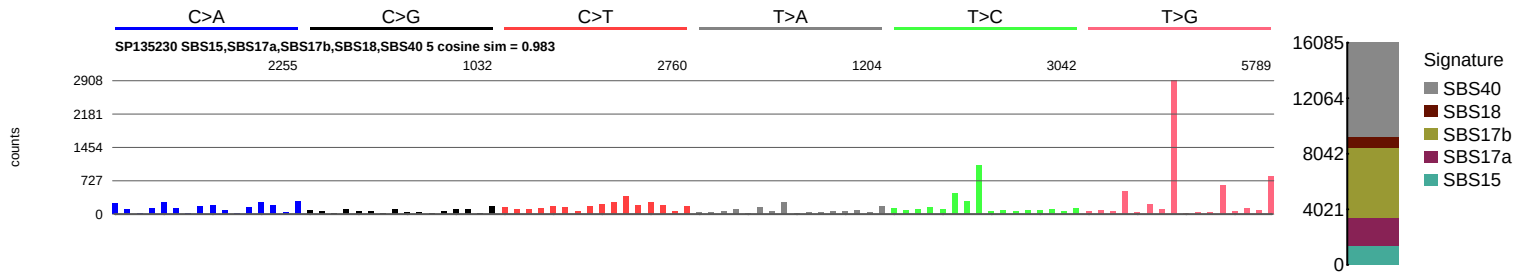
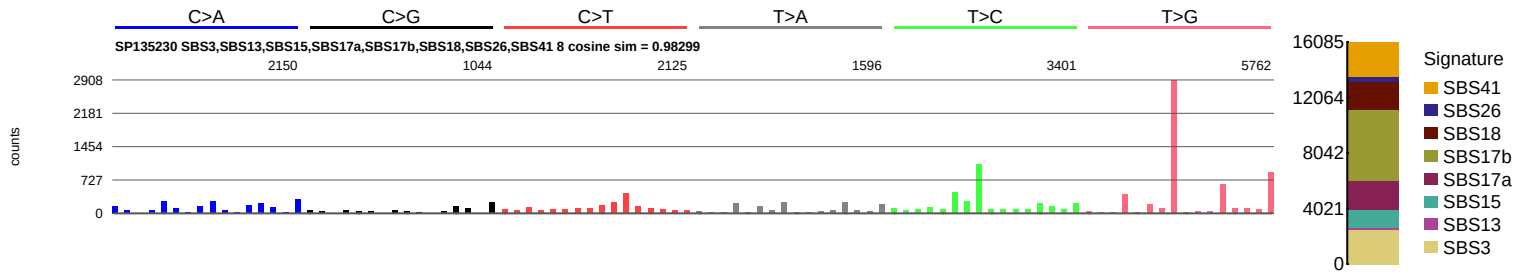
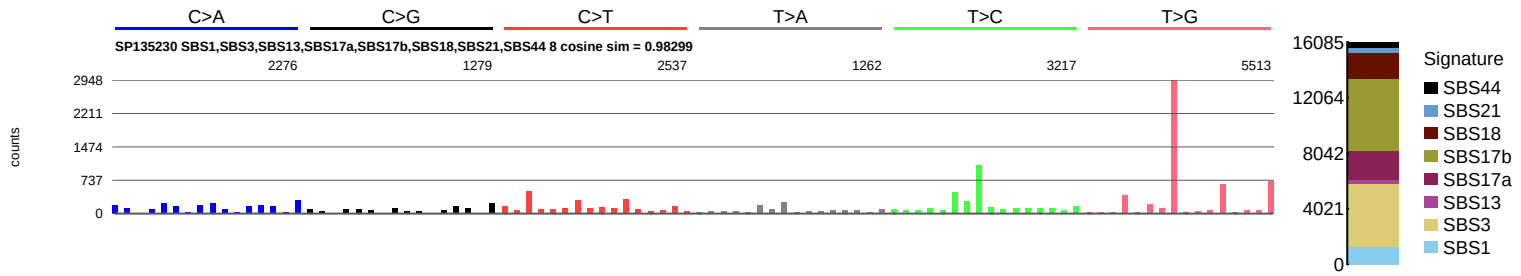
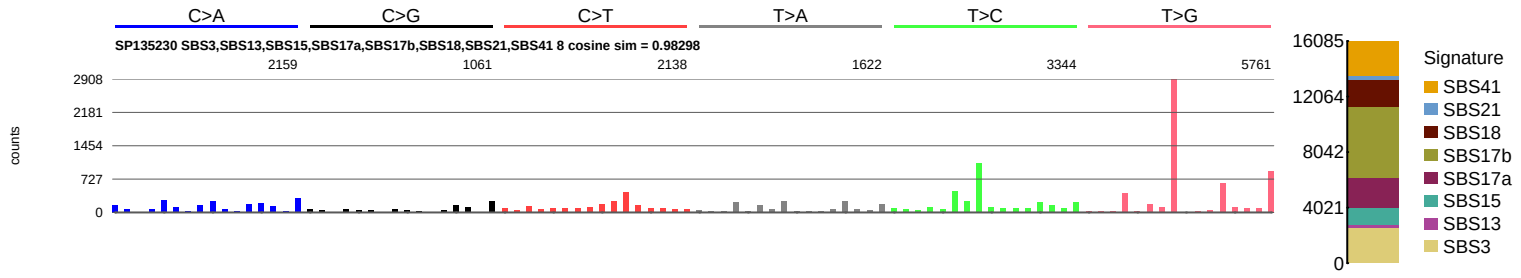
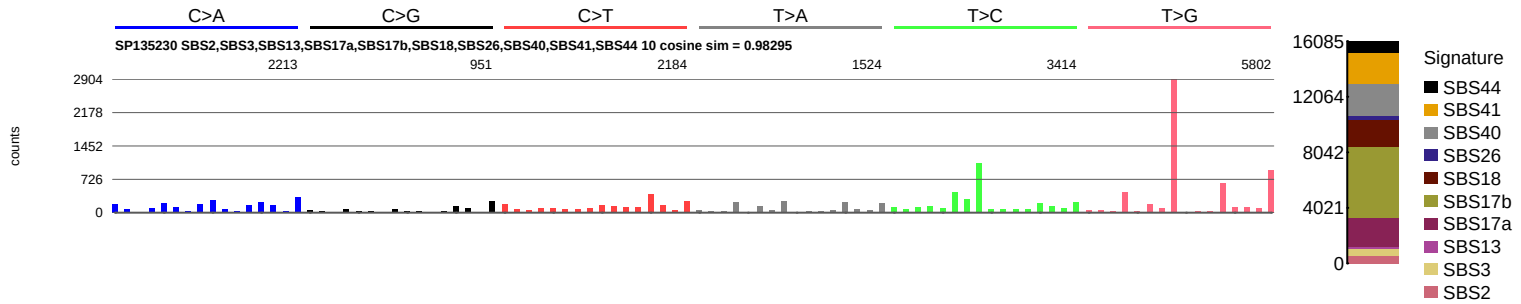
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

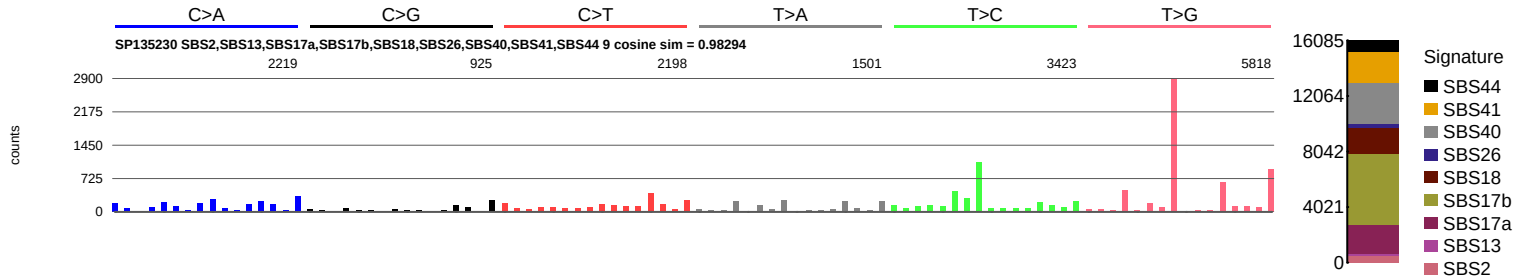

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.983
2255
1032
2760
1204
3042
5789
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15

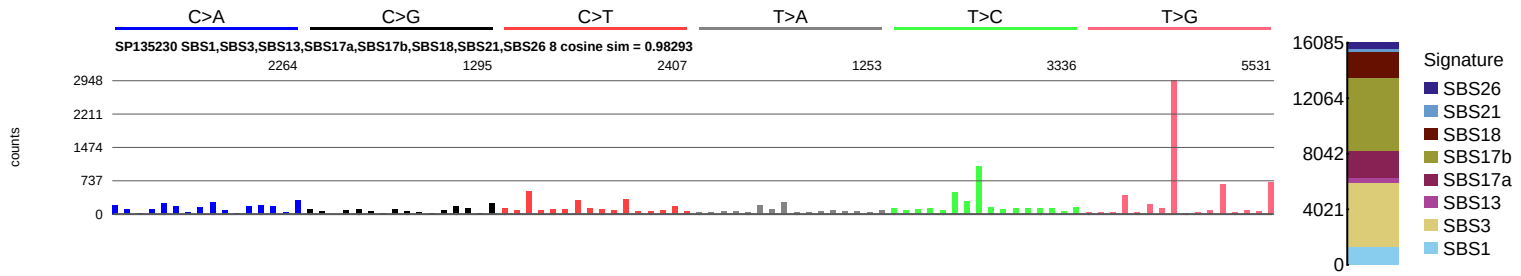

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98299
2150
1044
2125
1596
3401
5762
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

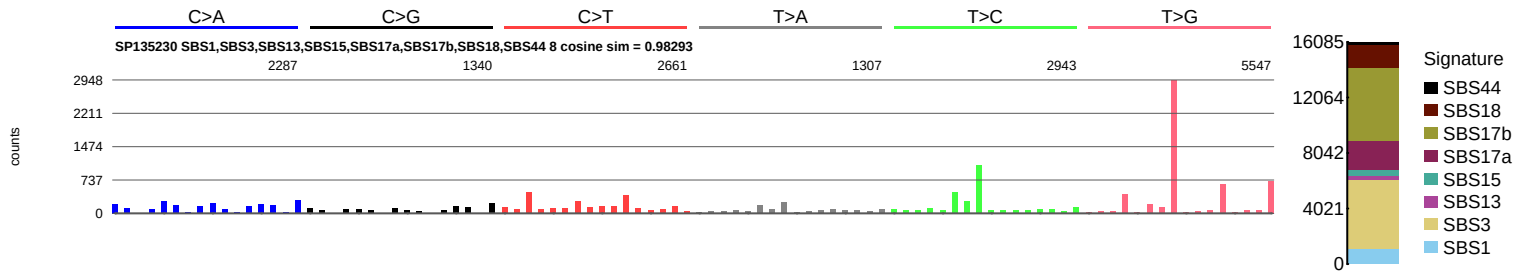

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98299
2276
1279
2537
1262
3217
5513
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98298
2159
1061
2138
1622
3344
5761
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98295
2213
951
2184
1524
3414
5802
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98294
2219
925
2198
1501
3423
5818
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98293
2264
1295
2407
1253
3336
5531
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98293
2287
1340
2661
1307
2943
5547
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
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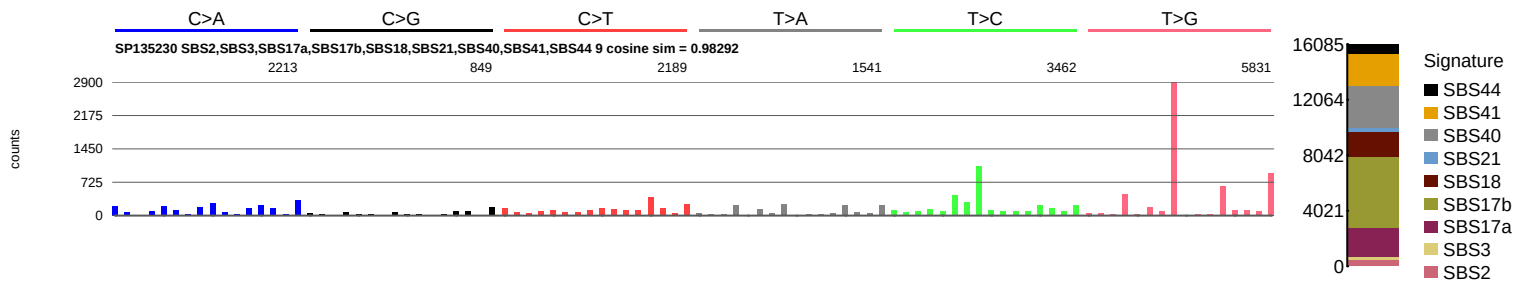
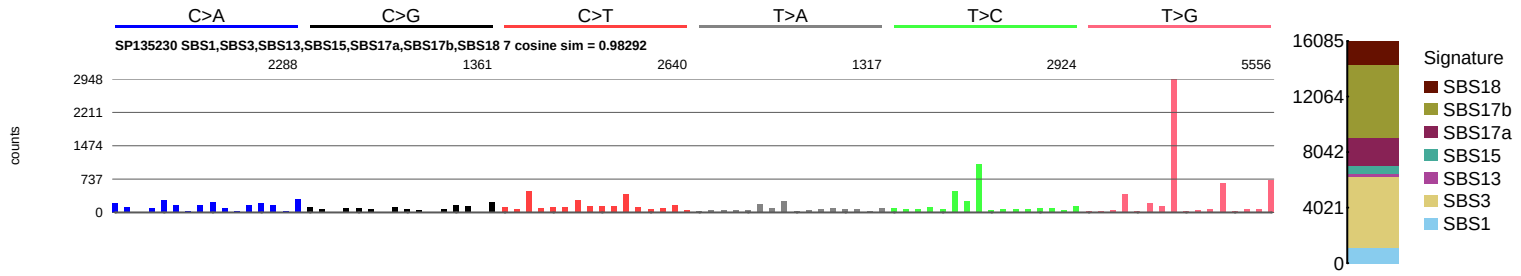
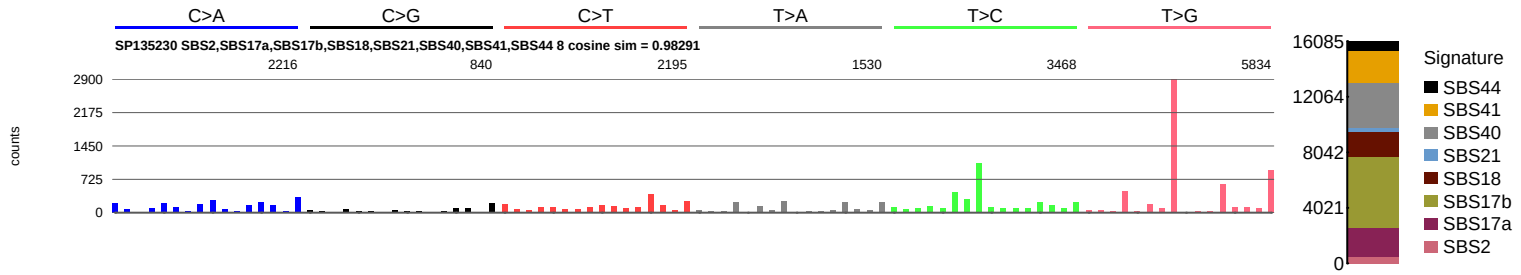
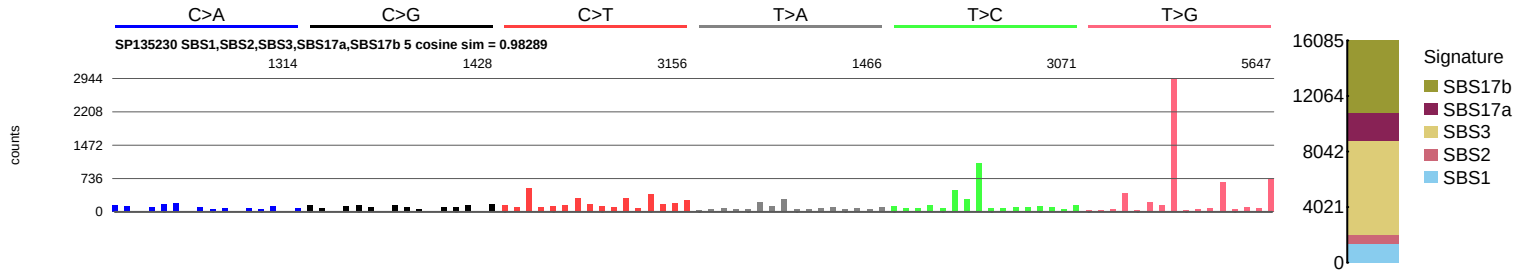
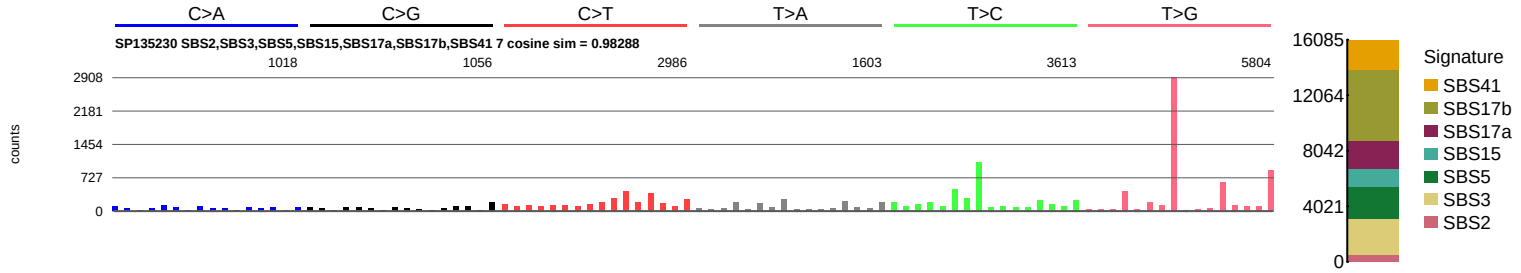
SBS13
SBS3
SBS1

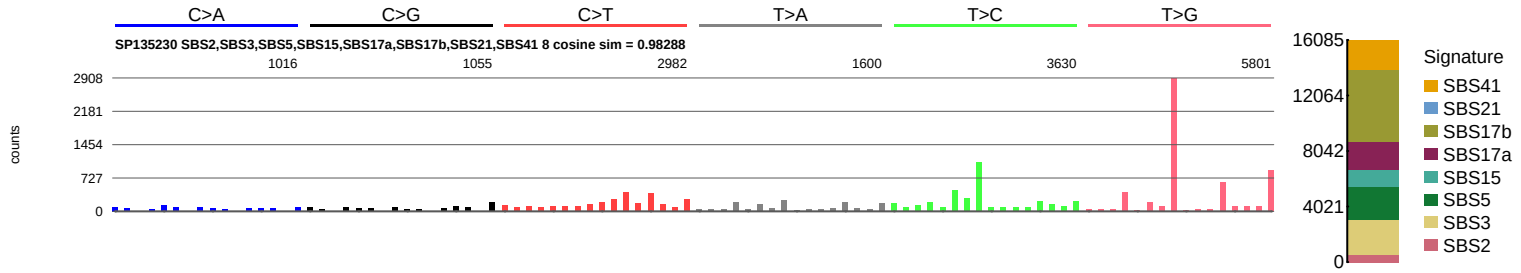

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98292
2213
849
2189
1541
3462
5831
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2

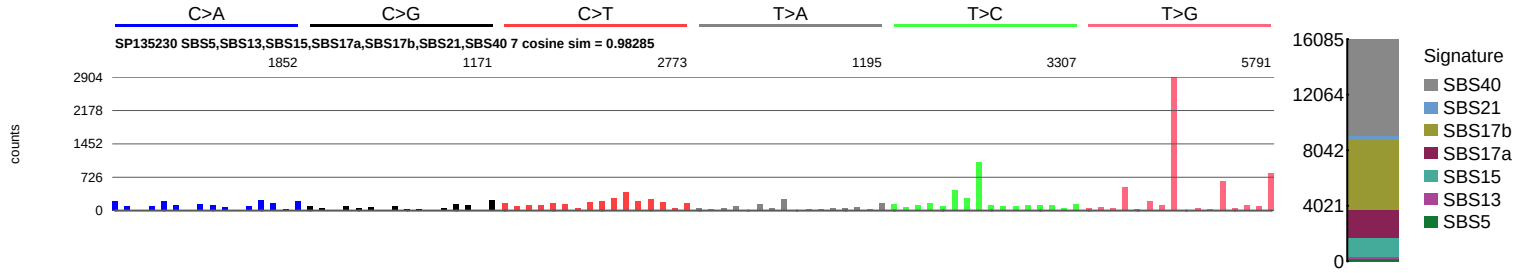

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98292
2288
1361
2640
1317
2924
5556
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

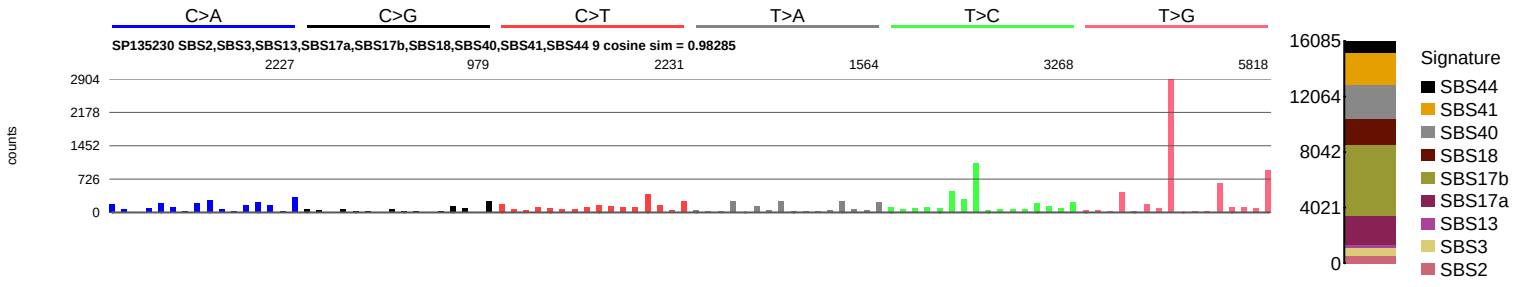

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98291
2216
840
2195
1530
3468
5834
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2

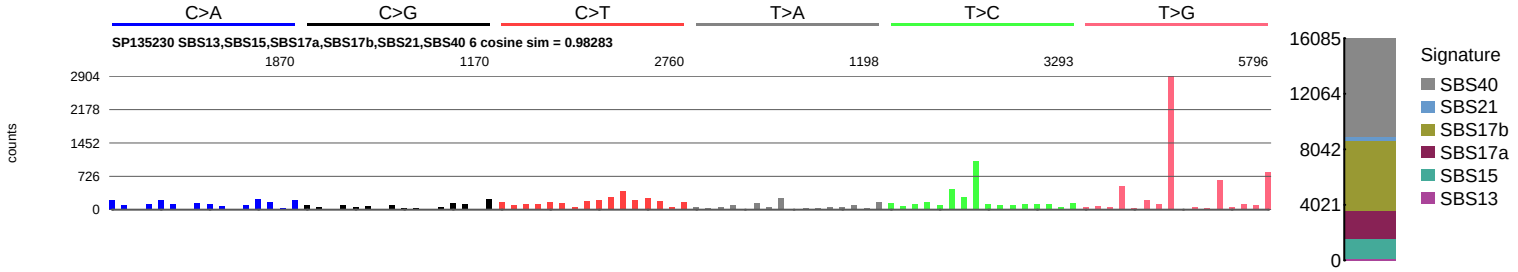

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS3,SBS17a,SBS17b 5 cosine sim = 0.98289
1314
1428
3156
1466
3071
5647
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS3
SBS2
SBS1

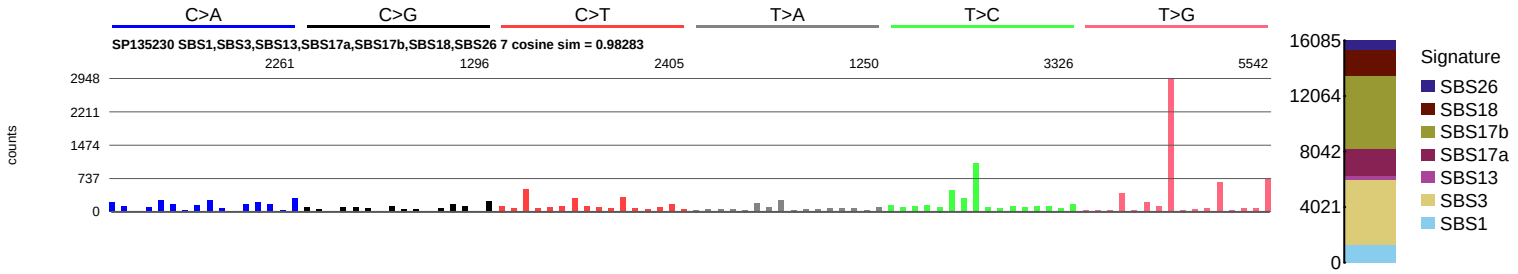

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98288
1018
1056
2986
1603
3613
5804
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

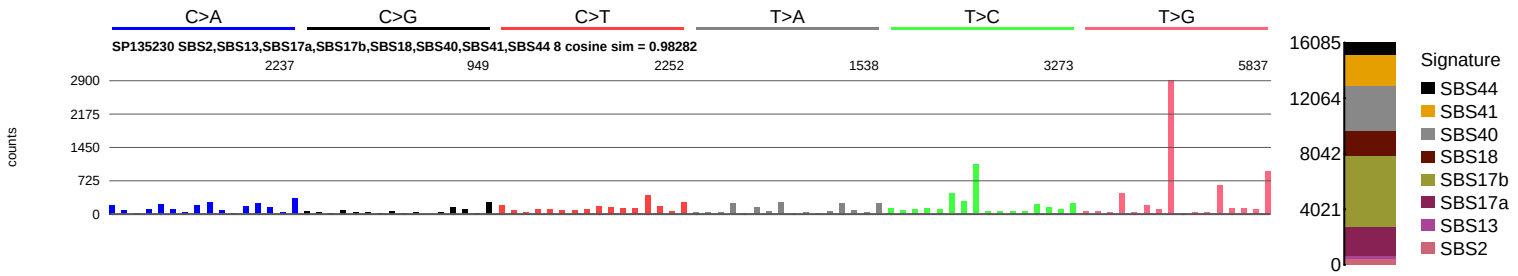

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98288
1016
1055
2982
1600
3630
5801
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

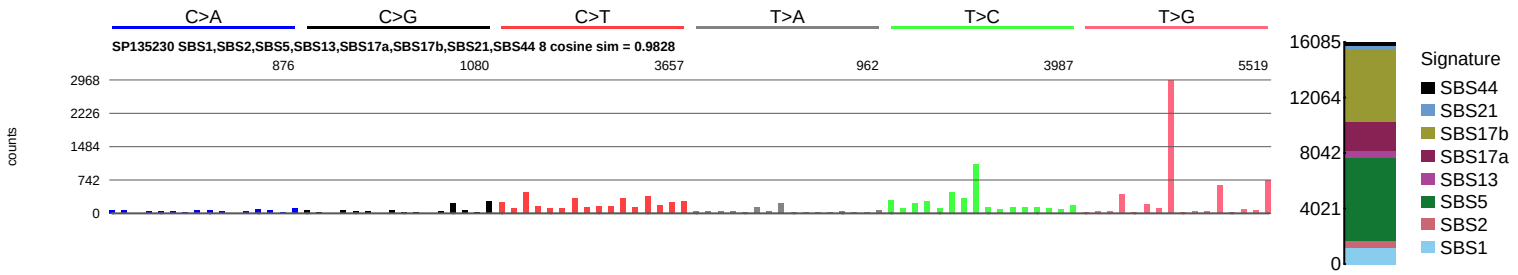

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98285
1852
1171
2773
1195
3307
5791
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98285
2227
979
2231
1564
3268
5818
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98283
1870
1170
2760
1198
3293
5796
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98283
2261
1296
2405
1250
3326
5542
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98282
2237
949
2252
1538
3273
5837
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.9828
876
1080
3657
962
3987
5519
counts
2968
2226
1484
742
0
16085
12064
8042
4021
0
Signature
SBS44
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SBS17b
SBS17a
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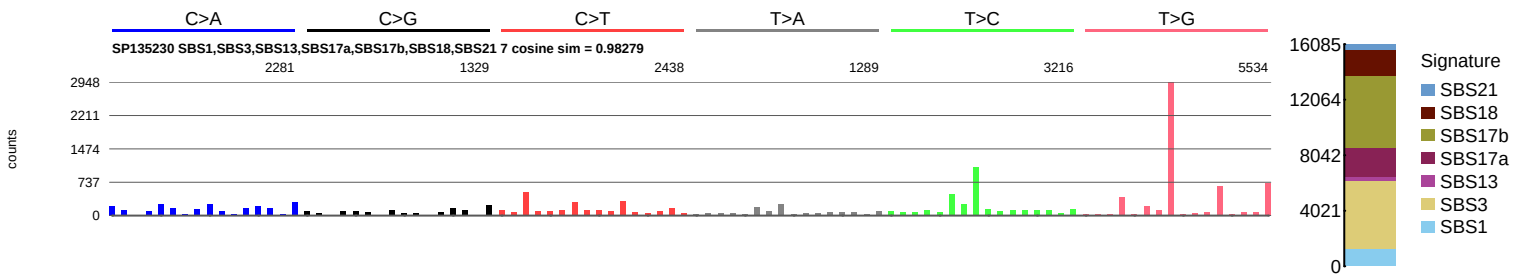
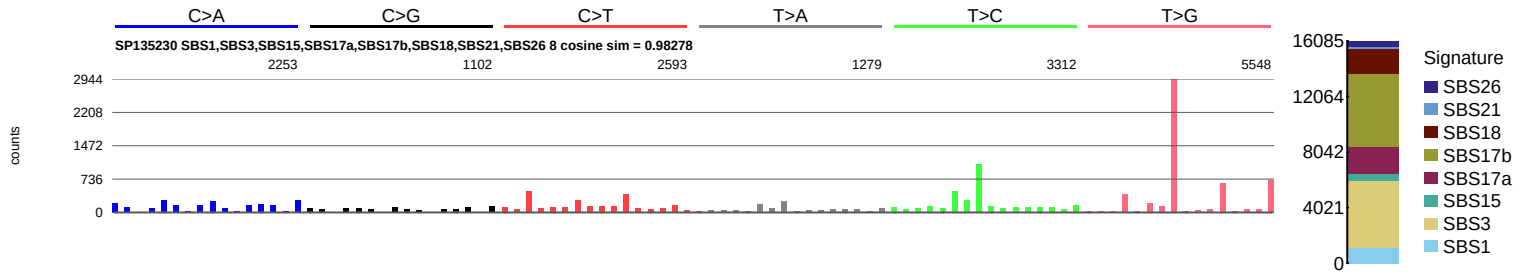
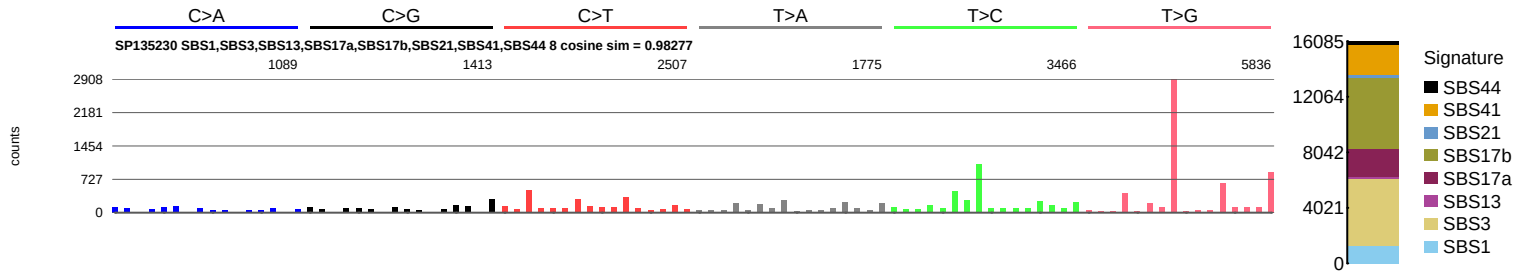
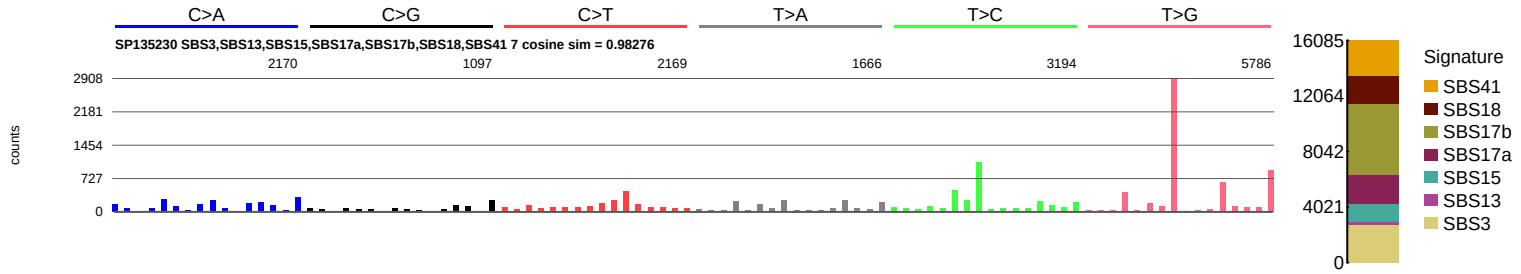
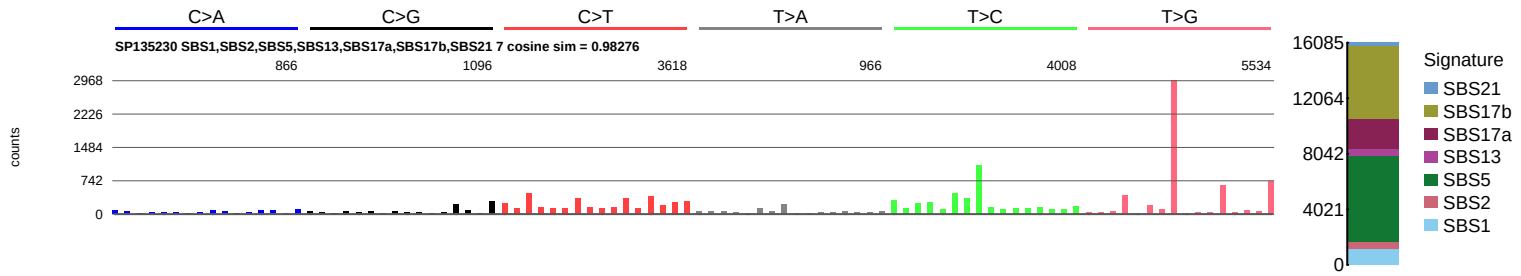
SBS5
SBS2
SBS1

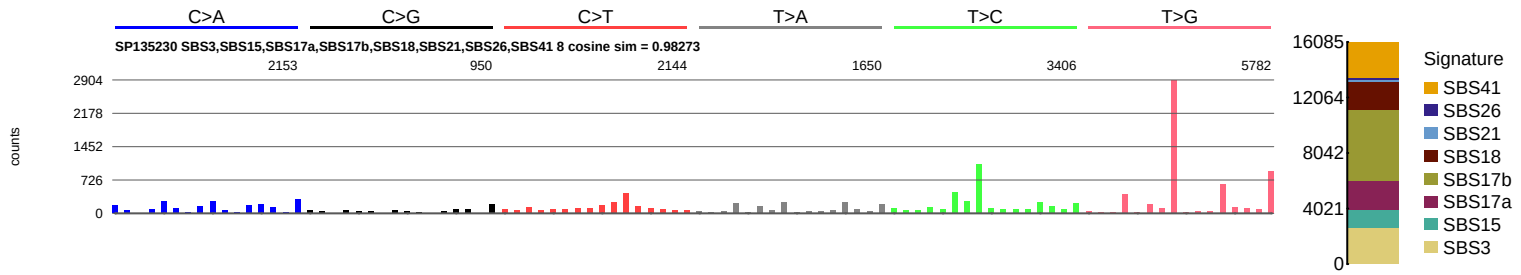

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98279
2281
1329
2438
1289
3216
5534
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

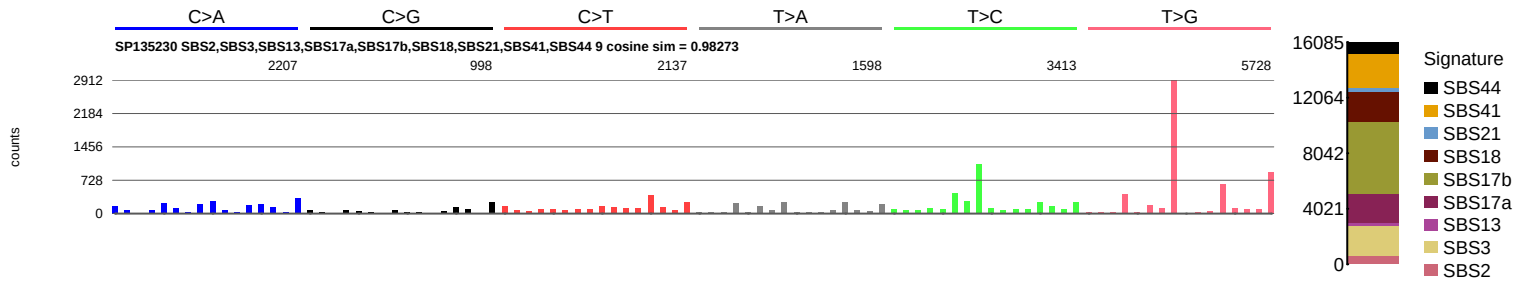

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98278
2253
1102
2593
1279
3312
5548
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

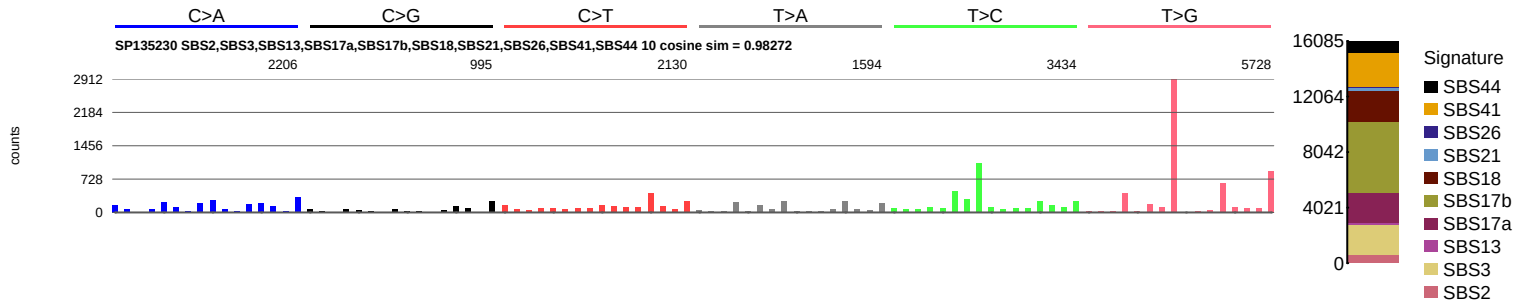

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98277
1089
1413
2507
1775
3466
5836
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1

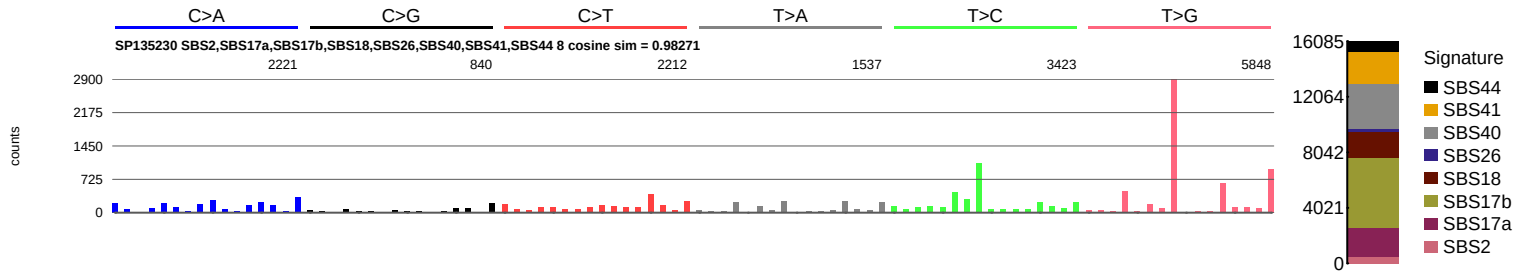

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98276
2170
1097
2169
1666
3194
5786
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3

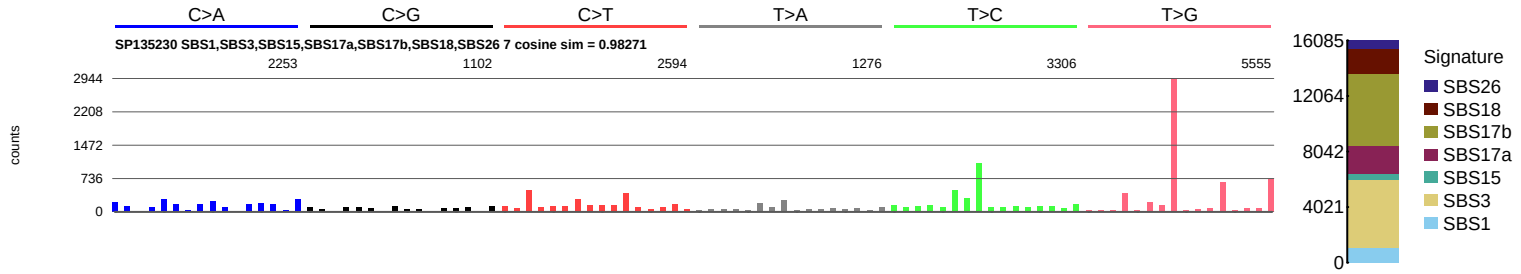

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98276
866
1096
3618
966
4008
5534
counts
2968
2226
1484
742
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

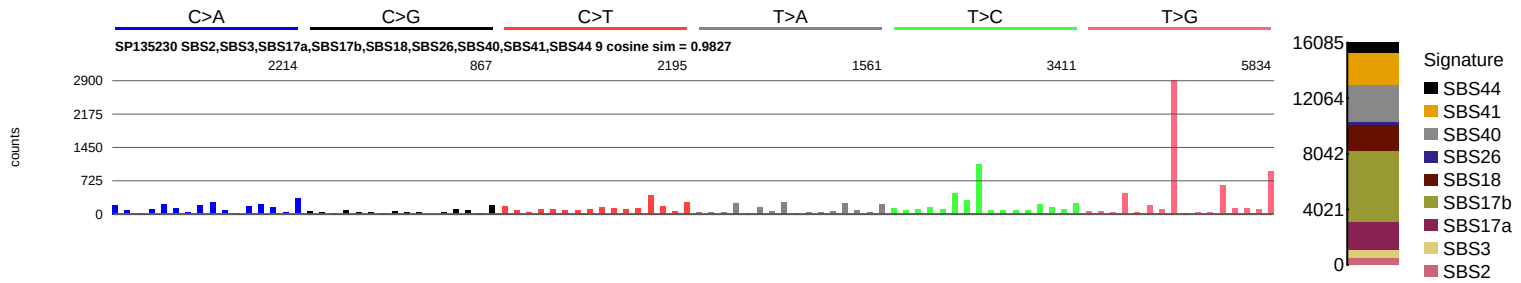

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98273
2153
950
2144
1650
3406
5782
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3

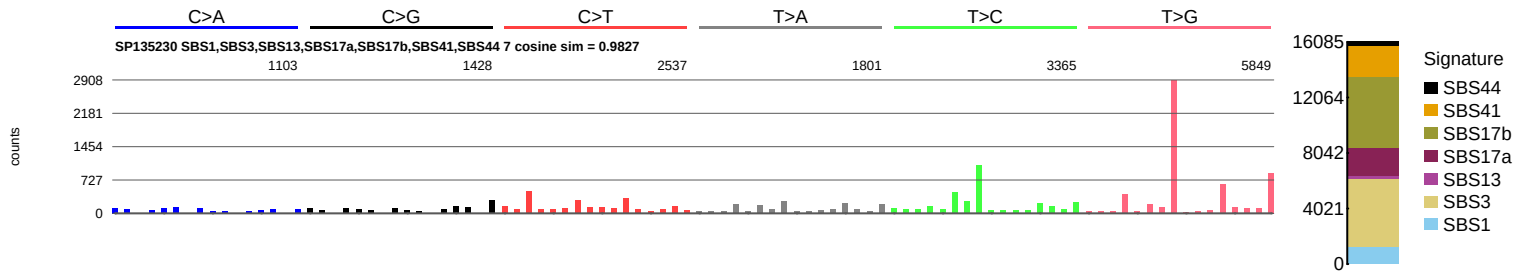

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98273
2207
998
2137
1598
3413
5728
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

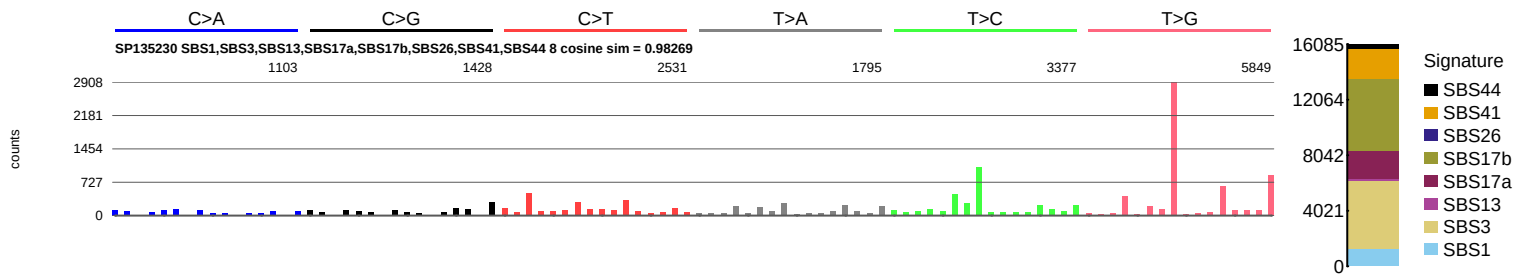

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98272
2206
995
2130
1594
3434
5728
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

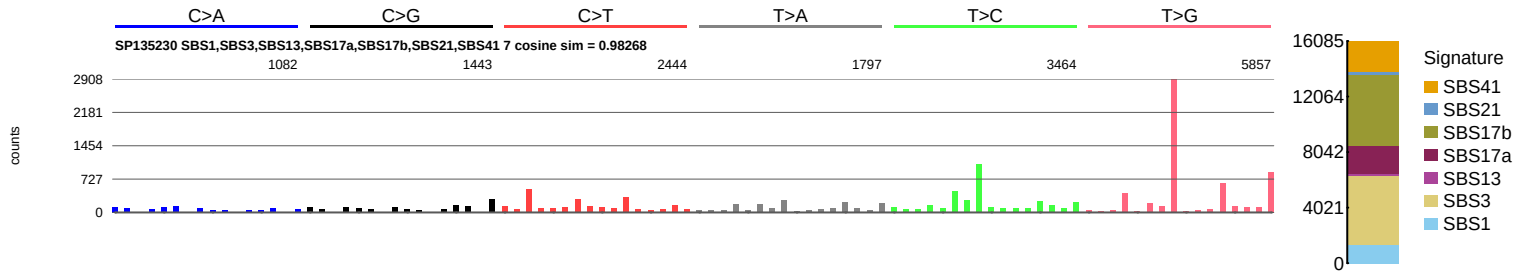

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98271
2221
840
2212
1537
3423
5848
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2

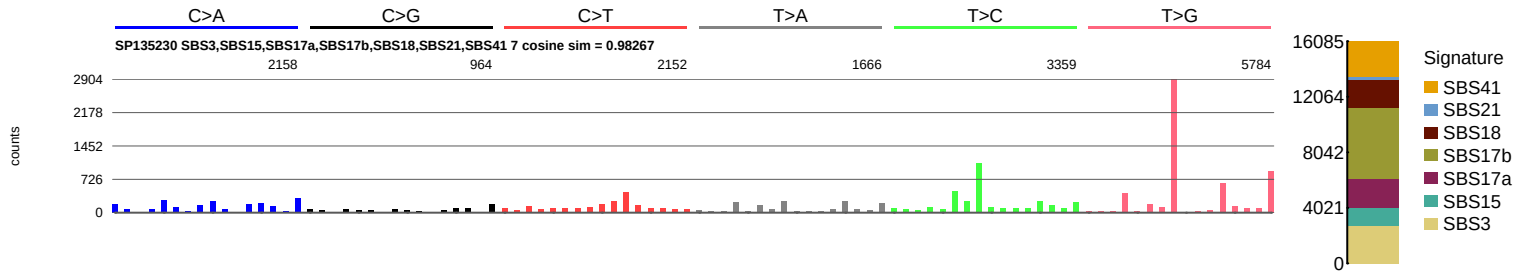

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98271
2253
1102
2594
1276
3306
5555
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

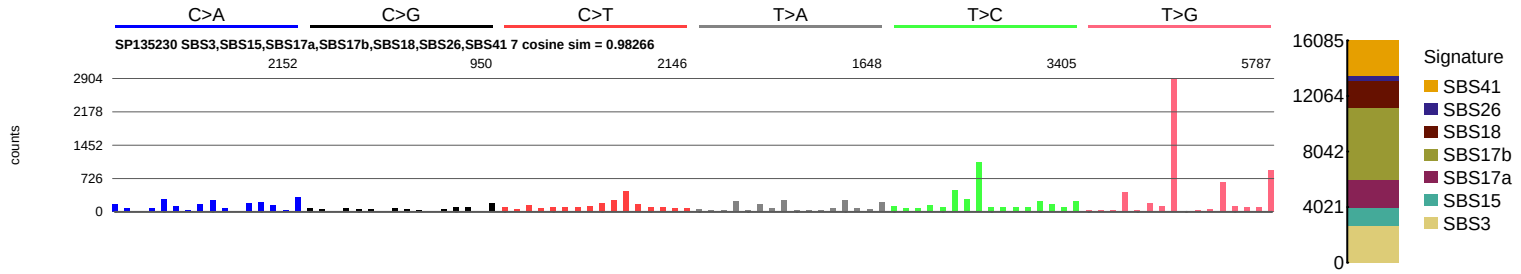

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9827
2214
867
2195
1561
3411
5834
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2

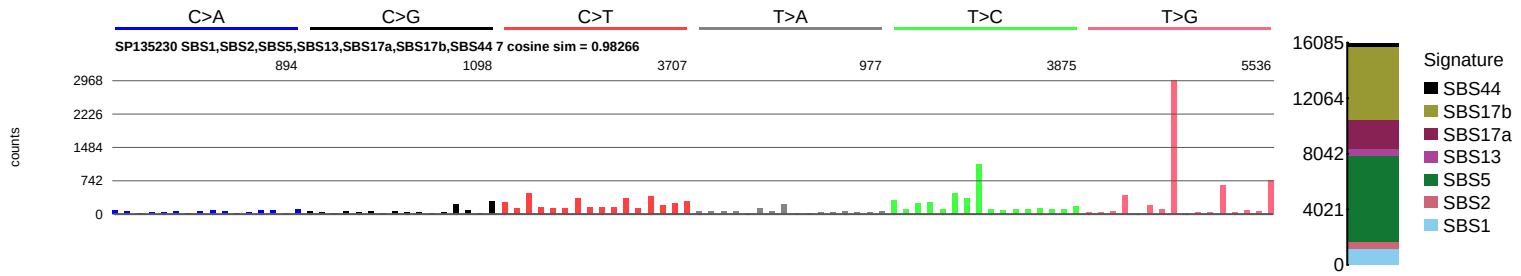

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.9827
1103
1428
2537
1801
3365
5849
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS1

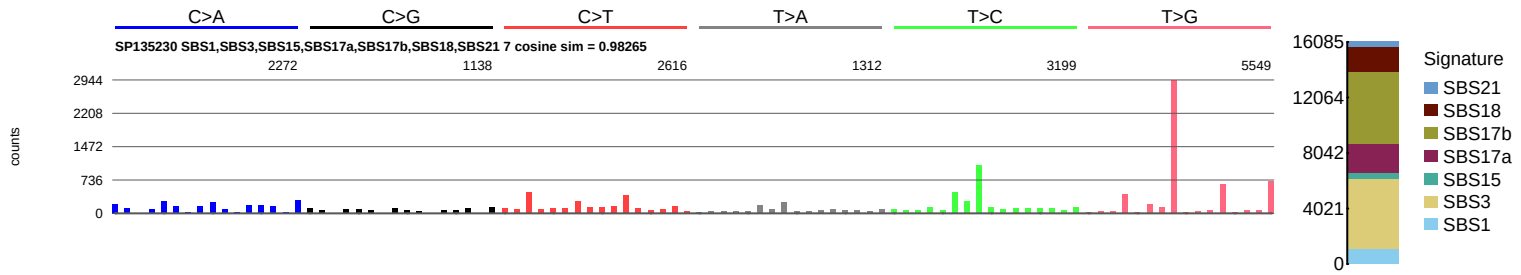

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98269
1103
1428
2531
1795
3377
5849
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1

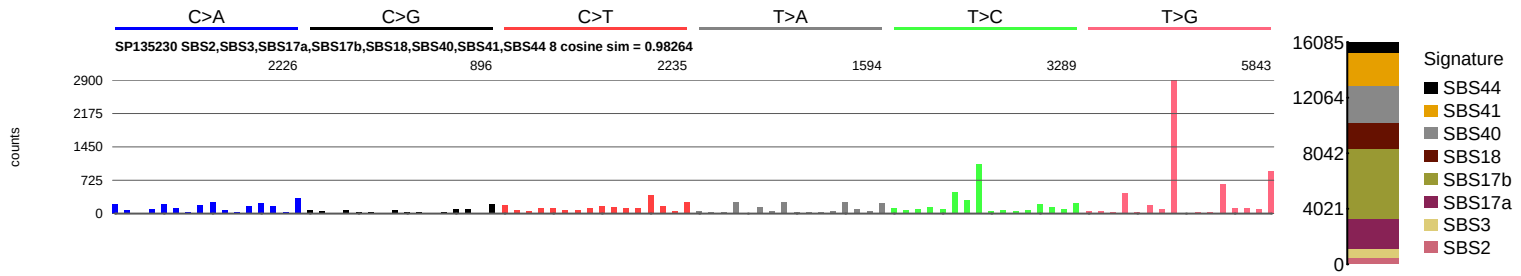

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98268
1082
1443
2444
1797
3464
5857
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98267
2158
964
2152
1666
3359
5784
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98266
2152
950
2146
1648
3405
5787
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98266
894
1098
3707
977
3875
5536
counts
2968
2226
1484
742
0
16085
12064
8042
4021
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98265
2272
1138
2616
1312
3199
5549
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98264
2226
896
2235
1594
3289
5843
counts
2900
2175
1450
725
0
16085
12064
8042
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Signature
SBS44
SBS41
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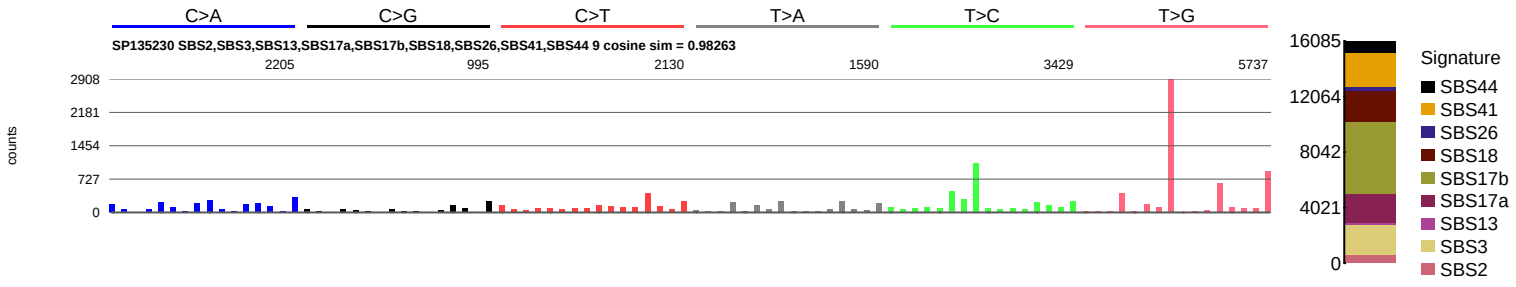
SBS3
SBS2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SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98263
2205
995
2130
1590
3429
5737
counts
2908
2181
1454
727
0
16085
12064
8042
4021
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SBS18
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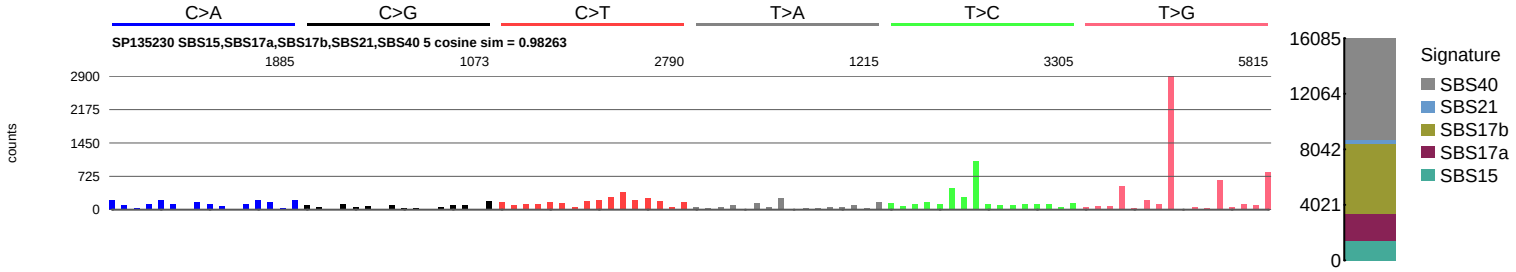
SBS3
SBS2


C>A
C>G
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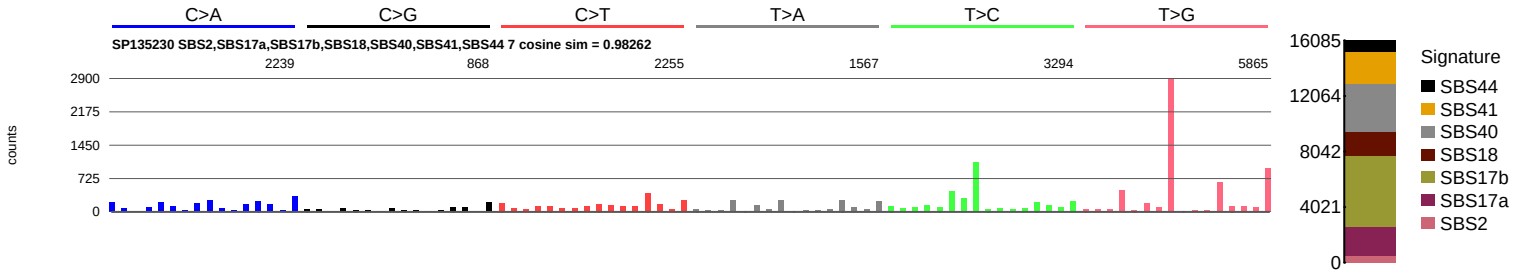
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98263
1885
1073
2790
1215
3305
5815
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15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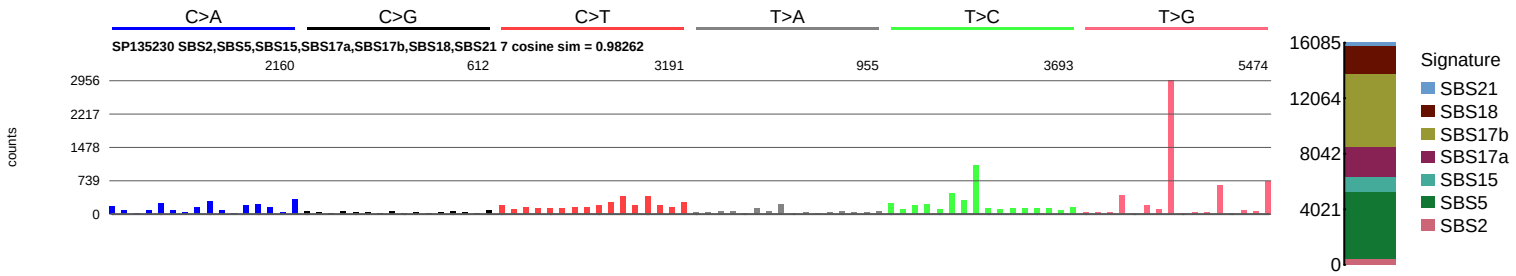
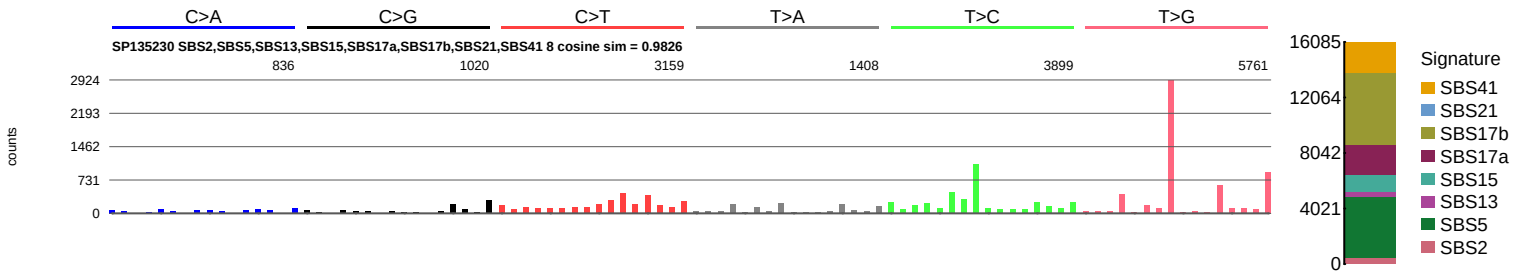
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98262
2239
868
2255
1567
3294
5865
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98262
2160
612
3191
955
3693
5474
counts
2956
2217
1478
739
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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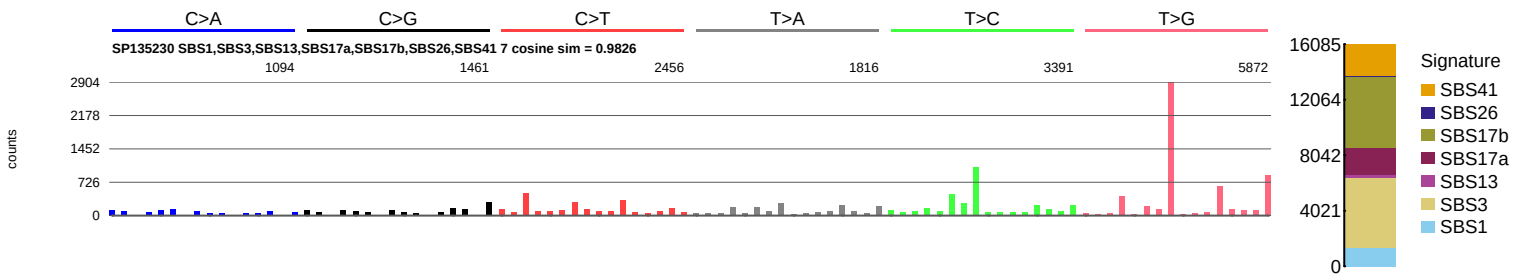
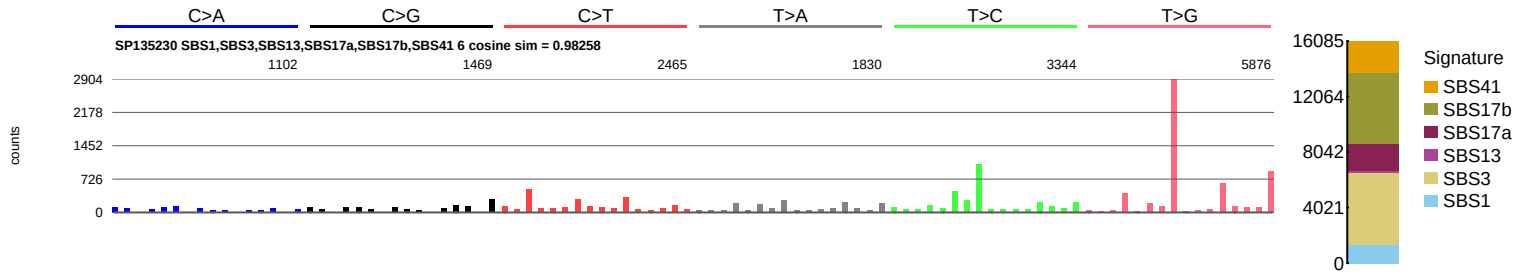
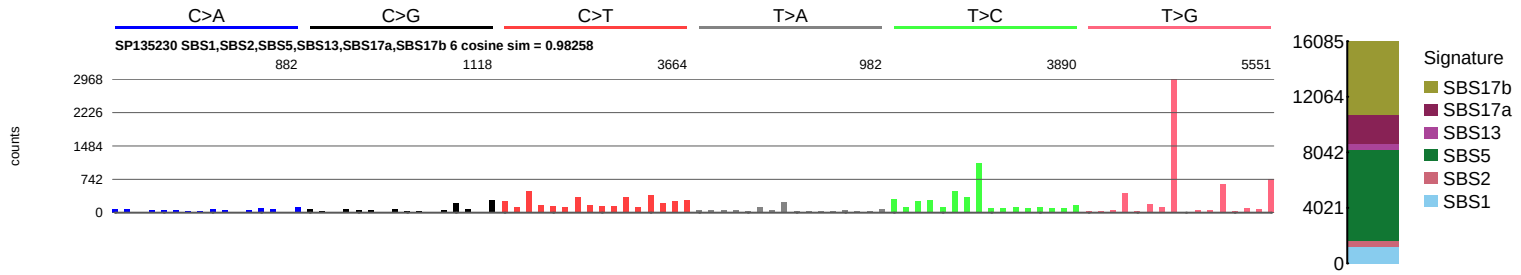
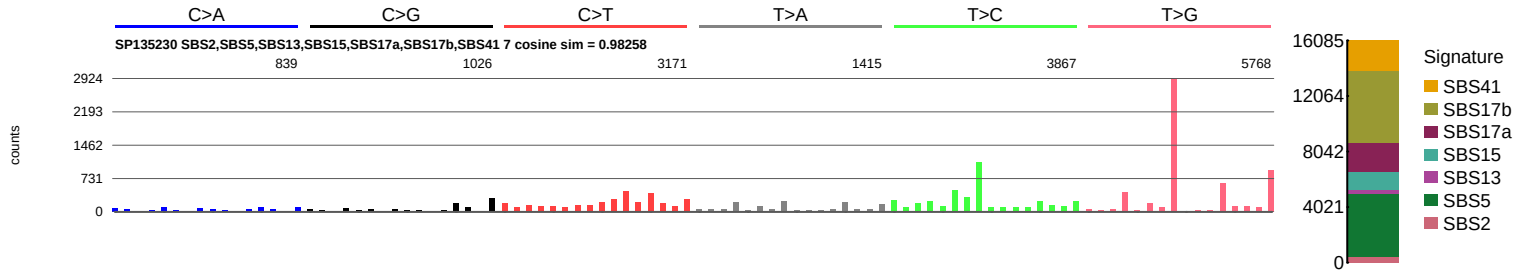
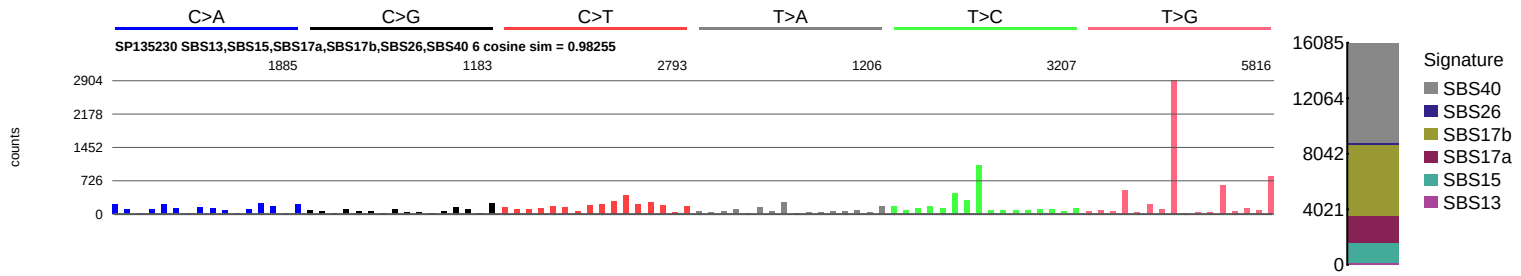
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9826
836
1020
3159
1408
3899
5761
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

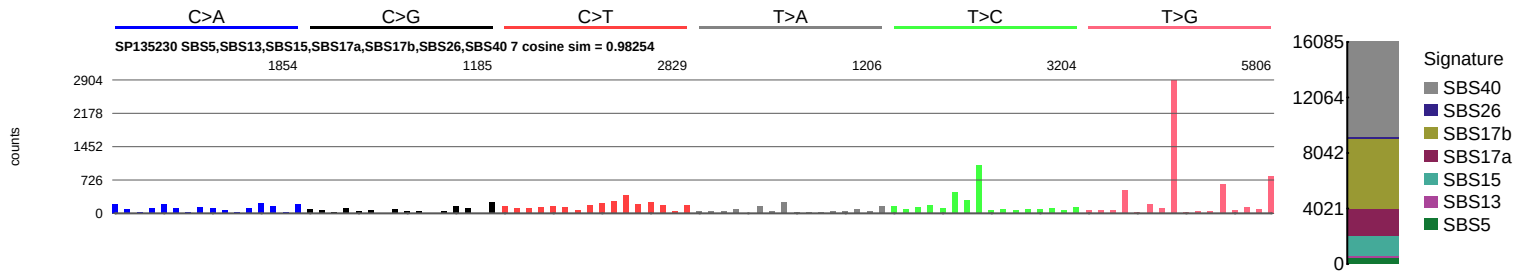

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.9826
1094
1461
2456
1816
3391
5872
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS1

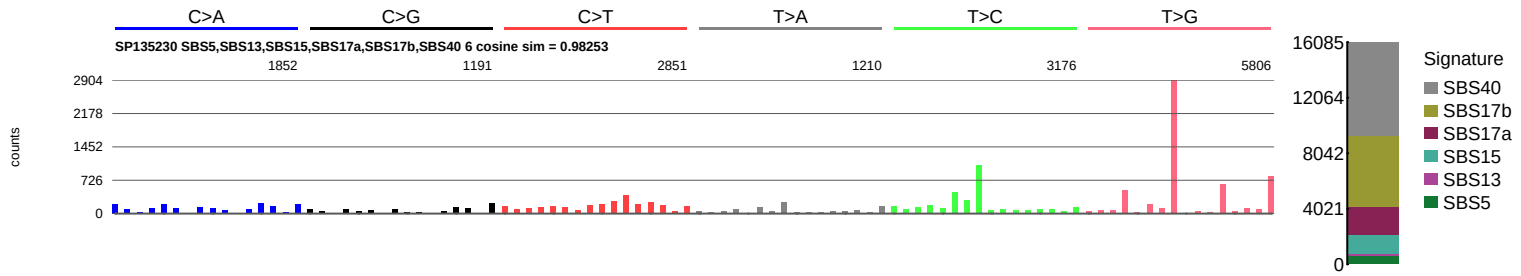

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98258
1102
1469
2465
1830
3344
5876
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS1

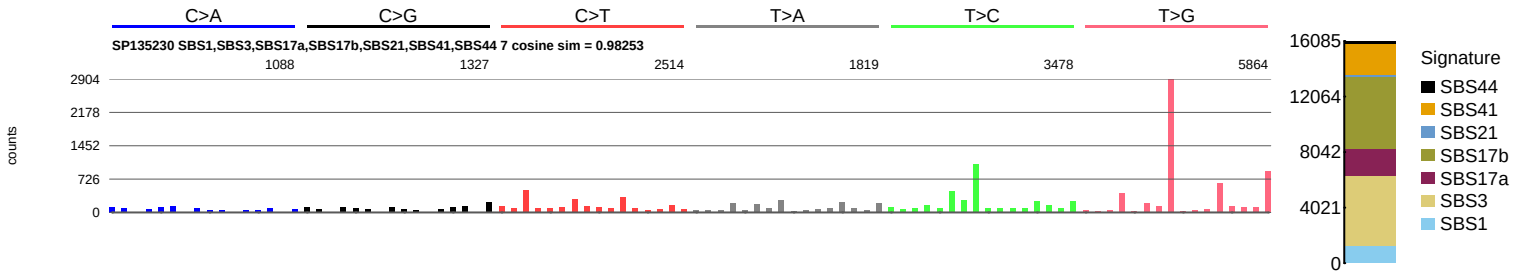

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98258
882
1118
3664
982
3890
5551
counts
2968
2226
1484
742
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

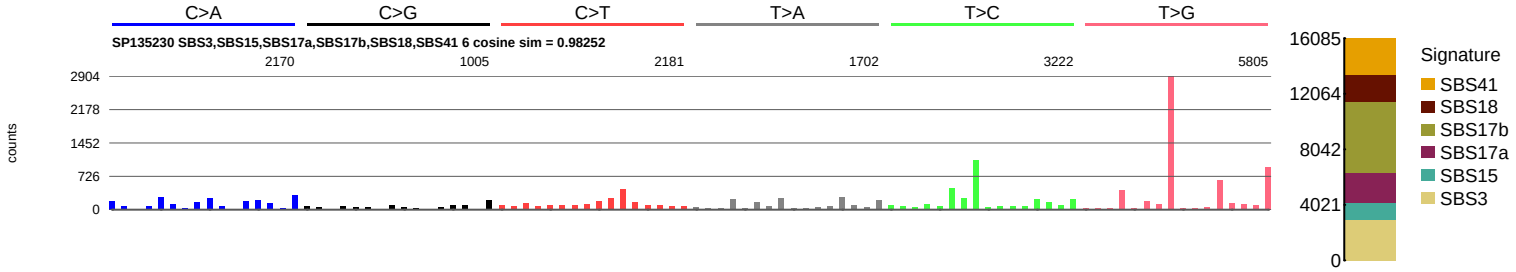

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98258
839
1026
3171
1415
3867
5768
counts
2924
2193
1462
731
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

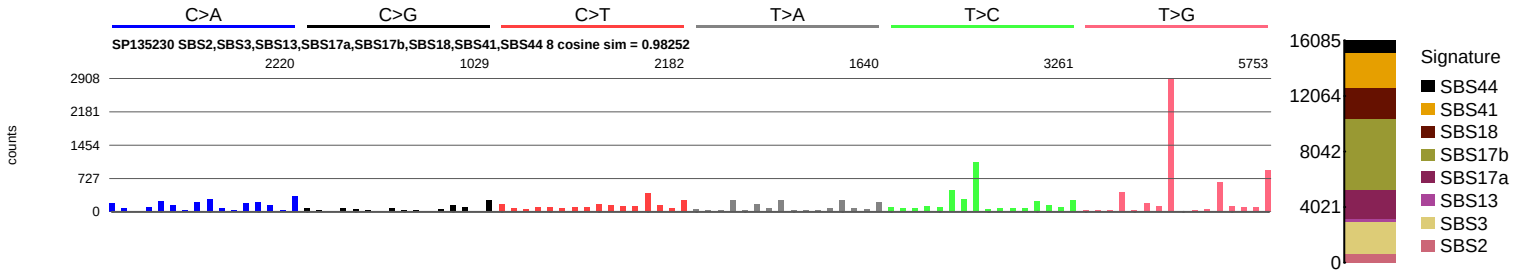

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98255
1885
1183
2793
1206
3207
5816
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98254
1854
1185
2829
1206
3204
5806
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98253
1852
1191
2851
1210
3176
5806
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98253
1088
1327
2514
1819
3478
5864
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98252
2170
1005
2181
1702
3222
5805
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3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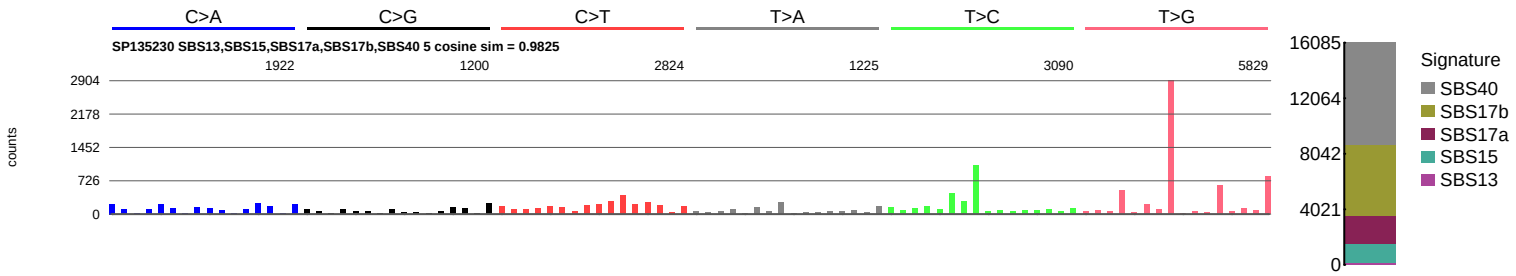
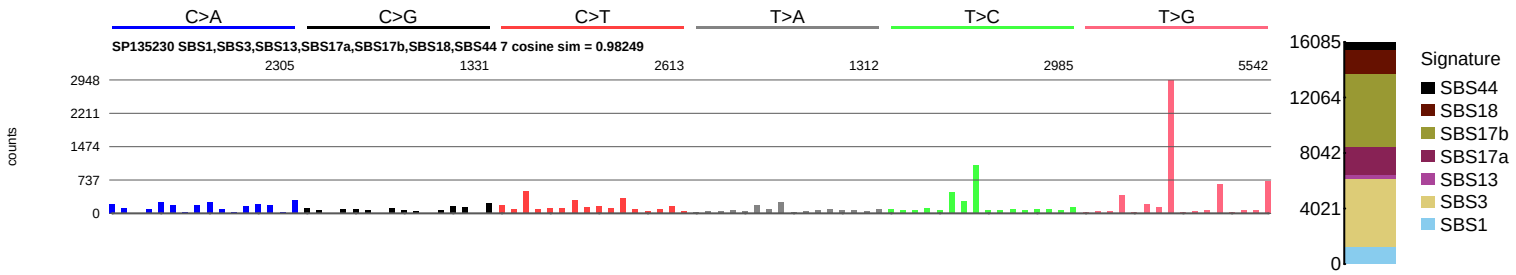
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98252
2220
1029
2182
1640
3261
5753
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.9825
1922
1200
2824
1225
3090
5829
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13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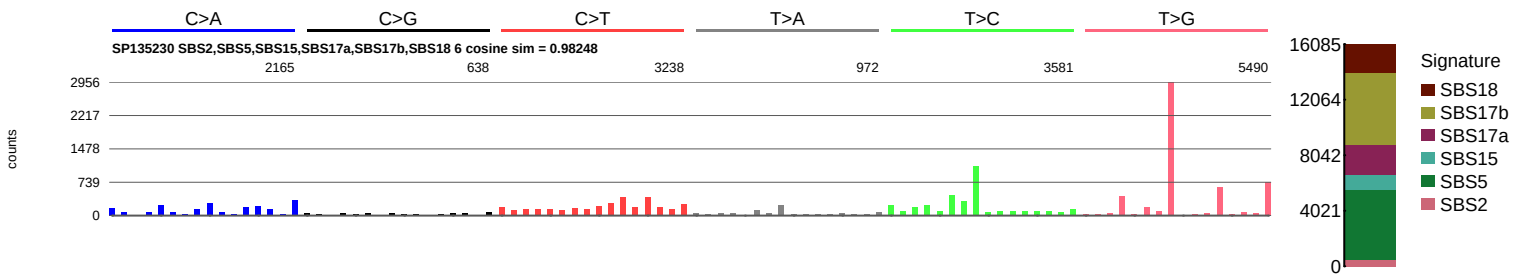
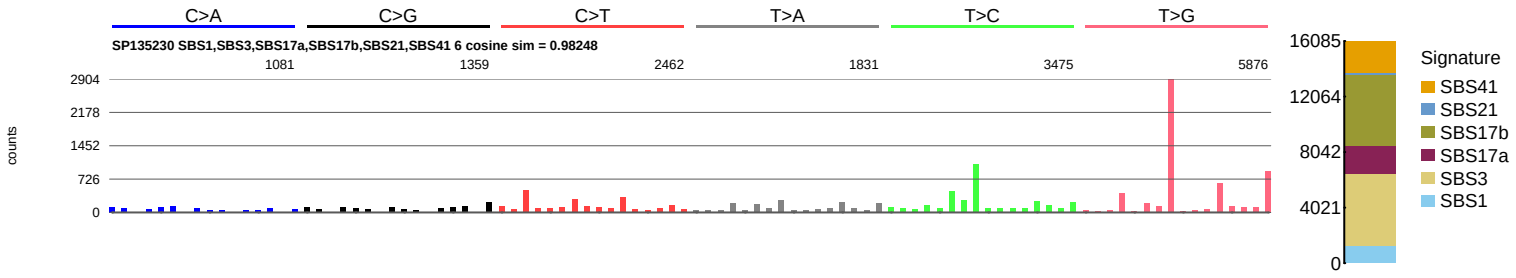
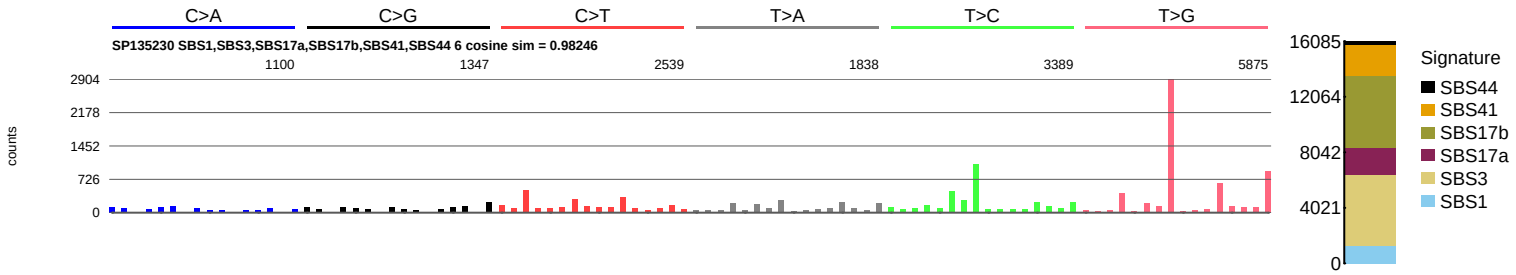
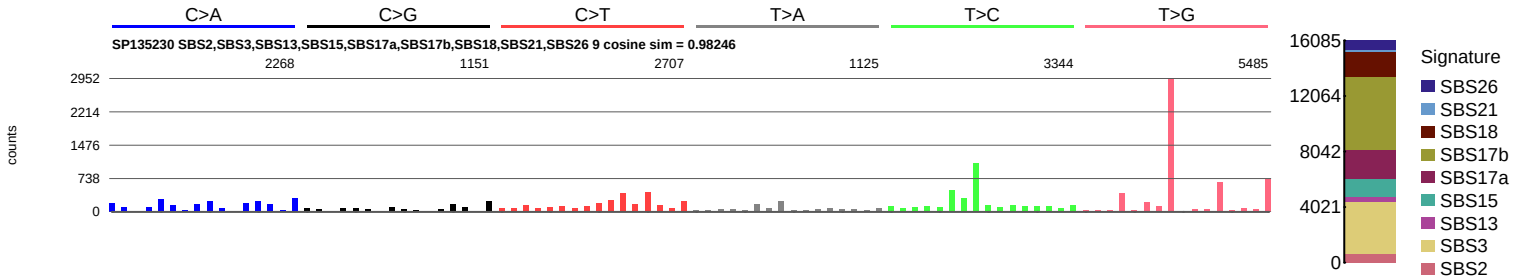
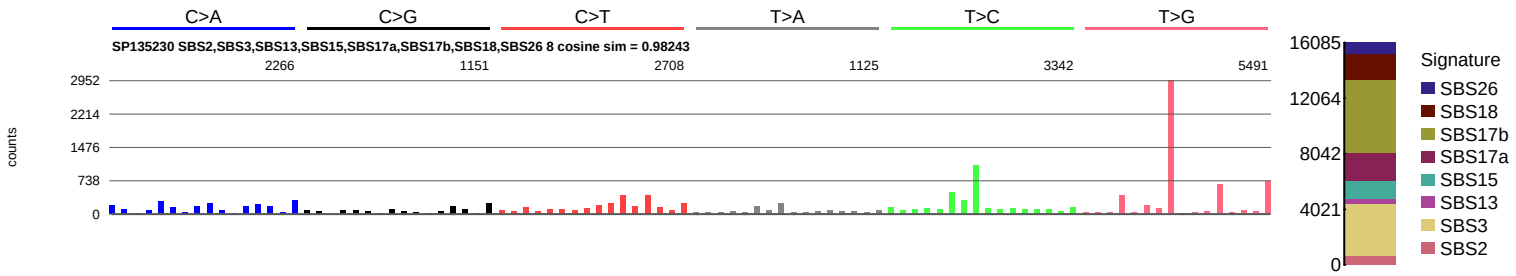
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98249
2305
1331
2613
1312
2985
5542
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

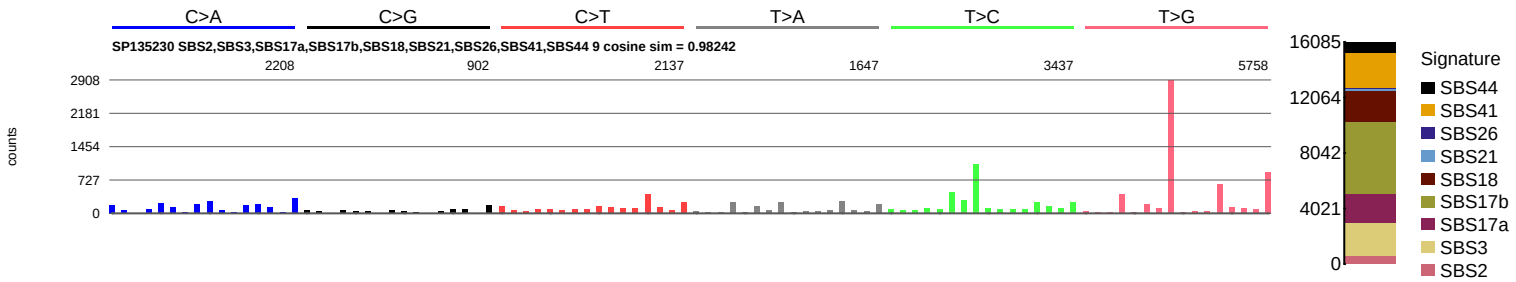

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98248
2165
638
3238
972
3581
5490
counts
2956
2217
1478
739
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

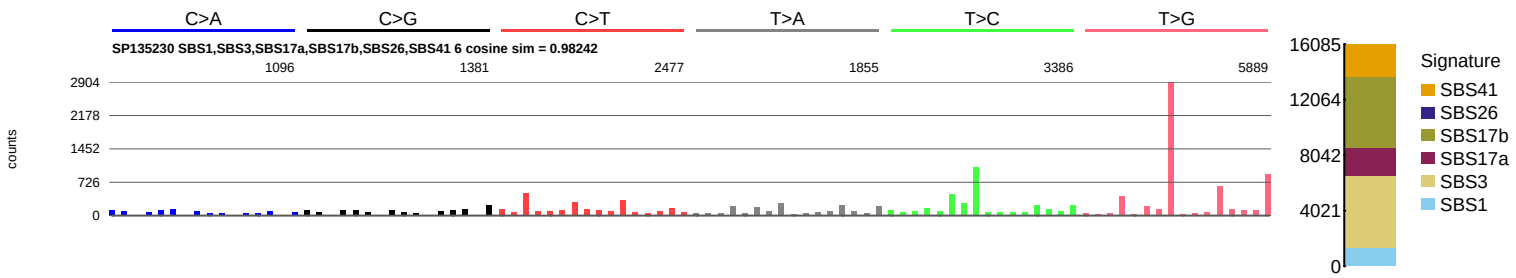

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98248
1081
1359
2462
1831
3475
5876
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1

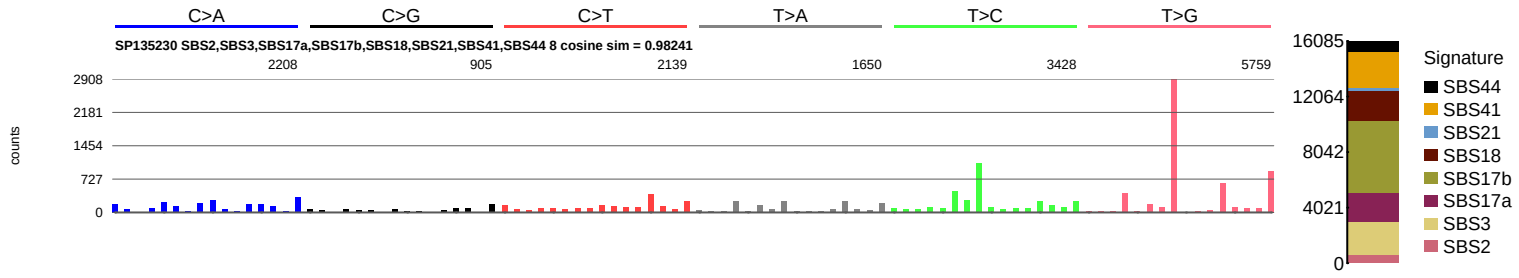

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98246
1100
1347
2539
1838
3389
5875
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS1

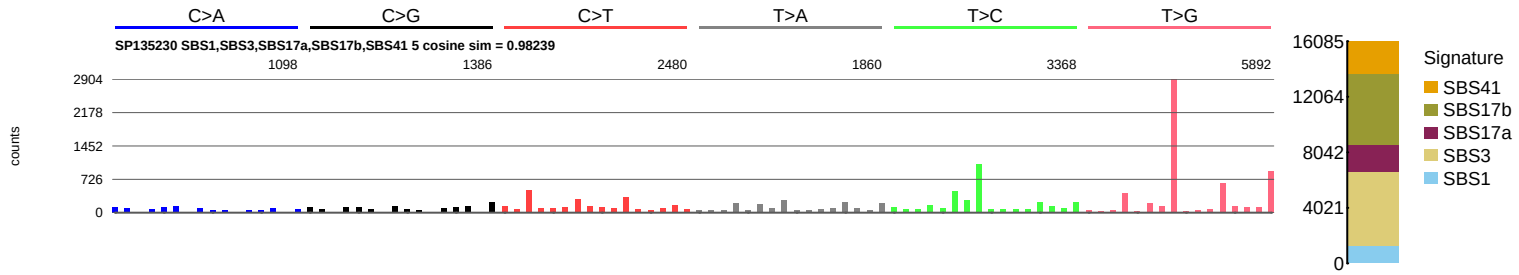

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98246
2268
1151
2707
1125
3344
5485
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98243
2266
1151
2708
1125
3342
5491
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98242
2208
902
2137
1647
3437
5758
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98242
1096
1381
2477
1855
3386
5889
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98241
2208
905
2139
1650
3428
5759
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2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C>G
C>T
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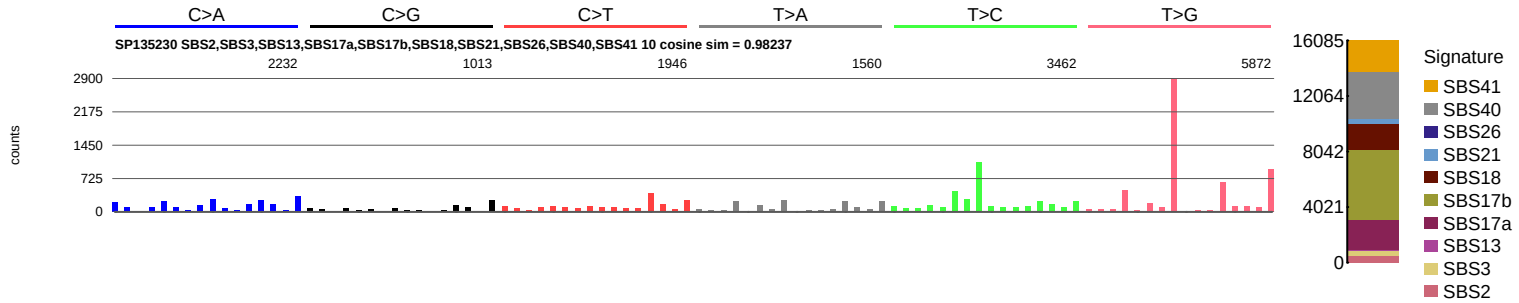
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98239
1098
1386
2480
1860
3368
5892
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS1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C>G
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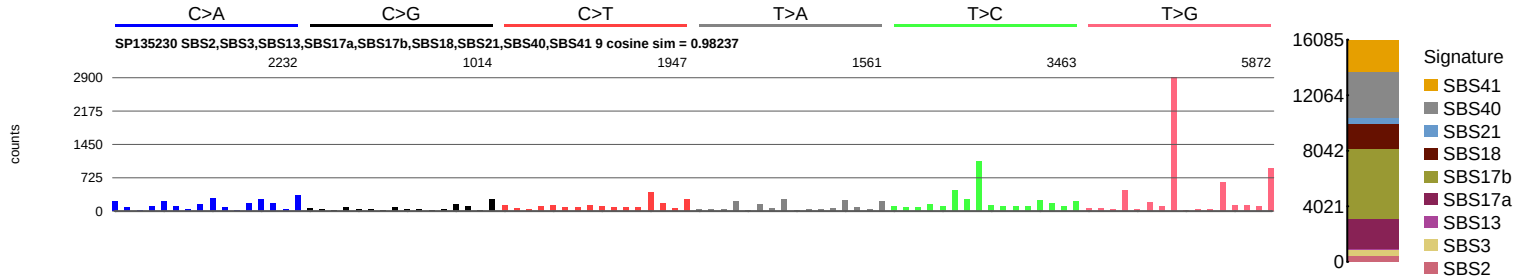
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98237
2232
1013
1946
1560
3462
5872
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2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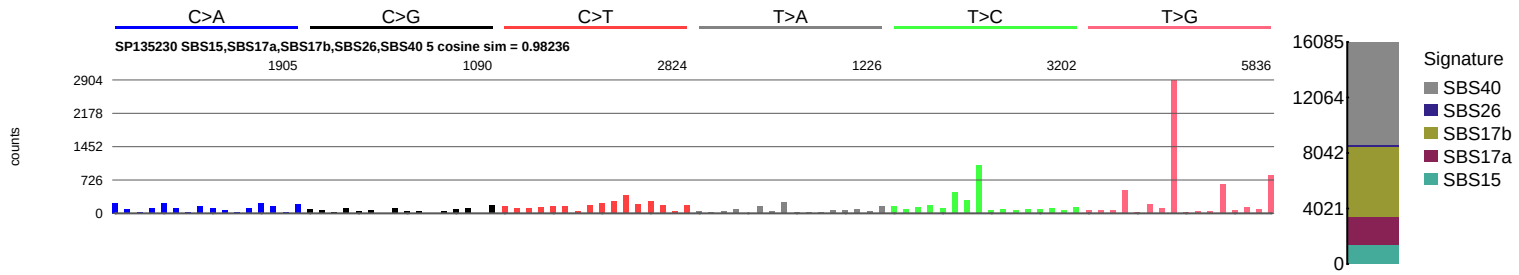
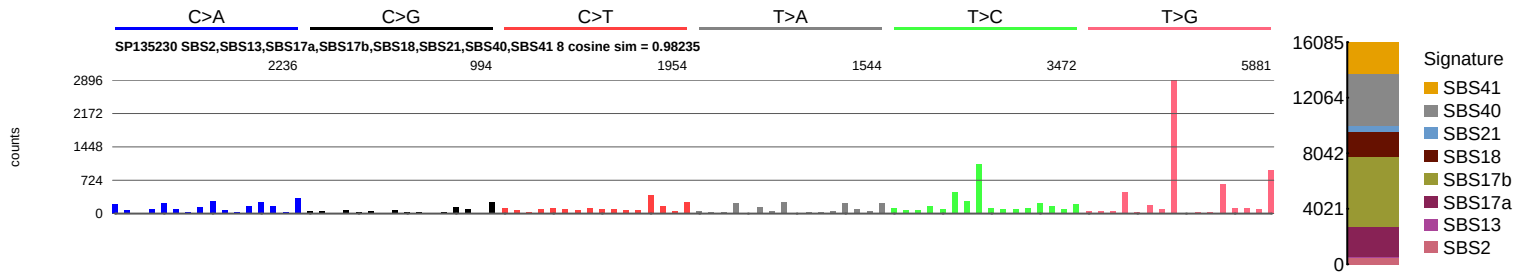
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98237
2232
1014
1947
1561
3463
5872
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98236
1905
1090
2824
1226
3202
5836
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15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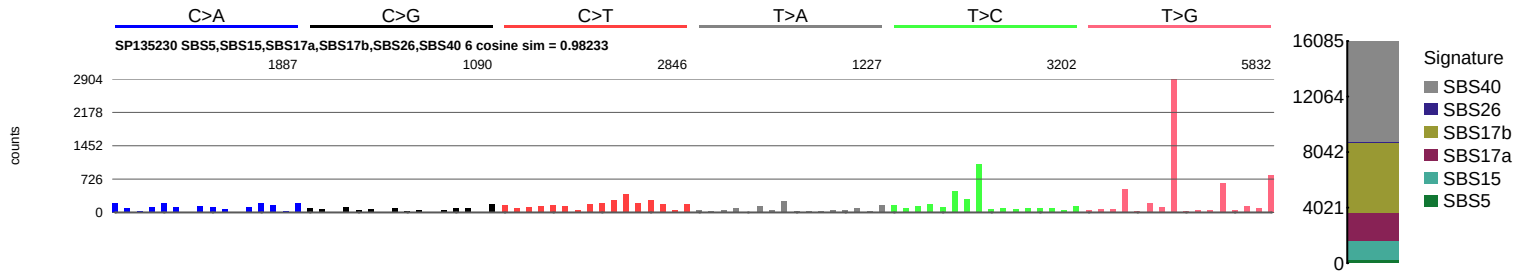
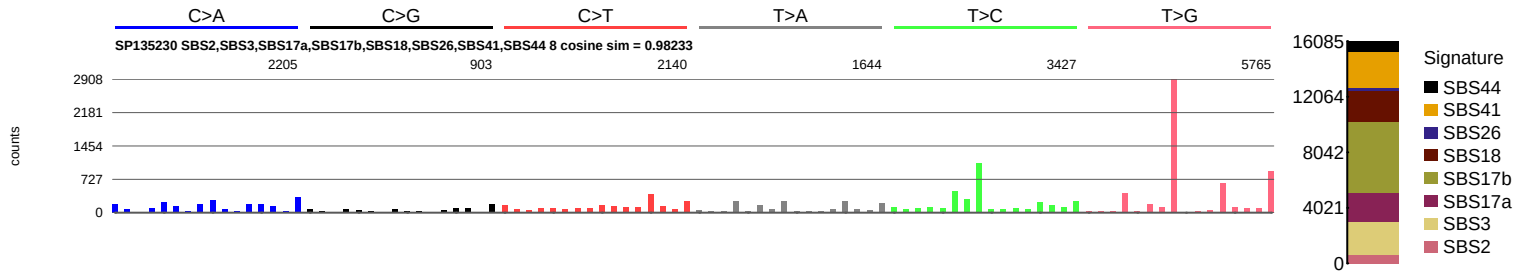
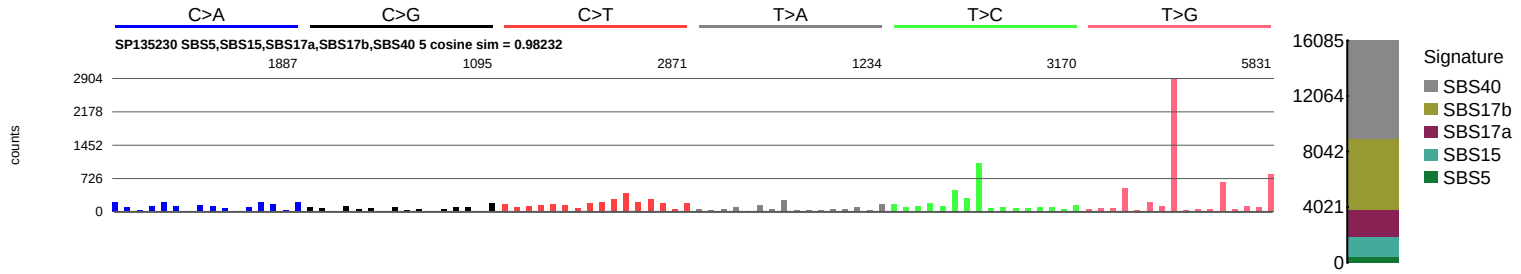
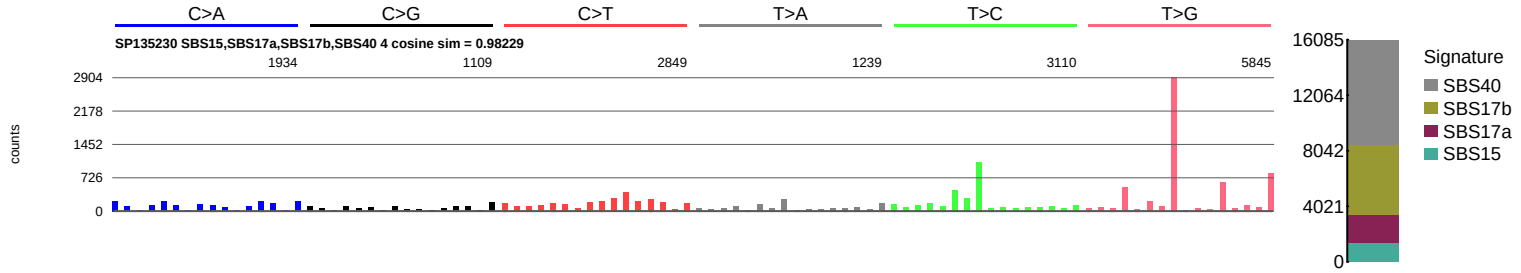
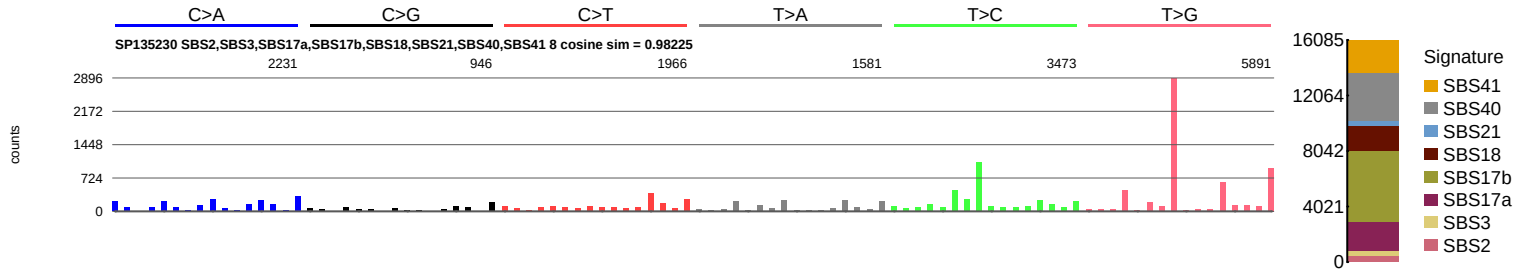
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98235
2236
994
1954
1544
3472
5881
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2

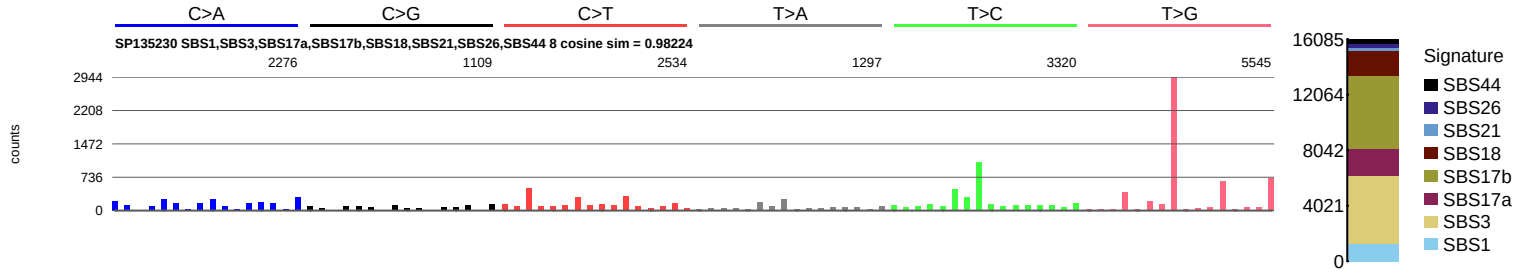

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98233
1887
1090
2846
1227
3202
5832
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5

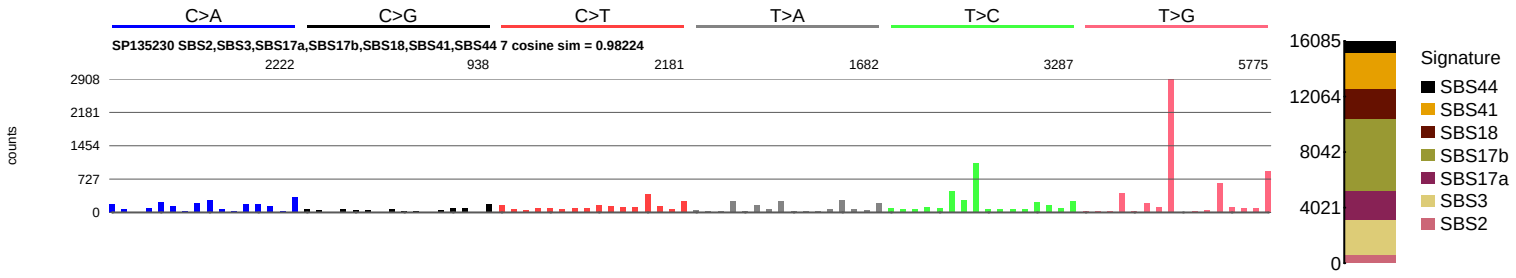

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98233
2205
903
2140
1644
3427
5765
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2

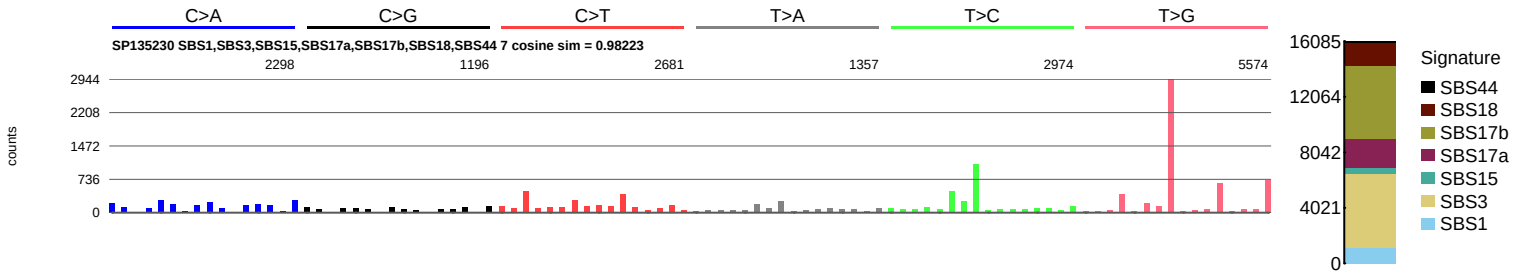

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98232
1887
1095
2871
1234
3170
5831
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5

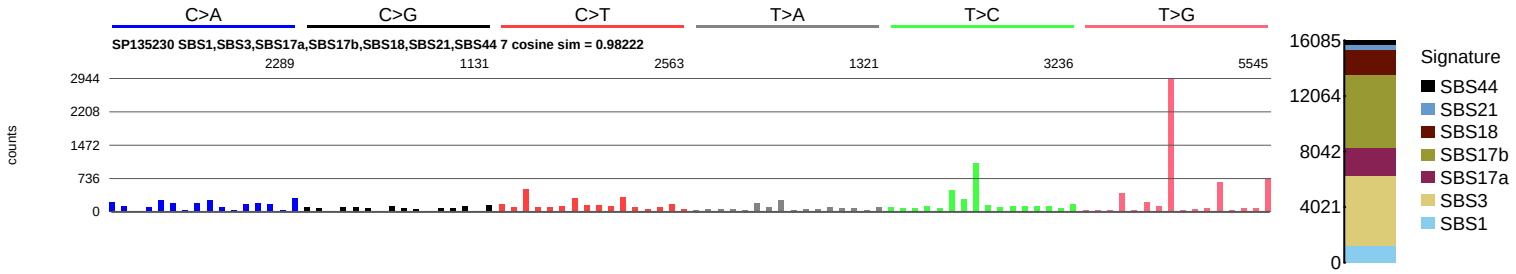

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98229
1934
1109
2849
1239
3110
5845
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS40
SBS17b
SBS17a
SBS15

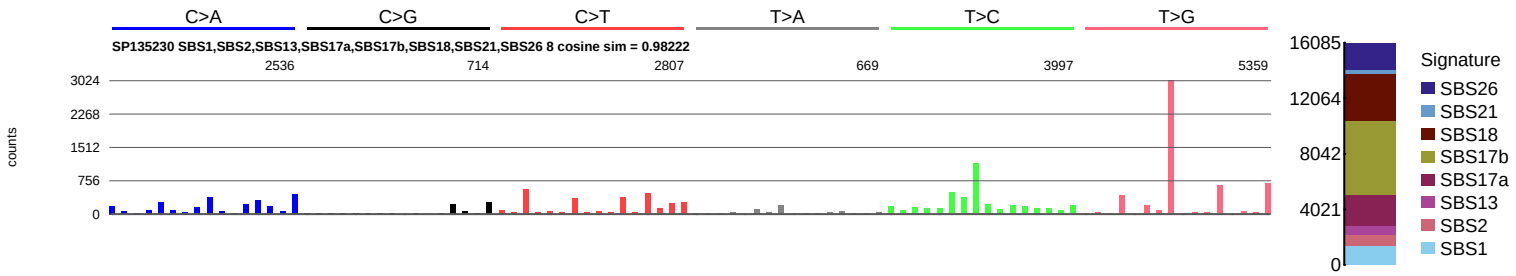

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98225
2231
946
1966
1581
3473
5891
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2

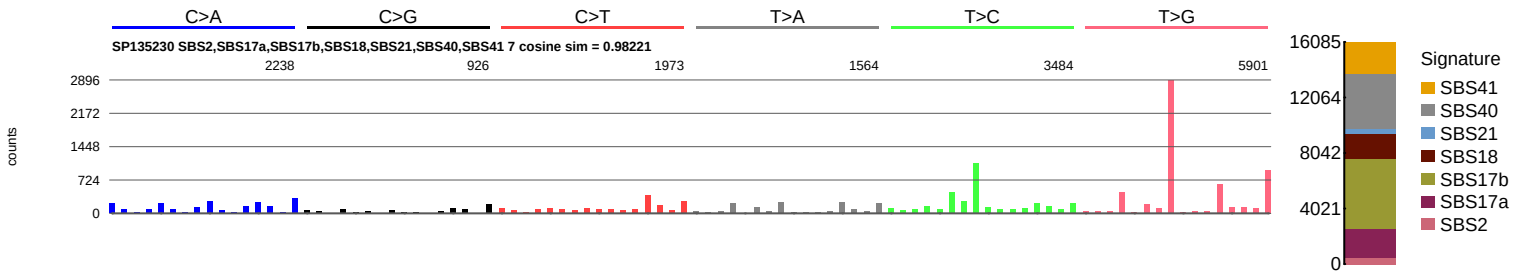

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98224
2276
1109
2534
1297
3320
5545
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98224
2222
938
2181
1682
3287
5775
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98223
2298
1196
2681
1357
2974
5574
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98222
2289
1131
2563
1321
3236
5545
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98222
2536
714
2807
669
3997
5359
counts
3024
2268
1512
756
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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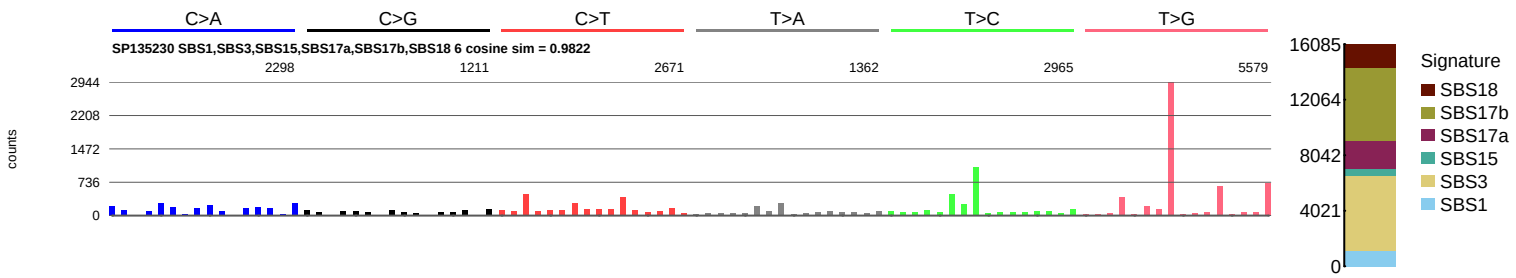
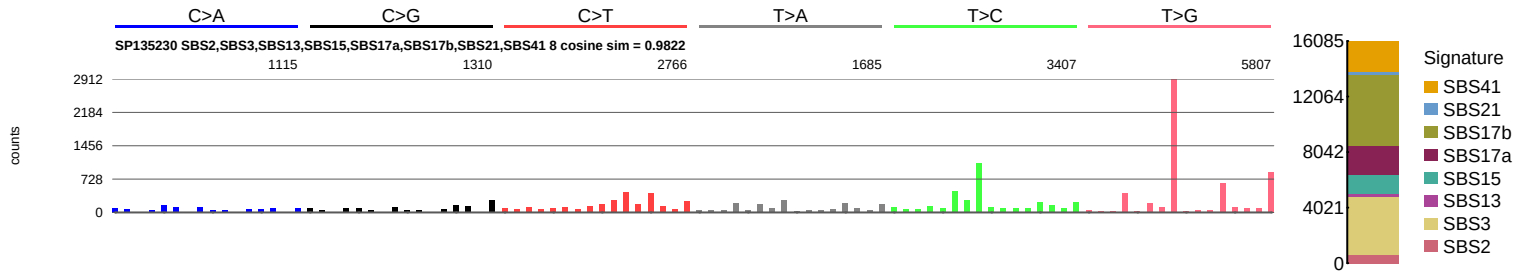
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98221
2238
926
1973
1564
3484
5901
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.9822
2298
1211
2671
1362
2965
5579
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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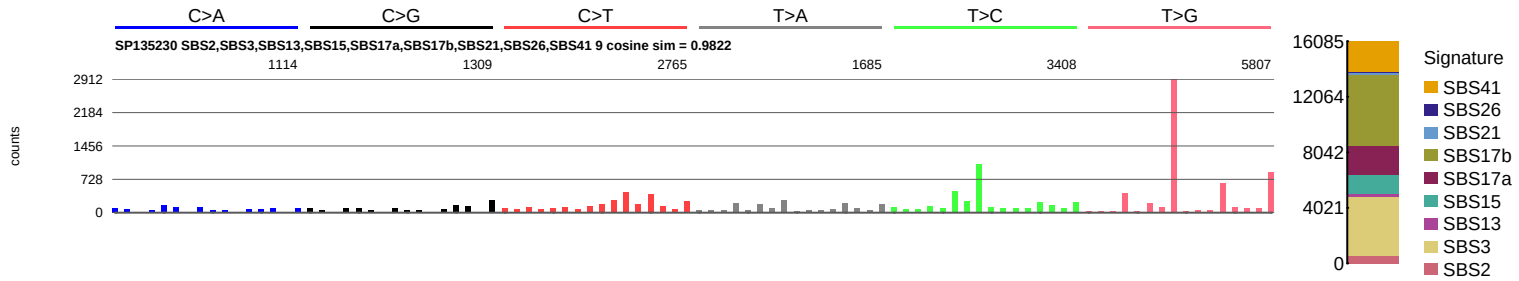
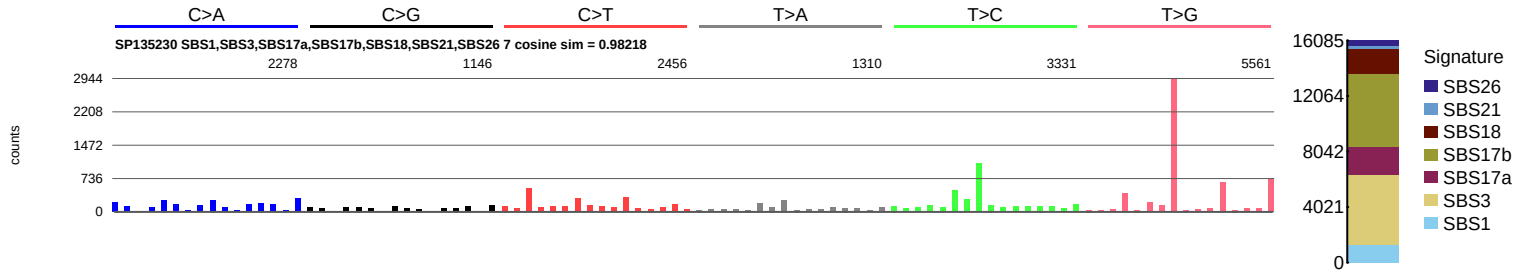
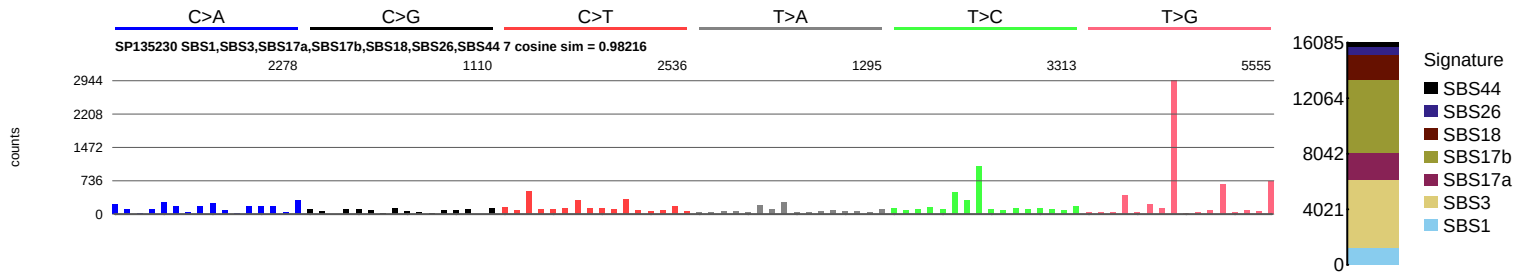
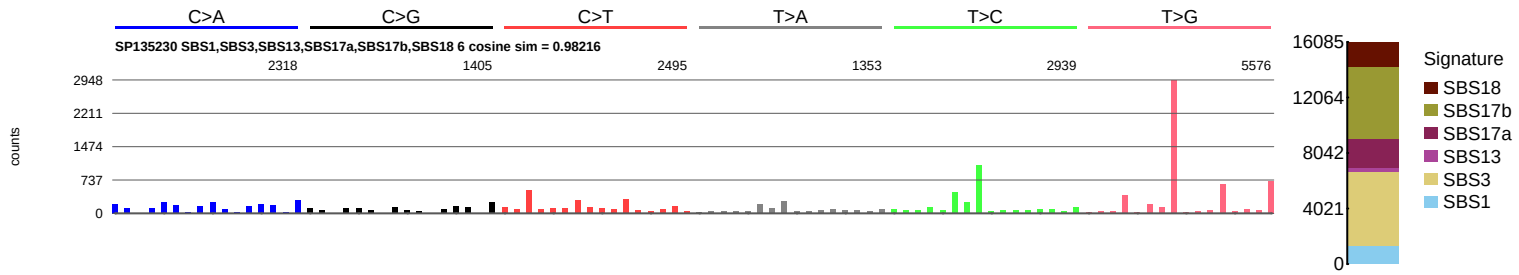
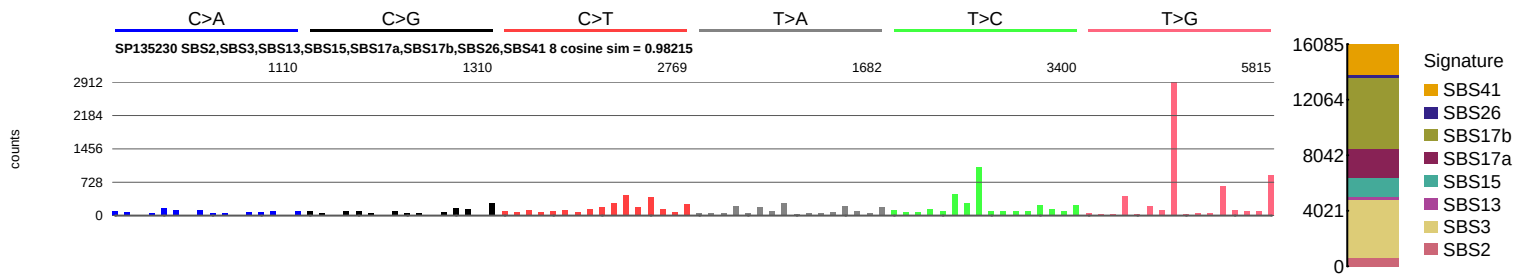
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9822
1115
1310
2766
1685
3407
5807
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

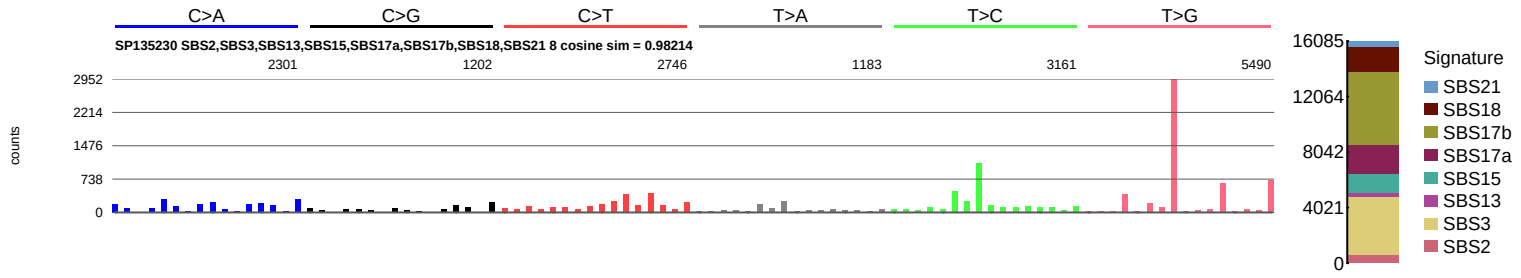

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.9822
1114
1309
2765
1685
3408
5807
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

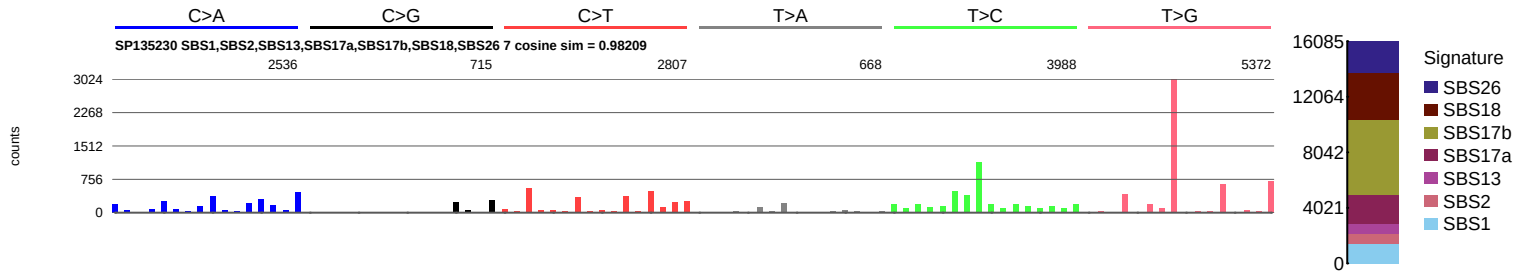

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98218
2278
1146
2456
1310
3331
5561
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

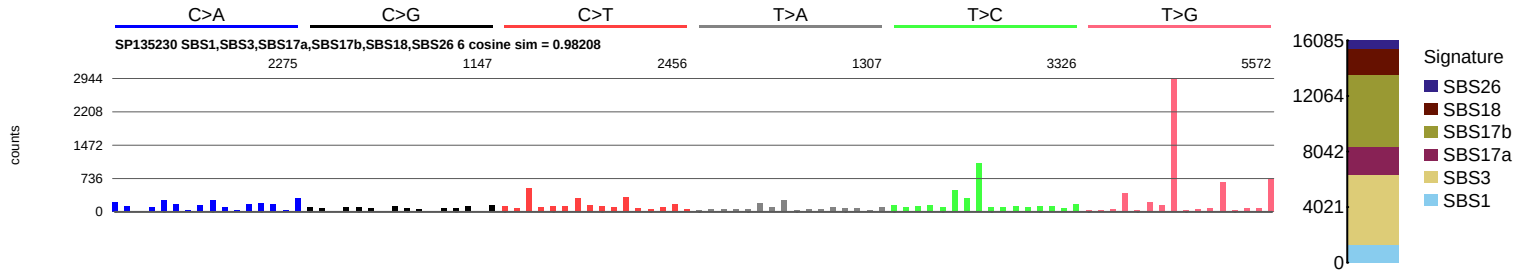

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98216
2278
1110
2536
1295
3313
5555
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98216
2318
1405
2495
1353
2939
5576
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98215
1110
1310
2769
1682
3400
5815
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98214
2301
1202
2746
1183
3161
5490
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98209
2536
715
2807
668
3988
5372
counts
3024
2268
1512
756
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
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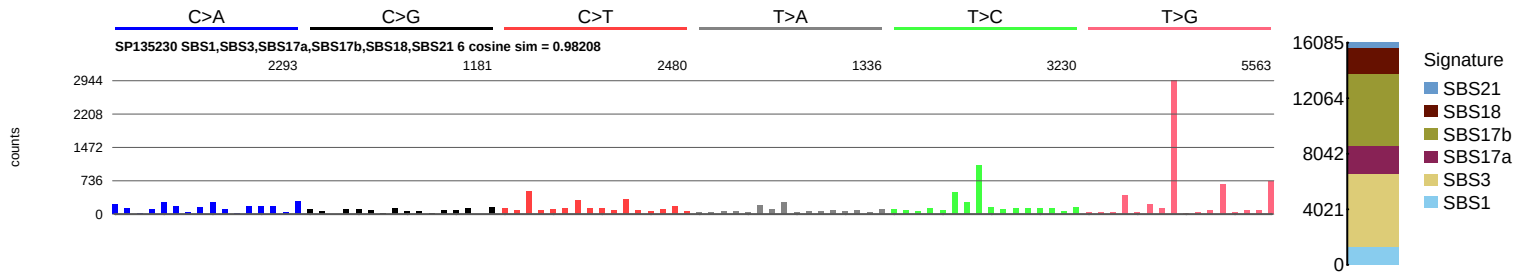
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98208
2275
1147
2456
1307
3326
5572
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1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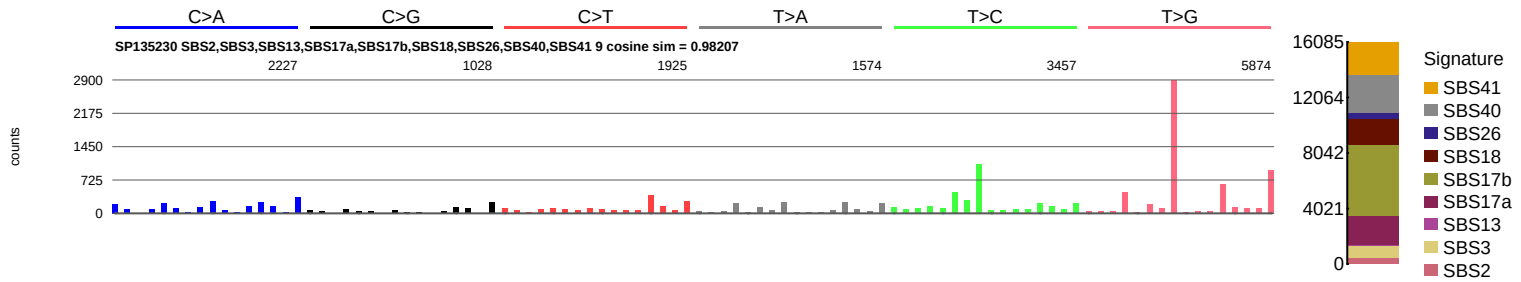
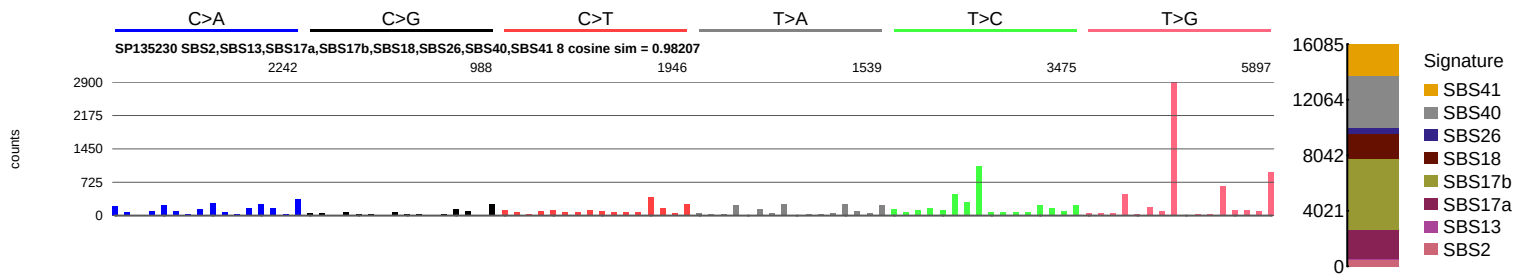
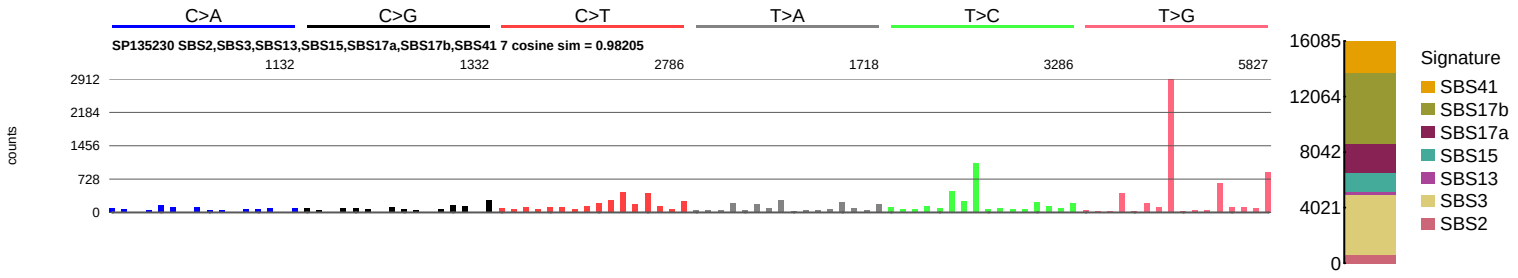
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98208
2293
1181
2480
1336
3230
5563
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98207
2227
1028
1925
1574
3457
5874
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98207
2242
988
1946
1539
3475
5897
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2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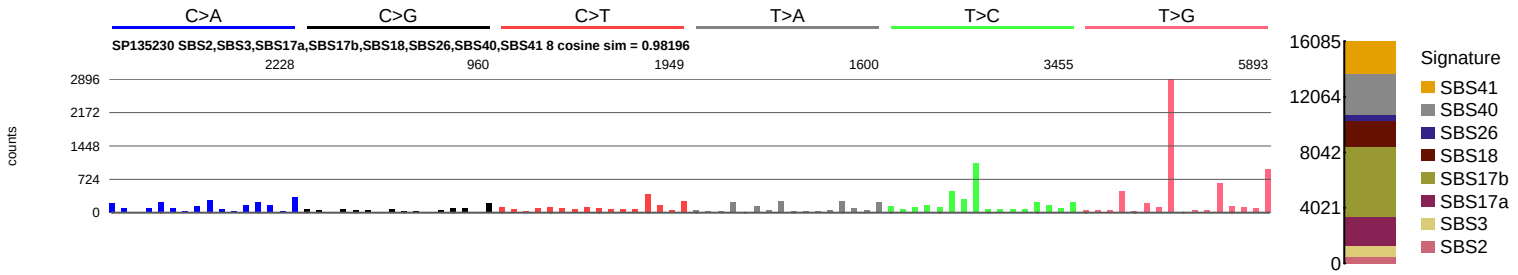
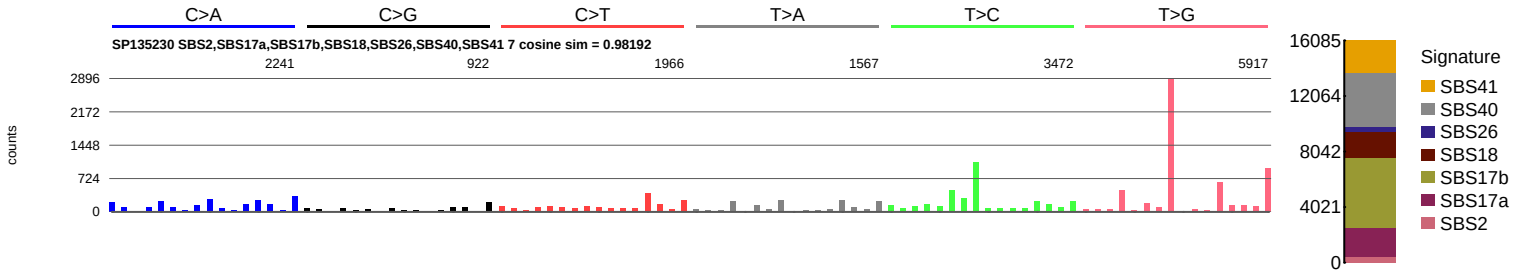
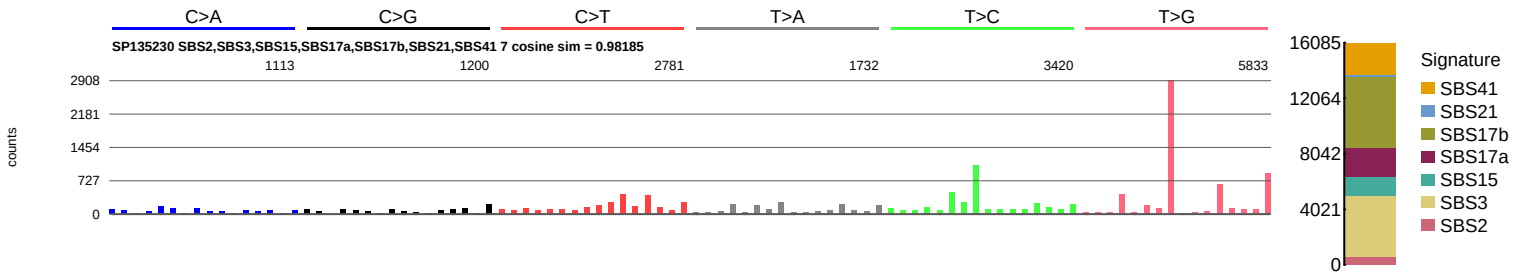
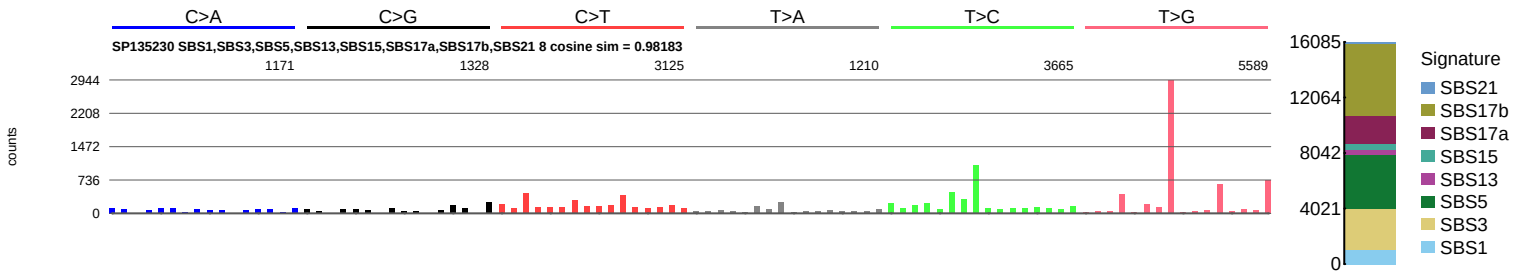
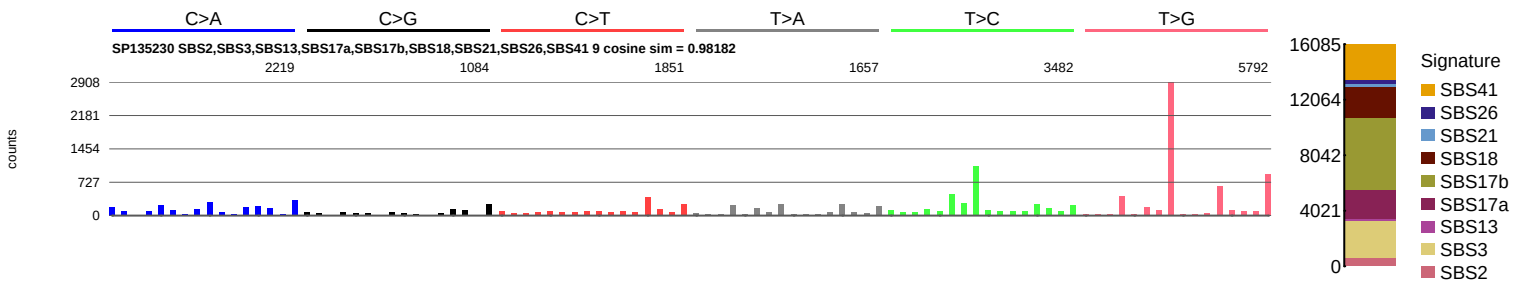
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98205
1132
1332
2786
1718
3286
5827
counts
2912
2184
1456
728
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

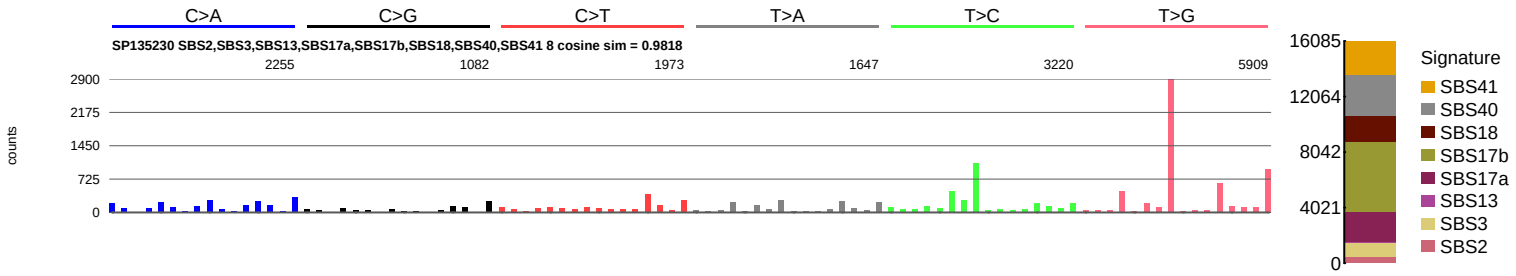

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98196
2228
960
1949
1600
3455
5893
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2

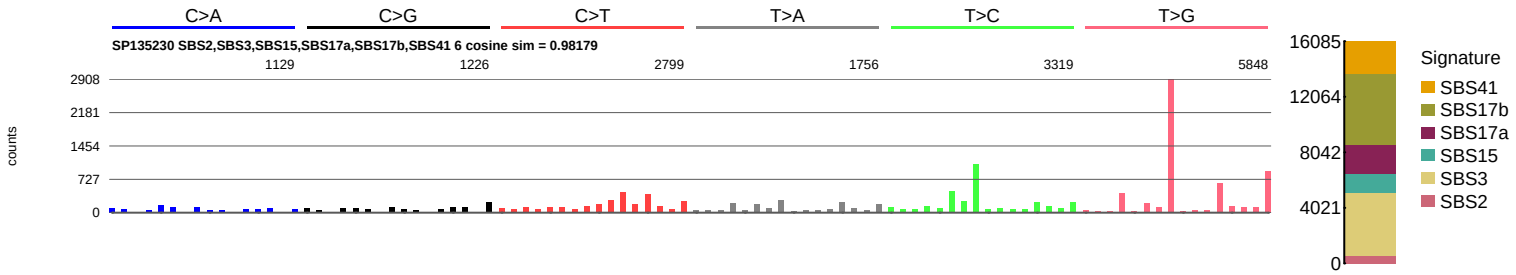

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98192
2241
922
1966
1567
3472
5917
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2

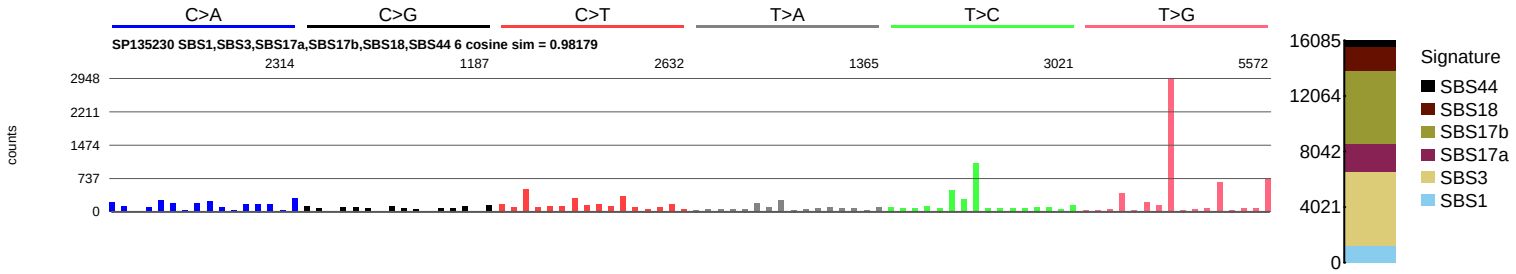

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98185
1113
1200
2781
1732
3420
5833
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2

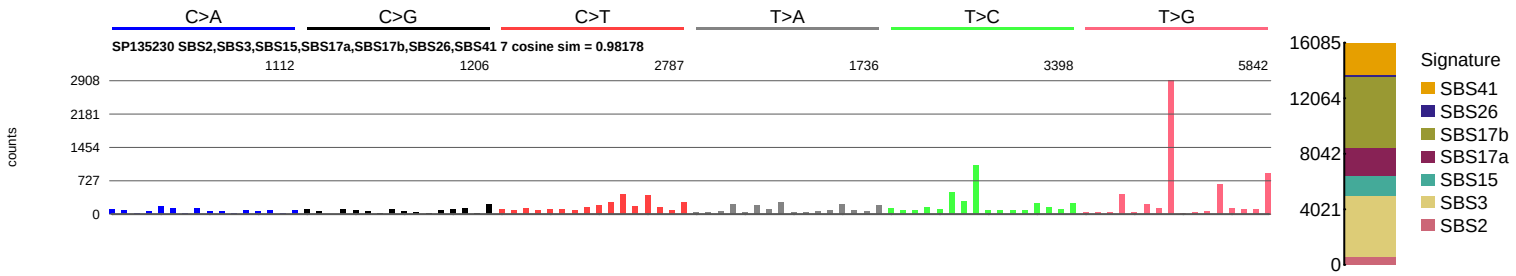

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98183
1171
1328
3125
1210
3665
5589
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

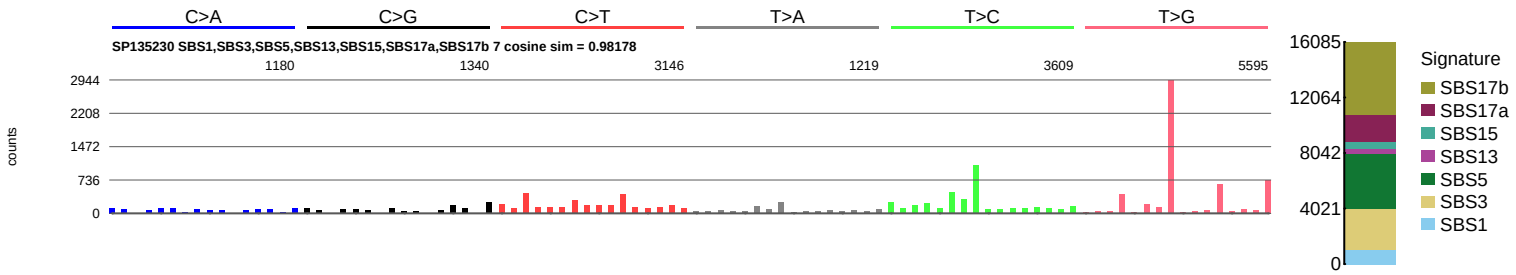

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98182
2219
1084
1851
1657
3482
5792
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

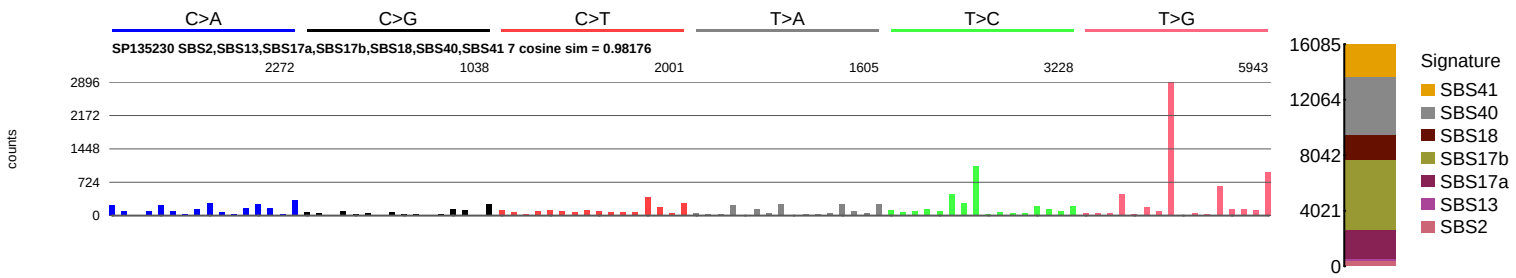

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9818
2255
1082
1973
1647
3220
5909
counts
2900
2175
1450
725
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

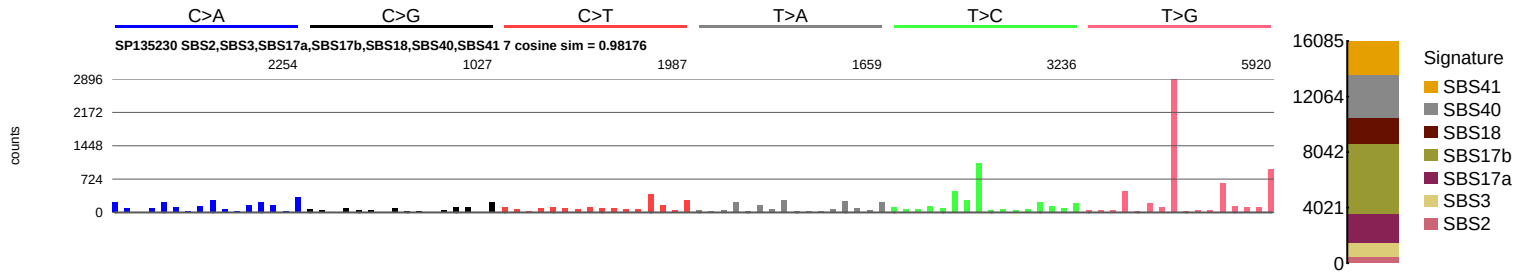

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98179
1129
1226
2799
1756
3319
5848
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2

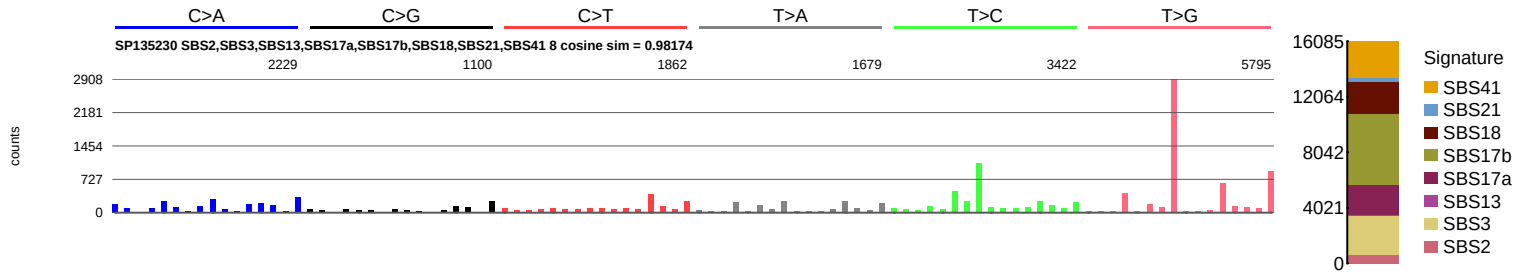

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98179
2314
1187
2632
1365
3021
5572
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS1

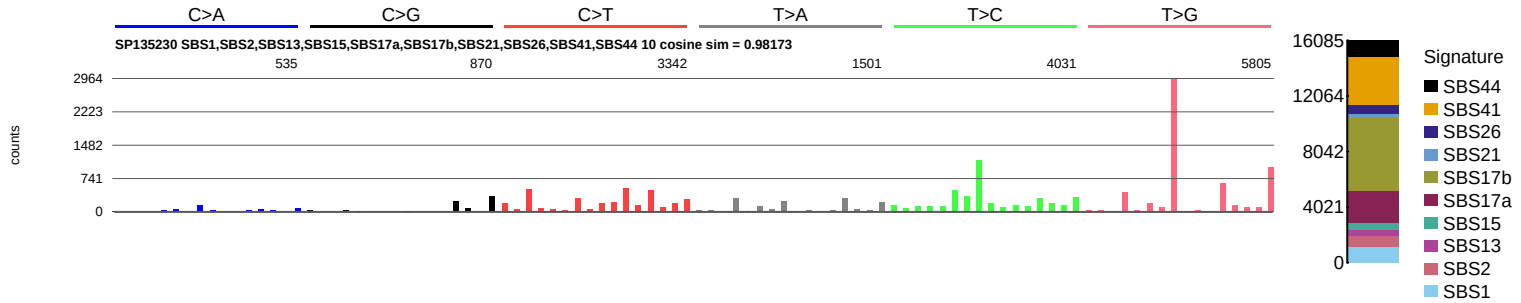

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98178
1112
1206
2787
1736
3398
5842
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2

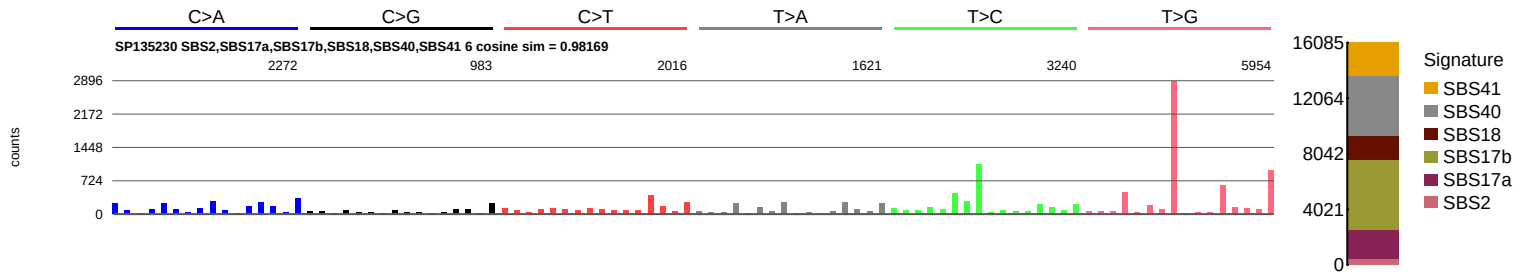

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98178
1180
1340
3146
1219
3609
5595
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

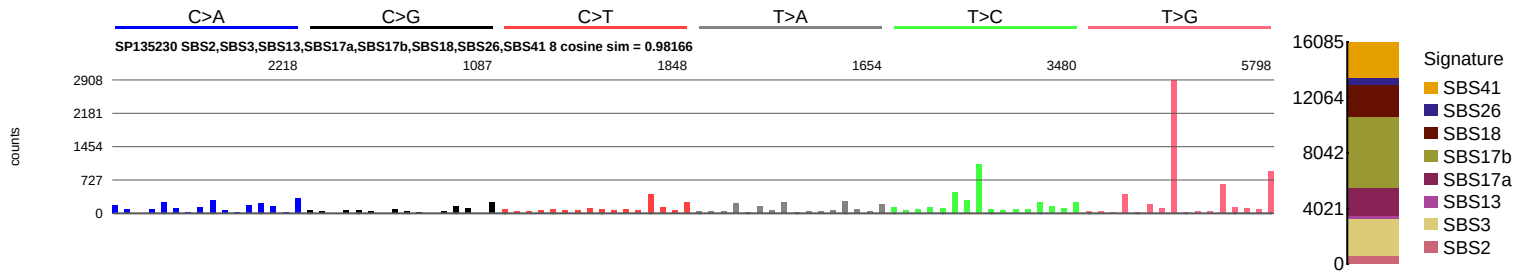

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98176
2272
1038
2001
1605
3228
5943
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2

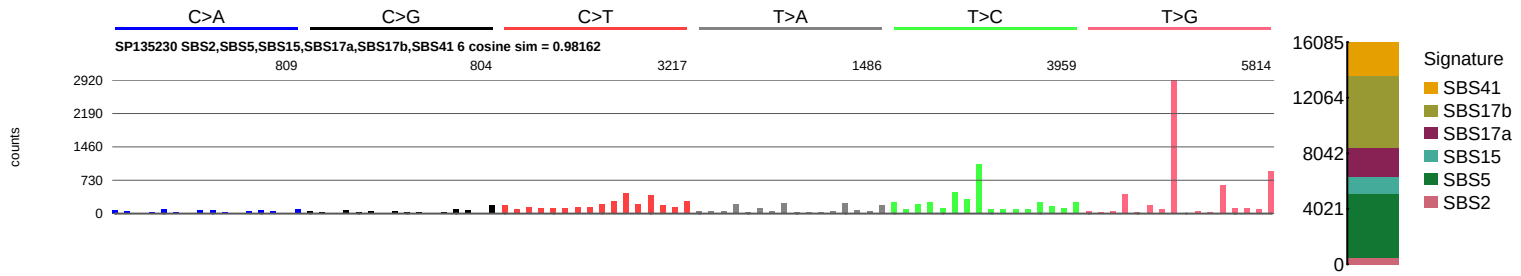

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98176
2254
1027
1987
1659
3236
5920
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2

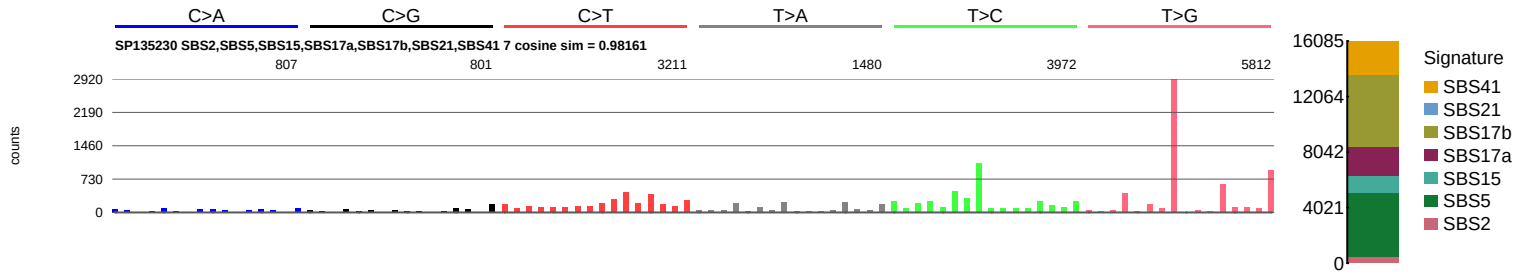

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98174
2229
1100
1862
1679
3422
5795
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

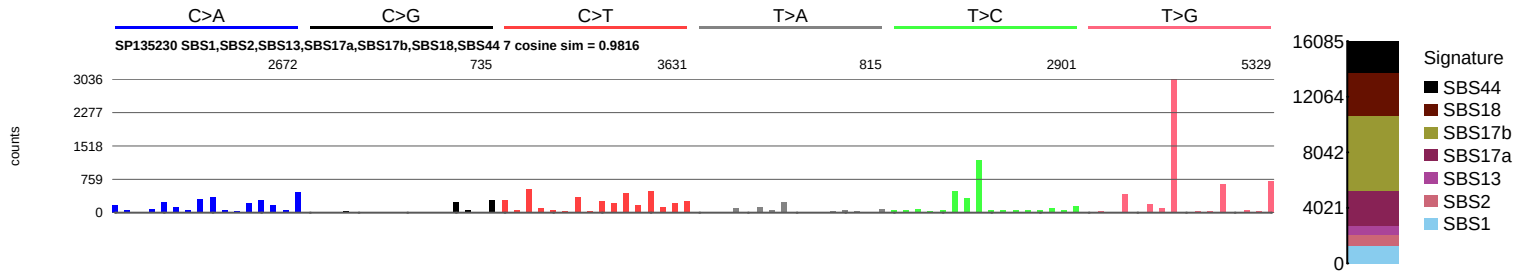

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98173
535
870
3342
1501
4031
5805
counts
2964
2223
1482
741
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

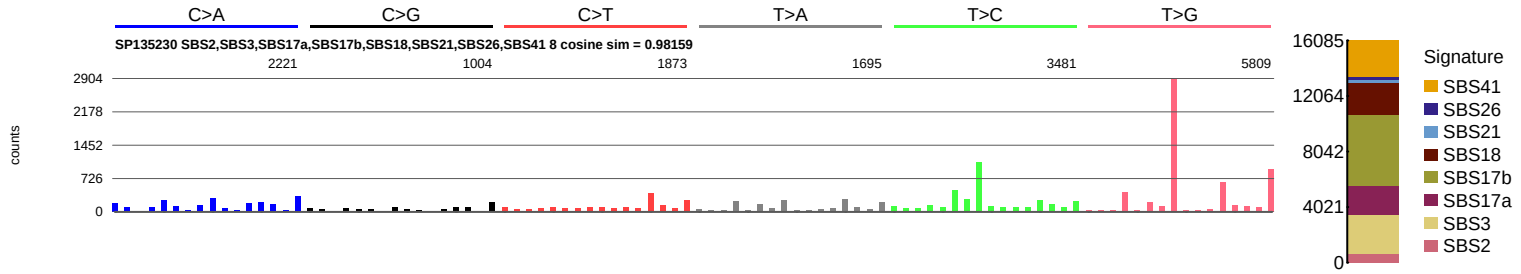

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98169
2272
983
2016
1621
3240
5954
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2

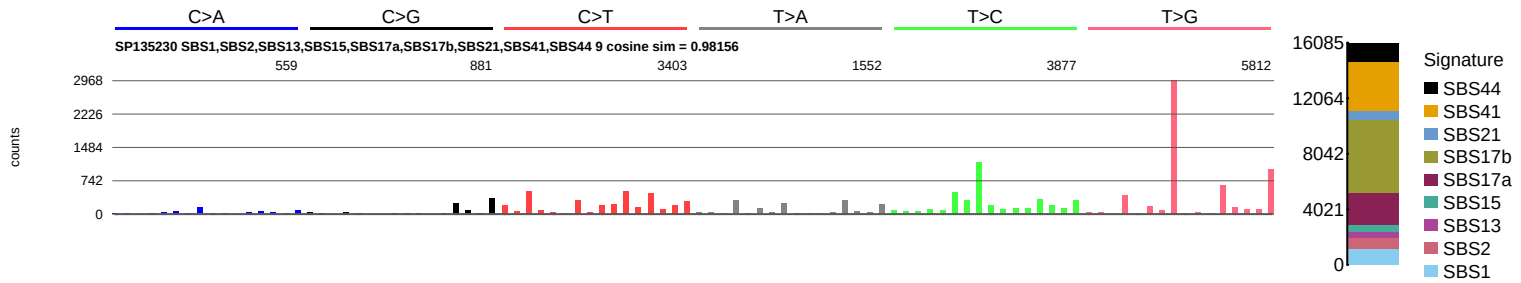

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98166
2218
1087
1848
1654
3480
5798
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

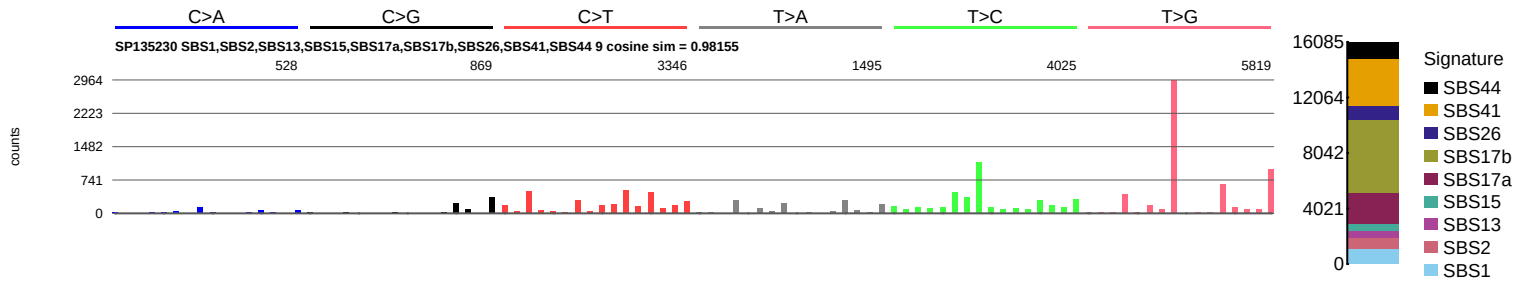

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98162
809
804
3217
1486
3959
5814
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98161
807
801
3211
1480
3972
5812
counts
2920
2190
1460
730
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.9816
2672
735
3631
815
2901
5329
counts
3036
2277
1518
759
0
16085
12064
8042
4021
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98159
2221
1004
1873
1695
3481
5809
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98156
559
881
3403
1552
3877
5812
counts
2968
2226
1484
742
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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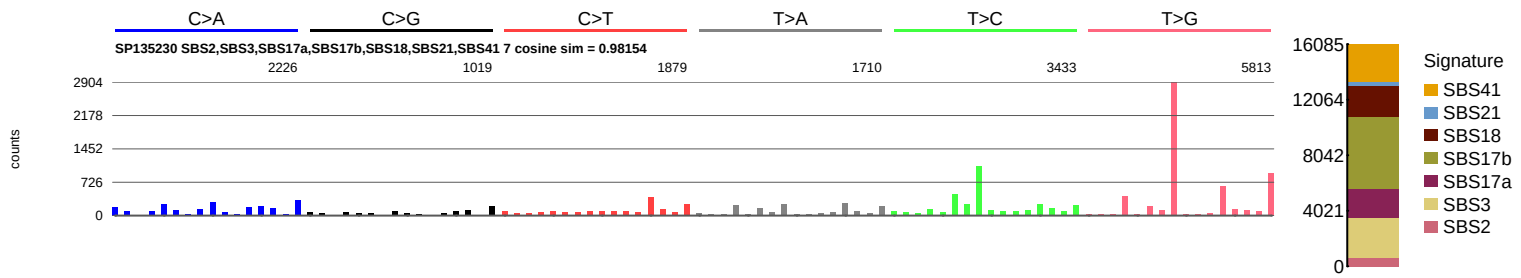
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98155
528
869
3346
1495
4025
5819
counts
2964
2223
1482
741
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
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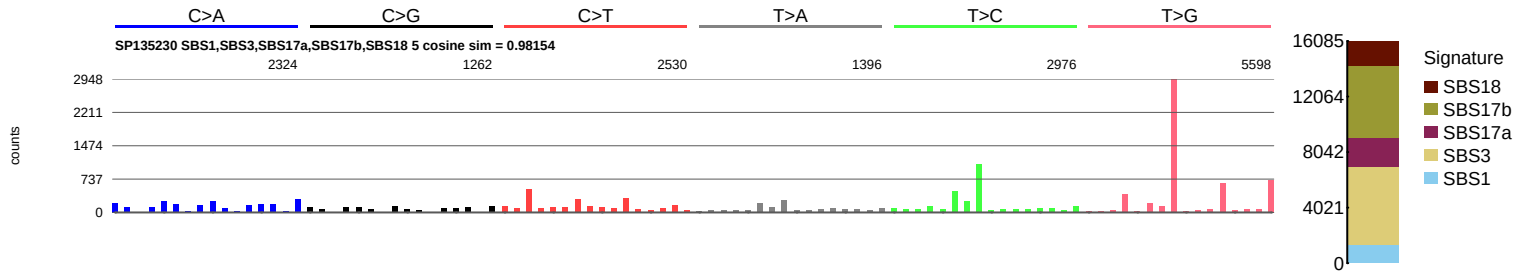
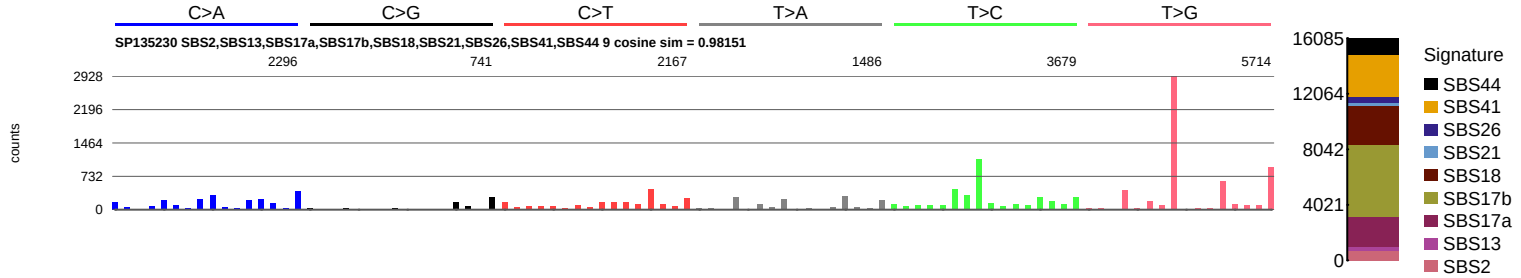
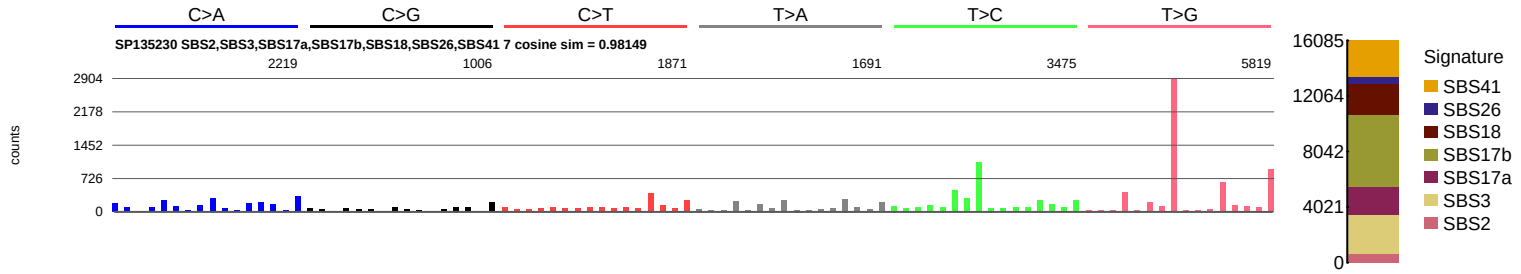
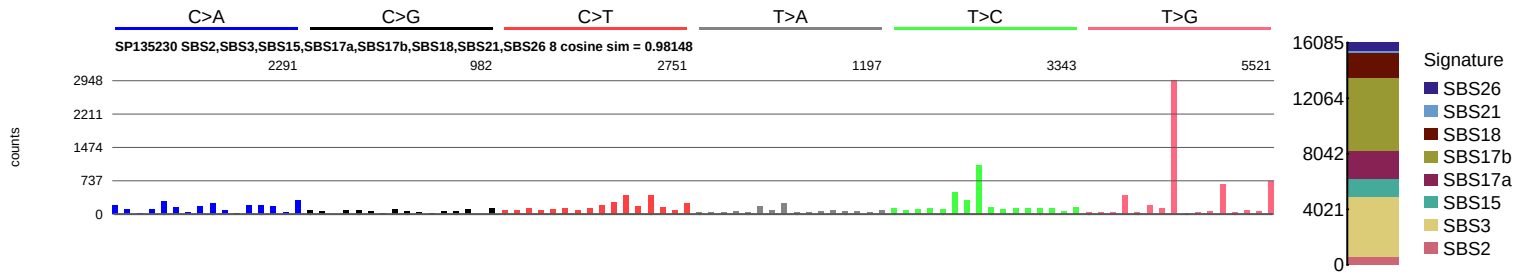
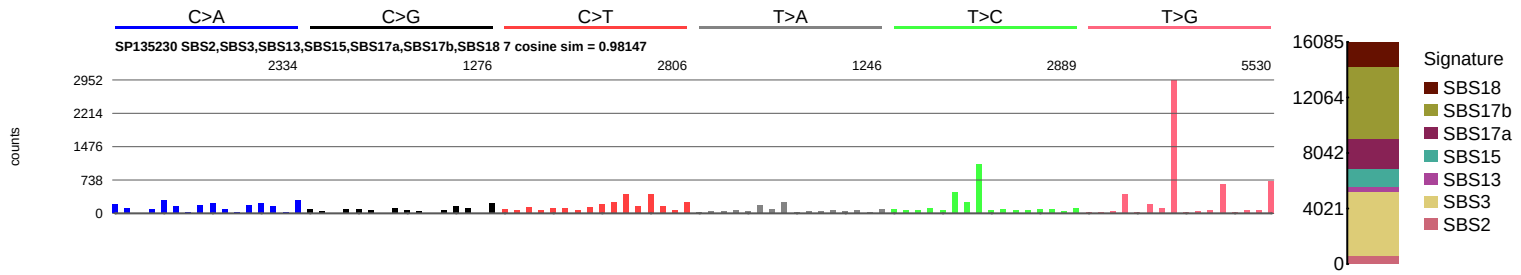
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98154
2226
1019
1879
1710
3433
5813
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2

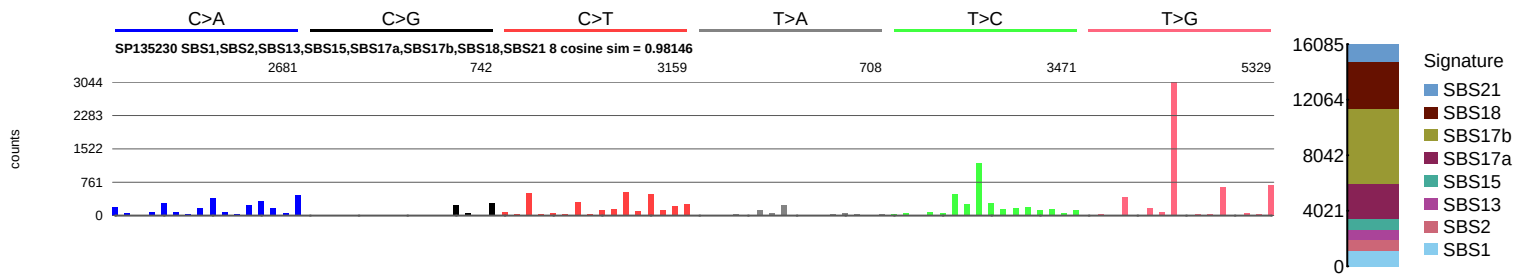

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98154
2324
1262
2530
1396
2976
5598
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS1

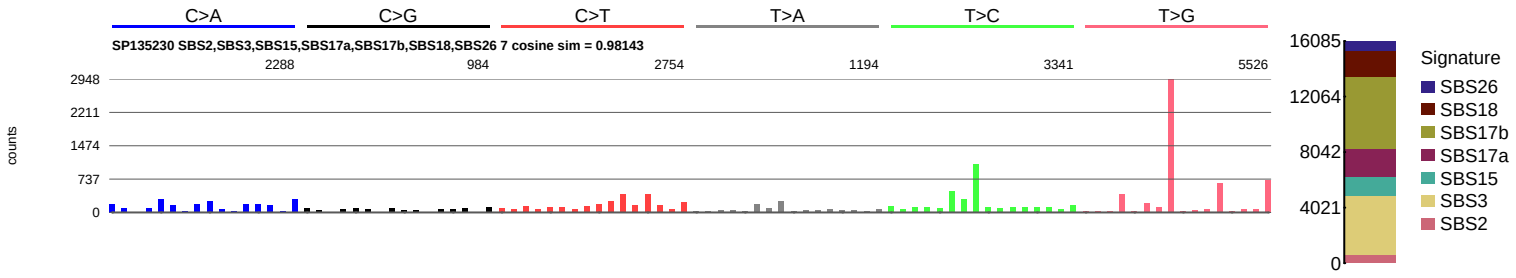

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98151
2296
741
2167
1486
3679
5714
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2

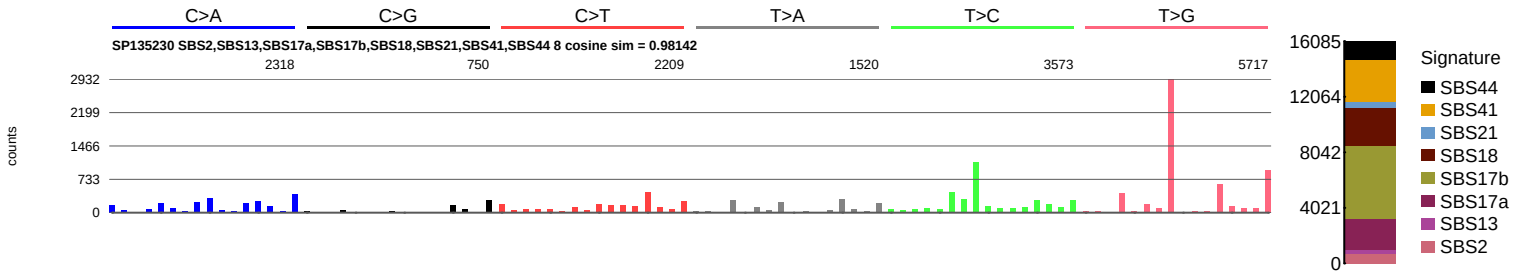

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98149
2219
1006
1871
1691
3475
5819
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2

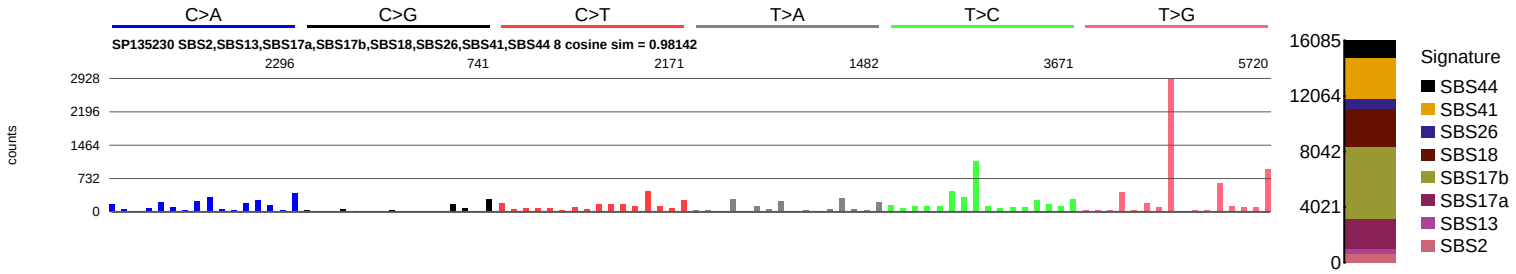

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98148
2291
982
2751
1197
3343
5521
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

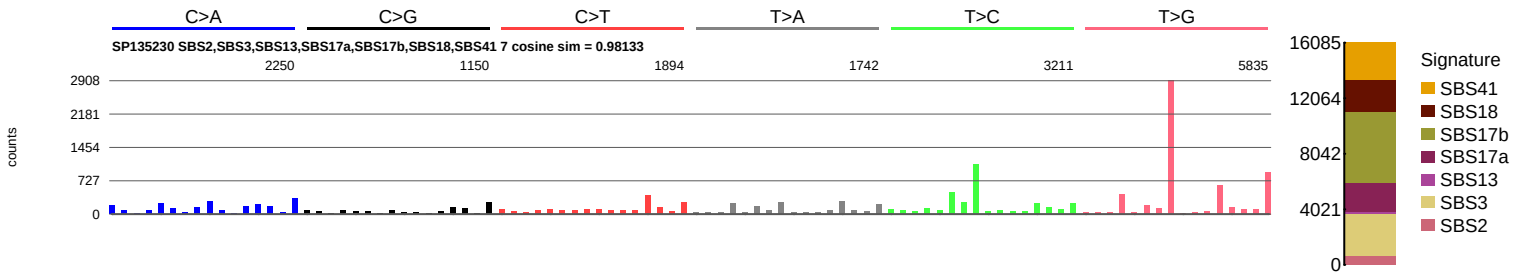

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98147
2334
1276
2806
1246
2889
5530
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

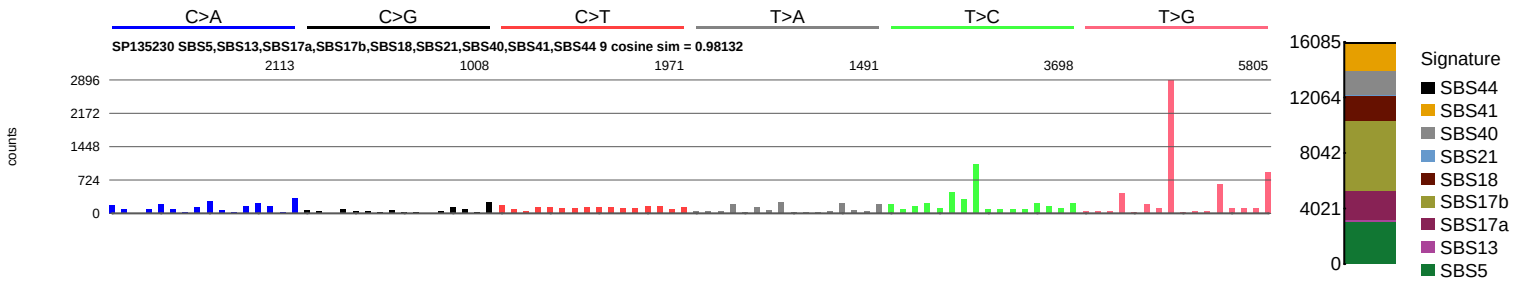

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98146
2681
742
3159
708
3471
5329
counts
3044
2283
1522
761
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

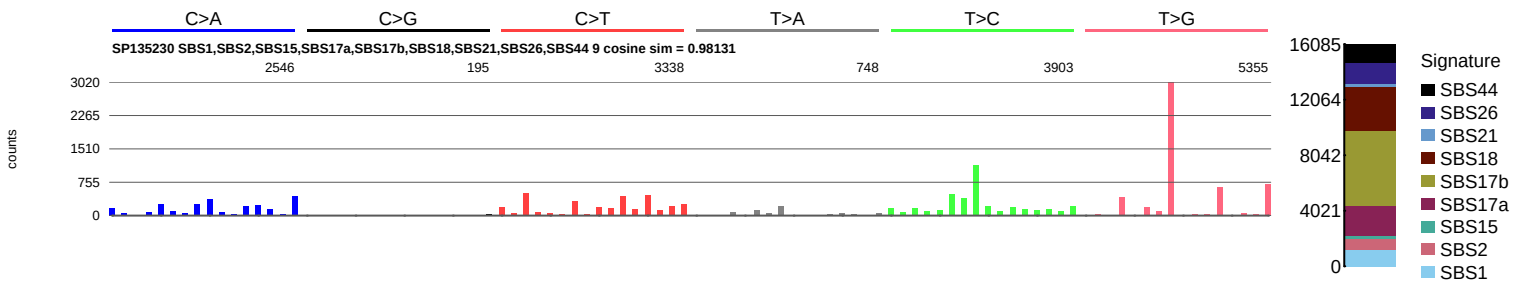

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98143
2288
984
2754
1194
3341
5526
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

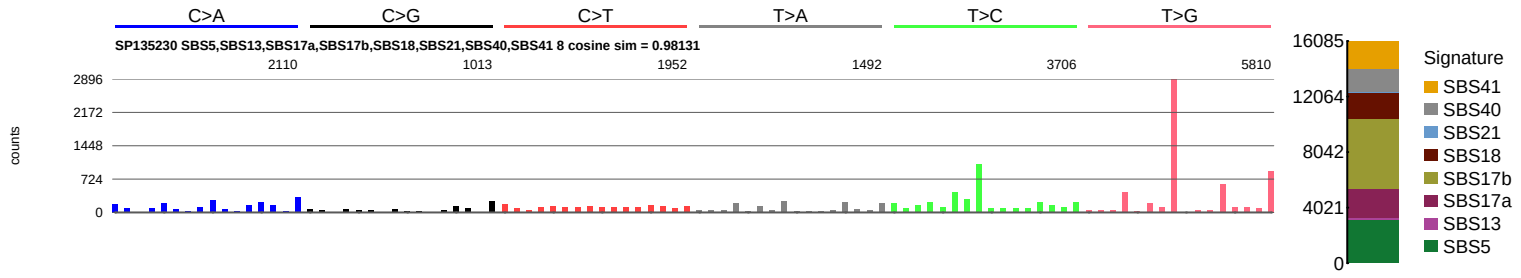

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98142
2318
750
2209
1520
3573
5717
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98142
2296
741
2171
1482
3671
5720
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98133
2250
1150
1894
1742
3211
5835
counts
2908
2181
1454
727
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98132
2113
1008
1971
1491
3698
5805
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98131
2546
195
3338
748
3903
5355
counts
3020
2265
1510
755
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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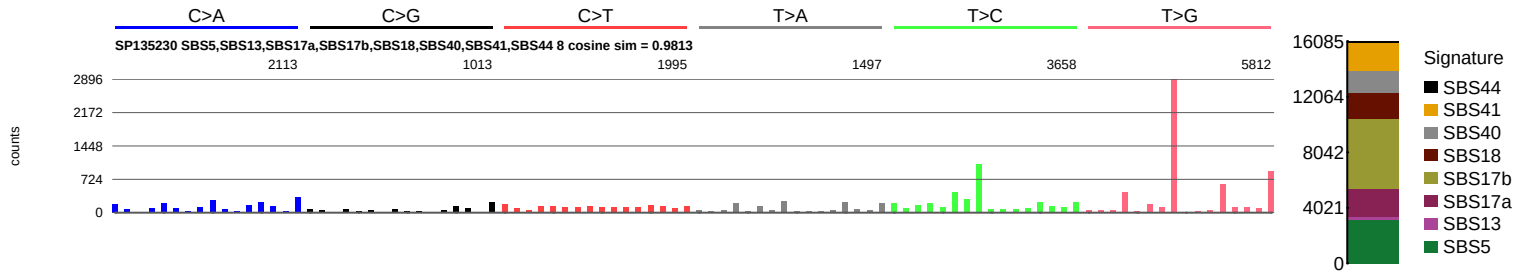
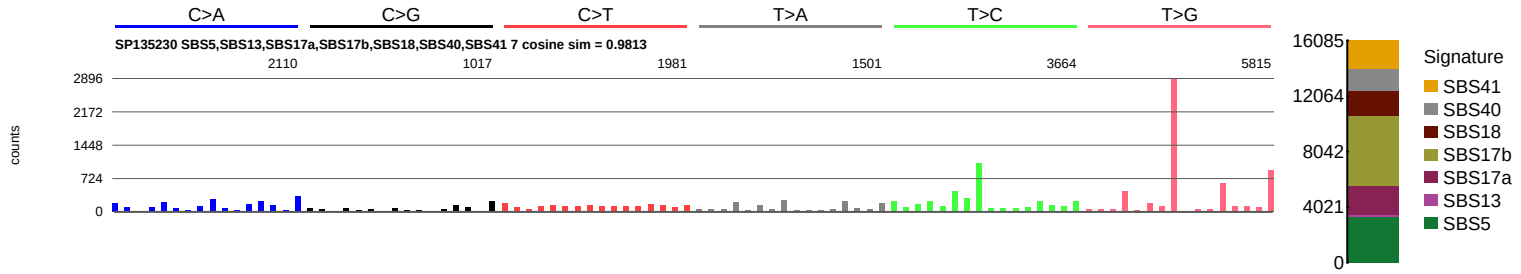
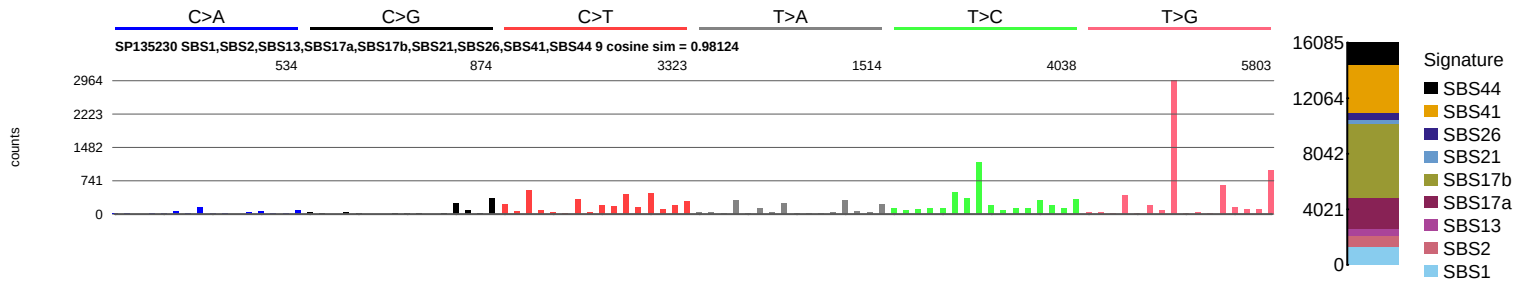
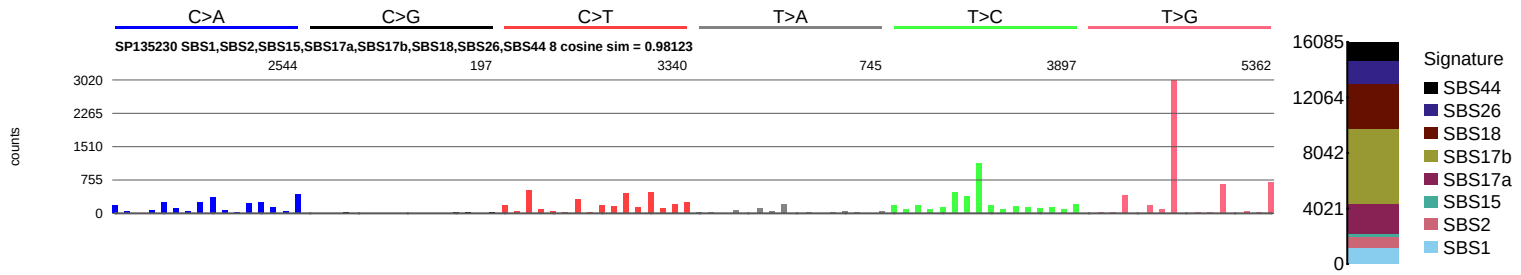
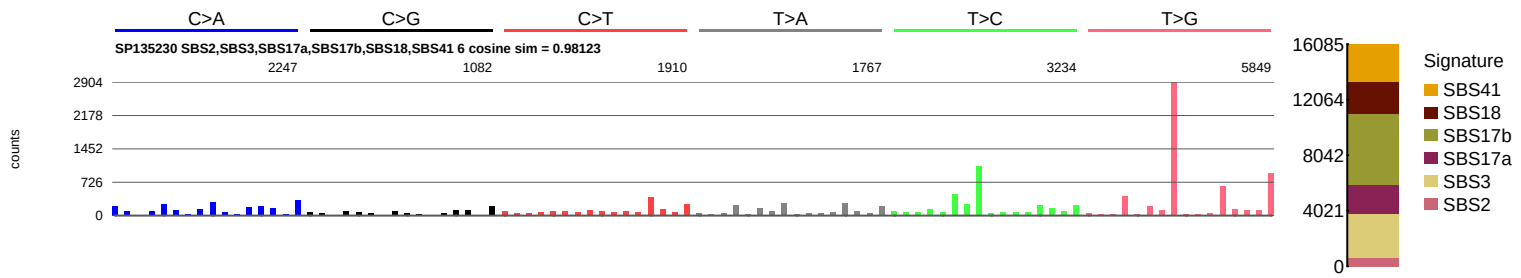
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98131
2110
1013
1952
1492
3706
5810
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5

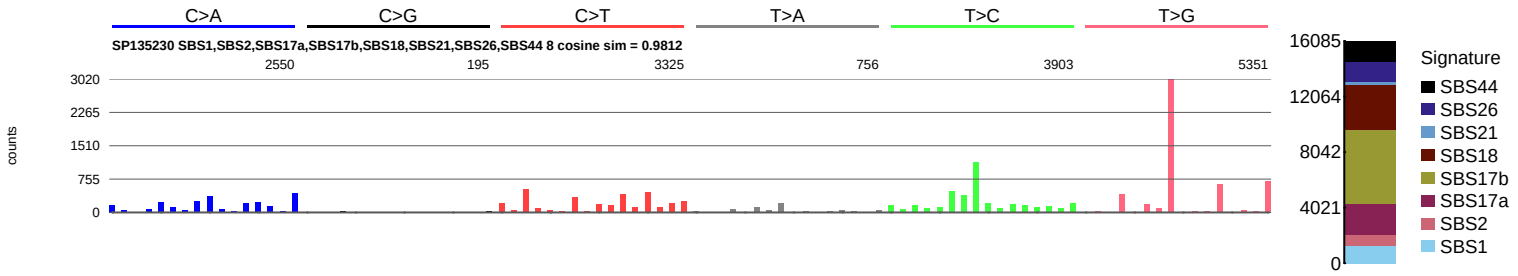

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9813
2113
1013
1995
1497
3658
5812
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5

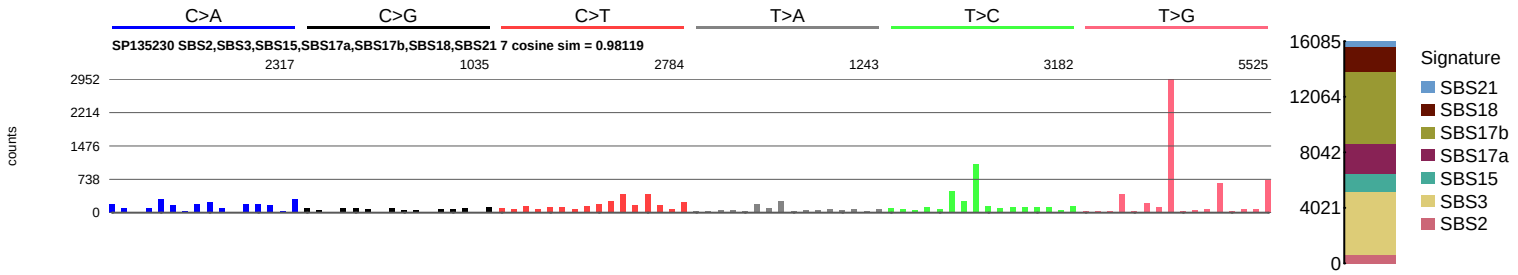

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.9813
2110
1017
1981
1501
3664
5815
counts
2896
2172
1448
724
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5

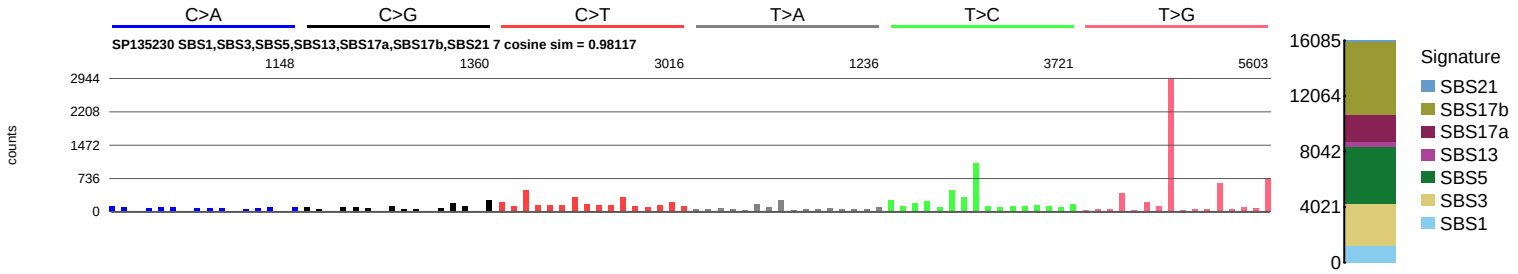

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98124
534
874
3323
1514
4038
5803
counts
2964
2223
1482
741
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

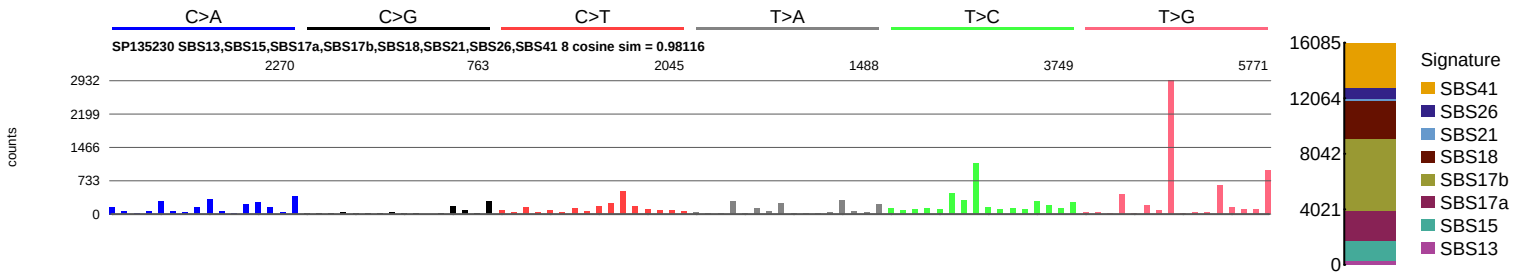

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98123
2544
197
3340
745
3897
5362
counts
3020
2265
1510
755
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98123
2247
1082
1910
1767
3234
5849
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.9812
2550
195
3325
756
3903
5351
counts
3020
2265
1510
755
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98119
2317
1035
2784
1243
3182
5525
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98117
1148
1360
3016
1236
3721
5603
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98116
2270
763
2045
1488
3749
5771
counts
2932
2199
1466
733
0
16085
12064
8042
4021
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SBS21
SBS18
SBS17b
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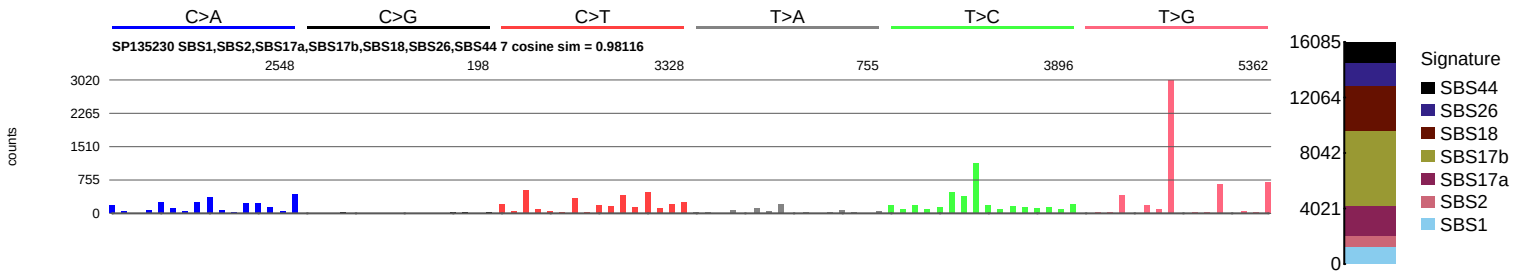
SBS15
SBS13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C>G
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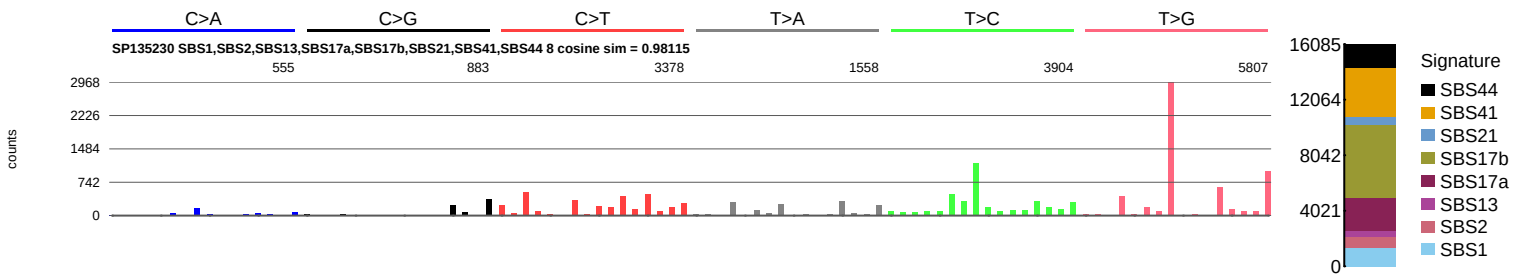
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98116
2548
198
3328
755
3896
5362
counts
3020
2265
1510
755
0
16085
12064
8042
4021
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>T
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T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98115
555
883
3378
1558
3904
5807
counts
2968
2226
1484
742
0
16085
12064
8042
4021
0
Signature
SBS44
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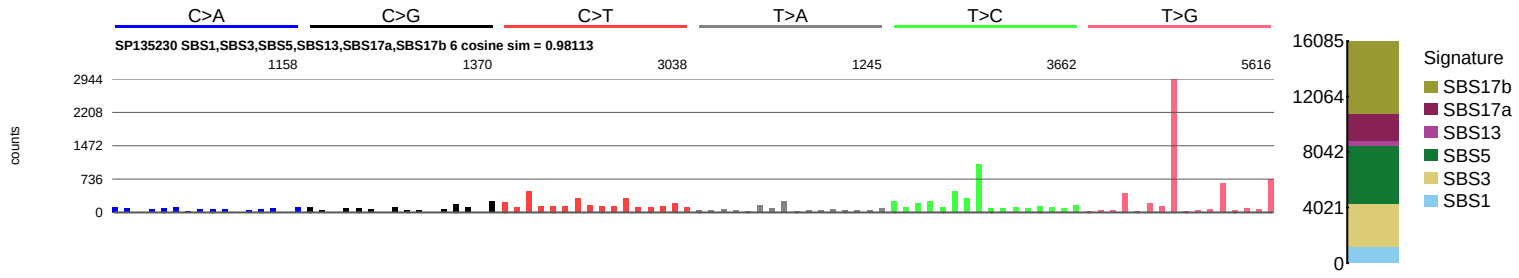
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1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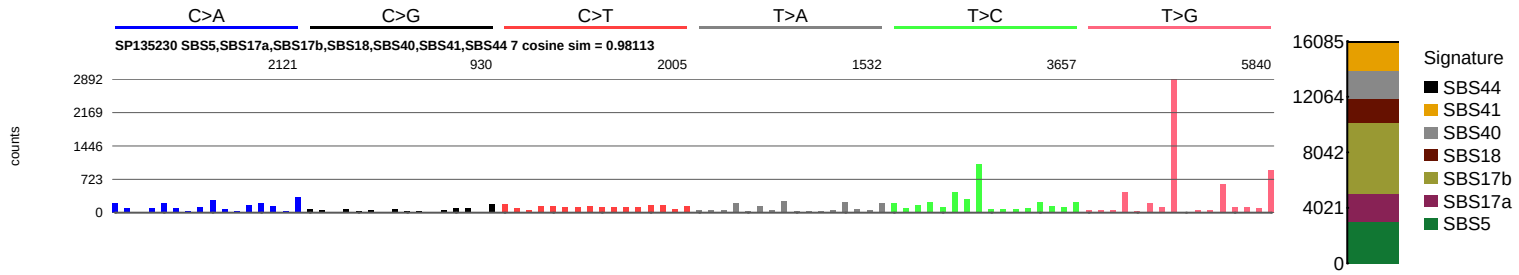
T>G
SP135230 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98113
1158
1370
3038
1245
3662
5616
counts
2944
2208
1472
736
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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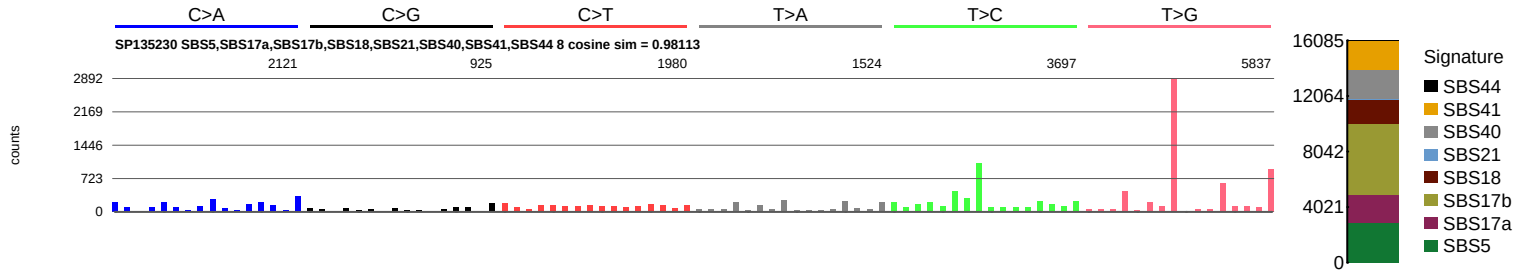
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98113
2121
930
2005
1532
3657
5840
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5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C>T
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T>C
T>G
SP135230 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98113
2121
925
1980
1524
3697
5837
counts
2892
2169
1446
723
0
16085
12064
8042
4021
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Signature
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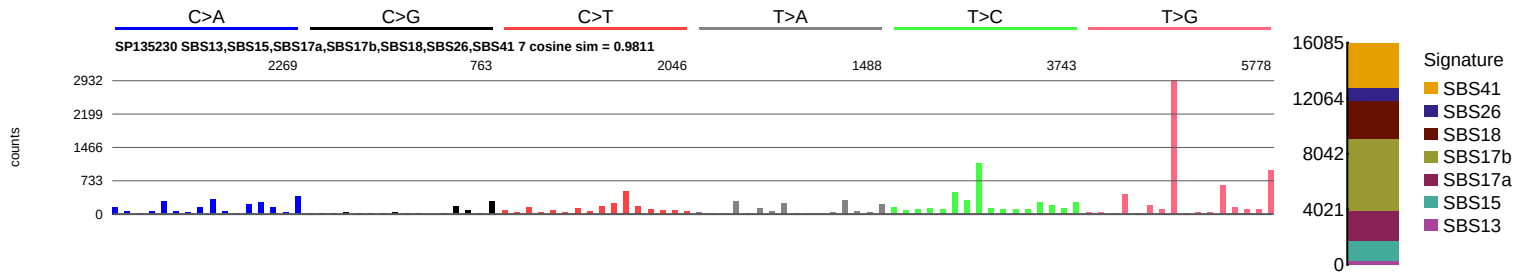
SBS17a
SBS5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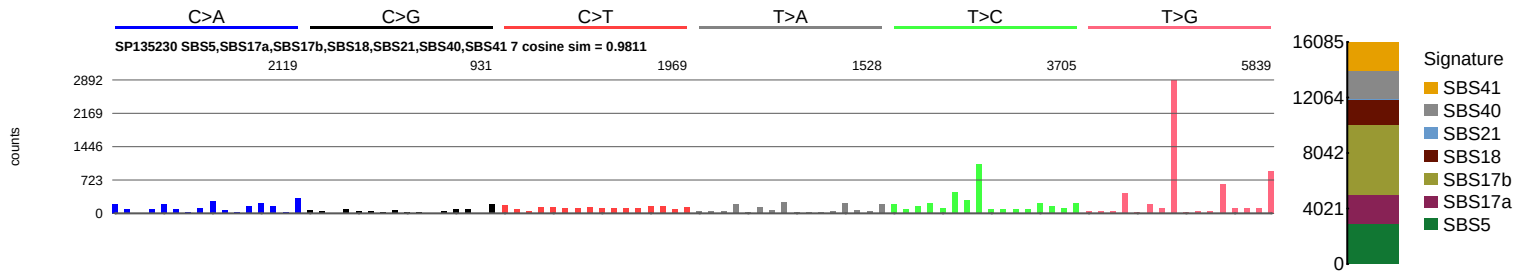
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.9811
2269
763
2046
1488
3743
5778
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13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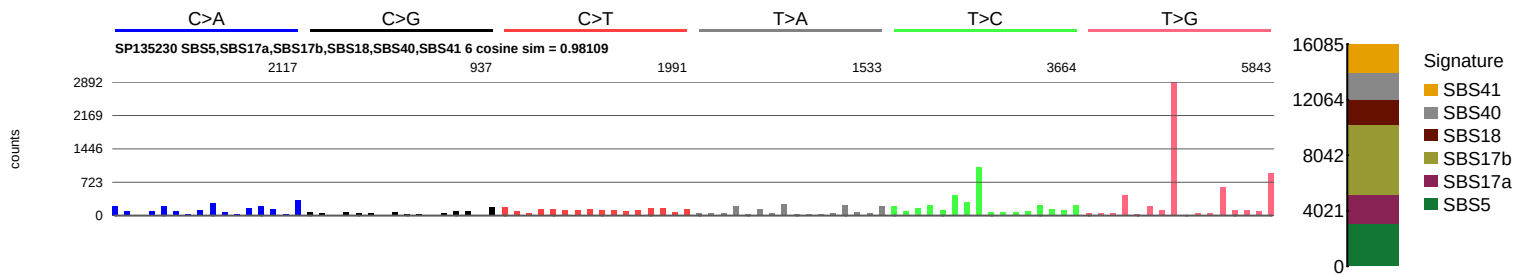
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.9811
2119
931
1969
1528
3705
5839
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5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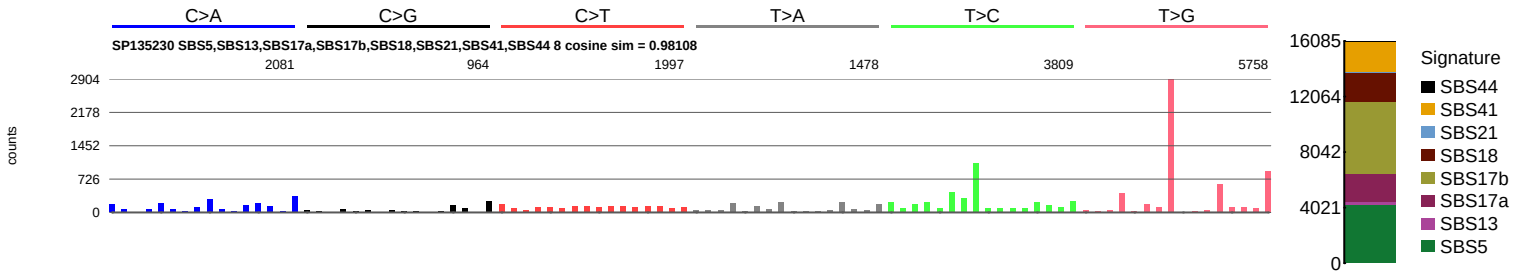
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98109
2117
937
1991
1533
3664
5843
counts
2892
2169
1446
723
0
16085
12064
8042
4021
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5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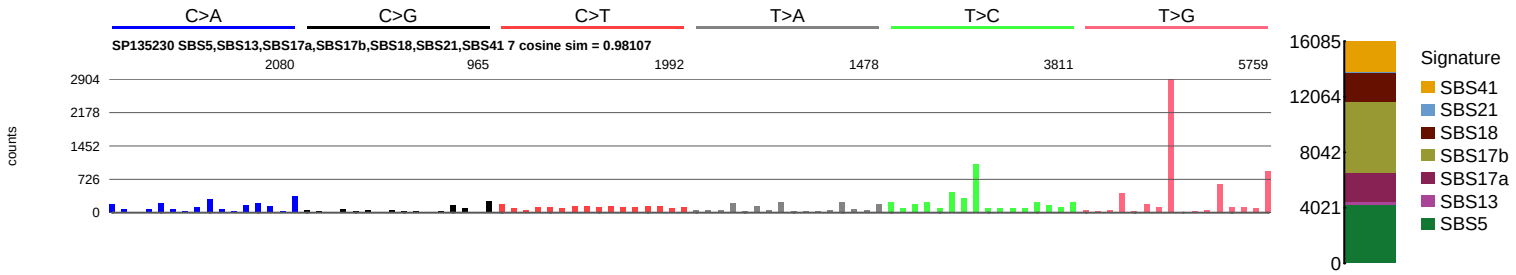
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98108
2081
964
1997
1478
3809
5758
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5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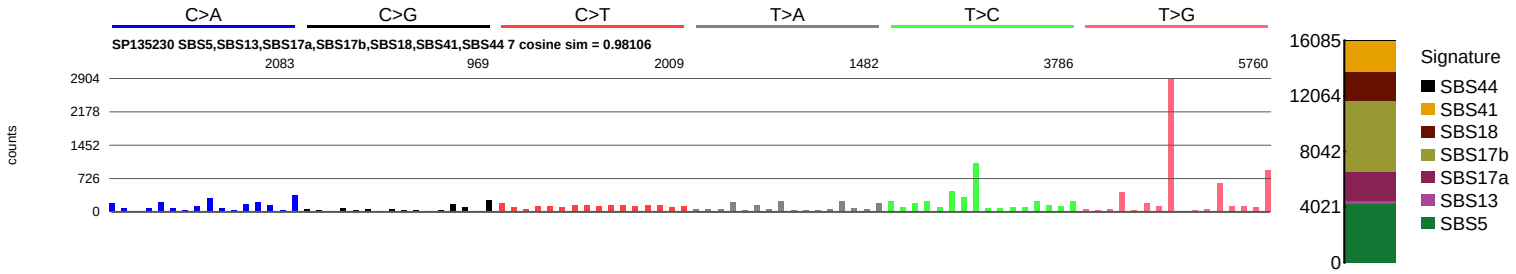
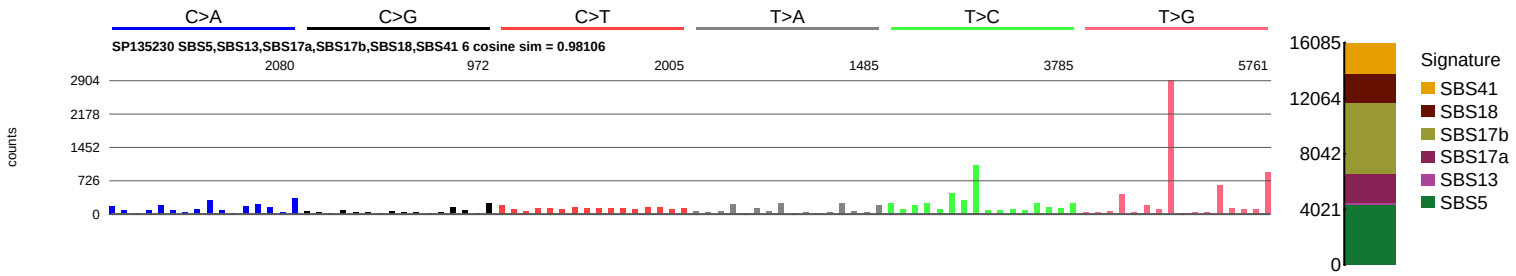
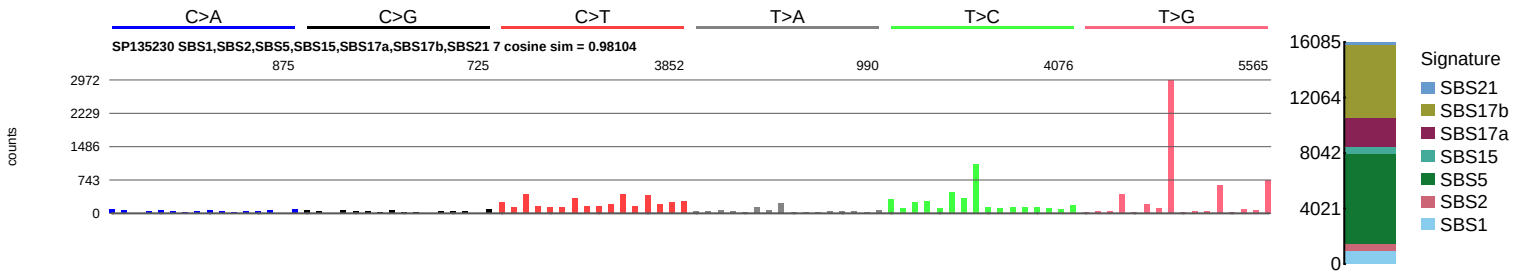
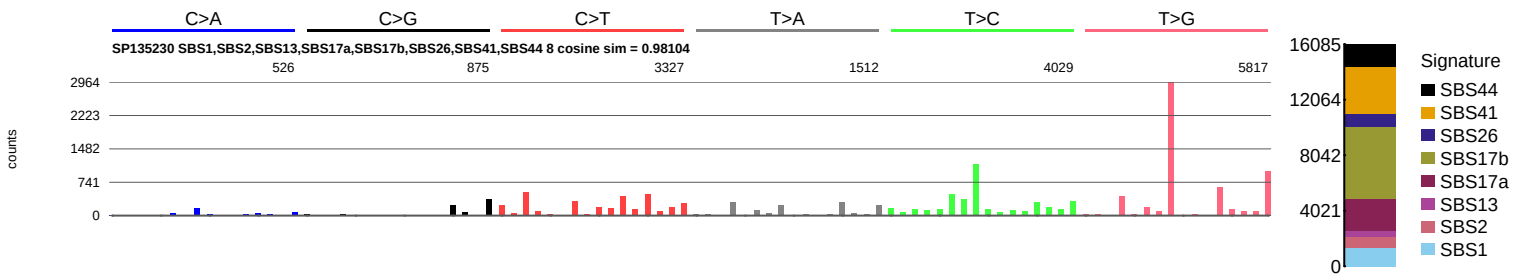
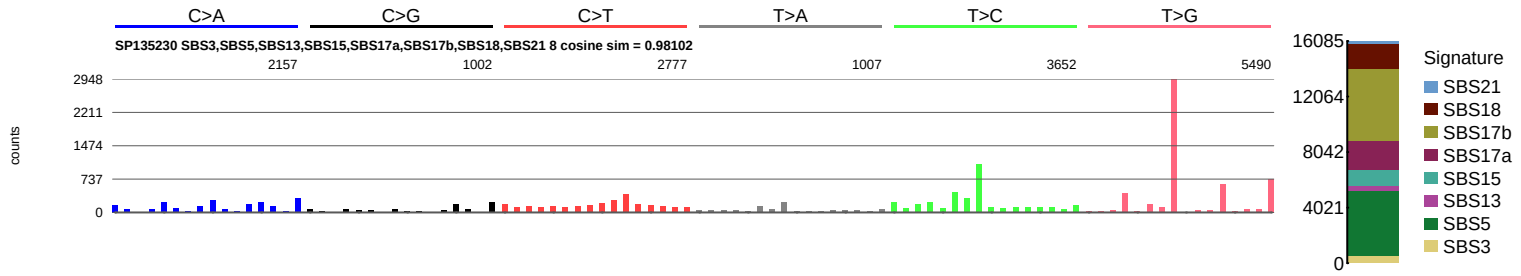
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98107
2080
965
1992
1478
3811
5759
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5

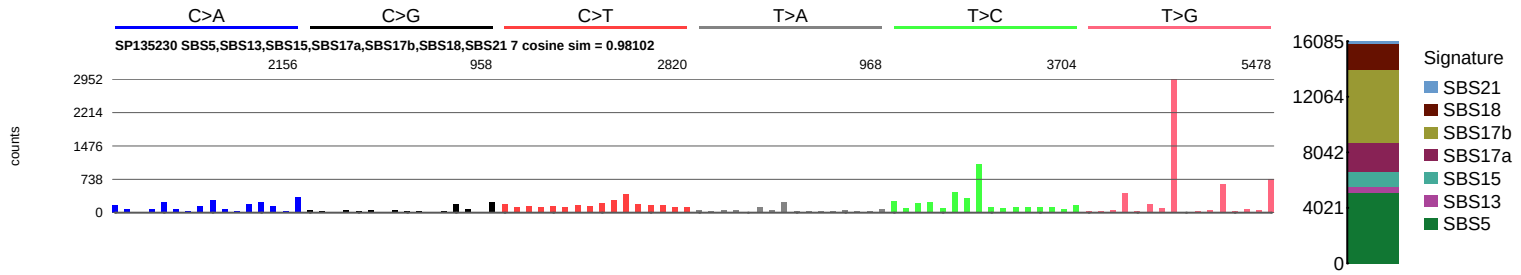

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98106
2083
969
2009
1482
3786
5760
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5

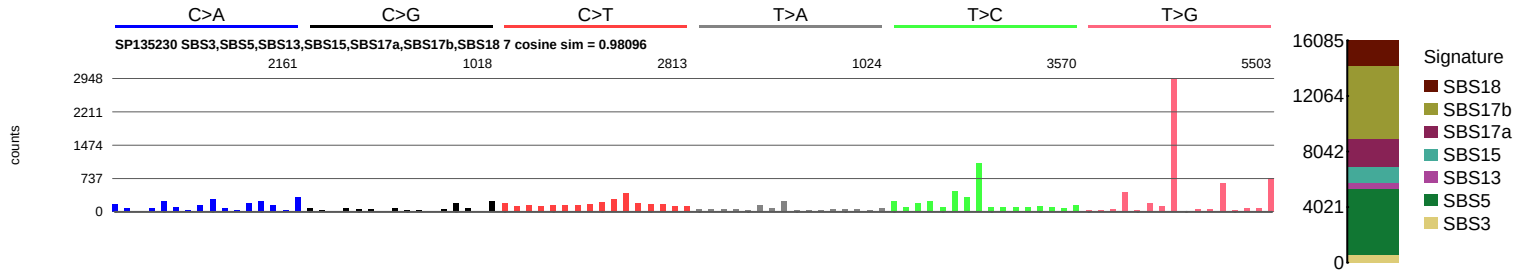

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98106
2080
972
2005
1485
3785
5761
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5

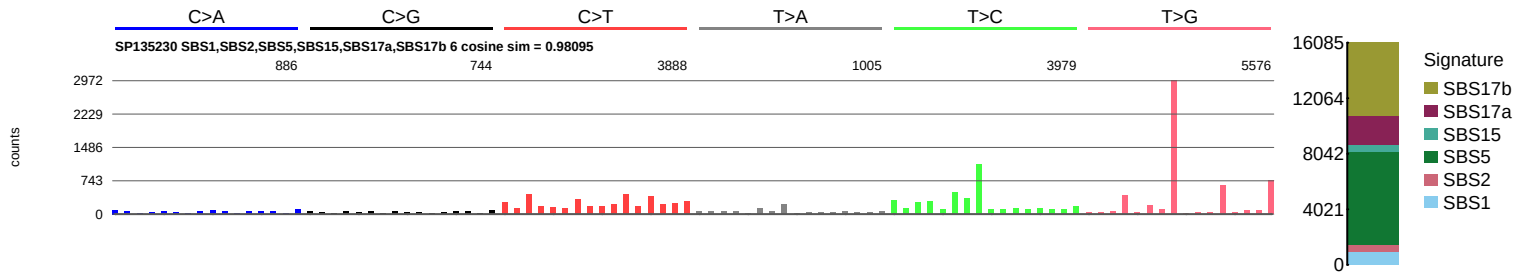

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98104
875
725
3852
990
4076
5565
counts
2972
2229
1486
743
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98104
526
875
3327
1512
4029
5817
counts
2964
2223
1482
741
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98102
2157
1002
2777
1007
3652
5490
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98102
2156
958
2820
968
3704
5478
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98096
2161
1018
2813
1024
3570
5503
counts
2948
2211
1474
737
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98095
886
744
3888
1005
3979
5576
counts
2972
2229
1486
743
0
16085
12064
8042
4021
0
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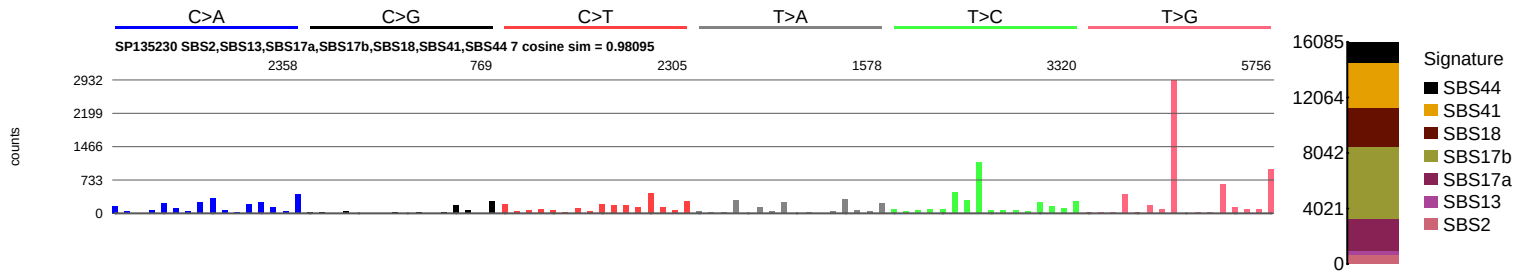
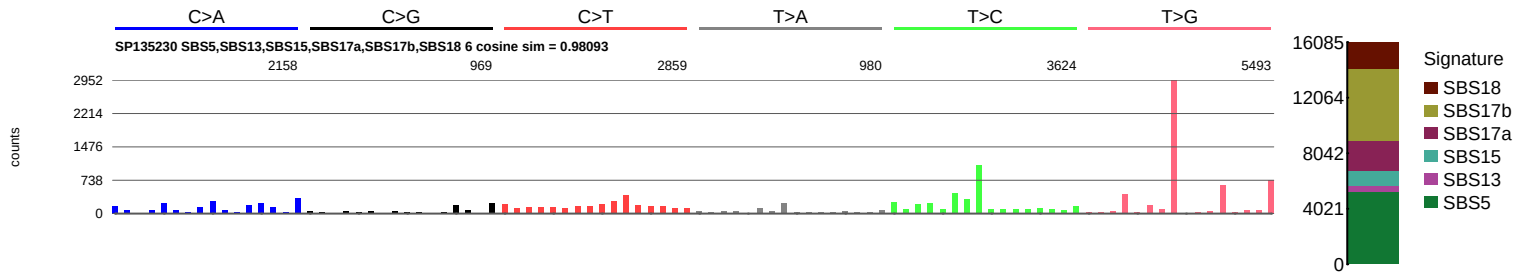
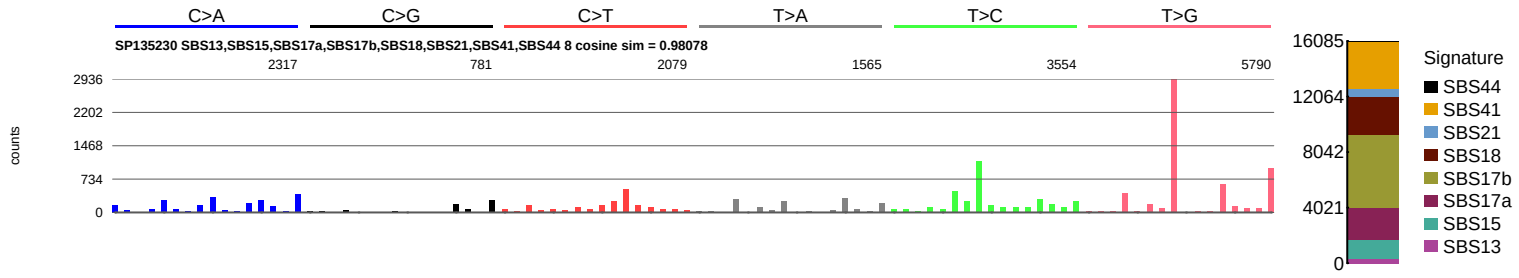
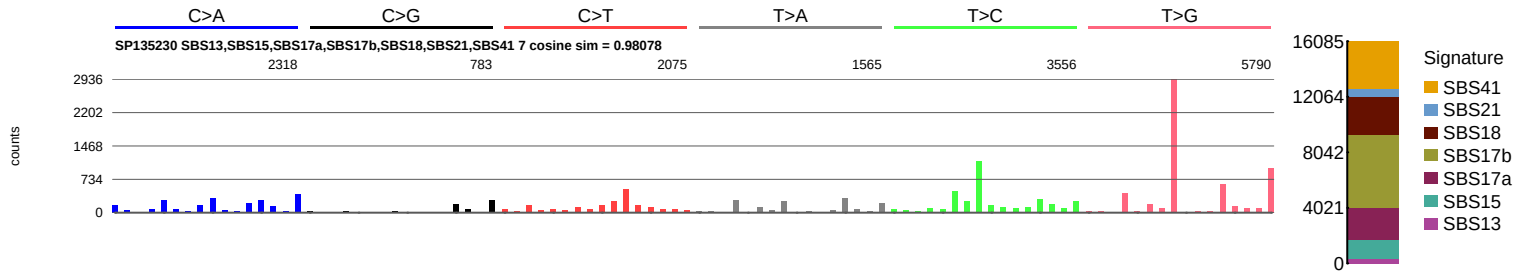
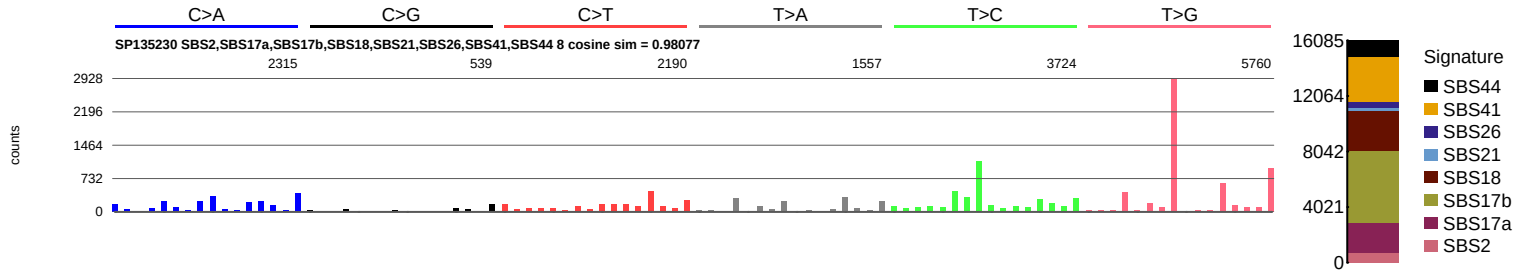
SBS2
SBS1

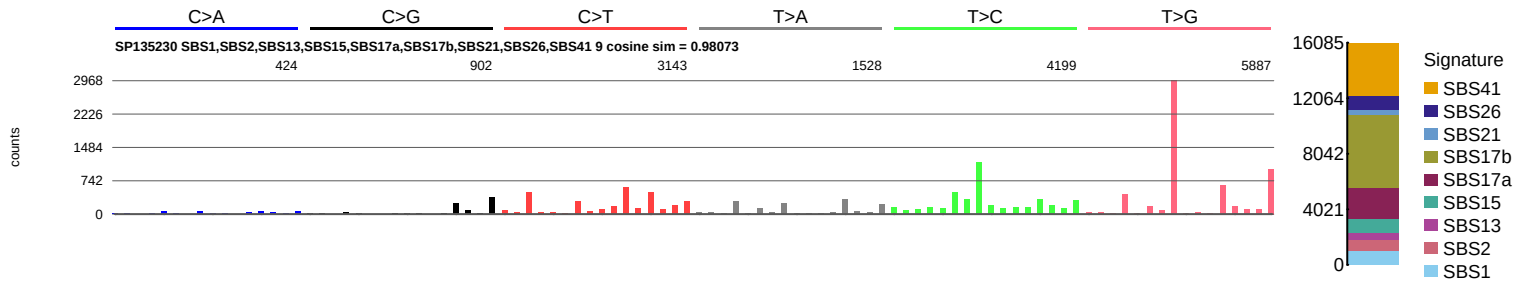

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98095
2358
769
2305
1578
3320
5756
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98093
2158
969
2859
980
3624
5493
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98078
2317
781
2079
1565
3554
5790
counts
2936
2202
1468
734
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98078
2318
783
2075
1565
3556
5790
counts
2936
2202
1468
734
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98077
2315
539
2190
1557
3724
5760
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.98073
424
902
3143
1528
4199
5887
counts
2968
2226
1484
742
0
16085
12064
8042
4021
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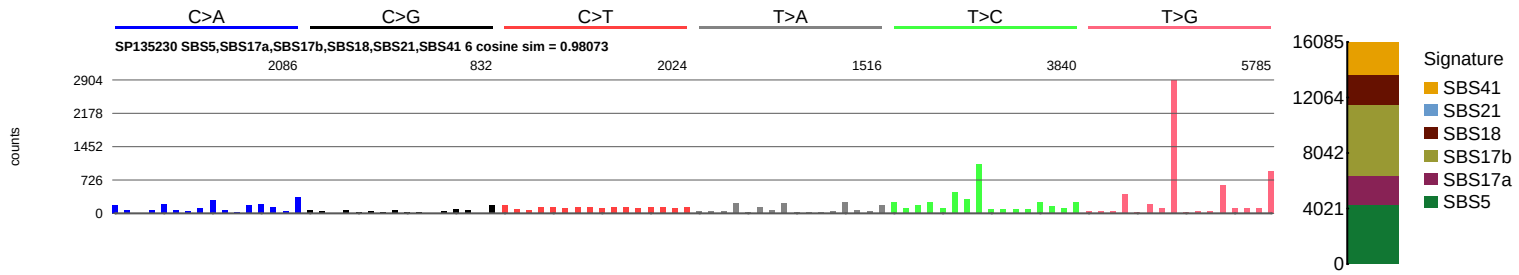
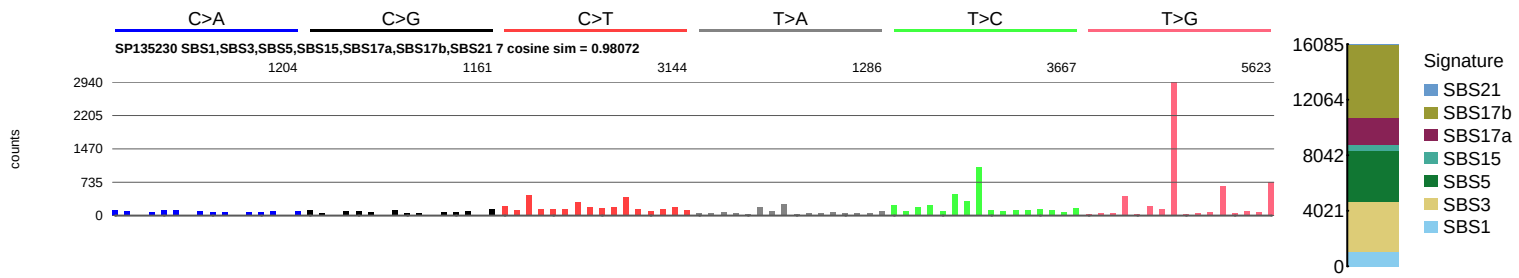
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98073
2086
832
2024
1516
3840
5785
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5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C>G
C>T
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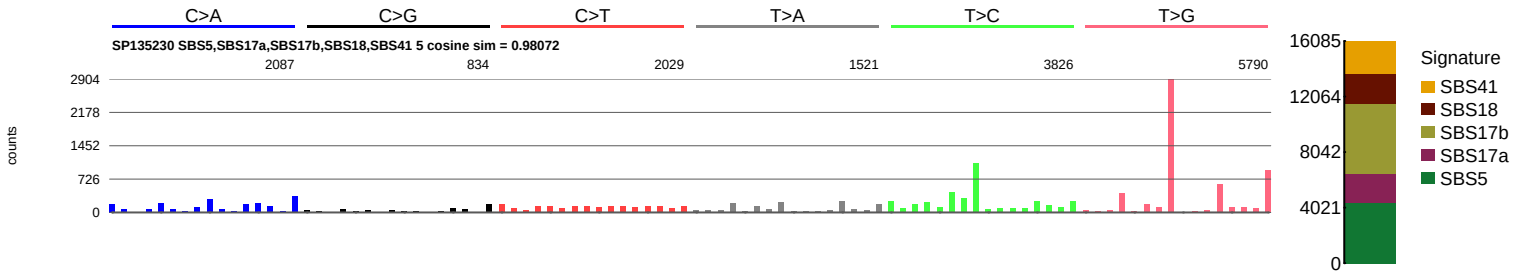
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98072
1204
1161
3144
1286
3667
5623
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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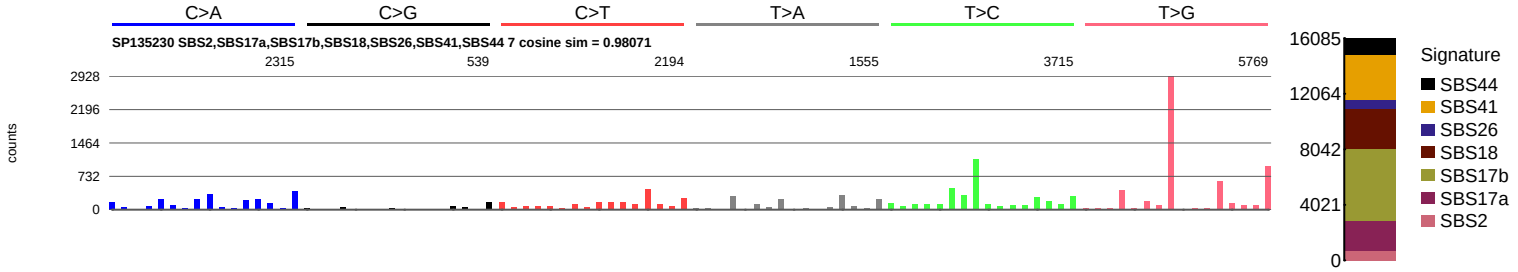
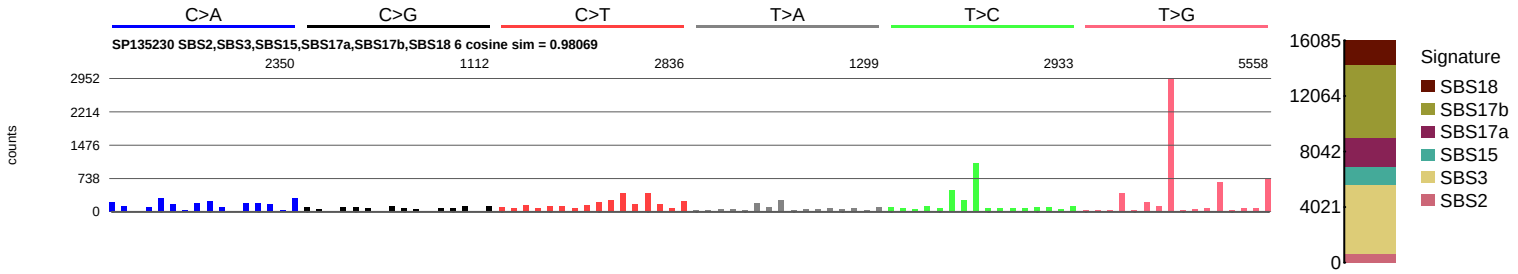
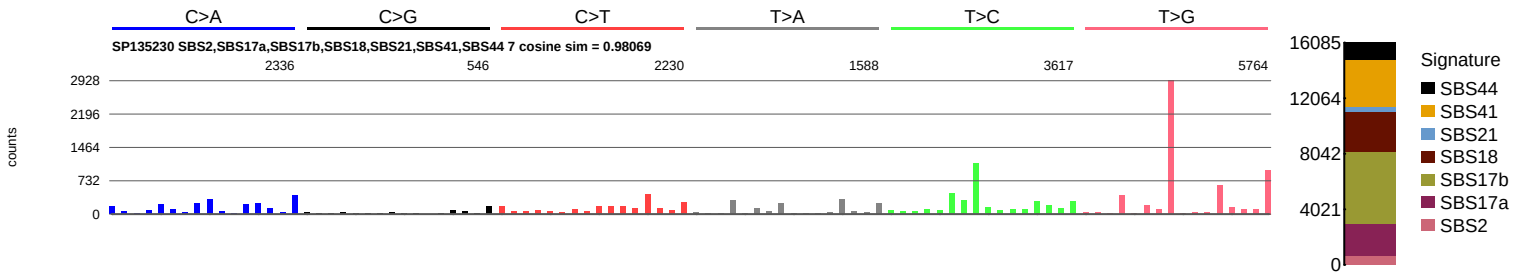
C>G
C>T
T>A
T>C
T>G
SP135230 SBS5,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98072
2087
834
2029
1521
3826
5790
counts
2904
2178
1452
726
0
16085
12064
8042
4021
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98071
2315
539
2194
1555
3715
5769
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98069
2350
1112
2836
1299
2933
5558
counts
2952
2214
1476
738
0
16085
12064
8042
4021
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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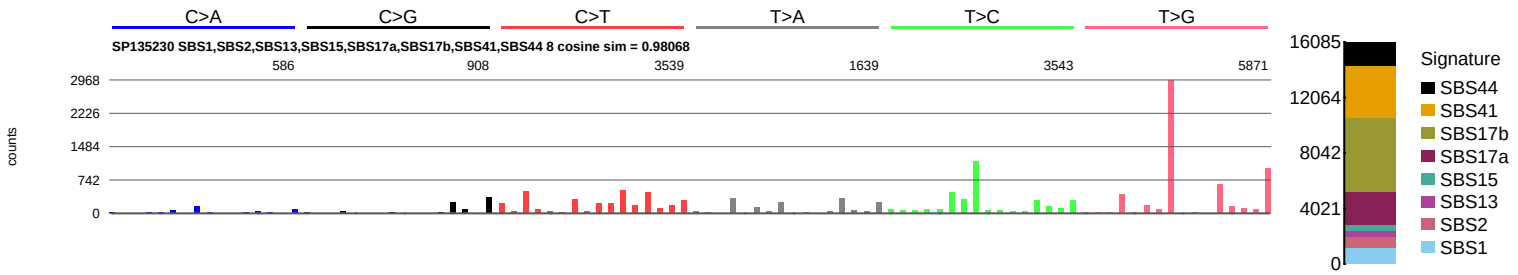
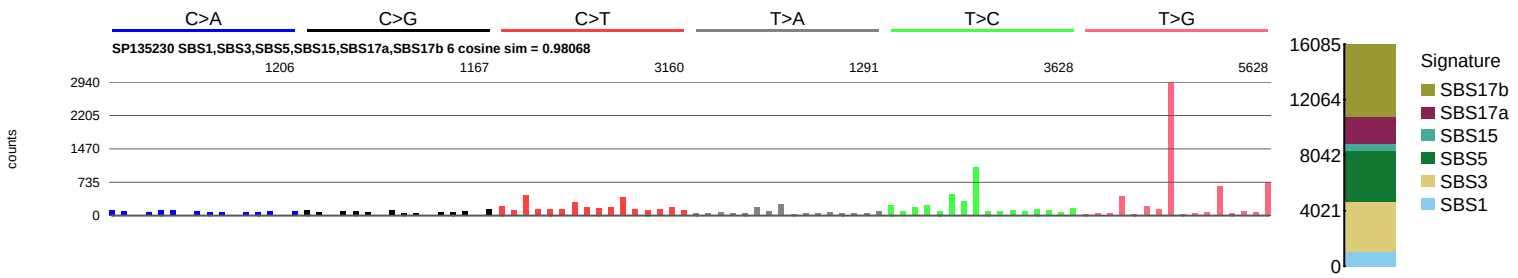
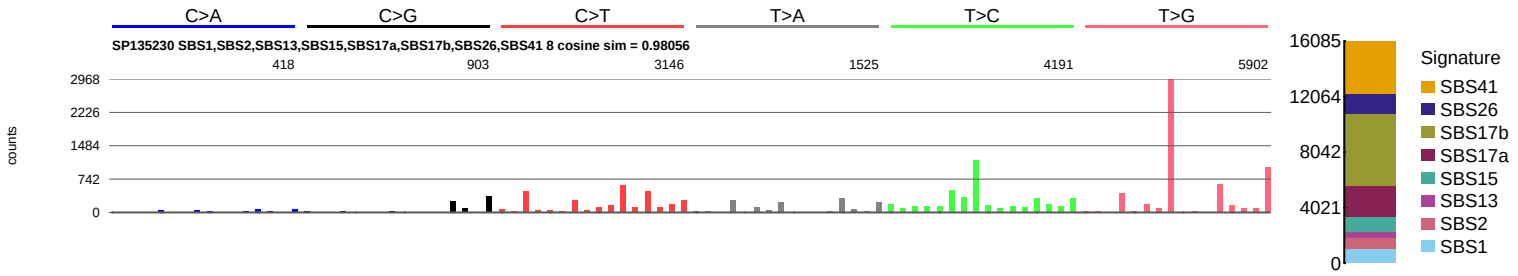
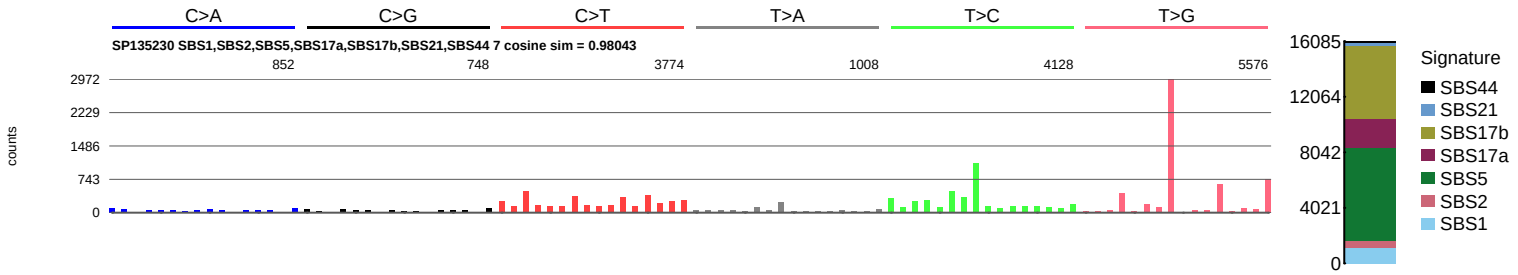
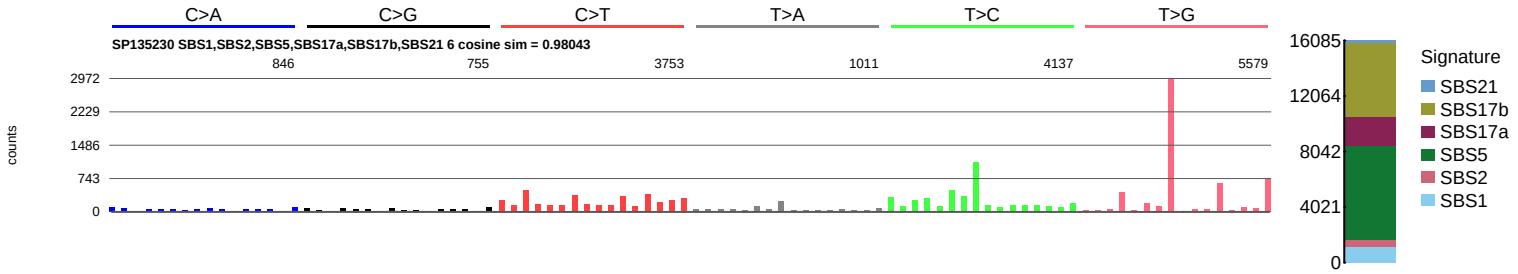
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98069
2336
546
2230
1588
3617
5764
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2

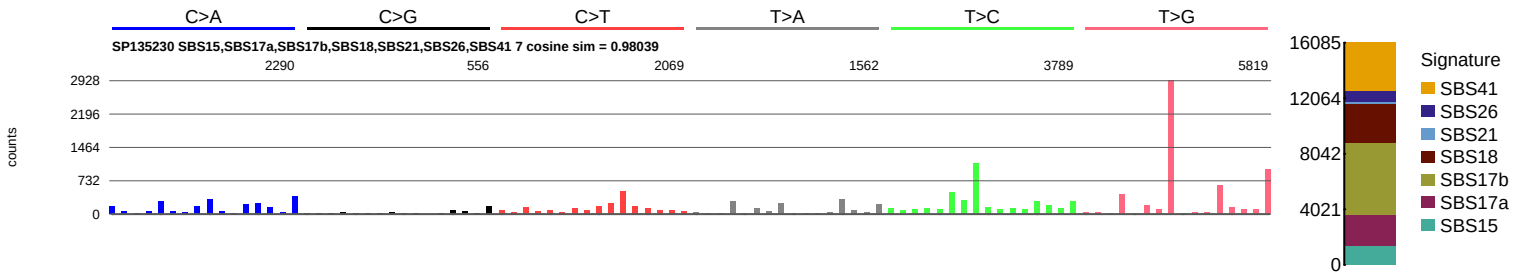

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98068
586
908
3539
1639
3543
5871
counts
2968
2226
1484
742
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

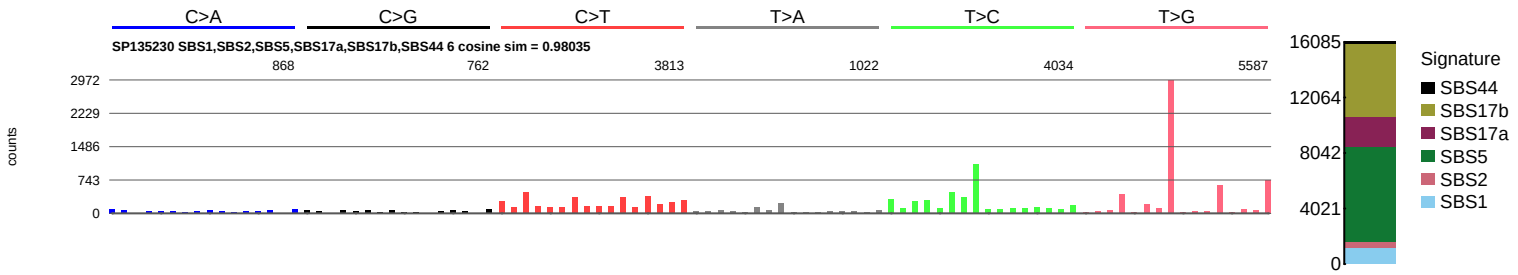

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98068
1206
1167
3160
1291
3628
5628
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

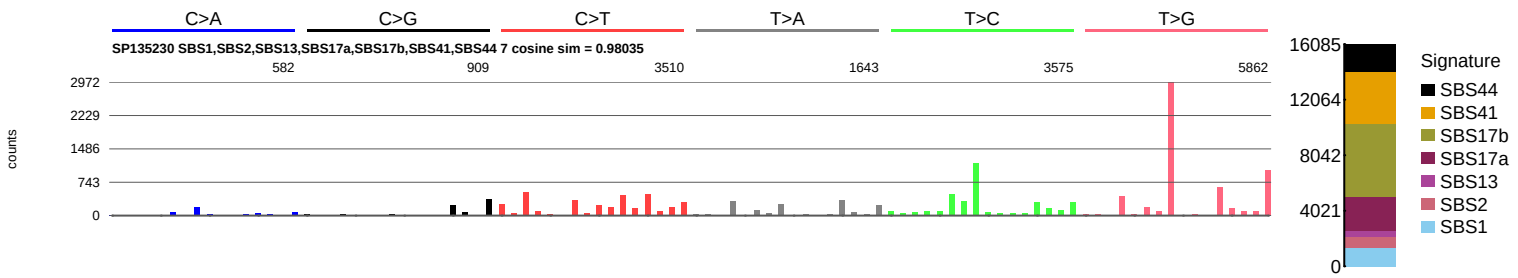

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98056
418
903
3146
1525
4191
5902
counts
2968
2226
1484
742
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

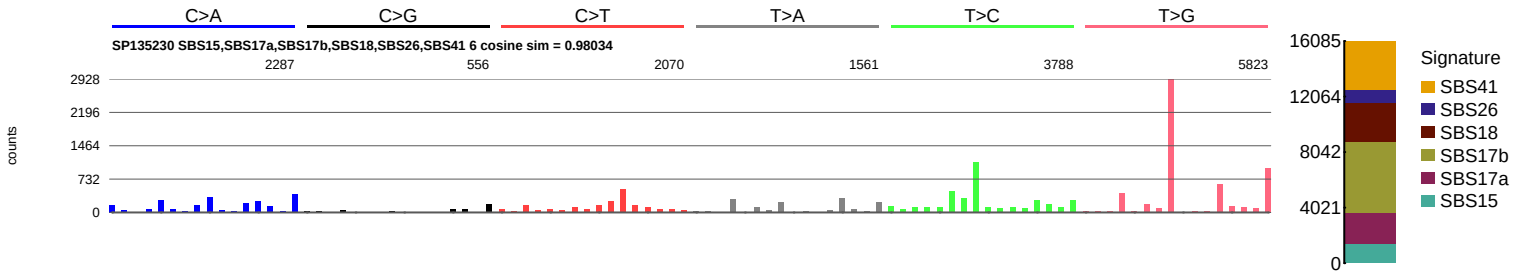

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98043
852
748
3774
1008
4128
5576
counts
2972
2229
1486
743
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

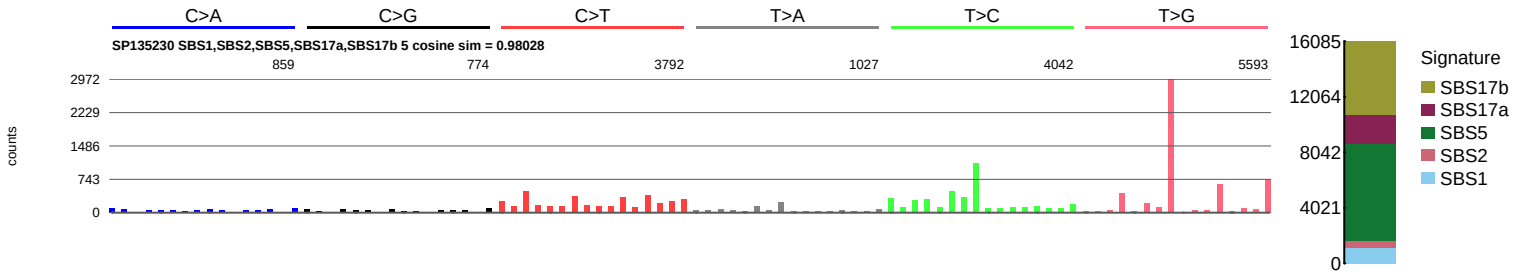

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98043
846
755
3753
1011
4137
5579
counts
2972
2229
1486
743
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

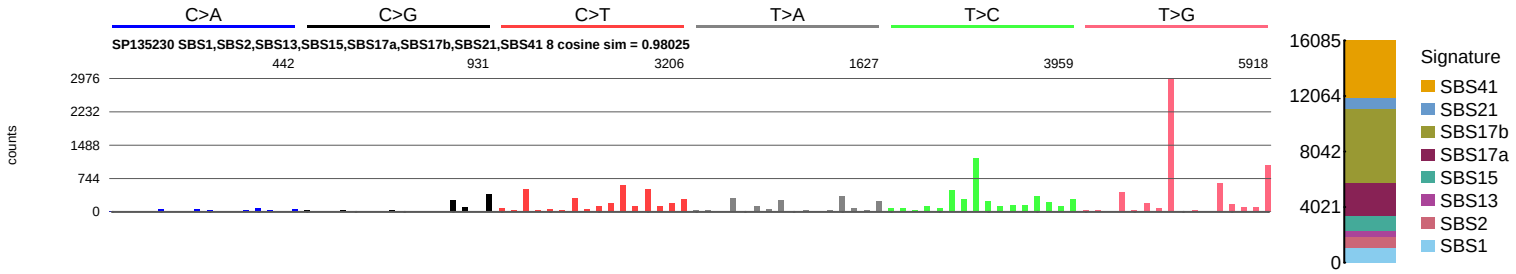

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.98039
2290
556
2069
1562
3789
5819
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15

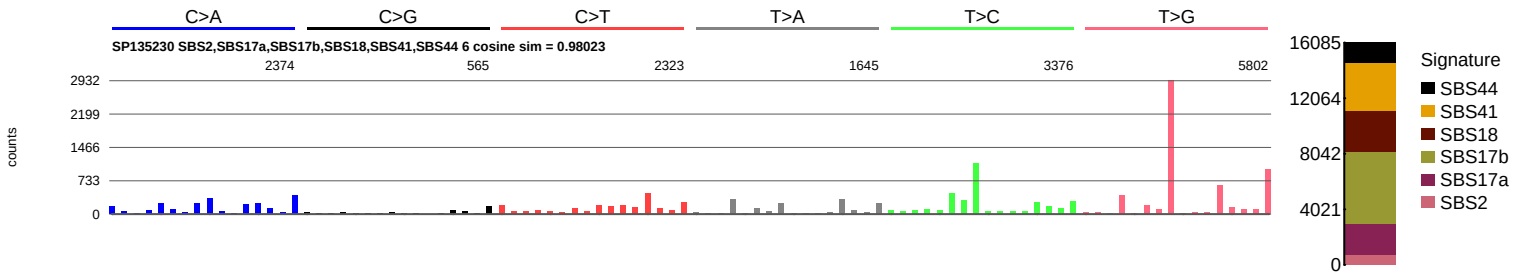

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98035
868
762
3813
1022
4034
5587
counts
2972
2229
1486
743
0
16085
12064
8042
4021
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS2
SBS1

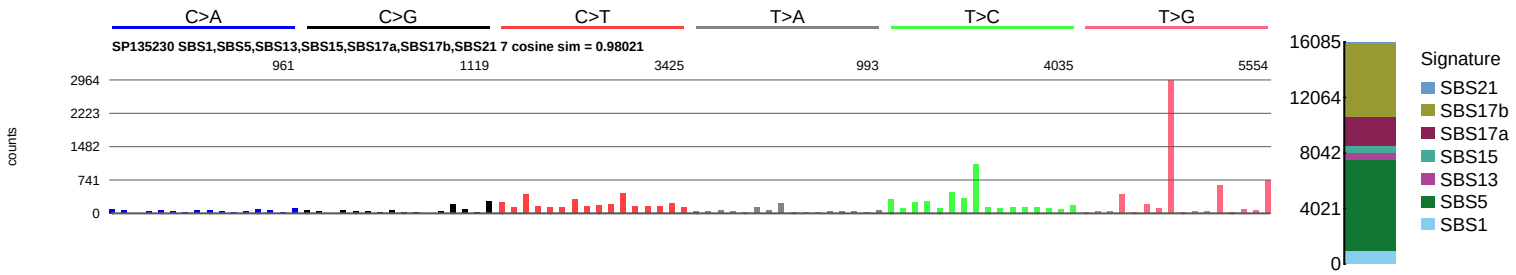

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98035
582
909
3510
1643
3575
5862
counts
2972
2229
1486
743
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS2
SBS1

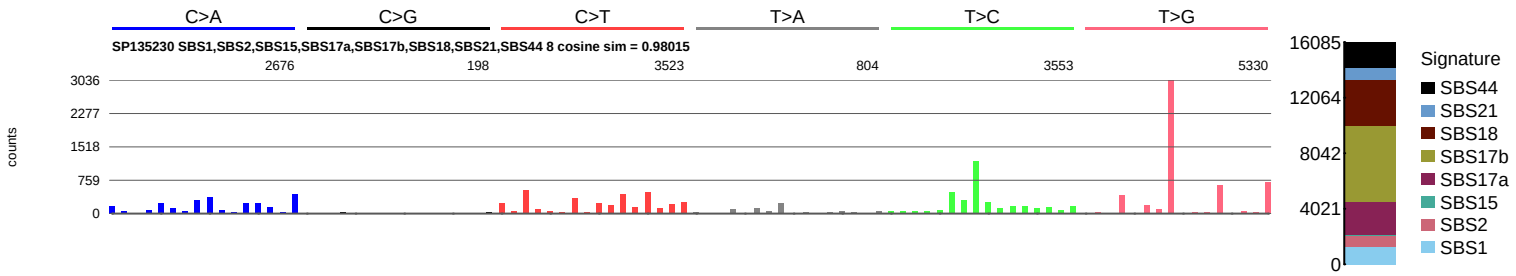

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98034
2287
556
2070
1561
3788
5823
counts
2928
2196
1464
732
0
16085
12064
8042
4021
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15

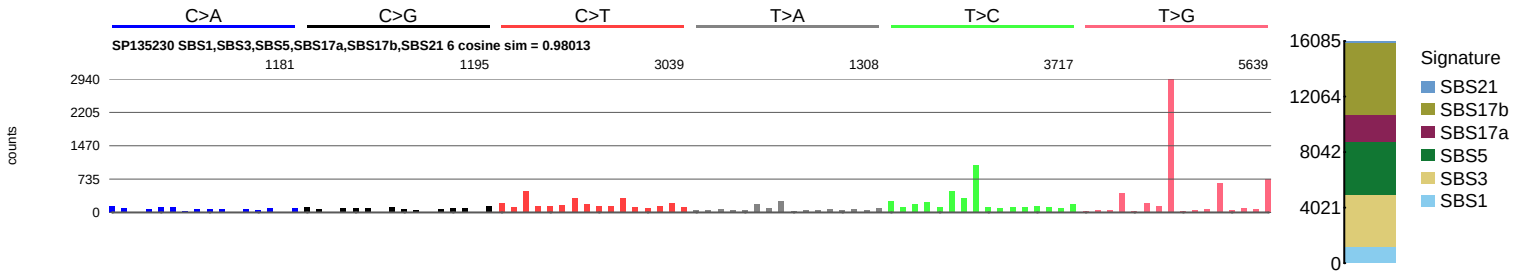

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS5,SBS17a,SBS17b 5 cosine sim = 0.98028
859
774
3792
1027
4042
5593
counts
2972
2229
1486
743
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS5
SBS2
SBS1

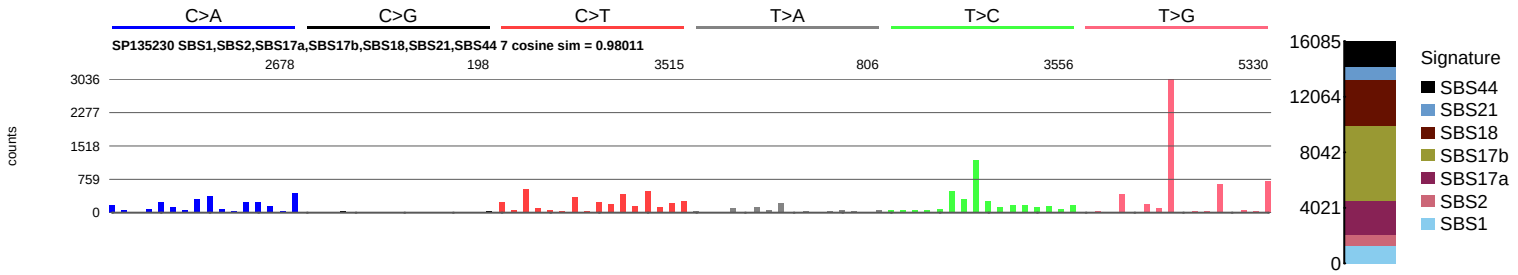

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98025
442
931
3206
1627
3959
5918
counts
2976
2232
1488
744
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

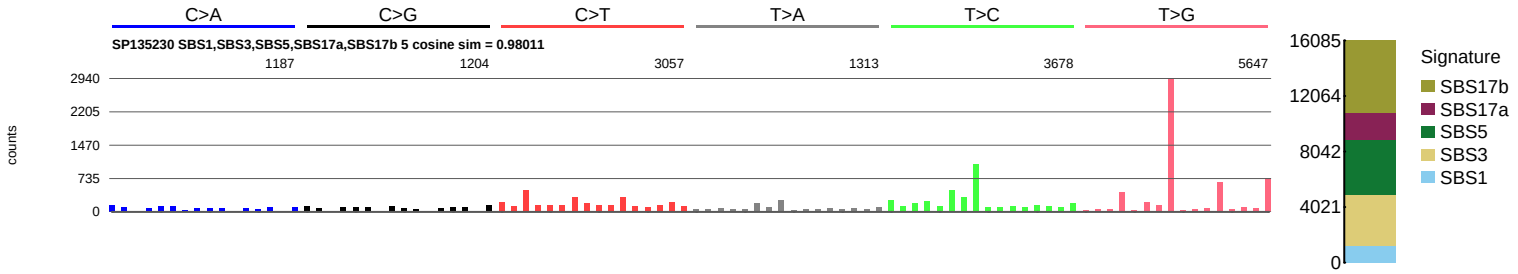

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98023
2374
565
2323
1645
3376
5802
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2

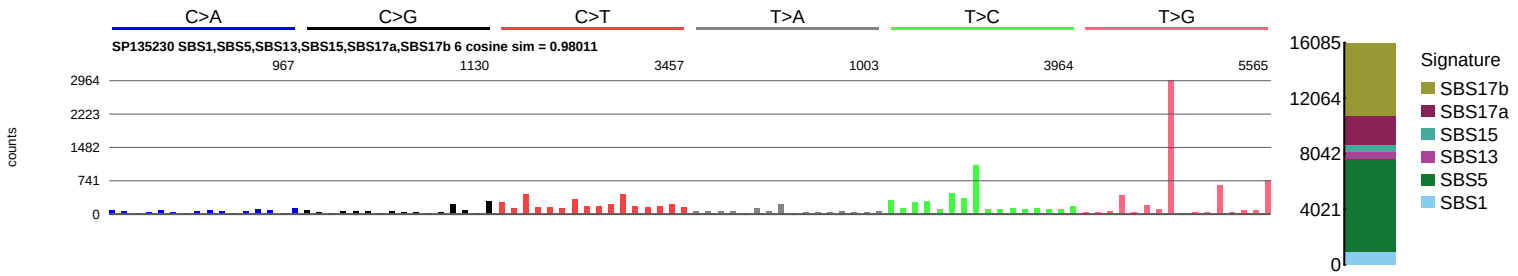

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98021
961
1119
3425
993
4035
5554
counts
2964
2223
1482
741
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

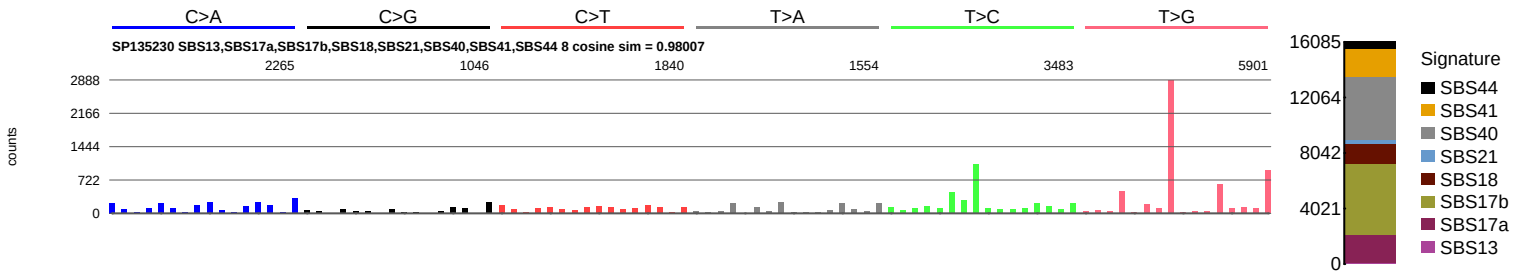

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98015
2676
198
3523
804
3553
5330
counts
3036
2277
1518
759
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

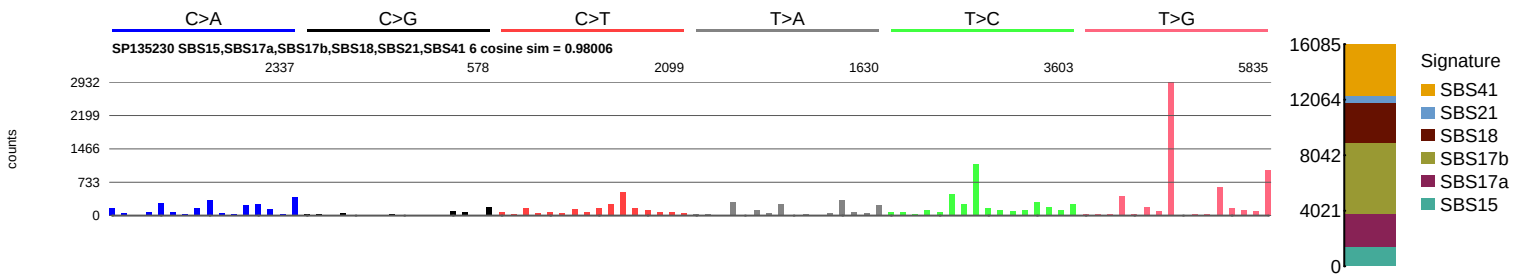

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98013
1181
1195
3039
1308
3717
5639
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

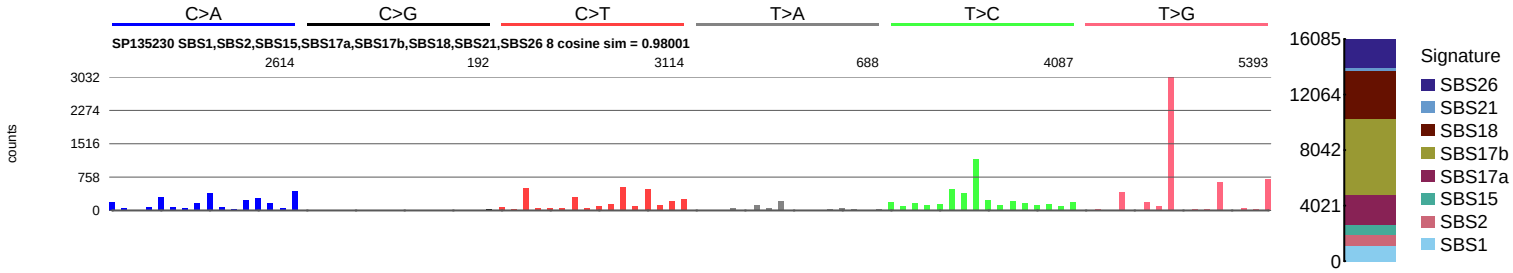

C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98011
2678
198
3515
806
3556
5330
counts
3036
2277
1518
759
0
16085
12064
8042
4021
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS3,SBS5,SBS17a,SBS17b 5 cosine sim = 0.98011
1187
1204
3057
1313
3678
5647
counts
2940
2205
1470
735
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98011
967
1130
3457
1003
3964
5565
counts
2964
2223
1482
741
0
16085
12064
8042
4021
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98007
2265
1046
1840
1554
3483
5901
counts
2888
2166
1444
722
0
16085
12064
8042
4021
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98006
2337
578
2099
1630
3603
5835
counts
2932
2199
1466
733
0
16085
12064
8042
4021
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP135230 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98001
2614
192
3114
688
4087
5393
3032
2274
1516
758
0
counts
16085
12064
8042
4021
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1